FOLDER ZERO

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- L2 - Head losses in straight tubes – Friction factor for steel pipes
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- M1 - Measurements of heads
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O - COMPRESSORS
- O1 - Dynamic and positive displacement compressors – General areas of application
- O2 - Centrigual compressor polytropic efficiency estimation
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P - PROPERTIES RELATIVE TO THE PETROLEUM CUTS
- P1 - Mean boiling point temperatures for petroleum cuts
- P2 - Characterization factor or hydrocarbons and petroleum cuts (K_{UOP})
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Q - DISTILLATION - ABSORPTION – STRIPPING
- Q1 - Gilliland's correlation
- Q2 - Tray efficiency – For distillation columns and absorption columns
- Q3 - Absorption (or stripping) ratio versus - Absorption (or stripping) factor

S - SAFETY
- S1 - Flammability

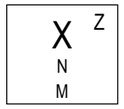
T - PHYSICAL QUANTITIES AND SYSTEMS OF UNITS
- T2 - Temperature conversion table

Y - SPECIFICATIONS OF PETROLEUM PRODUCTS
- Y - Specifications of Jet A-1

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PERIODIC TABLE OF ELEMENTS

Z : Atomic number
X : Symbol
N : Name
M : Atomic mass



(...) : Masses in parentheses are mass numbers of the most stable or best-known isotopes

Period	I A	II A	III B	IV B	V B	VI B	VII B	VIII			I B	II B	III A	IV A	V A	VI A	VII A	VIII A
1	H 1 Hydrogen 1,01																	He 2 Helium 4,0
2	Li 3 Lithium 6,94	Be 4 Beryllium 9,01											B 5 Boron 10,8	C 6 Carbon 12,0	N 7 Nitrogen 14,0	O 8 Oxygen 16,0	F 9 Fluorine 19,0	Ne 10 Neon 20,2
3	Na 11 Sodium 23,0	Mg 12 Magnesium 24,3											Al 13 Aluminum 27,0	Si 14 Silicon 28,1	P 15 Phosphorus 31,0	S 16 Sulfur 32,1	Cl 17 Chlorine 35,5	Ar 18 Argon 39,9
4	K 19 Potassium 39,1	Ca 20 Calcium 40,1	Sc 21 Scandium 45,0	Ti 22 Titanium 47,9	V 23 Vanadium 50,9	Cr 24 Chromium 52,0	Mn 25 Manganese 54,9	Fe 26 Iron 55,8	Co 27 Cobalt 58,9	Ni 28 Nickel 58,7	Cu 29 Copper 63,5	Zn 30 Zinc 65,4	Ga 31 Gallium 69,7	Ge 32 Germanium 72,6	As 33 Arsenic 74,9	Se 34 Selenium 79,0	Br 35 Bromine 79,9	Kr 36 Krypton 83,6
5	Rb 37 Rubidium 85,5	Sr 38 Strontium 87,5	Y 39 Yttrium 88,9	Zr 40 Zirconium 91,2	Nb 41 Niobium 92,9	Mo 42 Molybdenum 95,9	Tc 43 Technetium (99,0)	Ru 44 Ruthenium 101,1	Rh 45 Rhodium 102,9	Pd 46 Palladium 106,4	Ag 47 Silver 107,9	Cd 48 Cadmium 112,4	In 49 Indium 114,8	Sn 50 Tin 118,7	Sb 51 Antimony 121,6	Te 52 Tellurium 127,5	I 53 Iodine 126,9	Xe 54 Xenon 131,3
6	Cs 55 Cesium 132,9	Ba 56 Barium 137,3	57 à 71 Lanthanides	Hf 72 Hafnium 178,5	Ta 73 Tantalum 180,9	W 74 Wolfram 183,9	Re 75 Rhenium 186,2	Os 76 Osmium 190,2	Ir 77 Iridium 192,2	Pt 78 Platinum 195,1	Au 79 Gold 197,0	Hg 80 Mercury 200,6	Tl 81 Thallium 204,4	Pb 82 Lead 207,2	Bi 83 Bismuth 209,9	Po 84 Polonium (210)	At 85 Astatine (210)	Rn 86 Radon (222)
7	Fr 87 Francium (223)	Ra 88 Radium 226,1	89 à 103 Actinides	Rf 104 Rutherfordium (261)	Ha 105 Hahnium (262)	106 — (263)	107 —											

Transition elements

Lanthanides	La 57 Lanthanum 138,9	Ce 58 Cerium 140,1	Pr 59 Praseodymium 140,9	Nd 60 Neodymium 144,2	Pm 61 Promethium (145)	Sm 62 Samarium 150,4	Eu 63 Europium 152,0	Gd 64 Gadolinium 157,3	Tb 65 Terbium 158,9	Dy 66 Dysprosium 162,5	Ho 67 Holmium 164,9	Er 57 Erbium 167,8	Tm 69 Thulium 168,9	Yb 70 Ytterbium 173,0	Lu 71 Lutetium 176,0
Actinides	Ac 89 Actinium 227	Th 90 Thorium 232,0	Pa 91 Protactinium 231	U 92 Uranium 238,0	Np 93 Neptunium 237	Pu 94 Plutonium (242)	Am 95 Americium (243)	Cm 96 Curium (247)	Bk 97 Berkelium (247)	Cf 98 Californium (249)	Es 99 Einsteinium (254)	Fm 100 fermium (255)	Md 101 Mendelevium (256)	No 102 Nobelium (254)	Lr 103 Lawrencium (257)

D CHI 3042 D

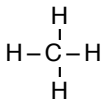
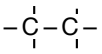
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PHYSICAL CHARACTERISTICS OF HYDROCARBONS AND OF SULFUR, NITROGEN AND OXYGEN COMPOUNDS

LINEAR ALKANES or n-paraffins 1
BRANCHED ALKANES or isoparaffins 2
ALKENES or olefins 4
DI-ALKENES or diolefins 6
ALKYNS 7
CYCLOALKANES or naphthenes 9
AROMATICS 11
POLYAROMATICS 13
NITROGEN COMPOUNDS 14
SULFUR COMPOUNDS 16
OXYGEN COMPOUNDS 18
Alcohols 18
Acids 19
Phenols 21

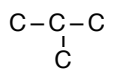
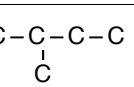
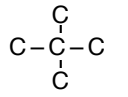
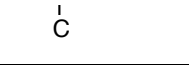
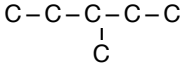
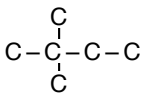
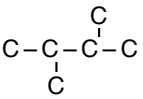
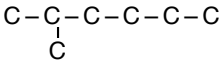
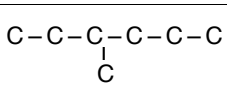
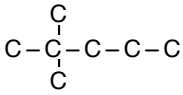
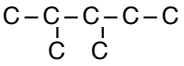
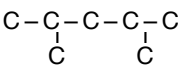
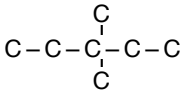
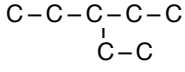
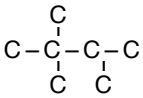
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LINEAR ALKANES
or n-paraffins

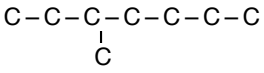
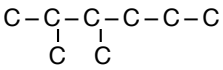
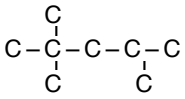
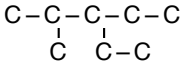
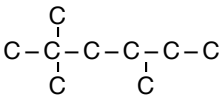
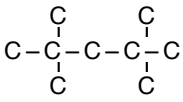
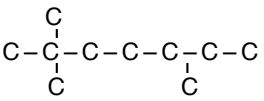
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Methane	CH_4		16.043	- 182.5	- 161.5	425 (-161.5)
Ehane	C_2H_6		30.070	- 183.3	- 88.7	548 (- 90)
Propane	C_3H_8	C – C – C	44.097	- 187.7	- 42.1	582 (- 42)
n-butane	C_4H_{10}	C – C – C – C	58.124	- 138.4	- 0.5	579 (20)
n-pentane	C_5H_{12}	C – C – C – C – C	72.151	- 129.8	36	626 (20)
n-hexane	C_6H_{14}	C – C – C – C – C – C	86.176	- 95.4	68.7	659 (20)
n-heptane	C_7H_{16}	C – C – C – C – C – C – C	100.205	- 90.6	98.4	684 (20)
n-octane	C_8H_{18}	C – C – C – C – C – C – C – C	114.232	- 56.8	125.6	703 (20)
n-nonane	C_9H_{20}	C – C – C – C – C – C – C – C – C	128.259	- 53.5	150.8	718 (20)
n-decane	$C_{10}H_{22}$	C – C – C – C – C – C – C – C – C – C	142.286	- 29.7	174.1	730 (20)
n-undecane	$C_{11}H_{24}$	C – C – C …… C – C – C (11)	156.313	- 25.6	195.9	740 (20)
n-dodecane	$C_{12}H_{26}$	C – C – C …… C – C – C (12)	170.340	- 9.6	216.3	748 (20)
n-tridecane	$C_{13}H_{28}$	C – C – C …… C – C – C (13)	184.367	- 5.4	235.4	756 (20)
n-tetradecane	$C_{14}H_{30}$	C – C – C …… C – C – C (14)	198.394	+ 5.8	253.5	763 (20)
pentadecane	$C_{15}H_{32}$	C – C – C …… C – C – C (15)	212.421	9.8	270.6	769 (20)
n-hexadecane	$C_{16}H_{34}$	C – C – C …… C – C – C (16)	226.448	17.8	286.8	773 (20)
n-heptadecane	$C_{17}H_{36}$	C – C – C …… C – C – C (17)	240.475	21.8	302	778 (20)
n-octadecane	$C_{18}H_{38}$	C – C – C …… C – C – C (18)	254.502	28.8	316.3	777 (28)
n-nonadecane	$C_{19}H_{40}$	C – C – C …… C – C – C (19)	268.529	31.8	329.9	789 (32)
n-eicosane	$C_{20}H_{42}$	C – C – C …… C – C – C (20)	282.556	36.8	343.8	775 (40)
Triacontane	$C_{30}H_{62}$	C – C – C …… C – C – C (30)	422.82	65.8	449.7	775 (78)
Tetracontane	$C_{40}H_{82}$	C – C – C …… C – C – C (40)	563	81.3		
Pentacontane	$C_{50}H_{102}$	C – C – C …… C – C – C (50)	703	92.3		

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BRANCHED
or isoparaffins

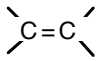
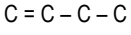
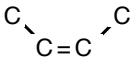
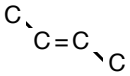
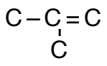
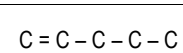
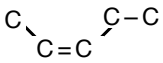
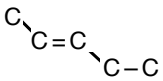
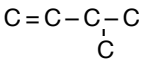
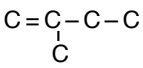
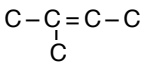
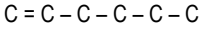
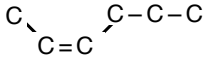
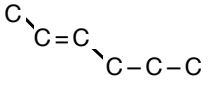
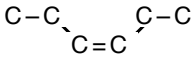
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Isobutane	C_4H_{10}		58.124	- 159.6	- 11.9	557 (20)
2-methyl butane (isopentane)	C_5H_{12}		72.151	- 159.9	27.8	620 (20)
2,2-dimethyl propane (neopentane)	C_5H_{12}		72.151	- 16.5	9.5	591 (20)
2-methyl pentane	C_6H_{14}		86.178	- 153.7	60.2	653 (20)
3-methyl pentane	C_6H_{14}		86.178	- 118.2	63.2	664 (20))
2,2-dimethyl butane	C_6H_{14}		86.178	- 99.9	49.7	649 (20)
2,3-dimethyl butane	C_6H_{14}		86.178	- 128.6	58	662 (20)
2-methyl hexane	C_7H_{16}		100.205	- 118.3	90	679 (20)
3-methyl hexane	C_7H_{16}		100.205	- 173.2	91.9	687 (20)
2,2-dimethyl pentane	C_7H_{16}		100.205	- 123.8	79.2	674 (20)
2,3-dimethyl pentane	C_7H_{16}		100.205		89.7	695 (20)
2,4-dimethyl pentane	C_7H_{16}		100.205	- 119.2	80.5	673 (20)
3,3-dimethyl pentane	C_7H_{16}		100.205	- 134.5	86	693 (20)
3-ethyl pentane	C_7H_{16}		100.205	- 118.6	93.4	698 (20)
2,2,3-trimethyl butane	C_7H_{16}		100.205	- 24.9	80.8	690 (20)

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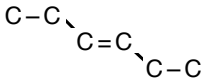
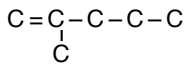
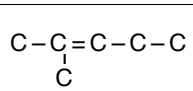
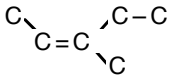
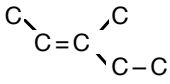
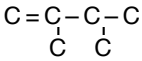
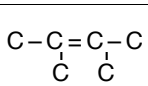
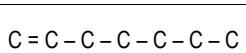
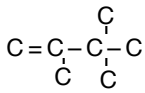
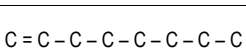
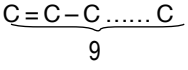
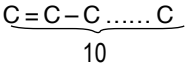
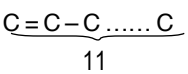
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
3-methyl heptane	C_8H_{18}		114.232	- 120.5	118.9	706 (20)
2,3-dimethyl hexane	C_8H_{18}		114.232		115.6	712 (20)
2,2,4-trimethyl pentane	C_8H_{18}		114.232	- 107.4	99.2	692 (20)
3-ethyl - 2-methyl pentane	C_8H_{18}		114.232	- 115	115.6	719 (20)
2,2,4-trimethyl hexane	C_9H_{20}		128.259	-120.2	126.5	720 (16)
2,2,4,4-tetramethyl pentane	C_9H_{20}		128.259	- 67.2	122.2	719 (20)
2,2.5-trimethyl heptane	$C_{10}H_{22}$		142.286		155.6	725 (20)

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ALKENES
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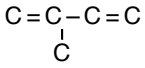
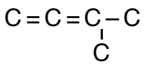
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Ethylene	C_2H_4		28	- 169.1	- 103.7	577 (- 110)
Propylene	C_3H_6		42.1	- 185.2	- 47.7	612 (-50)
1-Butene	C_4H_8		56.1	- 185.3	- 6.3	595 (20)
2-butene (cis)	C_4H_8		56.1	- 138.8	3.7	621 (20)
2-butene (trans)	C_4H_8		56.1	- 105.5	0.8	604 (20)
isobutene	C_4H_8		56.1	- 140.3	- 6.8	594 (20)
1-pentene	C_5H_{10}		70.1	- 165.2	30	640 (20)
2-pentene (cis)	C_5H_{10}		70.1	- 151.3	36.9	656 (20)
2-pentene (trans)	C_5H_{10}		70.1	- 140.2	36.3	649 (20)
3-methyl 1-butene	C_5H_{10}		70.1	- 163.4	20.1	627 (20)
2-methyl 1-butene	C_5H_{10}		70.1	- 137.5	31.1	650 (20)
2-methyl 2-butene	C_5H_{10}		70.1	- 133.7	38.5	662 (20)
1-hexene	C_6H_{12}		84.1	- 139.8	62.8	673 (20)
2-hexene (cis)	C_6H_{12}		84.1	- 141.1	68.8	687 (20)
2-hexene (trans)	C_6H_{12}		84.1	- 133.1	67.85	678 (20)
3-hexene (cis)	C_6H_{12}		84.1	- 137.8	66.4	680 (20)

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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
3-hexene (trans)	C_6H_{12}		84.1	- 113.4	67.1	677 (20)
2-methyl 1-pentene	C_6H_{12}		84.1	- 135.7	60.7	680 (20)
2-methyl 2-pentene	C_6H_{12}		84.1	- 135	67.3	691 (16)
3-methyl 2-pentene (cis)	C_6H_{12}		84.1	- 134.8	67.7	694 (20)
3-methyl 2-pentene (trans)	C_6H_{12}		84.1	- 138.4	70.4	698 (20)
2,3-dimethyl 1-butene	C_6H_{12}		84.1	- 157.2	55.6	678 (20)
2,3-dimethyl 2-butene	C_6H_{12}		84.1	- 74.2	73.2	708 (20)
1-heptene	C_7H_{14}		98.2	- 118.8	93.6	697 (20)
2,3,3-trimethyl 1-butene	C_7H_{14}		98.2	- 109.8	77.8	705 (20)
1-octene	C_8H_{18}		112.2	- 101.7	121.2	715 (20)
1-nonene	C_9H_{18}		126.2	- 81.3	146.8	745 (0)
1-decene	$C_{10}H_{20}$		140.3	- 66.2	170.5	741 (20)
1-undecene	$C_{11}H_{22}$		154.3	- 49.1	192.6	751 (20)

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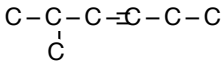
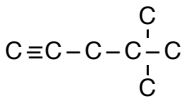
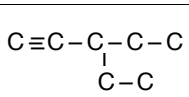
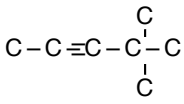
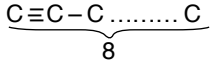
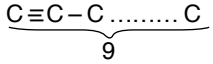
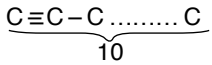
DI-ALCENES
or diolefines

Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Propadiene (allene)	C_3H_4	C = C = C	40.1	- 136	- 34.5	
1,2-butadiene	C_4H_6	C = C = C – C	54.1	- 136.2	10.8	652 (20)
1,3-butadiene	C_4H_6	C = C – C = C	54.1	- 108.9	- 4.4	621 (20)
2-methyl 1,3-butadiene (or isoprene)	C_5H_8		68.1	- 145.9	34.0	681 (20)
3-methyl 1,2-butadiene	C_5H_8		68.1	- 113.6	40.8	686 (20)
1,2-pentadiene	C_5H_8	C = C = C – C – C	68.1	- 137.2	44.8	693 (20)
1,3-pentadiene	C_5H_8	C = C – C = C – C	68.1	- 87.4	42.0	676 (20)
1,4-pentadiene	C_5H_8	C = C – C – C = C	68.1	- 148.2	25.9	661 (20)
2,3-pentadiene	C_5H_8	C – C = C = C – C	68.1	- 125.6	48.2	695 (20)

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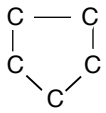
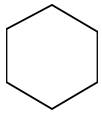
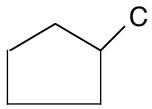
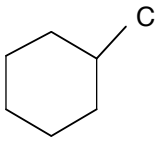
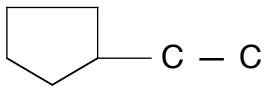
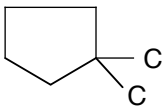
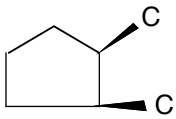
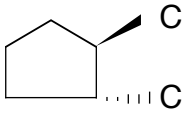
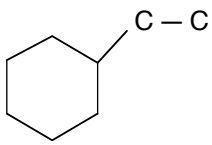
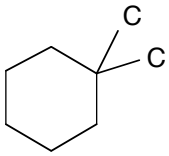
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Acetylene	C_2H_2	$C \equiv C$	26	- 80.7	- 84	615 (- 84)
Propyne (methyl acetylene)	C_3H_4	$C \equiv C - C$	40.1	- 102.6	- 23.1	706 (-50)
1-butyne	C_4H_6	$C \equiv C - C - C$	54.1	– 125.7	8.0	650 (16)
2-butyne (dimethyl acetylene)	C_4H_6	$C - C \equiv C - C$	54.1	- 32.2	27.0	691 (20)
1-pentyne	C_5H_8	$C \equiv C - C - C - C$	68.1	- 105.6	40.1	690 (20)
2-pentyne	C_5H_8	$C - C \equiv C - C - C$	68.1	- 101	56	710 (20)
3-methyl 1-butyne	C_5H_8	$C \equiv C - C(C) - C$	68.1	- 89.7	29.5	666 (20)
1-hexyne (or butyl acetylene)	C_6H_{10}	$C \equiv C - C - C - C - C$	82.1	- 131.9	71.3	715 (20)
Hexyne-2	C_6H_{10}	$C - C \equiv C - C - C - C$	82.1	- 89.6	84	731 (20)
Hexyne-3 (or diethyl acetylene)	C_6H_{10}	$C - C - C \equiv C - C - C$	82.1	- 103	81.5	723 (20)
3-methyl 1-pentyne	C_6H_{10}	$C \equiv C - C(C) - C - C$	82.1			
4-methyl 1-pentyne	C_6H_{10}	$C \equiv C - C - C(C) - C$	82.1	- 105	61	709 (15)
4-methyl 2-pentyne	C_6H_{10}	$C - C \equiv C - C(C) - C$	82.1	- 110.4	72.5	716 (19)
3,3-dimethyl 1-butyne	C_6H_{10}	$C \equiv C - C(C)(C) - C$	82.1	- 81.2	39	669 (20)
1-heptyne	C_7H_{12}	$C \equiv C - C - C - C - C - C$	96.2	- 81	99.7	733 (20)
2-heptyne	C_7H_{12}	$C - C \equiv C - C - C - C - C$	96.2	-	112	748 (20)
3-heptyne	C_7H_{12}	$C - C - C \equiv C - C - C - C$	96.2	-	105	753 (20)
5-methyl 2-hexyne	C_7H_{12}	$C - C \equiv C - C - C(C) - C$	96.2	- 92.9	102.5	738 (20)

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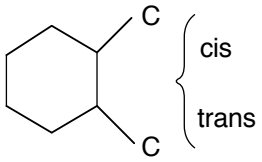
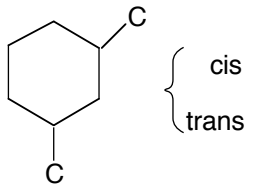
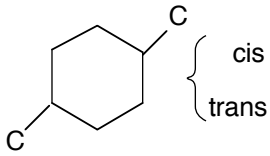
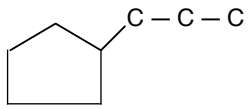
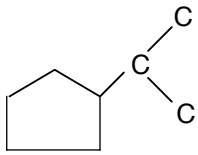
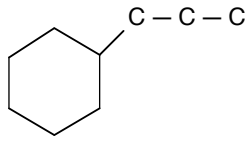
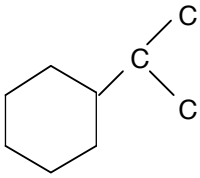
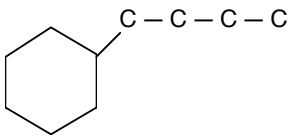
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
2-methyl 3-hexyne	C_7H_{12}		96.2	- 116.7	95.2	726 (20)
4,4-dimethyl 1-pentyne	C_7H_{12}		96.2	- 75.7	76.1	714 (20)
3-ethyl 1-pentyne	C_7H_{12}		96.2	-	88	725 (25)
4,4-dimethyl 2-pentyne	C_7H_{12}		96.2	- 82.4	83	717 (20)
1-octyne	C_8H_{14}		110.2	- 79.3	125.2	746 (20)
1-nonyne	C_9H_{16}		124.2	- 50	150.8	757 (20)
1-decyne	$C_{10}H_{18}$		138.2	- 36	174.6	765 (20)

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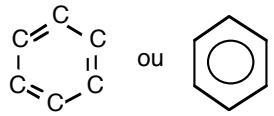
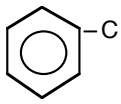
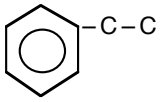
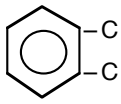
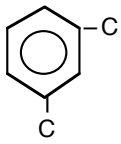
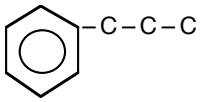
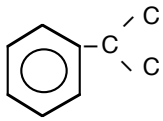
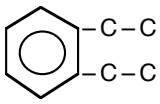
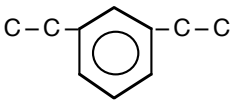
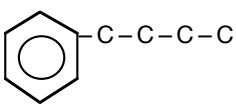
CYCLOALKANES
or naphthenes

Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Cyclobutane	C_4H_8		56.1	– 90.7	12.5	694 (20)
Cyclopentane	C_5H_{10}		70.1	– 93.8	49.2	745 (20)
Cyclohexane	C_6H_{12}		84.2	6.5	80.7	779 (20)
Methyl cyclopentane	C_6H_{12}		84.2	– 142.4	71.8	754 (16)
Cycloheptane	C_7H_{14}		98.2	– 8.1	118.7	810 (20)
Methyl cyclohexane	C_7H_{14}		98.2	– 126.5	100.9	774 (16)
Ethyl-cyclopentane	C_7H_{14}		98.2	– 138.4	103.4	771 (16)
1,1-dimethyl cyclopentane	C_7H_{14}		98.2	– 69.7	87.8	759 (16)
1,2-dimethyl cyclopentane (cis)	C_7H_{14}		98.2	– 53.8	99.5	777 (16)
1,2-dimethyl cyclopentane (trans)	C_7H_{14}		98.2	– 117.5	91.8	756 (16)
Ethyl cyclohexane	C_8H_{16}		112.2	– 111.3	131.7	788 (20)
1,1-dimethyl cyclohexane	C_8H_{16}		112.2	– 33.4	119.5	785 (16)

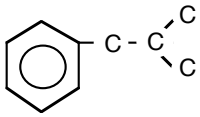
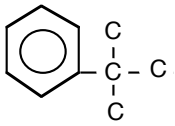
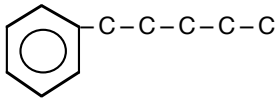
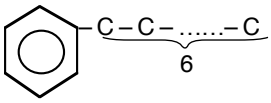
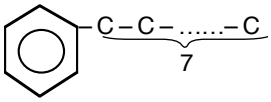
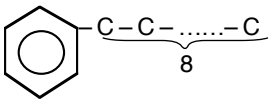
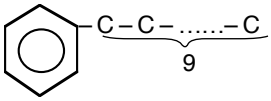
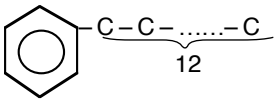
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
1,2-dimethyl cyclohexane	C_8H_{16}		112.2 112.2	– 50.0 – 88.1	129.7 123.4	796 (20) 776 (20)
1,3-dimethyl cyclohexane	C_8H_{16}		112.2 112.2	– 75.5 – 90.1	120.1 124.4	766 (20) 785 (20)
1,4-dimethyl cyclohexane	C_8H_{16}		112.2 112.2	– 87.4 – 36.9	124.3 119.3	783 (20) 763 (20)
n-propyl cyclopentane	C_8H_{16}		112.2	– 117.3	130.9	781 (16)
iso-propyl cyclopentane	C_8H_{16}		112.2	– 112.6	126.4	776 (20)
n-propyl cyclohexane	C_9H_{18}		126.2	– 94.4	156.8	865 (20)
iso-propyl cyclohexane	C_9H_{18}		126.2	– 89.7	154.6	793 (20)
n-butyl cyclohexane	$C_{10}H_{20}$		140.3	– 74.7	181	799 (20)

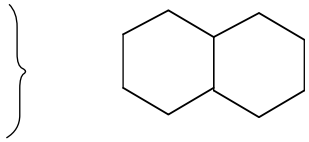
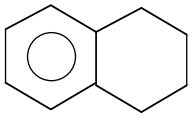
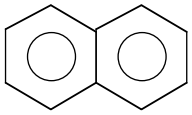
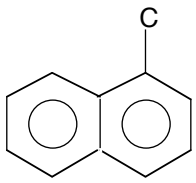
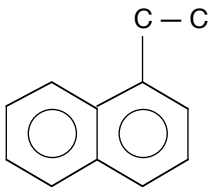
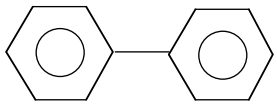
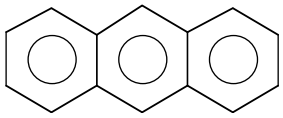
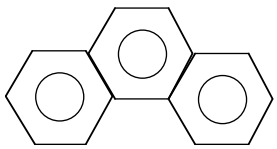
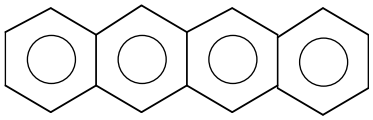
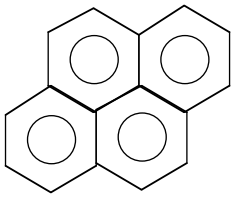
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Benzene	C_6H_6		78.1	5.6	80.2	885 (16)
Toluene	C_7H_8		92.1	- 95.2	110.7	867 (20)
Ethyl-benzene	C_8H_{10}		106.2	- 95	136.2	867 (20)
Ortho-xylene	C_8H_{10}		106.2	- 25.2	144.5	880 (20)
meta-xylene	C_8H_{10}		106.2	- 47.9	139.2	864 (20)
para-xylene	C_8H_{10}		106.2	13.3	138.4	861 (20)
n-propyl benzene	C_9H_{12}		120.2	- 99.5	159.3	862 (20)
iso-propyl benzene (cumene)	C_9H_{12}		120.2	- 96.1	152.5	862 (20)
ortho	$C_{10}H_{14}$		134.2	- 31.2	183.4	880 (20)
diethyl benzene meta			134.2	- 83.9	181	860 (20)
para			134.2	- 42.8	183.8	862 (20)
n-butyl benzene	$C_{10}H_{14}$		134.2	- 88	183.3	860 (20)

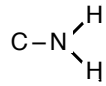
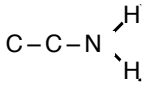
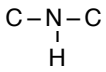
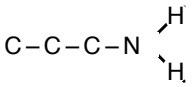
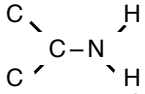
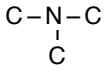
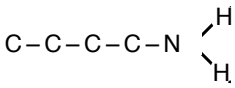
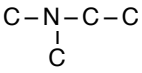
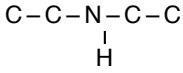
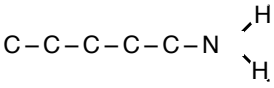
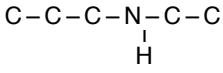
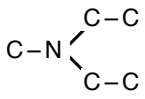
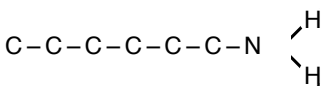
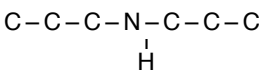
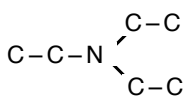
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
iso-butyl benzene	$C_{10}H_{14}$		134.2	- 51.5	172.8	853 (20)
tert-butyl benzene	$C_{10}H_{14}$		134.2	- 57.9	169.2	867 (20)
pentyl benzene (1-phenyl pentane)	$C_{11}H_{16}$		148.2	- 75	205	858 (20)
Hexyl benzene	$C_{12}H_{18}$		162.3	- 62	227	861 (20)
Heptyl benzene	$C_{13}H_{20}$		176.3	- 48	245.5	857 (20)
Octyl benzene	$C_{14}H_{22}$		190.3	- 7	264	858 (20)
Nonyl benzene	$C_{15}H_{24}$		204.4		280	858 (20)
Dodecyl benzene	$C_{18}H_{30}$		246.4	- 7	331	855 (20)

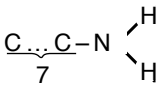
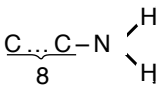
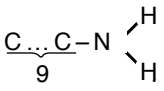
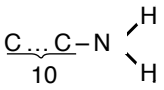
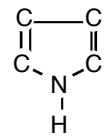
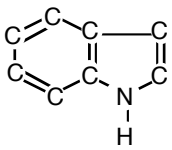
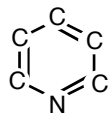
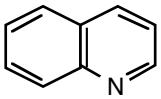
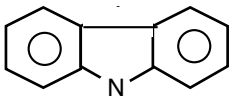
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Decaline (cis)	$C_{10}H_{18}$		138.2	– 43.2	195.8	897 (20)
Decaline (trans)	$C_{10}H_{18}$		138.2	– 30.4	187.3	870 (20)
Tetraline (1,2,3,4-tetrahydro naphtalene)	$C_{10}H_{12}$		132.2	– 31.2	207.6	973 (20°)
Naphtalene	$C_{10}H_8$		128.2	80.4	218.0	971 (90)
Methyl-α Naphtalene	$C_{11}H_{10}$		142.2	– 30.5	244.7	1020 (20)
Ethyl-α Naphtalene	$C_{12}H_{12}$		156.2	– 13.9	258.6	1008 (20)
Diphenyl	$C_{12}H_{10}$		154.2	69.3	255.3	990 (74)
Anthracene	$C_{14}H_{10}$		178.2	215	340	1283 (25)
Phenantrene	$C_{14}H_{10}$		178.2	101	340	980 (20)
Naphtacene	$C_{18}H_{12}$		228.3	—	357	—
Pyrene	$C_{16}H_{10}$		202.3	156	393	1271 (23)

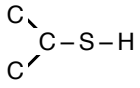
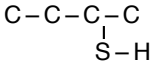
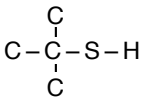
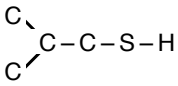
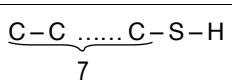
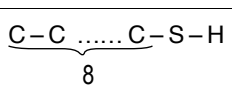
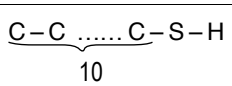
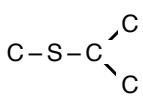
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
AMINES						
Methyl amine	CH_5N		31.1	- 93.5	- 6.3	703 (-13)
Ethyl amine	C_2H_7N		45.1	- 81	16.6	683 (20)
Dimethyl amine	C_2H_7N		45.1	- 92.1	6.9	656 (20)
n-propyl amine (1-amino propane)	C_3H_9N		59.1	- 83.1	48.7	717 (20)
iso-propyl amine	C_3H_9N		59.1	- 95	32.5	688 (20)
trimethyl amine	C_3H_9N		59.1	- 117.1	3	633 (20)
n-butyl amine (1-amino butane)	$C_4H_{11}N$		73.1	- 49.0	77.5	739 (20)
Dimethyl ethyl amine	$C_4H_{11}N$		73.1	- 36	36	675 (20)
Diethyl amine	$C_4H_{11}N$		73.1	- 49.7	55.5	707 (20)
n-pentyl amine (1-amino pentane)	$C_5H_{13}N$		87.2	- 55	104.4	755 (20)
Ethyl propyl amine	$C_5H_{13}N$		87.2	-	61	720 (17)
Diethyl methyl amine	$C_5H_{13}N$		87.2	-	66	703 (25)
n-hexyl amine (1-amino hexane)	$C_6H_{15}N$		101.2	- 19	130	766 (20)
Dipropyl amine	$C_6H_{15}N$		101.2	- 63.1	109.3	738 (20)
Triethyl amine	$C_6H_{15}N$		101.2	- 114.7	89.6	728 (20)

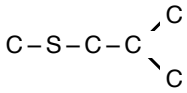
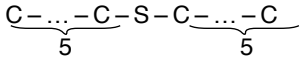
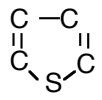
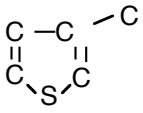
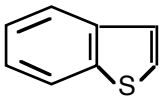
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
n-heptyl amine (1-amino heptane)	$C_7H_{17}N$		115.2	- 18	156.9	775 (20)
n-octyl amine (1-amino octane)	$C_8H_{19}N$		129.2	0	179.6	783 (20)
n-nonyl amine (1-amino nonane)	$C_9H_{21}N$		143.3	- 1	202.2	789 (20)
n-decyl amine (1-amino decane)	$C_{10}H_{23}N$		157.3	17	220.5	794 (20)
CYCLIC COMPOUNDS						
Pyrrole	C_4H_5N		67.1	-	130	969 (20)
Benzopyrrole (Indole)	C_8H_7N		117.4	52.5	254	1220 (20)
Pyridine	C_5H_5N		79.1	- 42	115.5	982 (20)
Quinoline (Benzopyridine)	C_9H_7N		129.1	- 15.6	238	1093 (20)
Carbazole (Dibenzopyrrole)	$C_{12}H_9N$		167.2	247	355	

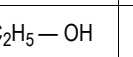
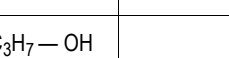
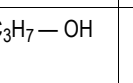
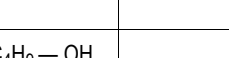
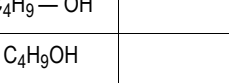
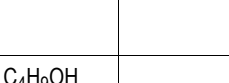
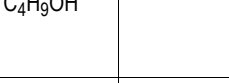
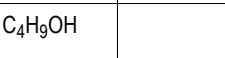
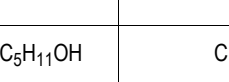
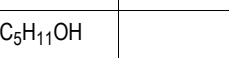
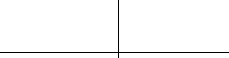
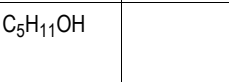
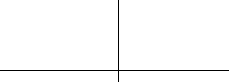
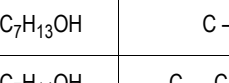
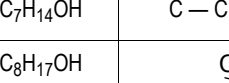
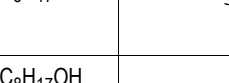
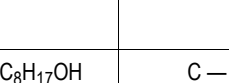
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Hydrogen sulfide	H_2S	H – S – H	34.1	- 85.6	- 60.4	993 (- 60)
MERCAPTANS						
Methyl mercaptan	CH_3SH	C – S – H	48.1	- 123.1	6	866 (20)
Ethyl mercaptan	C_2H_5SH	C – C – S – H	62.1	- 147.8	35.1	839 (20)
n-propyl mercaptan	C_3H_7SH	C – C – C – S – H	76.2	- 113.3	67	841 (20)
iso-propyl mercaptan	C_3H_7SH		76.2	- 130.5	52.5	814 (20)
n-butyl mercaptan (1- butanethiol)	C_4H_9SH	C – C – C – C – S – H	90.2	- 115.7	98.4	834 (20)
sec-butyl mercaptan (2- butanethiol)	C_4H_9SH		90.2	-	84	830 (20)
ter-butyl mercaptan	C_4H_9SH		90.2	1.1	64.2	800 (20)
iso-butyl mercaptan	C_4H_9SH		90.2	- 7.0	88.7	834 (20)
n-pentyl mercaptan (1- pentanethiol)	$C_5H_{11}SH$	C – C – C – C – C –S – H	104.2	- 75.7	126.6	842 (20)
n-hexyl mercaptan	$C_6H_{13}SH$	C – C – C– C – C – C – S – H	118.2	- 81	151	842 (20)
n-heptyl mercaptan	$C_7H_{15}SH$		132.3	- 43	177	843 (20)
n-octyl mercaptan	$C_8H_{17}SH$		146.3	- 49.2	199.1	843 (20)
n-decyl mercaptan	$C_{10}H_{21}SH$		174.3	- 26	240.6	844 (20)
SULFUR COMPOUNDS						
Dimethyl sulfide	C_2H_6S	C – S – C	62.1	- 98.3	37.3	848 (20)
Ethyl methyl sulfide	C_3H_8S	C – S – C – C	76.2	- 105.9	66.6	842 (20)
Diethyl sulfide	$C_4H_{10}S$	C – C – S – C – C	90.2	- 103.8	92.1	836 (20)
Methyl n-propyl sulfide	$C_4H_{10}S$	C – S – C – C – C	90.2	- 113	95.5	842 (20)
Methyl isopropyl sulfide	$C_4H_{10}S$		90.2	- 101.5	84.7	829 (20)

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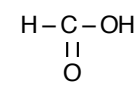
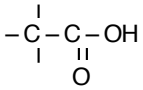
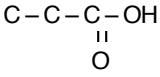
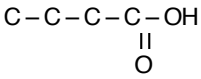
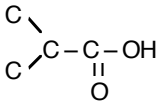
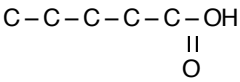
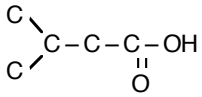
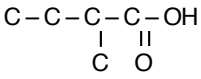
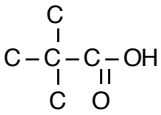
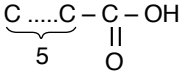
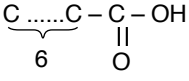
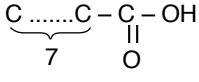
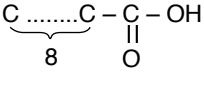
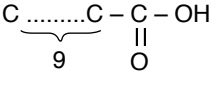
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Ethyl n-propyl sulfide	$C_5H_{12}S$	C – C – S – C – C – C	104.2	- 117	118.5	837 (20)
n-butyl methyl sulfide	$C_5H_{12}S$	C – S – C – C – C – C	104.2	- 97.8	123.2	843 (20)
iso-butyl methyl sulfide	$C_5H_{12}S$		104.2	-	112.5	833 (20)
Dipropyl sulfide	$C_6H_{14}S$	C – C – C – S – C – C – C	118.2	- 102.5	142.4	838 (20)
Dibutyl sulfide	C_8H_8S	C – C – C – C – S – C – C – C – C	146.3	- 79.7	185	839 (20)
Dipentyl sulfide	$C_{10}H_{22}S$		174.3	- 51.3	230.8	841 (20)
DISULFIDES						
Dimethyl disulfide	$C_2H_6S_2$	C – S – S – C	94.2	- 84.7	109.7	1062 (20)
Diethyl disulfide	$C_4H_{10}S_2$	C – C – S – S – C – C	122.2	- 101.5	154	993 (20)
Dipropyl disulfide	$C_6H_{14}S_2$	C – C – C – S – S – C – C – C	150.3	-	193.5	960 (20)
Dibutyl disulfide	$C_8H_{18}S_2$	C – C – C – C – S – S – C – C – C – C	178.3	-	226.8	938 (20)
THIOCYCLIC COMPOUNDS						
Thiophene	C_4H_4S		84.1	- 38.2	84.2	1065 (20)
3-methyl thiophene	C_5H_6S		98.2	- 69	115.4	1022 (20)
Benzothiophene (thionaphtene)	C_8H_6S		134.2	32	221	1148 (32)

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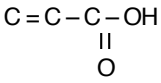
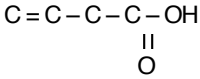
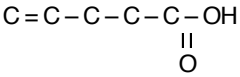
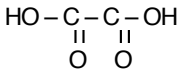
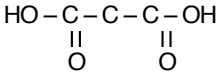
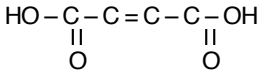
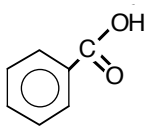
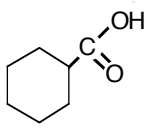
Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Methanol	CH_3 — OH		32.042	– 97.7	64.6	791 (20)
Ethanol	C_2H_5 — OH		46.069	– 114.1	78.3	789 (20)
1-propanol	C_3H_7 — OH		60.096	– 126.3	97.2	804 (20)
2-propanol (isopropanol)	C_3H_7 — OH		60.096	– 88.5	82.2	786 (20)
1-butanol	C_4H_9 — OH		74.123	– 89.1	117.7	810 (20)
2-butanol	C_4H_9OH		74.123	– 114.7	99.5	807 (20)
2-methyl 1-propanol	C_4H_9OH		74.123	– 108	107.8	802 (20)
2-methyl 2-propanol (tertiobutylique alcohol)	C_4H_9OH		74.123	25.6	82.4	787 (20)
1-pentanol	$C_5H_{11}OH$		88.15	– 78.2	137.8	815 (20)
2-methyl 1-butanol	$C_5H_{11}OH$		88.15	– 70.2	128.7	819 (20)
2,2-dimethyl 1-1-propanol	$C_5H_{11}OH$		88.15	+ 53.8	113.1	783 (53.8)
1-hexanol	$C_7H_{13}OH$		102.177	– 44	157	819 (20)
1-heptanol	$C_7H_{14}OH$		116.204	– 34	176.30	822 (20)
1-octanol	$C_8H_{17}OH$		130.231	– 15.5	195.2	826 (20)
2-octanol	$C_8H_{17}OH$		130.231	– 32	179.7	821 (20)
2-ethyl hexanol	$C_8H_{17}OH$		130.231	– 70	184.6	833 (20)
1-decanol	$C_{10}H_{21}OH$		158.285	+ 6.9	230.3	830 (20)
Dodecanol	$C_{12}H_{25}OH$		186.339	+ 23.9	259.9	835 (20)
Heptadecanol	$C_{17}H_{41}OH$		256.474	+ 53.8	323.8	848 (53.8)
1-eicosanol	$C_{20}H_{41}OH$		298.455	65.8	355.8	840 (20)

From "The properties of gases and liquids" by REID. PRAUSNITZ. SHERWOOD

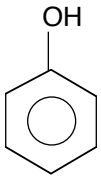
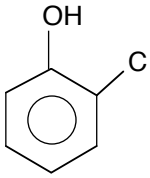
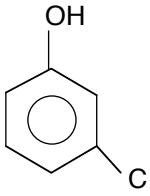
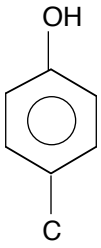
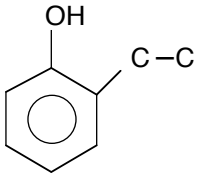
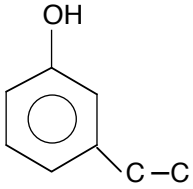
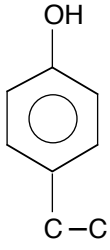
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Methanoic acid (formic acid)	CH_2O_2		46	8.4	100.7	1226 (15)
Ethanoic acid (or acetic)	$C_2H_4O_2$		60	16.7	118	1049 (20)
Propanoic acid (propionic acid)	$C_3H_6O_2$		74.1	- 20.6	141	993 (20)
n-butanoic acid (butyric acid)	$C_4H_8O_2$		88.1	- 5.2	163.3	958 (20)
2-methyl propanoic acid (isobutyric acid)	$C_4H_80_2$		88.1	- 45.9	154.8	968 (20)
n-pentanoic acid (n-valeric acid)	$C_5H_{10}O_2$		102.1	- 34.1	185.6	939 (20)
3-methyl butanoic acid (isovaleric acid)	$C_5H_{10}O_2$		102.1	- 29.3	176.7	929 (20)
2-methyl butanoic acid (2-methyl butyric acid)	$C_5H_{10}O_2$		102.1	< - 80	177	941 (20)
2,2-dimethyl propanoic acid (pivalic acid)	$C_5H_{10}O_2$		102.1	35	164	905 (50)
n-hexanoic acid (caproic acid)	$C_6H_{12}0_3$		116.2	- 2	205	927 (20)
n-heptanoic acid (enanthic acid)	$C_7H_{14}O_2$		130.2	- 7.5	223	920 (20)
n-octanoic acid (caprylic acid)	$C_8H_{16}O_2$		144.2	16.5	239.3	909 (20)
n-nonanoic acid (perlagonic acid)	$C_9H_{18}O_2$		158.2	15	255	906 (20)
n-decanoic acid (capric acid)	$C_{10}H_{20}O_2$		172.3	31.5	270	886 (40)

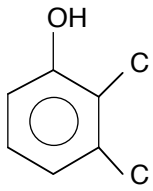
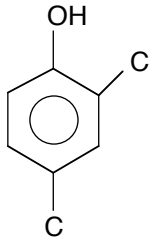
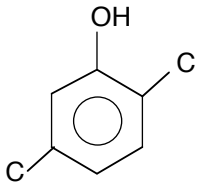
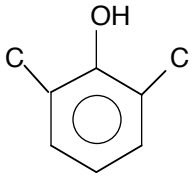
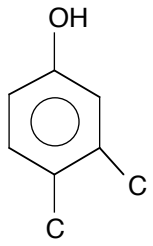
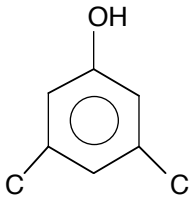
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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Acrylic acid	$C_3H_4O_2$		72.1	11.9	141	1051 (20)
3-butenoic acid (vinyl acetic acid)	$C_4H_6O_2$		86.1	- 35	169	1009 (20)
4-pentanoic acid (allyl acetic acid)	$C_5H_8O_2$		100.1	- 22.5	188	981 (20)
Ethanodioic acid (oxalic acid)	$C_2H_2O_4$		90			1900 (17)
Propanedioic acid (malonic acid)	$C_3H_40_4$		104.1	135.6	140	1620 (16)
Butenedioic acid (maleic acid)	$C_4H_4O_4$		116.1	139		1590 (20)
Benzoic acid	$C_7H_6O_2$		122.1	122.5	250	1075 (130)
Naphthenic acid (cyclohexanoic acid)	$C_7H_{12}O_2$		128.17	31	232	1033 (22)

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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
Phenol	C_6H_5–OH		94.113	40.8	181.8	1.059 (40)
ortho cresol	C_7H_7OH		108.14	30.9	191	1.028 (40)
meta cresol	C_7H_7OH		108.14	12.2	202.2	1.034 (20)
para cresol	C_7H_7OH		108.14	34.7	201.9	1.019 (20)
ortho ethylphenol	C_8H_9OH		122.167	– 3.4	204.5	1.037 (0)
meta ethylphenol	C_8H_9OH		122.167	– 4.2	218.4	1.025 (0)
para ethylphenol	C_8H_9OH		122.167	44.8	217.8	

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Compound	Condensed formula	Detailed formula	M (g/mole)	mp (°C)	t_{bp} (°C)	Density kg/m^3 at (t°C)
2,3-dimethyl phenol (2,3-xylenol)	C_8H_9OH		122.167	74.8	216.9	
2,4-dimethyl phenol (2,4-xylenol)	C_8H_9OH		122.167	24.8	210.8	965 (20)
2,5-dimethyl phenol (2,5-xylenol)	C_8H_9OH		122.167	74.8	211.1	
2,6-dimethyl phenol (2,6-xylenol)	C_8H_9OH		122.167	48.8	200.9	
3,4-dimethyl phenol	C_8H_9OH		122.167	64.8	226.8	983 (20)
3,5dimethyl phenol	C_8H_9OH		122.167	63.8	221.6	968 (20)

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight	Melting point temperature	Boiling point temperature	Density	T_C		P_C	Z_C	Acentric factor
		(kg/kmol)	(°C)	(°C)	(g/cm^3 à °C)	K	°C	(atm)		ω
Ar	Argon	39.95	– 189.4	– 185.95	1.373 (– 180)	150.8	– 122.4	48.1	0.291	– 0.004
Br_2	Bromine	159.81	– 7.2	58.7	3.12 (20)	584	310.8	102.0	0.270	0.132
Cl_2	Chlorine	70.91	– 101.0	– 34.5	1.563 (– 34)	417	143.8	76.0	0.275	0.073
F_2	Fluor	38.00	– 219.7	– 188.2	1.51 (– 193)	144.3	– 128.9	51.5	0.288	0.048
HCl	Hydrochloric gas	36.46	– 114.2	– 85.1	1.193 (– 85)	324.6	51.4	82.0	0.249	0.12
HF	Hydrogen fluoride	20.01	– 83.2	19.5	0.967 (+ 20)	461	187.8	64	0.12	0.372
H_2	Hydrogen	2.02	– 259.8	– 252.8	0.071 (– 253)	33.2	– 240	12.8	0.305	– 0.22
H_2O	Water	18.015	0.0	100.0	0.998 (+ 20)	647.3	374.1	217.6	0.229	0.344
H_2S	Hydrogen sulfide	34.08	– 85.6	– 60.4	0.998 (– 60)	372.2	100.0	88.2	0.284	0.100
NH_3	Ammonia	17.03	– 77.8	– 33.5	0.639 (0)	405.6	132.4	111.3	0.242	0.250
N_2	Nitrogen	28.01	– 209.9	– 195.8	0.804 (– 195)	126.2	– 147	33.5	0.290	0.040
O_2	Oxygen	32.00	– 218.8	– 183.0	1.149 (– 183)	154.6	– 118.6	49.8	0.288	0.021
CCl_4	Carbon tetrachloride	153.82	– 23.2	76.5	1.584 (25)	556.4	283.2	45.0	0.272	0.194
CO	Carbon monoxide	28.01	– 205.1	– 191.5	0.803 (– 192)	132.9	– 140.3	34.5	0.295	0.049
CO_2	Carbon dioxide	44.01	– 56.6	NC	0.777 (20)	304.2	31	72.8	0.274	0.225
CS_2	Carbon sulfide	76.13	– 11.9	– 46.2	1.293 (0)	552	278.8	78.0	0.293	0.115
HCN	Cyanidric acid	27.03	– 13.3	25.7	0.688 (20)	456.8	183.6	53.2	0.197	0.407
CH_2Cl_2	Dichloromethane	84.93	– 95.1	39.8	1.317 (25)	510.0	236.8	60.0	0.277	0.193
CH_2O	Aldehyde formic	30.03	– 117.2	– 19.2	0.815 (– 20)	408.0	134.8	65.0		0.253
CH_2O_2	Formic acid	46.03	8.3	100.6	1.226 (15)	580.0	306.8			
CH_4	Methane	16.04	– 182.5	– 161.5	0.425 (– 161.5)	190.6	– 82.6	45.4	0.288	0.008

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
CH_4O	Methanol	32.04	– 97.7	64.6	0.791 (20)	512.6	239.4	79.9	0.224	0.559
CH_4S	Methylmercaptan	48.11	– 123.2	5.9	0.866 (20)	470.0	196.8	71.4	0.268	0.155
C_2H_2	Acetylene	26.04	– 80.8	– 84.0	0.615 (– 84)	308.3	35.1	60.6	0.271	0.184
C_2H_3Cl	Vinyl chloride	62.50	– 153.8	– 13.4	0.969 (– 14)	429.7	156.5	55.3	0.265	0.122
C_2H_3N	Acetonitrile	41.05	– 43.9	81.6	0.782 (20)	548.0	274.8	47.7	0.184	0.321
C_2H_4	Ethylene	28.06	– 169.2	– 103.8	0.577 (– 110)	282.4	9.2	49.7	0.276	0.085
$C_2H_4Cl_2$	1,1-dichloroethane	98.96	– 97	57.2	1.168 (25)	523	249.8	50.0	0.28	0.248
$C_2H_4Cl_2$	1,2-dichloroethane	98.96	– 35.7	83.4	1.250 (16)	561	287.8	53.0	0.25	0.286
$C_2H_4F_2$	1,1-difluoroethane	66.05	– 117.0	– 24.8		386.6	113.4	44.4	0.253	0.266
C_2H_4O	Acetaldehyde	44.05	– 123.0	20.4	0.778 (20)	461	187.8	55.0	0.22	0.303
C_2H_4O	Ethylene oxide	44.05	– 112.2	10.3	0.899 (0)	469	195.8	71.0	0.258	0.200
$C_2H_4O_2$	Acetic acid	60.05	16.6	117.9	1.049 (20)	594.4	321.2	57.1	0.200	0.454
C_2H_6	Ethane	30.07	– 183.3	– 88.7	0.548 (– 90)	305.4	32.2	48.2	0.285	0.098
C_2H_6O	Dimethyleher	46.07	– 141.5	– 24.4	0.667 (20)	400	126.8	53.0	0.287	0.192
C_2H_6O	Ethanol	46.07	– 114.1	78.3	0.789 (20)	516.2	243	63.0	0.248	0.635
$C_2H_6O_2$	Ethylene glycol	62.07	– 13	197.2	1.114 (20)	645	371.8	76.0	0.27	
C_2H_6S	Ethylmercaptan	62.13	– 147.9	35	0.834 (20)	499	225.8	54.2	0.274	0.190
C_2H_7NO	Monoethanolamine	61.08	10.3	170.3	1.016 (20)	614	340.8	44.0	0.17	
C_3H_3N	Acrylonitrile	53.06	– 83.7	77.3	0.806 (20)	536	262.8	45.0	0.21	0.35
C_3H_4	Propadiene	40.06	– 136.3	– 34.5	0.658 (– 35)	393	119.8	54.0	0.271	0.313
C_3H_4	Methylacetylene	40.06	– 102.7	– 23.2	0.706 (– 50)	402.4	129.2	55.5	0.276	0.218

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
C_3H_4O	Acrolein	56.06	– 87.2	52.8	0.839 (20)	506.0	232.8	51.0		0.33
$C_3H_4O_2$	Acrylic acid	72.06	11.8	140.8	1.051 (20)	615	341.8	56.0	0.23	0.56
C_3H_6	Cyclopropane	42.08	– 127.5	– 32.8	0.563 (15)	397.8	124.6	54.2	0.282	0.264
C_3H_6	Propylene	42.08	– 185.3	– 47.8	0.612 (– 50)	365	91.8	45.6	0.275	0.148
$C_3H_6Cl_2$	1,2-dichloropropane	112.99	– 100.5	96.3	1.15 (20)	577	303.8	44.0	0.21	0.24
C_3H_6O	Acetone	58.8	– 95	56.2	0.790 (20)	508	234.9	46.4	0.232	0.309
C_3H_6O	Propionaldehyde	58.8	– 80.2	47.8	0.797 (20)	496	222.8	47.0	0.260	0.313
C_3H_6O	Propylene oxide	58.8	– 112.2	34.3	0.829 (20)	482.2	209	48.6	0.228	0.269
$C_3H_6O_2$	Methyl acetate	74.08	– 98.2	56.9	0.934 (20)	506.8	233.6	46.3	0.254	0.324
C_3H_8	Propane	44.10	– 187.7	– 42.1	0.582 (– 42)	369.8	96.6	41.9	0.281	0.152
C_3H_8O	1-propanol	60.10	– 126.3	97.2	0.804 (20)	536.7	263.5	51.0	0.253	0.624
C_3H_8O	Isopropylic alcohol	60.10	– 88.5	82.2	0.786 (20)	508.3	235.1	47.0	0.248	
C_3H_8O	Methylethylether	60.10	– 139.2	7.3	0.700 (20)	437.8	164.6	43.4	0.267	0.236
$C_3H_8O_3$	Glycerol	92.095	17.8	289.8	1.261 (20)	726	452.8	66.0	0.28	
C_3H_8S	Methylethylsulfide	76.16	– 106.0	66.6	0.837 (20)	533	259.8	42.0		
$C_4H_2O_3$	Anhydride maleic	98.06	52.8	199.6	1.310 (60)					
C_4H_4	Vinylacetylene	52.08	– 45.6	4.9	0.710 (0)	455	181.8	49.0	0.26	0.092
C_4H_4O	Furan	68.075	– 85.7	31.3	0.938 (20)	490.2	217.0	54.3	0.294	0.204
C_4H_4S	Thiophene	84.136	– 38.3	84.1	1.071 (16)	579.4	306.2	56.2	0.259	0.200
C_4H_5N	Pyrrole	67.09		129.8	0.967 (21)	640	367.0			
C_4H_6	1-Butyne	54.09	– 125.8	8.0	0.650 (16)	463.7	190.5	46.5	0.27	0.050

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
C_4H_6	2-butyne	54.09	– 32.3	27	0.691 (20)	488.6	215.4	50.2	0.277	0.124
C_4H_6	1,2-butadiene	54.09	– 136.2	10.8	0.652 (20)	443.7	170.5	44.4	0.267	0.255
C_4H_6	1,3-butadiene	54.09	– 108.9	4.5	0.621 (20)	425	151.8	42.7	0.270	0.195
$C_4H_6O_2$	Vinyl acetate	86.9	– 100.2	72.8	0.932 (20)	525	251.8	43.0	0.260	0.34
$C_4H_6O_3$	Acetic anhydride	102.09	– 74.2	138.8	1.087 (20)	569	295.8	46.2	0.287	
$C_4H_7O_2$	Methylacrylate	86.09	– 76.5	80.3	0.956 (20)	536	262.8	42.0	0.25	0.35
C_4H_8	1-butene	56.11	– 185.4	– 6.3	0.595 (20)	419.6	146.4	39.7	0.277	0.187
C_4H_8	2-butene (cis)	56.11	– 138.9	3.7	0.621 (20)	435.6	162.4	41.5	0.272	0.202
C_4H_8	2-butene (trans)	56.11	– 105.6	0.8	0.604 (20)	428.6	155.4	40.5	0.274	0.214
C_4H_8	Cyclobutane	56.11	– 90.8	12.5	0.694 (20)	459.9	186.7	49.2	0.274	0.209
C_4H_8	Isobutene	56.11	– 140.4	– 6.9	0.594 (20)	417.9	144.7	39.5	0.275	0.190
C_4H_8O	Methylethylketone	72.11	– 86.7	79.6	0.805 (20)	535.6	262.4	41.0	0.249	0.329
C_4H_8O	Tetrahydrofuran	72.11	– 108.5	65.9	0.889 (20)	540.2	267.0	51.2	0.259	
C_4H_{10}	n-butane	58.12	– 138.4	– 0.5	0.579 (20)	425.2	152	37.5	0.274	0.193
C_4H_{10}	isobutane	58.12	– 159.6	– 11.9	0.557 (20)	408.1	134.9	36.0	0.283	0.176
$C_4H_{10}O$	n-butanol	74.12	– 89.3	117.7	0.810 (20)	562.9	289.7	43.6	0.259	0.590
$C_4H_{10}O$	2-butanol	74.12	– 114.7	99.5	0.807 (20)	536	262.8	41.4	0.252	0.576
$C_4H_{10}O$	Isobutanol	74.12	– 108.0	107.8	0.802 (20)	547.7	274.5	42.4	0.257	0.588
$C_4H_{10}O$	Ethyl ether	74.12	– 116.3	34.5	0.713 (20)	466.7	193.5	35.9	0.262	0.281
$C_4H_{10}O_3$	Diethylene glycol	106.12	– 8.2	245.8	1.116 (20)	681	407.8	46.0	0.260	
$C_4H_{10}S$	Diethylsulfure	90.18	– 104.0	92.1	0.837 (20)	557	283.8	39.1	0.272	0.300

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
$C_4H_{10}S_2$	Diethyldisulfide	122.24	– 101.5		0.998 (20)	642	368.8			
C_5H_5N	Pyridine	79.10	– 41.7	115.3	0.983 (20)	620	346.8	55.6	0.277	0.240
C_5H_8	Cyclopentene	68.12	– 135.1	44.2	0.772 (20)	506	232.8			
C_5H_{10}	Cyclopentane	70.135	– 93.9	49.2	0.745 (20)	511.6	238.4	44.5	0.276	0.192
C_5H_{10}	1-pentene	70.135	– 165.3	29.9	0.640 (20)	464.7	191.5	40.0	0.31	0.245
$C_5H_{10}O$	Diethylketone	86.13	– 39.0	101.9	0.814 (20)	561.0	287.8	36.9	0.269	0.347
$C_5H_{11}N$	Piperidine	85.15	– 10.5	106.5	0.862 (20)	594	320.8	47.0	0.280	0.25
C_5H_{12}	n-pentane	72.15	– 129.8	36.0	0.626 (20)	469.6	196.4	33.3	0.262	0.251
C_5H_{12}	2-methylbutane	72.15	– 159.9	27.8	0.620 (20)	460.4	187.2	33.4	0.271	0.227
$C_5H_{12}O$	1-pentanol	88.15	– 78.0	138	0.815 (20)	586	313	38	0.26	0.58
$C_5H_{12}O$	Ethylpropylether	88.15	– 126.6	63.8	0.733 (20)	500.6	227.6	32.1		0.331
C_6H_6	Benzene	78.11	5.5	80.1	0.885 (16)	562.1	289.1	48.3	0.271	0.212
C_6H_6O	Phenol	94.11	40.8	181.8	1.059 (40)	694.2	421.1	60.5	0.24	0.440
C_6H_7N	Aniline	93.13	– 6.1	184.4	1.022 (20)	699	425.9	52.4	0.247	0.382
C_6H_{12}	Cyclohexane	84.16	6.6	80.8	0.779 (20)	553.4	280.3	40.2	0.273	0.213
C_6H_{12}	Methylcyclopentane	84.16	– 142.4	71.9	0.754 (16)	532.7	259.6	37.4	0.273	0.239
$C_6H_{12}O$	Methylisobutylketone	100.16	– 84.1	116.5	0.801 (20)	571	297.9	32.3	0.260	0.400
C_6H_{14}	n-hexane	86.18	– 95.3	68.8	0.659 (20)	507.4	234.3	29.3	0.260	0.296
$C_6H_{14}O$	1-hexanol	102.18	– 43.9	157.1	0.819 (20)	610	336.9	40.0	0.30	0.59
$C_6H_{14}O$	Diisopropylether	102.18	– 85.4	68.4	0.724 (20)	500.0	226.9	28.4	0.267	0.34
$C_6H_{15}N$	Dipropylamine	101.19	– 63.1	109.3	0.738 (20)	550	276.9	31.9	0.28	0.455

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
$C_7H_6O_2$	Benzoic acid	122.12	122.5	249.9	1.075 (130)	752	478.9	45	0.25	0.62
C_7H_8	Toluene	92.14	– 95.1	110.7	0.867 (20)	591.7	318.6	40.6	0.264	0.257
C_7H_8O	o-cresol	108.14	31	191.1	1.028 (40)	697.6	424.5	49.4	0.24	0.443
C_7H_8O	m-cresol	108.14	12.3	202.3	1.034 (20)	705.8	432.7	45.0	0.241	0.464
C_7H_8O	p-cresol	108.14	34.8	202	1.019 (40)	704.6	431.5	50.8		0.515
C_7H_{14}	Ethylcyclopentane	98.19	– 138.4	103.5	0.771 (16)	569.5	296.4	33.5	0.269	0.283
C_7H_{14}	Methylcyclohexane	98.19	– 126.5	101.0	0.774 (16)	572.1	299.0	34.3	0.269	0.233
C_7H_{16}	n-heptane	100.21	– 90.5	98.0	0.684 (20)	540.2	267.1	27.0	0.263	0.351
$C_7H_{16}O$	1-heptanol	116.20	– 33.9	176.4	0.822 (20)	633	359.9	30.0	0.25	0.56
$C_8H_4O_3$	Anhydridphtalic	148.12	130.9	286.9		810	536.9	47.0	0.26	
C_8H_8	Styrene	104.15	– 30.6	145.2	0.906 (20)	647	373.9	39.4		0.257
C_8H_{10}	o-xylene	106.17	– 25.1	144.5	0.880 (20)	630.2	357.1	36.8	0.263	0.314
C_8H_{10}	m-xylene	106.17	– 47.8	139.2	0.864 (20)	617.0	343.9	35.0	0.260	0.331
C_8H_{10}	p-xylene	106.17	13.3	138.4	0.861 (20)	616.2	343.1	34.7	0.260	0.324
C_8H_{10}	Ethylbenzene	106.17	– 94.9	136.2	0.867 (20)	617.1	344	35.6	0.263	0.301
C_8H_{16}	Ethylcyclohexane	112.22	– 111.3	131.8	0.788 (20)	609	335.9	29.9	0.27	0.243
C_8H_{18}	n-octane	114.23	– 56.7	125.7	0.703 (20)	568.8	295.7	24.5	0.259	0.394
C_9H_{12}	n-propylbenzene	120.20	– 99.4	159.3	0.862 (20)	638.3	365.2	31.6	0.265	0.344
C_9H_{12}	Isopropylbenzene	120.20	– 96	152.5	0.862 (20)	631.0	357.9	31.7	0.260	0.335
C_9H_{20}	n-nonane	128.26	– 53.4	150.9	0.718 (20)	594.6	321.5	22.8	0.26	0.444
$C_{10}H_8$	Naphtalene	128.17	80.4	218	0.971 (90)	748.4	475.3	40.0	0.267	0.302

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PHYSICAL CHARACTERISTICS OF PURE SUBSTANCES

		Molecular weight (kg/kmol)	Melting point temperature (°C)	Boiling point temperature (°C)	Density (g/cm^3 à °C)	T_C K	T_C °C	P_C (atm)	Z_C	Acentric factor ω
$C_{10}H_{22}$	n-decane	142.29	– 29.6	174.2	0.730 (20)	617.6	344.5	20.8	0.247	0.490
$C_{11}H_{10}$	1-methyl naphtalene	142.20	– 30.4	244.7	1.020 (20)	772	498.9	35.2	0.25	0.334
$C_{11}H_{10}$	2-methyl naphtalene	142.20	34.6	241.1	0.990 (40)	761	487.9	34.6	0.26	0.382
$C_{12}H_{10}$	Diphenyle	154.21	69.3	255.3	0.990 (74)	789	515.9	38	0.295	0.364
$C_{11}H_{24}$	n-undecane	156.31	– 25.5	196.0	0.740 (20)	638.8	365.7	19.4	0.24	0.535
$C_{12}H_{26}$	n-dodecane	170.34	– 9.5	216.4	0.748 (20)	658.3	385.2	18.0	0.24	0.562
$C_{13}H_{28}$	n-tridecane	184.37	– 5.3	235.5	0.756 (20)	675.8	402.7	17.0	0.24	0.623
$C_{14}H_{20}$	Anthrakene	188.31	216.6	341.3		883	609.9			
$C_{14}H_{10}$	Phenanthrene	178.23	100.6	339.5		878	604.9			
$C_{14}H_{30}$	n-tetradecane	198.34	+ 5.9	253.6	0.763 (20)	694	420.9	16.0	0.23	0.679
$C_{15}H_{32}$	n-pentadecane	212.42	+ 9.9	270.7	0.769 (20)	707	433.9	15.0	0.23	0.706
$C_{16}H_{34}$	n-hexadecane	226.45	+ 17.9	286.9	0.773 (20)	717	443.9	14.0		0.742
$C_{17}H_{36}$	n-heptadecane	220.47	+ 21.9	302.1	0.778 (20)	733	459.9	13.0	0.22	0.770
$C_{18}H_{38}$	n-octadecane	254.50	+ 28.2	316.4		745	471.9	11.9		0.790
$C_{19}H_{40}$	n-nonadecane	268.53	+ 31.9	330.0	0.789 (20)	756	482.9	11.0		0.827
$C_{20}H_{42}$	n-eicosane	282.56	+ 36.9	343.9	0.775 (40)	767	493.9	11.0		0.907

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□A□□□

VAPOR PRESSURE THROUGH ANTOINES EQUATION

$$\ln P^S = A - \frac{B}{C + T}$$

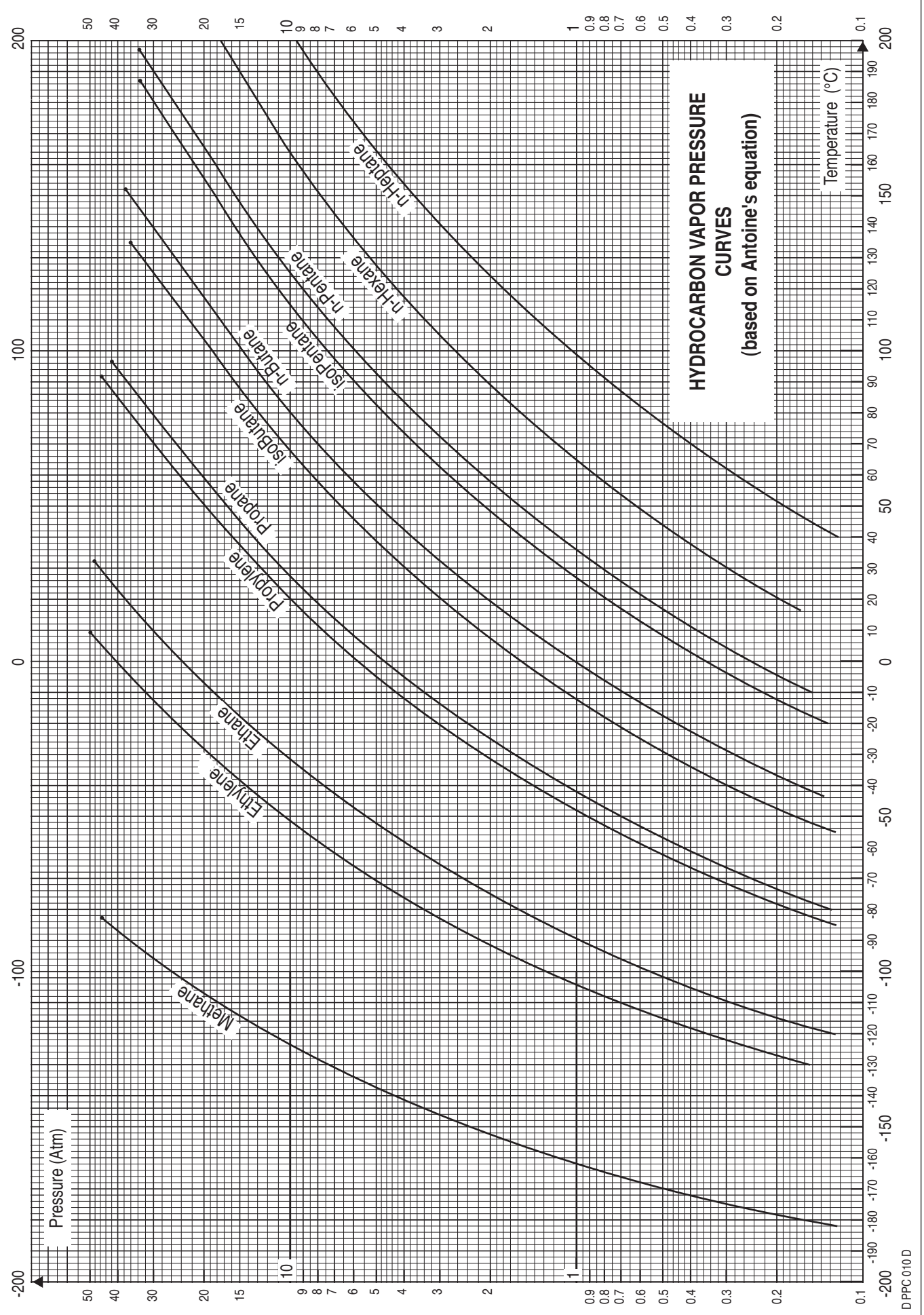
ln is natural logarithm
P^S in mmHg
T in Kelvin

		A	B	C	T_{max}	T_{min}
Ar	Argon	15.2330	700.51	– 5.84	94	81
Br_2	Bromine	15.8441	2582.32	– 51.56	354	259
Cl_2	Chlorine	15.9610	1978.32	– 27.01	264	172
F_2	Fluorine	15.6700	714.10	– 6.00	91	59
HCl	Hydrochloric acid	16.5040	1714.25	– 14.45	200	137
HF	Hydrofluoric acid	17.6958	3404.49	– 15.06	313	206
H_2	Hydrogen	13.6333	164.90	– 3.19	25	14
H_2O	Water	18.3036	3816.44	– 46.13	441	284
H_2S	Hydrogen sulfide	16.1040	1768.69	– 26.06	230	190
NH_3	Ammonia	16.9481	2132.50	– 32.98	261	179
N_2	Nitrogen	14.9542	588.72	– 6.60	90	54
O_2	Oxygen	15.4075	734.55	– 6.45	100	63
CCl_4	Carbon tetrachloride	15.8742	2808.19	– 45.99	374	253
CO	Carbon monoxide	14.3686	530.22	– 13.15	108	63
CO_2	Carbon dioxide	22.5898	3103.39	– 0.16	204	154
CS_2	Carbon disulfide	15.9844	2690.85	– 31.62	342	228
HCN	Cyanhydric acid	16.5138	2585.80	– 37.15	330	234
CH_2Cl_2	Dichloromethane	16.3029	2622.44	– 41.70	332	229
CH_2O	Fomic aldehyde	16.4775	2204.13	– 30.15	271	185
CH_2O_2	Fomic acid	16.9882	3599.58	– 26.09	409	271
CH_4	Methane	15.2243	897.84	– 7.16	120	93
CH_4O	Methanol	18.5875	3626.55	– 34.29	364	257
CH_4S	Methylmercaptan	16.1909	2338.38	– 34.44	300	200
C_2H_2	Acetylene	16.3481	1637.14	– 19.77	202	194
C_2H_3Cl	Vinyl chloride	14.9601	1803.84	– 43.15	290	185
C_2H_3N	Acetonitrile	16.2874	2945.47	– 49.15	390	260
C_2H_4	Ethylene	15.5368	1347.01	– 18.15	182	120
$C_2H_4Cl_2$	1,1-dichloroethane	16.0842	2697.29	– 45.03	352	242
$C_2H_4Cl_2$	1,2-dichloroethane	16.1764	2927.17	– 50.22	373	240
$C_2H_4F_2$	1,1-difluoroethane	16.1871	2095.35	– 29.16	273	238
C_2H_4O	Acetaldehyde	16.2481	2465.15	– 37.15	320	210
C_2H_4O	Ethylene oxide	16.7400	2567.61	– 29.01	310	200
$C_2H_4O_2$	Acetic acid	16.8080	3405.57	– 56.34	430	290
C_2H_6	Ethane	15.6637	1511.42	– 17.16	199	130
C_2H_6O	Dimethylether	16.8467	2361.44	– 17.10	265	179
C_2H_6O	Ethanol	18.9119	3803.98	– 41.68	369	270
$C_2H_6O_2$	Ethylene glycol	20.2501	6022.18	– 28.25	494	364
C_2H_6S	Ethylmercaptan	16.0077	2497.23	– 41.77	330	224
C_2H_7NO	Monoethanolamine	17.8174	3988.33	– 86.93	477	344
C_3H_3N	Acrylonitrile	15.9253	2782.21	– 51.15	385	255
C_3H_4	Propadiene	13.1563	1054.72	– 77.08	257	174
C_3H_4	Methylacetylene	15.6227	1850.66	– 44.07	267	183
C_3H_4O	Acrolein	15.9057	2606.53	– 45.15	360	235
$C_3H_4O_2$	Acrylic acid	16.5617	3319.18	– 80.15	450	315
C_3H_6	Cyclopropane	15.8599	1971.04	– 26.65	245	180
C_3H_6	Propylene	15.7027	1807.53	– 26.15	240	160
$C_3H_6Cl_2$	1,2-dichloropropane	16.0385	2985.07	– 52.16	408	288
C_3H_6O	Acetone	16.6513	2940.46	– 35.93	350	241
C_3H_6O	Propionaldehyde	16.2315	2659.02	– 44.15	350	235
C_3H_6O	Propylene oxide	15.3227	2107.58	– 64.87	340	225
$C_3H_6O_2$	Methyl acetate	16.1295	2601.92	– 56.15	360	245
C_3H_8	Propane	15.7260	1872.46	– 25.16	249	164
C_3H_8O	1-propanol	17.5439	3166.38	– 80.15	400	285
C_3H_8O	Isopropylic alcohol	18.6929	3640.20	– 53.54	374	273
C_3H_8O	Methylethylether	13.5435	1161.63	– 112.4	310	205
$C_3H_8O_3$	Glycerol	17.2392	4487.04	– 140.2	600	440
C_3H_8S	Metylethylsulfid	15.9765	2722.95	– 48.37	360	250
$C_4H_2O_3$	Maleic anhydride	16.2747	3765.65	– 82.15	516	352
C_4H_4	Vinylacetylene	16.0100	2203.57	– 43.15	305	200
C_4H_4O	Furan	16.0612	2442.70	– 45.41	363	238
C_4H_4S	Thiophene	16.0243	2869.07	– 51.80	380	260
C_4H_5N	Pyrrole	16.7966	3457.47	– 62.73	440	330

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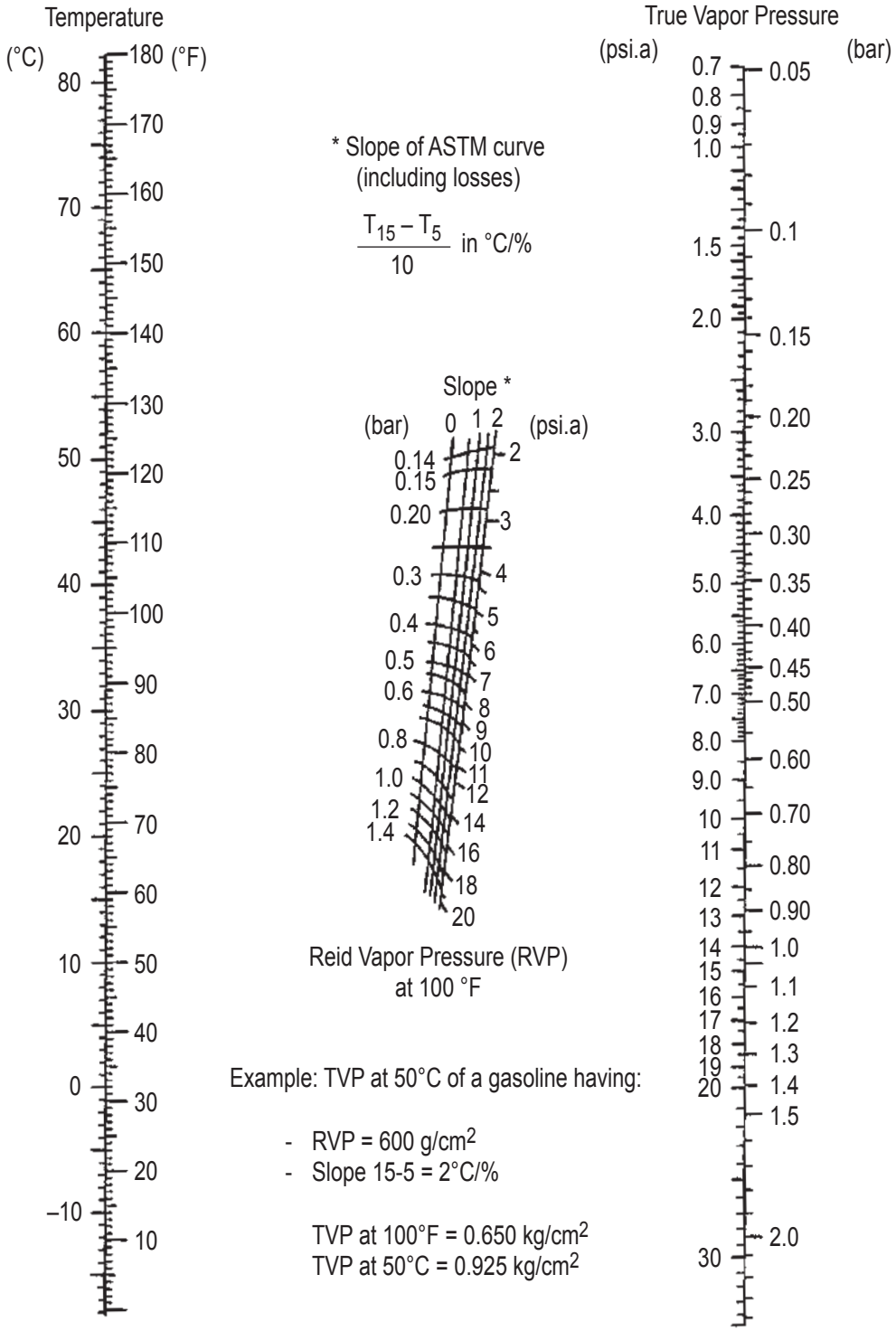
C_4H_6	1-butyne	16.0605	2271.42	– 40.30	300	200
C_4H_6	2-butyne	16.2871	2536.78	– 37.34	320	240
C_4H_6	1,2-butadiene	16.1039	2397.26	– 30.88	305	245
C_4H_6	1,3-butadiene	15.7727	2142.66	– 34.30	290	215
$C_4H_6O_2$	Vinyl acetate	16.1003	2744.68	– 56.15	379	255
$C_4H_6O_3$	Acetic anhydride	16.3982	3287.56	– 75.11	437	308
$C_4H_7O_2$	Methyl acrylate	16.1088	2788.43	– 59.15	390	260
C_4H_8	1-butene	15.7564	2132.42	– 33.15	295	190
C_4H_8	2-butene (cis)	15.8171	2210.71	– 36.15	305	200
C_4H_8	2-butene (trans)	15.8177	2212.32	– 33.15	300	200
C_4H_8	Cyclobutane	15.9254	2359.09	– 31.78	290	200
C_4H_8	Isobutene	15.7528	2125.75	– 33.15	290	190
C_4H_8O	Methylethylcetone	16.5986	3150.42	– 36.65	376	257
C_4H_8O	Tetrahydrofuran	16.1069	2768.38	– 46.90	370	270
C_4H_{10}	n-butane	15.6782	2154.90	– 34.42	290	195
C_4H_{10}	Isobutane	15.5381	2032.73	– 33.15	280	187
$C_4H_{10}O$	n-butanol	17.2160	3137.02	– 94.43	404	288
$C_4H_{10}O$	2-butanol	17.2102	3026.03	– 86.65	393	298
$C_4H_{10}O$	Isobutanol	16.8712	2874.73	– 100.3	388	293
$C_4H_{10}O$	Ethyl ether	16.0828	2511.29	– 41.95	340	225
$C_4H_{10}O_3$	Diethylene glycol	17.0326	4122.52	– 122.5	560	402
$C_4H_{10}S$	Diethylsulfide	15.9531	2896.27	– 54.49	390	260
$C_4H_{10}S_2$	Diethyldisulfide	16.0607	3421.57	– 64.19	455	312
C_5H_5N	Pyridine	16.0910	3095.13	– 61.15	425	285
C_5H_8	Cyclopentene	2583.07	2583.07	– 39.70	378	244
C_5H_{10}	Cyclopentane	15.8574	2588.48	– 41.79	345	230
C_5H_{10}	1-pentane	15.7646	2405.96	– 39.63	325	220
$C_5H_{10}O$	Diethylcetone	16.8138	3410.51	– 40.15	400	275
$C_5H_{11}N$	Piperidine	16.1004	3015.46	– 61.15	416	280
C_5H_{12}	n-pentane	15.8333	2477.07	– 39.94	330	220
C_5H_{12}	2-methylbutane	15.6338	2348.67	– 40.05	322	216
$C_5H_{12}O$	1-pentanol	16.5270	3026.89	– 105.0	411	310
$C_5H_{12}O$	Ethylpropylether	15.4539	2423.41	– 62.28	360	246
C_6H_6	Benzene	15.9008	2788.51	– 52.36	377	280
C_6H_6O	Phenol	16.4279	3490.89	– 98.59	481	345
C_6H_7N	Aniline	16.6748	3857.52	– 73.15	500	340
C_6H_{12}	Cyclohexane	15.7527	2766.63	– 50.50	380	280
C_6H_{12}	Methylcyclopentane	15.8023	2731.00	– 47.11	375	250
$C_6H_{12}O$	Methylisobutylcetone	15.7165	2893.66	– 70.75	425	285
C_6H_{14}	n-hexane	15.8366	2697.55	– 48.78	370	245
$C_6H_{14}O$	1-hexanol	18.0994	4055.45	– 76.49	430	308
$C_6H_{14}O$	Diisopropylether	16.3417	2895.73	– 43.15	364	249
$C_6H_{15}N$	Diproylamine	16.5939	3259.08	– 55.15	422	302
$C_7H_6O_2$	Benzoic acid	17.1634	4190.70	– 125.2	560	405
C_7H_8	Toluene	16.0137	3096.52	– 53.67	410	280
C_7H_8O	o-cresol	15.9148	3305.37	– 108.0	480	370
C_7H_8O	m-cresol	17.2878	4274.42	– 74.09	480	370
C_7H_8O	p-cresol	16.1989	3479.39	– 111.3	480	370
C_7H_{14}	Ethylcyclopentane	15.8581	2990.13	– 52.47	402	270
C_7H_{14}	Methylcyclohexane	15.7105	2926.04	– 51.75	400	270
C_7H_{16}	n-heptane	15.8737	2911.32	– 56.51	400	270
$C_7H_{16}O$	1-heptanol	15.3068	2626.42	– 146.6	449	333
$C_8H_4O_3$	Phthalic anhydride	15.9984	4467.01	– 83.15	615	409
C_8H_8	Styrene	16.0193	3328.57	– 63.72	460	305
C_8H_{10}	o-xylene	16.1156	3395.57	– 59.46	445	305
C_8H_{10}	m-xylene	16.1390	3366.99	– 58.04	440	300
C_8H10	p-xylene	16.0963	3346.65	– 57.84	440	300
C_8H_{10}	Ethylbenzene	16.0195	3279.47	– 59.95	450	300
C_8H_{16}	Ethylcyclohexane	15.8125	3183.25	– 58.15	433	293
C_8H_{18}	n-octane	15.9426	3120.29	– 63.63	425	292
C_9H_{12}	n-propylbenzene	16.0092	3433.84	– 66.01	461	316
C_9H_{12}	Isopropylbenzene	15.9722	3363.60	– 63.37	454	311
C_9H_{20}	n-nonane	15.9671	3291.45	– 71.33	452	312
$C_{10}H_8$	Naphthalene	16.1426	3992.01	– 71.29	525	360
$C_{10}H_{22}$	n-decane	16.0114	3456.80	– 78.67	476	330
$C_{11}H_{24}$	n-undecane	16.0541	3614.07	– 85.45	498	348
$C_{12}H_{26}$	n-dodecane	16.1134	3774.56	– 91.31	520	364
$C_{13}H_{28}$	n-tridecane	16.1355	3892.91	– 98.93	540	380
$C_{14}H_{30}$	n-tetradecane	16.1480	4008.52	– 105.4	560	394

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CORRELATION BETWEEN REID VAPOR PRESSURE (RVP) AND TRUE VAPOR PRESSURE



Example: TVP at 50°C of a gasoline having:

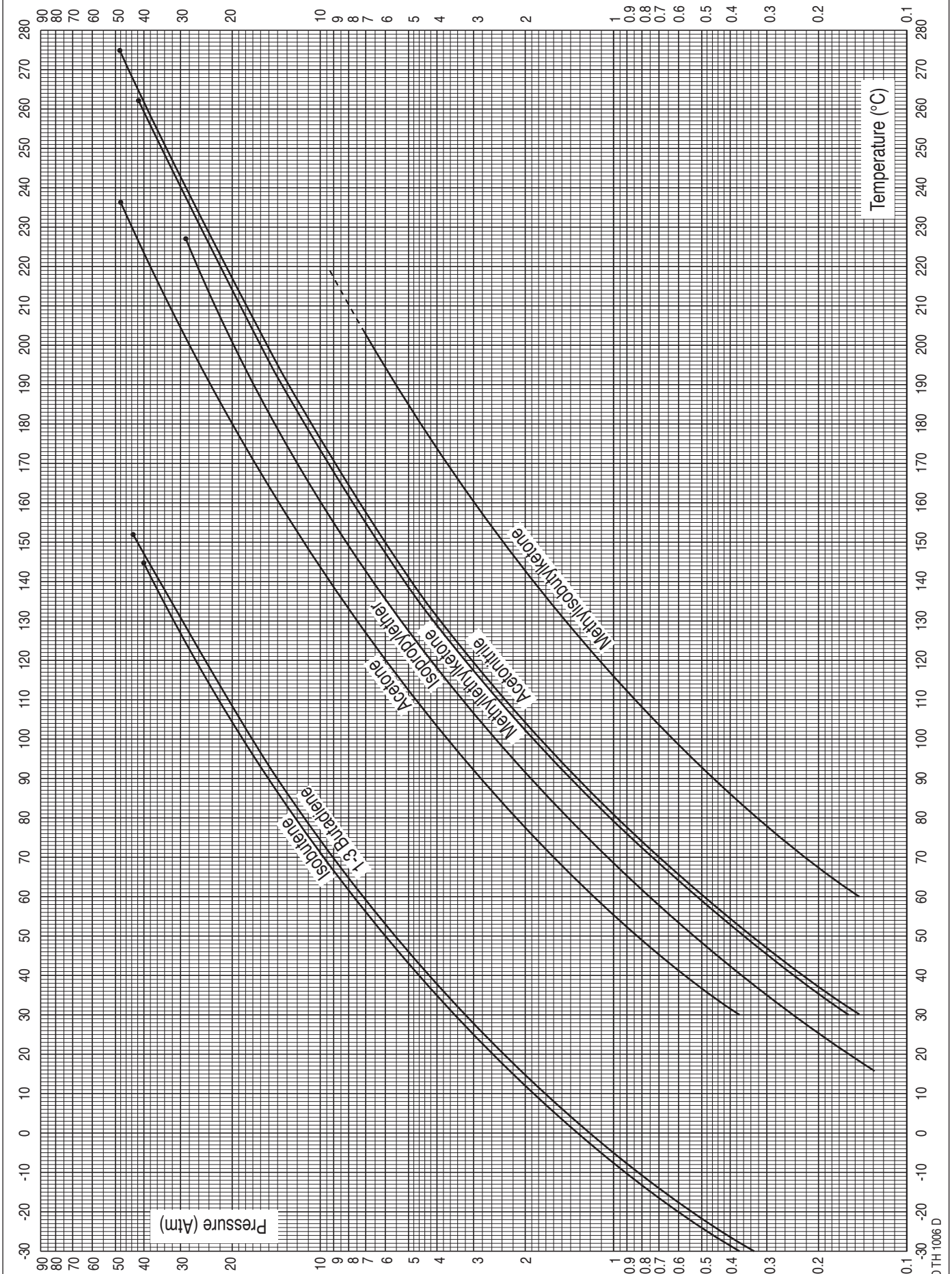
- RVP = 600 g/cm^2
- Slope 15-5 = 2°C/%

TVP at 100°F = 0.650 kg/cm^2
TVP at 50°C = 0.925 kg/cm^2

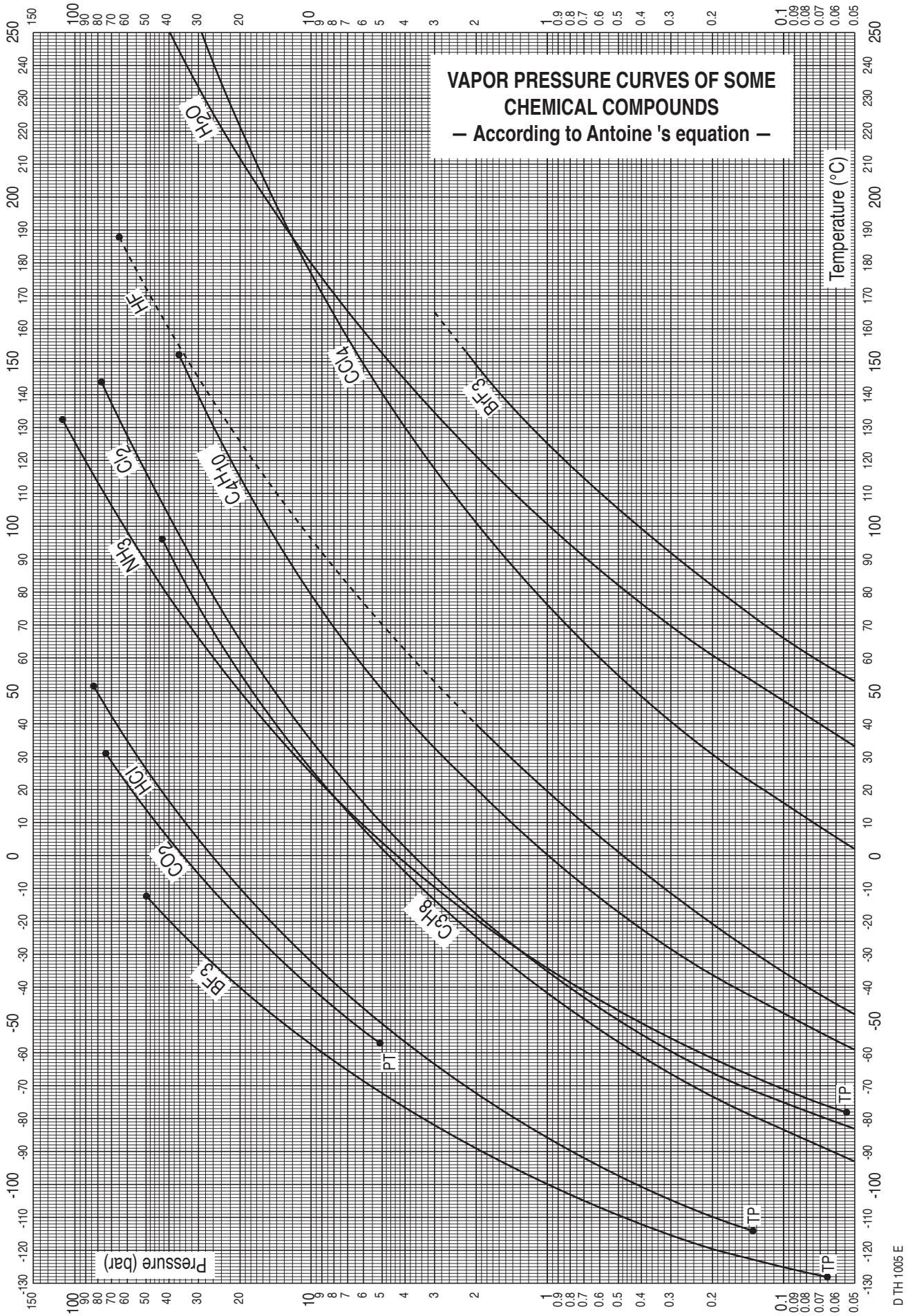
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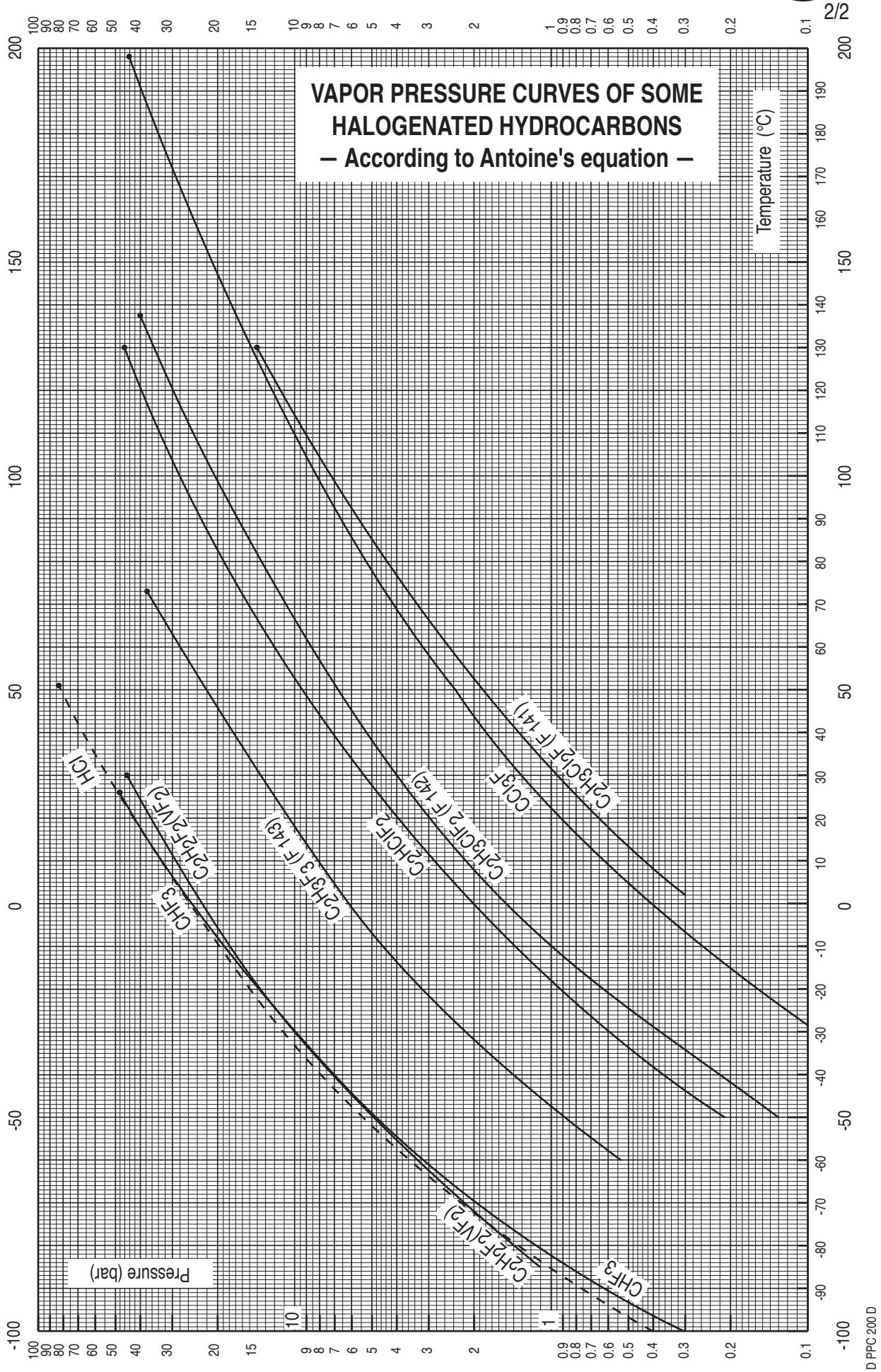
VAPOR PRESSURE CURVES OF SOME SOLVENTS
— According to Antoine's equation —



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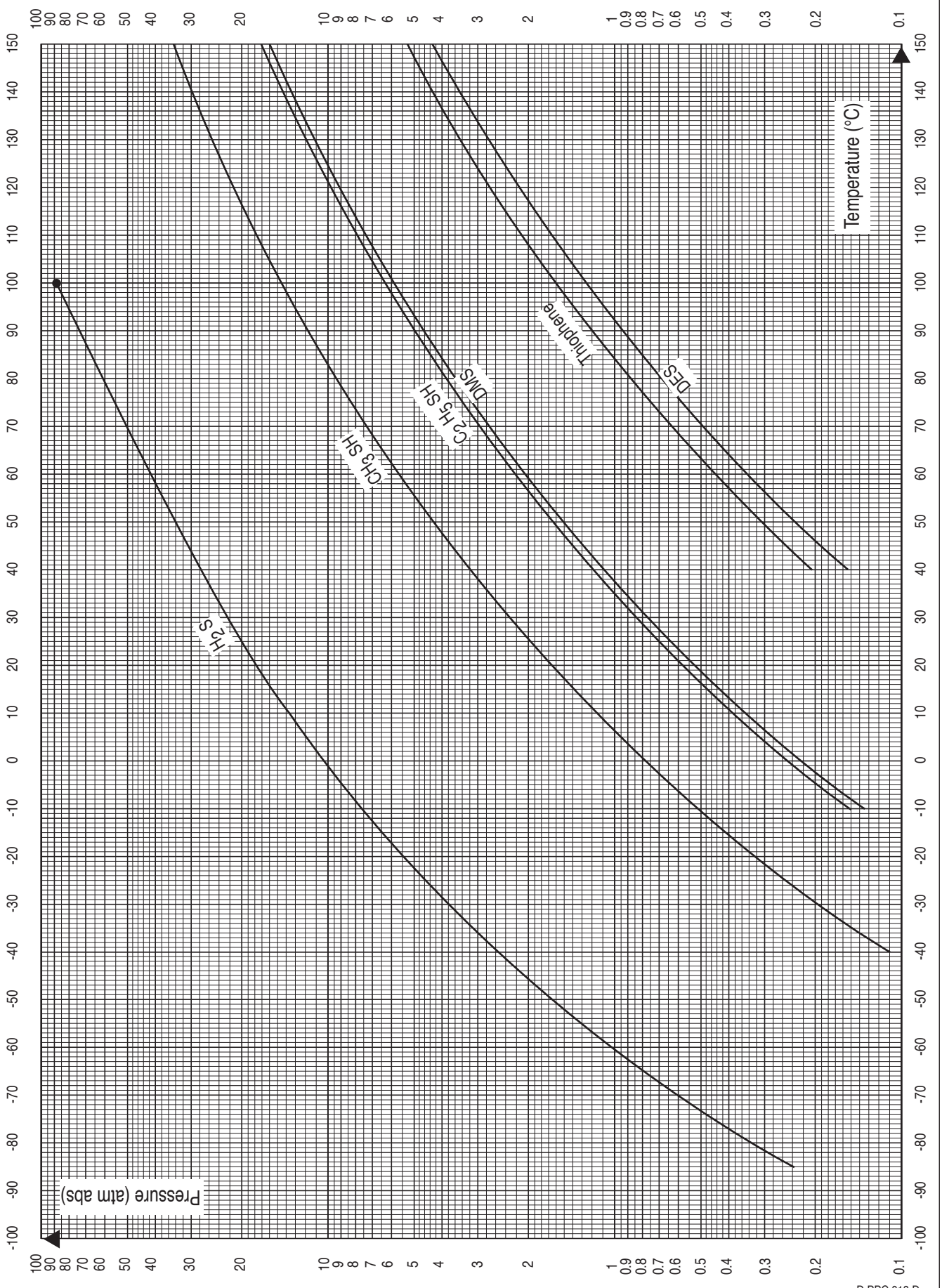


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VAPOR PRESSURE CURVES OF SOME SULFUR COMPOUNDS



D PPC 018 D

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VAPOR PRESSURES HARLACHER COEFFICIENTS*

$$\ln P^S = A' + \frac{B'}{C' + T} + C' \ln T + \frac{D' P^S}{T^2}$$

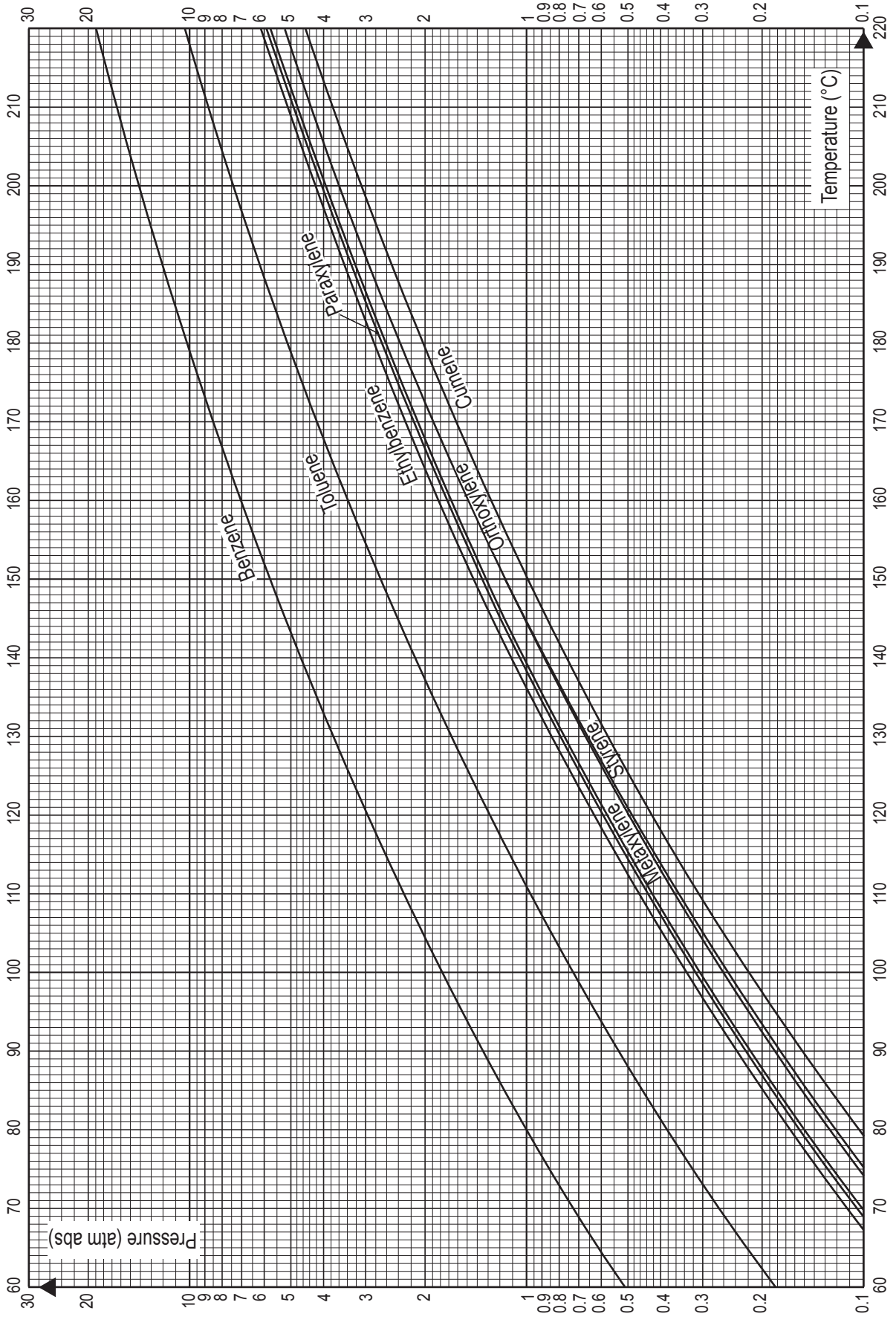
P^S in mercury mm - T in Kelvin

		A' B' C'			D'
Ar	Argon	31.173	– 1039.64	– 2.832	0.264
Cl_2	Chlorine	42.217	– 3412.28	– 3.894	1.27
H_2 H	ydrogen	12.050	– 114.95	0.048	0.048
H_2O	Water	55.336	– 6869.50	– 5.115	1.05
H_2S	Hydrogen sulfide	42.687	– 3132.31	– 3.985	0.871
HF	Hydrofluoric acid	26.160	– 3496.52	– 1.338	1.84
NH_3	Ammonia	51.947	– 4104.67	– 5.146	0.820
N_2	Nitrogen	31.927	– 924.86	– 3.075	0.264
O_2	Oxygen	31.041	– 1082.52	– 2.761	0.265
COS	Carbonyl sulfide	41.853	– 6137.78	– 3.914	1.30
CO_2	Carbon dioxide	52.703	– 3146.64	– 5.572	0.705
CH_4	Methane	30.715	– 1300.61	– 2.641	0.442
CH_4O	Methanol	72.268	– 7064.20	– 7.68	1.86
CH_4S	Methylmercaptan	46.610	– 4233.88	– 4.408	1.71
C_2H_2	Acetylene	46.122	– 2891.04	– 4.612	0.863
C_2H_3Cl	Vinyl chloride	48.672	– 3955.89	– 4.823	1.85
C_2H_3N	Acetonitrile	47.394	– 5392.43	– 4.357	3.49
C_2H_4	Ethylene	38.961	– 2282.37	– 3.678	0.881
C_2H_6 E	thane	38.759	– 2464.42	3.601	1.073
C_2H_6O	Ethanol	83.319	– 7994.90	– 9.201	2.35
C_2H_6S	Ethylmercaptan	51.954	– 490.34	– 5.139	2.55
C_3H_6	Propylene	44.794	– 3260.31	– 4.379	1.63
C_3H_8	Propane	43.492	– 3266.92	– 4.179	1.81
C_3H_8O	1-propanol	101.82	– 9416.25	– 11.79	3.13
C_3H_8O	Methylethylether	74.838	– 5631.77	– 8.549	2.45
C_4H_8	1-butene	48.333	– 3996.80	– 4.788	2.46
C_4H_8	2-butene (cis)	49.609	– 4217.05	– 4.938	2.58
C_4H_8	2-butene (trans)	50.137	– 4174.56	– 5.041	2.66
C_4H_8	Isobutene	50.832	– 4104.56	– 5.157	2.46
C_4H_8O	Methylethylketone	47.683	– 5328.22	– 4.426	3.88
C_4H_{10}	n-butane	48.334	– 4065.57	– 4.781	2.68
C_4H_{10}	Isobutane	46.141	– 3771.21	– 4.509	2.57
C_5H_{12}	n-pentane	52.682	– 4827.08	– 5.313	3.68
C_5H_{12}	Isopentane	50.428	– 4565.64	– 5.021	3.55
C_6H_6	Benzene	52.100	– 5557.61	– 5.072	3.61
C_6H_6O	Phenol	72.558	– 9072.60	– 7.516	4.42
C_6H_6N	Aniline	65.881	– 8442.37	– 6.662	5.18
C_6H_{14}	n-hexane	57.279	– 5587.42	– 5.885	4.778
C_7H_8	Toluene	56.785	– 6283.50	– 5.681	4.84
C_7H_{14}	Methylcyclohexane	52.902	– 5797.19	– 5.199	5.23
C_7H_{16}	n-heptane	61.276	– 6303.87	– 6.373	6.00
C_8H_{10}	o-xylene	61.763	– 7149.21	– 6.302	6.11
C_8H_{10}	m-xylene	55.493	– 6666.23	– 5.436	6.08
C_8H_{10}	p-xylene	56.175	– 6673.70	– 5.543	6.19
C_8H_{10}	Ethylbenzene	58.100	– 6792.54	– 5.802	5.75
C_8H_{18}	n-octane	66.639	– 7100.69	– 7.053	7.31
C_9H_{20}	n-nonane	73.133	– 7969.42	– 73.890	8.69
$C_{10}H_{22}$	n-decane	75.475	– 8563.64	– 8.149	10.2

According to "The properties of gases liquids" Reid-Prausnitz-Sherwood

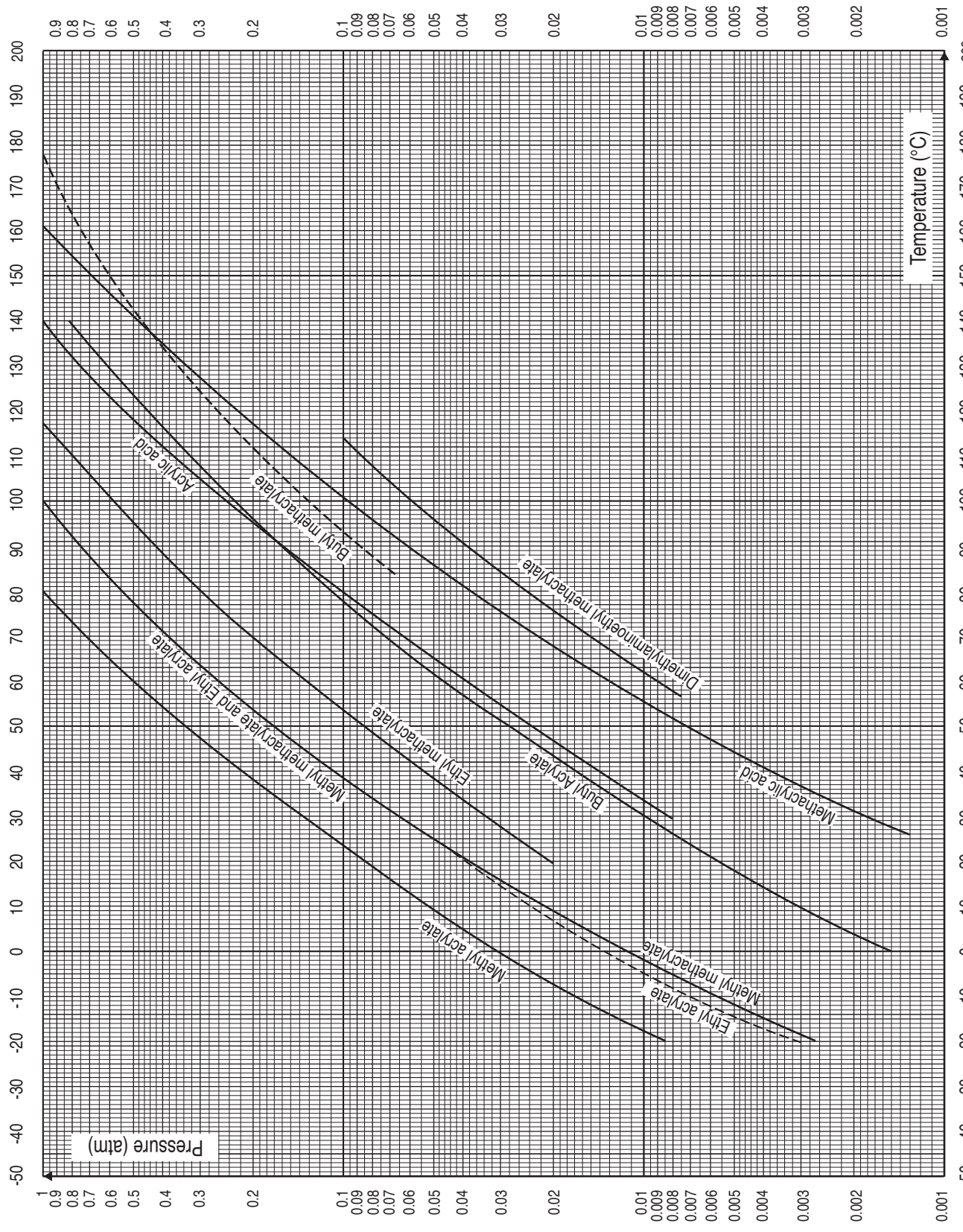
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AROMATIC HYDROCARBONS
— Vapor pressure curves from Antoine's equation —



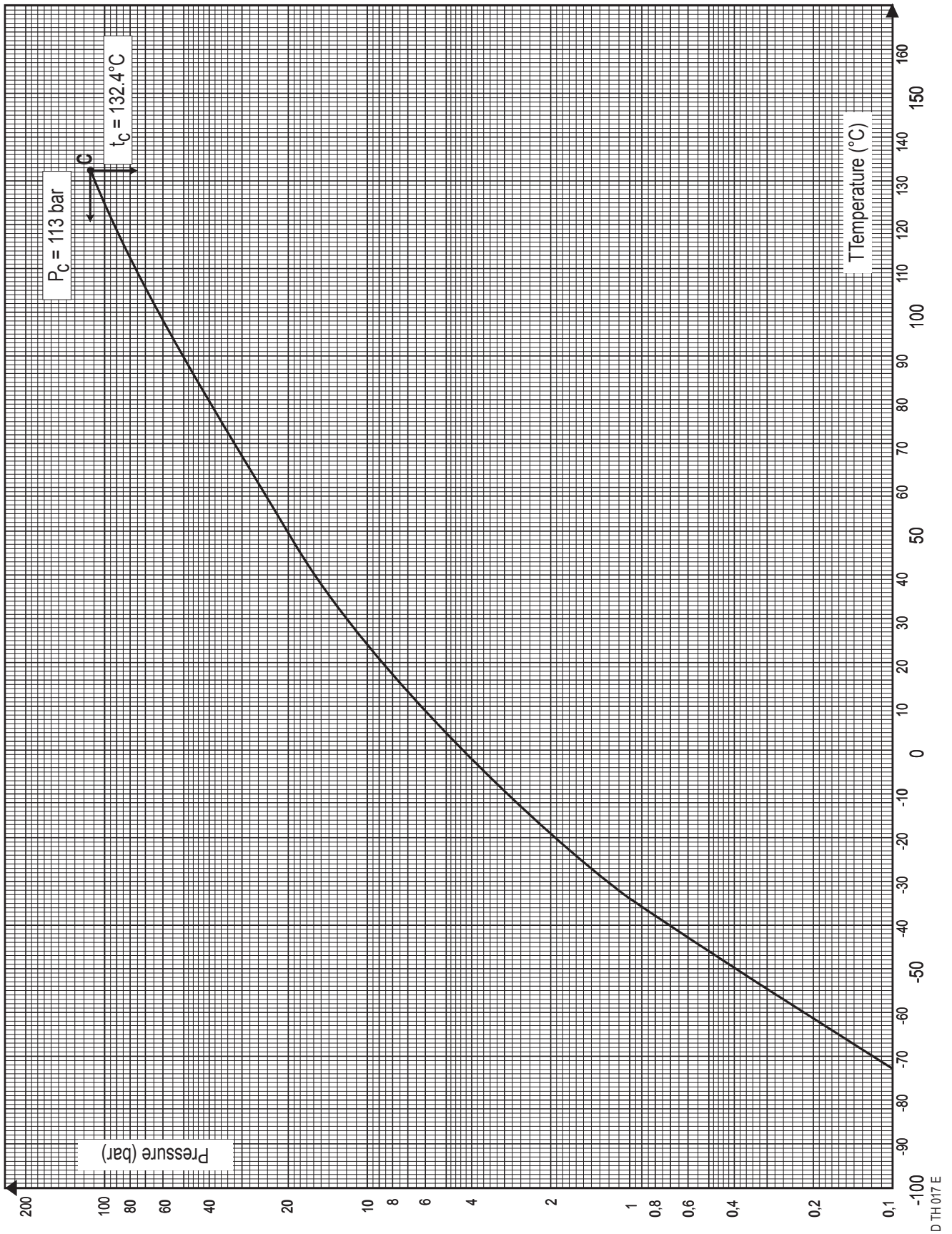
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VAPOR PRESSURE CURVES OF ACRYLIC AND METHACRYLIC ACIDS, AND DERIVATED ESTERS, AT PRESSURE LOWER THAN 1 ATM



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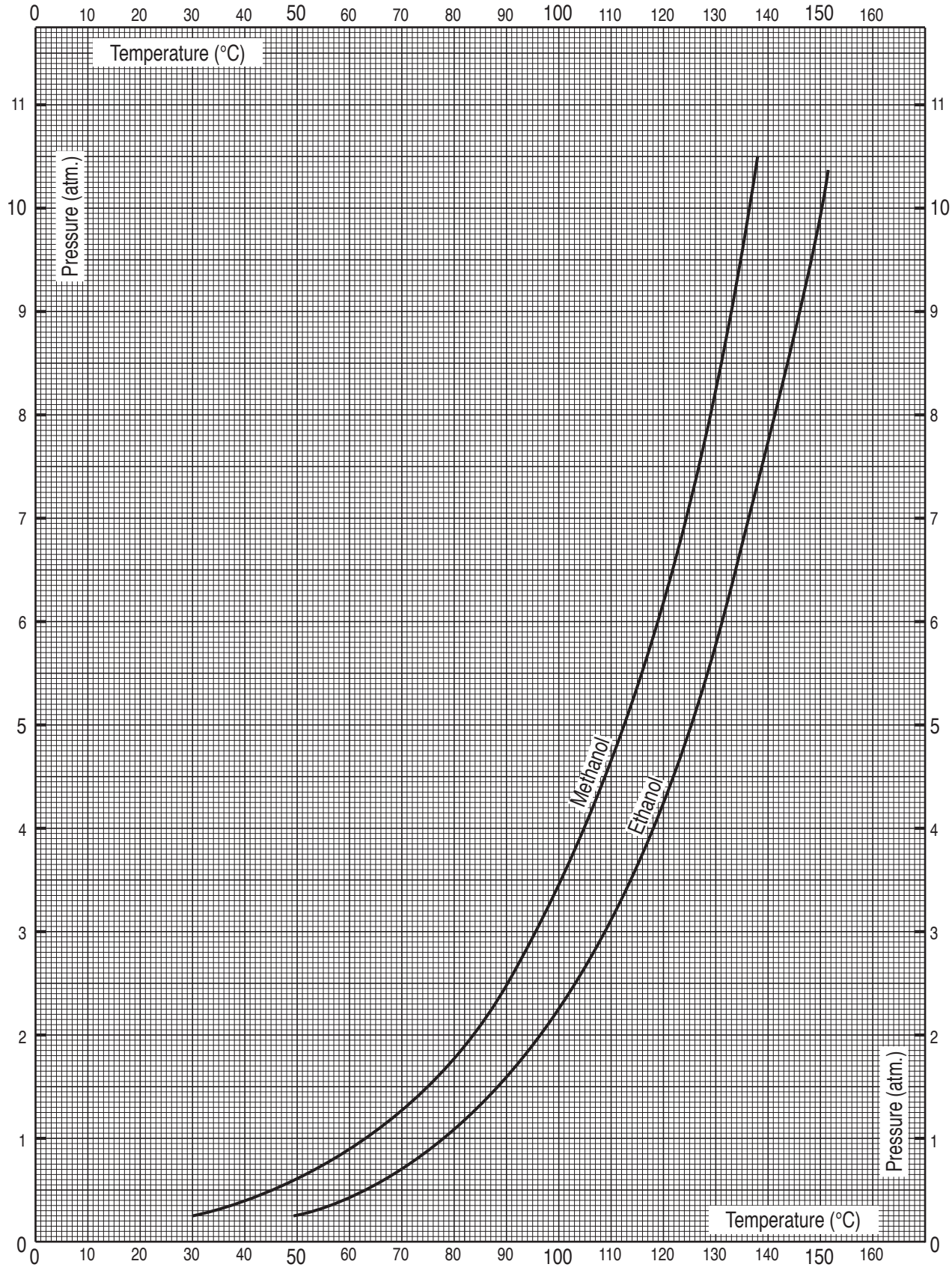
VAPOR PRESSURE CURVE OF AMMONIA
REF : PERRY



D TH 017 E

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VAPOR PRESSURE CURVES OF METHANOL AND ETHANOL



D TH 1067 D

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VAPOR PRESSURE AND ENTHALPY FOR SATURATED WATER AND STEAM — From 0 to 100°C —

Temperature	Vapor pressure		Enthalpy of saturated			
			Liquid		Vapor	
°C	mmHg	bar	kcal/kg	kJ/kg	kcal/kg	kJ/kg
0	4.6	0.006	0	0	597.3	2500
5	6.5	0.009	5.0	21.1	599.5	2509
10	9.2	0.012	10.0	42.0	601.7	2518
15	12.8	0.017	15.0	62.9	603.9	2528
20	17.5	0.023	20.0	83.9	606.0	2536
25	23.8	0.032	25.0	104.8	608.2	2546
30	31.8	0.042	30.0	125.6	610.4	2555
35	42.2	0.056	35.0	146.5	612.6	2564
40	55.3	0.074	40.0	167.5	614.7	2573
45	71.9	0.096	45.0	188.3	616.8	2582
50	92.5	0.123	50.0	209.2	619.0	2591
55	118.1	0.157	55.0	230.1	621.1	2600
60	149.4	0.199	60.0	251.0	623.2	2608
65	187.6	0.250	65.0	272.0	625.2	2617
70	233.8	0.312	70.0	292.9	627.3	2626
75	289.1	0.385	75.0	313.9	629.3	2634
80	355.2	0.474	80.0	334.8	631.3	2642
85	433.5	0.578	85.0	355.9	633.3	2651
90	525.9	0.701	90.0	376.9	635.2	2659
95	634.0	0.845	95.1	397.9	637.2	2667
100	760.0	1.013	100.1	419.0	639.1	2675

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VAPOR PRESSURE AND ENTHALPY FOR SATURATED WATER AND STEAM — From 100 to 200°C —

Temperature	Vapor pressure		Enthalpy of saturated			
			Liquid		Vapor	
°C	atm.	bar	kcal/kg	kJ/kg	kcal/kg	kJ/kg
100	1.000	1.013	100.1	419.0	639.1	2675
105	1.192	1.208	105.1	440.1	640.9	2682
110	1.414	1.433	110.2	461.2	642.8	2690
115	1.668	1.691	115.2	482.4	644.6	2698
120	1.959	1.985	120.3	503.5	646.4	2706
125	2.290	2.321	125.4	524.9	648.1	2713
130	2.666	2.701	130.5	546.2	649.8	2720
135	3.089	3.130	135.6	567.6	651.4	2726
140	3.566	3.614	140.7	588.9	653.0	2733
145	4.101	4.155	145.8	610.2	654.5	2739
150	4.698	4.760	151.0	632.0	656.0	2746
155	5.362	5.433	156.2	653.8	657.5	2752
160	6.099	6.180	161.3	675.1	658.7	2757
165	6.916	7.008	166.5	696.9	660.0	2762
170	7.816	7.920	171.8	719.1	661.3	2768
175	8.808	8.925	177.0	740.8	662.4	2772
180	9.896	10.03	182.3	763.0	663.6	2777
185	11.09	11.23	187.6	785.2	664.6	2782
190	12.39	12.55	192.9	807.4	665.5	2785
195	13.81	13.99	198.2	829.6	666.3	2789
200	15.35	15.55	203.6	852.2	667.1	2792

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VAPOR PRESSURE AND ENTHALPY FOR SATURATED WATER AND STEAM — From 200 to 300°C —

Temperature	Vapor pressure		Enthalpy of saturated			
			Liquid		Vapor	
°C	atm.	bar	kcal/kg	kJ/kg	kcal/kg	kJ/kg
200	15.35	15.55	203.6	852.2	667.1	2792
205	17.02	17.24	209.0	874.8	667.7	2795
210	18.83	19.08	214.4	897.4	668.3	2797
215	20.79	21.06	219.9	920.4	668.8	2799
220	22.90	23.20	225.4	943.4	669.1	2801
225	25.17	25.50	230.9	966.4	669.3	2801
230	27.61	27.98	236.5	989.9	669.5	2802
235	30.23	30.63	242.2	1014	669.7	2803
240	33.04	33.48	247.8	1037	669.5	2802
245	36.05	36.52	253.6	1061	669.4	2802
250	39.26	39.78	259.3	1085	669.0	2800
255	42.68	43.25	265.2	1110	668.5	2798
260	46.33	46.94	271.1	1135	667.9	2795
265	50.20	50.87	277.1	1160	667.3	2793
270	54.33	55.05	283.1	1185	666.3	2789
275	58.71	59.49	289.2	1210	665.2	2784
280	63.35	64.19	295.4	1236	663.9	2779
285	68.27	69.18	301.7	1263	662.4	2772
290	73.48	74.45	308.1	1290	660.7	2765
295	78.98	80.02	314.6	1317	658.8	2757
300	84.79	85.92	321.2	1344	656.6	2748

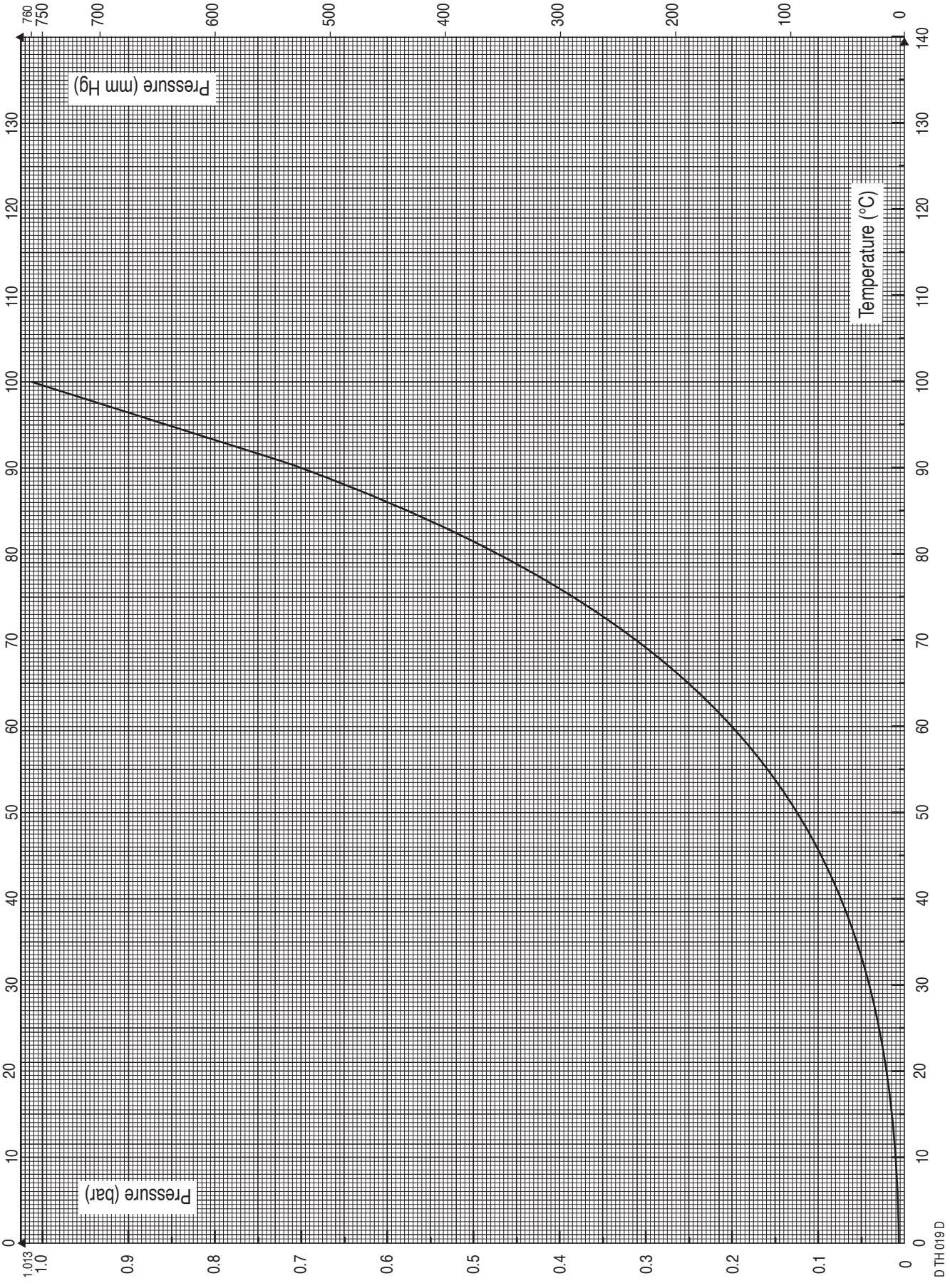
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VAPOR PRESSURE AND ENTHALPY FOR SATURATED WATER AND STEAM — From 300 to Critical Point —

Temperature	Vapor pressure		Enthalpy of saturated			
			Liquid		Vapor	
°C	atm.	bar	kcal/kg	kJ/kg	kcal/kg	kJ/kg
300	84.79	85.92	321.2	1344	656.6	2748
305	90.93	92.13	328.0	1373	654.2	2738
310	97.40	98.69	334.9	1402	651.4	2726
315	104.2	105.6	342.0	1431	648.3	2713
320	111.4	112.9	349.2	1462	644.9	2699
325	119.0	120.6	356.7	1493	641.0	2683
330	127.0	128.6	364.5	1526	636.7	2665
335	135.4	137.1	372.5	1559	631.8	2644
340	144.2	146.1	380.9	1594	626.2	2621
345	153.4	155.5	389.8	1632	619.9	2595
350	163.2	165.4	399.2	1671	612.5	2564
355	173.5	175.8	409.4	1714	603.6	2526
360	184.3	186.7	420.7	1761	592.6	2480
365	195.7	198.3	434.1	1817	578.2	2420
370	207.8	210.5	452.0	1892	556.7	2330
374.15	218.39	221.29	501.5	2099	501.5	2099

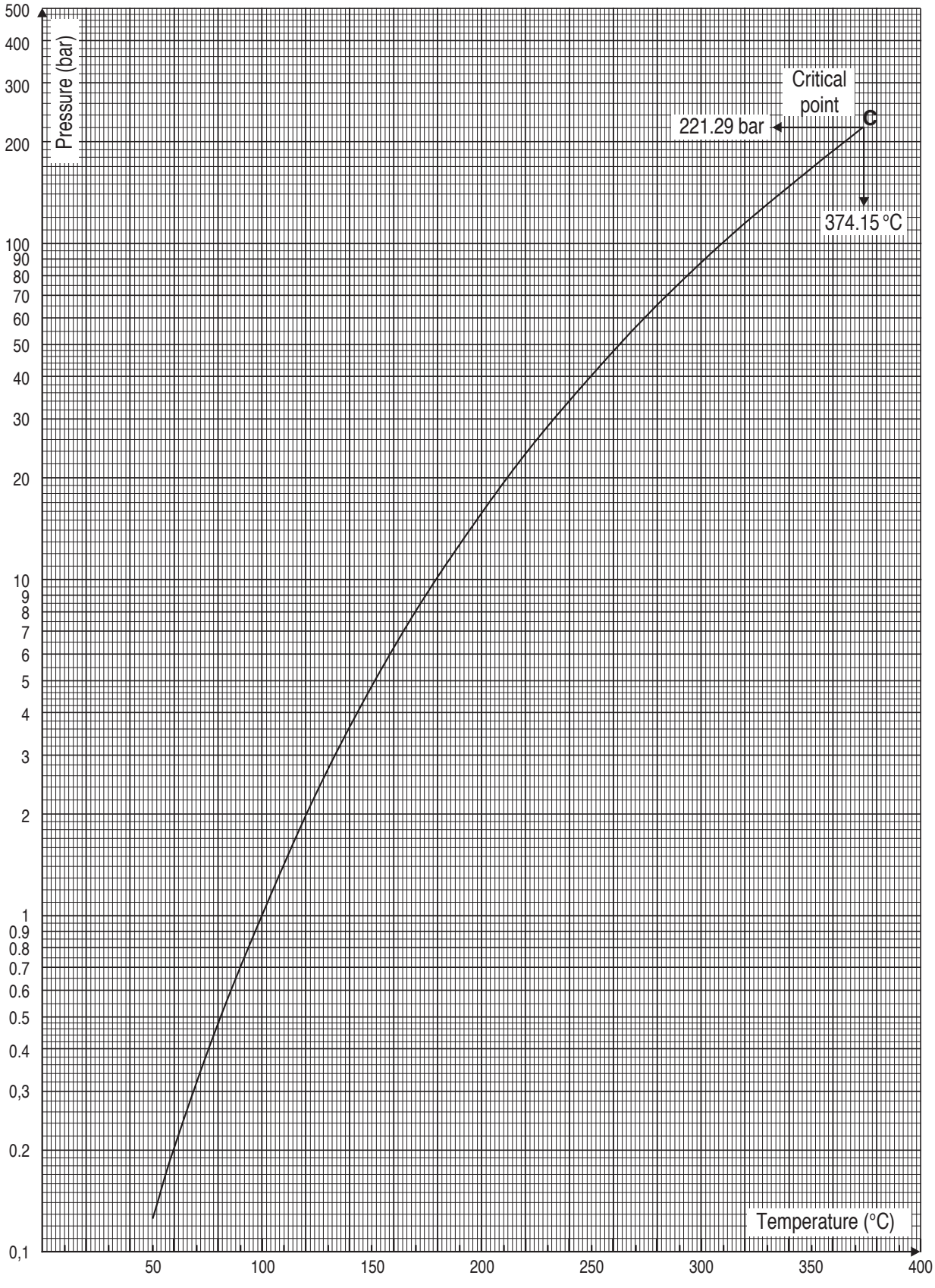
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VAPOR PRESSURE OF WATER FROM 0°C TO 100°C



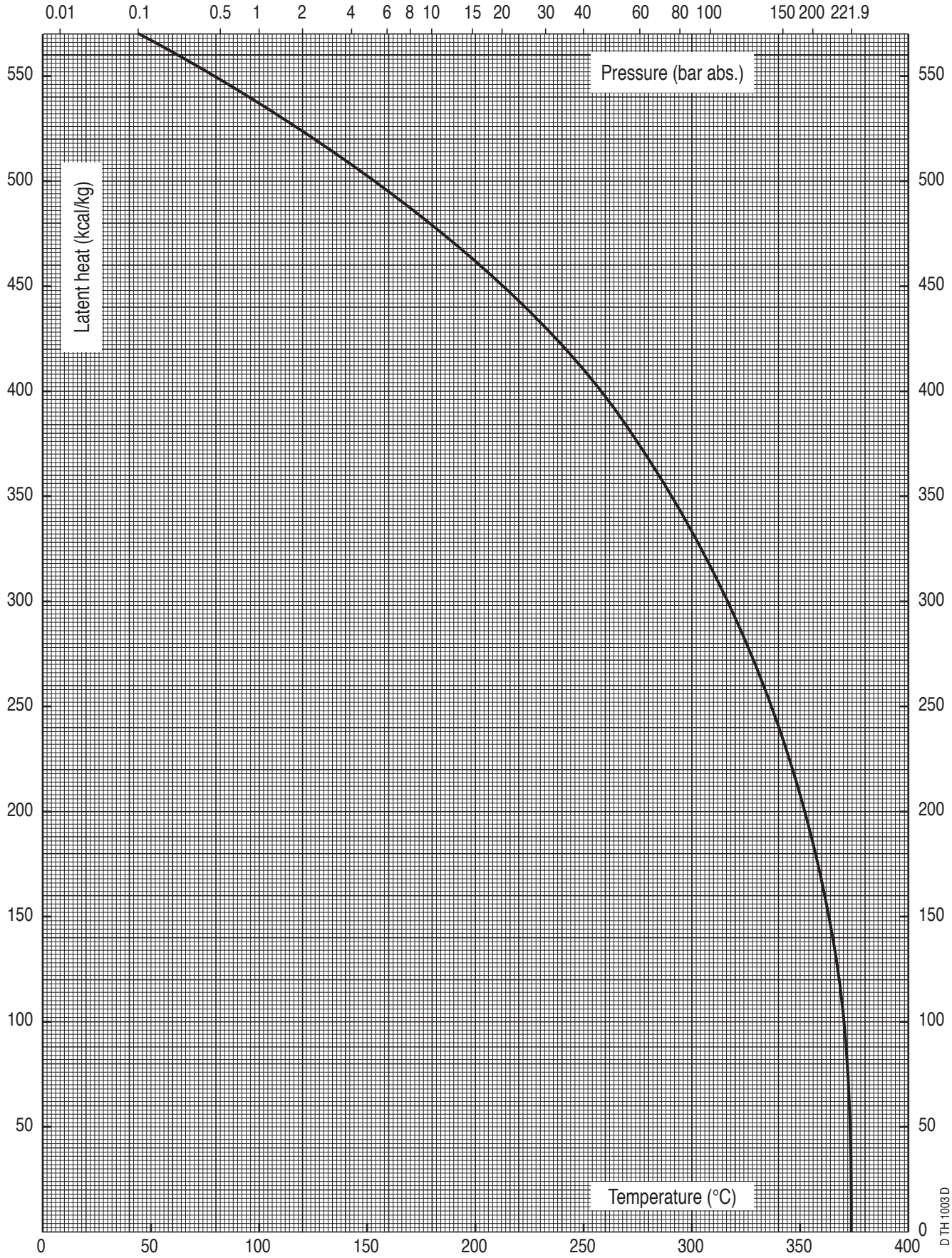
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VAPOR PRESSURE OF WATER FROM 50°C TO CRITICAL POINT



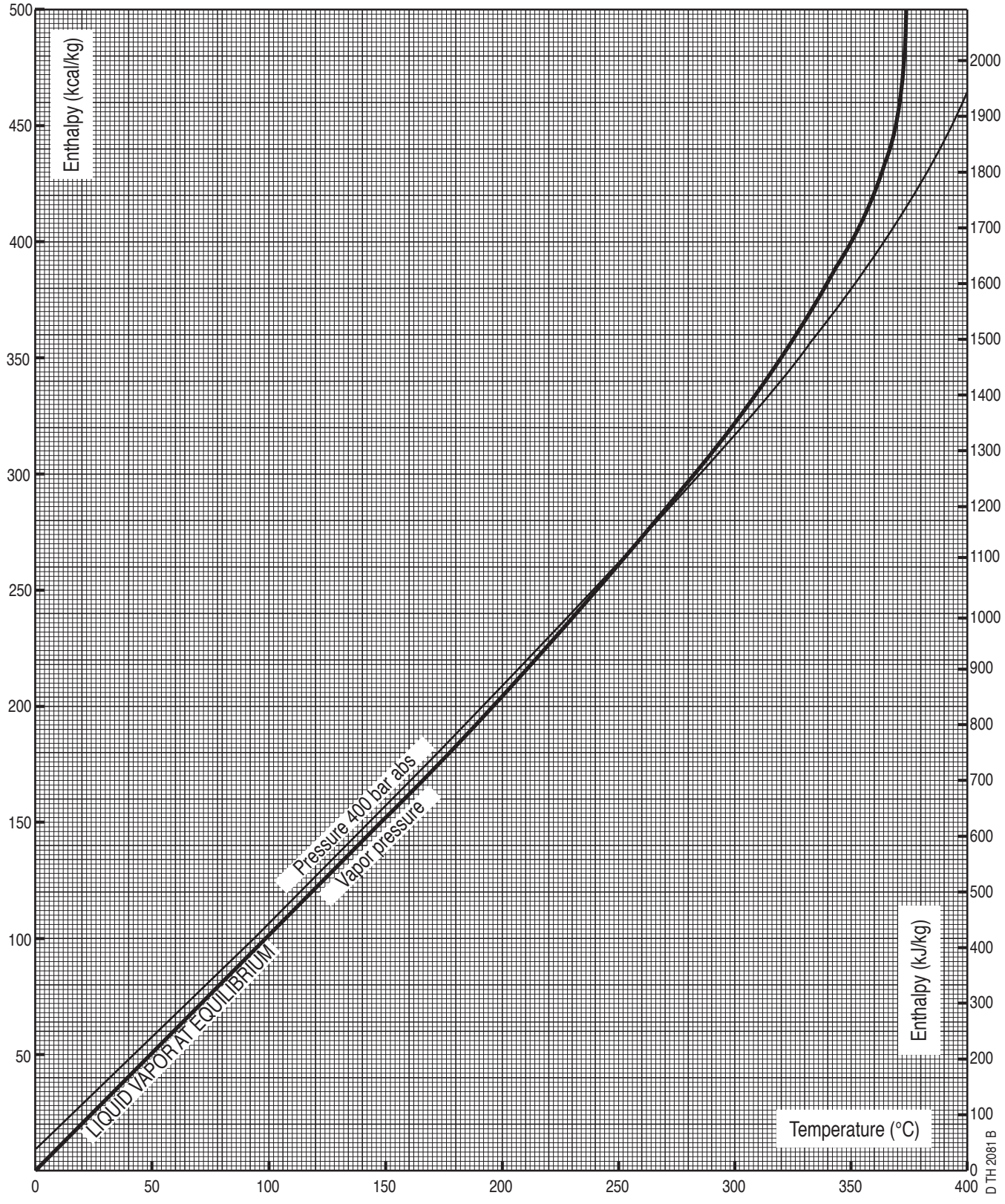
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WATER LATENT HEAT OF VAPORIZATION VARIATION WITH TEMPERATURE



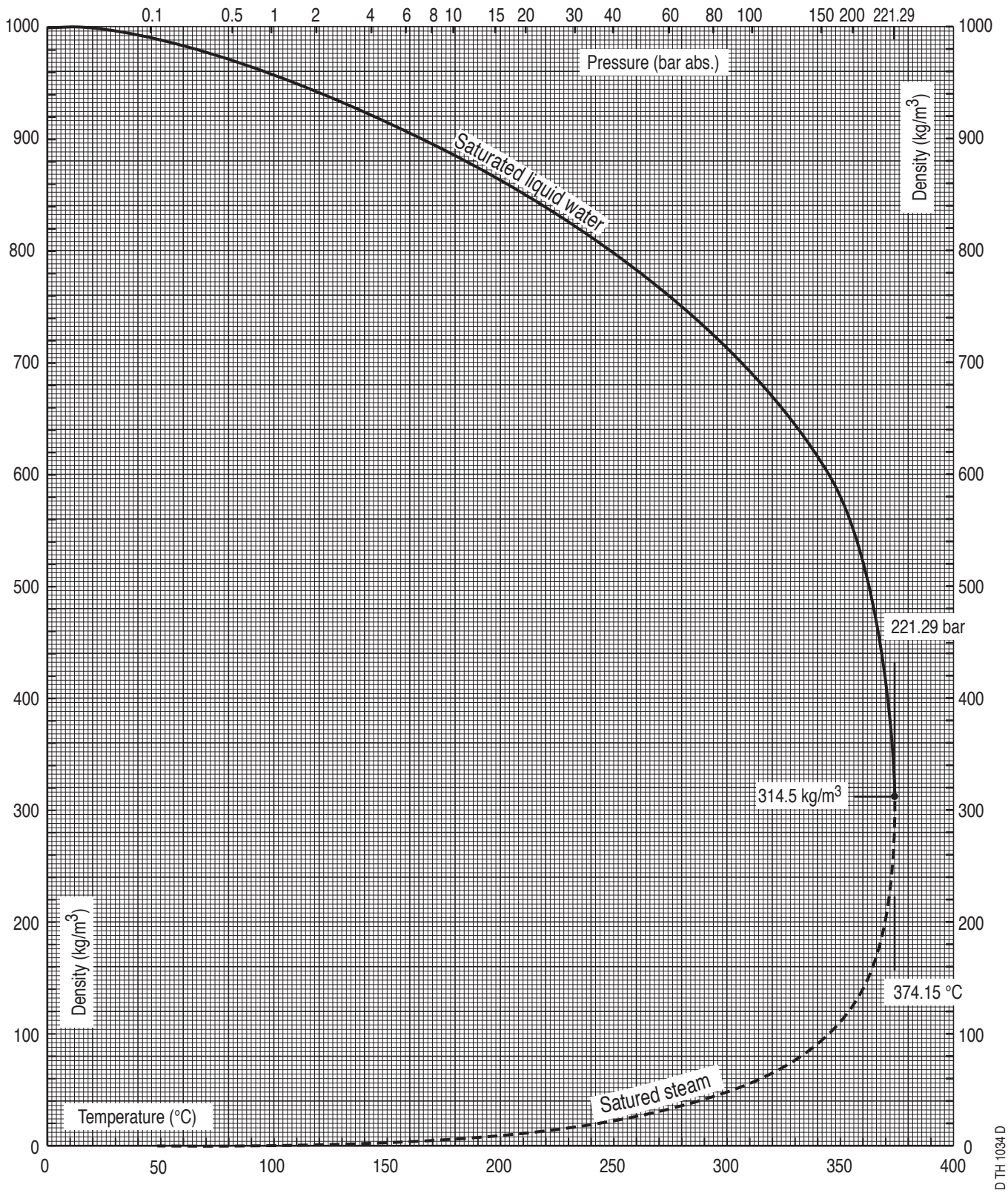
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ENTHALPY OF PRESSURIZED LIQUID WATER



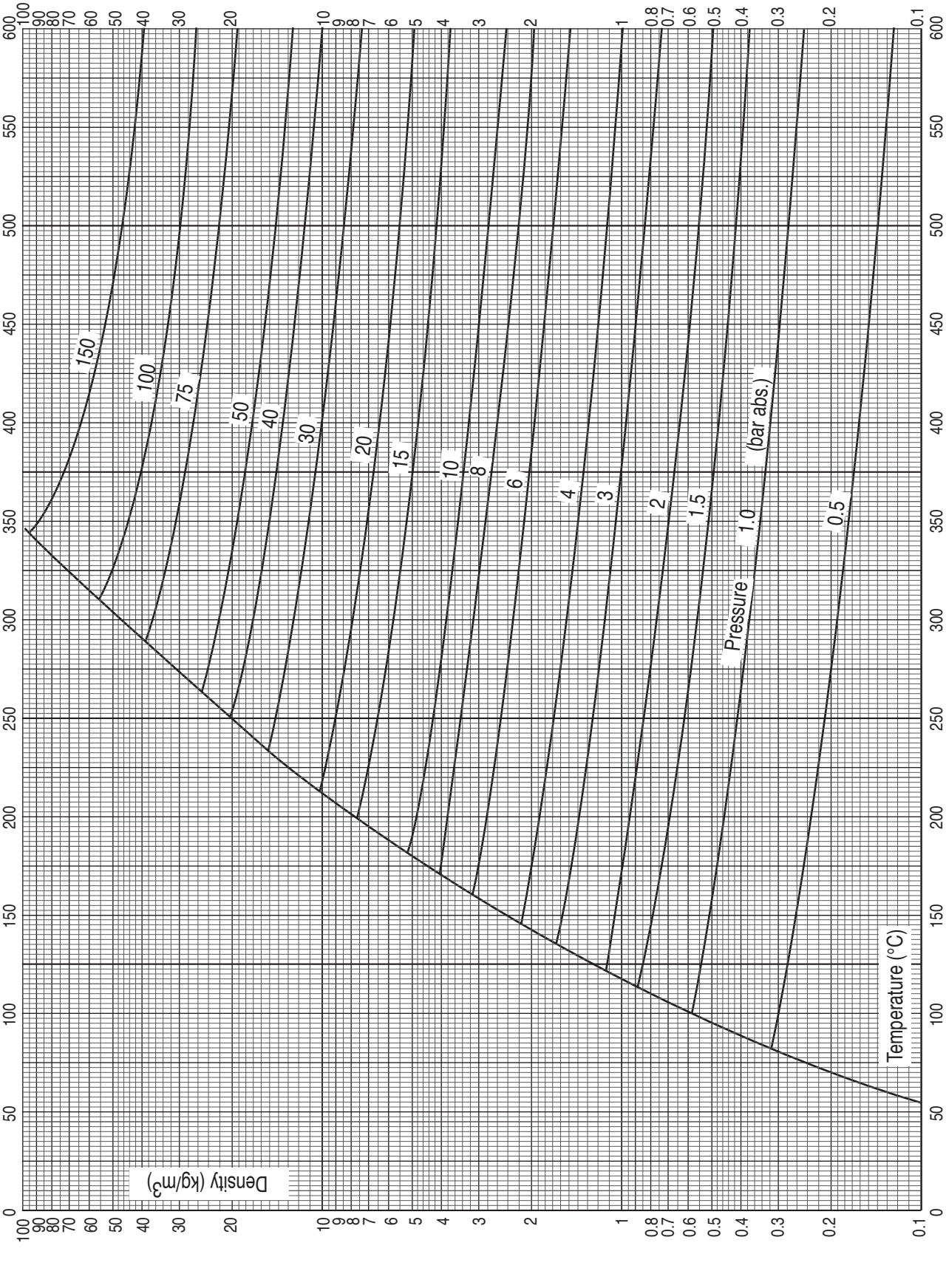
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DENSITY OF SATURATED LIQUID WATER AND STEAM



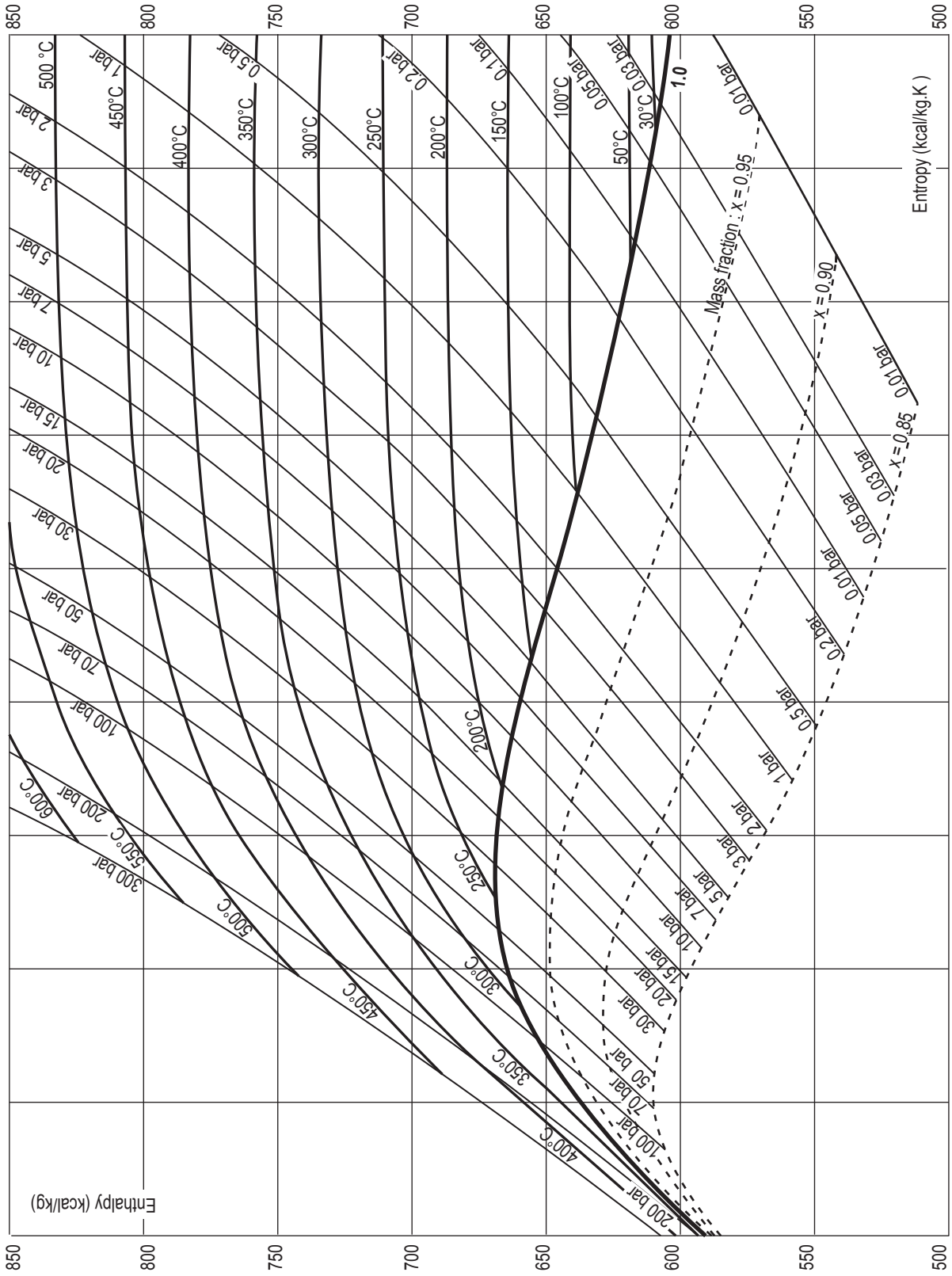
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DENSITY OF STEAM



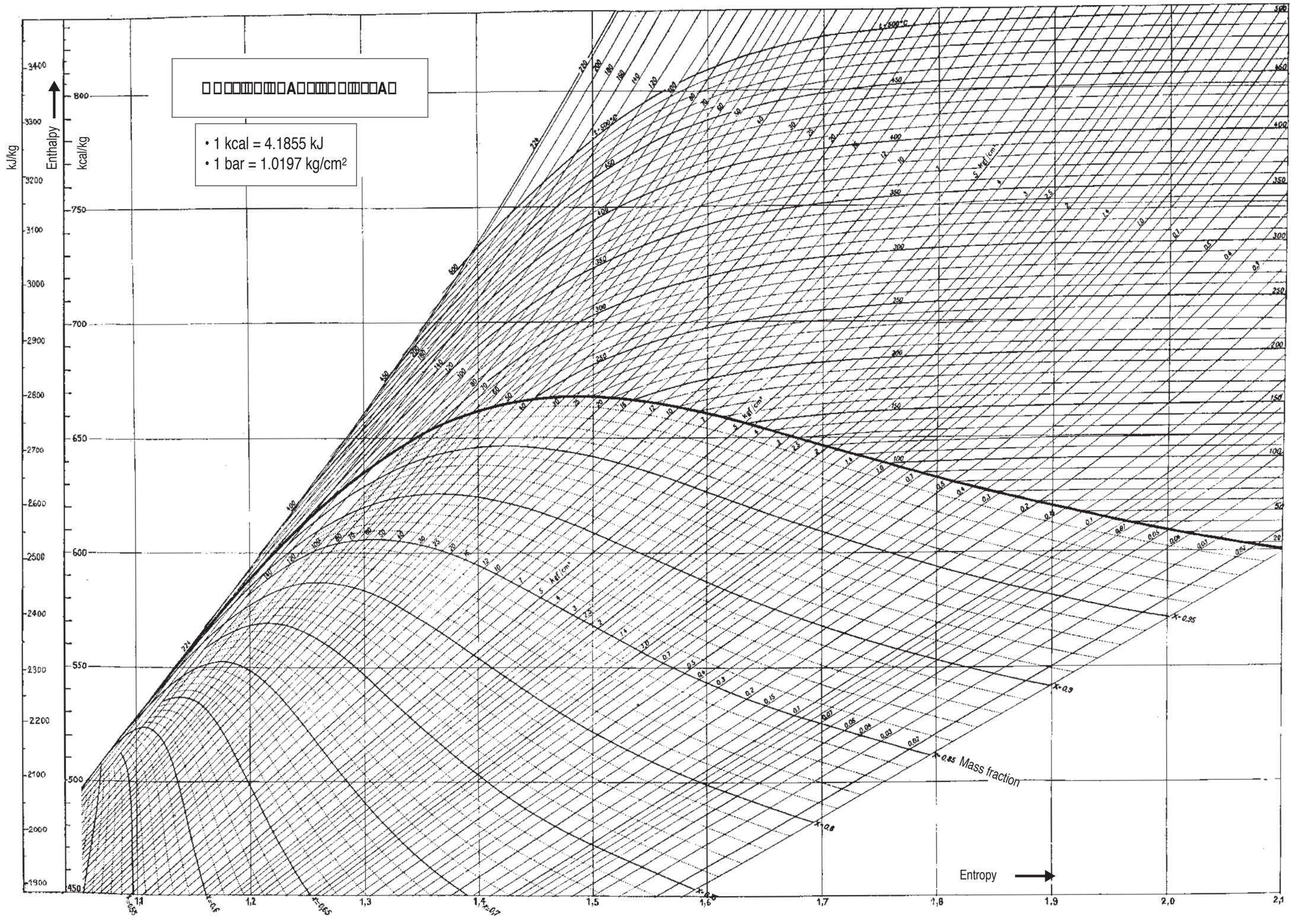
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MOLLIER CHART FOR STEAM



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MOLAR SPECIFIC HEAT OF IDEAL GASES

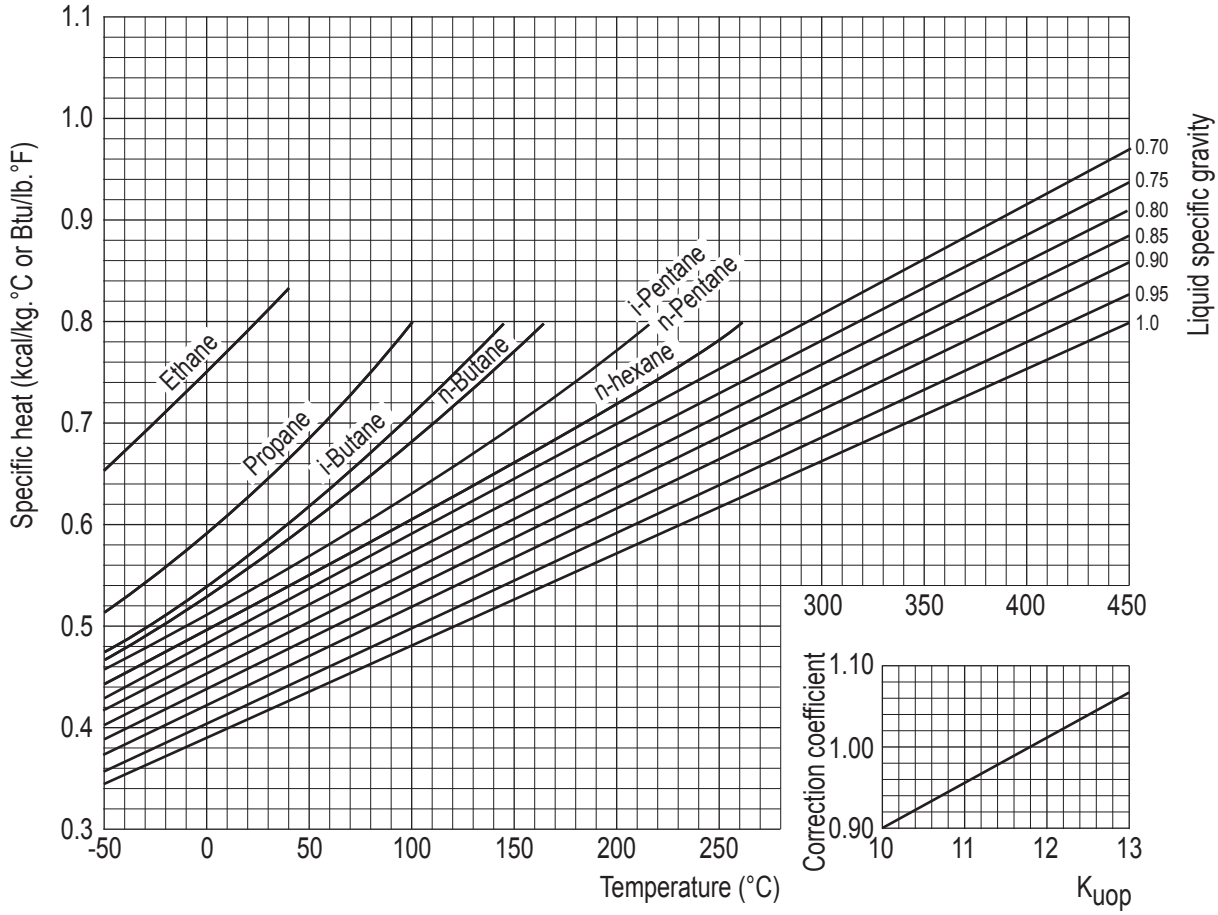
$Cp = a + b \cdot T + c \cdot T^2 + d \cdot T^3$ Cp in cal/mol . K T in K

	a	b . 10^2	c . 10^5	d . 10^9	Temperature range
Organic compounds					
Methane	4.75	1.2	0.303	– 2.63	273 - 1500 K
Ethane	1.648	4.124	– 1.53	1.74	
Propane	– 0.966	7.279	– 3.755	7.58	
n-butane	0.945	8.873	– 4.38	8.36	
Isobutane	– 1.89	9.936	– 5.495	11.92	
n-pentane	1.618	10.85	– 5.365	10.10	
Isopentane	– 2.273	12.434	– 7.097	15.86	
Néopentane	– 3.865	13.305	– 8.018	18.83	
n-hexane	1.657	13.19	– 6.844	13.78	
Ethylene	0.944	3.735	– 1.993	4.22	
Propylene	0.753	5.691	– 2.91	5.88	
1-butene	– 0.24	8.65	– 5.11	12.07	
Isobutene	1.65	7.702	– 3.981	8.02	
2-butene (cis)	– 1.778	8.078	– 4.074	7.89	
2-butene (trans)	2.34	7.22	– 3.403	6.07	
Cyclopropane	– 6.481	8.206	– 5.577	15.61	
Cyclopentane	– 12.957	13.087	– 7.447	16.41	
Methylcyclopentane	– 12.114	15.38	– 8.915	20.03	
Cyclohexane	– 15.935	16.454	– 9.203	19.27	
Methylcyclohexane	– 15.07	18.972	– 10.989	24.09	
Benzene	– 8.65	11.578	– 7.54	18.54	
Toluene	– 8.213	13.357	– 8.23	19.2	
Ethylbenzene	– 8.398	15.935	– 10.003	23.95	
Styrene	– 5.968	14.354	– 9.15	22.03	
Cumene	– 9.452	18.686	– 11.869	28.8	
o-xylene	– 3.789	14.291	– 8.354	18.8	
m-xylene	– 6.533	14.905	– 8.831	20.05	
p-xylene	– 5.334	14.22	– 7.984	17.03	
Acetylene	5.21	2.2008	– 1.559	4.349	
Methylacetylene	4.21	4.073	– 2.192	4.713	
Dimethylacetylene	3.54	5.838	– 2.76	4.974	
Propadiene	2.43	4.693	– 2.781	6.484	
1,3-butadiene	– 1.29	8.35	– 5.582	14.24	
Isoprene	– 0.44	10.418	– 6.762	16.93	

	a	b . 10^2	c . 10^5	d . 10^9	Temperature range
Inorganic compounds					
Nitrogen	7.440	– 0.324	0.640	– 2.79	273 - 1800 K
Oxygen	6.713	$-0.879.10^{-4}$	0.417	– 2.544	273 - 1800 K
Argon	4.969	$-0.767.10^{-3}$	$-1.234.10^{-3}$	0	273 - 1800 K
Hydrogen	6.483	0.221	– 0.333	1.826	273 - 1800 K
Carbon monoxide	7.373	– 0.307	0.666	– 3.037	273 - 1800 K
Carbon dioxide	4.728	1.754	– 1.338	4.097	273 - 1800 K
Steam	7.701	0.0459	0.252	– 0.859	273 - 1800 K
Sulfur biatomic (S_2)	6.499	52.98	– 38.88	9.52	273 - 1800 K
Sulfur oxide (SO_2)	6.157	138.4	– 91.03	20.57	273 - 1800 K
Sulfur trioxide (SO_3)	3.918	348.3	– 267.5	77.44	273 - 1500 K
Hydrogen sulfide (H_2S)	7.07	31.28	13.64	– 7.867	273 - 1800 K
Carbon sulfide	7.39	148.9	– 109.6	27.6	273 - 1800 K
Carbon oxysulfide	6.222	153.6	– 105.8	25.6	273 - 1800 K
Fluorine	6.115	58.64	– 41.86	9.797	273 - 2000 K
Chlorine	6.8214	57.095	– 51.07	115.47	273 - 1500 K
Bromine	8.051	24.62	– 21.28	6.406	273 - 1500 K
Iodine	8.504	13.135	– 10.684	3.125	273 - 1800 K
HF	7.201	– 11.78	15.76	– 3.76	273 - 2000 K
HCl	7.244	– 18.2	31.7	– 10.36	273 - 1500 K
HBr	7.169	– 16.04	33.14	11.61	273 - 1500 K
HI	6.702	4.546	12.16	– 4.813	273 - 1900 K
NO	6.461	23.58	– 7.705	0.8729	
N_2O	5.758	140.04	– 85.08	25.26	
NO_2	5.48	136.5	– 84.1	18.8	
N_2O_4	7.9	446	– 271	0	
Cyanogen	9.82	148.58	– 65.71		273 - 1500 K
Hydrocyanic acid	6.34	83.75	– 26.11		273 - 1000 K

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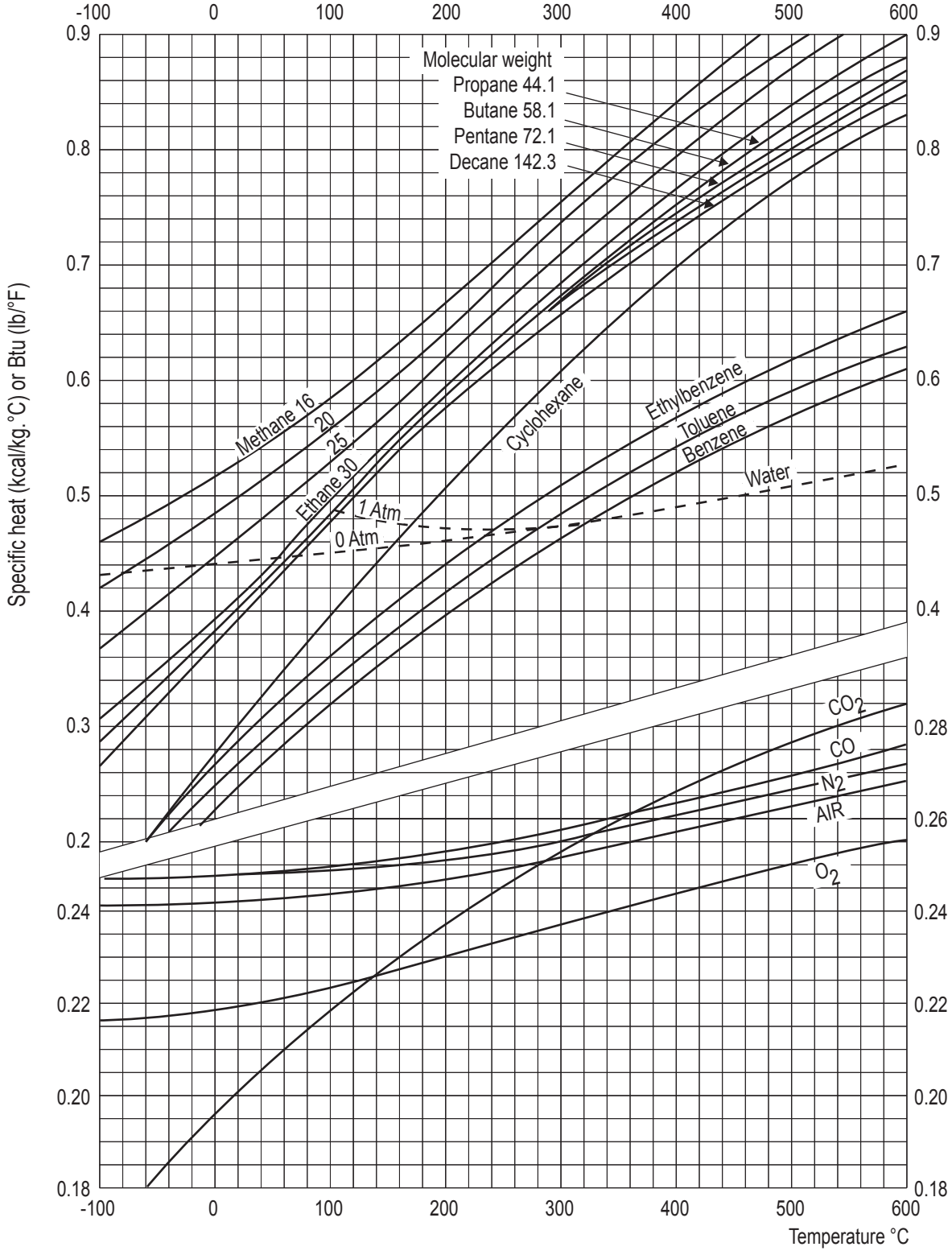
SPECIFIC HEAT OF LIQUID HYDROCARBONS



DTH2071D

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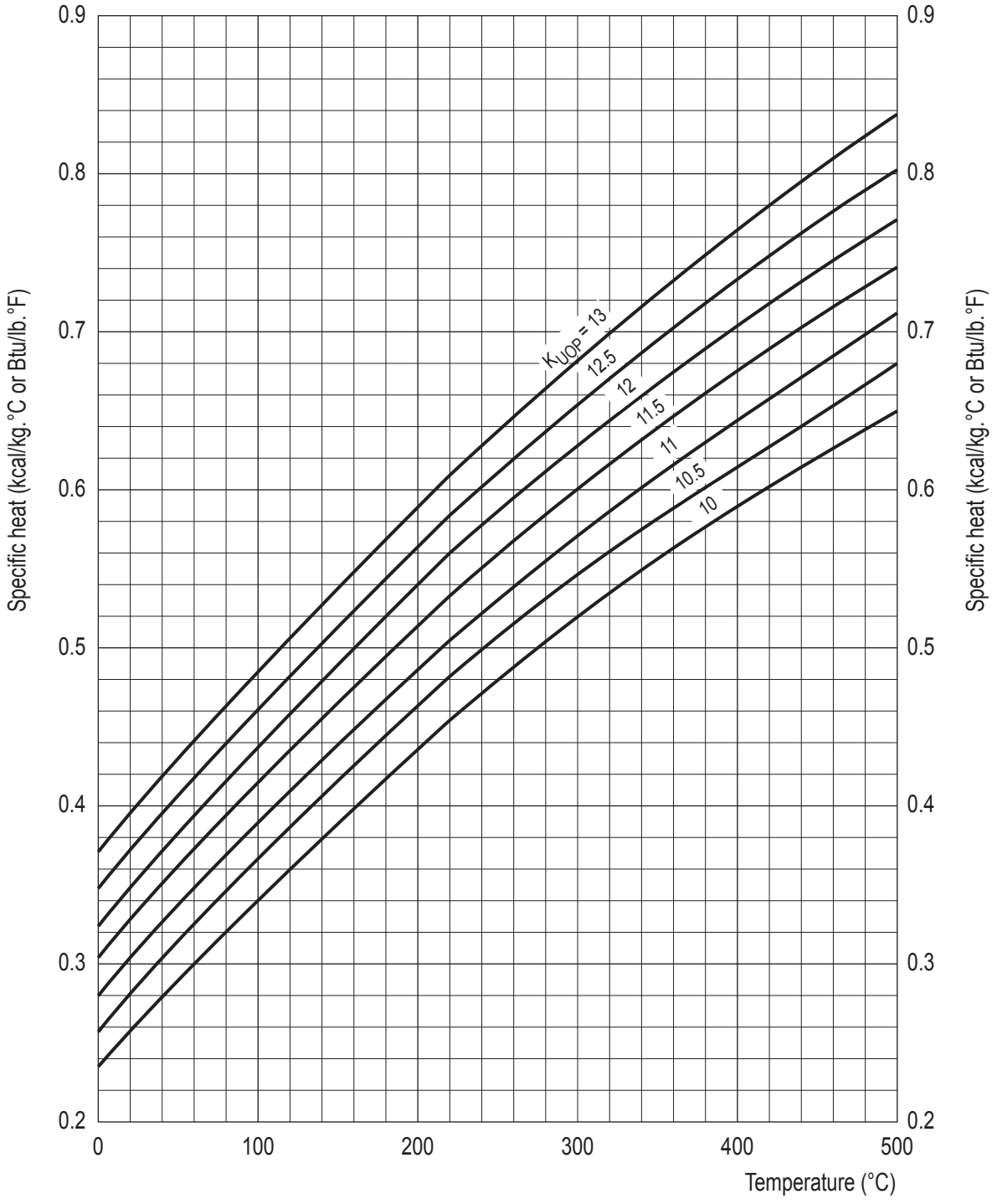
SPECIFIC HEAT OF GAZEOUS HYDROCARBONS AND SOME GASES AT ATMOSPHERIC PRESSURE



D T 2049 E

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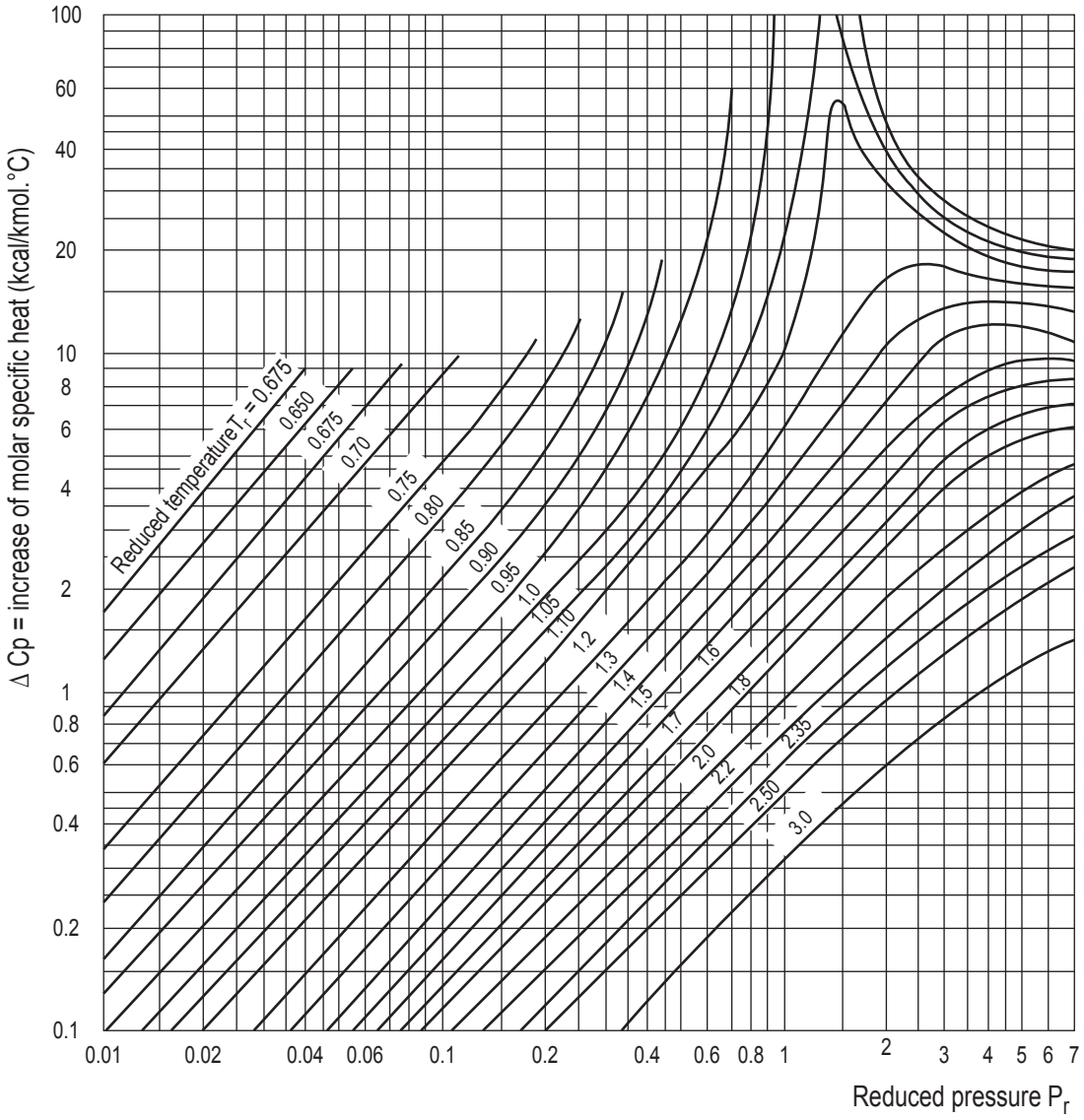
MASS SPECIFIC OF GASEOUS PETROLEUM CUTS AT ATMOSPHERIC AND VACUUM CONDITIONS



DTH 1402 D

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MOLAR SPECIFIC HEAT CORRECTION
— As a function of reduced pressure and temperature —



D TH 1401 D

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AVERAGE MOLAR HEAT FOR SOME GASES
— between 0° and t°C —
(cal/mol.k)

t°C	H_2	N_2	O_2	CO	H_2O_v	CO_2
0	6.90	6.93	6.96	6.92	8.00	8.63
20	6.90	6.94	6.98	6.93	8.02	8.73
50	6.90	6.95	7.02	6.95	8.03	8.88
100	6.92	6.97	7.05	6.97	8.03	9.17
200	6.95	7.00	7.15	7.00	8.12	9.65
300	6.97	7.04	7.26	7.06	8.22	10.06
400	6.98	7.09	7.38	7.12	8.34	10.40
500	6.99	7.15	7.49	7.19	8.47	10.75
600	7.01	7.21	7.59	7.27	8.60	11.03
700	7.03	7.27	7.68	7.34	8.74	11.28
800	7.06	7.35	7.77	7.43	8.89	11.50
900	7.09	7.42	7.85	7.50	9.04	11.70
1000	7.12	7.49	7.92	7.57	9.18	11.88
1100	7.15	7.56	7.98	7.64	9.32	12.05
1200	7.20	7.62	8.04	7.70	9.45	12.19
1300	7.24	7.67	8.11	7.76	9.58	12.32
1400	7.28	7.73	8.16	7.81	9.72	12.45
1500	7.32	7.78	8.20	7.85	9.84	12.56
1600	7.36	7.82	8.24	7.90	9.96	12.66
1700	7.40	7.86	8.28	7.94	10.09	12.75
1800	7.45	7.91	8.33	7.98	10.20	12.84
1900	7.49	7.94	8.38	8.02	10.30	12.92
2000	7.53	7.98	8.42	8.05	10.41	12.99
2100	7.57	8.01	8.45	8.09	10.52	13.06
2200	7.62	8.05	8.48	8.12	10.61	13.13
2300	7.66	8.08	8.52	8.15	10.71	13.19
2400	7.70	8.10	8.56	8.18	10.79	13.24
2500	7.74	8.14	8.59	8.21	10.87	13.30

t°C	CH_4	C_2H_6	C_3H_8	nC_4H_{10}	iC_4H_{10}	air
0	8.20	11.80	16.26	21.69	21.39	6.92
20	8.33	12.13	16.80	22.54	22.31	6.93
50	8.53	12.62	17.57	23.51	23.13	6.94
100	8.86	13.40	18.81	25.04	24.77	6.95

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DETERMINATION OF MOLAR HEAT FOR LIQUIDS AT 20 °C BY THE CONTRIBUTION GROUP METHOD

The molecule must be split into groups. For each group corresponds a value. **The molar heat** of a compound is obtained by the addition of values associated to each groups. Then some corrections have to be done.

– Add 4.5 for each carbon group linked by a simple bond to another carbon group linked itself by a double or triple bond to a third carbon group. If a carbon group target the above criteria more than once. add 4.5 as much as nécessary.

– Exceptions not inclued in the rule for :

• – CH_3 groups : not addition of 4.5.
• – CH_2 – groups : add 2.5 instead of 4.5 if it targets the criteria above only once ;
if it targets the criteria many times. add 2.5 the first time them 4.5 for the other times.

Group	Value
Alcane	
$-CH_3$	8.80
$-CH_2-$	7.26
$-CH-$	5.00
$-C-$	1.76
Alcene	
$=CH_2$	5.20
$=C-H$	5.10
$=C-$	3.80
Alcyne	
$-C\equiv H$	5.90
$-C\equiv$	5.90
In a cycle	
$-CH-$	4.4
$-C=$ ou $-C-$	2.9

Group	Value
$-CH=$	5.3
$-CH_2-$	6.2
Halogen	
$-Cl$ (first or second on a carbon)	8.6
$-Cl$ (third or fourth on a carbon)	6.0
$-Br$	9.0
$-F$	4.0
$-I$	8.6
Hydrogen	
H— (for formic acid, formates. hydrogen cyanide..)	3.5
Oxygen	
$-O-$	8.4
$>C=O$	12.66
$-C=O$	12.66
$-C(=O)-O-$	19.1

Group	Value
$-C(=O)-OH$	14.5
$-CH_2OH$	17.5
$-CHOH$	18.2
$-COH$	26.6
$-OH$	10.7
$-ONO_2$	28.5
Nitrogen	
$H-N(H)-$	14.0
$-N(-)-$	10.5
$-N-$	7.5
$-N=$ (in a ring)	4.5
$-C\equiv N$	13.9
Surfur	
$-SH$	10.7
$-S-$	8.0

Example : Pentadien –1.4

$$CH2 = CH - CH_2 - CH = CH_2$$

Contributions : 5.20 + 5.10 + 7.26 + 5.10 + 5.20 = 27.9

Corrections : for the group – CH_2 –. must be added 2.5 + 4.5
(it targets two times the criteria)

$$C_{p\ molaire} = 27.9 + 2.5 + 4.5 = 34.9\ cal/mol.K$$

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INTRODUCTION TO THE MAXWELL ENTHALPY DIAGRAMS

PURE COMPOUNDS

In the "DATA BOOK OF HYDROCARBONS", JB. MAXWELL provides the enthalpic diagram of the following pure compounds.

C_1 C_2 C_3 nC_4 nC_5 nC_6 nC_7 nC_8

iC_4 iC_5

$C_2 = C_3 = iC_4 = C_4 = 1$ Cis $C_4 = 2$ trans $C_4 = 2$

and petroleum cuts:

- $t_{mav} = 200°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 250°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 300°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 400°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 500°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 600°F$ and $K_{UOP} = 11$ or 12
- $t_{mav} = 800°F$ and $K_{UOP} = 11$ or 12

The enthalpy reference is H = 0 BTU/lb for the saturated liquid at – 200°F.

MIXTURES

For the mixture, he uses the following weighting rules:

- **liquid phase enthalpy h_m**

$$H_m = \sum_i x_i h_i^*$$

x_i mass fraction of component i in the liquid

h_i^* enthalpy of the pure component i given by the diagram

For compounds at superficial conditions, use the dashed line.

- **vapor phase enthalpy H_m**

1st step:

$$H_m = \sum_i y_i h_i^*$$

y_i mass fraction of component i in the gas phase

h_i^* enthalpy of the pure component i given by the diagram at 0-1 atmosphere

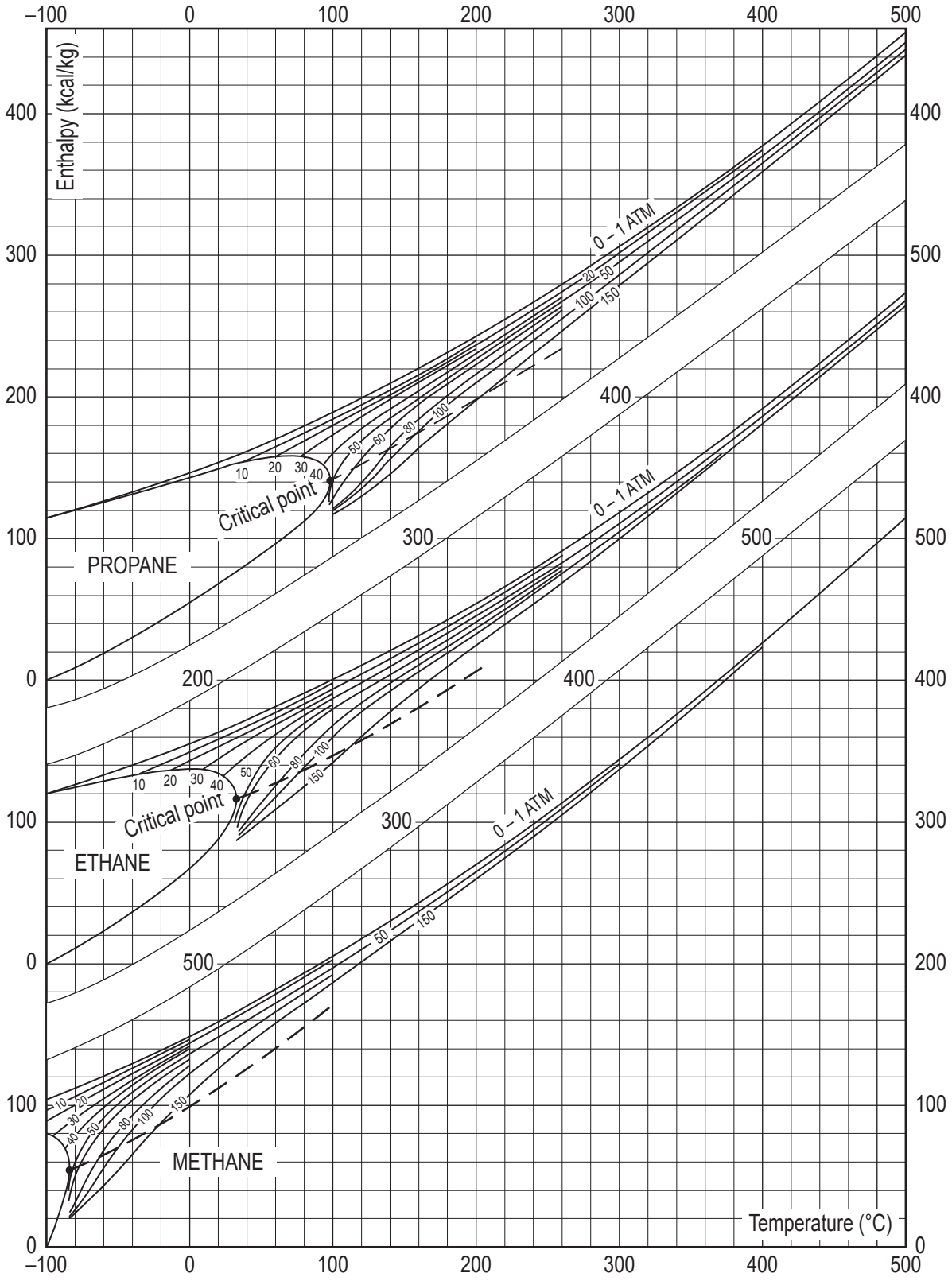
2nd step:

A pressure correction is applied, determined on the diagram of a pure component having the same molecular weight than the mixture.

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ENTHALPY DIAGRAM OF METHANE, ETHANE AND PROPANE

(for dissolved gases at supercritical conditions,
read enthalpy using the dashed line)
(J.B. MAXWELL)

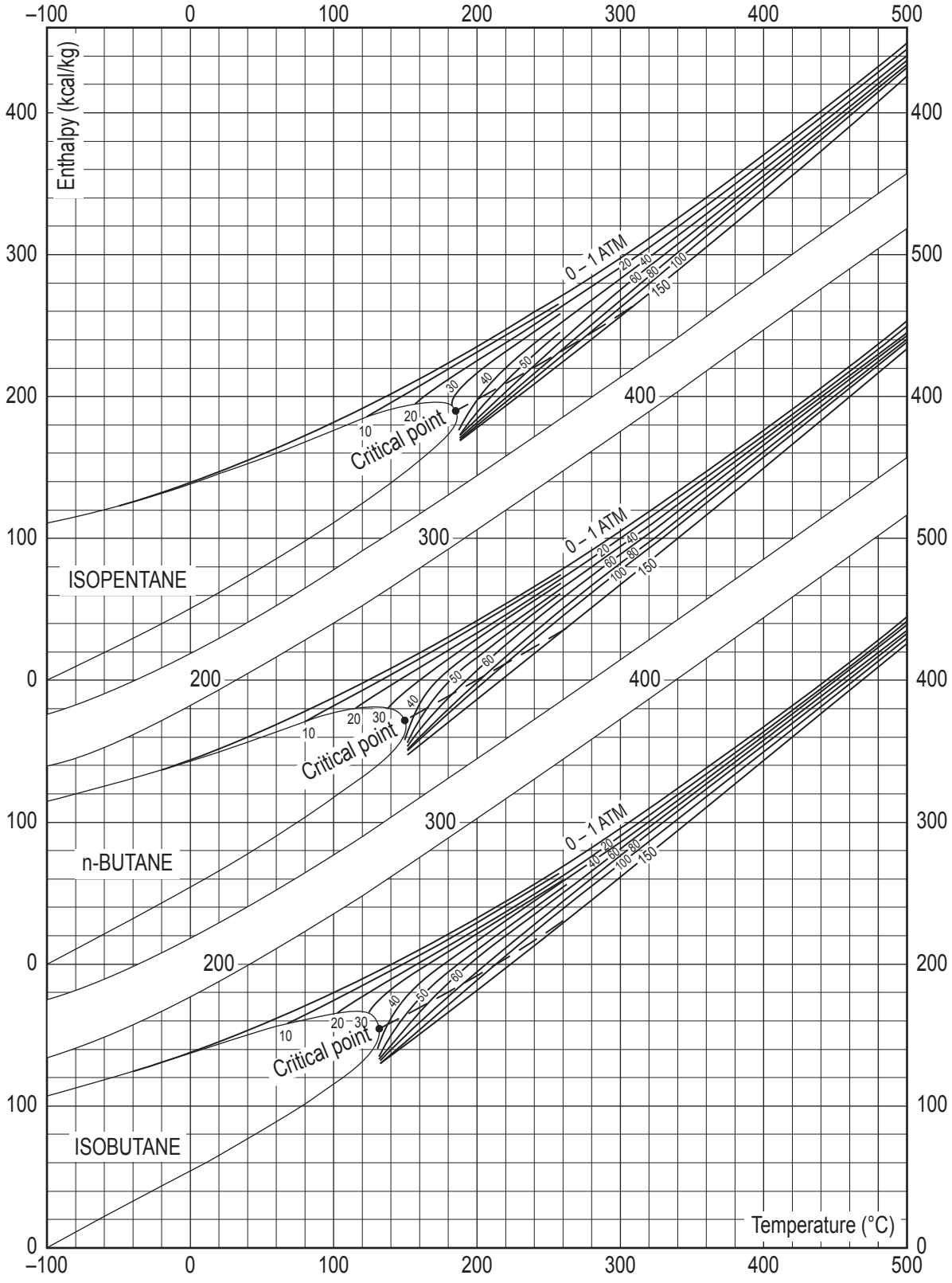


D TH 1400 D

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ENTHALPY DIAGRAM OF ISOBUTANE, n-BUTANE AND ISOPENTANE

(for dissolved gases at supercritical conditions,
read enthalpy using the dashed line)
(J.B. MAXWELL)

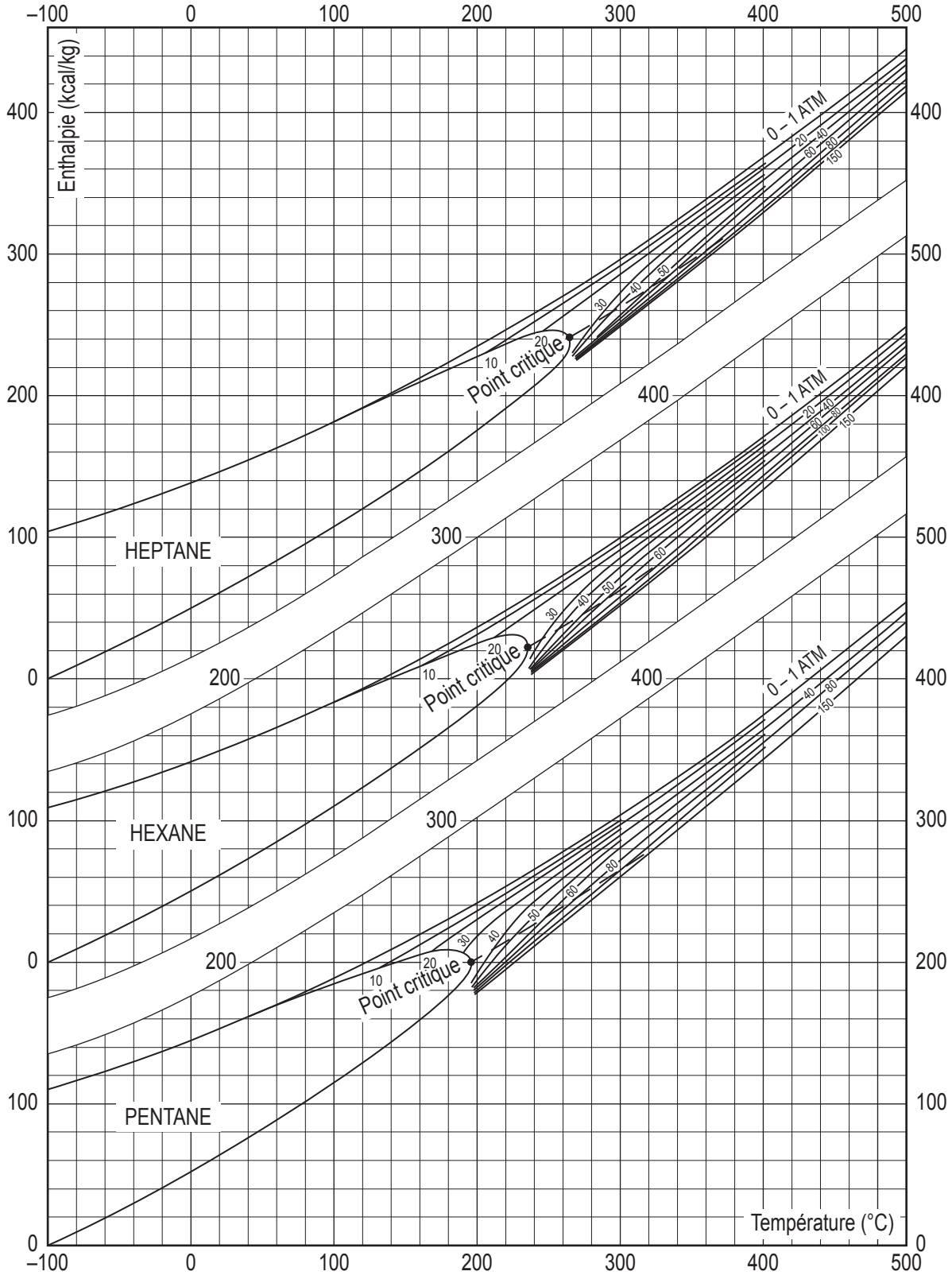


D TH 1399 D

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ENTHALPIE DU PENTANE, DE L'HEXANE ET DE L'HEPTANE

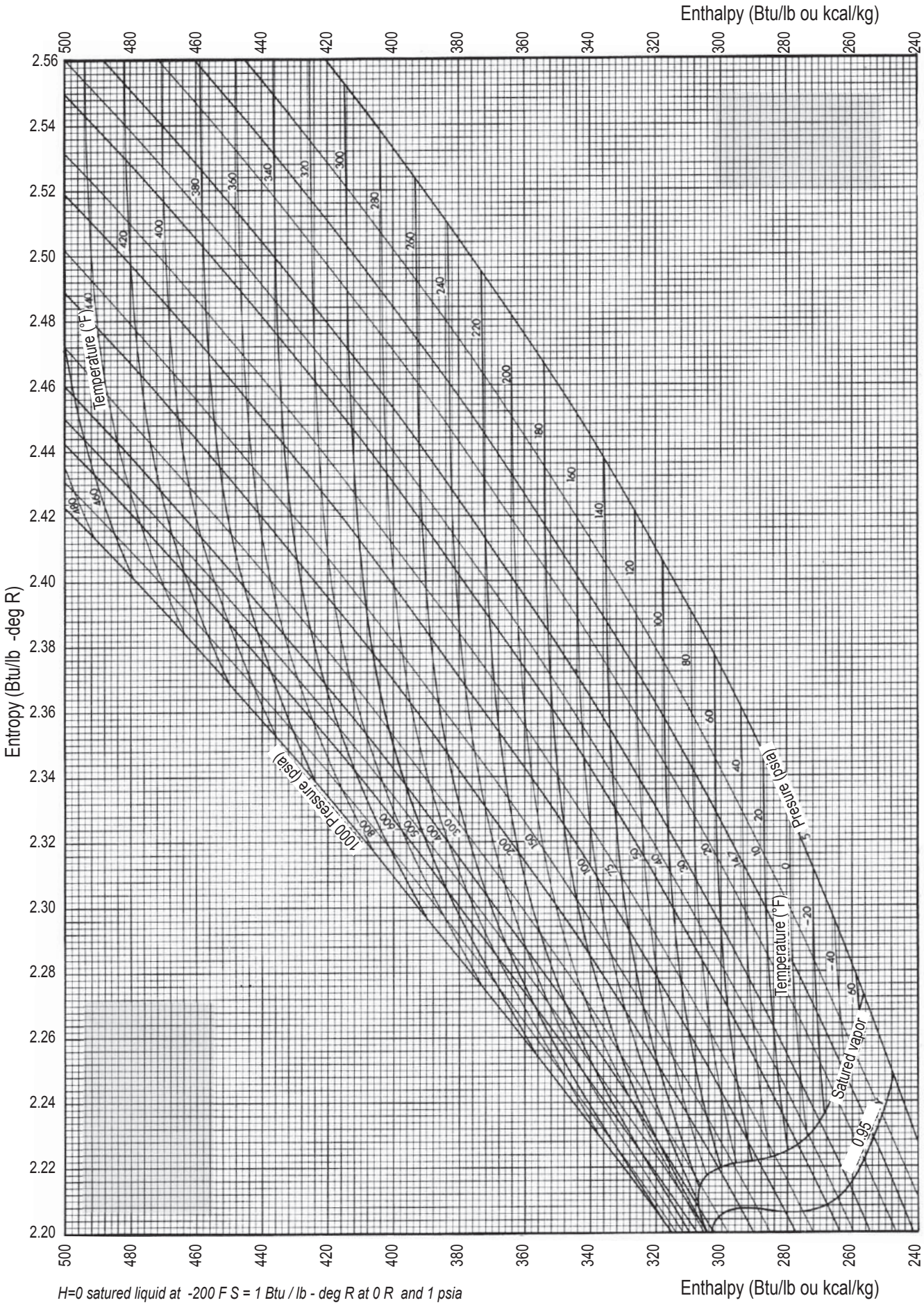
(pour les gaz en solution au-dessus de leur point critique,
lire l'enthalpie sur les courbes en tirets)



D TH 1398 D

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MOLLIER DIAGRAM FOR PROPANE



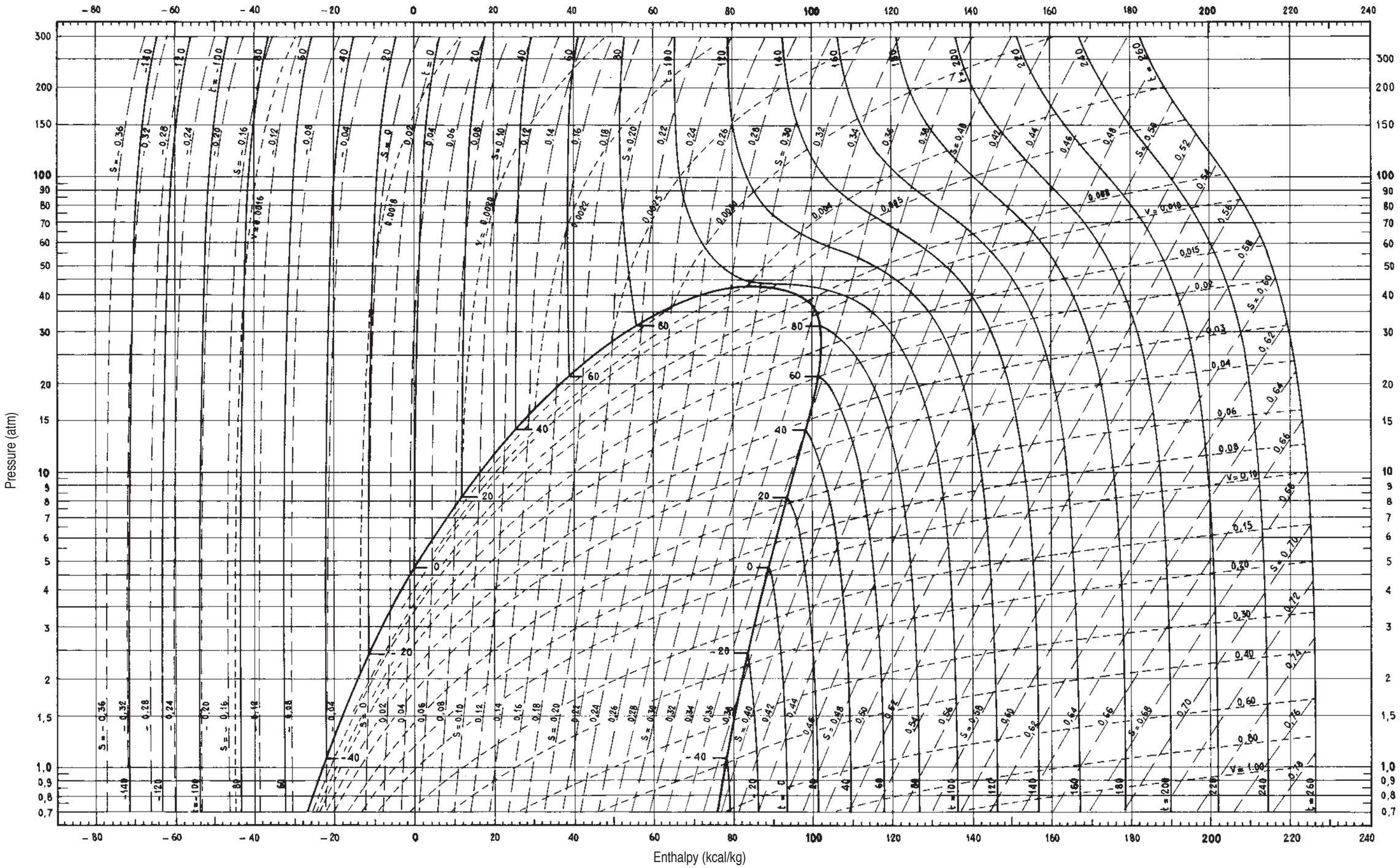
H=0 satured liquid at -200 F S = 1 Btu / lb - deg R at 0 R and 1 psia

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s : specific entropy (kcal/kg.K)
t : temperature (°C)
v : specific volume (m^3/kg)



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Example

Sp.Gr= 0.800	; K_{uop} =11
Temperature : 420°C ; Pressure	= 32 kg/cm²
Vapor enthalpy (1 atm.; K_{uop} = 12)	= 300 kcal/kg
Correction of K_{uop}	= -15 kcal/kg
Correction of pressure	= -10 kcal/kg
Vapor enthalpy	= 275 kcal/kg

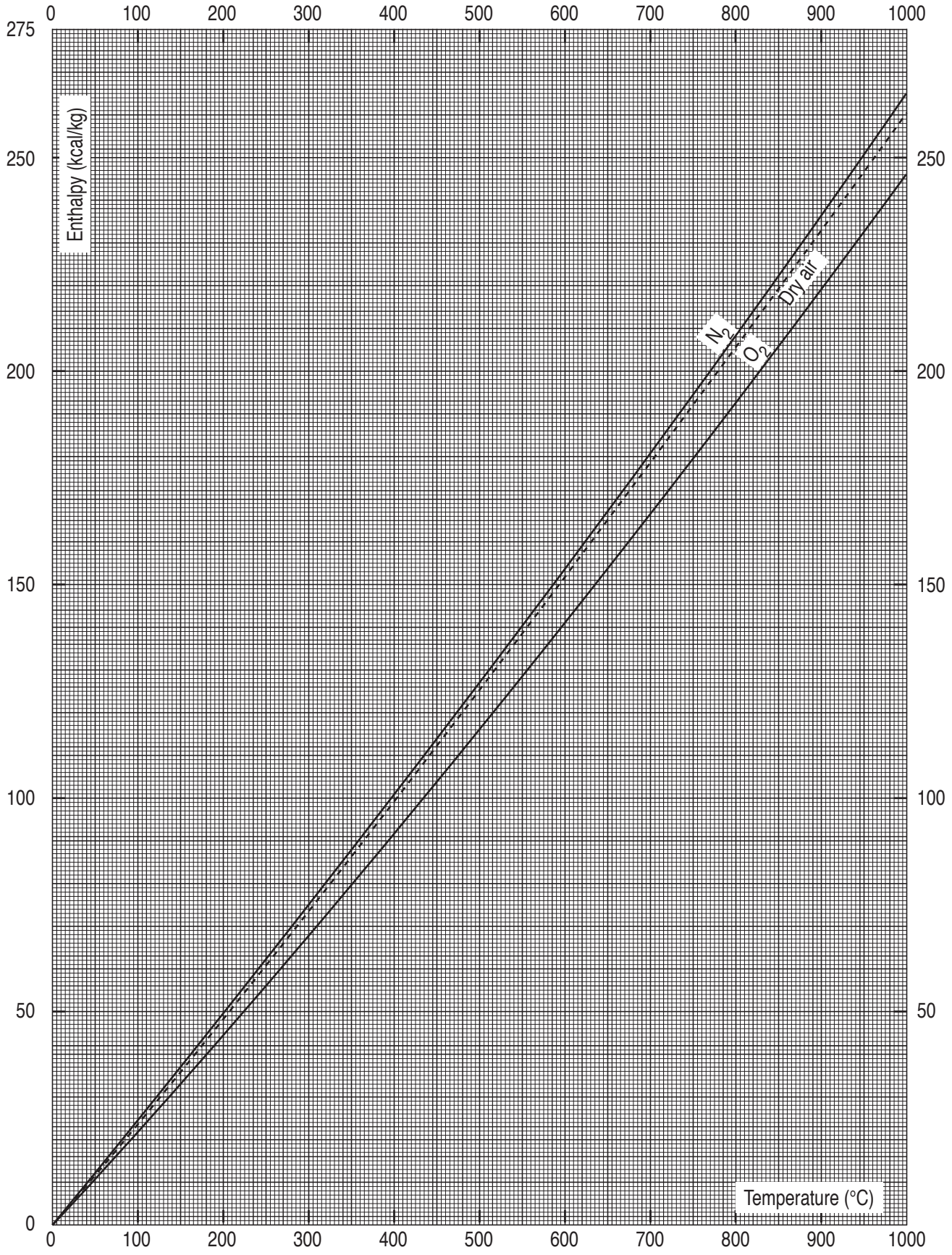
Example

Sp.Gr	= 0.8 ; K_{uop}=11
Temperature : 240°C	
Enthalpy liq (K_{uop} = 12)	= 140 kcal/kg
Correction factor	= 0.946
Enthalpy liquid (K_{uop} = 11)	= 132.4 kcal/kg

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D TH 1065 D

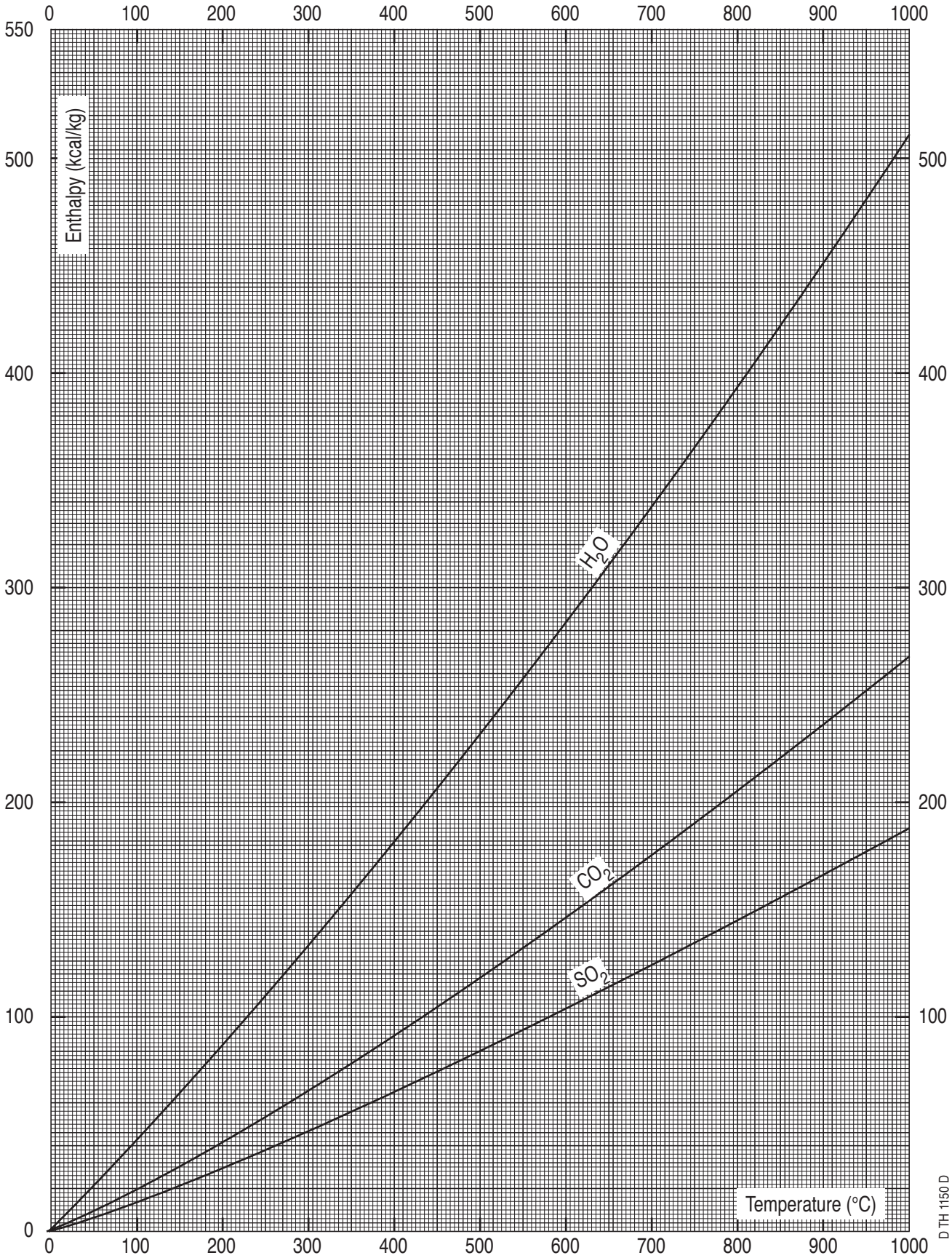
ENTHALPY OF O_2 AND N_2
— Reference 0°C vapor —



D TH 1149 D

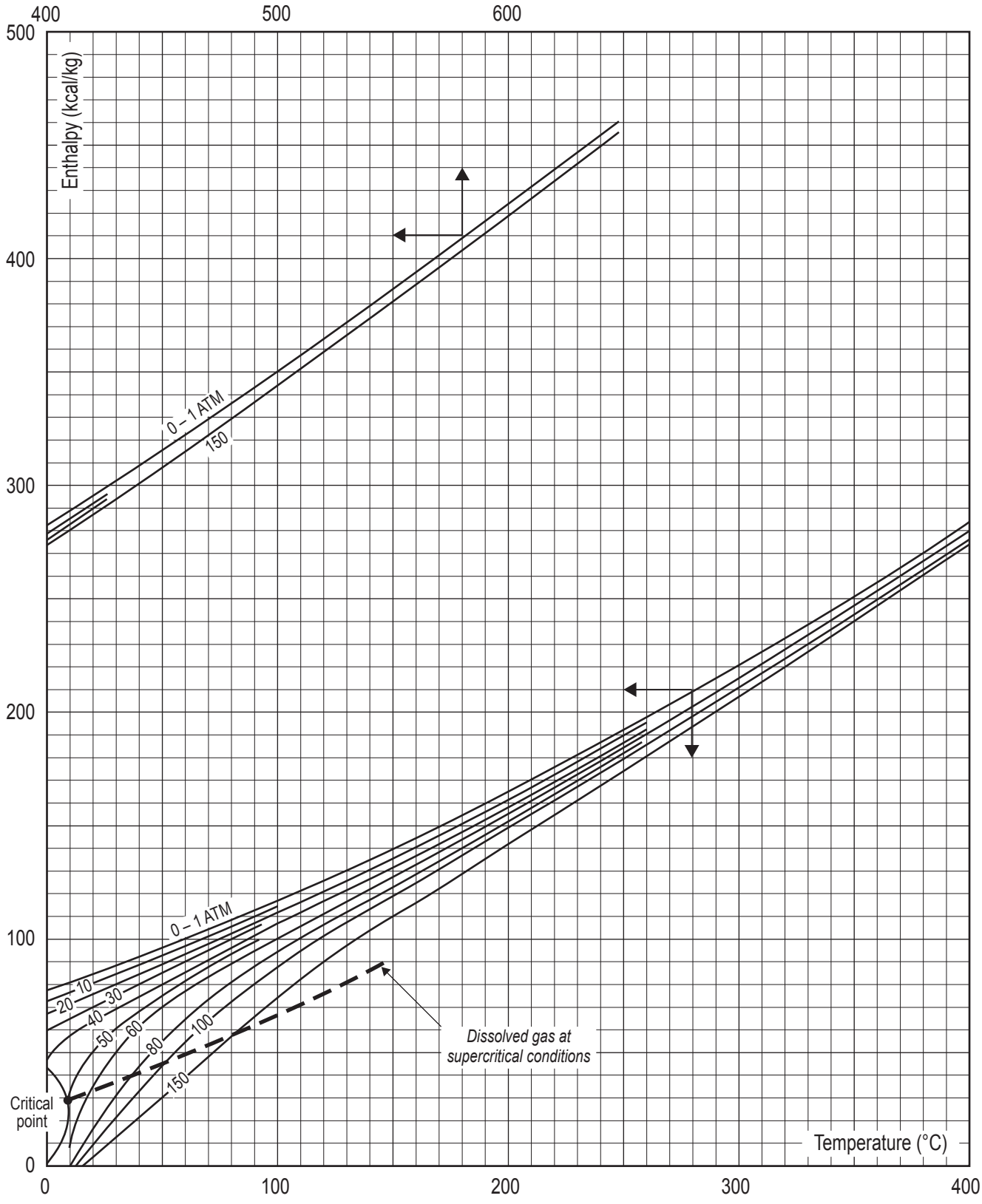
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ENTHALPY OF H_2O,CO_2 AND SO_2
— Reference 0°C vapor —



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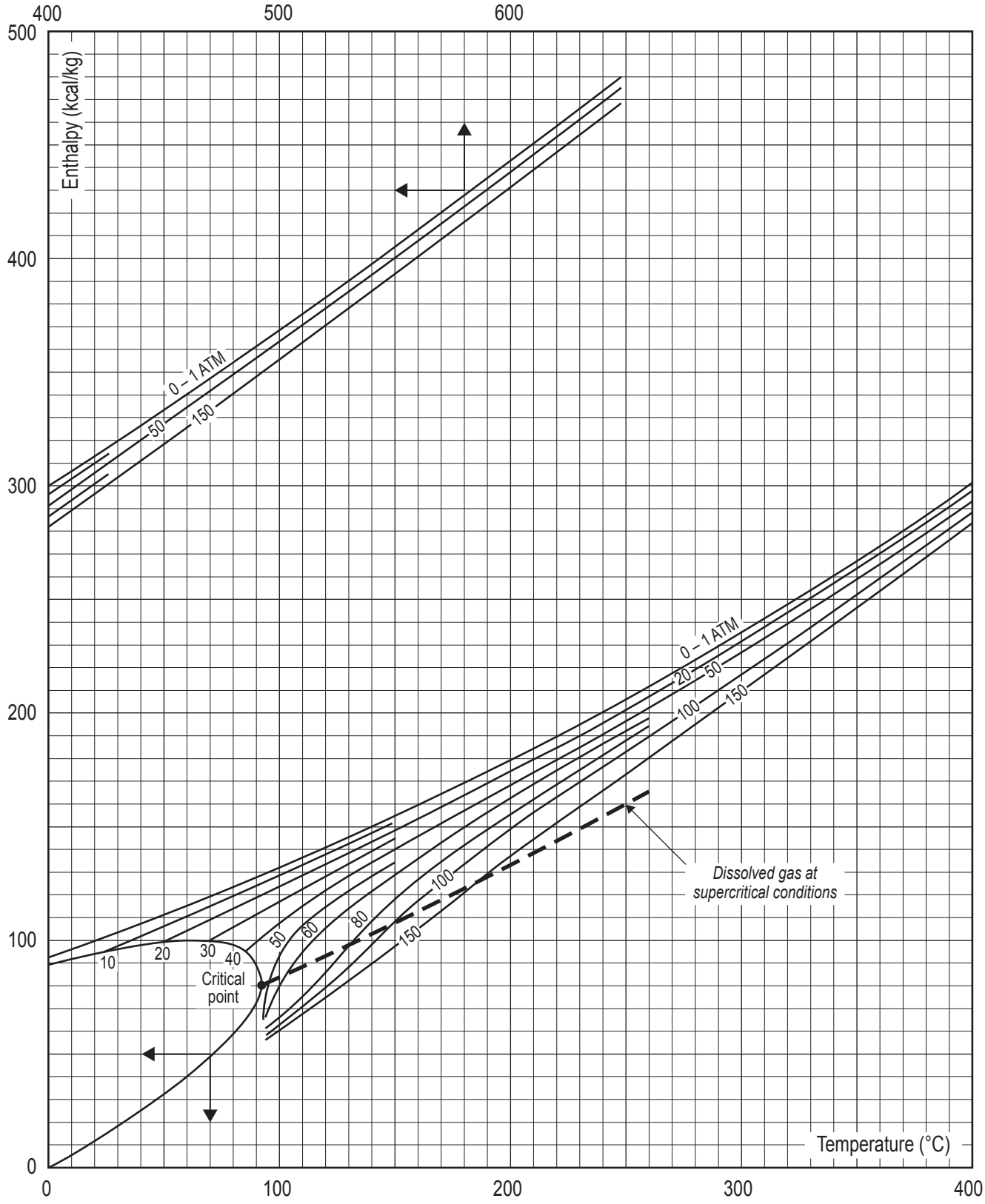
ENTHALPY DIAGRAM OF ETHYLENE ($C_2^=$)



D TH 3061 D

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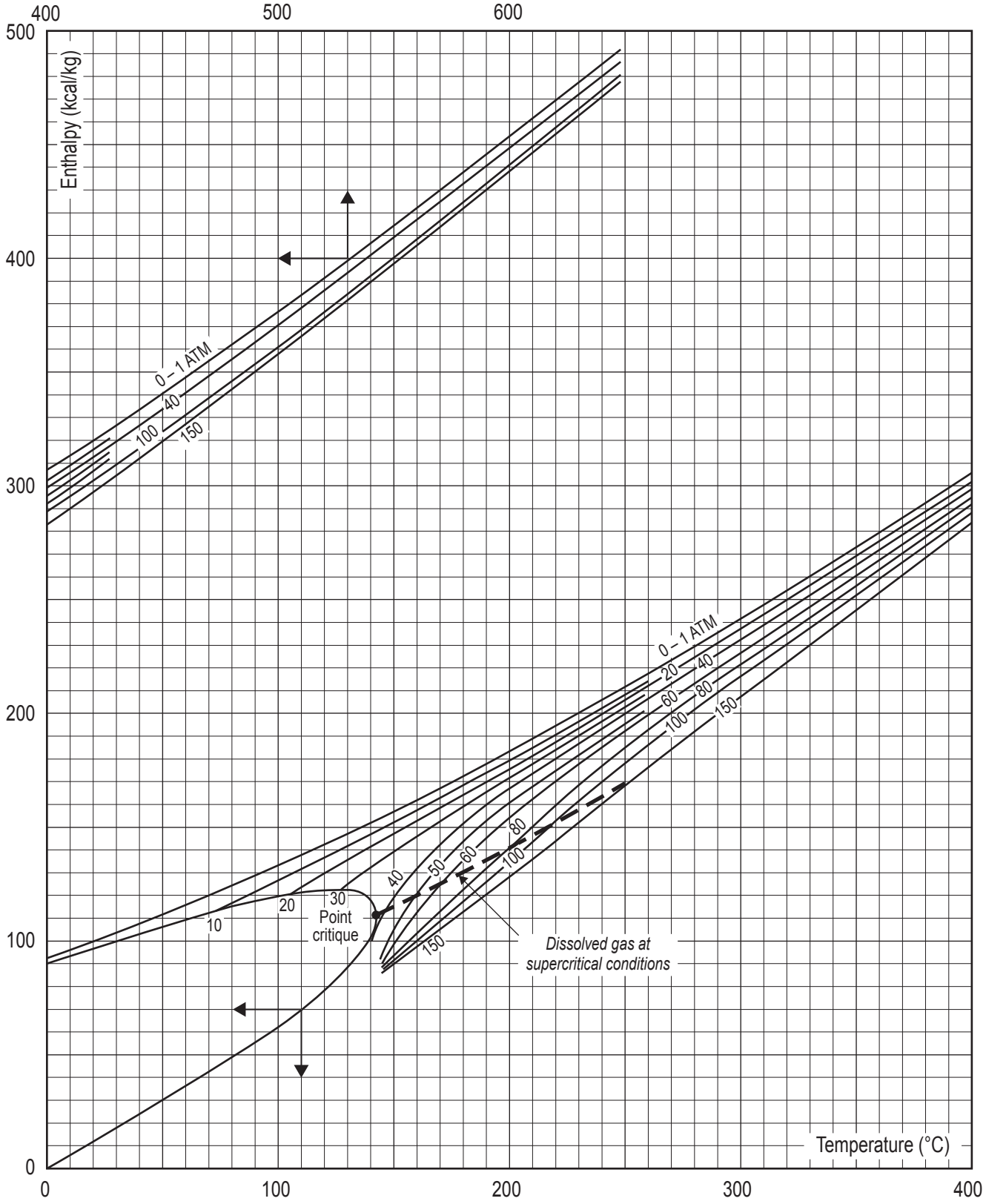
ENTHALPY DIAGRAM OF PROPYLENE ($C_3^=$)



D TH 3062 D

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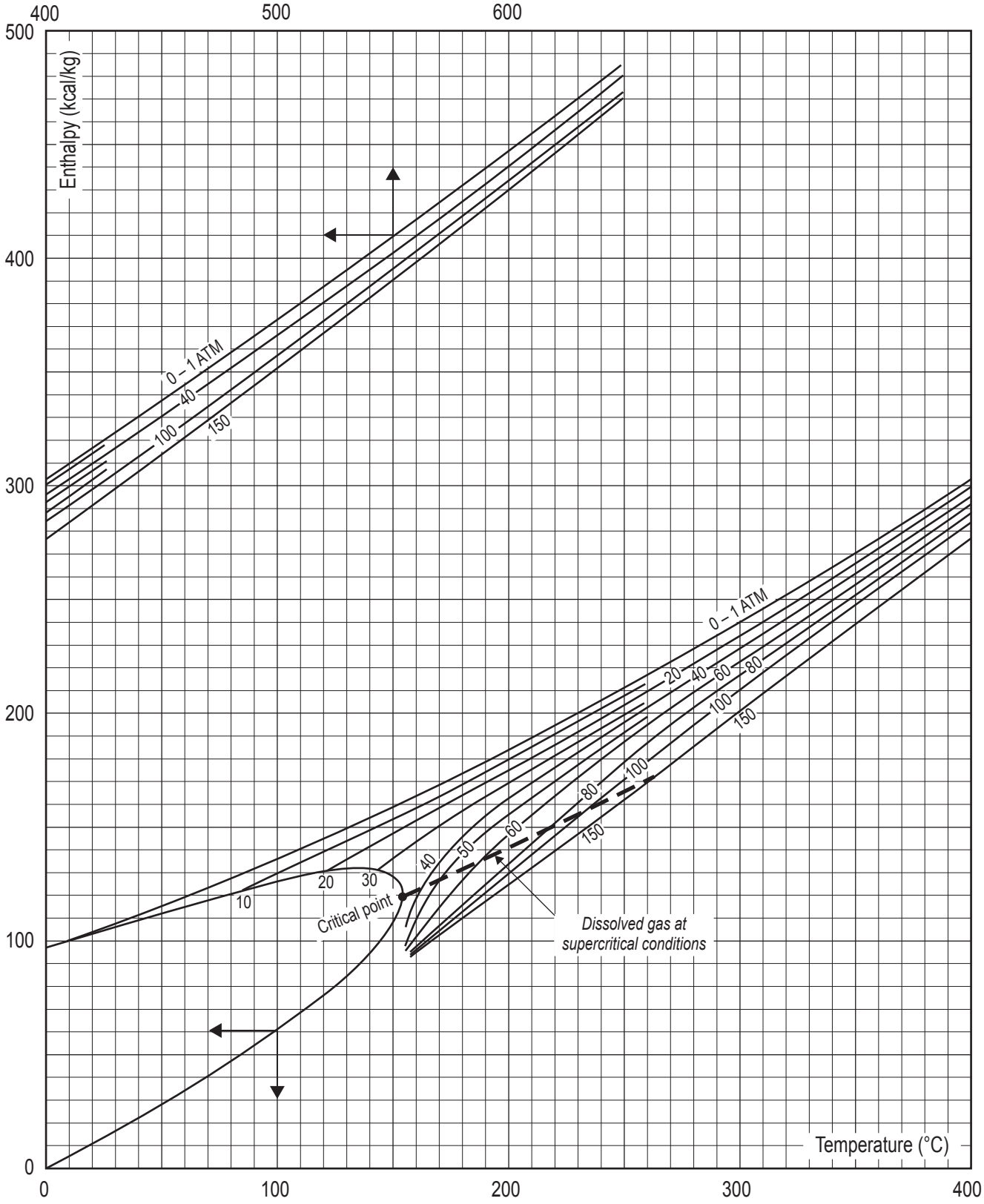
ENTHALPY DIAGRAM OF ISOBUTENE (iso $C_4^=$) AND BUTENE 1 (1 – $C_4^=$)



D TH 3063 D

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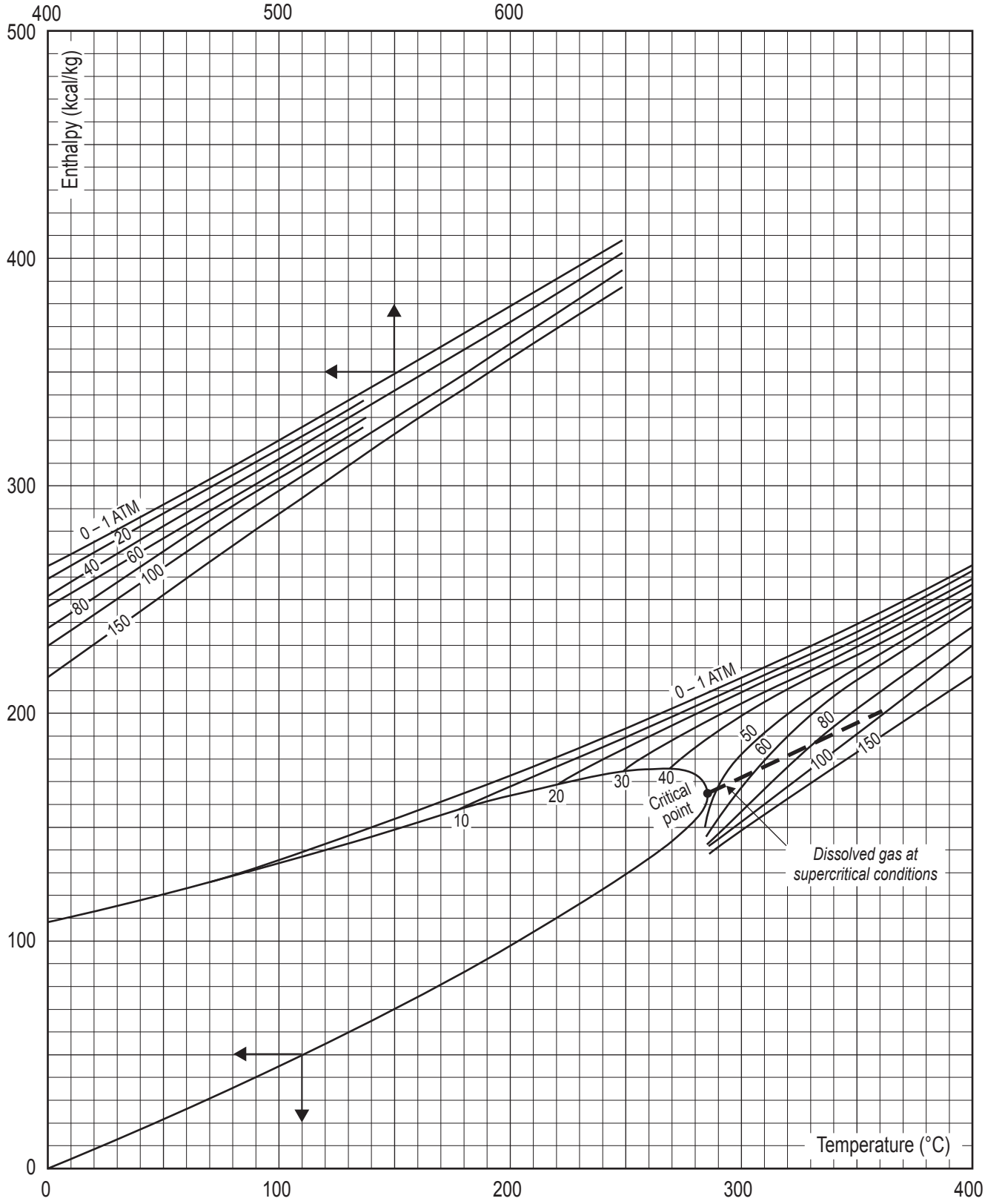
ENTHALPY DIAGRAM OF BUTENE 2 Cis AND BUTENE Trans



D TH 3064 D

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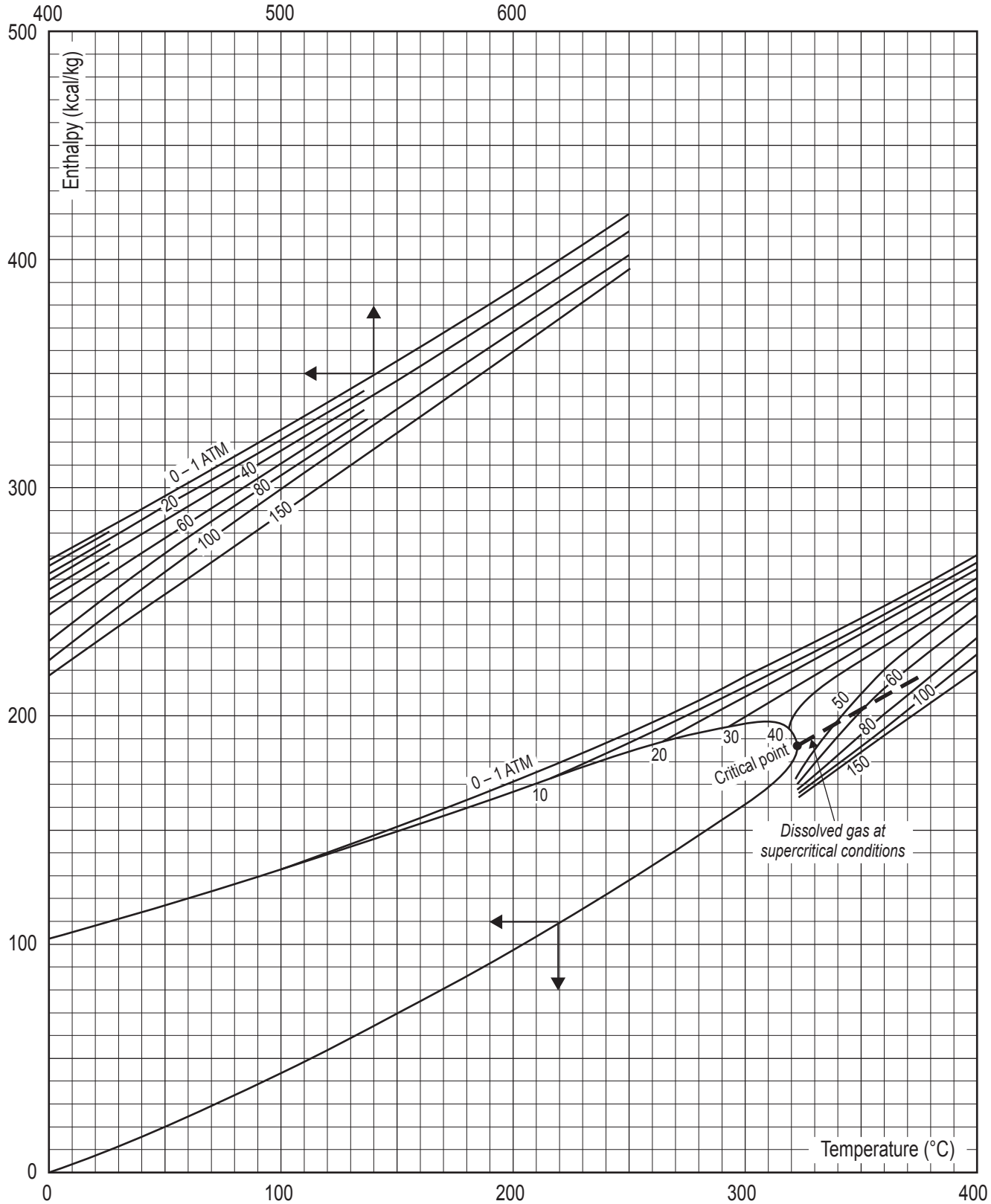
ENTHALPY DIAGRAM OF BENZENE



D TH 3065 D

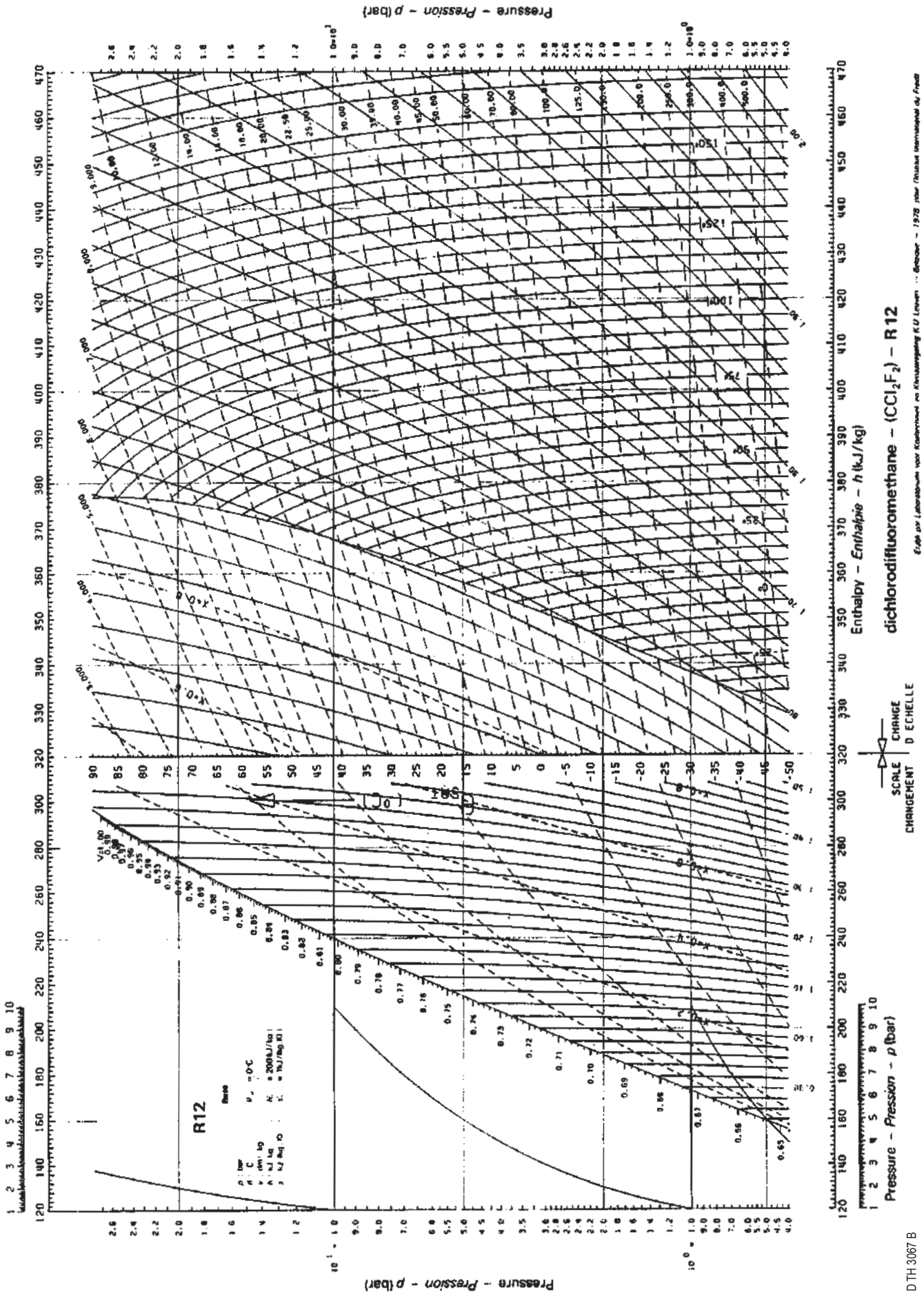
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ENTHALPY DIAGRAM OF TOLUENE



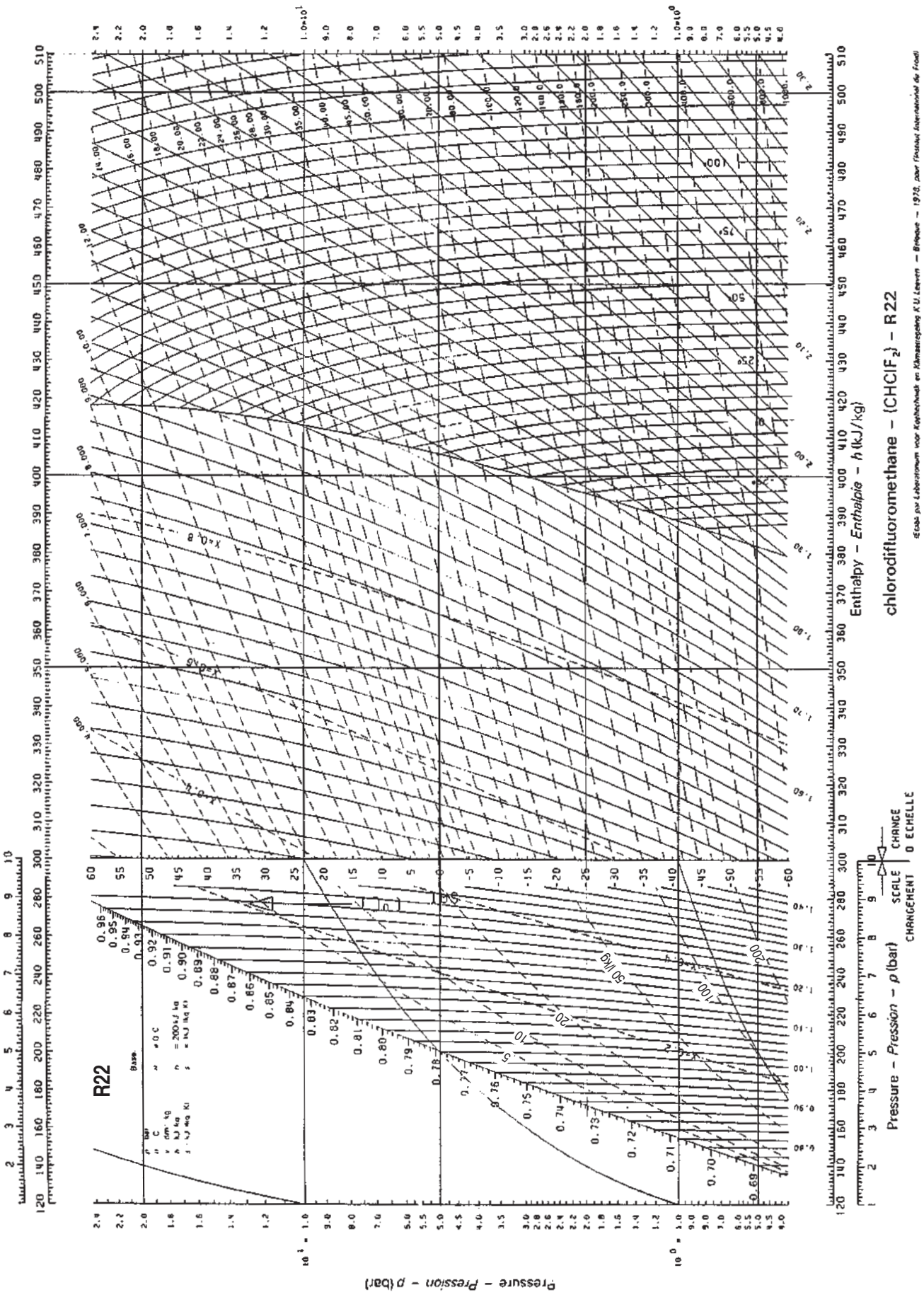
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ENTHALPY-PRESSURE DIAGRAM OF R12 CCl_2F_2



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ENTHALPY-PRESSURE DIAGRAM OF R22 $CHClF_2$



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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta H_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298°K	400°K	600°K	800°K	1000°K
Water	– 57 107	– 57 798	– 58 040	– 58 493	– 58 882	– 59 214
Nitric oxide	21 477	21 600	21 610	21 620	21 625	21 633
Carbon monoxide	– 27 202	– 26 416	– 26 317	– 26 330	– 26 511	– 26 768
Carbon dioxide	– 93 969	– 94 052	– 94 069	– 94 123	– 94 215	– 94 318
Methane	– 15 987	– 17 889	– 18 629	– 19 893	– 20 823	– 21 430
Ethane	– 16 517	– 20 236	– 21 419	– 23 280	– 24 530	– 25 280
Propane	– 19 482	– 24 820	– 26 358	– 28 660	– 30 120	– 30 900
Butane	– 23 670	– 30 150	– 32 000	– 34 730	– 36 420	– 37 260
Isobutane	– 25 300	– 32 150	– 34 000	– 36 670	– 38 300	– 39 120
Pentane	– 27 230	– 35 000	– 37 180	– 40 370	– 42 290	– 43 210
Isopentane	– 28 810	– 36 920	– 39 120	– 42 280	– 44 150	– 44 970
Neopentane	– 31 300	– 39 670	– 41 770	– 44 670	– 46 290	– 49 960
Hexane	– 30 910	– 39 960	– 42 460	– 46 110	– 48 280	– 49 270
Heptane	– 34 550	– 44 890	– 47 710	– 51 820	– 54 230	– 55 290
Octane	– 38 200	– 49 820	– 52 960	– 57 530	– 60 180	– 61 310
Nonane	– 41 840	– 54 740	– 58 220	– 63 230	– 66 120	– 67 330
Decane	– 45 490	– 59 670	– 63 470	– 68 940	– 72 070	– 73 350
Increment (CH_2)	– 3 644	– 4 926	– 5 252	– 5 708	– 5 948	– 6 020
Ethylene	14 522	12 496	11 766	10 600	9 760	9 205
Propylene	8 468	4 879	3 758	1 980	760	30
1-butene	4 960	– 30	– 149	– 3700	– 5 150	– 5 950
2-butene (cis)	3 480	– 1 670	– 3 310	– 5 820	– 7 480	– 8 450
2-butene (trans)	2 240	– 2 670	– 4 120	– 6 380	– 7 890	– 8 730
Isobutene	980	– 4 040	– 5 440	– 7 580	– 9 010	– 9 820
1-pentene	1 130	– 5 000	– 6 770	– 9 410	– 11 100	– 11 950
2-pentene (cis)	– 180	– 6 710	– 8 650	– 11 530	– 13 360	– 14 330
2-pentene (trans)	– 1 250	– 7 590	– 9 040	– 12 190	– 13 930	– 14 870
2-methyl - 1-butene	– 2 300	– 8 680	– 10 390	– 12 920	– 14 530	– 15 360
3-methyl - 1-butene	– 680	– 6 920	– 8 470	– 10 750	– 12 220	– 12 940
2-methyl - 2-butene	– 3 630	– 10 170	– 12 060	– 14 940	– 16 810	– 17 830

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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta H_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298°K	400°K	600°K	800°K	1000°K
1-hexene	– 2 540	– 9 960	– 12 060	– 15 160	– 17 080	– 18 010
1-heptene	– 6 180	– 14 890	– 17 310	– 20 870	– 23 030	– 24 030
1-octene	– 9 830	– 19 820	– 11 450	– 26 580	– 28 980	– 30 050
1-nonene	– 13 470	– 24 740	– 27 820	– 32 280	– 34 920	– 36 070
1-decene	– 17 120	– 29 670	– 33 070	– 37 990	– 40 870	– 42 090
Cyclopentane	– 10 680	– 18 460	– 20 800	– 24 120	– 26 060	– 26 910
Methylcyclopentane	– 16 620	– 25 500	– 28 070	– 31 680	– 33 740	– 34 580
Cyclohexane	– 20 010	– 29 430	– 31 700	– 35 570	– 37 190	– 37 410
Methylcyclohexane	– 26 300	– 36 990	– 39 790	– 43 460	– 45 100	– 42 250
1,3-butadiene	29 780	26 330	25 380	24 110	23 250	22 750
Isoprene	22 980	18 100	16 900	15 240	14 180	13 540
Cyclopentane	13 760	7 870	5 970	3 200	1 490	620
Cyclohexane	5 760	– 1 700	– 3 700	– 6 360	– 7 740	– 8 190
Acetylene	54 329	54 194	54 134	53 931	53 627	53 304
Propyne	46 017	44 319	43 775	42 829	42 069	41 530
1-butyne	42 740	39 480	38 570	37 100	36 040	35 380
2-butyne	38 090	34 970	33 950	33 230	30 980	30 190
Benzene	24 000	19 820	18 554	16 711	15 510	14 818
Toluene	17 500	11 950	10 327	7 932	6 399	5 564
Ethylbenzene	13 917	7 120	5 218	2 488	798	– 61
o-xylene	11 096	4 540	2 711	– 62	– 1 858	– 2 799
m-xylene	10 926	4 120	2 175	– 738	– 2 598	– 3 567
p-xylene	11 064	4 290	2 317	– 665	– 2 586	– 3 607
n-propylbenzene	9 810	1 870	– 310	– 3 440	– 5 340	– 6 300
Cumene	9 250	940	– 1280	– 4 370	– 6 240	– 7 130
Styrene	40 340	35 220	33 830	31 810	30 510	29 830

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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta H_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- SULFUR RHOMBOEDRIC EXCEPT FOR H_2S
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298°K	400°K	600°K	800°K	1000°K
S_2 gas		29 860				
SO_2		– 70 960				
SO_3		– 94 450				
H_2S (S_2 gas)	– 19 530	– 20 530	– 20 560	– 20 950	– 21 270	– 21 450
C S_2		27 640	27 640	27 760	27 820	27 850
Br (liquid)		– 3 670				
I (solid)		– 7 438				
HF		– 64 200	– 64 250	– 64 460	– 64 860	– 65 520
HCl		– 22 060	– 22 130	– 22 280	– 22 440	– 22 560
HBr		– 12 330	– 12 410	– 12 560	– 12 690	– 12 780
HI		– 1 240	– 1 330	– 1 480	– 1 600	– 1 670
CH_3Cl		– 19 580	– 20 220	– 21 230	– 21 930	– 22 370
CH_2Cl_2		– 21 000				
$CHCl_3$		– 24 000				
CCl_4		– 25 500	– 25 310	– 24 880	– 24 430	– 24 000
Ammonia		– 11 040	– 11 540	– 12 330	– 12 890	– 13 230
Hydrocyanic acid		31 200	31 160	31 050	30 940	30 840
Nitrogen monoxide (NO)		19 490	19 410	19 420	19 590	19 830
Nitrogen dioxide (NO_2)		8 090	7 960	7 840	7 840	7 900
Nitrogen tetraoxide		2 310	2 200	2 460		
Methanol	– 45 510	– 48 110	– 48 980	– 50 380	– 51 340	– 51 970
Ethanol	– 52 070	– 56 230	– 57 430	– 59 210	– 60 320	– 60 980
n-propanol	– 56 110	– 61 730	– 63 260	– 65 460	– 66 800	– 67 470
Isopropanol	– 59 460	– 65 130	– 66 600	– 68 700	– 69 930	– 70 560
Formol	– 27 700	– 28 620	– 29 030	– 29 780	– 30 370	– 30 780
Acetaldehyde	– 37 300	– 39 810	– 40 630	– 41 920	– 42 790	– 43 320
Acetone		– 51 720	– 52 900	– 54 750	– 55 980	– 56 660
Ethylene oxyd		– 12 190	– 13 110	– 14 460	– 15 300	– 15 750
Allylic alcohol		– 29 730	– 30 860	– 32 550	– 33 650	– 34 280
Phenol	18 630	– 23 000	– 24 200	– 25 900	– 26 900	– 27 400
Formic acid		– 90 490	– 90 960	– 91 630	– 92 040	– 92 260
Acetic acid		– 104 530	– 105 360	– 106 540	– 107 250	– 107 600

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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta G_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298 K	400 K	600 K	800 K	1000 K
Water	– 57 107	– 54 635.1	– 53 518	– 51 154	– 48 643	– 46 030
Nitric oxide	21 477	20 719	20 418	19 820	19 218	18 614
Carbon monoxide	– 27 201.9	– 32 807.9	– 35 007	– 39 358	– 43 677	– 41 942
Carbon dioxide	– 93 968.6	– 94 259.8	– 94 325	– 94 444	– 94 539	– 94 610
Methane	– 15 987	– 12 140	– 10 048	– 5 490	– 550	4 610
Ethane	– 16 517	– 7 860	– 3 447	+ 5 970	15 920	26 130
Propane	– 19 482	– 5 614	+ 1 191	+ 15 500	30 450	45 680
Butane	– 23 670	– 4 100	5 090	24 280	44 210	64 500
Isobutane	– 25 300	– 5 000	4 570	24 480	45 120	66 090
Pentane	– 27 230	– 2 000	9 600	33 760	58 770	84 180
Isopentane	– 28 810	– 3 540	8 220	32 630	57 880	83 510
Neopentane	– 31 300	– 3 640	8 990	35 080	61 930	89 080
Hexane	– 30 910	– 70	13 950	43 020	73 080	103 570
Heptane	– 34 550	1 940	18 340	52 350	87 450	123 050
Octane	– 38 200	3 950	22 770	61 730	101 900	142 620
Nonane	– 41 840	5 960	27 200	71 110	116 350	162 180
Decane	– 45 490	7 970	31 630	80 500	130 800	181 750
Increment (CH_2)	– 3 644	+ 2 009	4 431	9 382	14 451	19 566
Ethylene	14 522	16 282	17 675	20 918	24 490	28 249
Propene	8 468	14 990	18 610	26 460	34 810	43 430
1-butene	4 960	17 090	23 230	36 070	49 450	63 070
2-butene (cis)	3 480	15 740	21 920	35 150	49 080	63 350
2-butene (trans)	2 240	15 050	21 310	34 560	48 440	62 660
Isobutene	980	13 880	20 210	33 550	47 490	61 720
1-pentene	1 130	18 960	27 490	45 210	63 580	82 240
2-pentene (cis)	– 180	17 170	25 620	43 480	62 110	81 100
2-pentene (trans)	– 1 250	16 760	25 420	43 440	62 140	81 150
2-methyl - 1-butene	– 2 300	15 510	24 030	41 850	60 370	79 200
3-methyl - 1-butene	– 680	17 870	26 620	44 630	63 340	82 350
2-methyl - 2-butene	– 3 670	14 260	22 920	41 080	60 050	79 400

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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta G_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298 K	400 K	600 K	800 K	1000 K
1-hexene	– 2 540	20 940	31 900	54 570	78 020	101 810
1-heptene	– 6 180	22 950	36 330	63 950	92 470	121 380
1-octene	– 9 830	24 960	40 760	73 330	106 920	140 940
1-nonene	– 13 470	26 970	45 190	82 720	121 370	160 510
1-decene	– 17 120	28 980	49 620	92 100	135 820	180 070
Cyclopentane	– 10 680	9 230	19 060	39 780	61 400	83 380
Methylcyclopentane	– 16 620	8 550	20 590	45 780	71 930	98 460
Cyclohexane	– 20 010	7 590	20 660	47 860	75 960	104 300
Methylcyclohexane	– 26 300	6 520	21 840	53 550	86 160	119 030
1,3-butadiene	29 780	36 010	39 460	46 780	54 480	62 360
Isoprene	22 980	34 870	40 780	53 100	65 890	78 890
Cyclopentane	13 760	26 480	33 140	47 370	62 370	77 720
Cyclohexane	5 760	25 120	34 600	54 380	74 860	95 600
Acetylene	54 329	50 000	48 577	45 835	43 178	40 604
Propyne	46 017	46 313	47 287	49 255	51 514	53 950
1-butyne	42 740	48 300	51 450	58 230	65 440	72 880
2-butyne	38 090	44 320	47 680	54 940	62 700	70 740
Benzene	24 000	30 989	35 008	43 663	52 838	62 270
Toluene	17 500	29 228	35 390	48 477	62 236	76 320
Ethylbenzene	13 917	31 208	39 741	57 646	76 302	95 303
o-xylene	11 096	29 117	37 883	56 103	75 110	94 494
m-xylene	10 926	28 405	37 008	55 099	74 006	93 307
p-xylene	11 064	28 952	37 688	56 060	75 275	94 897
n-propylbenzene	9 810	32 810	43 730	66 480	90 090	114 080
Cumene	9 250	32 738	43 960	67 290	91 480	116 060
Styrene	40 340	51 100	56 740	68 670	81 160	93 920

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STANDARD ENTHALPY OF FORMATION

UNITS = $\left(\Delta G_T^\circ\right)_f$ CALORIES/MOLE

REFERENCES
- CARBON: GRAPHITE
- SULFUR RHOMBOEDRIC EXCEPT FOR H_2S
- OTHER SUBSTANCES ARE CONSIDERED AS PERFECT GAS

	0	298 K	400 K	600 K	800 K	1000 K
S_2 gas		19 440				
SO_2		– 71 790				
SO_3		– 88 520				
H_2S (S_2 gas)	– 19 530	– 17 530	– 16 480	– 14 360	– 12 110	– 9 800
C S_2		15 550	11 440	3 300	– 4 860	– 13 040
Br (liquid)		– 375				
I (solid)		– 2 315				
HF		– 64 700	– 64 860	– 65 130	– 65 310	– 65 350
HCl		– 22 770	– 23 000	– 23 410	– 23 760	– 24 070
HBr		– 13 120	– 13 370	– 13 820	– 14 220	– 14 590
HI		– 2 010	– 2 250	– 2 680	– 3 060	– 3 420
CH_3Cl		– 13 960	– 11 940	– 7 570	– 2 900	1 910
CH_2Cl_2		– 14 000	– 11 540	– 6 460	– 1 230	4 060
$CHCl_3$		– 13 660	– 10 980	– 5 630	– 270	5 090
CCl_4		– 15 350	– 11 910	– 5 300	+ 1 160	7 500
Ammonia		– 3 860	– 1 330	3 950	9 470	15 110
Hydrocyanic acid		28 700	27 850	26 220	24 630	23 060
Nitrogen monoxide (NO)		24 760	26 430	29 050	30 990	32 400
Nitrogen dioxide (NO_2)		12 390	13 880	16 880	19 890	22 900
Nitrogen tetraoxide		23 490	30 750	44 990		
Methanol	– 45 510	– 38 870	– 35 580	– 28 500	– 21 120	– 13 390
Ethanol	– 52 070	– 40 330	– 34 730	– 22 940	– 10 690	1 780
n-propanol	– 56 110	– 39 080	– 31 080	– 14 500	2 710	20 190
Isopropanol	– 59 460	– 41 500	– 33 070	– 16 010	1 770	19 790
Formol	– 27 700	– 27 190	– 26 640	– 25 250	– 23 490	– 21 950
Acetaldehyde	– 37 300	– 31 920	– 29 090	– 22 990	– 16 590	– 9 880
Acetone		– 36 300	– 30 850	– 29 390	– 7 400	4 830
Ethylene oxyd		– 2 790	570	7 720	15 260	22 960
Allylic alcohol		– 15 220	– 10 080	700	11 960	23 440
Phenol	– 18 630	– 7 880	– 2 510	+ 8 610	20 130	31 820
Formic acid		– 83 890	– 81 560	– 76 700	– 71 650	– 66 530
Acetic acid		– 75 410	– 65 330	– 45 040	– 24 420	– 3 670

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VISCOSITY : DEFINITIONS - UNITS

The viscosity is generally defined as a characteristic of flow resistance for a fluid ; it has to be considered in all phenomena in relation with fluid flows :

- pressure drops in pipes
- flows in valves, safety valves and all types of restrictions
- pumping
- filtration, settling

The viscosity is all the larger as the flow resistance is high and, due to the various methods used to determine it, two types of viscosity are defined: **dynamic** viscosity and **cinematic** viscosity.

1 - DYNAMIC VISCOSITY

The dynamic viscosity μ (Mu), also called **absolute viscosity**, is the expression of the molecular interactions within the moving fluid. Traditionally, the dynamic viscosity was expressed in **poises** (P) or **centipoises** (cP). It is now measured in International System of Units in pascal x seconde (Pa.s). In practice the correspondence between the two systems is very simple as the centipoise corresponds exactly to the millipascal seconde (mPa.s).

$$1\ cP = 1\ mPa.s = 0{,}001\ Pa.s$$

The dynamic viscosity μ of liquids can be measured using for example **rotary viscometers**, the working principle of which is based on the measurement of resisting couple when a mobile is rotating inside the sample to be analysed.

2 - CINEMATIC VISCOSITY

The cinematic viscosity υ (Nu), currently used for liquids, takes also into account the density ρ of the fluid at the same temperature. This one has to be considered each time that the viscosity is measured through the determination of a **flow time** under charge, this is to say: the time required for a liquid to pass through an orifice under the action of its own weight. We intuitively understand indeed that the time required depends not only from the own viscosity of the liquid but as well from its density.

The cinematic viscosity is obtained from the dynamic viscosity using the following formula:

$$\text{Cinematic viscosity } \upsilon = \frac{\text{dynamic viscosity } \mu}{\text{density } \rho}$$

$$\upsilon = \frac{\mu}{\rho}$$

Usually expressed in **stokes** (St) and **centistokes** (cSt), the cinematic viscosity in International System must be expressed in square meters per second (m^2/s).

The correspondence with usual units is as follows :

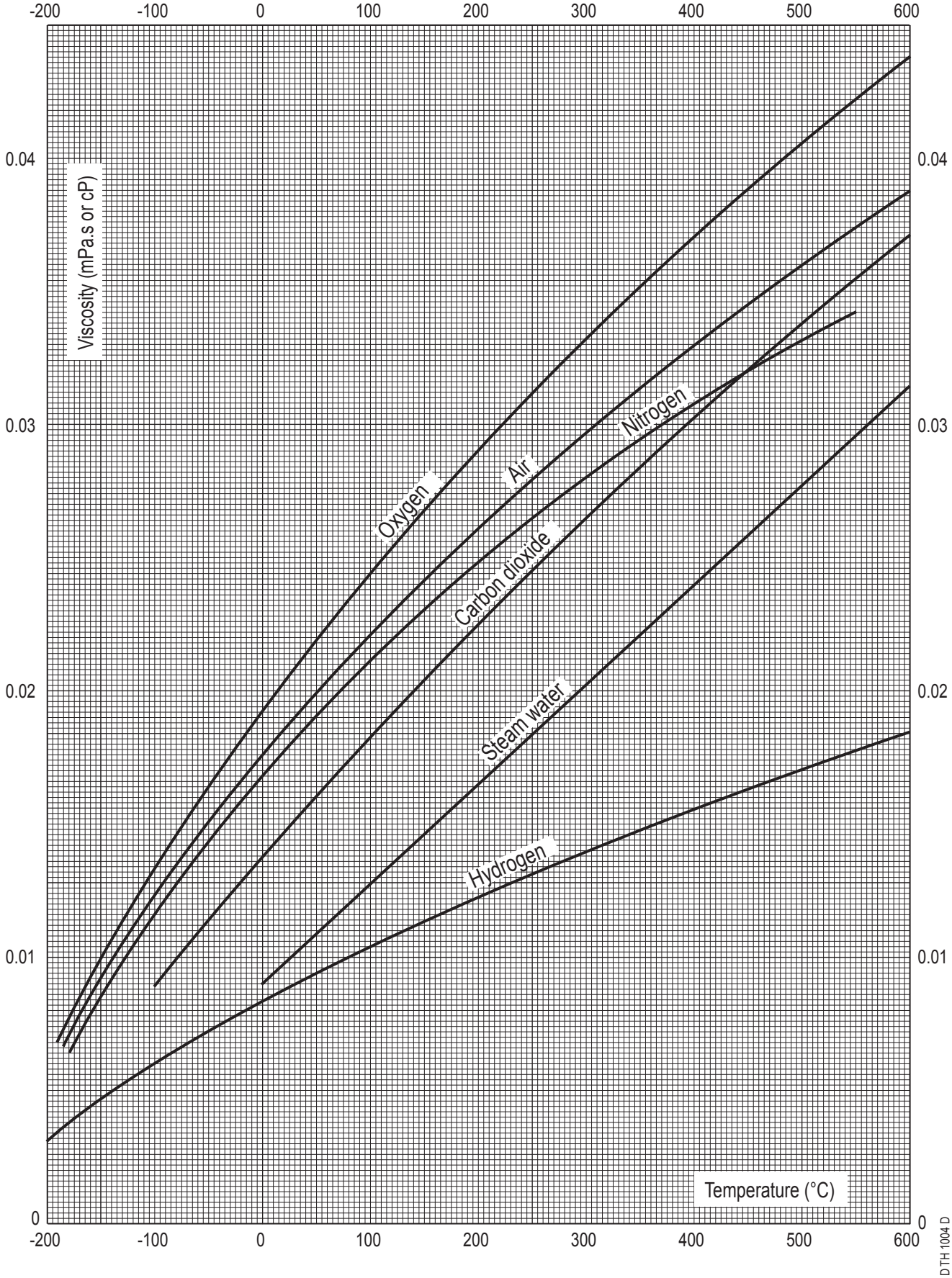
$$1\ St = 10^{-4}\ m^2/s$$

$$1\ cSt = 0{,}01\ St = 10^{-6}\ m^2/s = 1\ mm^2/s$$

$$1\ cSt = 1\ mm^2/s$$

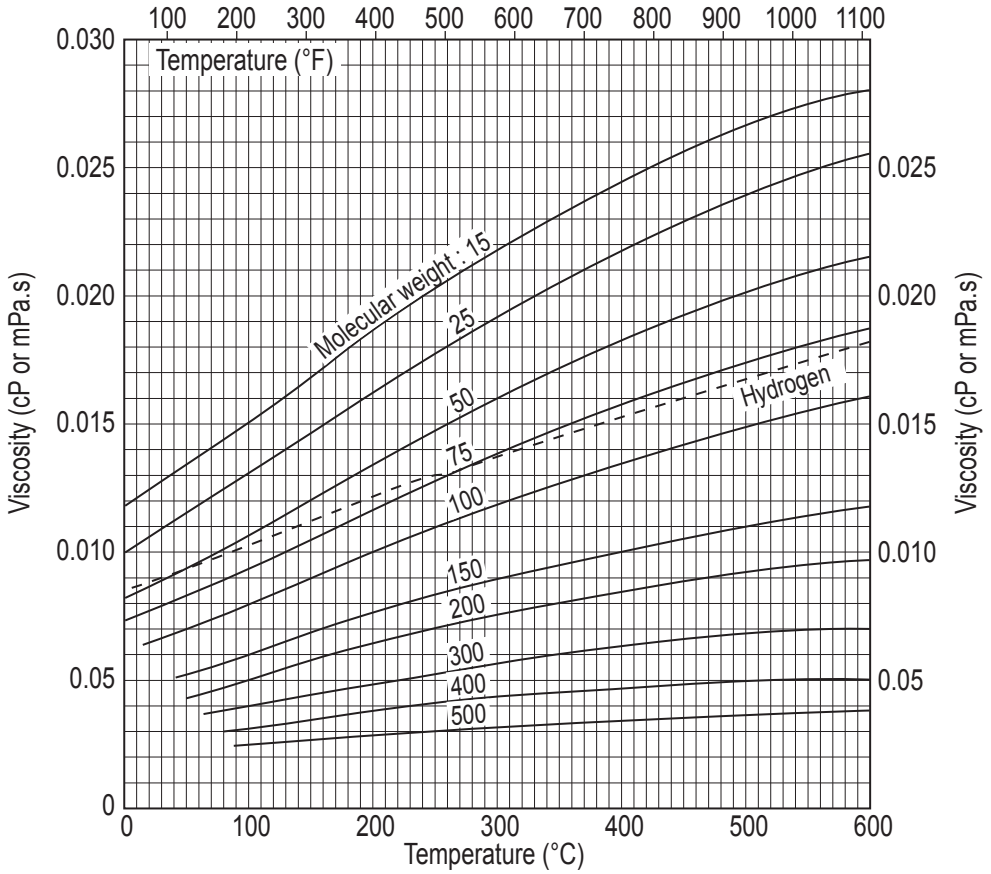
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VISCOSITY OF SOME GASES AT ATMOSPHERIC PRESSURE

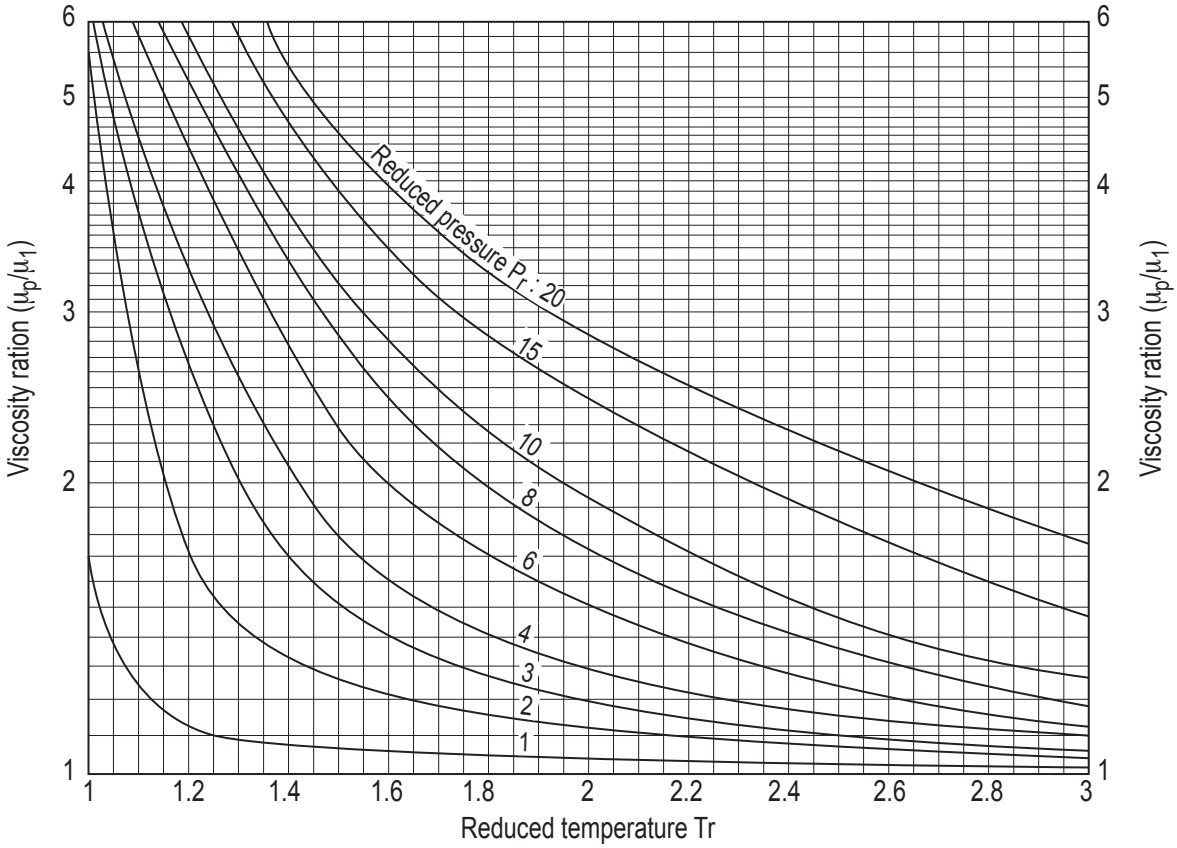


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VISCOSITY OF GAZEOUS HYDROCARBONS
— At atmospheric pressure —



VISCOSITY CORRECTION WITH PRESSURE
— μ_P viscosity at P ; μ_1 viscosity at 1 atm. a —



D TH 1396 D

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VISCOSITY OF GASES

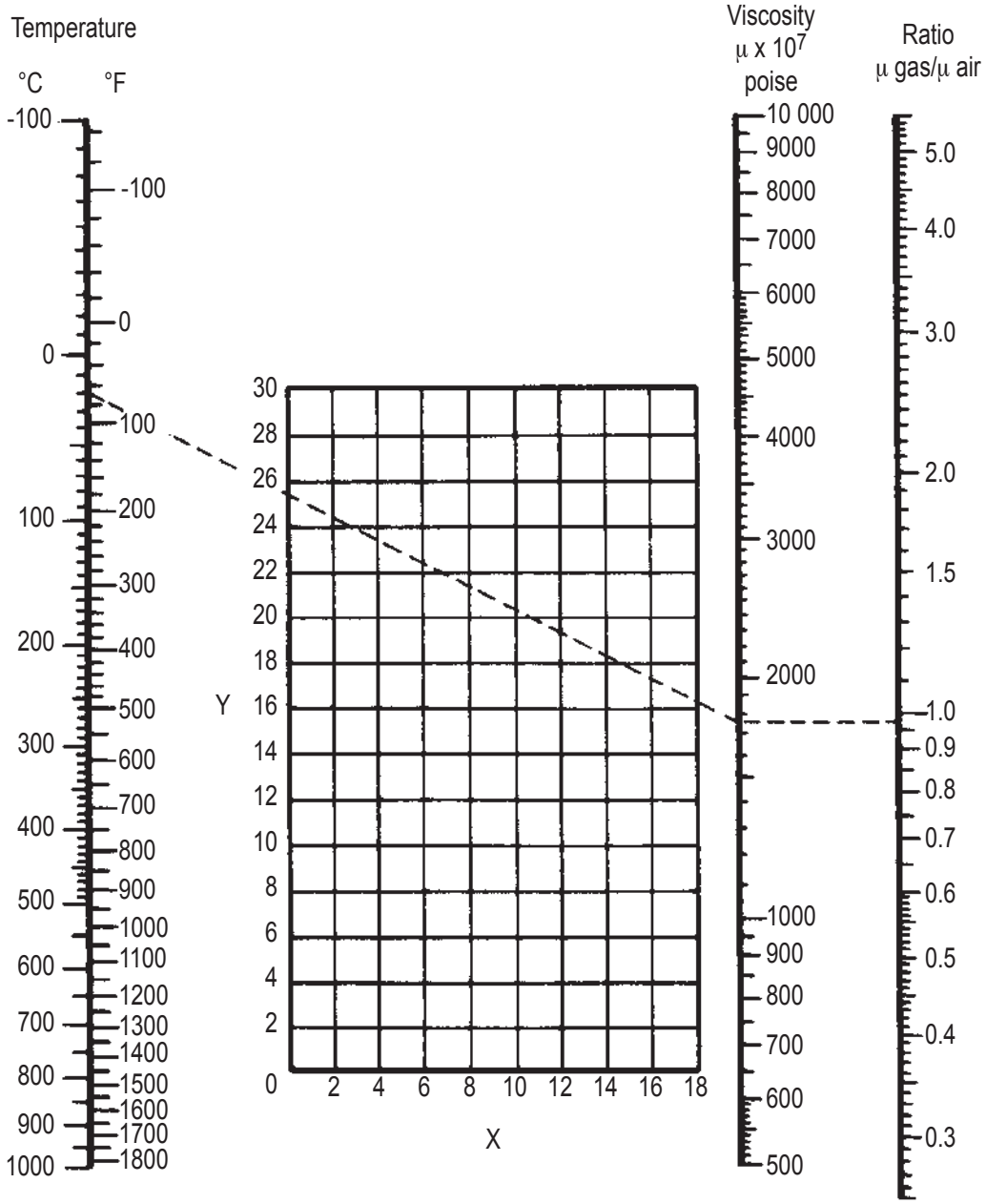
Gas	X	Y	$\mu . 10^7$ p*	Gas	X	Y	$\mu . 10^7$ p*
Acetic acid	7.0	14.6	825 (50°C)	Hydrogen-sulfur dioxide			
Acetone	8.4	13.2	735	10% H_2, 90% SO_2	8.7	18.1	1259 (17)
Acetylene	9.3	15.5	1017	20% H_2, 80% SO_2	8.6	18.2	1277 (17)
Air	10.4	20.4	1812	50% H_2, 50% SO_2	8.9	18.3	1332 (17)
Ammonia	8.4	16.0	1000	80% H_2, 20% SO_2	9.7	17.7	1306 (17)
Amylene (β)	8.6	12.2	676	Hydrogen bromide	8.4	21.6	1843
Argon	9.7	22.6	2215	Hydrogen chloride	8.5	19.2	1425
Arsine	8.6	20.0	1576	Hydrogen cyanide	7.1	14.5	737
Benzene	8.7	13.2	746	Hydrogen iodide	8.5	21.5	1830
Bromine	8.8	19.4	1495	Hydrogen sulfide	8.4	18.0	1265
Butane (n)	8.6	13.2	735	Iodine	8.7	18.7	1730 (100)
Butane (iso)	8.6	13.2	744	Krypton	9.4	24.0	2480
Butyl acetate (iso)	5.7	16.3	778	Mercury	7.4	24.9	4500 (200)
Butylene (α)	8.4	13.5	761	Mercuric bromide	8.5	19.0	2253
Butylene (β)	8.7	13.1	746	Mercuric chloride	7.7	18.7	2200 (200)
Butylene (iso)	8.3	13.9	786	Mercuric iodide	8.4	18.0	2045 (200)
Butyl formate (iso)	6.6	16.0	840	Mesitylene	9.5	10.2	660 (50)
Cadmium	7.8	22.5	5690 (500)	Methane	9.5	15.8	1092
Carbon dioxide	8.9	19.1	1463	Methane (deuterated)	9.5	17.6	1290
Carbon disulfide	8.5	15.8	990	Methanol	8.3	15.6	935
Carbon monoxide	10.5	20.0	1749	Methyl acetate	8.4	14.0	870 (50)
Carbon oxysulfide	8.2	17.9	1220	Methyl acetylene	8.9	14.3	867
Carbon tetrachloride	8.0	15.3	966	3-Methyl-1-butene	8.0	13.3	716
Chlorine	8.8	18.3	1335	Methyl butyrate (iso)	6.6	15.8	824
Chloroform	8.8	15.7	1000	Methyl bromide	8.1	18.7	1327
Cyanogen	8.2	16.2	1002	Methyl chloride	8.5	16.5	1062
Cyclohexane	9.0	12.2	701	3-Methylene-1-butene	8.0	13.3	716
Cyclopropane	8.3	14.7	870	Methylene chloride	8.5	15.8	989
Deuterium	11.0	16.2	1240	Methyl formate	5.1	18.0	923
Diethyl ether	8.8	12.7	730	Neon	11.1	25.8	3113
Dimethyl ether	9.0	15.0	925	Nitric oxide	10.4	20.8	1899
Diphenyl ether	8.6	10.4	610 (50)	Nitrogen	10.6	20.0	1766
Diphenyl methane	8.0	10.3	605 (50)	Nitrous oxide	9.0	19.0	1460
Ethane	9.0	14.5	915	Nonane (n)	9.2	8.9	554 (50)
Ethanol	8.2	14.5	835	Octane (n)	8.8	9.8	586 (50)
Ethyl acetate	8.4	13.4	743	Oxygen	10.2	21.6	2026
Ethyl chloride	8.5	15.6	978	Pentane (n)	8.5	12.3	668
Ethylene	9.5	15.2	1010	Pentane (iso)	8.9	12.1	685
Ethyl propionate	12.0	12.4	890	Phosphene	8.8	17.0	1150
Fluorine	7.3	23.8	2250	Propane	8.9	13.5	800
Freon-11	8.6	16.2	1298 (93)	Propanol (n)	8.4	13.5	770
Freon-12	9.0	17.4	1496 (93)	Propanol (iso)	8.4	13.6	774
Freon-14	9.5	20.4	1716	Propyl acetate	8.0	14.3	797
Freon-21	9.0	16.7	1389 (93)	Propylene	8.5	14.4	840
Freon-22	9.0	17.7	1554 (93)	Pyridine	8.6	13.3	830 (50)
Freon-113	11.0	14.0	1166 (93)	Silane	9.0	16.8	1148
Freon-114	9.4	16.4	1364 (93)	Stannic chloride	9.1	16.0	1330 (100)
Helium	11.3	20.8	1946	Stannic bromide	9.0	16.7	142 (100)
Heptane (n)	8.6	10.6	618 (50)	Sulfur dioxide	8.4	18.2	1250
Hexane (n)	8.4	12.0	644	Thiazole	10.0	14.4	958
Hydrogen	11.3	12.4	880	Thiophene	8.3	14.2	901 (50)
Hydrogen-helium				Toluene	8.6	12.5	686
10% H_2, 90% He	11.0	20.5	1780 (0)	2.2.3-Trimethylbutane	10.0	10.4	691 (50)
25% H_2, 75% He	11.0	19.4	1603 (0)	Trimethylethane	8.0	13.0	686
40% H_2, 60% He	10.7	18.4	1431 (0)	Water	8.0	16.0	1250 (100)
60% H_2, 40% He	10.8	16.7	1227 (0)	Xenon	9.3	23.0	2255
81% H_2, 19% He	10.5	15.0	1016 (0)	Zinc	8.0	22.0	5250 (500)

To find the viscosity of a gas, use the chart at the following page.
Plot the point X and Y, draw a line from this point to the temperature and read the viscosity on the appropriate scale.
**** Viscosity at 20°C, except if other indication.***

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VISCOSITY OF GAS

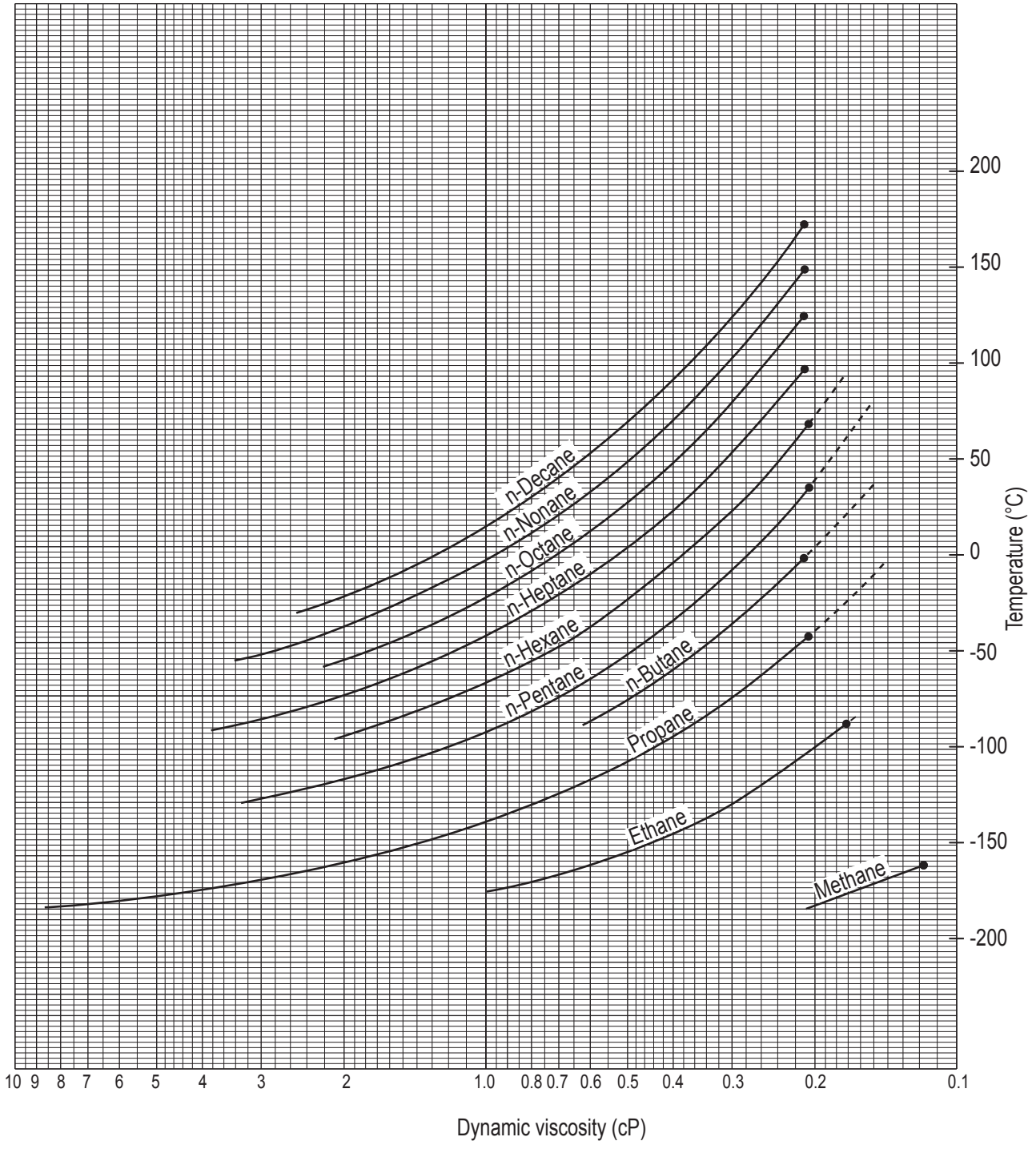
Chart



D TH 1623 D

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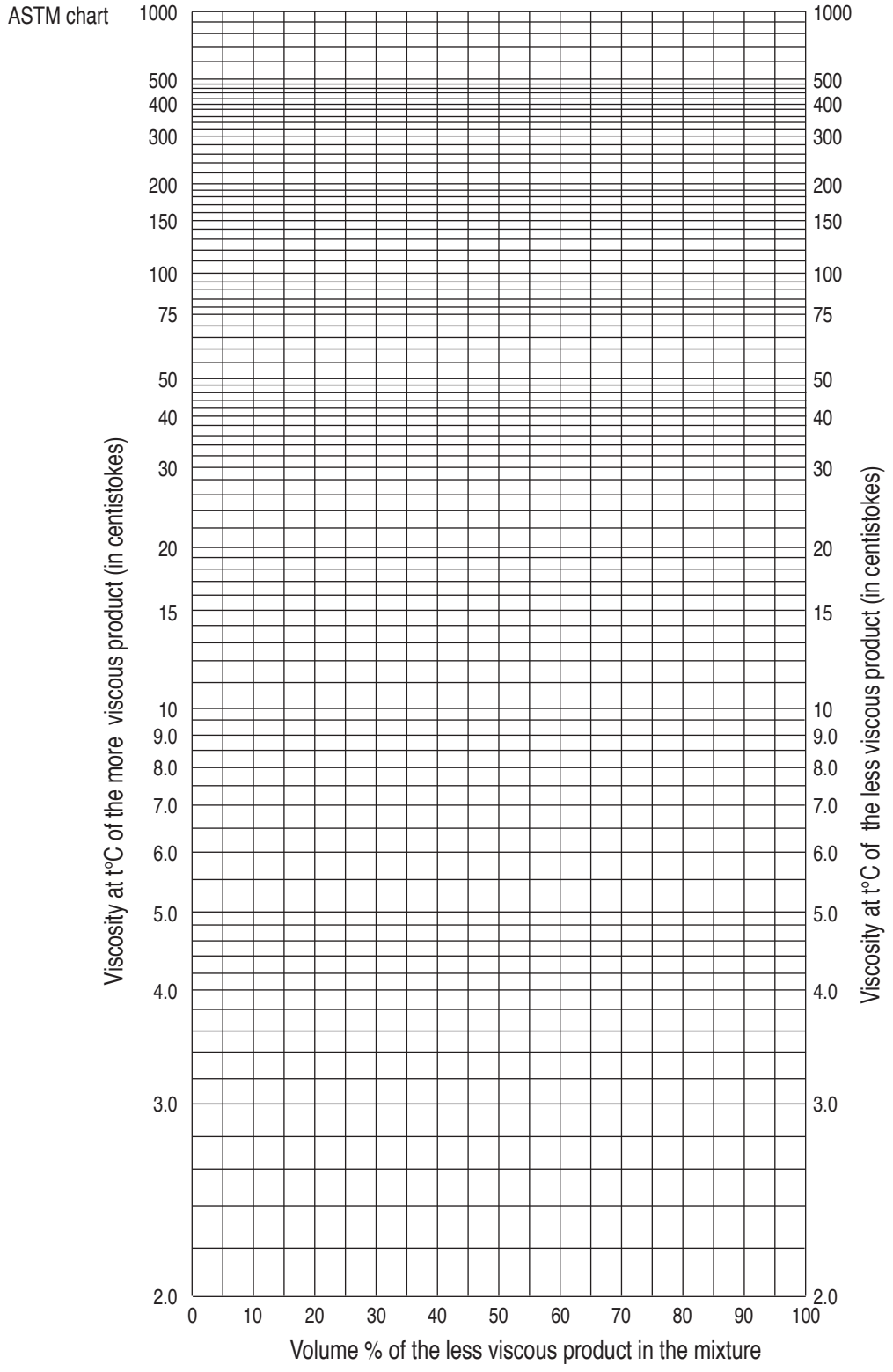
DYNAMIC VISCOSITY OF LIQUID HYDROCARBONS



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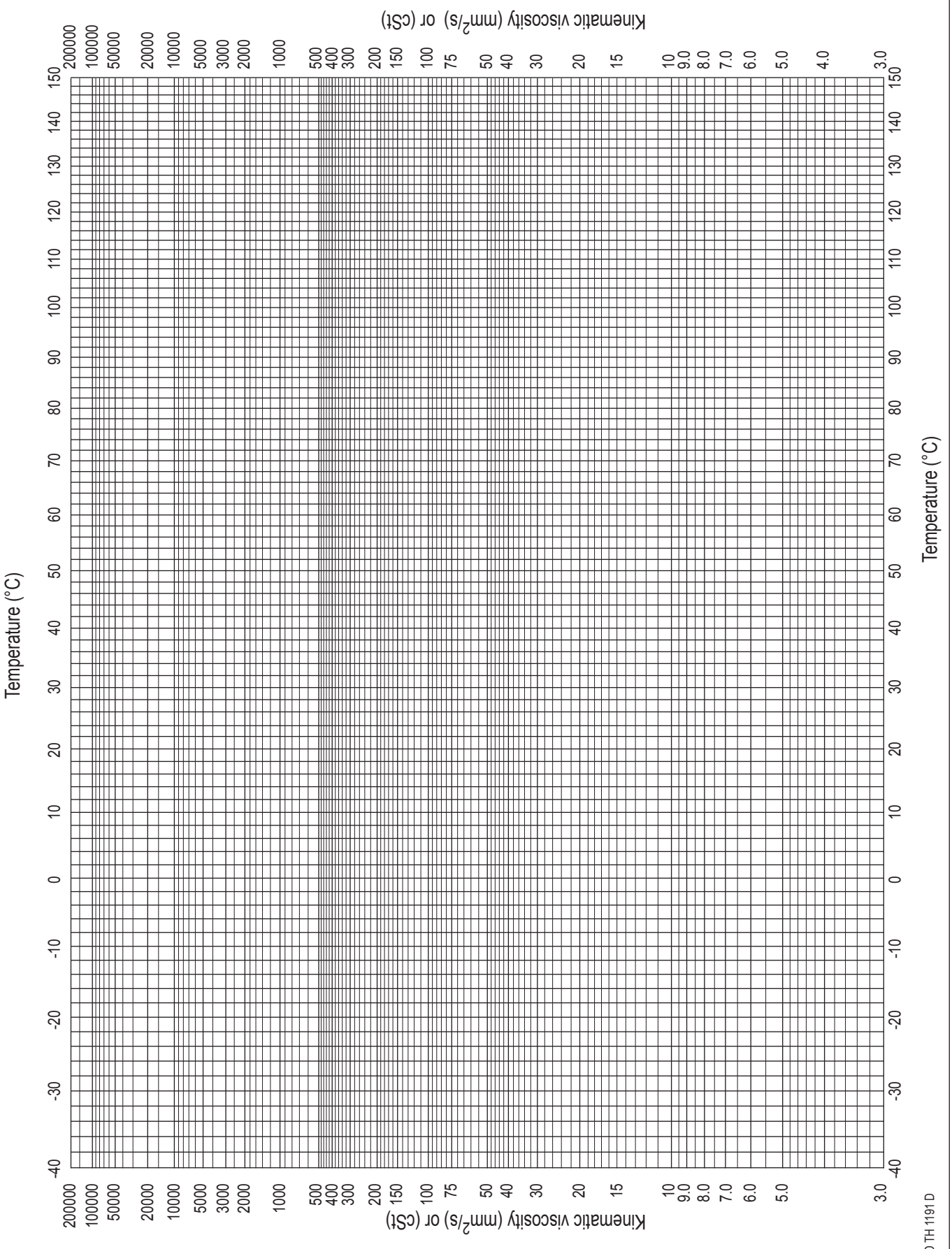
GRAPHIC DETERMINATION OF VISCOSITY AT T°C OF MIXTURE OF TWO LIQUID PETROLEUM PRODUCTS



D PPC 041 D

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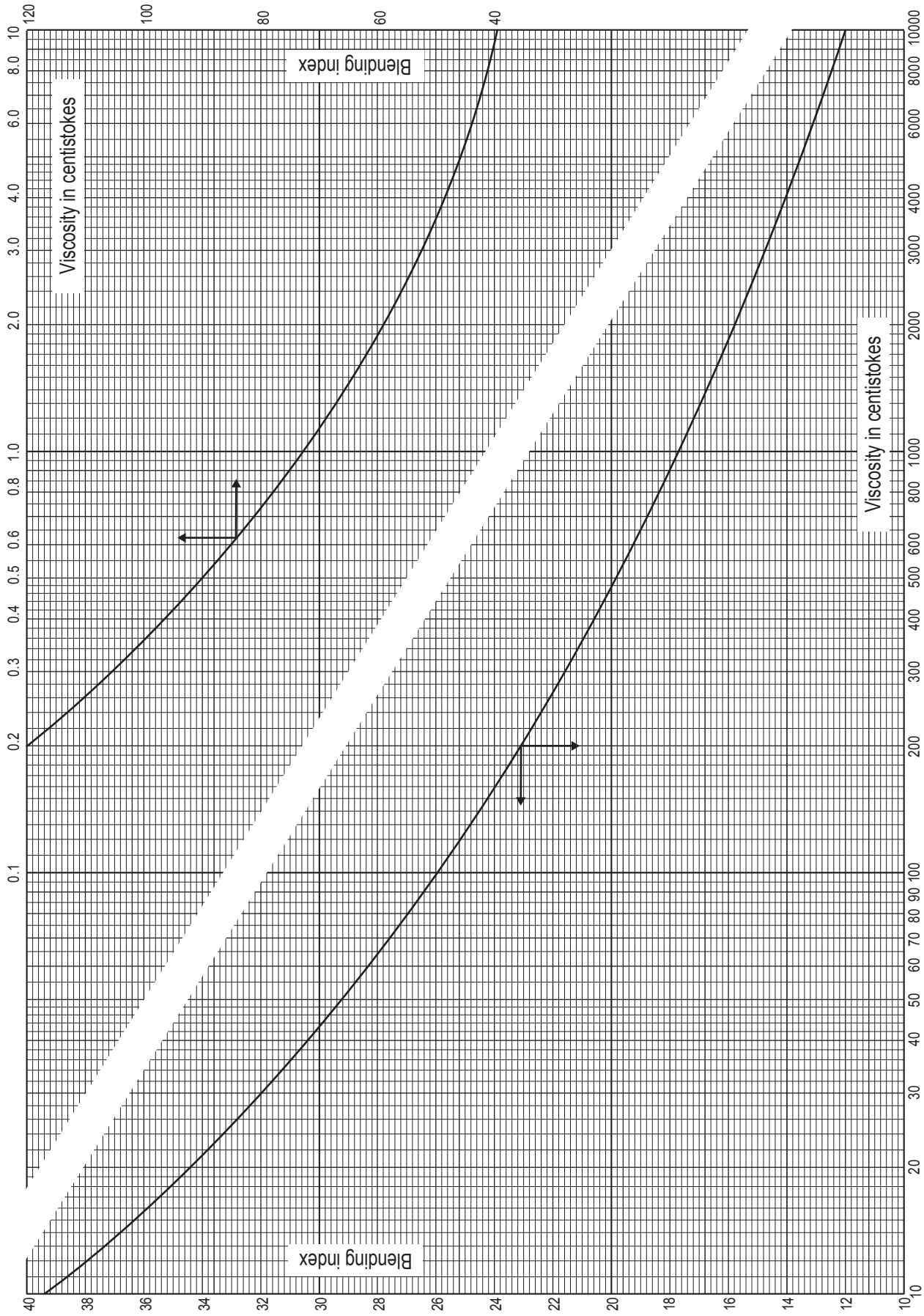
STANDARD VISCOSITY-TEMPERATURE CHART FOR LIQUID PETROLEUM PRODUCTS



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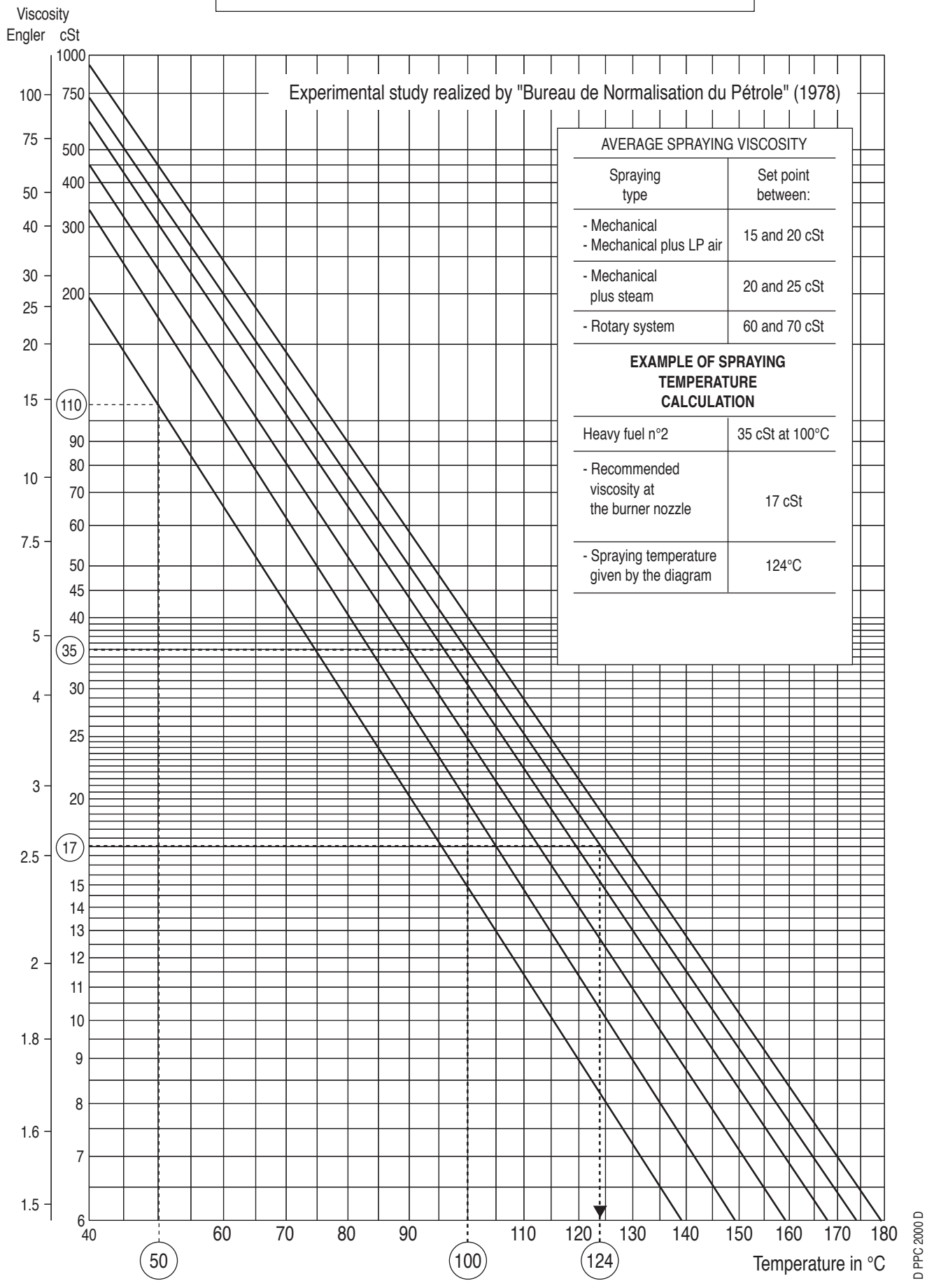
VISCOSITY OF MIXTURES

Volume blending rules : $V_M \cdot I_M = \Sigma V_i \cdot I_i$



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VISCOSITY DIAGRAM OF HEAVY FUELS N°2



AVERAGE SPRAYING VISCOSITY	
Spraying type	Set point between:
- Mechanical - Mechanical plus LP air	15 and 20 cSt
- Mechanical plus steam	20 and 25 cSt
- Rotary system	60 and 70 cSt

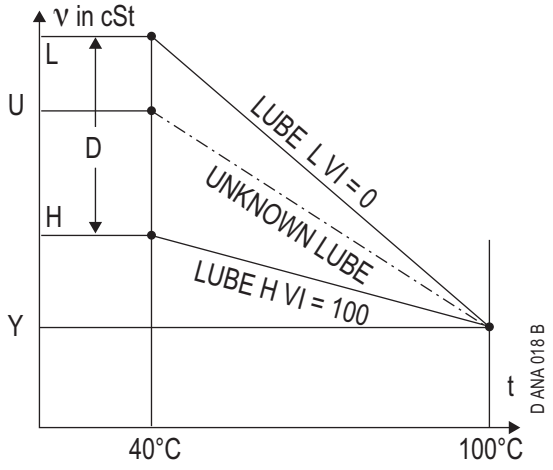
EXAMPLE OF SPRAYING TEMPERATURE CALCULATION

Heavy fuel n°2	35 cSt at 100°C
- Recommended viscosity at the burner nozzle	17 cSt
- Spraying temperature given by the diagram	124°C

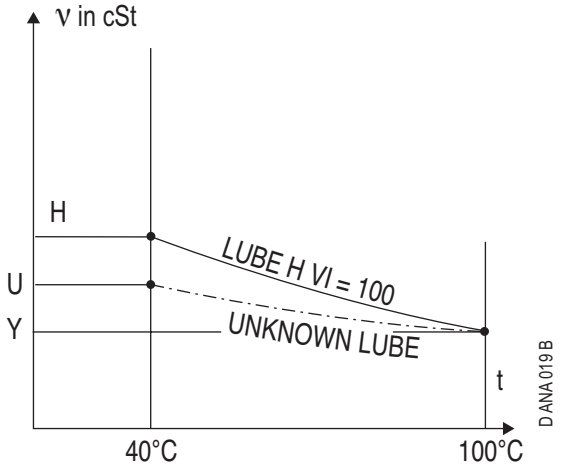
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CALCULATION OF VISCOSITY INDEX
NF T 60-136

DETERMINATION OF VI (<100)



DETERMINATION OF VI (≥100)



Y : Viscosity **at "hot T"** (100°C) of unknown lube in cSt (or mm^2/s)

U : Viscosity **at "cold T"** (40°C) of unknown lube in cSt

L : Viscosity at "cold T" in cSt of a lube in family L (VI = 0) with same "hot" viscosity as unknown lube

H : Viscosité at "cold T" in cSt of a lube in family H (VI = 100) with same "hot" viscosity as unknown lube

For Y comprised between 2 and 70 cSt

Value of L and H are read in following tables, as well as D, with:

$D = L - H$

$$VI = \frac{L - U}{D} \times 100$$

Value of H is read in following tables and N is calculated with:

$$N = \frac{\log H - \log U}{\log Y}$$

$$VI = \frac{10^N - 1}{0{,}00715} + 100$$

The viscosity index (VI) is rounded at the nearest integer (in cSt) (even integer if the decimal is exactly 0.5)

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VALUES OF L, D AND H AS FUNCTIONS OF CINEMATIC VISCOSITY

TABLE

Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*	Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*
2.00	7.994	1.600	6.394	7.50	88.85	34.87	53.98
2.10	8.640	1.746	6.894	7.60	91.04	35.94	55.09
2.20	9.309	1.898	7.410	7.70	93.20	37.01	56.20
2.30	10.00	2.056	7.944	7.80	95.43	38.12	57.31
2.40	10.71	2.219	8.496	7.90	97.72	39.27	58.45
2.50	11.45	2.390	9.063	8.00	100.0	40.40	59.60
2.60	12.21	2.567	9.647	8.10	102.3	41.57	60.74
2.70	13.00	2.748	10.25	8.20	104.6	42.72	61.89
2.80	13.80	2.937	10.87	8.30	106.9	43.85	63.05
2.90	14.63	3.132	11.50	8.40	109.2	45.01	64.18
3.00	15.49	3.334	12.15	8.50	111.5	46.19	65.32
3.10	16.36	3.540	12.82	8.60	113.9	47.40	66.48
3.20	17.26	3.753	13.51	8.70	116.2	48.57	67.64
3.30	18.18	3.971	14.21	8.80	118.5	49.75	68.79
3.40	19.12	4.196	14.93	8.90	120.9	50.96	69.94
3.50	20.09	4.428	15.66	9.00	123.3	52.20	71.10
3.60	21.08	4.665	16.42	9.10	125.7	53.40	72.27
3.70	22.09	4.909	17.19	9.20	128.0	54.61	73.42
3.80	23.13	5.157	17.97	9.30	130.4	55.84	74.57
3.90	24.19	5.415	18.77	9.40	132.8	57.10	75.73
4.00	25.32	5.756	19.56	9.50	135.3	58.36	76.91
4.10	26.50	6.129	20.37	9.60	137.7	59.68	78.08
4.20	27.75	6.546	21.21	9.70	140.1	60.87	79.27
4.30	29.07	7.017	22.05	9.80	142.7	62.22	80.46
4.40	30.48	7.560	22.92	9.90	145.2	63.54	81.67
4.50	31.96	8.156	23.81	10.0	147.7	64.86	82.87
4.60	33.52	8.806	24.71	10.1	150.3	66.22	84.08
4.70	35.13	9.499	25.63	10.2	152.9	67.56	85.30
4.80	36.79	10.22	26.57	10.3	155.4	68.90	86.51
4.90	38.50	10.97	27.53	10.4	158.0	70.25	87.72
5.00	40.23	11.74	28.49	10.5	160.6	71.63	88.95
5.10	41.99	12.53	29.46	10.6	163.2	73.00	90.19
5.20	43.76	13.32	30.43	10.7	165.8	74.42	91.40
5.30	45.53	14.13	31.40	10.8	168.5	75.86	92.65
5.40	47.31	14.94	32.37	10.9	171.2	77.33	93.92
5.50	49.09	15.75	33.34	11.0	173.9	78.75	95.19
5.60	50.07	16.55	34.32	11.1	176.6	80.20	96.45
5.70	52.64	17.36	35.29	11.2	179.4	81.65	97.71
5.80	54.42	18.16	36.26	11.3	182.1	83.13	98.97
5.90	56.20	18.97	37.23	11.4	184.9	84.63	100.2
6.00	57.97	19.78	38.19	11.5	187.6	86.10	101.5
6.10	59.74	20.57	39.17	11.6	190.4	87.61	102.8
6.20	61.52	21.38	40.15	11.7	193.3	89.18	104.1
6.30	63.32	22.19	41.13	11.8	196.2	90.75	105.4
6.40	65.18	23.03	42.14	11.9	199.0	92.30	106.7
6.50	67.12	23.94	43.18	12.0	201.9	93.87	108.0
6.60	69.16	24.92	44.24	12.1	204.8	95.47	109.4
6.70	71.29	25.96	45.33	12.2	207.8	97.07	110.7
6.80	73.48	27.04	46.44	12.3	210.7	98.66	112.0
6.90	75.72	28.21	47.51	12.4	213.6	100.3	113.3
7.00	78.00	29.43	48.57	12.5	216.6	101.9	114.7
7.10	80.25	30.63	49.61	12.6	219.6	103.6	116.0
7.20	82.39	31.70	50.69	12.7	222.6	105.3	117.4
7.30	84.53	32.74	51.78	12.8	225.7	107.0	118.7
7.40	86.66	33.79	52.88	12.9	228.8	108.7	120.1

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TABLE

Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*	Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*
13.0	231.9	110.4	121.5	18.0	408.6	212.4	196.2
13.1	236.0	112.1	122.9	18.1	412.6	214.8	197.8
13.2	238.1	113.8	124.2	18.2	416.7	217.3	199.4
13.3	241.2	115.6	125.6	18.3	420.7	219.7	201.0
13.4	244.3	117.3	127.0	18.4	424.9	222.2	202.6
13.5	247.4	119.0	128.4	18.5	429.0	224.7	204.3
13.6	250.6	120.8	129.8	18.6	433.2	227.2	205.9
13.7	253.8	122.6	131.2	18.7	437.3	229.7	207.6
13.8	257.0	124.4	132.6	18.8	441.5	232.3	209.3
13.9	260.1	126.2	134.0	18.9	445.7	234.7	211.0
14.0	263.3	128.0	135.4	19.0	449.9	237.3	212.7
14.1	266.6	129.8	136.8	19.1	454.2	239.8	214.4
14.2	269.8	131.6	138.2	19.2	458.4	242.3	216.1
14.3	273.0	133.5	139.6	19.3	462.7	245.0	217.7
14.4	276.3	135.3	141.0	19.4	467.0	247.6	219.4
14.5	279.6	137.2	142.4	19.5	471.3	250.2	221.1
14.6	283.0	139.1	143.9	19.6	475.7	252.9	222.8
14.7	286.4	141.1	145.3	19.7	479.7	255.2	224.5
14.8	289.7	142.9	146.8	19.8	483.9	257.8	226.2
14.9	293.0	144.8	148.2	19.9	488.6	260.9	227.7
15.0	296.5	146.8	149.7	20.0	493.2	263.7	229.5
15.1	300.0	148.8	151.2	20.2	501.5	268.5	233.0
15.2	303.4	150.8	152.6	20.4	510.8	274.4	236.4
15.3	306.9	152.8	154.1	20.6	519.9	279.8	240.1
15.4	310.3	154.8	155.6	20.8	528.8	285.3	243.5
15.5	313.9	156.9	157.0	21.0	538.4	291.3	247.1
15.6	317.5	158.9	158.6	21.2	547.5	296.8	250.7
15.7	321.1	161.0	160.1	21.4	556.7	302.6	254.2
15.8	324.6	163.0	161.6	21.6	566.4	308.6	257.8
15.9	328.3	165.2	163.1	21.8	575.6	314.1	261.5
16.0	331.9	167.3	164.6	22.0	585.2	320.2	264.9
16.1	336.5	169.4	166.1	22.2	595.0	326.4	268.6
16.2	339.2	171.5	167.7	22.4	604.3	332.0	272.3
16.3	342.9	173.7	169.2	22.6	614.2	338.4	275.8
16.4	346.6	175.8	170.7	22.8	624.1	344.5	279.6
16.5	350.3	178.1	172.3	23.0	633.6	350.3	283.3
16.6	354.1	180.3	173.8	23.2	643.4	356.6	286.8
16.7	358.0	182.5	175.4	23.4	653.8	363.3	290.5
16.8	361.7	189.7	177.0	23.6	663.3	369.0	294.4
16.9	365.6	187.0	178.6	23.8	673.7	375.7	297.9
17.0	369.4	189.2	180.2	24.0	683.9	382.1	301.8
17.1	373.3	191.5	181.7	24.2	694.5	388.9	305.6
17.2	377.1	193.8	183.3	24.4	704.2	394.8	309.4
17.3	381.0	196.1	184.9	24.6	714.9	401.9	313.0
17.4	384.9	198.4	186.5	24.8	725.7	408.8	317.0
17.5	388.9	200.8	188.1	25.0	736.5	415.6	320.9
17.6	392.7	203.0	189.7	25.2	747.2	422.4	324.9
17.7	396.7	205.3	191.3	25.4	758.2	429.5	328.8
17.8	400.7	207.7	192.9	25.6	769.3	436.6	332.7
17.9	404.6	210.0	194.6	25.8	779.7	443.0	336.7

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TABLE

Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*	Cinematic viscosity at 100 °C (mm^2/s)	*L*	*D = (L – H)*	*H*
26.0	790.4	449.8	340.5	45.0	2152	1371	780.9
26.2	801.6	457.2	344.4	45.5	2197	1403	794.5
26.4	812.8	464.4	348.4	46.0	2243	1434	808.2
26.6	824.1	471.8	352.3	46.5	2288	1466	821.9
26.8	835.5	479.1	356.4	47.0	2333	1498	835.5
27.0	847.0	486.6	360.5	47.5	2380	1530	849.2
27.2	857.5	492.9	364.6	48.0	2426	1563	863.0
27.4	869.0	500.6	368.3	48.5	2473	1596	876.9
27.6	880.6	508.3	372.3	49.0	2521	1630	890.9
27.8	892.3	515.9	376.4	49.5	2570	1665	905.3
28.0	904.1	523.5	380.6	50.0	2618	1699	919.6
28.2	915.8	531.2	384.6	50.5	2667	1733	933.6
28.4	927.6	538.8	388.8	51.0	2717	1769	948.2
28.6	938.6	545.7	393.0	51.5	2767	1804	962.9
28.8	951.2	554.5	396.6	52.0	2817	1839	977.5
29.0	963.4	562.3	401.1	52.5	2867	1875	992.1
29.2	975.4	570.1	405.3	53.0	2918	1911	1007
29.4	987.1	577.6	409.5	53.5	2969	1947	1021
29.6	998.9	585.3	413.5	54.0	3020	1984	1036
29.8	1011	593.4	417.6	54.5	3073	2022	1051
30.0	1023	601.6	421.7	55.0	3126	2060	1066
30.5	1055	622.3	432.4	55.5	3180	2098	1082
31.0	1086	643.2	443.2	56.0	3233	2136	1097
31.5	1119	664.5	454.0	56.5	3286	2174	1112
32.0	1151	686.0	464.9	57.0	3340	2213	1127
32.5	1184	708.0	475.9	57.5	3396	2253	1143
33.0	1217	730.2	487.0	58.0	3452	2293	1159
33.5	1251	752.8	498.1	58.5	3507	2332	1175
34.0	1286	776.8	509.6	59.0	3563	2372	1190
34.5	1321	799.9	521.1	59.5	3619	2413	1206
35.0	1356	823.4	532.5	60.0	3676	2454	1222
35.5	1391	847.2	544.0	60.5	3734	2496	1238
36.0	1427	871.2	555.6	61.0	3792	2538	1254
36.5	1464	896.5	567.1	61.5	3850	2579	1270
37.0	1501	921.8	579.3	62.0	3908	2621	1286
37.5	1533	946.8	591.3	62.5	3966	2664	1303
38.0	1575	972.3	603.1	63.0	4026	2707	1319
38.5	1613	998.3	615.0	63.5	4087	2751	1336
39.0	1651	1024	627.1	64.0	4147	2795	1392
39.5	1691	1052	639.2	64.5	4207	2858	1369
40.0	1730	1079	651.8	65.0	4268	2882	1386
40.5	1770	1106	664.2	65.5	4329	2927	1402
41.0	1810	1133	676.6	66.0	4392	2973	1419
41.5	1851	1162	689.1	66.5	4455	3018	1436
42.0	1892	1191	701.9	67.0	4517	3064	1454
42.5	1935	1220	714.9	67.5	4580	3110	1471
43.0	1978	1250	728.2	68.0	4645	3157	1488
43.5	2021	1280	741.3	68.5	4709	3204	1506
44.0	2064	1310	754.4	69.0	4773	3250	1523
44.5	2108	1340	767.6	69.5	4839	3298	1541
				70.0	4905	3346	1558

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VISCOSITY GRADES FOR MOTOR LUBES
— J 300 2002-2003 —

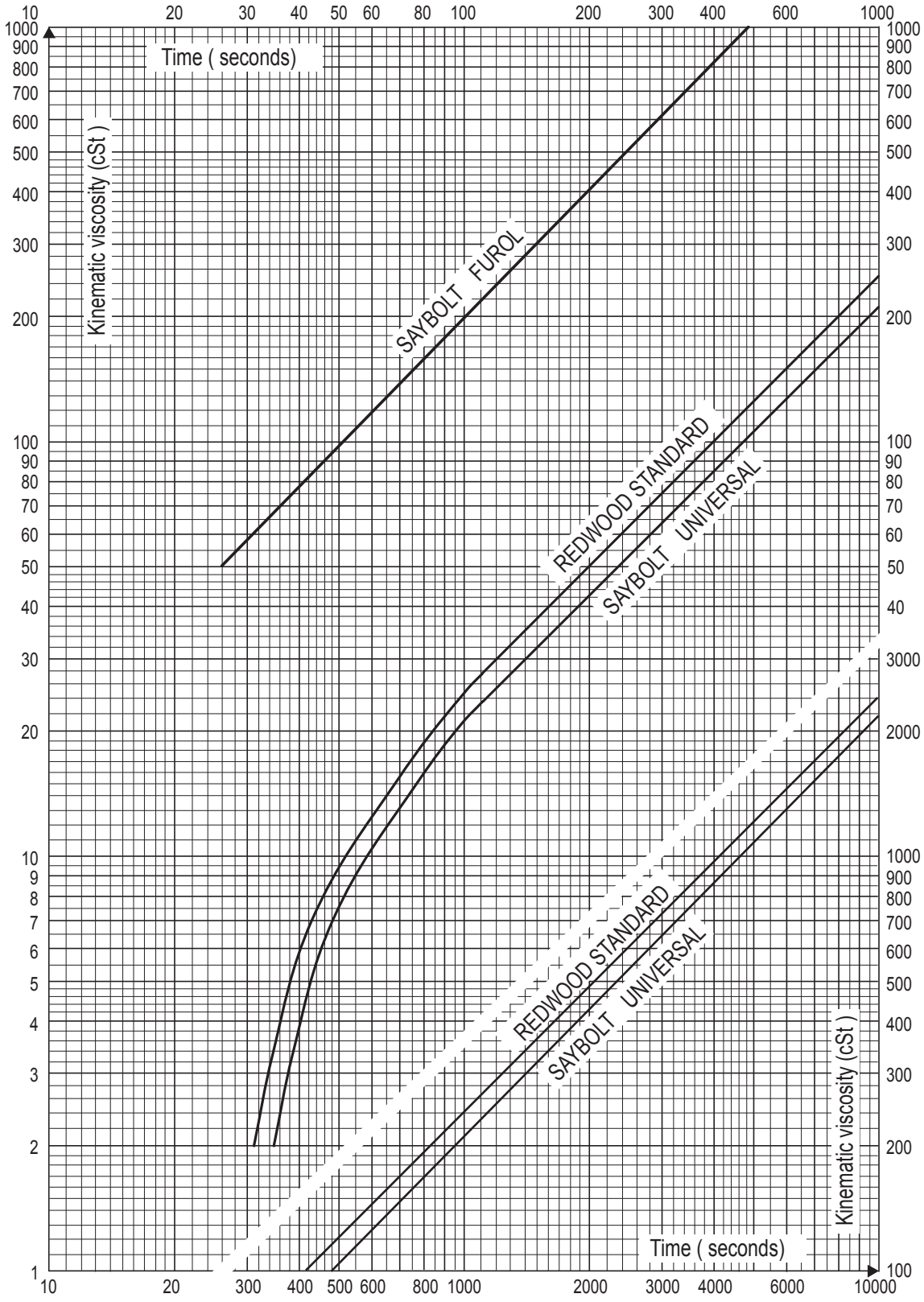
Viscosity grade SAE	Cold starting up viscosity mPa.s at T in °C	Cold pumpability viscosity mPa.s at T in °C	Viscosity under low shear rate at 100°C (mm^2/s)		Viscosity under high shear rate at 150°C and $10^6\ s^{-1}$ (mPa.s)
	maximum	maximum	minimum	maximum	minimum
0W	6200 at -35	60000 at -40	3.8	-	-
5W	6600 at -30	60000 at -35	3.8	-	-
10W	7000 at -25	6000 at -30	4.1	-	-
15W	7000 at -20	60000 at -25	5.6	-	-
20W	9500 at -15	60000 at -20	5.6	-	-
25W	13000 at -10	60000 at -15	9.3	-	-
20	-	-	5.6	< 9.3	2.6
30	-	-	9.3	<12.5	2.9
40	-	-	12.5	< 16.3	2.9 (0W-40, 5W-40, 10W-40)
40	-	-	12.5	< 16.3	3.7 (15W-40, 20W-40, 25W-40, 40)
50	-	-	16.3	< 21.9	3.7
60	-	-	21.9	< 26.1	3.7
	ASTM D 5293	ASTM D 4684	ISO 3104		CEC L-36-A-90

Notes : *- 1 cP = 1 m Pa.s ; 1 cSt = 1 mm^2/s*

- shear rate is calculated as a velocity (m/s) divided by a distance (between rotor and stator), i.e. expressed in s^{-1}

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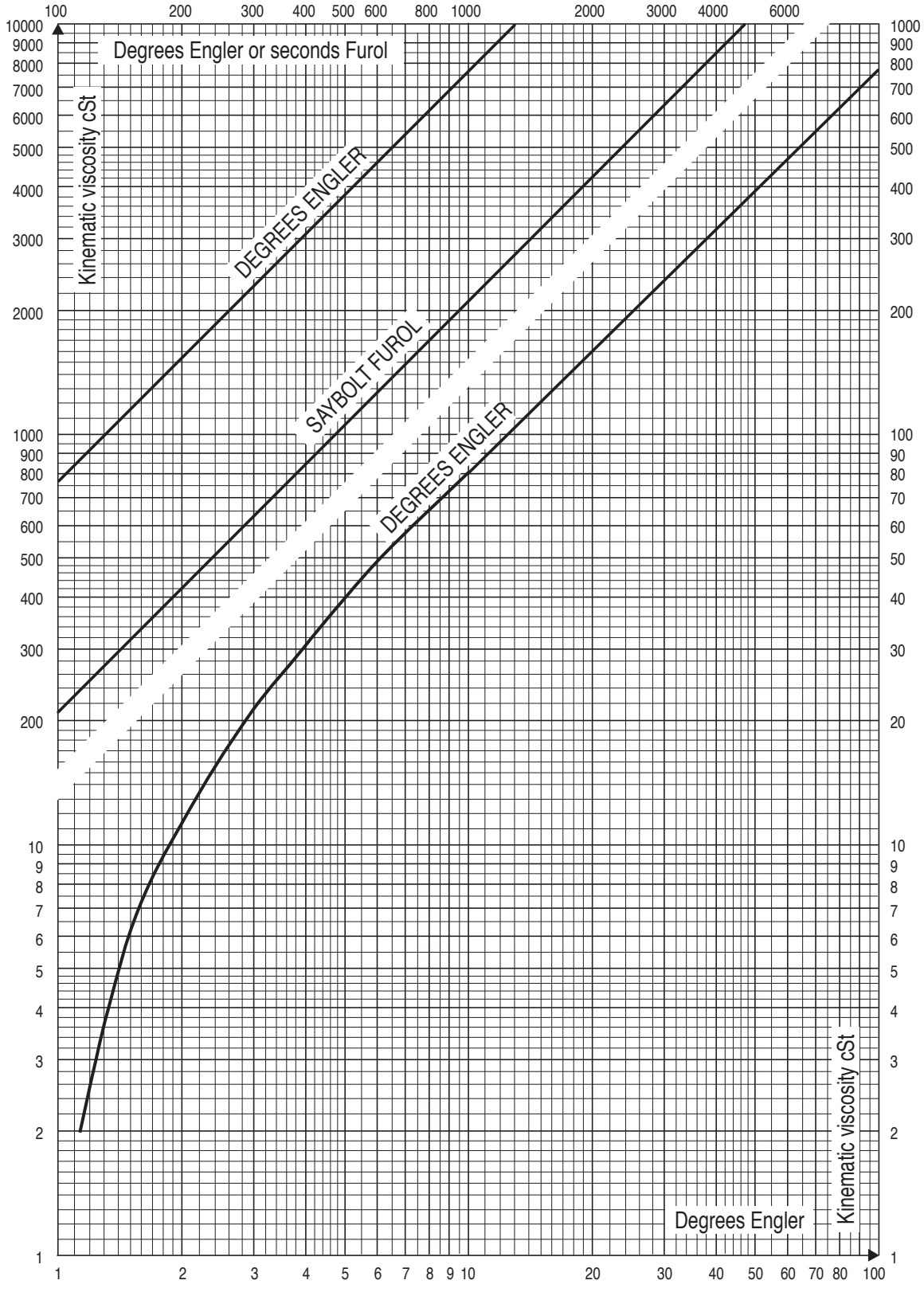
CONVERSION INTO CENTISTOKES



D PPC 045 B

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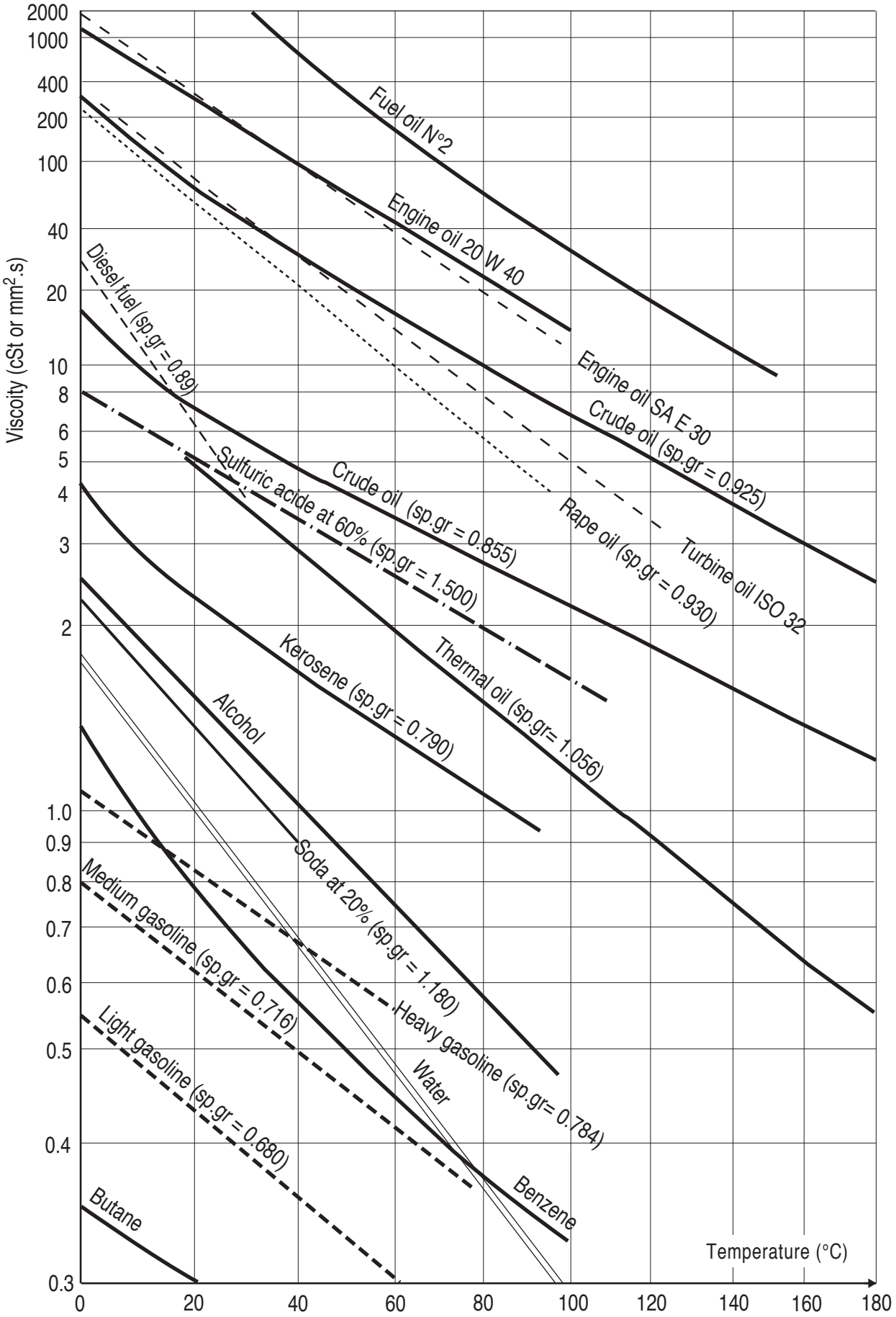
CONVERSION INTO CENTISTOKES



D PPC 044 D

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VISCOSITY OF SOME LIQUIDS



D PPC 046 D

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VISCOSITY OF LIQUID CHEMICAL PRODUCTS

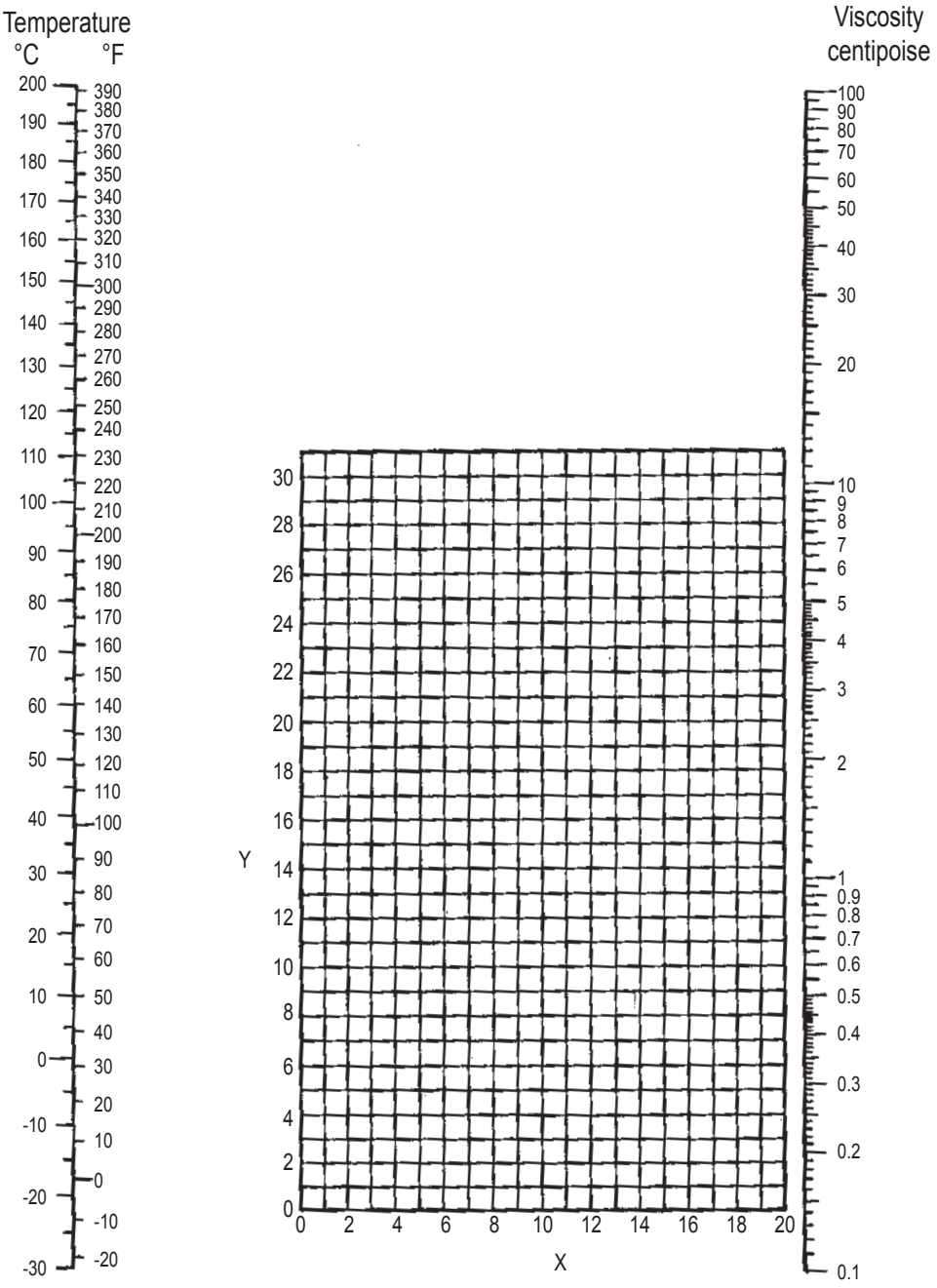
Liquid	X	Y	Liquid	X	Y
Acetaldehyde	15.2	4.8	Freon-113	12.5	11.4
Acetic acid. 100%	12.1	14.2	Glycerol. 100%	2.0	30.0
Acetic acid. 70%	9.5	17.0	Glycerol. 50%	6.9	19.6
Acetic anhydride	12.7	12.8	Heptane	14.1	8.4
Acetone. 100%	14.5	7.2	Hexane	14.7	7.0
Acetone. 35%	7.9	15.0	Hydrochloric acid. 31.5%	13.0	16.6
Acetonitrile	14.4	7.4	Iodobenzene	12.8	15.9
Acrylic acid	12.3	13.9	Isobutyl alcohol	7.1	18.0
Allyl alcohol	10.2	14.3	Isobutyric acid	12.2	14.4
Allyl bromide	14.4	9.6	Isopropyl alcohol	8.2	16.0
Allyl iodide	14.0	11.7	Isopropyl bromide	14.1	9.2
Ammonia. 100%	12.6	2.0	Isopropyl chloride	13.9	7.1
Ammonia. 26%	10.1	13.9	Isopropyl iodide	13.7	11.2
Amyl acetate	11.8	12.5	Kerosene	10.2	16.9
Amyl alcohol	7.5	18.4	Linseed oil. raw	7.5	27.2
Aniline	8.1	18.7	Mercury	18.4	16.4
Anisole	12.3	13.5	Methanol. 100%	12.4	10.5
Arsenic trichloride	13.9	14.5	Methanol. 90%	12.3	11.8
Benzene	12.5	10.9	Methanol. 40%	7.8	15.5
Brine. $CaCl_2$. 25%	6.6	15.9	Methyl acetate	14.2	8.2
Brine. NaCl. 25%	10.2	16.6	Methyl acrylate	13.0	9.5
Bromine	14.2	13.2	Methyl i-butyrate	12.3	9.7
Bromotoluene	20.0	15.9	Methyl n-butyrate	13.2	10.3
Butyl acetate	12.3	11.0	Methyl chloride	15.0	3.8
Butyl acrylate	11.5	12.6	Methyl ethyl ketone	13.9	8.6
Butyl alcohol	8.6	17.2	Methyl formate	14.2	7.5
Butyric acid	12.1	15.3	Methyl iodide	14.3	9.3
Carbon dioxide	11.6	0.3	Methyl propionate	13.5	9.0
Carbon disulfide	16.1	7.5	Methyl propyl ketone	14.3	9.5
Carbon tetrachloride	12.7	13.1	Methyl sulfide	15.3	6.4
Chlorobenzene	12.3	12.4	Naphtalene	7.9	18.1
Chloroform	14.4	10.2	Nitric acid. 95%	12.8	13.8
Chlorosulfonic acid	11.2	18.1	Nitric acid. 60%	10.8	17.0
Chlorotoluene, ortho	13.0	13.3	Nitrobenzene	10.6	16.2
Chlorotoluene, meta	13.3	12.5	Nitrogen dioxide	12.9	8.6
Chlorotoluene, para	13.3	12.5	Nitrotoluene	11.0	17.0
Cresol, meta	2.5	20.8	Octane	13.7	10.0
Cyclohexanol	2.9	24.3	Octyl alcohol	6.6	21.1
Cyclohexane	9.8	12.9	Pentachloroethane	10.9	17.3
Dibromomethane	12.7	15.8	Pentane	14.9	5.2
Dichloroethane	13.2	12.2	Phenol	6.9	20.8
Dichloromethane	14.6	8.9	Phosphorus tribromide	13.8	16.7
Diethyl ketone	13.5	9.2	Phosphorus trichloride	16.2	10.9
Diethyl oxalate	11.0	16.4	Propionic acid	12.8	13.8
Diethylene glycol	5.0	24.7	Propyl acetate	13.1	10.3
Diphenyl	12.0	18.3	Propyl alcohol	9.1	16.5
Dipropyl ether	13.2	8.6	Propyl bromide	14.5	9.6
Dipropyl oxalate	10.3	17.7	Propyl chloride	14.4	7.5
Ethyl acetate	13.7	9.1	Propyl formate	13.1	9.7
Ethyl acrylate	12.7	10.4	Propyl iodide	14.1	11.6
Ethyl alcohol. 100%	10.5	13.8	Sodium	16.4	13.9
Ethyl alcohol. 95%	9.8	14.3	Sodium hydroxide. 50%	3.2	25.8
Ethyl alcohol. 40%	6.5	16.6	Stannic chloride	13.5	12.8
Ethyl benzene	13.2	11.5	Succinonitrile	10.1	20.8
Ethyl bromide	14.5	8.1	Sulfur dioxide	15.2	7.1
2-Ethyl butyl acrylate	11.2	14.0	Sulfuric acid. 110%	7.2	27.4
Ethyl chloride	14.8	6.0	Sulfuric acid. 100%	8.0	25.1
Ethyl ether	14.5	5.3	Sulfuric acid. 98%	7.0	24.8
Ethyl formate	14.2	8.4	Sulfuric acid. 60%	10.2	21.3
2-Ethyl hexyl acrylate	9.0	15.0	Sulfuryl chloride	15.2	12.4
Ethyl iodide	14.7	10.3	Tetrachloroethane	11.9	15.7
Ethyl propionate	13.2	9.9	Thiophene	13.2	11.0
Ethyl propyl ether	14.0	7.0	Titanium tetrachloride	14.4	12.3
Ethyl sulfide	13.8	8.9	Toluene	13.7	10.4
Ethylene bromide	11.9	15.7	Trichloroethylene	14.8	10.5
Ethylene chloride	12.7	12.2	Triethylene glycol	4.7	24.8
Ethylene glycol	6.0	23.6	Turpentine	11.5	14.9
Ethylidene chloride	14.1	8.7	Vinyl acetate	14.0	8.8
Fluorobenzene	13.7	10.4	Vinyl toluene	13.4	12.0
Formic acid	10.7	15.8	Water	10.2	13.0
Freon-11	14.4	9.0	Xylene, ortho	13.5	12.1
Freon-12	16.8	5.6	Xylene, meta	13.9	10.6
Freon-21	15.7	7.5	Xylene, para	13.9	10.9
Freon-22	17.2	4.7			

To find the viscosity of one chemical product, use the chart at the following page.
Plot the point X and Y, draw a line from this point to the temperature and read the viscosity on the appropriate scale.

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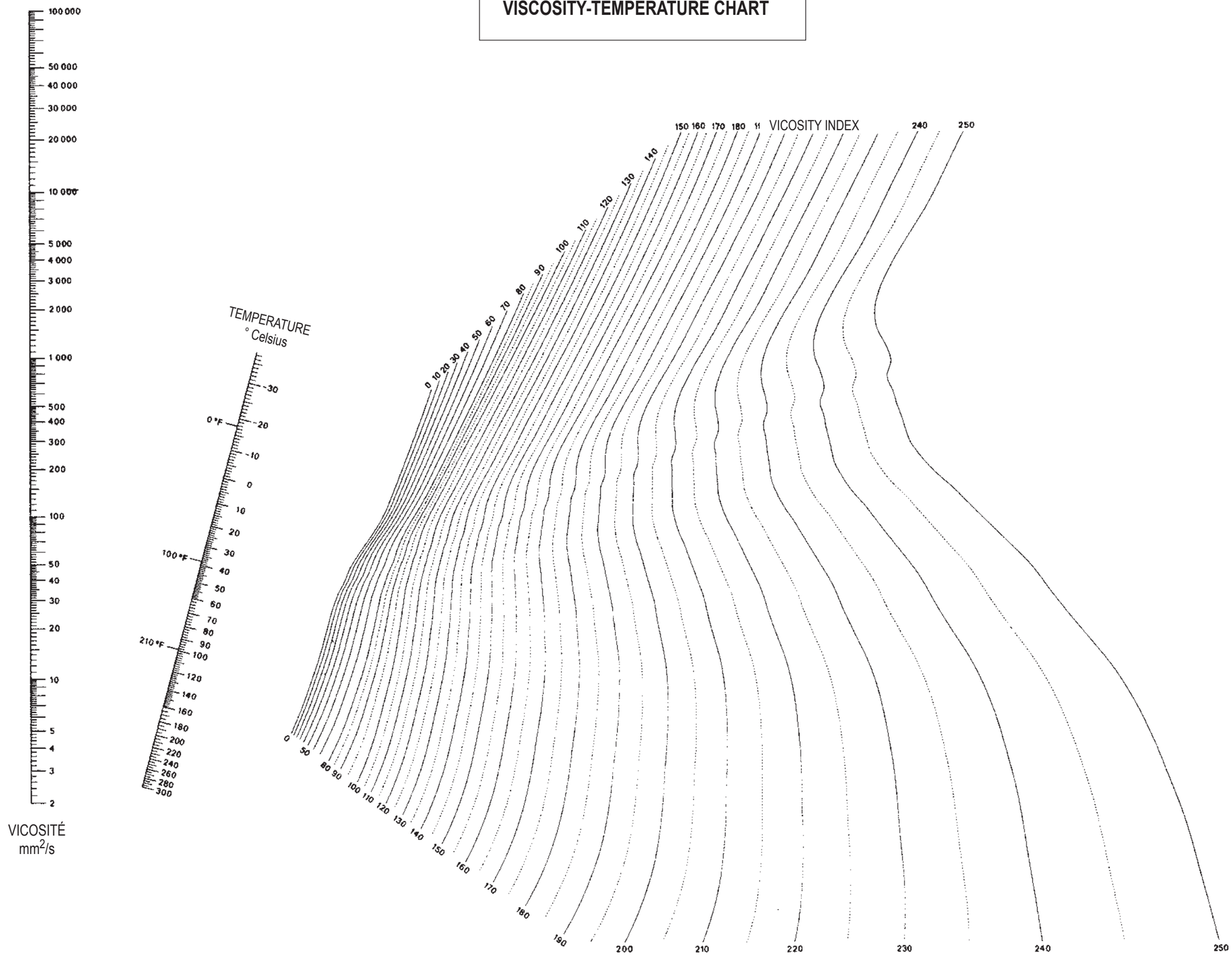
VISCOSITY OF LIQUID CHEMICAL PRODUCTS

Chart



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VISCOSITY-TEMPERATURE CHART



J. Groff's viscosity-temperature chart

D TH 1193 D

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THERMAL CONDUCTIVITY OF METALS

	Thermal conductivity (kcal/h.m.°C)																
Temperature (°C)	20	50	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800
Material																	
Carbon steel	44.9	44.3	43.2	42.1	40.9	39.7	38.4	37.1	35.8	34.4	32.9	31.4	29.9	28.3	26.6	24.9	23.2
C 0.5 Mo	37.4	37.4	37.2	37.0	36.6	36.1	35.5	34.7	33.8	32.9	31.7	30.5	29.2	27.7	26.1	24.4	22.6
1 Cr 0.5 Mo	32.1	32.2	32.3	32.4	32.3	32.0	31.7	31.2	30.6	29.9	29.1	28.2	27.1	25.9	24.6	23.2	21.6
2 1/4 Cr 1 Mo	31.3	31.5	31.6	31.7	31.6	31.4	31.1	30.7	30.2	29.5	28.7	27.8	26.8	25.6	24.4	23.0	21.5
5 Cr 1/2 Mo	25.8	26.2	26.9	27.4	27.7	28.0	28.1	28.1	28.0	27.8	27.5	27.0	26.4	25.7	24.9	23.9	22.9
7 Cr 1/2 Mo	21.3	21.9	22.8	23.6	24.2	24.8	25.2	25.5	25.7	25.8	25.8	25.6	25.3	24.9	24.4	23.7	22.9
9 Cr 1 Mo	19.3	19.9	20.9	21.7	22.4	23.0	23.6	24.0	24.3	24.6	24.7	24.8	24.7	24.6	24.4	24.0	23.6
3 1/2 Ni	34.9	35.0	35.1	35.0	34.8	34.4	33.9	33.3	32.5	31.6	30.5	29.3	27.9	26.4	24.7	22.9	21.0
13 Cr	22.7	22.8	23.0	23.2	23.4	23.5	23.6	23.6	23.6	23.6	23.5	23.4	23.2	23.0	22.8	22.5	22.2
15 Cr	21.1	21.2	21.4	21.5	21.7	21.8	21.9	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0
17 Cr	18.9	18.9	19.1	19.2	19.4	19.5	19.7	19.8	20.0	20.1	20.3	20.4	20.6	20.8	20.9	21.1	21.3
TP 304	12.8	13.2	13.9	14.6	15.3	16.0	16.7	17.3	18.0	18.6	19.2	19.8	20.4	21.0	21.5	22.1	22.7
TP 316 & 317	11.5	11.9	12.6	13.3	14.0	14.7	15.4	16.1	16.7	17.4	18.0	18.6	19.2	19.8	20.4	21.0	21.5
TP 321 & 347	12.1	12.6	13.3	14.0	14.7	15.4	16.0	16.7	17.4	18.0	18.7	19.3	19.9	20.5	21.1	21.7	22.3
TP 310	10.9	11.3	12.0	12.7	13.4	14.1	14.8	15.5	16.2	16.9	17.6	18.2	18.9	19.6	20.2	20.9	21.5
Ni 200			57.7	54.6	51.9	49.8	48.1	46.9	46.2	45.9	46.2						
Ni Cu 400	18.8	19.7	21.0	22.4	23.7	25.1	26.4	27.7	29.0	30.3	31.5						
Ni Cr Fe 600	12.7	13.1	13.7	14.3	14.9	15.6	16.2	16.9	17.6	18.3	19.0	19.7	20.5	21.2	22.0	22.7	23.5
Ni Fe Cr 800	10.0	10.4	11.1	11.9	12.6	13.3	14.0	14.7	15.4	16.1	16.9	17.6	18.3	19.0	19.7	20.4	21.2
Ni Fe Cr Mo Cu 825			10.7	11.3	12.0	12.6	13.2	13.8	14.4	15.1	15.7	16.3	17.0	17.6	18.3	19.0	19.6
Ni Mo B		9.2	9.6	9.9	10.4	10.9	11.5	12.2	12.9	13.7	14.6	15.6	16.6				
Ni Mo Cr 276		8.9	9.7	10.4	11.2	11.9	12.7	13.4	14.2	14.9	15.7	16.4	17.2	17.9	18.7	19.4	
Al 3003	152.1	153.4	155.3	157.0	158.5												
Al 6061	142.9	144.8	147.6	150.0	151.8												
Titanium	18.8	18.4	17.9	17.5	17.2	17.0	16.9	16.9	16.9	17.1	17.4	17.7	18.2				
Amiralty			105.1	111.1	117.3	123.6	130.1										
Naval Brass			106.1	110.1	114.1	118.2	122.2										
Copper			334.6	334.2	333.7	333.0	332.1										
90 10 Cu Ni			42.8	47.4	51.9	56.3	60.7	64.9	69.0	73.0	76.9						
70 30 Cu Ni			27.3	29.3	31.7	34.6	38.0	41.9	46.3	51.2	56.5						
Muntz			105.6														
Zirconium			17.8														
Cr Mo XM 27			16.8														
20 Cb			11.3														
625	8.4	8.8	9.4	10.0		11.3		12.5		13.8		15.1		16.3		17.6	

From TEMA Standards

D MTE 1322 D

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THERMAL CONDUCTIVITY OF LIQUIDS

A linear variation with temperature may be assumed. The extreme values given constitute also the temperature limits over which the data are recommended

Thermal conductivity λ(kcal/h.m.°C)

Liquid	T (°C)	λ	Liquid	T (°C)	λ
Acetic acid	20	0.137	Formaldehyde	-80	0.275
	150	0.116		-20	0.196
Acetone	-20	0.138		20	0.172
	80	0.113	Glycerine	20	0.239
Acetyléne	-140	0.204		200	0.269
	-80	0.132	Heptane (n-)	10	0.11
	0	0.085		150	0.074
Acrylic acid	0	0.214	Hexane (n-)	10	0.107
	40	0.184		150	0.068
	160	0.128	Heptyl alcohol	20	0.114
Allyl alcohol	20	0.141		140	0.106
	100	0.137	Hexyl alcohol	20	0.114
Amyl alcohol	20	0.132		120	0.11
	100	0.126	Methylthyl ketone (MEK)	-20	0.132
Aniline	20	0.198		120	0.1
	150	0.132	Nonane (n-)	10	0.114
Benzene	20	0.126		150	0.083
	160	0.088	Octane	10	0.113
Bromobenzene	0	0.097		150	0.08
	200	0.088	Para Xylene	20	0.113
Butyl acetate (n-)	0	0.122		80	0.097
	160	0.083		200	0.07
Butyl alcohol (iso-)	-40	0.149	Pentane	10	0.103
	10	0.129		120	0.071
	70	0.114	Propyl alcohol (n-)	-40	0.158
	150	0.111		150	0.107
Butyl alcohol (n-)	-40	0.155	Propyl alcohol (iso-)	-40	0.137
	150	0.095		60	0.112
Carbon disulfide	-80	0.125		150	0.107
	20	0.107	Toluene	0	0.123
Carbon tetrachloride	-80	0.106		200	0.074
	100	0.077	Trichlorethylene	-40	0.125
Chlorobenzene	0	0.112		30	0.097
	200	0.101		150	0.068
Chloroform	-75	0.123	Vinyl acetate	0	0.131
	100	0.083		110	0.097
Cumene	0	0.112	Water	0	0.51
	200	0.074		40	0.54
Cyclohexane	5	0.132		95	0.57
	40	0.12		150	0.587
	120	0.089		215	0.559
Dichlorodifluoromethane	-60	0.098		325	0.409
	10	0.094	Xylene (ortho)	0	0.129
	60	0.086		80	0.101
Ethyl acetate	0	0.131		200	0.071
	110	0.097	Xylene (meta)	0	0.119
Ethyl alcohol	-40	0.164		80	0.092
	150	0.119		200	0.065
Ethyl benzene	0	0.119			
	200	0.0667			

From TEMA Standards

D MTE 1323 D

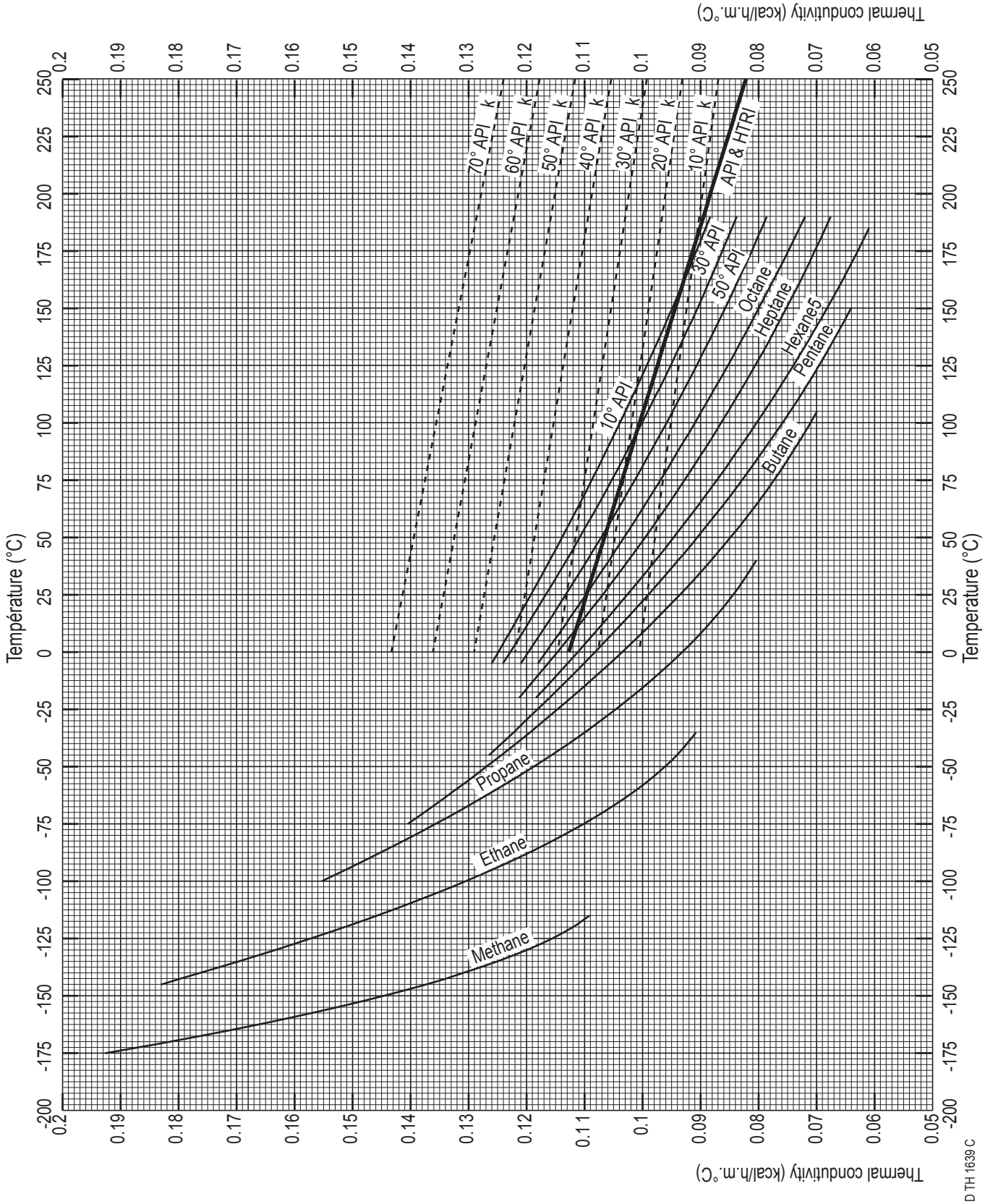
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THERMAL CONDUCTIVITY OF LIQUID PETROLEUM CUTS

From RW Gallant (Physical properties of hydrocarbons)

From D Kern (National Bureau Standards) $\lambda = \frac{0.0067}{sp.gr._4^{15}} (1 - 0.00054.t)$

From API & HTRI ($\lambda = 0.1128 - 0.000122.t$)



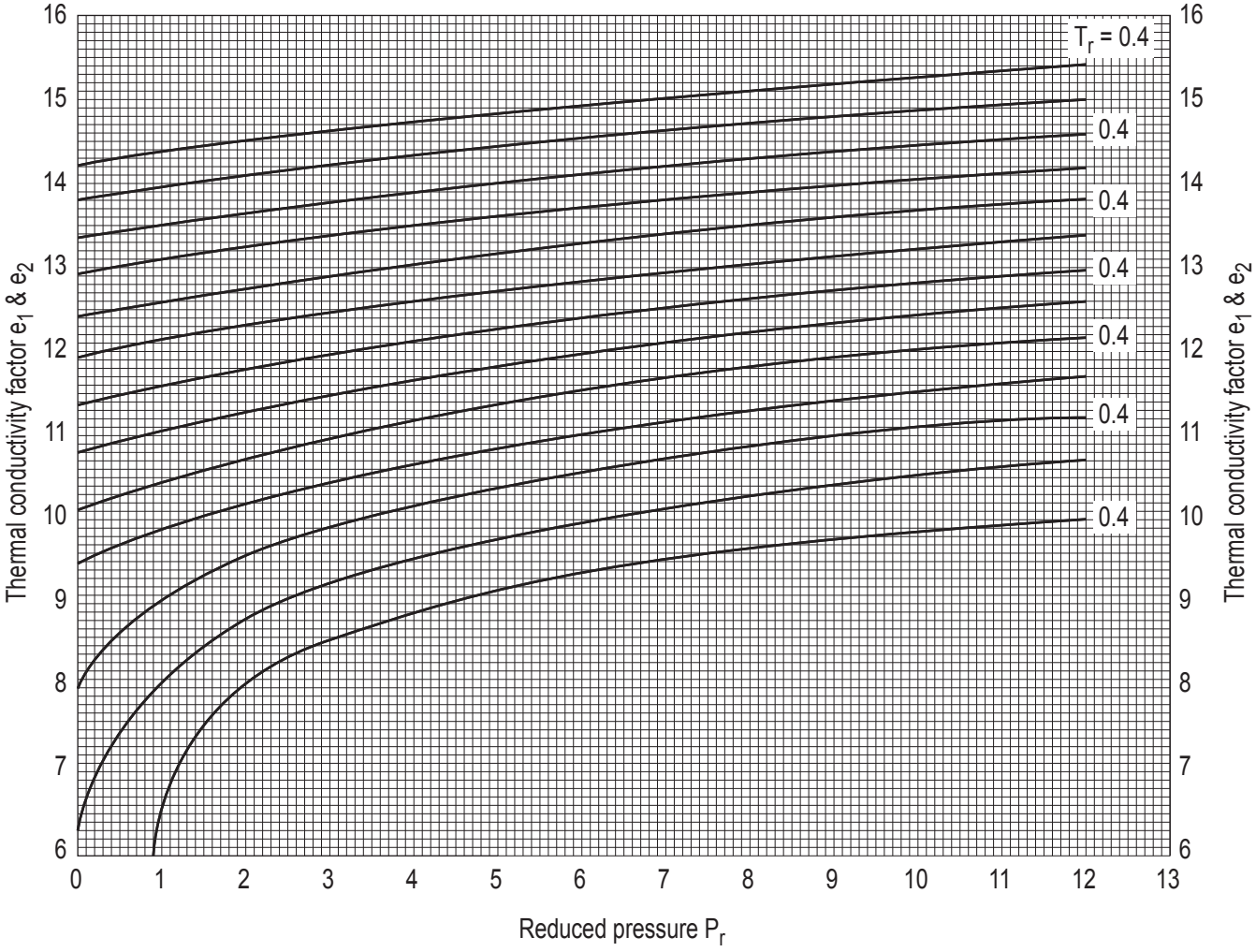
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THERMAL CONDUCTIVITY
— Liquid pressure correction - generalized correlation —

Note : To find thermal conductivity λ_2 at pressure P_2 and temperature T. multiply know value λ_1 by ratio $\left(\frac{e_2}{e_1}\right)$

$$\lambda_2 = \lambda_1 \left(\frac{e_2}{e_1}\right)$$

Where : λ_1 = Know thermal conductivity at any pressure P_1 and temperature T
λ_2 = Desired thermal conductivity at P_2 and T
e_1 = Thermal conductivity factor at $(Pr)_1$ and T_r
e_2 = Thermal conductivity factor at $(Pr)_2$ and T_r
P_1 and P_2 = Pressures. PSIA
P_c = Critical Pressure. PSIA
$(P_r)_1$ = P_1/P_c. Dimensionless
$(P_r)_2$ = P_2/P_c. Dimensionless
T = Temperature. °R (= 460 + °F)
T_c = Critical temperature. °R
T_r = T/T_c. Dimensionless

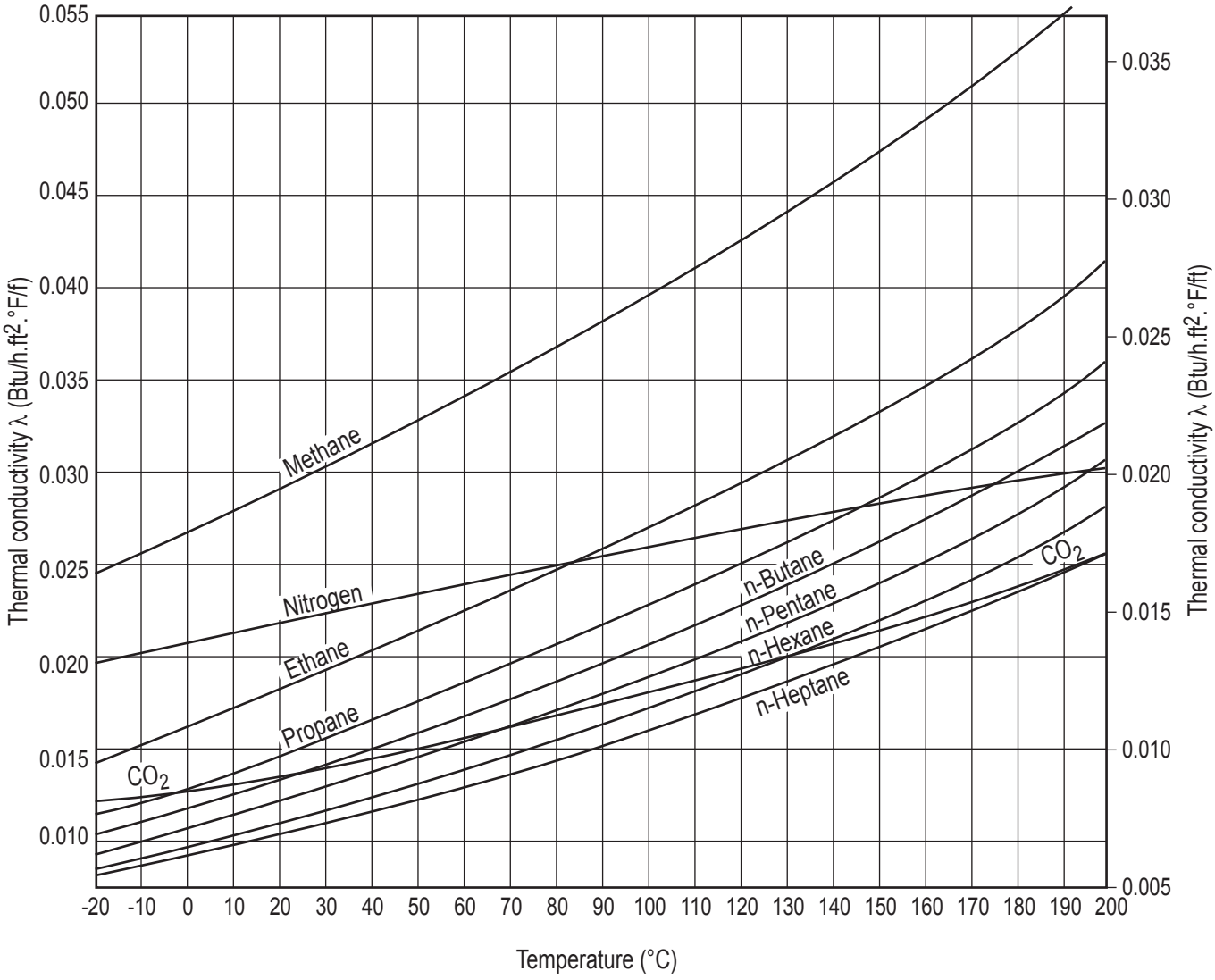


D TH 1626 D

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THERMAL CONDUCTIVITY
— In gazeous phase at 1 atm —

$$1 \quad \frac{\text{Btu}}{\text{hr.ft.°F}} = 1.487 \qquad \frac{\text{kcal}}{\text{h.m.°C}} = 1.729 \qquad \frac{\text{W}}{\text{m.K}}$$



D TH 1627 D

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THERMAL CONDUCTIVITY OF GASES

	Thermal conductivity (kcal/h.m.°C)							
Température (°C)	**-200**	**-100**	**0**	**50**	**100**	**200**	**300**	**400**
Compound								
Acetone			0.0085	0.0113	0.0147	0.0233		
Acetylene		0.0083	0.0161	0.0208	0.0256			
Air	0.0059	0.0135	0.0208		0.0274	0.0333	0.0387	
Ammonia		0.0144*	0.0187		0.0286	0.0416	0.0572	0.0757
Argon		0.0094	0.0141		0.0183	0.0220	0.0254	
Benzene			0.0077	0.0112	0.0153	0.0247		
Butane (n-)			0.0116		0.0201			
Butane (iso-)			0.0119		0.0207			
Carbon dioxide		0.0095*	0.0125		0.0190	0.0263	0.0341	
Carbon disulfide			0.0059					
Carbon monoxide	0.0055	0.0131	0.0199		0.0262			
Carbon tetrachloride				0.0062	0.0077	0.0101		
Chlorine			0.0064					
Chloroform			0.0057		0.0086	0.0120		
Cyclohexane					0.0140			
Dichlorodifluoromethane			0.0071	0.0095	0.0119	0.0171		
Ethane		0.0082	0.0158		0.0260			
Ethyl acetate				0.0110	0.0143	0.0223		
Ethyl alcohol			0.0120		0.0184			
Ethyl chloride			0.0082		0.0141	0.0216		
Ethyl ether			0.0114	0.0150	0.0195	0.0297		
Ethylene		0.0076	0.0150	0.0195	0.0239			
Helium	0.0503	0.0910	0.1216		0.1469			
Heptane (n-)					0.0153	0.0167		
Hexane (n-)			0.0107	0.0119**				
Hexene			0.0091		0.0162			
Hydrogen	0.0436	0.0970	0.1436		0.1844	0.2207	0.2535	
Hydrogen suffide			0.0113					
Mercury						0.0293		
Methane	0.0067	0.0162	0.0262		0.0379	0.0532	0.0729	
Methyl acetate			0.0088	0.0101**				
Methyl alcohol			0.0123		0.0190			
Methyl chloride			0.0079	0.0110	0.0140	0.0208		
Methylene chloride			0.0058	0.0074	0.0094	0.0135		
Neon			0.0039					
Nitric oxide		0.0132	0.0205	0.0239				
Nitrogen	0.0059	0.0135	0.0207		0.0269	0.0327	0.0379	0.0427
Nitrous oxide		0.0070	0.0131		0.0205			
Oxygen	0.0057	0.0135	0.0211	0.0247	0.0280			
Pentane (n-)			0.0110	0.0123**				
Pentane (iso-)			0.0107		0.0189			
Propane			0.0129		0.0225			
Sulfur dioxide			0.0074		0.0103			
Water vapor					0.0202	0.0271	0.0342	0.0415

* : value at -50°C

**:value at 20°C

From TEMA Standards

D MTE 1324 D

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THERMAL CONDUCTIVITY
— Gases pressure correction - generalized correlation —

Note : To find thermal conductivity at any desired temperature and pressure. multiply thermal conductivity at atmospheric pressure (14.7 PSIA) and desired temperature by thermal conductivity ratio $\left(\frac{\lambda}{\lambda^\circ}\right)$ at reduced pressure P_r and reduced temperature T_r

$$\lambda = \lambda^\circ \cdot \left(\frac{\lambda}{\lambda^\circ}\right)_{P_r, T_r}$$

Where : λ = Thermal conductivity at pressure (PSIA) and desired temperature T

λ° = Thermal conductivity at 14.7 PSIA and desired temperature T

$\left(\frac{\lambda}{\lambda^\circ}\right)_{P_r, T_r}$ = Thermal conductivity ratio at reduced pressure P_r and reduced temperature T_r ,dimensionless

T = Desired temperature, °R (= 460 + °F)

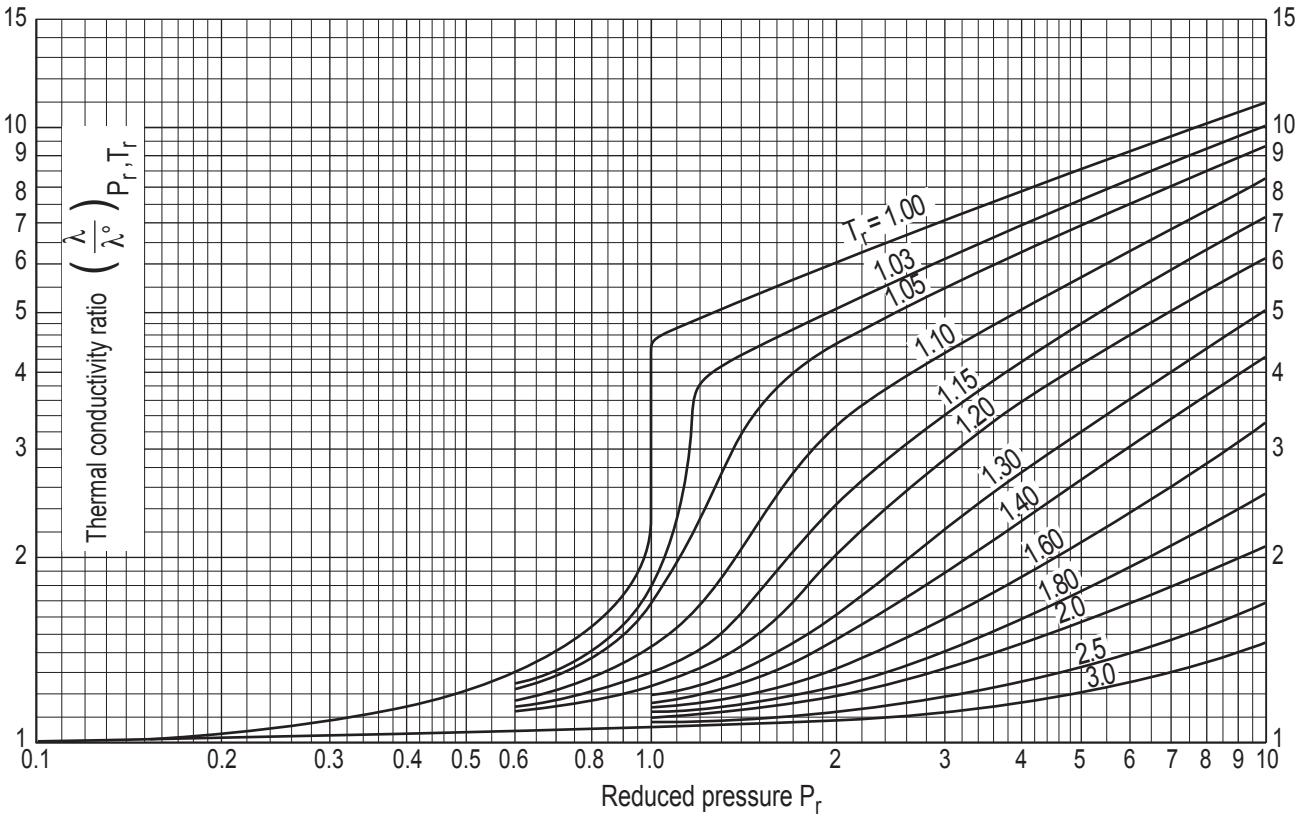
T_e = Critical temperaure, °R

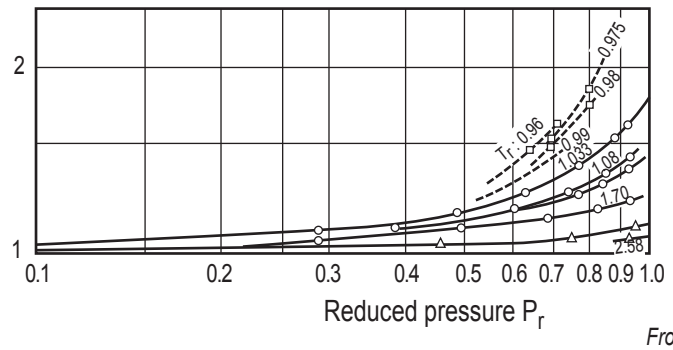
$T_r = \frac{T}{T_e}$, dimensionless

P = Pressure, PSIA

P_e = Critical presure, PSIA

$P_r = \frac{P}{P_e}$, dimensionless





□ Data of Sellschopp

○ Data of Lenoir and Comings

△ Data of Comings. Junk and Lenoir

From LENOIR, JUNK & COMINGS, Chem. Eng. Prog. 49, 539-542 (1935)
(TEMA standards)

D TH 1628 D

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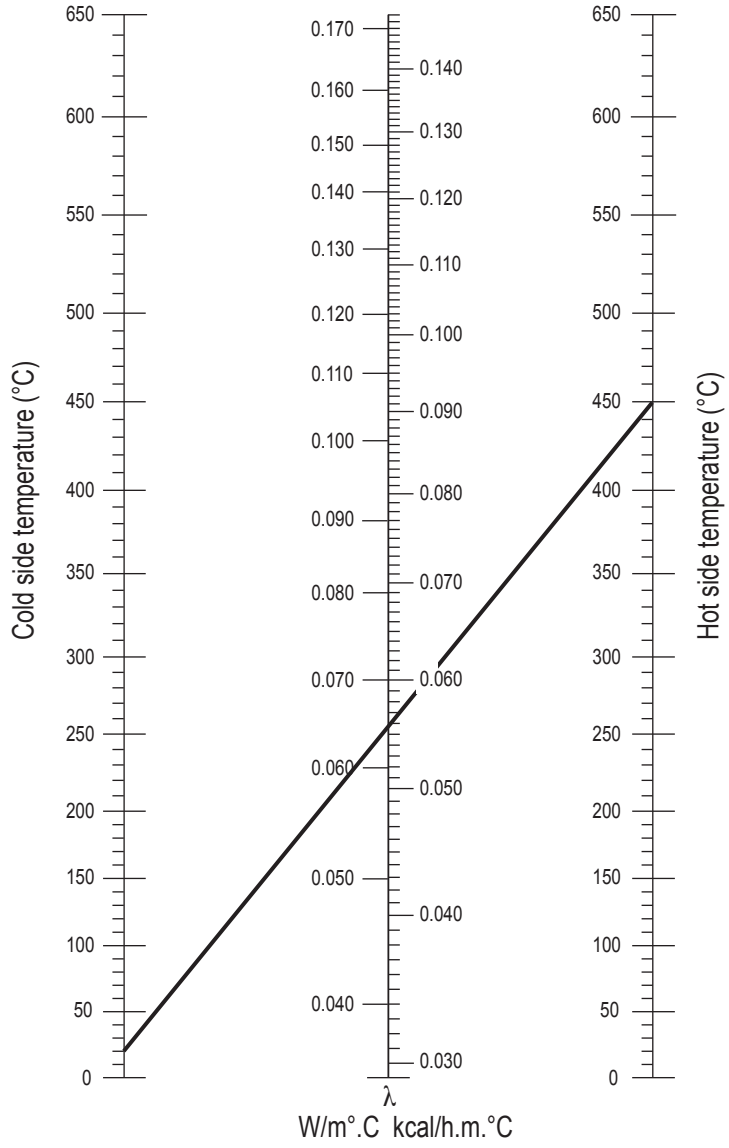
AVERAGE THERMAL CONDUCTIVITY OF A MATERIAL UNDER TEMPERATURE GRADIENT

• **Evaluation by calculation**, evolution of conductibility versus temperature $\lambda = f(t)$ being kown.

$$\lambda_{av} = \frac{\int_{t_1}^{t_2} \lambda(t) \, . \, \lambda t}{t_2 - t_1}$$

• **Use of correlation given by supplier**

Example of chart given for an insulating material by a supplier. This chart allows determination of average conductivity of the material from both side temperatures of the panel.



This chart is valid only for a given material

D MTE 188 D

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THERMAL CONDUCTIBILITY

	SYSTEMS OF UNITS		
	SI	USUAL	BRITISH
Units	W/m K	$\frac{\text{kcal}}{\text{hr} \cdot \text{m} \cdot °\text{C}}$	$\frac{\text{Btu}}{\text{hr} \cdot \text{ft} \cdot °\text{F}}$
W/m K	1	0.8601	0.5784
$\frac{\text{kcal}}{\text{hr} \cdot \text{m} \cdot °\text{C}}$	1.1626	1	0.6725
$\frac{\text{BTu}}{\text{hr} \cdot \text{ft} \cdot °\text{F}}$	1.7288	1.4870	1

— **characterization of hydrocarbon mixtures**

- **Liquid mixtures of identified hydrocarbons**

  littlest value is considered

  $\lambda_m = \Sigma \lambda_i x_i$ ↗ Molar fraction or weight fraction

- **Petroleum cuts** *(see figure G3)*

  HTRI : 7 % deviation (average)
  30 % deviation (maximum)

- **Gas mixtures**

$$\lambda_m = \frac{\sum y_i \lambda_i (M_i)^{1/3}}{\sum y_i (M_i)^{1/3}}$$

M_i Molar mass
y_i Molar fraction

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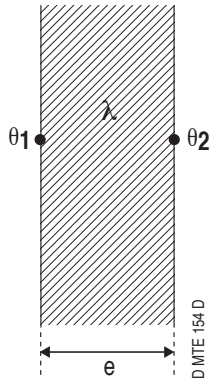
CONDUCTION RESISTANCE

RESISTANCE OF PLANE-PARALLEL WALLS

– Simple wall

If the material is homogeneous and isotropic, integration of Fourier's law gives

$$Q = \lambda A \frac{\theta_1 - \theta_2}{e}$$



or $$\boxed{\frac{Q}{A} = \Phi = \frac{\theta}{\frac{e}{\lambda_m}}}$$

and resistance $R = \frac{e}{\lambda_m}$

If thermal conductivity is temperature dependent, an average value λ_m should be used in the above equations.

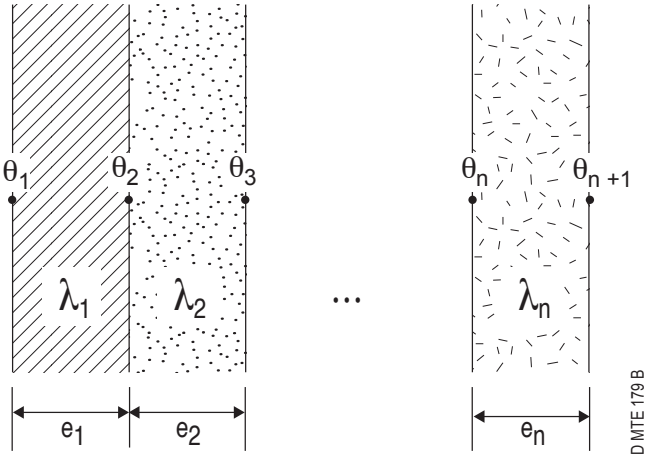
$$\lambda_m = \frac{1}{\theta_1 - \theta_2} \int_{\theta_1}^{\theta_2} \lambda(\theta)\, d\theta$$

Note: The temperature distribution in the wall is linear only if the thermal conductivity is constant.

– Composite walls

The total resistance is the sum of the individual resistances in series.

$$R_{TOTAL} = \Sigma \frac{e_i}{\lambda_i}$$



$$Q = \lambda_1 A \frac{\theta_1 - \theta_2}{e_1} = \lambda_2 A \frac{\theta_2 - \theta_3}{e_2} = \ldots\ldots\ldots = \lambda_n A \frac{\theta_n - \theta_{n+1}}{e_n}$$

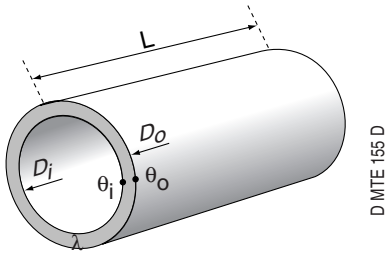
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CONDUCTION RESISTANCE

CYLINDRICAL WALLS

- For a simple wall

$$Q = \frac{2\pi\lambda L\left(\theta_i - \theta_o\right)}{Ln\left(D_o/D_i\right)}$$



D MTE 155 D

In heat loss calculations, the heat flow per unit length is usually used.

$$q = \frac{Q}{L} = \frac{2\pi\lambda\Delta\theta}{Ln\left(D_o/D_i\right)} = \frac{\pi\Delta\theta}{\frac{1}{2\lambda}Ln\left(D_o/D_i\right)}$$

To express the equation in terms of heat flux ($\Phi = D\theta/R$), it is necessary to define a reference heat transfer surface. In furnace and heat exchange calculations, it is conventional to use the external surface area of the tubes: $\pi D_o L$

$$\phi_e = \frac{Q}{\pi D_o L} = \frac{\theta_i - \theta_o}{\frac{D_o}{2\lambda}Ln\left(D_o/D_i\right)}$$

Therefore the resistance associated with the external area of the tubes is given by:

$$R_e = \frac{D_o}{2\lambda}Ln\left(D_o/D_i\right)$$

- **Composite walls**

Heat flow per unit length is constant through the composite wall:

$$q = \frac{Q}{L} = \frac{\pi\ (\theta_1 - \theta_2)}{\frac{1}{2\lambda}\ \ell n\ \frac{D_2}{D_1}} = \frac{\pi\ (\theta_2 - \theta_3)}{\frac{1}{2\lambda}\ \ell n\ \frac{D_3}{D_2}} = \ldots\ldots\ldots\ldots = \frac{\pi\ (\theta_n - \theta_{n+1})}{\frac{1}{2\lambda}\ \ell n\ \frac{D_{n+1}}{D_n}}$$

or:

$$q = \frac{\pi\ \Delta\ \theta}{\Sigma\ \frac{1}{2\ \lambda_i}\ \ell n\ \frac{D_{i+1}}{D_i}}$$

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SPECIFIC GRAVITY OF LIQUID ORGANIC COMPOUNDS

The hereafter described method allows to calculate the density of 700 chemical products and organic liquids, such as hydrocarbons and oxygen, nitrogen, halogen and sulfur compounds.

The density is calculated by means of "modified **Rackett Equation**", which is function of temperature:

$$DEN = A \cdot B^{-(1-TR)^{2/7}}$$

where:

DEN	density of saturated liquid
A and B	correlation constants, specific for each compound
TR	reduced temperature of component (in Kelvin); TR = $\frac{T}{TC}$, where TC: critical temperature of compound

A and B constants are given in the tables below, as well as critical temperature TC. Also figure:

T_{min}	minimum temperature at which results still valid (TC is maximum temperature)
den T, and T (°C)	liquid density at T, according to published values

Example

Let's calculate the density of chloromethane CH_3Cl at T = 100°C.

We find in the tables below: TC = 416.2 K
A = 0.3610
B = 0.268

$$DEN = 0.3610 \cdot 0.268^{-\left(1-\frac{373.16}{416.2}\right)^{\frac{2}{7}}}$$

DEN = 0.719 g/ml

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NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENSQT	T,C
1	CCL4	CARBON-TETRACHLORIDE	0.5566	0.272	250.0	556.5	1.584	25
2	CCL3F	TRICHLOROFLUOROMETHANE	0.5670	0.280	162.0	471.2	1.494	17
3	CCL2F2	DICHLORODIFLUOROMETHANE	0.5731	0.273	115.4	384.8	1.750	-115
4	CCLF3	CHLOROTRIFLUOROMETHANE	0.5816	0.278	92.0	302.0	1.298	-30
5	CF4	CARBON-TETRAFLUORIDE	0.6276	0.276	86.4	227.5	1.330	-80
6	CO	CARBON-MONOXIDE	0.3145	0.294	68.1	133.4	0.803	-192
7	CO2	CARBON-DIOXIDE	0.4664	0.275	216.6	304.2	0.713	25
8	COS	CARBONYL-SULFIDE	0.4369	0.279	134.3	378.0	1.274	-99
9	CS2	CARBON-DISULFIDE	0.4483	0.276	161.3	552.0	1.293	0
10	CHCL3	CHLOROFORM	0.5594	0.293	209.6	535.5	1.489	20
11	CHCL2F	DICHLOROFLUOROMETHANE	0.5143	0.271	138.0	451.6	1.380	9
12	CHCLF2	CHLORODIFLUOROMETHANE	0.5224	0.266	113.0	369.3	1.230	16
13	CHF3	TRIFLUOROMETHANE	0.5304	0.259	110.0	299.1	1.246	-34
14	CHI3	TRIIODOMETHANE	1.1266	0.281	396.2	794.6	3.194*	123
15	CHNS	ISOTHIOCYANIC-ACID	-	-	-	-	-	-
16	CH2CL2	DICHLOROMETHANE	0.4274	0.235	178.1	508.3	1.317	25
17	CH2CLF	CHLOROFLUOROMETHANE	0.4438	0.230	140.2	424.9	1.271	20
18	CH2F2	DIFLUOROMETHANE	0.4296	0.241	101.7	351.5	0.986*	25
19	CH2I2	DIIODOMETHANE	0.9487	0.235	279.3	740.9	3.325	20
20	CH2O	FORMALDEHYDE	0.2395	0.197	156.0	402.7	0.815	-20
21	CH2O2	FORMIC-ACID	0.2606	0.152	281.5	580.0	1.226	15
22	CH3BR	BROMOMETHANE	0.6096	0.262	179.5	464.0	1.737	-5
23	CH3CL	CHLOROMETHANE	0.3610	0.268	175.4	416.2	0.915	20
24	CH3F	FLUOROMETHANE	0.3080	0.251	131.4	317.9	0.843	-60
25	CH3I	IODOMETHANE	0.7125	0.231	206.7	528.0	2.279	20
26	CH3NO2	NITROMETHANE	0.3331	0.224	244.6	588.0	1.138	20
27	CH3NO2	METHYL-NITRITE	-	-	-	-	-	-
28	CH3NO3	METHYL-NITRATE	-	-	-	-	1.208	20
29	CH4	METHANE	0.1616	0.288	90.7	190.7	0.425	-161
30	CH4O	METHANOL	0.2444	0.224	175.5	513.0	0.791	20
31	CH4S	METHANETHIOL	0.3199	0.268	150.0	469.7	0.866	20
32	CH5N	METHYLAMINE	0.2148	0.213	179.7	429.6	0.703	-13
33	C2CL4	TETRACHLOROETHENE	0.5105	0.250	251.0	620.2	1.620	20
34	C2CL6	HEXACHLOROETHANE	0.5643	0.282	458.2	704.4	1.441	185
35	C2CL3F3	112TRICHLOROTRIFLUORDETHANE	0.5805	0.274	238.2	487.3	1.580	16
36	C2CL2F4	1201CHLOROTETRAFLUOROETHANE	0.5738	0.275	179.0	418.9	1.480	4
37	C2CLF5	CHLOROPENTAFLUOROETHANE	0.6041	0.277	167.0	353,2	1.260	30
38	C2F4	TETRAFLUOROETHENE	0.5679	0.267	130.7	306.4	1.519	-76
39	C2F6	HEXAFLUOROETHANE	0.6399	0.288	172.4	292.9	1.590	-78
40	C2N2	CYANOGEN	0.5063	0.431	245.3	399.9	0.954	-21
41	C2HCL3	TRICHLOROETHENE	0.4956	0.265	186.8	572.0	1.462	20
42	C2HCL5	PENTACHLOROETHANE	0.5681	0.276	244.0	646.0	1.671	25
43	C2HF3	TRIFLUOROETHENE	0.4495	0.286	94.5	347.2	0.920*	25
44	C2H2	ACETYLENE(ETHYNE)	0.2306	0.274	192.4	309.2	0.615	-81
45	C2H2CL2	1,1-DICHLOROETHENE	0.4427	0.252	155.9	489.0	1.270*	25
46	C2H2CL2	CIS-1,2-DICHLOROETHENE	0.4254	0.249	192.7	537.0	1.282	25
47	C2H2CL2	TRANS-1,2-DICHLOROETHENE	0.4332	0.258	223.0	513.0	1.255	20
48	C2H2CL4	1,1,2,2-TETRACHLOROETHANE	0.3891	0.188	237.0	661.2	1.600	20
49	C2H2F2	1,1-DIFLUOROETHENE	0.4040	0.273	129.0	303.0	0.617	24
50	C2H2F2	CIS-1,2-DIFLUOROETHENE	0.3917	0.238	107.9	394.7	1.023*	25
51	C2H2F2	TRANS-1,2-DIFLUOROETHENE	0.3917	0.238	107.9	394.7	1.023*	25
52	C2H2O	KETENE	0.2899	0.300	138.0	380.0	0.630*	25
53	C2H3BR	BROMOETHYLENE	0.5575	0.274	133.7	470.1	1.473	25
54	C2H3CL	CHLOROETHENE	0.3470	0.261	119.4	425.0	0.969	-14
55	C2H3CL3	1,1,2-TRICHLOROETHANE	0.4618	0.253	235.8	606.0	1.441	20
56	C2H3CLO	ACETYL-CHLORIDE	0.4079	0.280	160.2	508.0	1.104	20
57	C2H3F	FLUOROETHENE	0.3036	0.277	130.0	327.8	0.681	-10
58	C2H3F3	1,1,1-TRIFLUOROETHANE	0.4336	0.253	161.9	346.2	0.948*	25
59	C2H3N	ACETONITRILE	0.2006	0.184	229.3	547.9	0.782	20
60	C2H4	ETHYLENE	0.2131	0.280	104.0	282.8	0.577	-110
61	C2H4BR2	1,2-DIBROMOETHANE	0.8358	0.320	283.3	646.0	2.180	20
62	C2H4CL2	1,1-DICHLOROETHANE	0.4234	0.275	176.2	523.2	1.168	25
63	C2H4CL2	1,2-DICHLOROETHANE	0.4155	0.259	237.5	566.1	1.250	16
64	C2H4F2	1,1-DIFLUOROETHANE	0.3623	0.253	156.2	386.6	1.012	-26
65	C2H4I2	1,2-DIIODOETHANE	0.8713	0.245	356.2	749.9	2.807*	83
66	C2H4O	ETHYLENE-OXIDE	0.3137	0.259	161.0	468.9	0.899	0
67	C2H4O	ACETALDEHYDE	0.2501	0.220	150.2	461.1	0.778	20
68	C2H4O2	ACETIC-ACID	0.2798	0.201	289.8	594.4	1.049	20
69	C2H4O2	METHYL-FORMATE	0.3408	0.255	174.2	486.7	0.974	20
70	C2H4OS	THIOACETIC-ACID	0.4163	0.317	150.2	577.3	1.064	20
71	C2H4S	THIACYCLOPROPANE	0.2923	0.214	165.4	555.0	1.008	25
72	C2H5BR	BROMOETHANE	0.5997	0.320	154.6	505.7	1.451	25
73	C2H5CL	CHLOROETHANE	0.3398	0.274	136.8	460.4	0.896	20
74	C2H5F	FLUOROETHANE	0.2844	0.272	129.9	375.3	0.652*	25
75	C2H5I	IODOETHANE	0.6334	0.248	165.0	554.0	1.950	20
76	C2H5N	ETHYLENEIMINE	0.3022	0.215	195.2	524.0	1.012*	25
77	C2H5NO2	NITROETHANE	0.3220	0.238	223.2	595.0	1.045	25
78	C2H5NO3	ETHYL-NITRATE	-	-	-	-	1.108	20
79	C2H6	ETHANE	0.2084	0.285	89.9	305.4	0.548	-90
80	C2H6O	METHYL-ETHER	0.2966	0.307	131.7	400.1	0.667	20
81	C2H6O	ETHYL-ALCOHOL	0.2567	0.240	159.1	515.8	0.789	20
82	C2H6O2	ETHYLENE-GLYCOL	0.3074	0.230	260.2	790.0	1.114	20
83	C2H6S	METHYL-SULFIDE	0.3021	0.266	174.9	503.3	0.848	20
84	C2H6S	ETHANETHIOL	0.3071	0.274	125.3	498.6	0.839	20
85	C2H6S2	METHYL-DISULFIDE	0.3572	0.268	188.4	605.0	1.057	25
86	C2H7N	ETHYLAMINE	0.2573	0.270	192.0	456.3	0.683	20
87	C2H7N	DIMETHYLAMINE	0.2430	0.256	181.0	437.8	0.656	20
88	C3H3N	ACRYLONITRILE	0.2321	0.210	189.5	536.0	0.806	20
89	C3H4	ALLENE(PROPADIENE)	0.2440	0.271	136.9	385.4	0.658	-35
90	C3H4	PROPYNE(METHYLACETYLENE)	0.2537	0.275	170.5	401.0	0.706	-50
91	C3H4O2	ACRYLIC-ACID	0.3098	0.230	285.0	615.0	1.051	20
92	C3H5BR	3-BROMO-1-PROPENE	0.5079	0.282	153.8	540.2	1.398	20
93	C3H5CL	3-CHLORO-1-PROPENE	0.3252	0.260	138.7	513.8	0.937	20
94	C3H5CL3	1,2,3-TRICHLOROPROPANE	0.4317	0.250	258.5	651.0	1.389	20
95	C3H5I	3-IODO-1-PROPENE	0.5880	0.249	173.9	595.8	1.849	20
96	C3H5N	PROPIONITRILE	0.2165	0.205	180.3	562.5	0.782	20
97	C3H6	PROPENE	0.2328	0.282	87.9	365.0	0.612	-50
98	C3H6	CYCLOPROPANE	0.2285	0.272	145.7	398.0	0.563	15
99	C3H6BR2	1,2-DIBROMOPROPANE	0.7634	0.330	218.0	634.1	1.932	20
100	C3H6CL2	1,2-DICHLOROPROPANE	0.3215	0.210	172.7	577.0	1.150	20
101	C3H6CL2	1,3-DICHLOROPROPANE	0.3664	0.241	173.7	602.7	1.182	25
102	C3H6CL2	2,2-DICHLOROPROPANE	0.3863	0.266	239.4	539.5	1.114	20
103	C3H6I2	1,2-DIIODOPROPANE	0.7451	0.242	253.2	780.5	2.576	20
104	C3H6O	PROPYLENE-OXIDE	0.2683	0.229	161.0	482.2	0.829	20
105	C3H6O	ALLYL-ALCOHOL	0.2704	0.240	144.0	545.0	0.855	15
106	C3H6O	PROPIONALDEHYDE	0.2763	0.260	193.0	515.3	0.797	20
107	C3H6O	ACETONE	0.2519	0.232	178.2	508.4	0.790	20
108	C3H6S	THIACYCLOBUTANE	0.3004	0.228	200.0	603.0	1.020	20
109	C3H7BR	1-BROMOPROPANE	0.5031	0.289	163.2	535.5	1.354	20
110	C3H7BR	2-BROMOPROPANE	0.4979	0.293	184.2	522.5	1.314	20
111	C3H7CL	1-CHLOROPROPANE	0.3286	0.278	150.4	503.1	0.891	20
112	C3H7CL	2-CHLOROPROPANE	0.3122	0.269	156.0	494.1	0.862	20
113	C3H7F	1-FLUOROPROPANE	0.3071	0.263	114.2	422.0	0.796	20
114	C3H7F	2-FLUOROPROPANE	0.2881	0.262	139.8	415.7	0.733*	25
115	C3H7I	1-IODOPROPANE	0.5690	0.255	172.2	589.4	1.749	20

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NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENSITY	T,C
116	C3H7I	2-IODOPROPANE	0.5659	0.259	183.1	574.6	1.703	20
117	C3H7NO2	1-NITROPROPANE	0.3255	0.254	165.2	606.0	1.008	24
118	C3H7NO2	2-NITROPROPANE	0.3211	0.256	180.2	597.0	0.988	20
119	C3H7NO3	PROPYL-NITRATE	-	-	-	-	1.054	20
120	C3H7NO3	ISOPROPYL-NITRATE	-	-	-	-	1.036	19
121	C3H8	PROPANE	0.2218	0.279	85.5	369.7	0.582	-42
122	C3H8O	ETHYL-METHYL-ETHER	0.2673	0.267	134.0	437.8	0.700	20
123	C3H8O	PROPYL-ALCOHOL	0.2684	0.253	146.9	536.9	0.804	20
124	C3H8O	ISOPROPYL-ALCOHOL	0.2623	0.248	184.7	516.6	0.786	20
125	C3H8S	ETHYL-METHYL-SULFIDE	0.2882	0.262	167.2	532.8	0.837	20
126	C3H8S	1-PROPANETHIOL	0.3030	0.278	160.0	536.0	0.836	25
127	C3H8S	2-PROPANETHIOL	0.3032	0.284	142.6	512.0	0.809	25
128	C3H9N	PROPYLAMINE	0.2607	0.271	190.0	496.1	0.717	20
129	C3H9N	TRIMETHYLAMINE	0.2568	0.288	156.0	433.4	0.633	20
130	C4F8	OCTAFLUOROCYCLOBUTANE	0.6414	0.278	232.2	388.4	1.654	-20
131	C4H2	BUTADIYNE(DIACETYLENE)	0.2640	0.271	237.2	478.0	0.736	0
132	C4H4	1BUTEN3YNE(VINYLACETYLENE)	0.2518	0.260	227.6	455.0	0.710	0
133	C4H4O	FURAN	0.3583	0.286	187.5	487.0	0.938	20
134	C4H4S	THIOPHENE	0.3532	0.259	234.9	580.0	1.071	16
135	C4H6	1,2-BUTADIENE	0.2472	0.267	137.0	443.7	0.652	20
136	C4H6	1,3-BUTADIENE	0.2432	0.270	164.2	425.1	0.621	20
137	C4H6	1-BUTYNE(ETHYLACETYLENE)	0.2414	0.270	147.4	463.7	0.650	16
138	C4H6	2-BUTYNE(DIMETHYLACETYLENE)	0.2572	0.277	240.9	488.7	0.691	20
139	C4H6	CYCLOBUTENE	0.2767	0.278	153.8	446.3	0.704*	25
140	C4H6O3	ACETIC-ANHYDRIDE	0.3657	0.262	199.0	569.1	1.087	20
141	C4H7N	BUTYRONITRILE	0.2318	0.223	161.0	582.2	0.792	20
142	C4H7N	ISOBUTYRONITRILE	0.2312	0.227	201.7	566.0	0.761	30
143	C4H8	1-BUTENE	0.2391	0.277	87.8	419.6	0.595	20
144	C4H8	2-BUTENE,CIS	0.2412	0.271	134.3	433.1	0.621	20
145	C4H8	2-BUTENE,TRANS	0.2379	0.274	167.6	428.6	0.604	20
146	C4H8	2-METHYLPROPENE	0.2381	0.275	132.8	417.9	0.594	20
147	C4H8	CYCLOBUTANE	0.2652	0.278	182.4	463.5	0.694	20
148	C4H8BR2	1,2-DIBROMOBUTANE	0.6927	0.325	207.8	659.3	1.792	20
149	C4H8BR2	2,3-DIBROMOBUTANE	0.6875	0.324	238.7	657.0	1.775	25
150	C4H8I2	1,2-DIIODOBUTANE	0.7277	0.265	279.1	726.4	2.280	25
151	C4H8O	BUTYRALDEHYDE	0.2704	0.258	176.8	545.4	0.802	20
152	C4H8O	2-BUTANONE	0.2682	0.252	186.5	535.6	0.805	20
153	C4H8O2	P-DIOXANE	0.3353	0.254	285.0	588.0	1.033	20
154	C4H8O2	ETHYL-ACETATE	0.3027	0.252	189.6	524.1	0.901	20
155	C4H8S	THIACYCLOPENTANE	0.3039	0.241	177.0	632.0	1.000	20
156	C4H9BR	1-BROMOBUTANE	0.4648	0.289	160.8	569.5	1.276	20
157	C4H9BR	2-BROMOBUTANE	0.4651	0.292	161.3	558.7	1.259	20
158	C4H9BR	2-BROMO-2-METHYLPROPANE	0.4671	0.301	257.0	541.1	1.221	20
159	C4H9CL	1-CHLOROBUTANE	0.2966	0.255	150.1	542.0	0.886	20
160	C4H9CL	2-CHLOROBUTANE	0.3195	0.280	141.8	520.7	0.873	20
161	C4H9CL	1-CHLORO-2-METHYLPROPANE	0.3029	0.260	142.9	526.5	0.881	20
162	C4H9CL	2-CHLORO-2-METHYLPROPANE	0.3112	0.269	247.8	507.8	0.863*	25
163	C4H9I	2-IODO-2-METHYLPROPANE	0.5223	0.267	235.0	587.9	1.545	20
164	C4H9N	PYRROLIDINE	0.3208	0.300	213.2	568.6	0.852	22
165	C4H9NO2	1-NITROBUTANE	0.3163	0.260	191.8	624.0	0.968	25
166	C4H9NO2	2-NITROBUTANE	0.3248	0.264	141.2	615.0	0.985	17
167	C4H10	BUTANE	0.2290	0.274	134.8	425.6	0.579	20
168	C4H10	2-METHYLPROPANE(ISOBUTANE)	0.2314	0.283	113.6	407.7	0.557	20
169	C4H10O	ETHYL-ETHER	0.2597	0.262	156.9	466.8	0.713	20
170	C4H10O	METHYL-PROPYL-ETHER	0.2637	0.263	156.9	476.2	0.723	25
171	C4H10O	METHYL-ISOPROPYL-ETHER	0.2688	0.270	156.9	464.5	0.714	25
172	C4H10O	BUTYL-ALCOHOL	0.2818	0.259	165.2	561.6	0.837	25
173	C4H10O	SEC-BUTYL-ALCOHOL	0.2684	0.252	158.5	538.0	0.807	20
174	C4H10O	TERT-BUTYL-ALCOHOL	0.2695	0.259	298.8	508.0	0.769	26
175	C4H10S	ETHYLSULFIDE	0.2923	0.272	169.2	557.5	0.837	20
176	C4H10S	ISOPROPYL-METHYL-SULFIDE	0.2883	0.269	171.7	551.0	0.825	25
177	C4H10S	METHYL-PROPYL-SULFIDE	0.2879	0.266	160.2	563.0	0.837	25
178	C4H10S	1-BUTANETHIOL	0.2981	0.279	157.5	568.0	0.837	25
179	C4H10S	2-BUTANETHIOL	0.3011	0.284	133.0	551.0	0.825	25
180	C4H10S	2-METHYL-1-PROPANETHIOL	0.2990	0.281	128.3	557.0	0.829	25
181	C4H10S	2-METHYL-2-PROPANETHIOL	0.3049	0.295	274.3	521.0	0.795	25
182	C4H10S2	ETHYL-DISULFIDE	0.3414	0.279	171.7	642.1	0.998	20
183	C4H11N	BUTYLAMINE	0.2676	0.277	224.1	524.0	0.739	20
184	C4H11N	SEC-BUTYLAMINE	0.2691	0.283	188.0	511.0	0.722	22
185	C4H11N	TERT-BUTYLAMINE	0.2500	0.279	201.2	483.9	0.660*	25
186	C4H11N	DIETHYLAMINE	0.2563	0.270	223.4	496.2	0.707	20
187	C5H5N	PYRIDINE	0.3376	0.277	231.5	618.7	0.983	20
188	C5H6S	2-METHYLTHIOPHENE	0.3386	0.265	209.8	610.0	1.014	25
189	C5H6S	3-METHYLTHIOPHENE	0.3366	0.263	204.2	615.0	1.016	25
190	C5H8	1,2-PENTADIENE	0.2491	0.269	135.9	503.0	0.693	20
191	C5H8	1,3-PENTADIENE,CIS	0.2423	0.266	185.7	496.0	0.676	20
192	C5H8	1,3-PENTADIENE,TRANS	0.2398	0.262	185.7	494.3	0.671	25
193	C5H8	1,4-PENTADIENE	0.2388	0.263	124.9	478.0	0.661	20
194	C5H8	2,3-PENTADIENE	0.2507	0.271	147.5	505.8	0.690	25
195	C5H8	3-METHYL-1,2-BUTADIENE	0.2459	0.266	159.5	496.0	0.686	20
196	C5H8	2-METHYL-1,3-BUTADIENE	0.2340	0.248	127.2	483.3	0.681	20
197	C5H8	1-PENTYNE	0.2543	0.275	167.5	493.5	0.690	20
198	C5H8	2-PENTYNE	0.2523	0.270	163.9	522.0	0.706	25
199	C5H8	3-METHYL-1-BUTYNE	0.2585	0.289	183.5	476.0	0.660	25
200	C5H8	CYCLOPENTENE	0.2817	0.275	138.1	506.0	0.772	20
201	C5H8	SPIROPENTANE	0.2880	0.297	166.1	499.7	0.735*	25
202	C5H10	1-PENTENE	0.2443	0.278	107.9	464.7	0.640	20
203	C5H10	2-PENTENE,CIS	0.2459	0.275	121.7	474.9	0.656	20
204	C5H10	2-PENTENE,TRANS	0.2442	0.275	132.9	471.0	0.649	20
205	C5H10	2-METHYL-1-BUTENE	0.2487	0.279	135.6	464.8	0.650	20
206	C5H10	3-METHYL-1-BUTENE	0.2463	0.283	104.7	450.0	0.627	20
207	C5H10	2-METHYL-2-BUTENE	0.2486	0.274	139.3	470.0	0.662	20
208	C5H10	CYCLOPENTANE	0.2714	0.276	179.3	511.8	0.745	20
209	C5H10BR2	2,3-DIBROMO-2-METHYLBUTANE	0.5442	0.324	288.0	668.4	1.410*	25
210	C5H10O	VALERALDEHYDE	0.2733	0.260	182.0	554.0	0.810	20
211	C5H10O	2-PENTANONE	0.2518	0.238	196.0	562.5	0.806	20
212	C5H10S	THIACYCLOHEXANE	0.3076	0.252	292.1	657.1	0.981	25
213	C5H10S	CYCLOPENTANETHIOL	0.3291	0.276	155.4	629.0	0.961*	25
214	C5H11BR	1-BROMOPENTANE	0.4623	0.303	185.3	564.8	1.212	25
215	C5H11CL	1-CHLOROPENTANE	0.2924	0.257	174.2	552.0	0.868*	25
216	C5H11CL	1-CHLORO-3-METHYLBUTANE	0.2919	0.259	168.8	558.9	0.870	20
217	C5H11CL	2-CHLORO-2-METHYLBUTANE	0.2956	0.263	199.7	549.0	0.865	20
218	C5H12	PENTANE	0.2305	0.267	143.4	470.0	0.626	20
219	C5H12	2-METHYLBUTANE(ISOPENTANE)	0.2324	0.270	113.3	461.1	0.620	20
220	C5H12	2,2-DIMETHYPROPANE	0.2324	0.276	256.6	433.8	0.591	20
221	C5H12O	METHYL-TERT-BUTYL-ETHER	0.2675	0.269	164.6	497.1	0.735	25
222	C5H12O	PENTYL-ALCOHOL	0.2708	0.261	195.0	585.9	0.815	20
223	C5H12O	TERT-PENTYL-ALCOHOL	0.2930	0.280	327.0	549.0	0.783	54
224	C5H12S	BUTYL-METHYL-SULFIDE	0.2826	0.265	175.3	591.0	0.838	25
225	C5H12S	ETHYL-PROPYL-SULFIDE	0.2844	0.268	156.2	584.0	0.832	25
226	C5H12S	2-METHYL-2-BUTANETHIOL	0.3005	0.288	169.4	566.0	0.821	25
227	C5H12S	1-PENTANETHIOL	0.2914	0.277	197.5	601.3	0.838	25
228	C6CL6	HEXACHLOROBENZENE	0.5108	0.263	503.2	825.8	1.418*	230
229	C6F6	HEXAFLUOROBENZENE	0.5539	0.255	278.4	516.6	1.613	25
230	C6H4CL2	O-DICHLOROBENZENE	0.3864	0.244	256.1	729.0	1.306	20

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NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENST	T,C
231	C6H4CL2	M-DICHLOROBENZENE	0.4174	0.266	248.5	679.6	1.288	20
232	C6H4CL2	P-DICHLOROBENZENE	0.4067	0.265	326.3	680.9	1.224*	53
233	C6H4F2	M-DIFLUOROBENZENE	0.4007	0.265	249.2	552.9	1.169	20
234	C6H4F2	O-DIFLUOROBENZENE	0.3952	0.264	239.2	554.5	1.160	18
235	C6H4F2	P-DIFLUOROBENZENE	0.4012	0.266	260.2	556.0	1.169	20
236	C6H5BR	BROMOBENZENE	0.4814	0.263	242.3	670.1	1.495	20
237	C6H5CL	CHLOROBENZENE	0.3638	0.265	227.6	632.6	1.106	20
238	C6H5F	FLUOROBENZENE	0.3475	0.263	234.0	559.8	1.024	20
239	C6H5I	IODOBENZENE	0.5837	0.265	241.8	721.1	1.855	4
240	C6H6	BENZENE	0.3041	0.269	278.7	562.0	0.885	16
241	C6H6O	PHENOL	0.4104	0.244	314.0	692.9	1.345	41
242	C6H6S	BENZENETHIOL	0.3648	0.281	258.4	685.0	1.072	25
243	C6H7N	2-PICOLINE	0.3439	0.297	207.0	621.1	0.950	15
244	C6H7N	3-PICOLINE	0.3201	0.272	254.9	644.8	0.961	15
245	C6H7N	ANILINE	0.3119	0.250	267.0	698.9	1.022	20
246	C6H10	1-HEXYNE	0.2602	0.280	141.3	529.0	0.711	25
247	C6H10	CYCLOHEXENE	0.2823	0.271	169.7	560.4	0.816	16
248	C6H10	1-METHYLCYCLOPENTENE	0.2760	0.273	146.0	542.0	0.785	15
249	C6H10	3-METHYLCYCLOPENTENE	0.2679	0.269	130.2	535.7	0.759	25
250	C6H10	4-METHYLCYCLOPENTENE	0.2649	0.265	112.3	543.8	0.763	25
251	C6H10O	CYCLOHEXANONE	0.2950	0.248	242.0	629.1	0.951	15
252	C6H12	1-HEXENE	0.2416	0.270	133.3	508.5	0.673	20
253	C6H12	2-HEXENE,CIS	0.2448	0.270	132.0	518.0	0.687	20
254	C6H12	2-HEXENE,TRANS	0.2420	0.270	140.0	516.0	0.678	20
255	C6H12	3-HEXENE,CIS	0.2425	0.270	135.3	517.0	0.680	20
256	C6H12	3-HEXENE,TRANS	0.2339	0.260	159.7	519.9	0.677	20
257	C6H12	2-METHYL-1-PENTENE	0.2479	0.275	137.4	506.5	0.675	25
258	C6H12	3-METHYL-1-PENTENE	0.2471	0.277	120.2	495.3	0.663	25
259	C6H12	4-METHYL-1-PENTENE	0.2457	0.277	119.5	494.8	0.659	25
260	C6H12	2-METHYL-2-PENTENE	0.2450	0.270	138.1	518.0	0.691	16
261	C6H12	3-METHYL-2-PENTENE,CIS	0.2473	0.270	138.3	518.0	0.694	20
262	C6H12	3-METHYL-2-PENTENE,TRANS	0.2482	0.270	134.7	521.0	0.698	20
263	C6H12	4-METHYL-2-PENTENE,CIS	0.2439	0.270	139.0	490.0	0.669	20
264	C6H12	4-METHYL-2-PENTENE,TRANS	0.2432	0.270	132.0	493.0	0.669	20
265	C6H12	2-ETHYL-1-BUTENE	0.2493	0.273	141.6	510.4	0.685	25
266	C6H12	2,3-DIMETHYL-1-BUTENE	0.2448	0.270	115.9	501.0	0.678	20
267	C6H12	3,3-DIMETHYL-1-BUTENE	0.2380	0.270	158.0	490.0	0.653	20
268	C6H12	2,3-DIMETHYL-2-BUTENE	0.2512	0.270	198.9	524.0	0.708	20
269	C6H12	CYCLOHEXANE	0.2735	0.273	279.6	553.5	0.779	20
270	C6H12	METHYLCYCLOPENTANE	0.2669	0.273	130.7	533.2	0.754	16
271	C6H12O	CYCLOHEXANOL	0.3276	0.279	298.0	625.1	0.942	30
272	C6H12O	HEXANAL	0.2669	0.256	217.2	581.1	0.814	20
273	C6H12S	THIACYCLOHEPTANE	0.2969	0.322	292.1	640.1	0.766*	25
274	C6H14	HEXANE	0.2325	0.264	177.8	507.9	0.659	20
275	C6H14	2-METHYLPENTANE	0.2343	0.267	119.5	498.5	0.653	20
276	C6H14	3-METHYLPENTANE	0.2412	0.273	155.0	504.4	0.664	20
277	C6H14	2,2-DIMETHYLBUTANE	0.2383	0.272	173.3	488.2	0.649	20
278	C6H14	2,3-DIMETHYLBUTANE	0.2385	0.269	144.6	500.3	0.662	20
279	C6H14O	PROPYL-ETHER	0.2599	0.270	151.0	530.6	0.736	20
280	C6H14O	ISOPROPYL-ETHER	0.2556	0.262	187.7	500.2	0.724	20
281	C6H14O	HEXYL-ALCOHOL	0.2697	0.262	229.2	610.0	0.819	20
282	C6H14S	BUTYL-ETHYL-SULFIDE	0.2785	0.265	178.0	609.0	0.833	25
283	C6H14S	ISOPROPYL-SULFIDE	0.2859	0.274	170.5	585.7	0.823	25
284	C6H14S	METHYL-PENTYL-SULFIDE	0.2930	0.276	179.2	588.0	0.839	25
285	C6H14S	PROPYL-SULFIDE	0.2792	0.266	170.5	609.7	0.833	25
286	C6H14S	1-HEXANETHIOL	0.2853	0.273	192.7	622.0	0.838	25
287	C6H14S2	PROPYL-DISULFIDE	0.3223	0.277	187.7	673.0	0.955	25
288	C6H15N	TRIETHYLAMINE	0.2556	0.269	158.4	535.2	0.728	20
289	C7H5F3	A,A,A-TRIFLUOROTOLUENE	0.4187	0.276	244.1	562.6	1.188	20
290	C7H5N	BENZONITRILE	0.2974	0.241	260.0	699.4	1.010	15
291	C7H6O2	BENZOIC-ACID	0.3531	0.250	395.6	752.0	1.075	130
292	C7H7F	P-FLUOROTOLUENE	0.3313	0.262	216.4	590.5	1.001	16
293	C7H8	TOLUENE	0.2906	0.265	178.0	593.1	0.867	20
294	C7H8	1,3,5-CYCLOHEPTATRIENE	0.2980	0.266	193.7	593.9	0.888	19
295	C7H8O	M-CRESOL	0.3040	0.240	285.4	705.4	1.034	20
296	C7H8O	O-CRESOL	0.3131	0.244	304.1	695.3	1.028	40
297	C7H8O	P-CRESOL	0.3092	0.244	307.9	704.5	1.019	40
298	C7H12	1-HEPTYNE	0.2584	0.276	192.3	559.7	0.728	25
299	C7H14	1-HEPTENE	0.2523	0.280	154.3	537.2	0.697	20
300	C7H14	CYCLOHEPTANE	0.2736	0.269	265.0	602.0	0.810	20
301	C7H14	ETHYLCYCLOPENTANE	0.2638	0.269	134.7	569.5	0.771	16
302	C7H14	1,1-DIMETHYLCYCLOPENTANE	0.2634	0.270	203.4	550.0	0.759	16
303	C7H14	C-1,2-DIMETHYLCYCLOPENTANE	0.2673	0.270	219.3	565.0	0.777	16
304	C7H14	T-1,2-DIMETHYLCYCLOPENTANE	0.2619	0.270	155.6	553.0	0.756	16
305	C7H14	C-1,3-DIMETHYLCYCLOPENTANE	0.2595	0.270	139.5	551.0	0.740	25
306	C7H14	T-1,3-DIMETHYLCYCLOPENTANE	0.2606	0.270	139.2	553.0	0.744	25
307	C7H14	METHYLCYCLOHEXANE	0.2596	0.263	146.6	572.3	0.774	16
308	C7H14O	HEPTANAL	0.2655	0.250	229.9	638.5	0.850	20
309	C7H16	HEPTANE	0.2343	0.262	182.6	540.3	0.684	20
310	C7H16	2-METHYLHEXANE	0.2334	0.261	154.9	531.0	0.679	20
311	C7H16	3-METHYLHEXANE	0.2311	0.255	100.0	535.6	0.687	20
312	C7H16	3-ETHYLPENTANE	0.2434	0.268	154.6	540.7	0.698	20
313	C7H16	2,2-DIMETHYLPENTANE	0.2370	0.266	149.4	520.4	0.674	20
314	C7H16	2,3-DIMETHYLPENTANE	0.2341	0.256	149.4	537.6	0.695	20
315	C7H16	2,4-DIMETHYLPENTANE	0.2351	0.264	154.0	520.8	0.673	20
316	C7H16	3,3-DIMETHYLPENTANE	0.2460	0.273	138.7	536.3	0.693	20
317	C7H16	2,2,3-TRIMETHYLBUTANE	0.2409	0.266	248.3	530.4	0.690	20
318	C7H16O	ISOPROPYL-TERT-BUTYL-ETHER	0.2548	0.261	177.8	558.2	0.755	20
319	C7H16O	HEPTYL-ALCOHOL	0.2679	0.262	239.2	631.9	0.822	20
320	C7H16S	BUTYL-PROPYL-SULFIDE	0.2697	0.259	206.7	653.5	0.839	25
321	C7H16S	ETHYL-PENTYL-SULFIDE	0.2715	0.259	206.7	638.4	0.839	25
322	C7H16S	HEXYL-METHYL-SULFIDE	0.2715	0.259	206.7	638.4	0.839	25
323	C7H16S	1-HEPTANETHIOL	0.2775	0.267	229.9	645.0	0.839	25
324	C8H6	ETHYNYLBENZENE	0.3025	0.273	242.5	655.4	0.901	25
325	C8H8	STYRENE	0.2992	0.264	242.5	617.1	0.906	20
326	C8H8	1,3,5,7-CYCLOOCTATETRAENE	0.3015	0.268	266.2	642.6	0.908*	25
327	C8H10	ETHYLBENZENE	0.2898	0.268	178.2	617.9	0.867	20
328	C8H10	M-XYLENE	0.2806	0.259	225.3	619.0	0.864	20
329	C8H10	O-XYLENE	0.2869	0.262	248.0	631.1	0.880	20
330	C8H10	P-XYLENE	0.2807	0.260	286.4	617.4	0.861	20
331	C8H14	1-OCTYNE	0.2547	0.270	193.9	586.9	0.742	25
332	C8H16	1-OCTENE	0.2535	0.280	171.4	571.7	0.715	20
333	C8H16	CYCLOOCTANE	0.2779	0.271	287.6	647.2	0.834	20
334	C8H16	PROPYLCYCLOPENTANE	0.2472	0.250	155.8	603.0	0.781	16
335	C8H16	ETHYLCYCLOHEXANE	0.2661	0.270	161.8	609.0	0.788	20
336	C8H16	1,1-DIMETHYLCYCLOHEXANE	0.2664	0.270	239.7	591.0	0.785	16
337	C8H16	C-1,2-DIMETHYLCYCLOHEXANE	0.2692	0.270	223.1	606.0	0.796	20
338	C8H16	T-1,2-DIMETHYLCYCLOHEXANE	0.2637	0.270	185.0	596.0	0.776	20
339	C8H16	C-1,3-DIMETHYLCYCLOHEXANE	0.2610	0.270	197.6	591.0	0.766	20
340	C8H16	T-1,3-DIMETHYLCYCLOHEXANE	0.2534	0.254	183.0	598.0	0.785	20
341	C8H16	C-1,4-DIMETHYLCYCLOHEXANE	0.2658	0.270	185.7	598.0	0.783	20
342	C8H16	T-1,4-DIMETHYLCYCLOHEXANE	0.2604	0.270	236.2	587.7	0.763	20
343	C8H16O	OCTANAL	0.2635	0.256	245.9	621.1	0.816	25
344	C8H18	OCTANE	0.2343	0.259	216.4	569.1	0.703	20
345	C8H18	2-METHYLHEPTANE	0.2355	0.261	164.0	560.6	0.702	16
346	C8H18	3-METHYLHEPTANE	0.2308	0.252	152.7	564.1	0.706	20
347	C8H18	4-METHYLHEPTANE	0.2359	0.259	152.2	562.2	0.705	20

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NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENSAT	T,C
348	C8H18	3-ETHYLHEXANE	0.2334	0.252	152.2	566.1	0.718	16
349	C8H18	2,2-DIMETHYLHEXANE	0.2380	0.264	152.0	549.9	0.695	20
350	C8H18	2,3-DIMETHYLHEXANE	0.2409	0.263	172.0	564.5	0.712	20
351	C8H18	2,4-DIMETHYLHEXANE	0.2376	0.262	172.0	554.1	0.700	20
352	C8H18	2,5-DIMETHYLHEXANE	0.2359	0.262	181.9	549.9	0.693	20
353	C8H18	3,3-DIMETHYLHEXANE	0.2323	0.252	147.0	562.8	0.710	20
354	C8H18	3,4-DIMETHYLHEXANE	0.2433	0.264	147.1	569.7	0.719	20
355	C8H18	3-ETHYL-2-METHYLPENTANE	0.2348	0.254	158.2	567.0	0.718	16
356	C8H18	3-ETHYL-3-METHYLPENTANE	0.2466	0.266	182.3	576.5	0.727	20
357	C8H18	2,2,3-TRIMETHYLPENTANE	0.2357	0.254	160.9	563.5	0.716	20
358	C8H18	2,2,4-TRIMETHYLPENTANE	0.2394	0.266	165.8	543.8	0.692	20
359	C8H18	2,3,3-TRIMETHYLPENTANE	0.2489	0.269	172.5	573.5	0.726	20
360	C8H18	2,3,4-TRIMETHYLPENTANE	0.2445	0.265	163.9	566.4	0.719	20
361	C8H18	2,2,3,3-TETRAMETHYLBUTANE	0.2474	0.280	172.5	567.8	0.692*	25
362	C8H18O	BUTYL-ETHER	0.2544	0.259	175.0	580.0	0.768	20
363	C8H18O	SEC-BUTYL-ETHER	0.2601	0.266	171.7	562.0	0.756	25
364	C8H18O	TERT-BUTYL-ETHER	0.2716	0.268	206.5	550.0	0.779*	25
365	C8H18O	OCTYL-ALCOHOL	0.2635	0.258	257.7	652.5	0.826	20
366	C8H18S	BUTYL-SULFIDE	0.2711	0.260	209.9	650.0	0.840	25
367	C8H18S	ETHYL-HEXYL-SULFIDE	0.2629	0.252	209.9	660.7	0.840	25
368	C8H18S	HEPTYL-METHYL-SULFIDE	0.2629	0.252	209.9	660.7	0.840	25
369	C8H18S	PENTYL-PROPYL-SULFIDE	0.2629	0.252	209.9	660.7	0.840	25
370	C8H18S	1-OCTANETHIOL	0.2711	0.262	223.9	665.0	0.839	25
371	C8H18S2	BUTYL-DISULFIDE	0.3003	0.265	202.2	704.2	0.934	25
372	C9H10	ALPHA-METHYLSTYRENE	0.2916	0.260	249.9	654.0	0.905	25
373	C9H10	PROPENYLBENZENE,CIS	0.2884	0.258	211.5	664.6	0.905	25
374	C9H10	PROPENYLBENZENE,TRANS	0.2877	0.258	243.8	664.6	0.902	25
375	C9H10	M-METHYLSTYRENE	0.2864	0.254	186.8	658.2	0.908	25
376	C9H10	O-METHYLSTYRENE	0.2839	0.252	204.6	662.7	0.908	25
377	C9H10	P-METHYLSTYRENE	0.2883	0.253	239.0	659.7	0.917	25
378	C9H12	PROPYLBENZENE	0.2828	0.265	173.7	638.6	0.862	20
379	C9H12	CUMENE	0.2819	0.263	177.1	631.7	0.862	20
380	C9H12	M-ETHYLTOLUENE	0.2795	0.260	177.6	637.0	0.865	20
381	C9H12	O-ETHYLTOLUENE	0.2830	0.260	192.3	651.0	0.881	20
382	C9H12	P-ETHYLTOLUENE	0.2779	0.260	210.8	640.0	0.861	20
383	C9H12	1,2,3-TRIMETHYLBENZENE	0.2857	0.260	247.7	664.5	0.894	20
384	C9H12	1,2,4-TRIMETHYLBENZENE	0.2782	0.256	227.0	649.2	0.880	16
385	C9H12	MESITYLENE	0.2810	0.262	228.4	640.1	0.865	20
386	C9H16	1-NONYNE	0.2505	0.264	223.2	610.8	0.753	25
387	C9H18	1-NONENE	0.2564	0.280	191.8	592.0	0.745	0
388	C9H18	BUTYLCYCLOPENTANE	0.2516	0.256	165.2	625.1	0.781	25
389	C9H18	PROPYLCYCLOHEXANE	0.2527	0.256	178.7	639.0	0.793	20
390	C9H18	C-C-1351TRIMETHYLCYCLOHEXANE	0.2419	0.247	223.5	607.9	0.771	20
391	C9H18	C-T-1351TRIMETHYLCYCLOHEXANE	0.2258	0.245	188.8	602.2	0.722	20
392	C9H18O	NONANAL	0.2669	0.262	267.3	640.0	0.827	13
393	C9H20	NONANE	0.2314	0.253	219.7	594.8	0.718	20
394	C9H20	2-METHYLOCTANE	0.2315	0.254	192.8	587.0	0.713	20
395	C9H20	3-METHYLOCTANE	0.2322	0.252	165.6	590.0	0.717	25
396	C9H20	4-METHYLOCTANE	0.2339	0.254	160.0	586.7	0.716	25
397	C9H20	3-ETHYLHEPTANE	0.2358	0.254	160.0	587.5	0.723	25
398	C9H20	4-ETHYLHEPTANE	0.2375	0.255	160.0	585.0	0.724	25
399	C9H20	2,2-DIMETHYLHEPTANE	0.2344	0.257	160.0	576.8	0.711	20
400	C9H20	2,3-DIMETHYLHEPTANE	0.2339	0.252	160.2	589.6	0.722	25
401	C9H20	2,4-DIMETHYLHEPTANE	0.2322	0.252	160.2	576.8	0.712	25
402	C9H20	2,5-DIMETHYLHEPTANE	0.2335	0.254	160.2	581.1	0.713	25
403	C9H20	2,6-DIMETHYLHEPTANE	0.2329	0.256	170.3	577.9	0.705	25
404	C9H20	3,3-DIMETHYLHEPTANE	0.2339	0.252	170.3	588.4	0.722	25
405	C9H20	3,4-DIMETHYLHEPTANE	0.2354	0.252	170.3	591.9	0.728	25
406	C9H20	3,5-DIMETHYLHEPTANE	0.2344	0.253	170.3	583.2	0.719	25
407	C9H20	4,4-DIMETHYLHEPTANE	0.2328	0.250	170.3	585.4	0.722	25
408	C9H20	3-ETHYL-2-METHYLHEXANE	0.2342	0.249	160.2	588.1	0.730	25
409	C9H20	4-ETHYL-2-METHYLHEXANE	0.2336	0.251	160.2	580.0	0.720	25
410	C9H20	3-ETHYL-3-METHYLHEXANE	0.2362	0.250	160.2	597.5	0.737	25
411	C9H20	3-ETHYL-4-METHYLHEXANE	0.2357	0.249	160.2	593.7	0.736	25
412	C9H20	2,2,3-TRIMETHYLHEXANE	0.2366	0.254	153.2	588.0	0.729	20
413	C9H20	2,2,4-TRIMETHYLHEXANE	0.2338	0.253	153.0	573.5	0.720	16
414	C9H20	2,2,5-TRIMETHYLHEXANE	0.2358	0.256	167.4	568.0	0.717	16
415	C9H20	2,3,3-TRIMETHYLHEXANE	0.2379	0.253	156.4	596.0	0.735	25
416	C9H20	2,3,4-TRIMETHYLHEXANE	0.2376	0.252	156.4	594.5	0.735	25
417	C9H20	2,3,5-TRIMETHYLHEXANE	0.2355	0.254	145.4	579.2	0.718	25
418	C9H20	2,4,4-TRIMETHYLHEXANE	0.2336	0.251	159.8	581.5	0.720	25
419	C9H20	3,3,4-TRIMETHYLHEXANE	0.2401	0.254	172.0	602.3	0.741	25
420	C9H20	3,3-DIETHYLPENTANE	0.2374	0.249	240.1	610.0	0.752	20
421	C9H20	3-ETHYL-2,2-DIMETHYLPENTANE	0.2375	0.253	173.7	590.4	0.731	25
422	C9H20	3-ETHYL-2,3-DIMETHYLPENTANE	0.2426	0.254	173.7	606.8	0.751	25
423	C9H20	3-ETHYL-2,4-DIMETHYLPENTANE	0.2368	0.251	150.8	591.2	0.734	25
424	C9H20	2,2,3,3-TETRAMETHYLPENTANE	0.2368	0.246	263.0	607.5	0.757	20
425	C9H20	2,2,3,4-TETRAMETHYLPENTANE	0.2369	0.251	152.0	592.6	0.739	20
426	C9H20	2,2,4,4-TETRAMETHYLPENTANE	0.2374	0.257	206.0	574.5	0.719	20
427	C9H20	2,3,3,4-TETRAMETHYLPENTANE	0.2362	0.246	171.1	607.5	0.755	20
428	C9H20O	NONYL-ALCOHOL	0.2631	0.259	268.0	671.0	0.828	20
429	C9H20S	BUTYL-PENTYL-SULFIDE	0.2556	0.246	231.2	681.6	0.840	25
430	C9H20S	ETHYL-HEPTYL-SULFIDE	0.2556	0.246	231.2	681.6	0.840	25
431	C9H20S	HEXYL-PROPYL-SULFIDE	0.2556	0.246	231.2	681.6	0.840	25
432	C9H20S	METHYL-OCTYL-SULFIDE	0.2556	0.246	231.2	681.6	0.840	25
433	C9H20S	1-NONANETHIOL	0.2629	0.255	253.0	686.4	0.840	25
434	C10H8	NAPHTHALENE	0.3138	0.268	353.5	747.8	0.940*	80
435	C10H8	AZULENE	0.3130	0.248	173.7	773.5	1.053*	25
436	C10H14	BUTYLBENZENE	0.2761	0.261	185.2	660.7	0.860	20
437	C10H14	M-DIETHYLBENZENE	0.2734	0.257	189.2	663.6	0.860	25
438	C10H14	O-DIETHYLBENZENE	0.2769	0.256	241.9	669.6	0.876	25
439	C10H14	P-DIETHYLBENZENE	0.2672	0.250	231.0	657.9	0.862	20
440	C10H14	1,2,3,4-TETRAMETHYLBENZENE	0.2643	0.237	266.9	695.1	0.902	25
441	C10H14	1,2,3,5-TETRAMETHYLBENZENE	0.2652	0.241	249.0	679.0	0.890	20
442	C10H14	1,2,4,5-TETRAMETHYLBENZENE	0.2651	0.241	352.0	675.6	0.838	81
443	C10H18	1-DECYNE	0.2454	0.257	229.2	632.5	0.762	25
444	C10H18	DECAHYDRONAPHTHALENE,CIS	0.2734	0.250	230.0	702.2	0.897	20
445	C10H18	DECAHYDRONAPHTHALENE,TRANS	0.2703	0.254	242.8	687.0	0.870	20
446	C10H20	1-DECENE	0.2158	0.280	206.9	615.0	0.619*	25
447	C10H20	1-CYCLOPENTYLPENTANE	0.2463	0.250	190.2	647.5	0.787	25
448	C10H20	BUTYLCYCLOHEXANE	0.2498	0.255	198.4	667.0	0.796	20
449	C10H20O	DECANAL	0.2610	0.255	268.2	652.0	0.830	15
450	C10H22	DECANE	0.2282	0.247	243.5	616.1	0.730	20
451	C10H22	2-METHYLNONANE	0.2269	0.246	198.5	610.3	0.723	25
452	C10H22	3-METHYLNONANE	0.2273	0.244	188.4	613.4	0.730	25
453	C10H22	4-METHYLNONANE	0.2256	0.242	174.5	610.5	0.728	25
454	C10H22	5-METHYLNONANE	0.2259	0.242	185.5	609.6	0.729	25
455	C10H22	3-ETHYLOCTANE	0.2269	0.241	185.5	613.6	0.736	25
456	C10H22	4-ETHYLOCTANE	0.2237	0.237	185.5	609.6	0.734	25
457	C10H22	2,2-DIMETHYLOCTANE	0.2281	0.247	219.2	602.0	0.721	25
458	C10H22	2,3-DIMETHYLOCTANE	0.2279	0.243	219.2	613.2	0.734	25
459	C10H22	2,4-DIMETHYLOCTANE	0.2260	0.243	219.2	599.4	0.723	25
460	C10H22	2,5-DIMETHYLOCTANE	0.2275	0.244	219.2	603.0	0.726	25
461	C10H22	2,6-DIMETHYLOCTANE	0.2289	0.247	219.2	603.1	0.724	25
462	C10H22	2,7-DIMETHYLOCTANE	0.2278	0.247	219.2	602.9	0.720	25
463	C10H22	3,3-DIMETHYLOCTANE	0.2283	0.243	219.2	612.1	0.735	25
464	C10H22	3,4-DIMETHYLOCTANE	0.2292	0.242	219.2	614.0	0.741	25

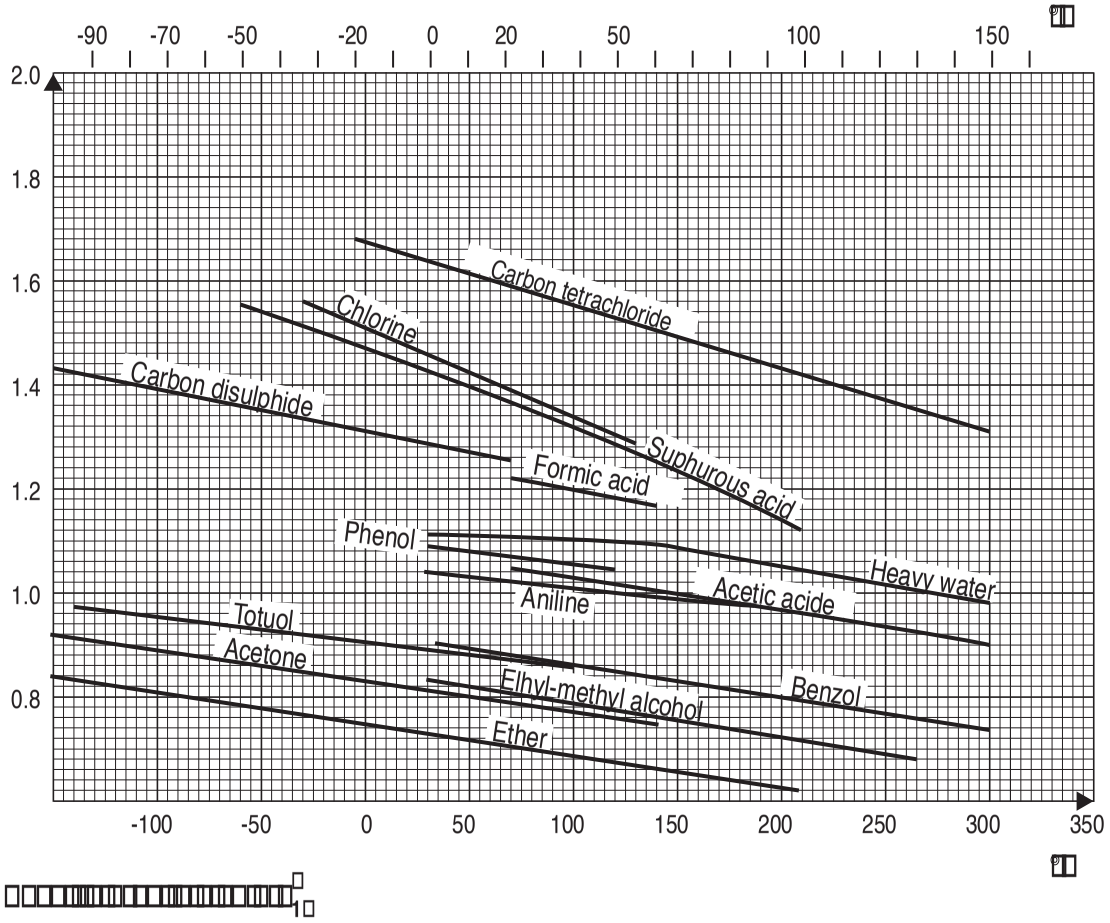
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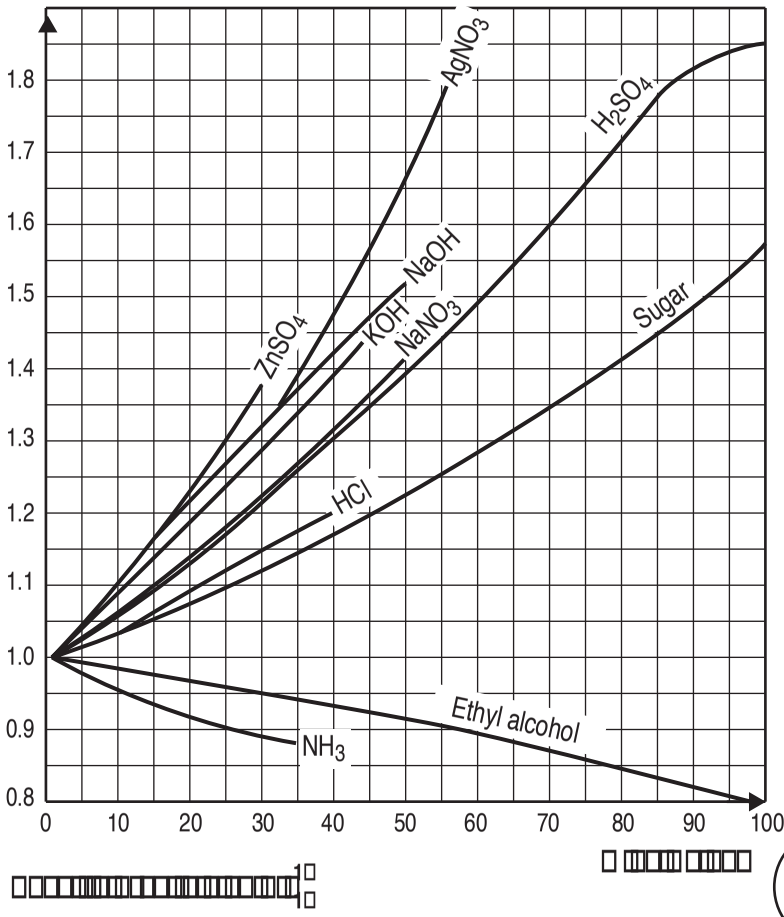
NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENSITY	T,C
465	C10H22	3,5-DIMETHYLOCTANE	0.2268	0.241	219.2	606.3	0.733	25
466	C10H22	3,6-DIMETHYLOCTANE	0.2280	0.243	219.2	608.3	0.732	25
467	C10H22	4,4-DIMETHYLOCTANE	0.2254	0.240	219.2	606.9	0.731	25
468	C10H22	4,5-DIMETHYLOCTANE	0.2285	0.240	219.2	612.2	0.743	25
469	C10H22	4-PROPYLHEPTANE	0.2249	0.238	219.2	601.0	0.732	25
470	C10H22	4-ISOPROPYLHEPTANE	0.2243	0.237	219.2	607.6	0.735	25
471	C10H22	3-ETHYL-2-METHYLHEPTANE	0.2268	0.239	219.2	610.9	0.740	25
472	C10H22	4-ETHYL-2-METHYLHEPTANE	0.2248	0.238	219.2	601.8	0.732	25
473	C10H22	5-ETHYL-2-METHYLHEPTANE	0.2264	0.241	219.2	606.7	0.732	25
474	C10H22	3-ETHYL-3-METHYLHEPTANE	0.2269	0.238	219.2	620.0	0.746	25
475	C10H22	4-ETHYL-3-METHYLHEPTANE	0.2269	0.237	219.2	614.3	0.747	25
476	C10H22	3-ETHYL-5-METHYLHEPTANE	0.2264	0.239	219.2	606.6	0.737	25
477	C10H22	3-ETHYL-4-METHYLHEPTANE	0.2268	0.237	219.2	615.5	0.747	25
478	C10H22	4-ETHYL-4-METHYLHEPTANE	0.2269	0.237	219.2	615.7	0.747	25
479	C10H22	2,2,3-TRIMETHYLHEPTANE	0.2302	0.244	219.2	611.7	0.739	25
480	C10H22	2,2,4-TRIMETHYLHEPTANE	0.2262	0.242	219.2	594.5	0.724	25
481	C10H22	2,2,5-TRIMETHYLHEPTANE	0.2275	0.244	219.2	598.0	0.724	25
482	C10H22	2,2,6-TRIMETHYLHEPTANE	0.2290	0.247	219.2	593.4	0.720	25
483	C10H22	2,3,3-TRIMETHYLHEPTANE	0.2308	0.243	219.2	617.5	0.745	25
484	C10H22	2,3,4-TRIMETHYLHEPTANE	0.2296	0.241	219.2	613.7	0.745	25
485	C10H22	2,3,5-TRIMETHYLHEPTANE	0.2278	0.240	219.2	612.8	0.741	25
486	C10H22	2,3,6-TRIMETHYLHEPTANE	0.2288	0.244	219.2	604.1	0.731	25
487	C10H22	2,4,4-TRIMETHYLHEPTANE	0.2269	0.241	219.2	600.3	0.731	25
488	C10H22	2,4,5-TRIMETHYLHEPTANE	0.2281	0.241	219.2	606.9	0.737	25
489	C10H22	2,4,6-TRIMETHYLHEPTANE	0.2275	0.245	219.2	590.3	0.719	25
490	C10H22	2,5,5-TRIMETHYLHEPTANE	0.2298	0.243	219.2	602.9	0.736	25
491	C10H22	3,3,4-TRIMETHYLHEPTANE	0.2310	0.241	219.2	622.1	0.753	25
492	C10H22	3,3,5-TRIMETHYLHEPTANE	0.2338	0.248	219.2	609.5	0.739	25
493	C10H22	3,4,4-TRIMETHYLHEPTANE	0.2314	0.241	219.2	620.9	0.754	25
494	C10H22	3,4,5-TRIMETHYLHEPTANE	0.2311	0.240	219.2	612.8	0.752	25
495	C10H22	3-ISOPROPYL-2-METHYLHEXANE	0.2226	0.234	219.2	623.4	0.744	25
496	C10H22	3,3-DIETHYLHEXANE	0.2278	0.236	219.2	627.8	0.758	25
497	C10H22	3,4-DIETHYLHEXANE	0.2250	0.235	219.2	618.8	0.747	25
498	C10H22	3-ETHYL-2,2-DIMETHYLHEXANE	0.2282	0.239	219.2	611.7	0.745	25
499	C10H22	4-ETHYL-2,2-DIMETHYLHEXANE	0.2282	0.242	219.2	594.6	0.730	25
500	C10H22	3-ETHYL-2,3-DIMETHYLHEXANE	0.2319	0.240	219.2	626.8	0.760	25
501	C10H22	4-ETHYL-2,3-DIMETHYLHEXANE	0.2297	0.239	219.2	617.3	0.752	25
502	C10H22	3-ETHYL-2,4-DIMETHYLHEXANE	0.2297	0.239	219.2	616.1	0.751	25
503	C10H22	4-ETHYL-2,4-DIMETHYLHEXANE	0.2311	0.241	219.2	620.9	0.753	25
504	C10H22	3-ETHYL-2,5-DIMETHYLHEXANE	0.2276	0.240	219.2	603.5	0.737	25
505	C10H22	4-ETHYL-3,3-DIMETHYLHEXANE	0.2312	0.239	219.2	625.7	0.760	25
506	C10H22	3-ETHYL-3,4-DIMETHYLHEXANE	0.2305	0.238	219.2	624.5	0.760	25
507	C10H22	2,2,3,3-TETRAMETHYLHEXANE	0.2342	0.242	219.2	623.0	0.761	25
508	C10H22	2,2,3,4-TETRAMETHYLHEXANE	0.2308	0.241	219.2	620.4	0.751	25
509	C10H22	2,2,3,5-TETRAMETHYLHEXANE	0.2307	0.245	219.2	601.3	0.734	25
510	C10H22	2,2,4,4-TETRAMETHYLHEXANE	0.2261	0.237	219.2	610.2	0.742	25
511	C10H22	2,2,4,5-TETRAMETHYLHEXANE	0.2289	0.243	219.2	598.5	0.732	25
512	C10H22	2,2,5,5-TETRAMETHYLHEXANE	0.2356	0.256	260.6	581.4	0.715	25
513	C10H22	2,3,3,4-TETRAMETHYLHEXANE	0.2345	0.242	260.6	633.1	0.766	25
514	C10H22	2,3,3,5-TETRAMETHYLHEXANE	0.2316	0.243	260.6	610.1	0.745	25
515	C10H22	2,3,4,4-TETRAMETHYLHEXANE	0.2385	0.241	260.6	626.6	0.779	25
516	C10H22	2,3,4,5-TETRAMETHYLHEXANE	0.2315	0.243	260.6	613.2	0.746	25
517	C10H22	3,3,4,4-TETRAMETHYLHEXANE	0.2371	0.242	260.6	646.7	0.779	25
518	C10H22	2&DIMETHYL3ISOPROPYLPENTANE	0.2309	0.239	191.5	614.4	0.755	25
519	C10H22	33-DIETHYL-2-METHYLPENTANE	0.2344	0.239	191.5	639.9	0.776	25
520	C10H22	3ETHYL-223TRIMETHYLPENTANE	0.2361	0.241	191.5	646.0	0.778	25
521	C10H22	3ETHYL-224TRIMETHYLPENTANE	0.2312	0.240	191.5	615.3	0.753	25
522	C10H22	3ETHYL-234TRIMETHYLPENTANE	0.2351	0.241	191.5	642.3	0.774	25
523	C10H22	22334-PENTAMETHYLPENTANE	0.2392	0.245	236.7	643.8	0.777	25
524	C10H22	22344-PENTAMETHYLPENTANE	0.2330	0.240	234.4	627.3	0.764	25
525	C10H22O	DECYL-ALCOHOL	0.2561	0.252	280.1	687.0	0.830	20
526	C10H22S	BUTYL-HEXYL-SULFIDE	0.2478	0.239	238.2	701.0	0.841	25
527	C10H22S	ETHYL-OCTYL-SULFIDE	0.2478	0.239	238.2	701.0	0.841	25
528	C10H22S	HEPTYL-PROPYL-SULFIDE	0.2478	0.239	238.2	701.0	0.841	25
529	C10H22S	METHYL-NONYL-SULFIDE	0.2478	0.239	238.2	701.0	0.841	25
530	C10H22S	PENTYL-SULFIDE	0.2478	0.239	238.2	701.0	0.841	25
531	C10H22S	1-DECANETHIOL	0.2564	0.249	247.6	702.2	0.841	25
532	C10H22S2	PENTYL-DISULFIDE	0.2816	0.253	214.2	726.9	0.918	25
533	C11H10	1-METHYLNAPHTHALENE	0.2872	0.234	242.7	772.0	1.020	20
534	C11H10	2-METHYLNAPHTHALENE	0.3107	0.260	307.7	764.3	0.990	40
535	C11H16	PENTYLBENZENE	0.2727	0.260	198.2	679.9	0.855	25
536	C11H16	PENTAMETHYLBENZENE	0.2728	0.228	327.5	719.0	0.945*	55
537	C11H20	1-UNDECYNE	0.2609	0.251	248.2	651.0	0.769	25
538	C11H22	1-UNDECENE	0.2269	0.240	224.0	637.0	0.751	20
539	C11H22	1-CYCLOPENTYLHEXANE	0.2400	0.243	200.2	667.7	0.793	25
540	C11H22	PENTYLCYCLOHEXANE	0.2425	0.244	215.7	674.0	0.800	25
541	C11H24	UNDECANE	0.2241	0.241	247.6	640.1	0.740	20
542	C11H24O	UNDECYL-ALCOHOL	0.2598	0.254	292.2	666.3	0.830	20
543	C11H24S	BUTYL-HEPTYL-SULFIDE	0.2402	0.232	254.7	717.9	0.841	25
544	C11H24S	DECYL-METHYL-SULFIDE	0.2402	0.232	254.7	717.9	0.841	25
545	C11H24S	ETHYL-NONYL-SULFIDE	0.2402	0.232	254.7	717.9	0.841	25
546	C11H24S	OCTYL-PROPYL-SULFIDE	0.2402	0.232	254.7	717.9	0.841	25
547	C11H24S	1-UNDECANETHIOL	0.2491	0.242	270.4	716.5	0.841	25
548	C12H10	BIPHENYL	0.3508	0.294	342.4	789.3	0.990	74
549	C12H12	1-ETHYLNAPHTHALENE	0.3036	0.253	259.4	774.9	1.004	25
550	C12H12	2-ETHYLNAPHTHALENE	0.2997	0.254	265.8	774.9	0.988	25
551	C12H12	1,2-DIMETHYLNAPHTHALENE	0.2960	0.243	272.2	775.3	1.014	25
552	C12H12	1,3-DIMETHYLNAPHTHALENE	0.2938	0.244	269.2	773.8	1.003	25
553	C12H12	1,4-DIMETHYLNAPHTHALENE	0.2956	0.243	280.8	776.8	1.013	25
554	C12H12	1,5-DIMETHYLNAPHTHALENE	0.2996	0.244	355.2	773.5	0.978*	82
555	C12H12	1,6-DIMETHYLNAPHTHALENE	0.2940	0.245	259.2	770.6	0.999	25
556	C12H12	1,7-DIMETHYLNAPHTHALENE	0.2940	0.245	260.2	770.6	0.999	25
557	C12H12	2,3-DIMETHYLNAPHTHALENE	0.2996	0.242	378.2	777.8	0.968*	105
558	C12H12	2,6-DIMETHYLNAPHTHALENE	0.2996	0.245	385.2	769.2	0.949*	112
559	C12H12	2,7-DIMETHYLNAPHTHALENE	0.2996	0.245	371.2	770.6	0.961*	98
560	C12H18	HEXYLBENZENE	0.2706	0.260	212.2	697.5	0.854	25
561	C12H18	1,2,3-TRIETHYLBENZENE	0.2644	0.246	206.7	684.4	0.874	20
562	C12H18	1,2,4-TRIETHYLBENZENE	0.2644	0.246	206.7	684.4	0.874	20
563	C12H18	1,3,5-TRIETHYLBENZENE	0.2707	0.247	206.7	682.3	0.887*	25
564	C12H18	HEXAMETHYLBENZENE	0.2707	0.216	438.7	758.0	0.896*	166
565	C12H22	1-DODECYNE	0.2354	0.244	254.2	668.2	0.775	25
566	C12H24	1-DODECENE	0.2214	0.233	238.0	657.0	0.758	20
567	C12H24	1-CYCLOPENTYLHEPTANE	0.2379	0.237	220.0	679.0	0.810	20
568	C12H24	1-CYCLOHEXYLHEXANE	0.2628	0.238	263.6	691.8	0.892*	25
569	C12H26	DODECANE	0.2222	0.238	263.6	659.5	0.748	20
570	C12H26O	DODECYL-ALCOHOL	0.2474	0.240	297.1	679.0	0.830	25
571	C12H26S	BUTYL-OCTYL-SULFIDE	0.2328	0.225	259.2	733.7	0.842	25
572	C12H26S	DECYL-ETHYL-SULFIDE	0.2328	0.225	259.2	733.7	0.842	25
573	C12H26S	HEXYL-SULFIDE	0.2328	0.225	259.2	733.7	0.842	25
574	C12H26S	METHYL-UNDECYL-SULFIDE	0.2328	0.225	259.2	733.7	0.842	25
575	C12H26S	NONYL-PROPYL-SULFIDE	0.2328	0.225	259.2	733.7	0.842	25
576	C12H26S	1-DODECANETHIOL	0.2429	0.236	265.4	729.8	0.842	25
577	C12H26S2	HEXYL-DISULFIDE	0.2653	0.241	225.2	747.1	0.908	25
578	C13H14	1-PROPYLNAPHTHALENE	0.2934	0.248	266.7	771.5	0.987	25
579	C13H14	2-PROPYLNAPHTHALENE	0.2893	0.248	270.2	772.4	0.973	25
580	C13H14	2ETHYL-3-METHYLNAPHTHALENE	0.2948	0.243	344.2	776.4	0.976*	71
581	C13H14	2ETHYL-6-METHYLNAPHTHALENE	0.2948	0.246	318.2	766.6	0.982*	45
582	C13H14	2ETHYL-7-METHYLNAPHTHALENE	0.2948	0.246	318.2	766.6	0.982*	45

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NO.	FORMULA	NAME	A	B	TMIN,K	TC,K	DENSAT	T,C
583	C13H20	1-PHENYLHEPTANE	0.2601	0.250	225.2	713.5	0.853	25
584	C13H24	1-TRIDECYNE	0.2299	0.237	268.2	684.1	0.781	25
585	C13H26	1-TRIDECENE	0.2165	0.226	250.1	674.0	0.766	20
586	C13H26	1-CYCLOPENTYLOCTANE	0.2292	0.231	229.2	702.1	0.801	25
587	C13H26	1-CYCLOHEXYLHEPTANE	0.2305	0.231	242.7	708.6	0.808	25
588	C13H28	TRIDECANE	0.2237	0.239	267.8	678.4	0.756	20
589	C13H28O	1-TRIDECANOL	0.2561	0.241	304.2	693.7	0.856*	31
590	C13H28S	BUTYL-NONYL-SULFIDE	0.2265	0.219	271.2	748.4	0.842	25
591	C13H28S	DECYL-PROPYL-SULFIDE	0.2265	0.219	271.2	748.4	0.842	25
592	C13H28S	DODECYL-METHYL-SULFIDE	0.2265	0.219	271.2	748.4	0.842	25
593	C13H28S	ETHYL-UNDECYL-SULFIDE	0.2265	0.219	271.2	748.4	0.842	25
594	C13H28S	1-TRIDECANETHIOL	0.2367	0.230	282.0	742.1	0.842	25
595	C14H16	1-BUTYLNAPHTHALENE	0.2845	0.244	253.4	781.5	0.973	25
596	C14H16	2-BUTYLNAPHTHALENE	0.2812	0.244	268.2	781.0	0.962	25
597	C14H22	1-PHENYLOCTANE	0.2587	0.250	237.2	728.0	0.853	25
598	C14H22	1,2,3,4-TETRAETHYLBENZENE	0.2541	0.233	285.0	708.2	0.888	20
599	C14H22	1,2,3,5-TETRAETHYLBENZENE	0.2535	0.234	284.2	707.5	0.882	20
600	C14H22	1,2,4,5-TETRAETHYLBENZENE	0.2527	0.234	283.2	706.9	0.879	20
601	C14H26	1-TETRADECYNE	0.2249	0.231	273.2	698.9	0.785	25
602	C14H28	1-TETRADECENE	0.2119	0.220	260.3	689.0	0.786	0
603	C14H28	1-CYCLOPENTYLNONANE	0.2238	0.225	244.2	717.0	0.805	25
604	C14H28	1-CYCLOHEXYLOCTANE	0.2249	0.225	253.5	723.6	0.810	25
605	C14H30	TETRADECANE	0.2193	0.233	279.0	696.8	0.759	25
606	C14H30O	1-TETRADECANOL	0.2557	0.245	312.7	674.9	0.830*	40
607	C14H30S	BUTYL-DECYL-SULFIDE	0.2202	0.213	276.2	762.2	0.843	25
608	C14H30S	DODECYL-ETHYL-SULFIDE	0.2202	0.213	276.2	762.2	0.843	25
609	C14H30S	HEPTYL-SULFIDE	0.2202	0.213	276.2	762.2	0.843	25
610	C14H30S	METHYL-TRIDECYL-SULFIDE	0.2202	0.213	276.2	762.2	0.843	25
611	C14H30S	PROPYL-UNDECYL-SULFIDE	0.2202	0.213	276.2	762.2	0.843	25
612	C14H30S	1-TETRADECANETHIOL	0.2297	0.223	279.3	753.8	0.842	25
613	C14H30S2	HEPTYL-DISULFIDE	0.2492	0.228	235.2	766.0	0.900	25
614	C15H18	1-PENTYLNAPHTHALENE	0.2744	0.238	251.2	793.3	0.962	25
615	C15H18	2-PENTYLNAPHTHALENE	0.2694	0.236	269.2	797.5	0.953	25
616	C15H24	1-PHENYLNONANE	0.2575	0.250	249.2	741.0	0.852	25
617	C15H28	1-PENTADECYNE	0.2200	0.225	283.2	711.4	0.789	25
618	C15H30	1-PENTADECENE	0.2071	0.214	269.4	704.0	0.791	0
619	C15H30	1-CYCLOPENTYLDECANE	0.2184	0.219	251.0	730.6	0.807	25
620	C15H30	1-CYCLOHEXYLNONANE	0.2193	0.219	263.0	737.8	0.813	25
621	C15H32	PENTADECANE	0.2160	0.229	283.1	717.6	0.765	25
622	C15H32O	1-PENTADECANOL	0.2554	0.226	317.0	722.5	0.901	44
623	C15H32S	BUTYL-UNDECYL-SULFIDE	0.2140	0.207	284.2	775.2	0.843	25
624	C15H32S	DODECYL-PROPYL-SULFIDE	0.2140	0.207	284.2	775.2	0.843	25
625	C15H32S	ETHYL-TRIDECYL-SULFIDE	0.2140	0.207	284.2	775.2	0.843	25
626	C15H32S	METHYL-TETRADECYL-SULFIDE	0.2140	0.207	284.2	775.2	0.843	25
627	C15H32S	1-PENTADECANETHIOL	0.2236	0.217	290.9	764.8	0.843	25
628	C16H26	1-PHENYLDECANE	0.2565	0.250	258.8	753.0	0.852	25
629	C16H26	PENTAETHYLBENZENE	0.2652	0.222	327.7	723.6	0.941*	55
630	C16H30	1-HEXADECYNE	0.2150	0.219	288.2	724.3	0.793	25
631	C16H32	1-HEXADECENE	0.2022	0.208	277.3	717.0	0.788	10
632	C16H32	1-CYCLOPENTYLUNDECANE	0.2130	0.213	263.2	743.3	0.810	25
633	C16H32	1-CYCLOHEXYLDECANE	0.2138	0.213	271.4	750.0	0.815	25
634	C16H34	HEXADECANE	0.2120	0.224	291.0	734.3	0.773	20
635	C16H34O	1-HEXADECANOL	0.2154	0.210	322.5	767.1	0.812	60
636	C16H34S	BUTYL-DODECYL-SULFIDE	0.2079	0.201	288.2	787.3	0.844	25
637	C16H34S	ETHYL-TETRADECYL-SULFIDE	0.1904	0.182	288.2	791.7	0.844	25
638	C16H34S	METHYL-PENTADECYL-SULFIDE	0.2079	0.201	288.2	787.3	0.844	25
639	C16H34S	OCTYL-SULFIDE	0.2079	0.201	288.2	787.3	0.844	25
640	C16H34S	PROPYL-TRIDECYL-SULFIDE	0.2079	0.201	288.2	787.3	0.844	25
641	C16H34S	1-HEXADECANETHIOL	0.2176	0.211	290.9	774.7	0.843	25
642	C16H34S2	OCTYL-DISULFIDE	0.2339	0.215	244.2	784.5	0.894	25
643	C17H28	1-PHENYLUNDECANE	0.2467	0.240	268.2	764.0	0.852	25
644	C17H32	1-HEPTADECYNE	0.2090	0.212	295.2	736.2	0.796	25
645	C17H34	1-HEPTADECENE	0.1996	0.205	284.4	732.4	0.782	25
646	C17H34	1-CYCLOPENTYLDODECANE	0.2075	0.207	268.2	755.2	0.812	25
647	C17H34	1-CYCLOHEXYLUNDECANE	0.2092	0.208	279.0	761.7	0.817	25
648	C17H36	HEPTADECANE	0.2066	0.216	295.0	749.3	0.778	25
649	C17H36O	1-HEPTADECANOL	0.2321	0.216	327.0	736.0	0.848	54
650	C17H36S	BUTYL-TRIDECYL-SULFIDE	0.2018	0.195	294.2	798.6	0.844	25
651	C17H36S	ETHYL-PENTADECYL-SULFIDE	0.2018	0.195	294.2	798.6	0.844	25
652	C17H36S	HEXADECYL-METHYL-SULFIDE	0.2018	0.195	294.2	798.6	0.844	25
653	C17H36S	PROPYL-TETRADECYL-SULFIDE	0.2018	0.195	294.2	798.6	0.844	25
654	C17H36S	1-HEPTADECANETHIOL	0.2617	0.205	300.4	786.0	1.041	27
655	C18H30	1-PHENYLDODECANE	0.2459	0.240	276.2	774.0	0.852	25
656	C18H30	HEXAETHYLBENZENE	0.2415	0.212	401.2	734.8	0.831	130
657	C18H34	1-OCTADECYNE	0.2491	0.207	300.2	747.3	0.971	27
658	C18H36	1-OCTADECENE	0.1933	0.197	290.8	739.0	0.789	20
659	C18H36	1-CYCOPENTYLTRIDECANE	0.2029	0.202	278.2	766.5	0.814	25
660	C18H36	1-CYCLOHEXYLDODECANE	0.2037	0.202	285.7	772.8	0.819	25
661	C18H38	OCTADECANE	0.2439	0.195	301.3	748.0	1.000	28
662	C18H38O	1-OCTADECANOL	0.2162	0.209	331.0	747.0	0.812	59
663	C18H38S	BUTYL-TETRADECYL-SULFIDE	0.2611	0.190	298.2	810.5	1.121	25
664	C18H38S	ETHYL-HEXADECYL-SULFIDE	0.2611	0.190	298.2	810.5	1.121	25
665	C18H38S	HEPTADECYL-METHYL-SULFIDE	0.2611	0.190	298.2	810.5	1.121	25
666	C18H38S	NONYL-SULFIDE	0.2611	0.190	298.2	810.5	1.121	25
667	C18H38S	PENTADECYL-PROPYL-SULFIDE	0.2611	0.190	298.2	810.5	1.121	25
668	C18H38S	1-OCTADECANETHIOL	0.2611	0.200	300.9	795.4	1.064	28
669	C18H38S2	NONYL-DISULFIDE	0.2210	0.204	252.2	802.3	0.889	25
670	C19H32	1-PHENYLTRIDECANE	0.2452	0.240	283.2	784.0	0.851	25
671	C19H36	1-NONADECYNE	0.2492	0.201	306.2	758.9	0.995	33
672	C19H38	1-NONADECENE	0.1913	0.195	296.6	755.1	0.789	25
673	C19H38	1-CYCLOPENTYLTETRADECANE	0.2022	0.201	282.0	772.0	0.820	20
674	C19H38	1-CYCLOHEXYLTRIDECANE	0.1990	0.197	291.7	783.4	0.821	25
675	C19H40	NONADECANE	0.2056	0.212	305.0	776.0	0.789	32
676	C19H40O	1-NONADECANOL	0.2544	0.199	334.9	775.3	1.005	62
677	C19H40S	BUTYL-PENTADECYL-SULFIDE	0.2606	0.185	303.2	821.8	1.144	30
678	C19H40S	ETHYL-HEPTADECYL-SULFIDE	0.2606	0.185	303.2	821.8	1.144	30
679	C19H40S	HEXADECYL-PROPYL-SULFIDE	0.2606	0.185	303.2	821.8	1.144	30
680	C19H40S	METHYL-OCTADECYL-SULFIDE	0.2606	0.185	303.2	821.8	1.144	30
681	C19H40S	1-NONADECANETHIOL	0.2606	0.194	307.0	805.3	1.089	34
682	C20H34	1-PHENYLTETRADECANE	0.2446	0.240	289.2	792.0	0.851	25
683	C20H38	1-EICOSYNE	0.2492	0.195	309.2	769.8	1.022	36
684	C20H40	1-EICOSENE	0.2409	0.190	301.8	765.4	1.016	29
685	C20H40	1-CYCLOPENTYLPENTADECANE	0.1932	0.191	290.0	780.0	0.821	20
686	C20H40	1-CYCLOHEXYLTETRADECANE	0.1943	0.192	297.2	792.8	0.822	25
687	C20H42	EICOSANE	0.1805	0.184	310.0	767.0	0.775	40
688	C20H42O	1-EICOSANOL	0.2542	0.197	339.0	770.0	1.007*	66
689	C20H42S	BUTYL-HEXADECYL-SULFIDE	0.2601	0.179	308.2	832.3	1.174	35
690	C20H42S	DECYL-SULFIDE	0.2601	0.179	308.2	832.3	1.174	35
691	C20H42S	ETHYL-OCTADECYL-SULFIDE	0.2601	0.179	308.2	832.3	1.174	35
692	C20H42S	HEPTADECYL-PROPYL-SULFIDE	0.2601	0.179	308.2	832.3	1.174	35
693	C20H42S	METHYL-NONADECYL-SULFIDE	0.2601	0.179	308.2	832.3	1.174	35
694	C20H42S	1-EICOSANETHIOL	0.2601	0.189	310.4	814.6	1.112	37
695	C20H42S2	DECYL-DISULFIDE	0.2075	0.192	259.2	820.1	0.885	25
696	C21H36	1-PHENYLPENTADECANE	0.2351	0.230	295.2	800.0	0.851	25
697	C21H42	1-CYCLOPENTYLHEXADECANE	0.1881	0.186	294.2	797.3	0.819	25
698	C21H42	1-CYCLOHEXYLPENTADECANE	0.2574	0.187	302.2	803.5	1.116	29
699	C22H38	1-PHENYLHEXADECANE	0.2521	0.230	300.2	808.0	0.913	27
700	C22H44	1-CYCLOHEXYLHEXADECANE	0.2571	0.182	306.8	813.4	1.139	34

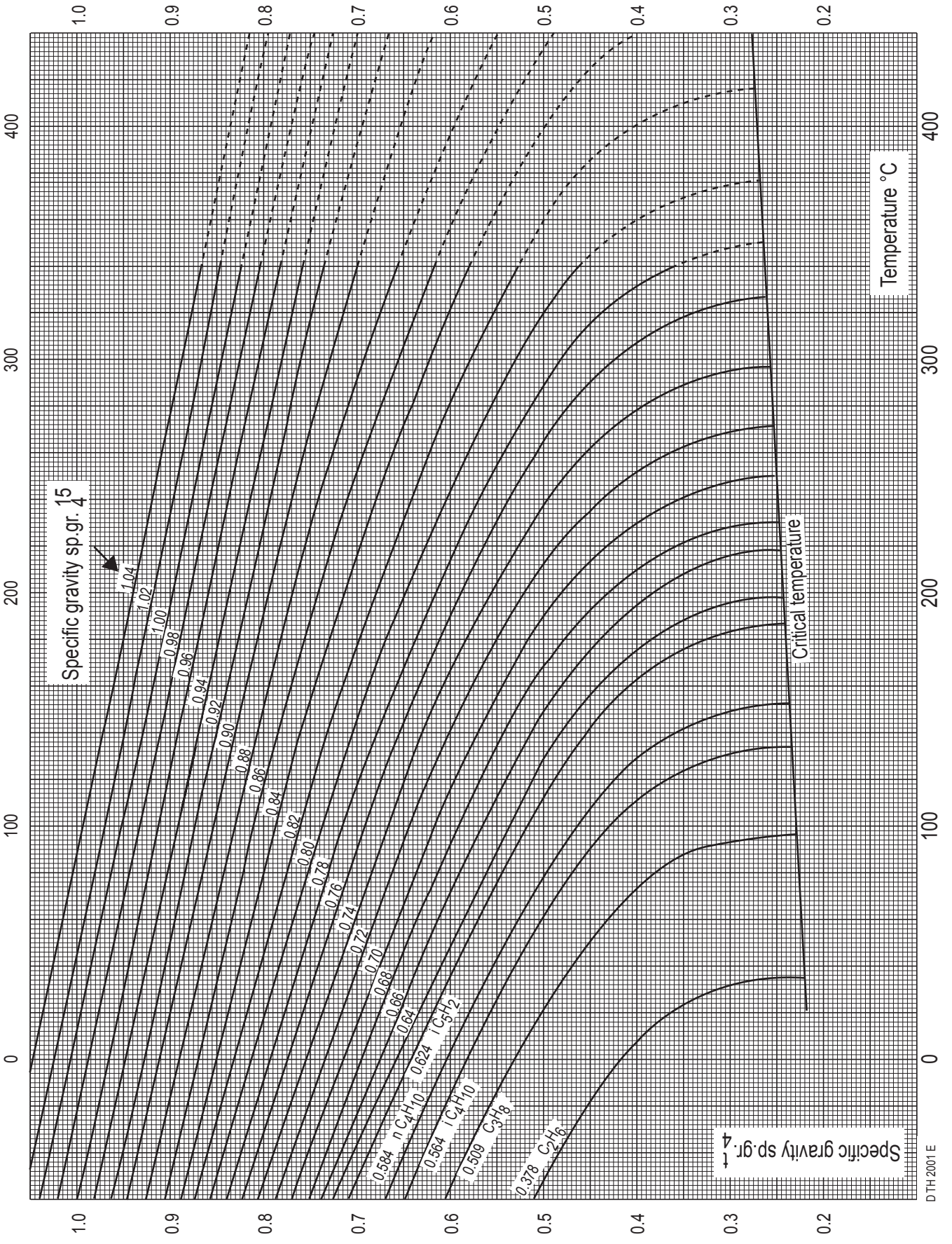
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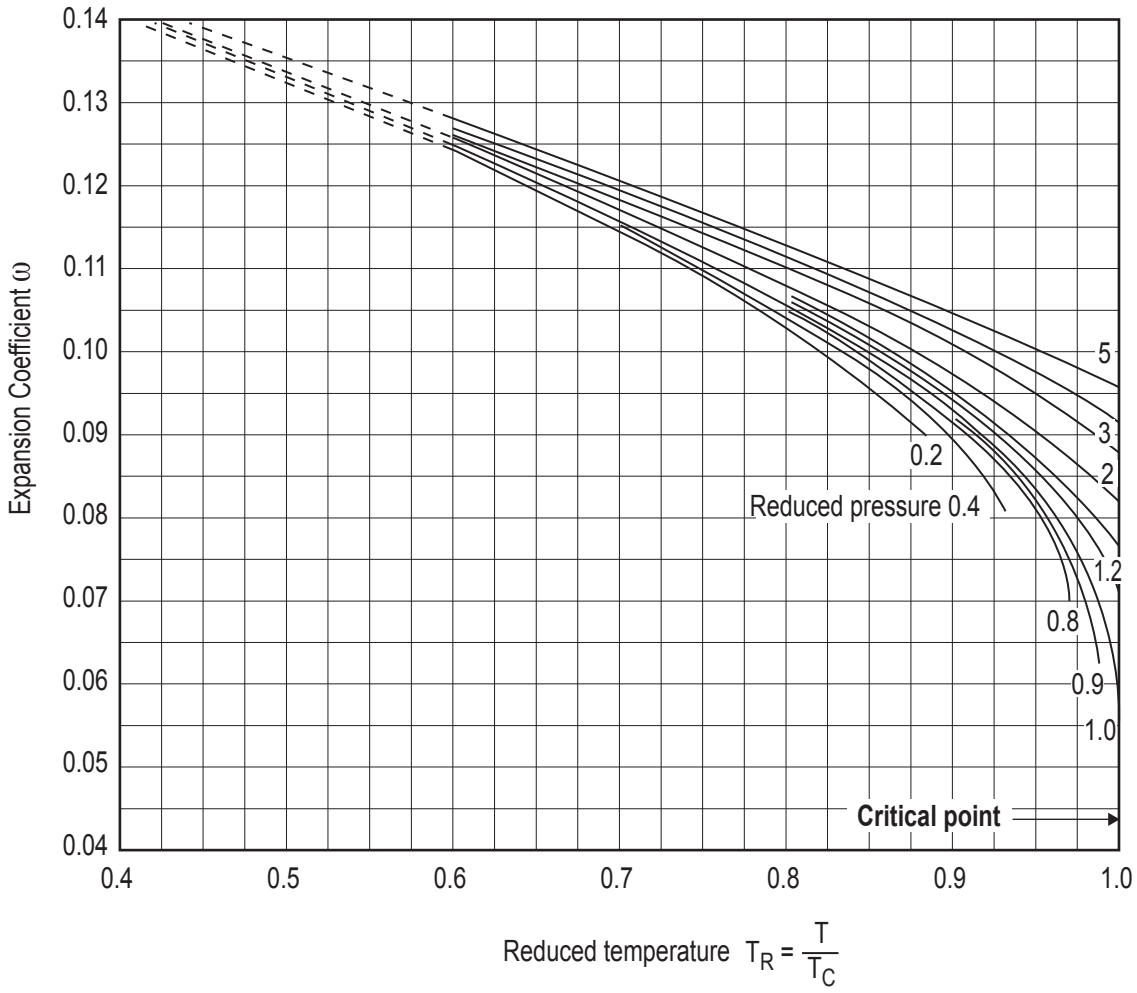
VARIATION OF LIQUID HYDROCARBON SPECIFIC GRAVITY VERSUS THE TEMPERATURE



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EXPANSION COEFFICIENT OF LIQUIDS VERSUS THE REDUCED COORDINATES

$$\frac{\text{Sp.gr.1}}{\omega 1} = \frac{\text{Sp.gr.2}}{\omega 2}$$



From Watson and Gamson Chart

DTH 1403 D

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ASTM TABLE EXTRACTIONS

Refined products
Conversion at 15°C of density

From 853.0 à 873.0 and 19.50 to 27.00°C

Temp. in °C	Observed density											Temp. in °C
	Density at 15°C											
	853.0	855.0	857.0	859.0	861.0	863.0	865.0	867.0	869.0	871.0	873.0	
19.50	856.1	858.1	860.1	862.1	864.1	866.1	868.1	870.1	872.1	874.1	876.1	19.50
19.75	856.3	858.3	860.3	862.3	864.3	866.2	868.2	870.2	872.2	874.2	876.2	19.75
20.00	856.4	858.4	860.4	862.4	864.4	866.4	868.4	870.4	872.4	874.4	876.4	20.00
20.25	856.6	858.6	860.6	862.6	864.6	866.6	868.6	870.6	872.6	874.6	876.6	20.25
20.50	856.8	858.8	860.8	862.8	864.8	866.8	868.8	870.8	872.7	874.7	876.7	20.50
20.75	856.9	858.9	860.9	862.9	864.9	866.9	868.9	870.9	872.9	874.9	876.9	20.75
21.00	857.1	859.1	861.1	863.1	865.1	867.1	869.1	871.1	873.1	875.1	877.1	21.00
21.25	857.3	859.3	861.3	863.3	865.3	867.3	869.3	871.3	873.3	875.2	877.2	21.25
21.50	857.5	859.5	861.4	863.4	865.4	867.4	869.4	871.4	873.4	875.4	877.4	21.50
21.75	857.6	859.6	861.6	863.6	865.6	867.6	869.6	871.6	873.6	875.6	877.6	21.75
22.00	857.8	859.8	861.8	863.8	865.8	867.8	869.8	871.8	873.8	875.8	877.8	22.00
22.25	858.0	860.0	862.0	864.0	866.0	868.0	869.9	871.9	873.9	875.9	877.9	22.25
22.50	858.1	860.1	862.1	864.1	866.1	868.1	870.1	872.1	874.1	876.1	878.1	22.50
22.75	858.3	860.3	862.3	864.3	866.3	868.3	870.3	872.3	874.3	876.3	878.3	22.75
23.00	858.5	860.5	862.5	864.5	866.5	868.5	870.5	872.5	874.5	876.4	878.4	23.00
23.25	858.7	860.7	862.7	864.6	866.6	868.6	870.6	872.6	874.6	876.6	878.6	23.25
23.50	858.8	860.8	862.8	864.8	866.8	868.8	870.8	872.8	874.8	876.8	878.8	23.50
23.75	859.0	861.0	863.0	865.0	867.0	869.0	871.0	873.0	875.0	877.0	878.9	23.75
24.00	859.2	861.2	863.2	865.2	867.1	869.1	871.1	873.1	875.1	877.1	879.1	24.00
24.25	859.3	861.3	863.3	865.3	867.3	869.3	871.3	873.3	875.3	877.3	879.3	24.25
24.50	859.5	861.5	863.5	865.5	867.5	869.5	871.5	873.5	875.5	877.5	879.5	24.50
24.75	859.7	861.7	863.7	865.7	867.7	869.7	871.6	873.6	875.6	877.6	879.6	24.75
25.00	859.9	861.8	863.8	865.8	867.8	869.8	871.8	873.8	875.8	877.8	879.8	25.00
25.25	860.0	862.0	864.0	866.0	868.0	870.0	872.0	874.0	876.0	878.0	880.0	25.25
25.50	860.2	862.2	864.2	866.2	868.2	870.2	872.2	874.2	876.2	878.1	880.1	25.50
25.75	860.4	862.4	864.4	866.4	868.4	870.3	872.3	874.3	876.3	878.3	880.3	25.75
26.00	860.5	862.5	864.5	866.5	868.5	870.5	872.5	874.5	876.5	878.5	880.5	26.00
26.25	860.7	862.7	864.7	866.7	868.7	870.7	872.7	874.7	876.7	878.7	880.6	26.25
26.50	860.9	862.9	864.9	866.9	868.9	870.9	872.8	874.8	876.8	878.8	880.8	26.50
26.75	861.1	863.1	865.0	867.0	869.0	871.0	873.0	875.0	877.0	879.0	881.0	26.75
27.00	861.2	863.2	865.2	867.2	869.2	871.2	873.2	875.2	877.2	879.2	881.2	27.00

* extrapoled value

From document Table ASTM 53 B page 186

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CONCENTRATION AND SPECIFIC GRAVITY OF SODIUM HYDROXIDE SOLUTIONS

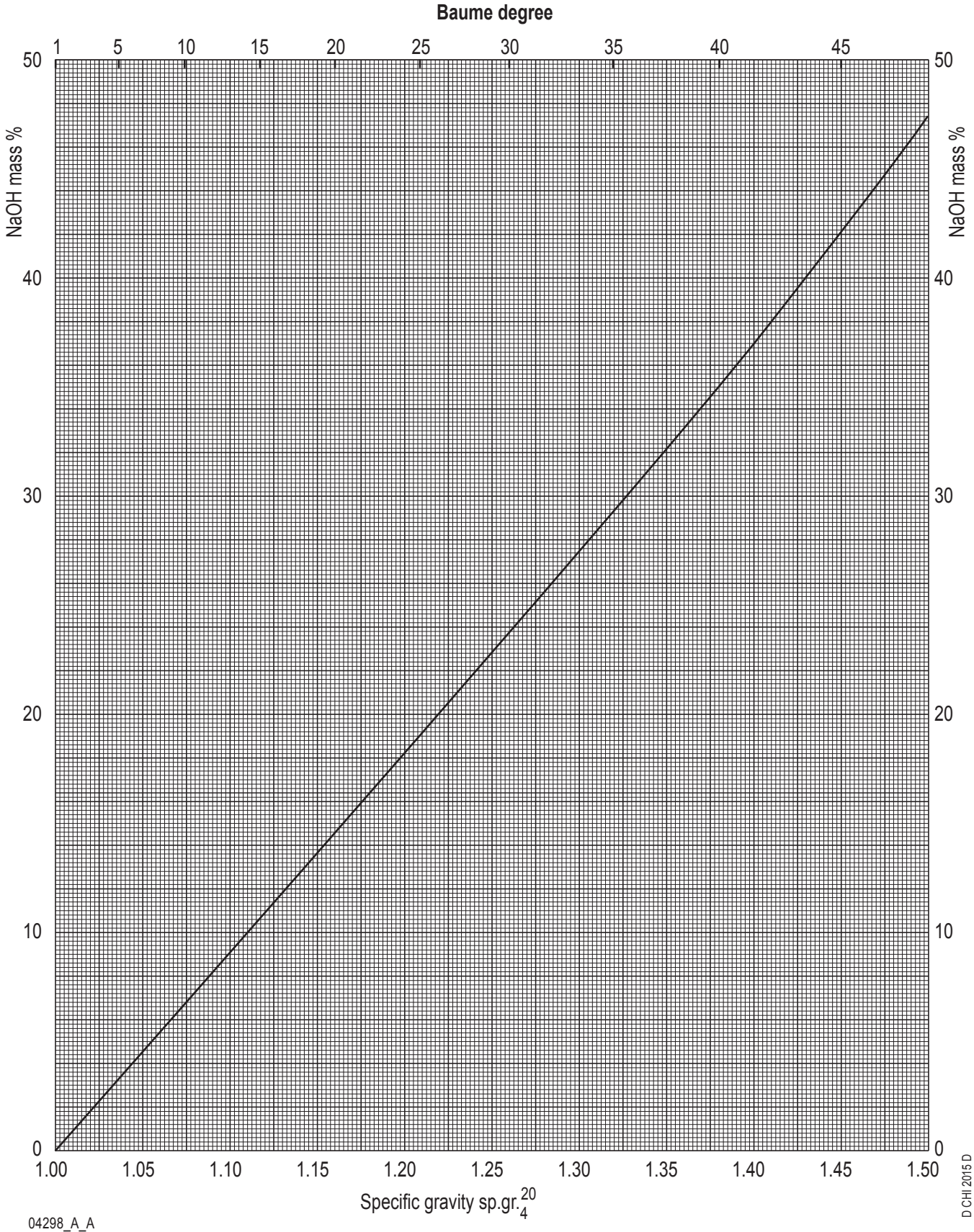
WEIGHT % OF NaOH	SPECIFIC GRAVITY $sp.gr._4^{20}$	BAUME DEGREE *	NaOH CONCENTRATION (g/l)
1	1.0095	1.4	10.10
2	1.0207	2.9	20.41
3	1.0318	4.5	30.95
4	1.0428	6.0	41.71
5	1.0538	7.4	52.69
6	1.0648	8.8	63.89
7	1.0758	10.2	75.31
8	1.0869	11.6	86.95
9	1.0979	12.9	98.91
10	1.1089	14.2	110.9
12	1.1309	16.8	135.7
14	1.1530	19.2	161.4
16	1.7510	21.6	188.0
18	1.1972	23.9	215.5
20	1.2191	26.1	243.8
22	1.2411	28.2	273.0
24	1.2629	30.2	303.1
26	1.2848	32.1	334.0
28	1.3064	34.0	365.8
30	1.3279	35.8	398.4
32	1.3490	37.5	431.7
34	1.3696	39.1	465.7
36	1.3900	40.7	500.4
38	1.4101	42.2	535.8
40	1.4300	43.6	572.0
42	1.4494	45.0	608.7
44	1.4685	46.3	646.1
46	1.4873	47.5	684.2
48	1.5065	48.8	723.1
50	1.5253	49.9	762.7

* BAUME degree : $^\circ B = 145 - \dfrac{145}{sp.gr._4^{20}}$ et $sp.gr._4^{20} = \dfrac{145}{145 - {}^\circ B}$

From "International Critical Tables"

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NaOH SOLUTIONS
— Specific gravity - NaOH mass % curve - Baume degree —



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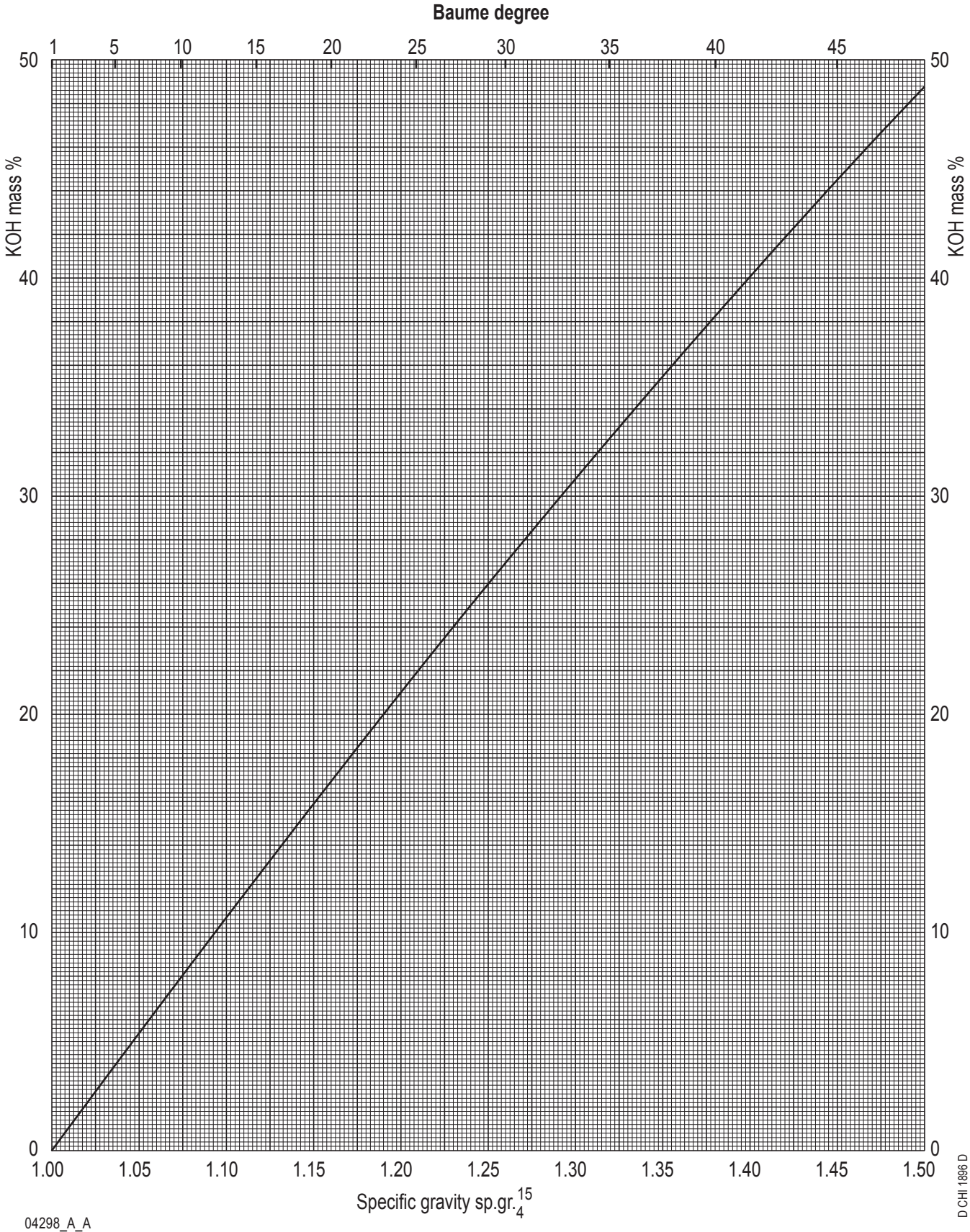
CONCENTRATION AND SPECIFIC GRAVITY OF POTASSIUM HYDROXIDE SOLUTIONS

WEIGHT % OF KOH	SPECIFIC GRAVITY $sp.gr._4^{15}$	BAUME DEGREE	KOH CONCENTRATION (g/l)
1	1.0083	1.2	10.08
2	1.0175	2.5	20.35
3	1.0267	3.8	30.80
4	1.0359	5.0	41.44
5	1.0452	6.3	52.26
6	1.0544	7.5	63.26
7	1.0637	8.7	74.46
8	1.0730	9.9	85.84
9	1.0824	11.0	87.42
10	1.0918	12.2	109.2
12	1.1108	14.5	133.3
14	1.1299	16.7	158.2
16	1.1493	18.8	183.9
18	1.1688	20.9	210.4
20	1.1884	23.0	237.7
22	1.2083	25.0	265.8
24	1.2285	27.0	294.8
26	1.2489	28.9	324.7
28	1.2695	30.8	355.5
30	1.2905	32.6	387.2
32	1.3117	34.5	419.7
34	1.3331	36.2	453.3
36	1.3549	38.0	487.8
38	1.3769	39.7	523.2
40	1.3991	41.4	559.6
42	1.4215	43.0	597.0
44	1.4443	44.6	635.5
46	1.4673	46.2	675.0
48	1.4907	47.7	715.5
50	1.5143	49.2	757.2

From "International Critical Tables"

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KOH SOLUTIONS
— Specific gravity - KOH mass % curve - Baume degree —



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CONCENTRATION AND SPECIFIC GRAVITY OF HYDROGEN CHLORIDE SOLUTIONS

WEIGHT % Of HCL	SPECIFIC GRAVITY at 20 °C	HCl CONCENTRATION (g/l)
0.50	1.0007	5.0
1.00	1.0031	10.0
1.50	1.0056	15.1
2.00	1.0081	20.2
2.50	1.0105	25.3
3.00	1.0130	30.4
3.50	1.0154	35.5
4.00	1.0179	40.7
4.50	1.0204	45.9
5.00	1.0228	51.1
5.50	1.0253	56.4
6.00	1.0278	61.7
6.50	1.0302	67.0
7.00	1.0327	72.3
7.50	1.0352	77.6
8.00	1.0377	83.0
8.50	1.0401	88.4
9.00	1.0426	93.8
9.50	1.0451	99.3
10.00	1.0476	104.8
11.00	1.0526	115.8
12.00	1.0576	126.9
13.00	1.0626	138.1
14.00	1.0676	149.5
15.00	1.0726	160.9
16.00	1.0777	172.4
17.00	1.0828	184.1
18.00	1.0878	195.8
19.00	1.0929	207.7
20.00	1.0980	219.6
22.00	1.1083	243.8
24.00	1.1185	268.4
26.00	1.1288	293.5
28.00	1.1391	318.9
30.00	1.1492	344.8
32.00	1.1594	371.0
34.00	1.1693	397.6
36.00	1.1791	424.5
38.00	1.1886	451.7
40.00	1.1977	479.1

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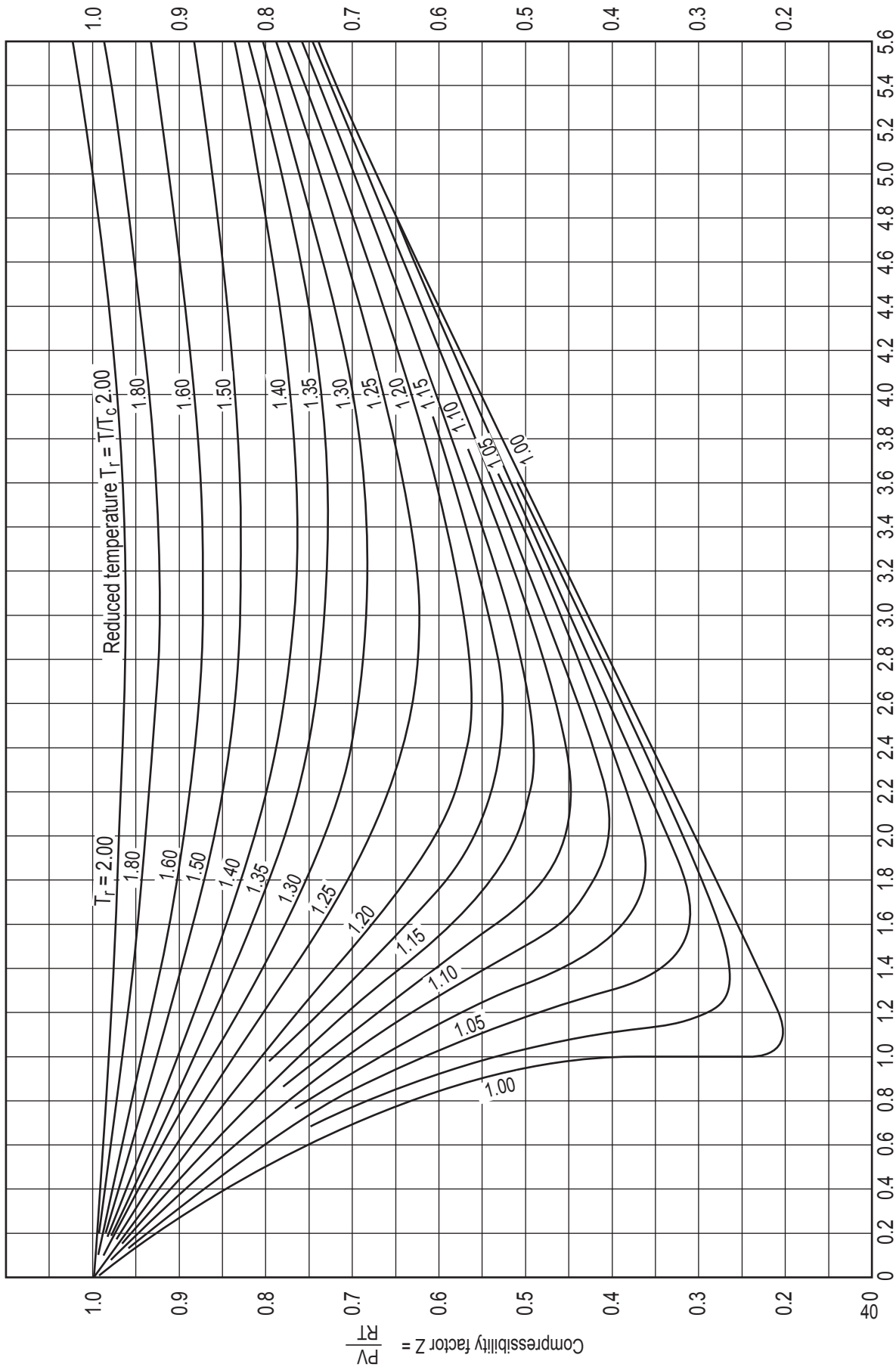
CONCENTRATION AND SPECIFIC GRAVITY OF SULFURIC ACID SOLUTIONS

SPECIFIC GRAVITY at 20 °C	WEIGHT % of H_2SO_4	g H_2SO_4/ litre
1.0051	1	10.05
1.0118	2	20.24
1.0184	3	30.55
1.0250	4	41.00
1.0317	5	51.59
1.0385	6	62.31
1.0453	7	73.17
1.0522	8	84.18
1.0591	9	95.32
1.0661	10	106.6
1.0731	11	118.0
1.0802	12	129.6
1.0874	13	141.4
1.0947	14	153.3
1.1020	15	165.3
1.1094	16	177.5
1.1168	17	189.9
1.1243	18	202.4
1.1318	19	215.0
1.1394	20	227.9
1.1471	21	240.9
1.1548	22	254.1
1.1626	23	267.4
1.1704	24	280.9
1.1783	25	294.6
1.1862	26	308.4
1.1942	27	322.4
1.2023	28	336.6
1.2104	29	351.0
1.2185	30	365.6
1.2267	31	380.3
1.2349	32	395.2
1.2432	33	410.3
1.2515	34	425.5
1.2599	35	441.0
1.2684	36	456.6
1.2769	37	472.5
1.2855	38	488.5
1.2941	39	504.7
1.3028	40	521.1
1.3116	41	537.8
1.3205	42	554.6
1.3294	43	571.6
1.3384	44	588.9
1.3476	45	606.4
1.3569	46	624.2
1.3663	47	642.2
1.3758	48	660.4
1.3854	49	678.8
1.3951	50	697.6

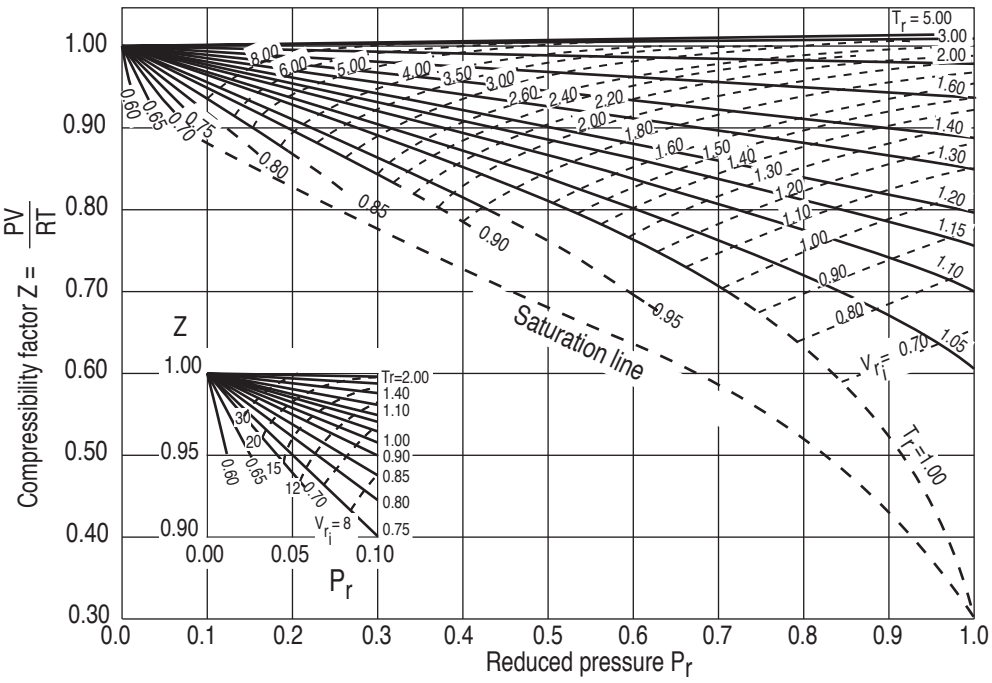
SPECIFIC GRAVITY at 20 °C	WEIGHT % of H_2SO_4	g H_2SO_4/ litre
1.4049	51	716.5
1.4148	52	735.7
1.4248	53	755.1
1.4350	54	774.9
1.4453	55	794.9
1.4557	56	815.2
1.4662	57	835.7
1.4768	58	856.5
1.4875	59	877.6
1.4983	60	899.0
1.5091	61	920.6
1.5200	62	942.4
1.5310	63	964.5
1.5421	64	986.9
1.5533	65	1010
1.5646	66	1033
1.5760	67	1056
1.5874	68	1079
1.5989	69	1103
1.6105	70	1127
1.6221	71	1152
1.6338	72	1176
1.6456	73	1201
1.6574	74	1226
1.6692	75	1252
1.6810	76	1278
1.6927	77	1303
1.7043	78	1329
1.7158	79	1355
1.7272	80	1382
1.7383	81	1408
1.7491	82	1434
1.7594	83	1460
1.7693	84	1486
1.7786	85	1512
1.7872	86	1537
1.7951	87	1562
1.8022	88	1586
1.8087	89	1610
1.8144	90	1633
1.8195	91	1656
1.8240	92	1678
1.8279	93	1700
1.8312	94	1721
1.8337	95	1742
1.8355	96	1762
1.8364	97	1781
1.8361	98	1799
1.8342	99	1816
1.8305	100	1831

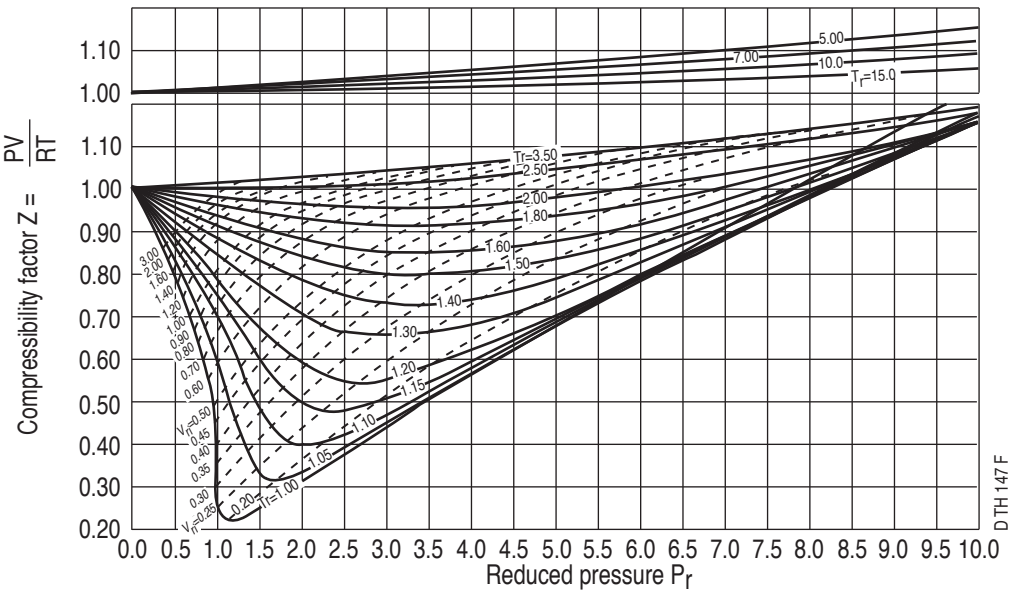
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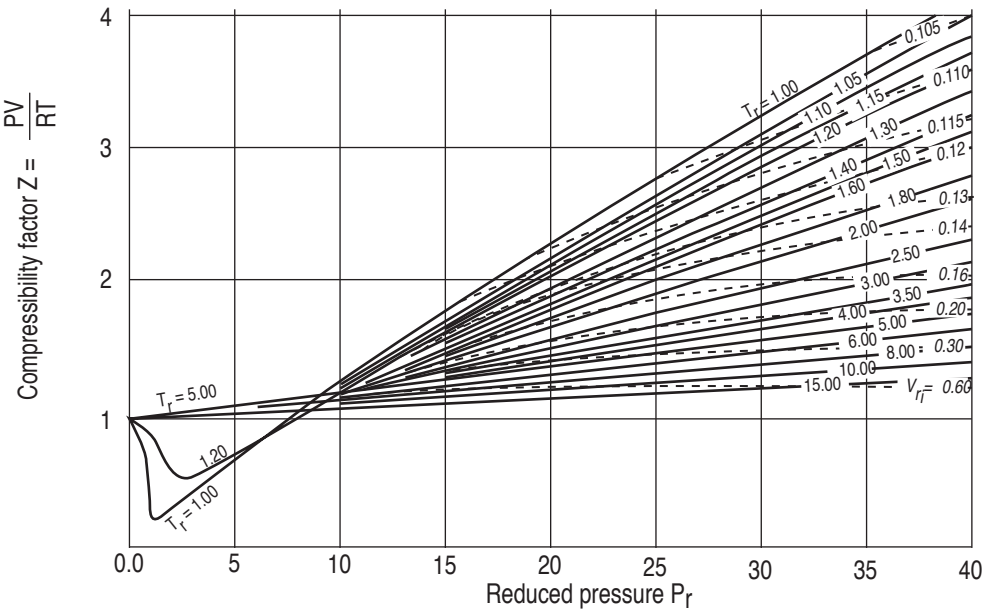
GENERALIZED COMPRESSIBILITY CHART



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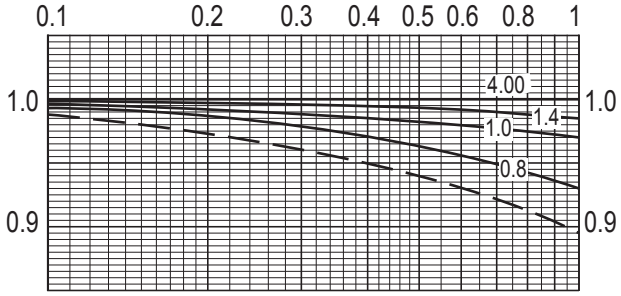


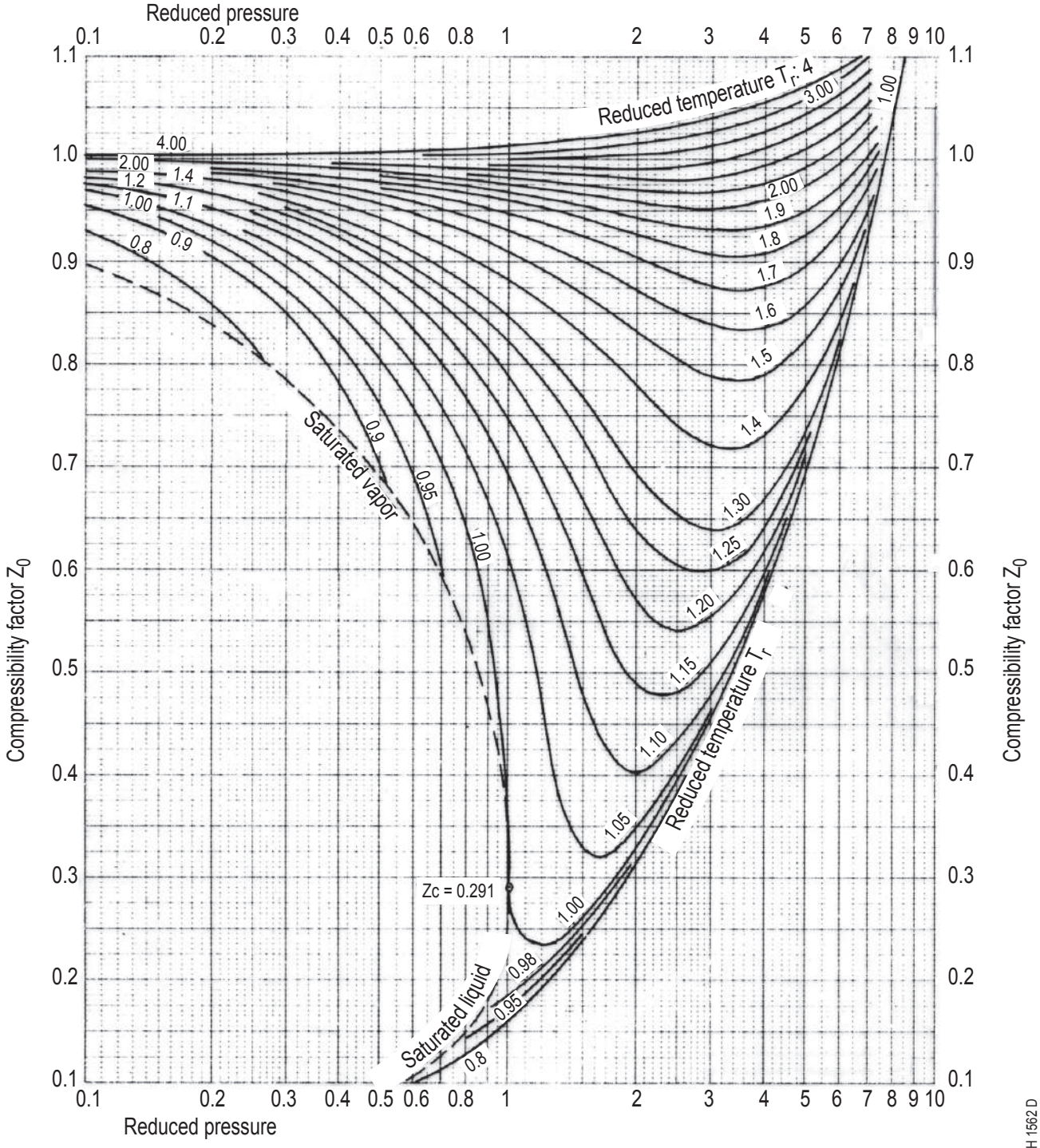


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DETERMINATION OF COMPRESSIBILITY FACTOR BY LAW OF CORRESPONDING STATE USING THE PITZER ACENTRIC FACTOR — Z_0 Charts —

COMPRESSIBILITY FACTOR Z_0
OF A SIMPLE FLUID
$Z = Z_0 + \omega Z_1$
(Pitzer K.S.)



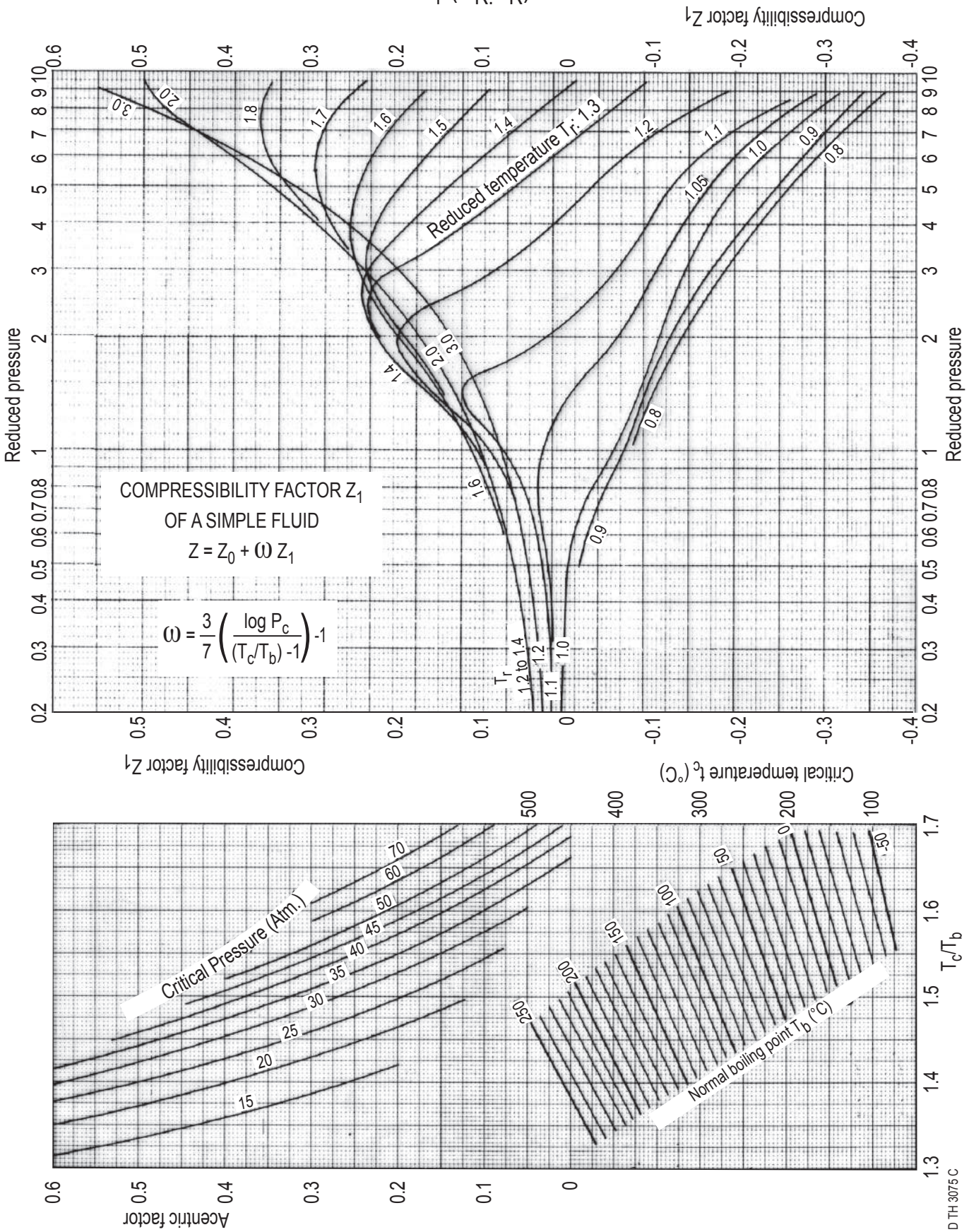


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DETERMINATION OF COMPRESSIBILITY FACTOR BY LAW OF CORRESPONDING STATE USING THE PITZER ACENTRIC FACTOR — Z_1 Charts —

$Z_1 (P_R, T_R)$



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CORRESPONDING STATE LAW WITH 3 PARAMETERS
— Lee and Kesler method —

Simple fluid $\omega = 0$
Heavy fluid for reference $\omega = 0{,}3978$

- SATURATION PROPERTIES: **Figure I4 (2/6)**

$$X(Tr) = X_{(\omega=0)}(Tr) + \frac{\omega}{0.3978}\left[X(Tr)X(Tr)\right]$$

$$X(Tr) = X_{\omega=0} + \omega\, X_{corr}(Tr)$$

Correction factor X_{corr}: **Figure I4 (2/6)**

- SINGLE PHASE PROPERTIES

From I8-3 to I8-10 : compressibility factor

$$Z = Z_0 + \frac{\omega}{\omega_r}(Zr - Z_0) = Z_0 + \omega\, Z_{corr}$$

$$Z(Tr, Pr) = Z_0(Tr, Pr) + \omega \,.\, Z_{corr}(Tr, Pr)$$

$Z_0(Tr, Pr)$: **Figure I4 (3/6)**

$Z_{corr} = \frac{Z_r - Z_0}{\omega_r}$: **Figures I4 (4/6)**

From I4 (5/6) to I4 (6/6) : residual enthalpy (dimensionless)

$$\frac{\Delta h^{res}}{RT_c} = \frac{h - h^{\#}}{RT_c}(Tr, Pr) = \left(\frac{h - h^{\#}}{RT_c}\right)_0 (Tr, Pr) + \omega \left(\frac{h - h^{\#}}{RT_c}\right)_{corr} (Tr, Pr)$$

$\left(\frac{h - h^{\#}}{RT_c}\right)_0$ simple fluid $\omega = 0$: **Figure I4 (5/6)**

$\left(\frac{h - h^{\#}}{RT_c}\right)_{corr}$ reference fluid $\omega = 0.3978$: **Figure I4 (6/6)**

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SATURATION PROPERTIES

Acentric factor $\omega = 0$

T_r	P_r sat	Z_{liquid}	Z_{vapor}	- ln (f/P)	$-\Delta h_{liquid}$	$-\Delta h_{vapor}$
0.50	0.004573	0.0009452	0.9882	0.01172	5.465	0.01744
0.55	0.01236	0.002412	0.9757	0.02405	5.315	0.03932
0.60	0.02794	0.005190	0.9567	0.04246	5.163	0.07606
0.65	0.05527	0.009852	0.9307	0.06720	5.010	0.1318
0.70	0.09879	0.01704	0.8972	0.09808	4.853	0.2106
0.75	0.1632	0.02747	0.8559	0.1347	4.688	0.3175
0.80	0.2536	0.04205	0.8059	0.1767	4.509	0.4592
0.85	0.3752	0.06211	0.7455	0.2237	4.308	0.6467
0.90	0.5339	0.08999	0.6703	0.2761	4.066	0.9021
0.92	0.6095	0.1044	0.6339	0.2987	3.949	1.034
0.94	0.6927	0.1216	0.5919	0.3225	3.814	1.192
0.96	0.7846	0.1433	0.5410	0.3479	3.647	1.393
0.98	0.8863	0.1741	0.4722	0.3753	3.410	1.679
1.00	1.0000	0.2918	0.2918	0.4064	2.574	2.574

Correction factor (Lee & Kesler (1975) A I Ch E J 21(3). 510-527)

T_r	P_r sat	Z_{liquid}	Z_{vapor}	- ln (f/P)	$-\Delta h_{liquid}$	$-\Delta h_{vapor}$
0.50	- 0.01065	- 0.002232	0.02590	- 0.02575	8.868	- 0.03709
0.55	- 0.02688	- 0.005388	0.04796	- 0.04734	8.213	- 0.07280
0.60	- 0.05514	- 0.01074	0.07394	- 0.07209	7.570	- 0.1163
0.65	- 0.09621	- 0.01848	0.09913	- 0.09478	6.952	- 0.1567
0.70	- 0.1472	- 0.02838	0.1190	- 0.1105	6.362	- 0.1803
0.75	- 0.2014	- 0.03980	0.1303	- 0.1158	5.800	- 0.1726
0.80	- 0.2484	- 0.05177	0.1310	- 0.1088	5.264	- 0.1181
0.85	- 0.2744	- 0.06308	0.1196	- 0.08869	4.745	0.002705
0.90	- 0.2620	- 0.07228	0.09498	- 0.05510	4.231	0.2217
0.92	- 0.2412	- 0.07502	0.08075	- 0.03763	4.021	0.3507
0.94	- 0.2086	- 0.07712	0.06338	- 0.01754	3.803	0.5161
0.96	- 0.1615	- 0.07865	0.04176	0.005714	3.570	0.7381
0.98	- 0.09571	- 0.08057	0.01249	0.03347	3.301	1.073
1.00	0.0000	- 0.07887	- 0.07887	0.07189	2.382	2.382

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COMPRESSIBILITY FACTOR Z_0 (T_r, P_r)
— Simple fluid —

Tr ↓ / Pr →	0.01	0.05	0.1	0.2	0.4	0.6	0.8	1	1.2	1.5	2	3	5	7	10
0.3	0.0029	0.0145	0.0290	0.0579	0.1158	0.1737	0.2315	0.2893	0.3470	0.4336	0.5776	0.8649	1.4368	2.0051	2.8511
0.35	0.0026	0.0130	0.0261	0.0522	0.1043	0.1564	0.2084	0.2604	0.3123	0.3901	0.5196	0.7776	1.2903	1.7988	2.5542
0.4	0.0024	0.0119	0.0239	0.0477	0.0954	0.1429	0.1905	0.2379	0.2853	0.3564	0.4744	0.7095	1.1759	1.6374	2.3213
0.45	0.0022	0.0110	0.0221	0.0442	0.0882	0.1322	0.1762	0.2201	0.2639	0.3295	0.4384	0.6552	1.0842	1.5078	2.1340
0.5	0.0021	0.0103	0.0207	0.0413	0.0825	0.1236	0.1647	0.2056	0.2465	0.3077	0.4093	0.6110	1.0094	1.4018	1.9802
0.55	0.9804	0.0098	0.0195	0.0390	0.0778	0.1166	0.1553	0.1939	0.2323	0.2899	0.3854	0.5747	0.9476	1.3137	1.8521
0.6	0.9849	0.0093	0.0186	0.0371	0.0741	0.1109	0.1476	0.1843	0.2208	0.2753	0.3657	0.5446	0.8960	1.2399	1.7440
0.65	0.9881	0.9377	0.0178	0.0356	0.0710	0.1063	0.1415	0.1765	0.2113	0.2634	0.3495	0.5197	0.8526	1.1773	1.6520
0.7	0.9904	0.9504	0.0172	0.0344	0.0687	0.1027	0.1366	0.1703	0.2038	0.2538	0.3364	0.4991	0.8161	1.1241	1.5730
0.75	0.9922	0.9598	0.9165	0.0336	0.0670	0.1001	0.1330	0.1656	0.1981	0.2464	0.3260	0.4823	0.7854	1.0787	1.5048
0.8	0.9935	0.9669	0.9319	0.8539	0.0661	0.0985	0.1307	0.1626	0.1942	0.2411	0.3183	0.4690	0.7598	1.0400	1.4457
0.85	0.9946	0.9725	0.9436	0.8810	0.0661	0.0984	0.1301	0.1614	0.1924	0.2382	0.3132	0.4591	0.7388	1.0071	1.3944
0.9	0.9954	0.9768	0.9528	0.9015	0.7800	0.1006	0.1321	0.1630	0.1935	0.2383	0.3114	0.4527	0.7220	0.9793	1.3497
0.93	0.9959	0.9790	0.9573	0.9115	0.8059	0.6636	0.1359	0.1664	0.1964	0.2405	0.3123	0.4507	0.7139	0.9649	1.3258
0.95	0.9961	0.9803	0.9600	0.9174	0.8207	0.6968	0.1410	0.1705	0.1999	0.2432	0.3138	0.4502	0.7093	0.9561	1.3109
0.97	0.9963	0.9815	0.9625	0.9228	0.8338	0.7240	0.5582	0.1780	0.2055	0.2474	0.3165	0.4504	0.7053	0.9481	1.2969
0.98	0.9965	0.9821	0.9637	0.9253	0.8398	0.7360	0.5888	0.1844	0.2097	0.2503	0.3182	0.4508	0.7035	0.9443	1.2901
0.99	0.9966	0.9826	0.9648	0.9277	0.8455	0.7471	0.6139	0.1961	0.2155	0.2539	0.3204	0.4514	0.7019	0.9407	1.2836
1	0.9967	0.9832	0.9659	0.9300	0.8509	0.7575	0.6354	0.3210	0.2238	0.2584	0.3230	0.4523	0.7005	0.9372	1.2773
1.01	0.9968	0.9837	0.9669	0.9322	0.8561	0.7671	0.6542	0.4652	0.2372	0.2641	0.3261	0.4534	0.6992	0.9339	1.2711
1.02	0.9969	0.9842	0.9679	0.9343	0.8610	0.7762	0.6711	0.5148	0.2632	0.2715	0.3297	0.4547	0.6981	0.9308	1.2651
1.05	0.9971	0.9855	0.9707	0.9401	0.8744	0.8002	0.7131	0.6027	0.4441	0.3133	0.3453	0.4605	0.6957	0.9223	1.2482
1.1	0.9975	0.9874	0.9747	0.9485	0.8930	0.8324	0.7649	0.6880	0.5986	0.4583	0.3955	0.4771	0.6951	0.9110	1.2233
1.15	0.9978	0.9891	0.9780	0.9554	0.9081	0.8576	0.8032	0.7443	0.6804	0.5799	0.4763	0.5044	0.6988	0.9034	1.2022
1.2	0.9981	0.9904	0.9808	0.9611	0.9205	0.8779	0.8330	0.7858	0.7364	0.6606	0.5607	0.5427	0.7071	0.8991	1.1845
1.3	0.9985	0.9926	0.9852	0.9702	0.9396	0.9083	0.8764	0.8439	0.8111	0.7625	0.6910	0.6347	0.7359	0.9000	1.1581
1.4	0.9988	0.9942	0.9884	0.9768	0.9534	0.9298	0.9063	0.8827	0.8595	0.8257	0.7754	0.7204	0.7764	0.9114	1.1420
1.5	0.9991	0.9954	0.9909	0.9818	0.9636	0.9456	0.9278	0.9104	0.8933	0.8689	0.8328	0.7888	0.8203	0.9299	1.1341
1.6	0.9993	0.9964	0.9928	0.9856	0.9714	0.9575	0.9440	0.9308	0.9181	0.9001	0.8739	0.8411	0.8619	0.9520	1.1322
1.7	0.9994	0.9971	0.9943	0.9886	0.9775	0.9667	0.9563	0.9463	0.9367	0.9234	0.9043	0.8810	0.8986	0.9748	1.1345
1.8	0.9995	0.9977	0.9955	0.9910	0.9823	0.9739	0.9659	0.9583	0.9511	0.9413	0.9276	0.9118	0.9298	0.9963	1.1394
1.9	0.9996	0.9982	0.9964	0.9929	0.9861	0.9796	0.9735	0.9678	0.9624	0.9553	0.9456	0.9360	0.9558	1.0159	1.1454
2	0.9997	0.9986	0.9972	0.9944	0.9892	0.9842	0.9796	0.9754	0.9715	0.9664	0.9599	0.9551	0.9773	1.0330	1.1518
2.2	0.9998	0.9992	0.9983	0.9967	0.9937	0.9910	0.9886	0.9865	0.9847	0.9826	0.9806	0.9828	1.0095	1.0602	1.1637
2.4	0.9999	0.9996	0.9991	0.9983	0.9969	0.9957	0.9948	0.9941	0.9936	0.9935	0.9945	1.0012	1.0314	1.0795	1.1730
2.6	1.0000	0.9998	0.9997	0.9994	0.9991	0.9990	0.9990	0.9993	0.9998	1.0010	1.0040	1.0137	1.0463	1.0928	1.1794
2.8	1.0000	1.0000	1.0001	1.0002	1.0007	1.0013	1.0021	1.0031	1.0042	1.0063	1.0106	1.0224	1.0565	1.1017	1.1832
3	1.0000	1.0002	1.0004	1.0008	1.0018	1.0030	1.0043	1.0058	1.0074	1.0101	1.0154	1.0285	1.0635	1.1076	1.1850
3.5	1.0001	1.0004	1.0008	1.0017	1.0035	1.0055	1.0075	1.0097	1.0120	1.0156	1.0221	1.0369	1.0724	1.1139	1.1835
4	1.0001	1.0005	1.0010	1.0021	1.0043	1.0066	1.0090	1.0115	1.0140	1.0179	1.0249	1.0401	1.0747	1.1137	1.1774

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COMPRESSIBILITY FACTOR Z_{corr} (T_r. P_r)
— Heavy fluid for reference —

Tr ↓ / Pr →	0.01	0.05	0.1	0.2	0.4	0.6	0.8	1	1.2	1.5	2	3	5	7	10
0.3	-0.00081106	-0.00405495	-0.00810904	-0.0162146	-0.03241536	-0.04860233	-0.0647756	-0.08093523	-0.0970813	-0.12127513	-0.16153135	-0.24179701	-0.40137019	-0.55971835	-0.79507224
0.35	-0.00092948	-0.00464645	-0.0092906	-0.01857201	-0.03710742	-0.05560661	-0.07406993	-0.09249773	-0.11089038	-0.13841416	-0.18411609	-0.27489663	-0.45409514	-0.63036851	-0.88980687
0.4	-0.00095708	-0.00478395	-0.00956424	-0.01911387	-0.03816973	-0.05716838	-0.07611059	-0.09499712	-0.1138287	-0.14197464	-0.18861988	-0.28095517	-0.46206881	-0.63887933	-0.89698764
0.45	-0.00094239	-0.00470995	-0.00941485	-0.01880964	-0.03753972	-0.05619166	-0.07476684	-0.0932666	-0.11169222	-0.13919443	-0.1846782	-0.27438455	-0.44918939	-0.61855394	-0.86387256
0.5	-0.00090878	-0.00454126	-0.00907598	-0.018126	-0.03614935	-0.05407239	-0.07189739	-0.08962653	-0.10726192	-0.13354364	-0.17690548	-0.26207434	-0.42685409	-0.58524782	-0.81290807
0.55	-2.46427864	-0.00433644	-0.00866463	-0.01729653	-0.03446405	-0.05150637	-0.06842708	-0.08522963	-0.10191731	-0.12674027	-0.16758006	-0.24741308	-0.4006452	-0.54670967	-0.75503331
0.6	-2.47572874	-0.00412616	-0.00824195	-0.01644286	-0.03272471	-0.04885166	-0.06482941	-0.08066338	-0.09635862	-0.11965179	-0.15784432	-0.23208454	-0.37331614	-0.50675324	-0.69562056
0.65	-0.01376047	-2.34225725	-0.00784235	-0.01563273	-0.03106286	-0.04630041	-0.06135459	-0.07623395	-0.09094638	-0.11271791	-0.14826593	-0.21690605	-0.34618604	-0.4672077	-0.63725252
0.7	-0.00934361	-2.37480233	-0.00748921	-0.01491086	-0.02956067	-0.04396668	-0.05814434	-0.07210757	-0.08586894	-0.10615721	-0.13910955	-0.20223257	-0.3198152	-0.42886622	-0.58107514
0.75	-0.00639159	-0.0339821	-2.27227795	-0.0143209	-0.02829005	-0.04193967	-0.05529744	-0.06838732	-0.08123023	-0.10007105	-0.13046889	-0.18815089	-0.29433878	-0.39196218	-0.52748073
0.8	-0.00436889	-0.0228562	-0.04878852	-2.08013419	-0.02735777	-0.04033453	-0.05292192	-0.06516495	-0.07710069	-0.09449438	-0.12232206	-0.17456111	-0.26962863	-0.35641317	-0.4764635
0.85	-0.00295404	-0.01526683	-0.03197388	-0.07171179	-0.02702313	-0.03939809	-0.05121793	-0.06257952	-0.07355458	-0.0894089	-0.11452519	-0.16117877	-0.24536123	-0.32194648	-0.42780974
0.9	-0.00194716	-0.00996446	-0.02056816	-0.04430984	-0.11217749	-0.0399744	-0.05075806	-0.06095586	-0.07070766	-0.08468861	-0.10670065	-0.14744579	-0.22102763	-0.28816306	-0.38120263
0.93	-0.00148355	-0.00755154	-0.01546974	-0.0326867	-0.07659346	-1.44981368	-0.05197384	-0.06082483	-0.06940408	-0.08183463	-0.10161299	-0.13862082	-0.20610446	-0.26801954	-0.35407181
0.95	-0.00122024	-0.00618939	-0.01261717	-0.02633954	-0.05921394	-0.11172847	-0.05478747	-0.06146712	-0.06870285	-0.07971355	-0.0977813	-0.13227045	-0.19590213	-0.25457112	-0.33628287
0.97	-0.00098796	-0.00499293	-0.01012702	-0.02088748	-0.04520802	-0.07755742	-0.16743431	-0.06346797	-0.06800354	-0.07705345	-0.09326922	-0.12535242	-0.18541683	-0.24106411	-0.31870553
0.98	-0.00088218	-0.00444973	-0.00900131	-0.01844956	-0.03918963	-0.06461605	-0.11164793	-0.0657373	-0.06746603	-0.07528126	-0.09060259	-0.12160971	-0.18004396	-0.23427676	-0.30998916
0.99	-0.00078272	-0.00393988	-0.0079475	-0.01618232	-0.03371786	-0.0535378	-0.08089076	-0.0707713	-0.06639197	-0.07293377	-0.08753009	-0.11763002	-0.17457035	-0.22746043	-0.30131733
1	-0.00068915	-0.00346111	-0.00696042	-0.01407177	-0.02872754	-0.04392215	-0.05977077	-0.28330419	-0.06347464	-0.06956686	-0.08390175	-0.11336955	-0.16898487	-0.22061028	-0.29268738
1.01	-0.0006011	-0.00301135	-0.00603534	-0.0121053	-0.02416376	-0.03548481	-0.04378834	-0.02662683	-0.05214948	-0.06428842	-0.07950416	-0.10877704	-0.16327589	-0.2137215	-0.28409672
1.02	-0.00051821	-0.00258866	-0.00516794	-0.01027154	-0.01997985	-0.02801653	-0.03101261	-0.00863662	0.01169407	-0.05519176	-0.07403178	-0.10379274	-0.15743131	-0.20678929	-0.27554286
1.05	-0.00029738	-0.00146589	-0.00287341	-0.00546819	-0.00933378	-0.01000701	-0.00374356	0.02081745	0.10253255	0.03931642	-0.04597728	-0.08575013	-0.13895762	-0.18568523	-0.25007836
1.1	-0.00000620	0.00000658	0.00011375	0.00067968	0.00366766	0.0103764	0.0232904	0.04694065	0.08854859	0.16018229	0.06588502	-0.03994151	-0.10426831	-0.14922965	-0.20816953
1.15	0.00021141	0.00110043	0.00231515	0.00512978	0.01265659	0.02359516	0.0394032	0.06206114	0.09362719	0.15339039	0.16380785	0.03006729	-0.063583	-0.11081947	-0.16675229
1.2	0.00037456	0.00191634	0.00394602	0.00837763	0.01898671	0.03250133	0.04974469	0.07160395	0.09868329	0.1468461	0.19714219	0.10652323	-0.01678995	-0.07033519	-0.12572109
1.3	0.00058877	0.00298053	0.00605507	0.01250176	0.02669696	0.0428383	0.0611474	0.08174644	0.1045397	0.14160215	0.19816352	0.20585477	0.08471678	0.0149168	-0.0450201
1.4	0.0007077	0.0035662	0.00720252	0.01469112	0.03056663	0.04768611	0.06606002	0.08562041	0.10618405	0.1381178	0.18885976	0.23837025	0.17128673	0.09816643	0.0322834
1.5	0.00077058	0.00387253	0.00779427	0.01578583	0.03235912	0.04969739	0.06774074	0.08638101	0.10545119	0.13433032	0.18032679	0.24245288	0.22893273	0.16928322	0.10319137
1.6	0.0007998	0.00401216	0.00805702	0.01624292	0.03298143	0.05016209	0.0677071	0.08551107	0.1034388	0.13019227	0.17268259	0.23752943	0.26161041	0.22334919	0.16481514
1.7	0.00080854	0.00405094	0.00812212	0.01632225	0.03293016	0.04976221	0.06674223	0.08377896	0.10076641	0.12589049	0.16568561	0.23005259	0.27762918	0.26103216	0.21559592
1.8	0.00080478	0.00402847	0.00806793	0.01617667	0.03248936	0.04887838	0.06527578	0.08160586	0.09778612	0.12158202	0.15920708	0.22206144	0.28369373	0.28555185	0.25555142
1.9	0.00079343	0.00396903	0.00794228	0.01589843	0.03182674	0.04773094	0.06355277	0.07923051	0.09469982	0.11736882	0.15317969	0.21422482	0.28409688	0.30042943	0.2857974
2	0.00077758	0.0038878	0.00777491	0.0155443	0.03104356	0.04645054	0.06171626	0.07679066	0.09162332	0.11331004	0.14756041	0.20676234	0.28135704	0.30858433	0.30792871
2.2	0.00073949	0.00369489	0.00738303	0.01473691	0.02933979	0.04377411	0.05800591	0.07200227	0.08573163	0.10576039	0.13741507	0.1931436	0.27157354	0.31270876	0.33420177
2.4	0.00069914	0.00349188	0.00697402	0.01390744	0.02763971	0.04117202	0.05448095	0.06754457	0.08034263	0.09900199	0.12853952	0.18117891	0.25990933	0.30839415	0.34488924
2.6	0.00066002	0.00329576	0.00658045	0.01311542	0.02603968	0.03875502	0.05124513	0.06349519	0.07549194	0.09298695	0.12073557	0.17064155	0.24821416	0.30047689	0.34674435
2.8	0.00062348	0.00311285	0.00621424	0.01238172	0.02456958	0.03655081	0.04831391	0.05984874	0.07114646	0.08763232	0.11383425	0.1613066	0.23710442	0.29116903	0.34370431
3	0.00058989	0.00294493	0.00587848	0.01171083	0.02323209	0.03455449	0.04566987	0.05657119	0.0672525	0.0828515	0.10769448	0.15298321	0.22675963	0.28149822	0.33800288
3.5	0.00051827	0.0025873	0.00516436	0.01028746	0.02040745	0.03035566	0.04012848	0.04972301	0.05913701	0.07291614	0.09496386	0.13564516	0.20422284	0.25829147	0.31915108
4	0.00046137	0.00230331	0.00459779	0.0091601	0.01817707	0.02704882	0.03577374	0.04435065	0.0527788	0.06514133	0.08500113	0.12197788	0.18572523	0.23781066	0.2992392

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DIMENSIONLESS RESIDUAL ENTHALPY — Simple fluid —

$$\left(\frac{h - h^{\#}}{RT_c}\right)_0$$

Tr ↓ / Pr →	0.01	0.05	0.1	0.2	0.4	0.6	0.8	1	1.2	1.5	2	3	5	7	10
0.3	-6.0551	-6.0528	-6.0498	-6.0439	-6.0320	-6.0202	-6.0083	-5.9965	-5.9846	-5.9668	-5.9370	-5.8774	-5.7576	-5.6372	-5.4555
0.35	-5.9138	-5.9114	-5.9083	-5.9021	-5.8898	-5.8774	-5.8650	-5.8526	-5.8402	-5.8216	-5.7905	-5.7282	-5.6027	-5.4763	-5.2852
0.4	-5.7688	-5.7662	-5.7631	-5.7567	-5.7440	-5.7313	-5.7186	-5.7059	-5.6931	-5.6739	-5.6419	-5.5775	-5.4478	-5.3168	-5.1185
0.45	-5.6196	-5.6170	-5.6138	-5.6074	-5.5945	-5.5816	-5.5687	-5.5557	-5.5427	-5.5233	-5.4906	-5.4250	-5.2923	-5.1580	-4.9540
0.5	-5.4688	-5.4662	-5.4630	-5.4566	-5.4437	-5.4308	-5.4178	-5.4049	-5.3919	-5.3723	-5.3395	-5.2733	-5.1389	-5.0023	-4.7942
0.55	-0.0317	-5.3154	-5.3122	-5.3059	-5.2933	-5.2806	-5.2678	-5.2550	-5.2421	-5.2227	-5.1901	-5.1242	-4.9894	-4.8516	-4.6408
0.6	-0.0265	-5.1647	-5.1617	-5.1557	-5.1435	-5.1312	-5.1189	-5.1065	-5.0940	-5.0751	-5.0432	-4.9783	-4.8445	-4.7067	-4.4944
0.65	-0.0226	-0.1184	-5.0101	-5.0046	-4.9933	-4.9818	-4.9703	-4.9586	-4.9468	-4.9288	-4.8983	-4.8356	-4.7044	-4.5676	-4.3552
0.7	-0.0196	-0.1014	-4.8550	-4.8502	-4.8404	-4.8303	-4.8201	-4.8095	-4.7988	-4.7824	-4.7542	-4.6950	-4.5682	-4.4338	-4.2227
0.75	-0.0171	-0.0882	-0.1834	-4.6882	-4.6810	-4.6732	-4.6651	-4.6565	-4.6476	-4.6336	-4.6089	-4.5552	-4.4353	-4.3046	-4.0962
0.8	-0.0152	-0.0776	-0.1601	-0.3447	-4.5084	-4.5048	-4.5003	-4.4950	-4.4891	-4.4791	-4.4599	-4.4143	-4.3044	-4.1792	-3.9749
0.85	-0.0135	-0.0689	-0.1414	-0.2995	-4.3095	-4.3141	-4.3166	-4.3174	-4.3167	-4.3134	-4.3032	-4.2701	-4.1743	-4.0566	-3.8582
0.9	-0.0122	-0.0618	-0.1261	-0.2640	-0.5955	-4.0746	-4.0943	-4.1084	-4.1184	-4.1278	-4.1330	-4.1195	-4.0436	-3.9358	-3.7451
0.93	-0.0114	-0.0580	-0.1182	-0.2462	-0.5448	-0.9595	-3.9203	-3.9533	-3.9767	-4.0002	-4.0208	-4.0245	-3.9643	-3.8639	-3.6786
0.95	-0.0110	-0.0558	-0.1134	-0.2354	-0.5158	-0.8847	-3.7636	-3.8256	-3.8656	-3.9046	-3.9400	-3.9585	-3.9108	-3.8159	-3.6349
0.97	-0.0106	-0.0536	-0.1089	-0.2254	-0.4897	-0.8239	-1.3558	-3.6583	-3.7322	-3.7966	-3.8529	-3.8901	-3.8567	-3.7680	-3.5915
0.98	-0.0104	-0.0526	-0.1068	-0.2207	-0.4776	-0.7972	-1.2724	-3.5435	-3.6521	-3.7363	-3.8065	-3.8548	-3.8294	-3.7441	-3.5699
0.99	-0.0102	-0.0516	-0.1047	-0.2162	-0.4660	-0.7725	-1.2060	-3.3755	-3.5577	-3.6704	-3.7579	-3.8188	-3.8020	-3.7201	-3.5484
1	-0.0100	-0.0506	-0.1027	-0.2118	-0.4550	-0.7496	-1.1503	-2.4093	-3.4406	-3.5975	-3.7066	-3.7819	-3.7743	-3.6960	-3.5270
1.01	-0.0098	-0.0497	-0.1008	-0.2076	-0.4445	-0.7281	-1.1021	-1.7944	-3.2823	-3.5157	-3.6525	-3.7441	-3.7464	-3.6720	-3.5056
1.02	-0.0097	-0.0488	-0.0989	-0.2035	-0.4344	-0.7080	-1.0596	-1.6266	-3.0368	-3.4219	-3.5950	-3.7054	-3.7184	-3.6479	-3.4843
1.05	-0.0092	-0.0462	-0.0936	-0.1920	-0.4066	-0.6542	-0.9549	-1.3585	-2.0325	-3.0290	-3.3975	-3.5827	-3.6327	-3.5753	-3.4207
1.1	-0.0084	-0.0424	-0.0857	-0.1750	-0.3669	-0.5812	-0.8270	-1.1198	-1.4869	-2.2015	-2.9641	-3.3530	-3.4845	-3.4530	-3.3155
1.15	-0.0078	-0.0391	-0.0788	-0.1605	-0.3337	-0.5227	-0.7321	-0.9679	-1.2384	-1.7187	-2.4783	-3.0901	-3.3288	-3.3286	-3.2110
1.2	-0.0072	-0.0361	-0.0728	-0.1478	-0.3053	-0.4743	-0.6572	-0.8566	-1.0755	-1.4422	-2.0779	-2.8065	-3.1657	-3.2019	-3.1069
1.3	-0.0062	-0.0312	-0.0627	-0.1268	-0.2595	-0.3985	-0.5446	-0.6982	-0.8596	-1.1155	-1.5597	-2.2729	-2.8242	-2.9421	-2.8989
1.4	-0.0054	-0.0272	-0.0547	-0.1102	-0.2240	-0.3414	-0.4626	-0.5876	-0.7162	-0.9148	-1.2528	-1.8563	-2.4851	-2.6784	-2.6913
1.5	-0.0048	-0.0240	-0.0481	-0.0968	-0.1957	-0.2967	-0.3997	-0.5047	-0.6114	-0.7737	-1.0458	-1.5483	-2.1742	-2.4199	-2.4856
1.6	-0.0043	-0.0213	-0.0427	-0.0856	-0.1725	-0.2606	-0.3497	-0.4398	-0.5305	-0.6673	-0.8941	-1.3173	-1.9030	-2.1758	-2.2844
1.7	-0.0038	-0.0190	-0.0381	-0.0763	-0.1533	-0.2309	-0.3089	-0.3873	-0.4658	-0.5833	-0.7768	-1.1386	-1.6717	-1.9518	-2.0908
1.8	-0.0034	-0.0171	-0.0342	-0.0684	-0.1371	-0.2060	-0.2750	-0.3439	-0.4126	-0.5151	-0.6828	-0.9961	-1.4756	-1.7500	-1.9072
1.9	-0.0031	-0.0154	-0.0308	-0.0616	-0.1232	-0.1848	-0.2462	-0.3074	-0.3682	-0.4584	-0.6054	-0.8795	-1.3090	-1.5700	-1.7354
2	-0.0028	-0.0139	-0.0279	-0.0557	-0.1113	-0.1666	-0.2216	-0.2762	-0.3303	-0.4105	-0.5405	-0.7823	-1.1665	-1.4102	-1.5760
2.2	-0.0023	-0.0115	-0.0230	-0.0460	-0.0916	-0.1368	-0.1815	-0.2257	-0.2694	-0.3337	-0.4373	-0.6288	-0.9368	-1.1421	-1.2941
2.4	-0.0019	-0.0096	-0.0192	-0.0383	-0.0762	-0.1135	-0.1504	-0.1867	-0.2223	-0.2747	-0.3586	-0.5127	-0.7605	-0.9286	-1.0569
2.6	-0.0016	-0.0081	-0.0161	-0.0321	-0.0638	-0.0949	-0.1255	-0.1555	-0.1849	-0.2280	-0.2966	-0.4216	-0.6210	-0.7558	-0.8573
2.8	-0.0014	-0.0068	-0.0136	-0.0270	-0.0536	-0.0796	-0.1051	-0.1300	-0.1544	-0.1900	-0.2464	-0.3481	-0.5080	-0.6137	-0.6882
3	-0.0012	-0.0057	-0.0115	-0.0228	-0.0451	-0.0668	-0.0881	-0.1089	-0.1291	-0.1585	-0.2048	-0.2875	-0.4145	-0.4949	-0.5440
3.5	-0.0007	-0.0037	-0.0074	-0.0147	-0.0289	-0.0427	-0.0560	-0.0689	-0.0813	-0.0992	-0.1267	-0.1739	-0.2388	-0.2692	-0.2632
4	-0.0005	-0.0023	-0.0045	-0.0089	-0.0175	-0.0256	-0.0334	-0.0408	-0.0478	-0.0576	-0.0722	-0.0947	-0.1158	-0.1096	-0.0601

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DIMENSIONLESS RESIDUAL ENTHALPY — Correction factor —

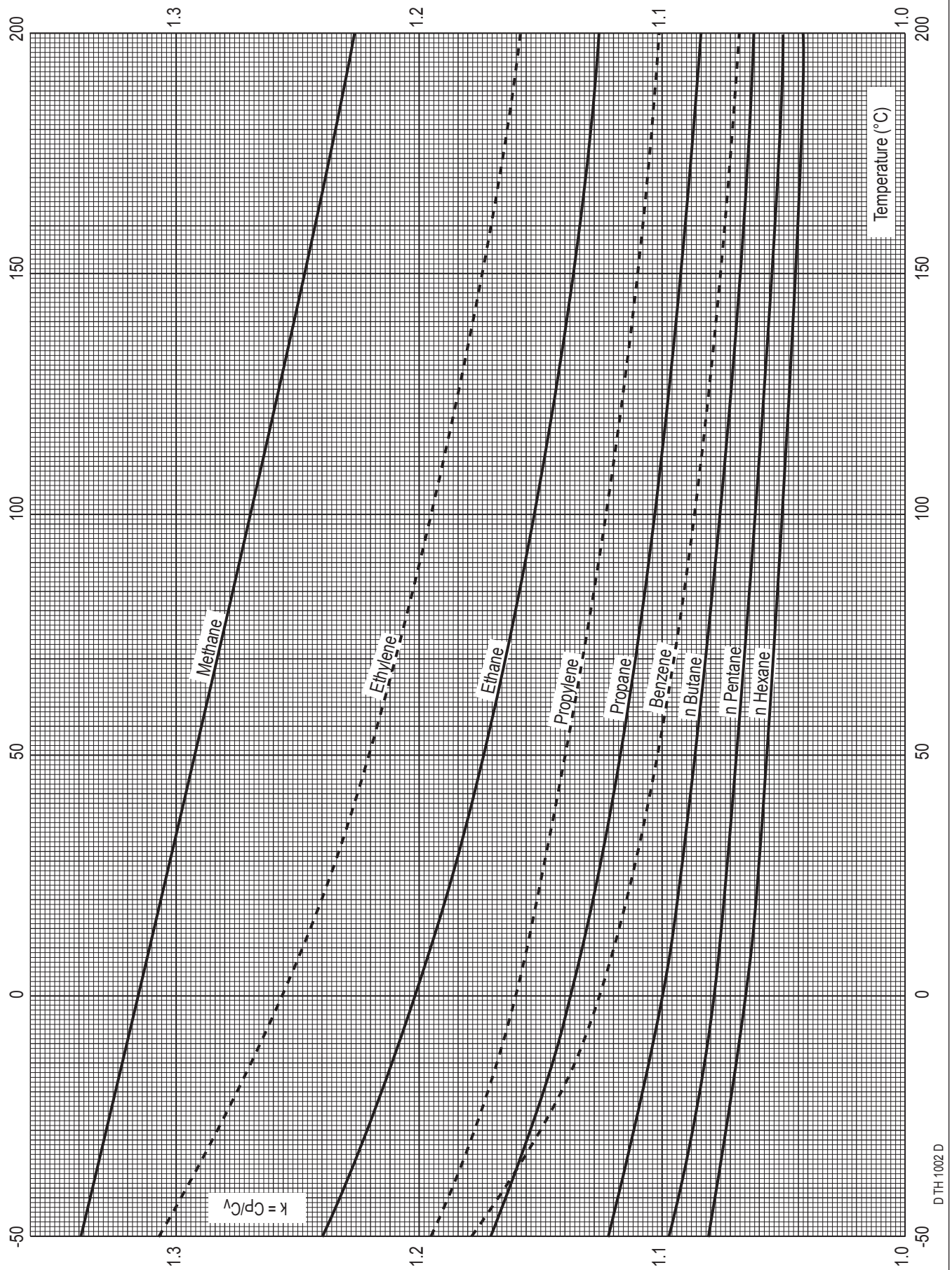
$$\left(\frac{h - h^{\#}}{RT_c}\right)_{corr}$$

Tr → / Pr →	0.01	0.05	0.1	0.2	0.4	0.6	0.8	1	1.2	1.5	2	3	5	7	10
0.3	-10.9589	-10.9575	-10.9558	-10.9523	-10.9455	-10.9386	-10.9318	-10.9250	-10.9182	-10.9081	-10.8913	-10.8582	-10.7933	-10.7303	-10.6389
0.35	-10.5567	-10.5561	-10.5553	-10.5538	-10.5508	-10.5479	-10.5449	-10.5420	-10.5392	-10.5349	-10.5279	-10.5143	-10.4890	-10.4657	-10.4344
0.4	-10.0552	-10.0552	-10.0551	-10.0551	-10.0551	-10.0551	-10.0552	-10.0552	-10.0554	-10.0556	-10.0561	-10.0577	-10.0630	-10.0706	-10.0859
0.45	-9.4697	-9.4701	-9.4707	-9.4718	-9.4740	-9.4763	-9.4786	-9.4810	-9.4834	-9.4871	-9.4934	-9.5068	-9.5359	-9.5678	-9.6197
0.5	-8.8380	-8.8388	-8.8398	-8.8417	-8.8457	-8.8497	-8.8538	-8.8579	-8.8621	-8.8684	-8.8793	-8.9019	-8.9500	-9.0012	-9.0825
0.55	-21.5017	-8.1940	-8.1953	-8.1979	-8.2031	-8.2084	-8.2138	-8.2192	-8.2248	-8.2332	-8.2476	-8.2776	-8.3411	-8.4082	-8.5136
0.6	-20.4985	-7.5577	-7.5592	-7.5622	-7.5683	-7.5745	-7.5809	-7.5874	-7.5939	-7.6040	-7.6212	-7.6572	-7.7336	-7.8142	-7.9400
0.65	-0.0447	-19.2650	-6.9441	-6.9473	-6.9540	-6.9608	-6.9678	-6.9750	-6.9823	-6.9936	-7.0131	-7.0541	-7.1418	-7.2344	-7.3780
0.7	-0.0344	-18.3282	-6.3562	-6.3594	-6.3661	-6.3731	-6.3804	-6.3880	-6.3958	-6.4080	-6.4293	-6.4747	-6.5729	-6.6766	-6.8367
0.75	-0.0269	-0.1418	-17.1474	-5.8001	-5.8061	-5.8127	-5.8199	-5.8275	-5.8355	-5.8483	-5.8711	-5.9207	-6.0296	-6.1444	-6.3203
0.8	-0.0213	-0.1106	-0.2340	-15.7583	-5.2727	-5.2778	-5.2840	-5.2911	-5.2990	-5.3120	-5.3362	-5.3905	-5.5114	-5.6381	-5.8300
0.85	-0.0170	-0.0875	-0.1823	-0.4021	-4.7639	-4.7645	-4.7678	-4.7732	-4.7803	-4.7932	-4.8191	-4.8800	-5.0161	-5.1567	-5.3655
0.9	-0.0137	-0.0700	-0.1441	-0.3082	-0.7525	-4.2691	-4.2623	-4.2627	-4.2676	-4.2803	-4.3097	-4.3820	-4.5402	-4.6978	-4.9255
0.93	-0.0121	-0.0614	-0.1258	-0.2655	-0.6139	-11.3205	-3.9593	-3.9504	-3.9524	-3.9656	-4.0003	-4.0851	-4.2621	-4.4323	-4.6726
0.95	-0.0111	-0.0564	-0.1152	-0.2412	-0.5431	-0.9967	-3.7598	-3.7327	-3.7311	-3.7464	-3.7882	-3.8859	-4.0792	-4.2590	-4.5083
0.97	-0.0102	-0.0518	-0.1056	-0.2196	-0.4839	-0.8395	-1.6270	-3.4970	-3.4893	-3.5115	-3.5673	-3.6843	-3.8978	-4.0885	-4.3474
0.98	-0.0098	-0.0497	-0.1011	-0.2097	-0.4578	-0.7780	-1.3307	-3.3653	-3.3527	-3.3840	-3.4518	-3.5822	-3.8076	-4.0042	-4.2681
0.99	-0.0094	-0.0477	-0.0969	-0.2003	-0.4336	-0.7243	-1.1595	-3.2134	-3.1947	-3.2456	-3.3317	-3.4788	-3.7177	-3.9205	-4.1897
1	-0.0091	-0.0458	-0.0928	-0.1914	-0.4111	-0.6767	-1.0378	-4.0878	-2.9914	-3.0908	-3.2057	-3.3740	-3.6279	-3.8375	-4.1120
1.01	-0.0087	-0.0439	-0.0890	-0.1830	-0.3902	-0.6341	-0.9433	-1.3948	-2.6565	-2.9099	-3.0720	-3.2674	-3.5384	-3.7550	-4.0351
1.02	-0.0083	-0.0421	-0.0853	-0.1750	-0.3707	-0.5957	-0.8658	-1.1909	-1.8208	-2.6854	-2.9284	-3.1588	-3.4490	-3.6731	-3.9589
1.05	-0.0074	-0.0373	-0.0752	-0.1533	-0.3191	-0.4994	-0.6938	-0.8821	-0.8941	-1.5403	-2.4086	-2.8180	-3.1814	-3.4307	-3.7349
1.1	-0.0061	-0.0304	-0.0611	-0.1235	-0.2513	-0.3816	-0.5092	-0.6198	-0.6777	-0.6300	-1.2872	-2.1873	-2.7356	-3.0371	-3.3755
1.15	-0.0050	-0.0248	-0.0498	-0.0997	-0.1995	-0.2967	-0.3867	-0.4611	-0.5063	-0.4931	-0.6182	-1.5171	-2.2911	-2.6559	-3.0327
1.2	-0.0041	-0.0203	-0.0405	-0.0806	-0.1588	-0.2325	-0.2980	-0.3505	-0.3834	-0.3851	-0.3693	-0.9504	-1.8559	-2.2882	-2.7059
1.3	-0.0027	-0.0133	-0.0264	-0.0520	-0.1000	-0.1425	-0.1779	-0.2042	-0.2196	-0.2203	-0.1818	-0.3093	-1.0807	-1.6063	-2.0998
1.4	-0.0017	-0.0084	-0.0165	-0.0321	-0.0602	-0.0834	-0.1010	-0.1121	-0.1164	-0.1097	-0.0725	-0.0499	-0.5154	-1.0251	-1.5605
1.5	-0.0010	-0.0048	-0.0094	-0.0180	-0.0323	-0.0426	-0.0485	-0.0498	-0.0465	-0.0331	0.0060	0.0737	-0.1511	-0.5672	-1.0934
1.6	-0.0004	-0.0022	-0.0041	-0.0075	-0.0120	-0.0132	-0.0111	-0.0057	0.0031	0.0218	0.0642	0.1483	0.0753	-0.2267	-0.7009
1.7	0.0000	-0.0002	-0.0002	0.0003	0.0031	0.0084	0.0163	0.0267	0.0394	0.0624	0.1081	0.2002	0.2181	0.0193	-0.3798
1.8	0.0003	0.0014	0.0029	0.0062	0.0145	0.0248	0.0370	0.0511	0.0668	0.0931	0.1419	0.2392	0.3123	0.1960	-0.1219
1.9	0.0005	0.0026	0.0052	0.0109	0.0233	0.0374	0.0529	0.0698	0.0879	0.1169	0.1684	0.2695	0.3776	0.3243	0.0835
2	0.0007	0.0035	0.0071	0.0145	0.0303	0.0473	0.0654	0.0845	0.1044	0.1356	0.1894	0.2938	0.4253	0.4192	0.2468
2.2	0.0010	0.0049	0.0098	0.0198	0.0403	0.0615	0.0833	0.1056	0.1283	0.1627	0.2204	0.3302	0.4906	0.5471	0.4826
2.4	0.0012	0.0058	0.0116	0.0233	0.0470	0.0710	0.0953	0.1198	0.1443	0.1811	0.2418	0.3560	0.5338	0.6283	0.6396
2.6	0.0013	0.0064	0.0128	0.0257	0.0517	0.0777	0.1037	0.1297	0.1557	0.1942	0.2572	0.3753	0.5655	0.6849	0.7500
2.8	0.0014	0.0069	0.0138	0.0275	0.0550	0.0825	0.1099	0.1370	0.1640	0.2040	0.2689	0.3904	0.5902	0.7275	0.8321
3	0.0014	0.0072	0.0144	0.0289	0.0576	0.0861	0.1145	0.1426	0.1704	0.2115	0.2781	0.4025	0.6105	0.7616	0.8962
3.5	0.0016	0.0078	0.0156	0.0310	0.0618	0.0922	0.1223	0.1520	0.1814	0.2246	0.2945	0.4254	0.6498	0.8258	1.0126
4	0.0016	0.0081	0.0162	0.0324	0.0644	0.0960	0.1273	0.1581	0.1885	0.2334	0.3058	0.4421	0.6799	0.8739	1.0960

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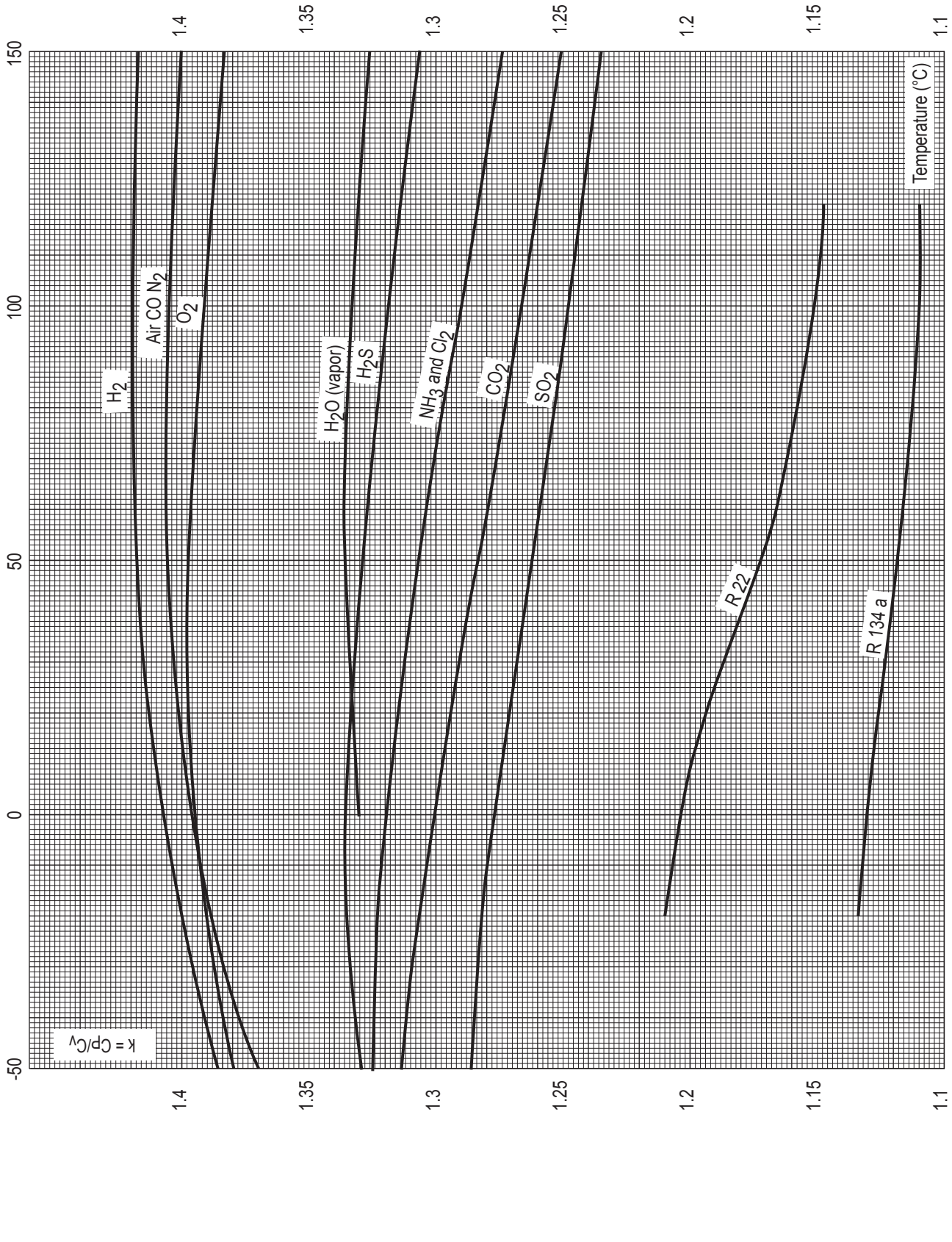
IDEAL GASES ISENTROPIC EXPONENT k
— From API technical data book —

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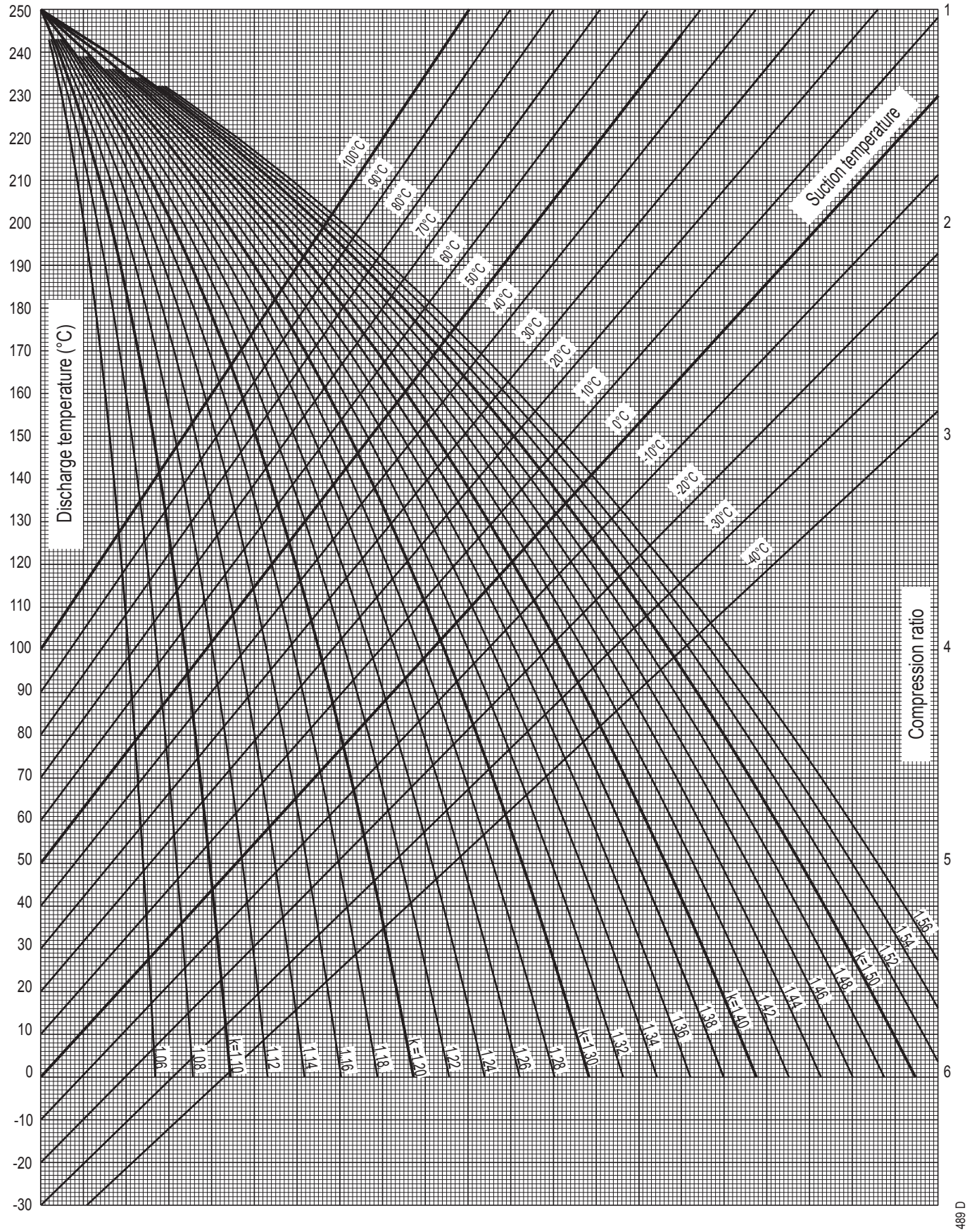
ISENTROPIC EXPONENT k FOR IDEAL GASES — From API technical data book —



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DISCHARCHE TEMPERATURE OF A COMPRESSOR
— Isentropic compression —

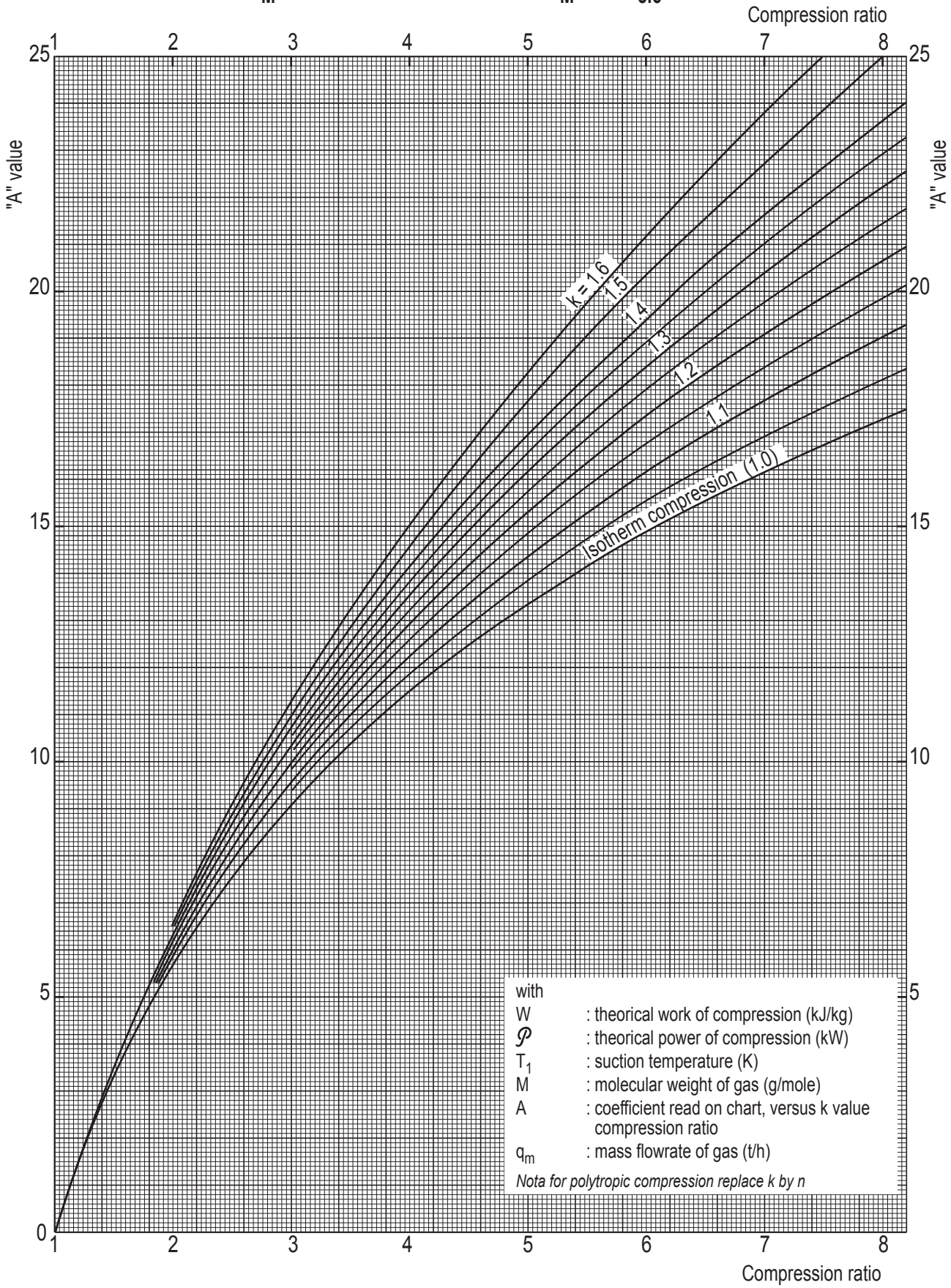


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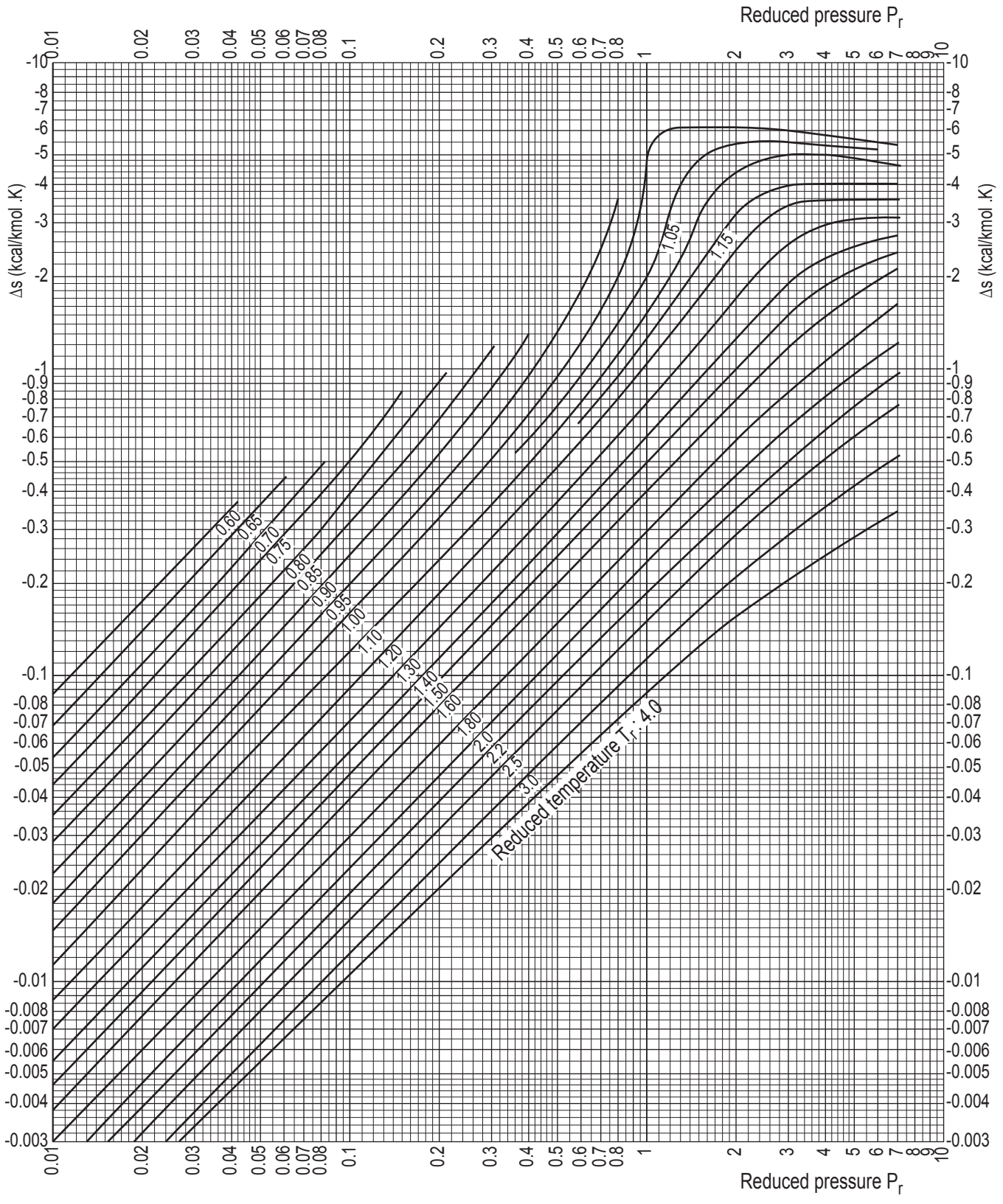
DETERMINATION OF COMPRESSION WORK AND POWER

Work : $W = \frac{T_1}{M} . A$ Power : $\mathcal{P} = \frac{T_1}{M} . A . \frac{q_m}{3.6}$



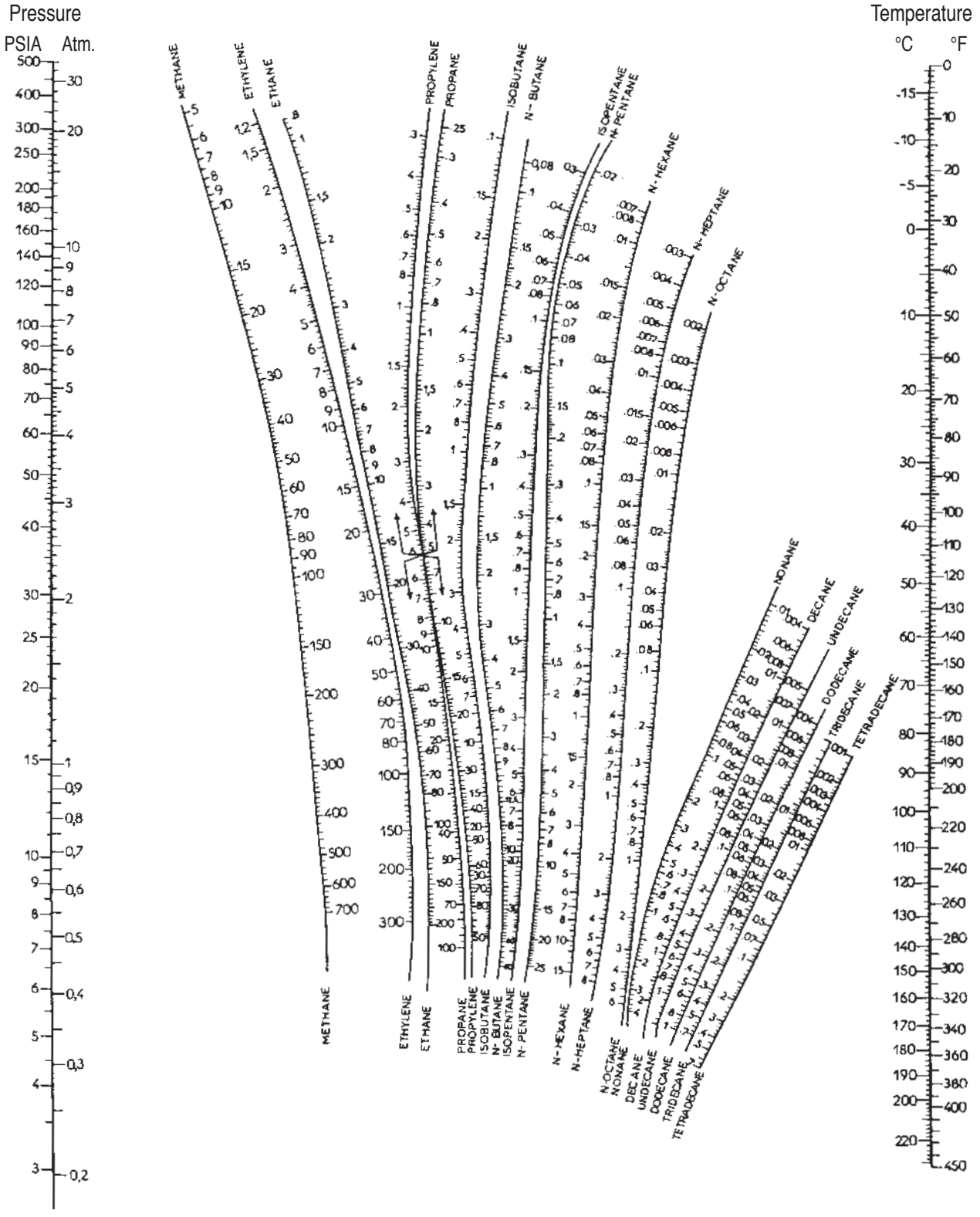
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CORRECTION OF ENTROPY OF IDEAL GAS
— Edmister method —



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EQUILIBRIUM COEFFICIENTS OF HYDROCARBONS
— Scheibel and Jenny chart —

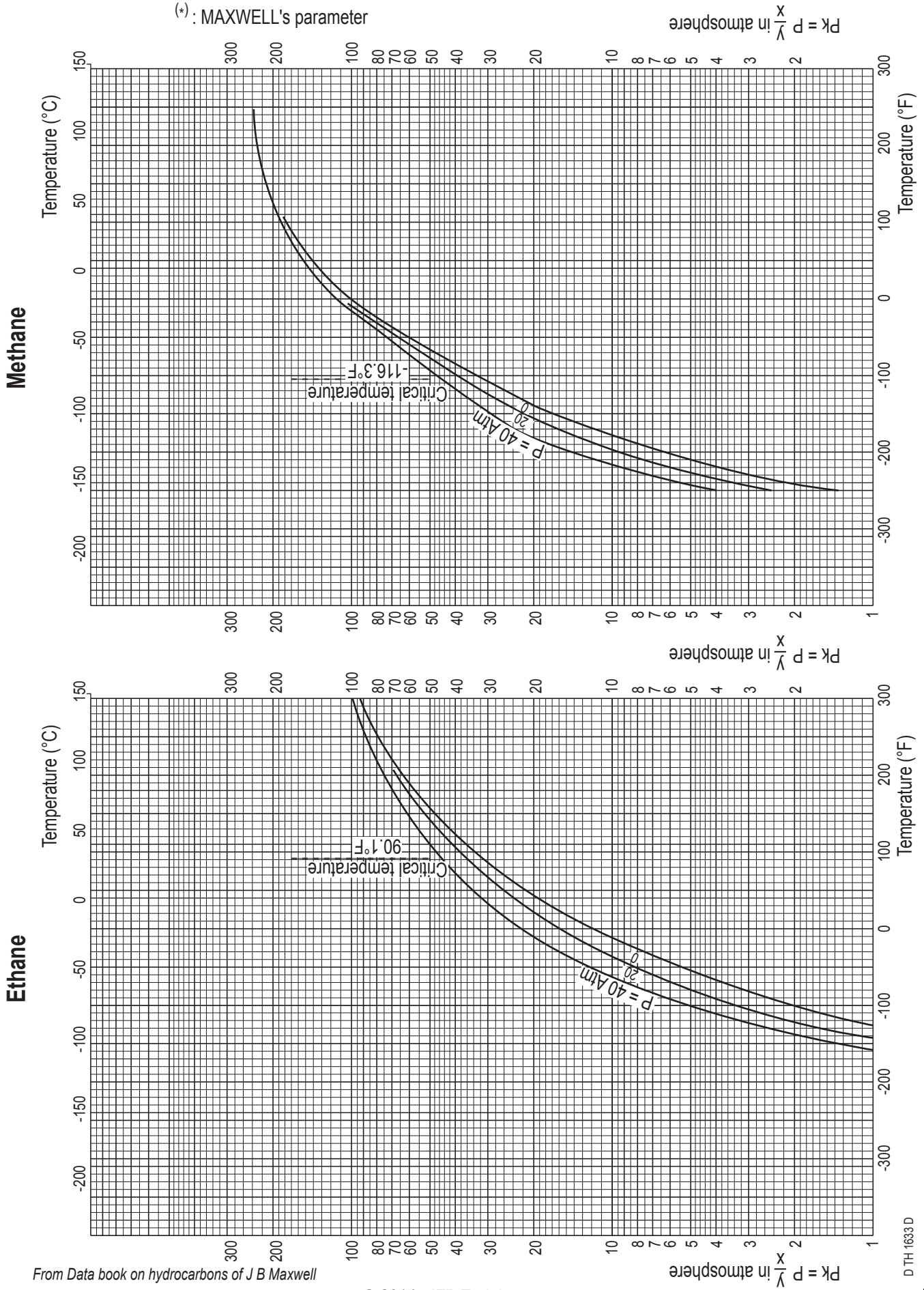


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FUGACITY FUNCTION (*) OF METHANE AND OF ETHANE

(*) : MAXWELL's parameter



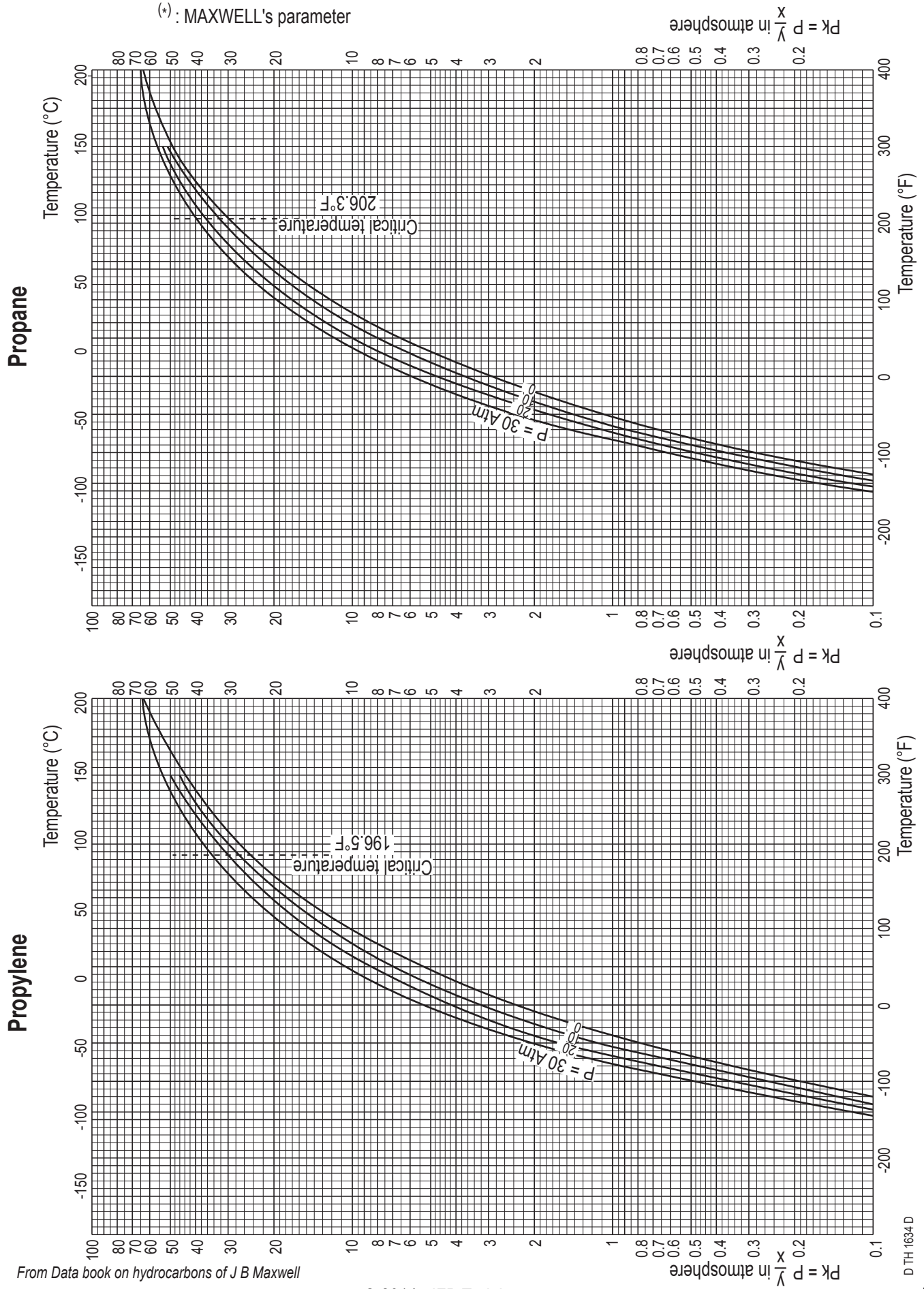
From Data book on hydrocarbons of J B Maxwell

D TH 1633 D

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FUGACITY FUNCTION (*) OF PROPANE AND OF PROPYLENE

(*) : MAXWELL's parameter



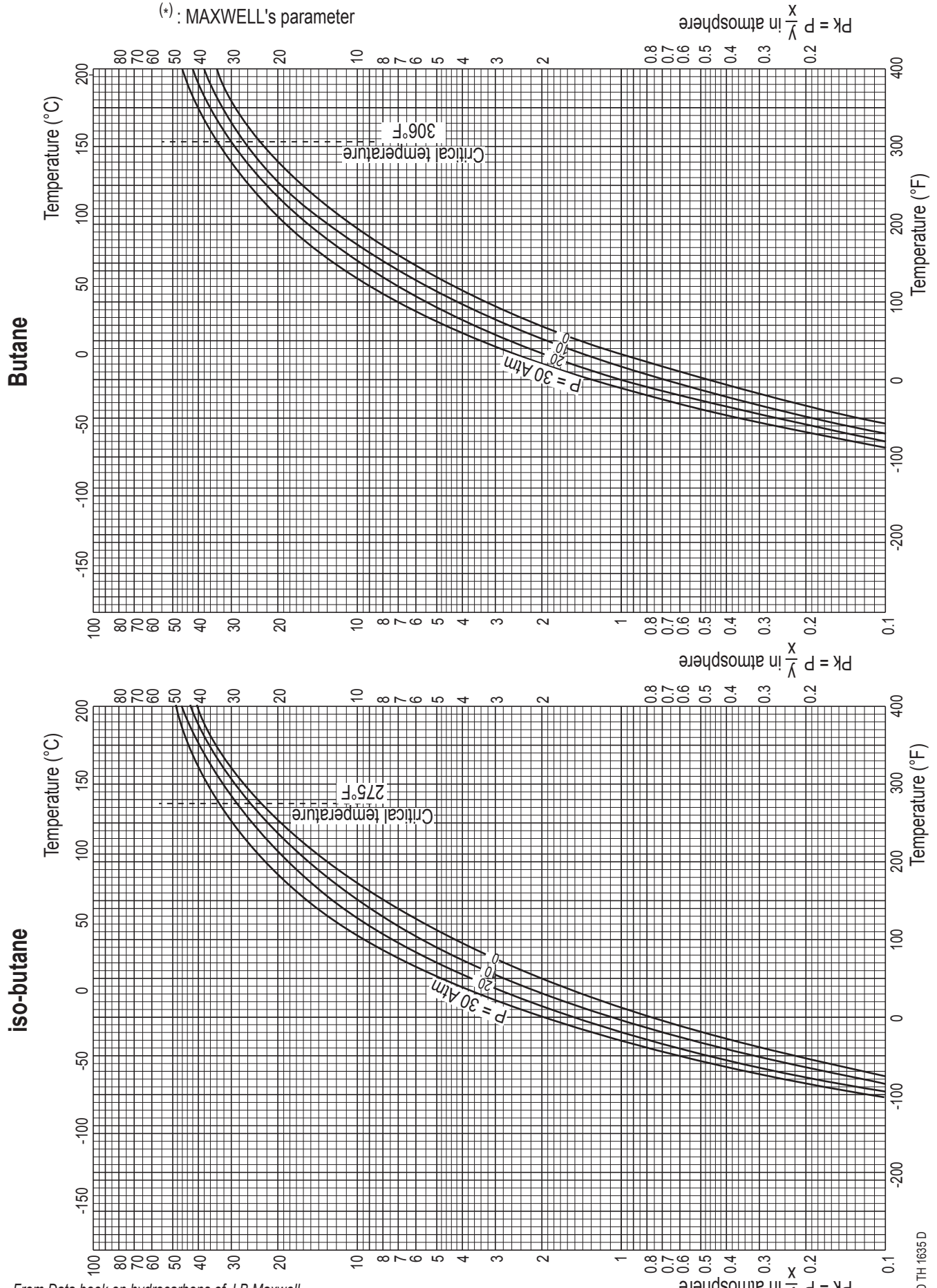
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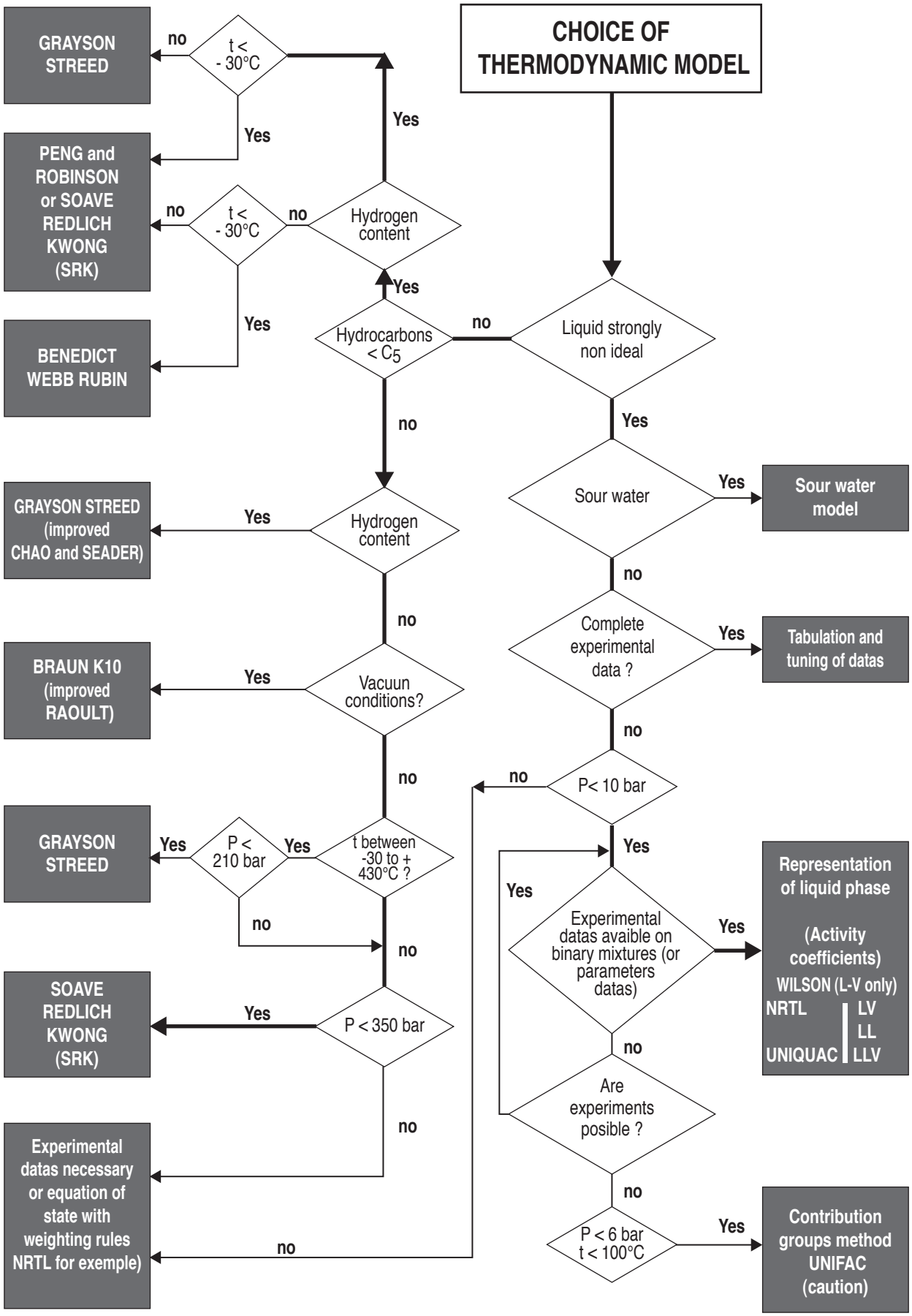
FUGACITY FUNCTION (*) OF BUTANE AND OF ISO-BUTANE

(*) : MAXWELL's parameter



From Data book on hydrocarbons of J B Maxwell

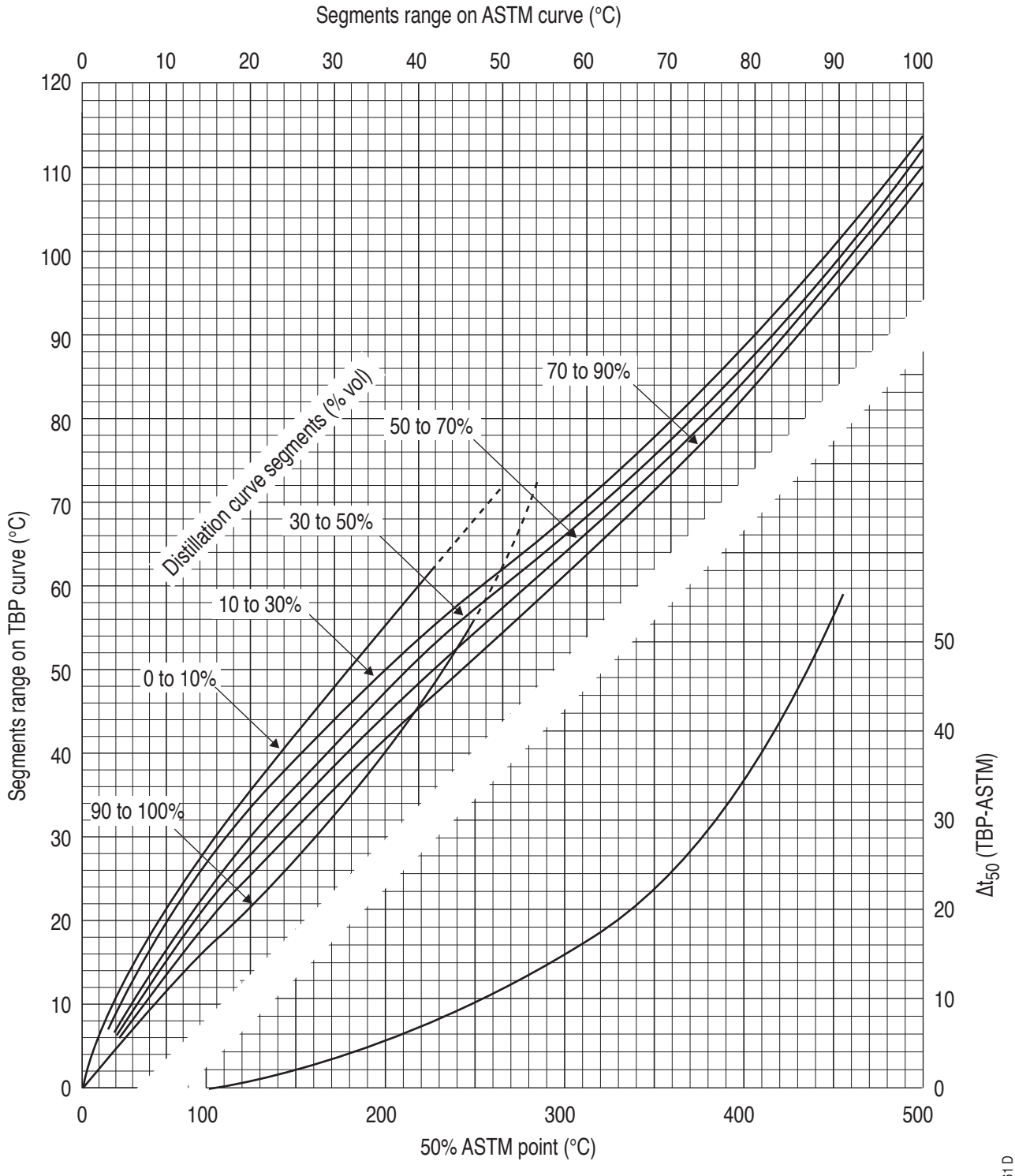
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DTH 1406 D

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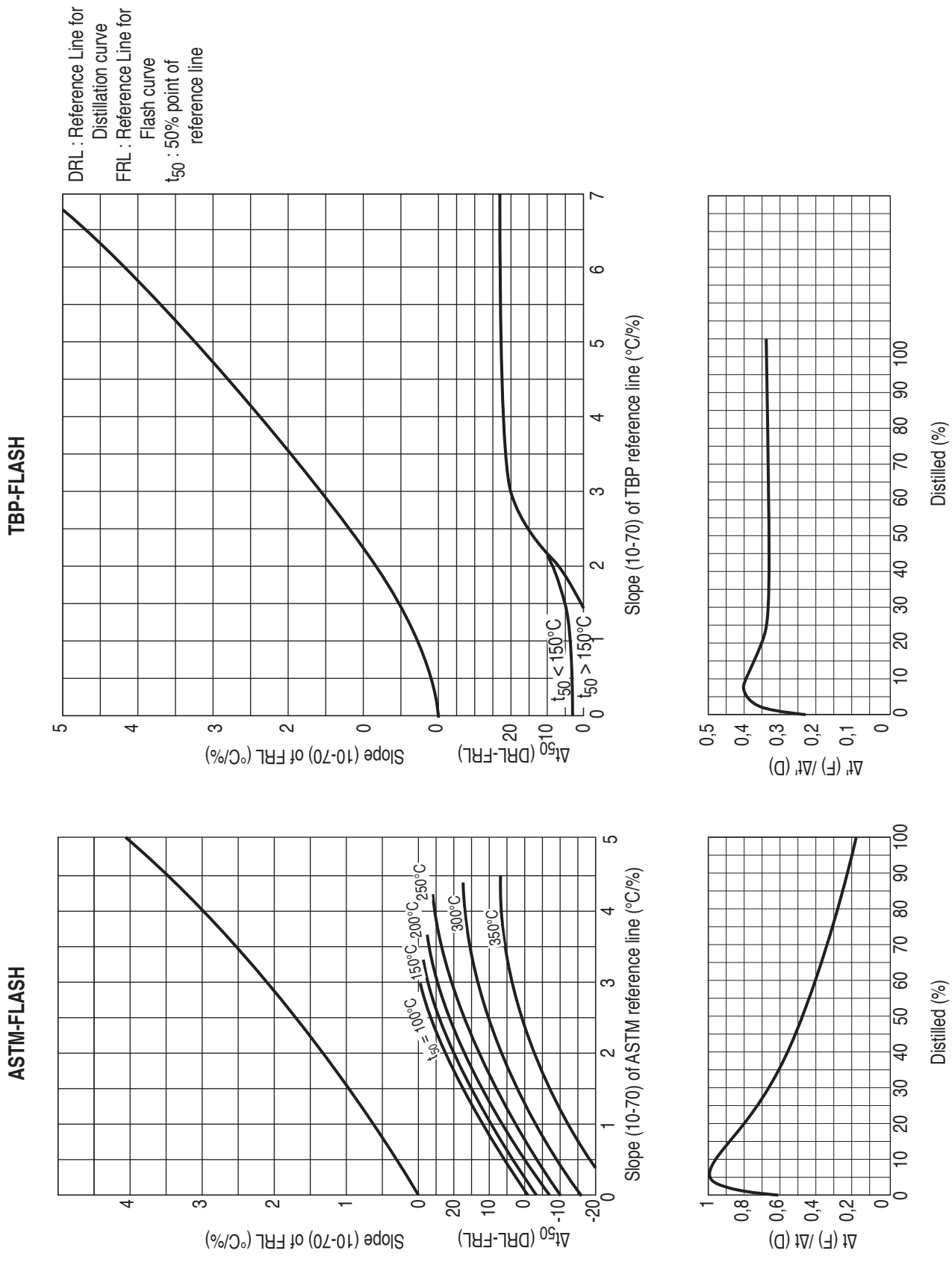
TRANSFORMATION OF TBP TO ASTM DISTILLATION CURVE
— Edmister correlation —



D TH 1161 D

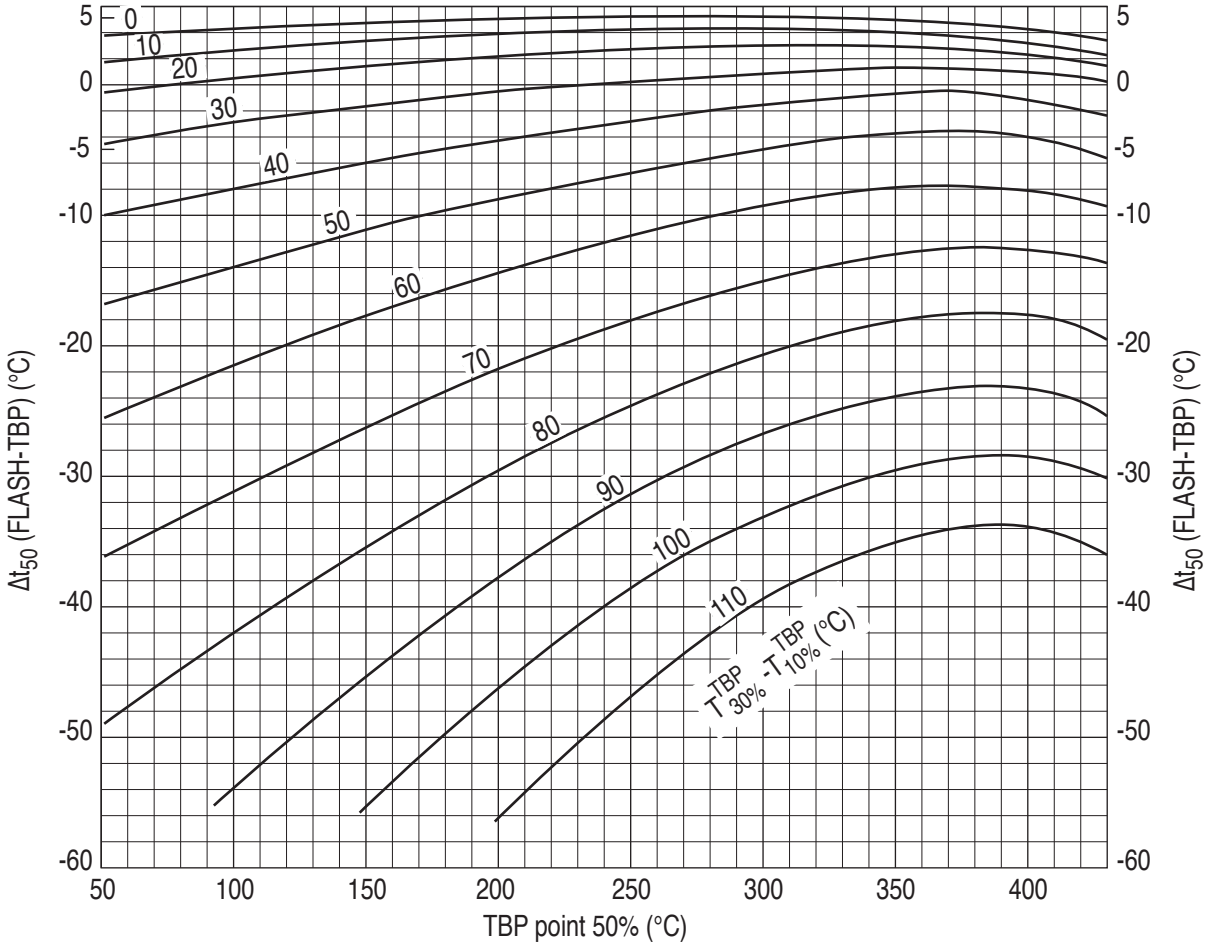
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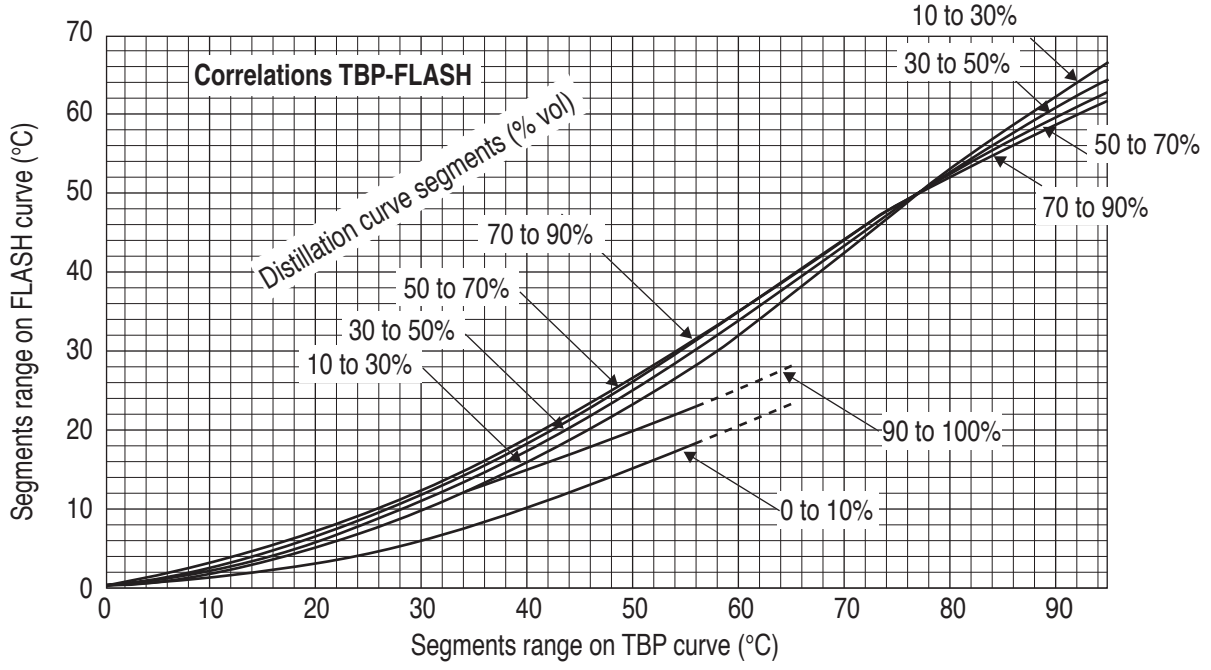
CALCULATION OF THE ATMOSPHERIC FLASH CURVE STARTING FROM THE TBP OR ASTM CURVE
— Maxwell method —



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CALCULATION OF THE ATMOSPHERIC FLASH CURVE STARTING FROM TBP CURVE
— Edmister method —

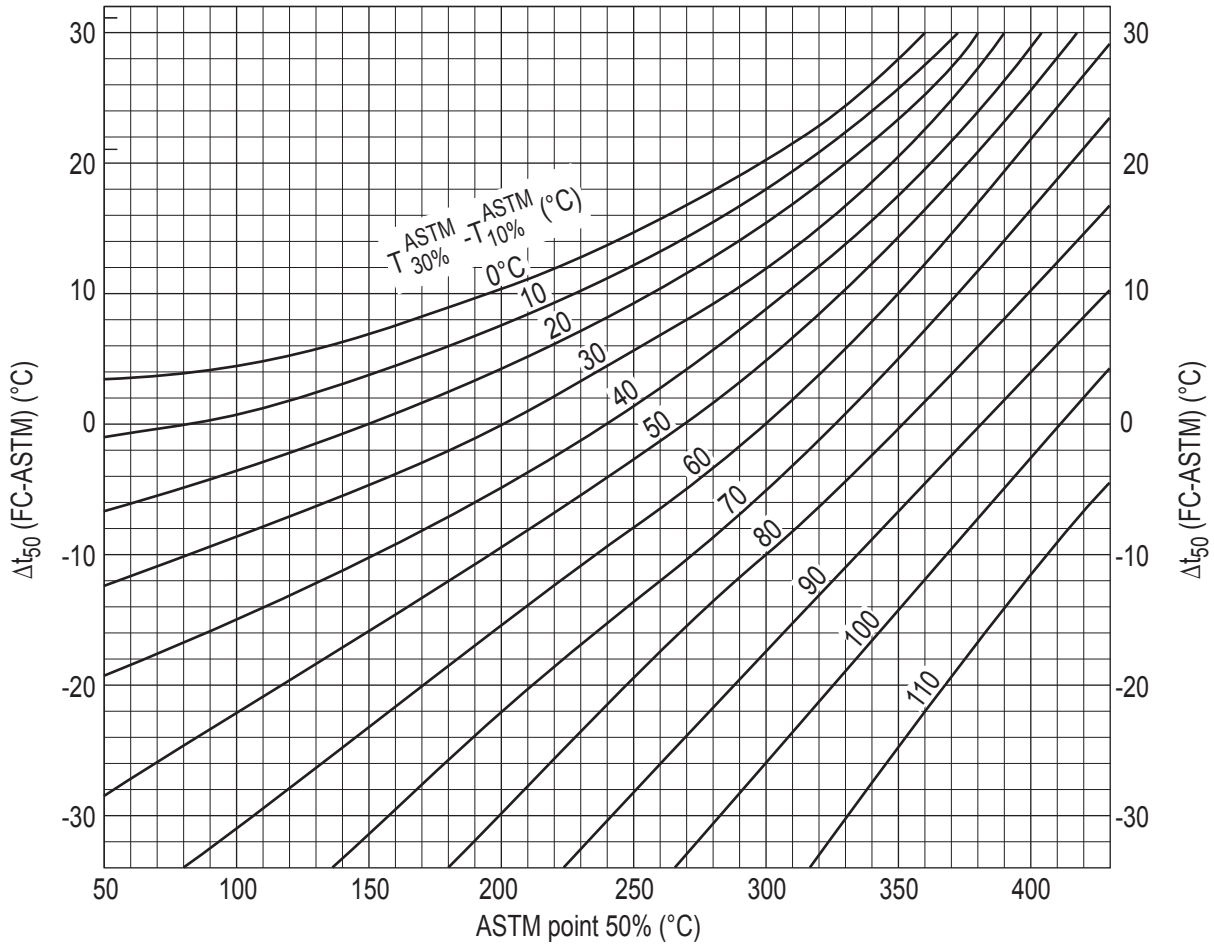


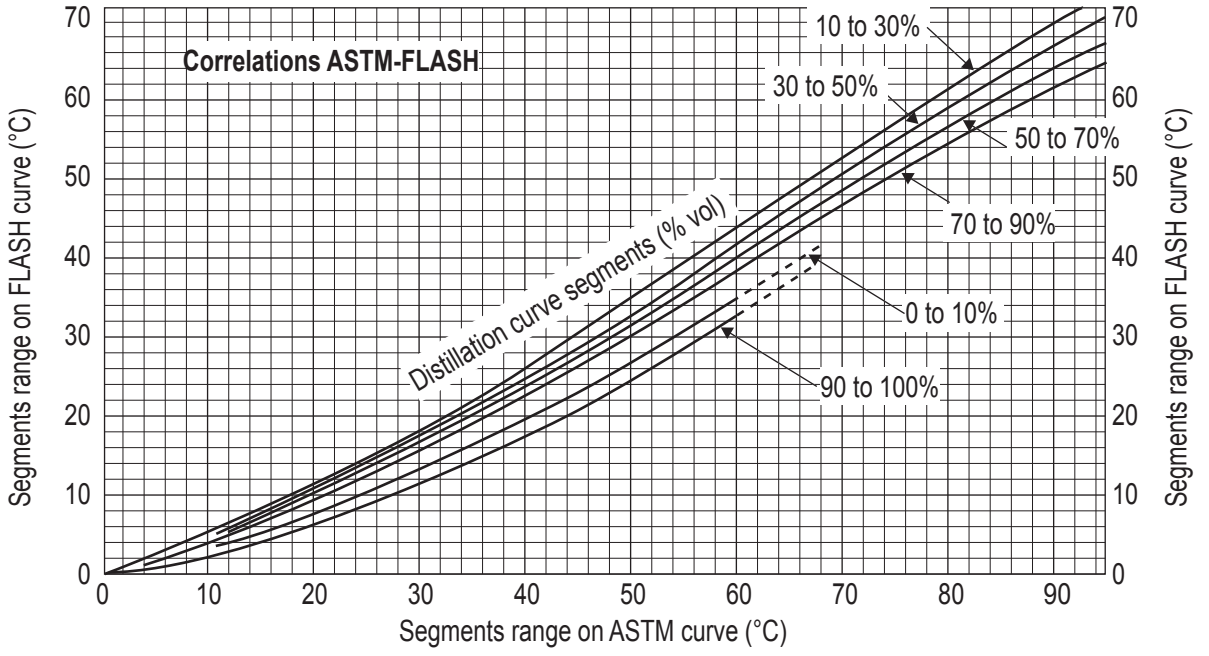


D TH 1163 D

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CALCULATION OF THE ATMOSPHERIC FLASH CURVE STARTING FROM ASTM CURVE
— Edmister method —

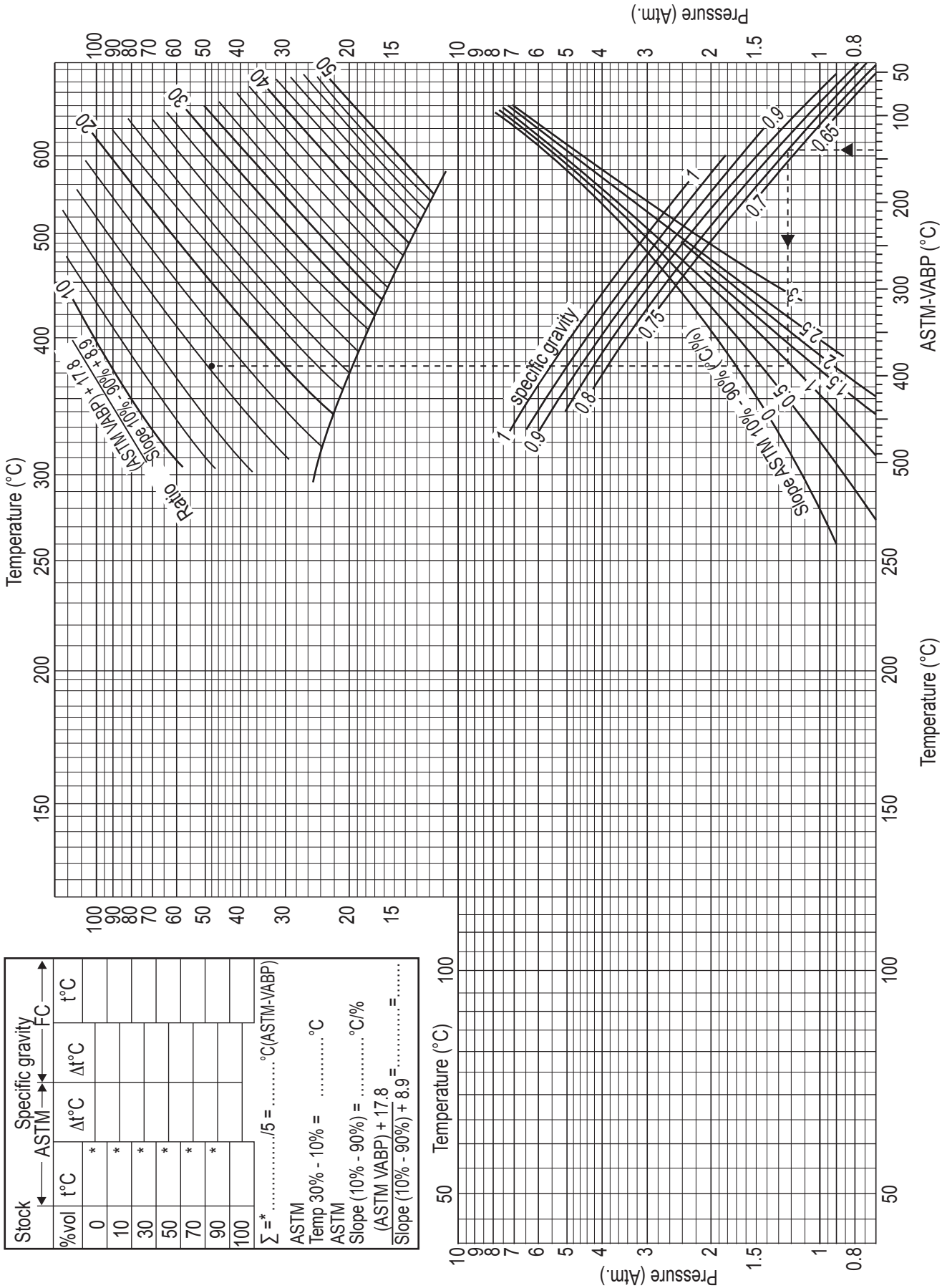




D TH 1561 D

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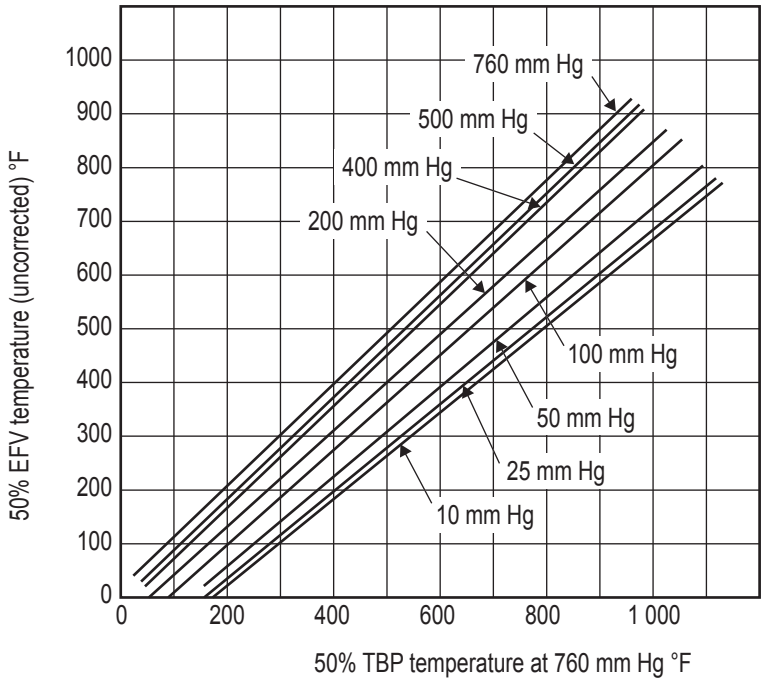
SHIFT OF FLASH CURVE
— Edmister method —

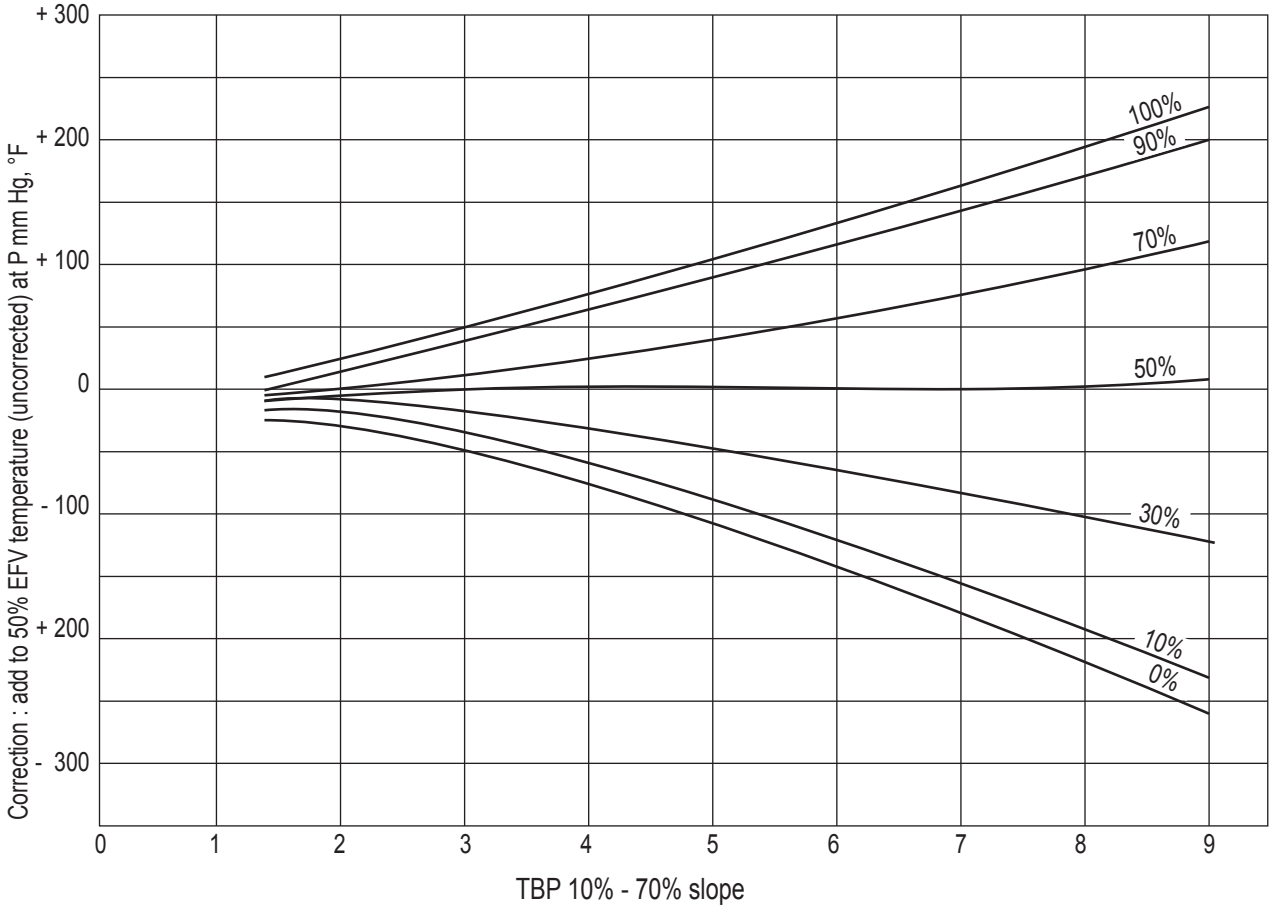


D TH 1638 D

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FLASH CURVE UNDER VACUUM–TBP CORRELATION
— Van Winkle method —

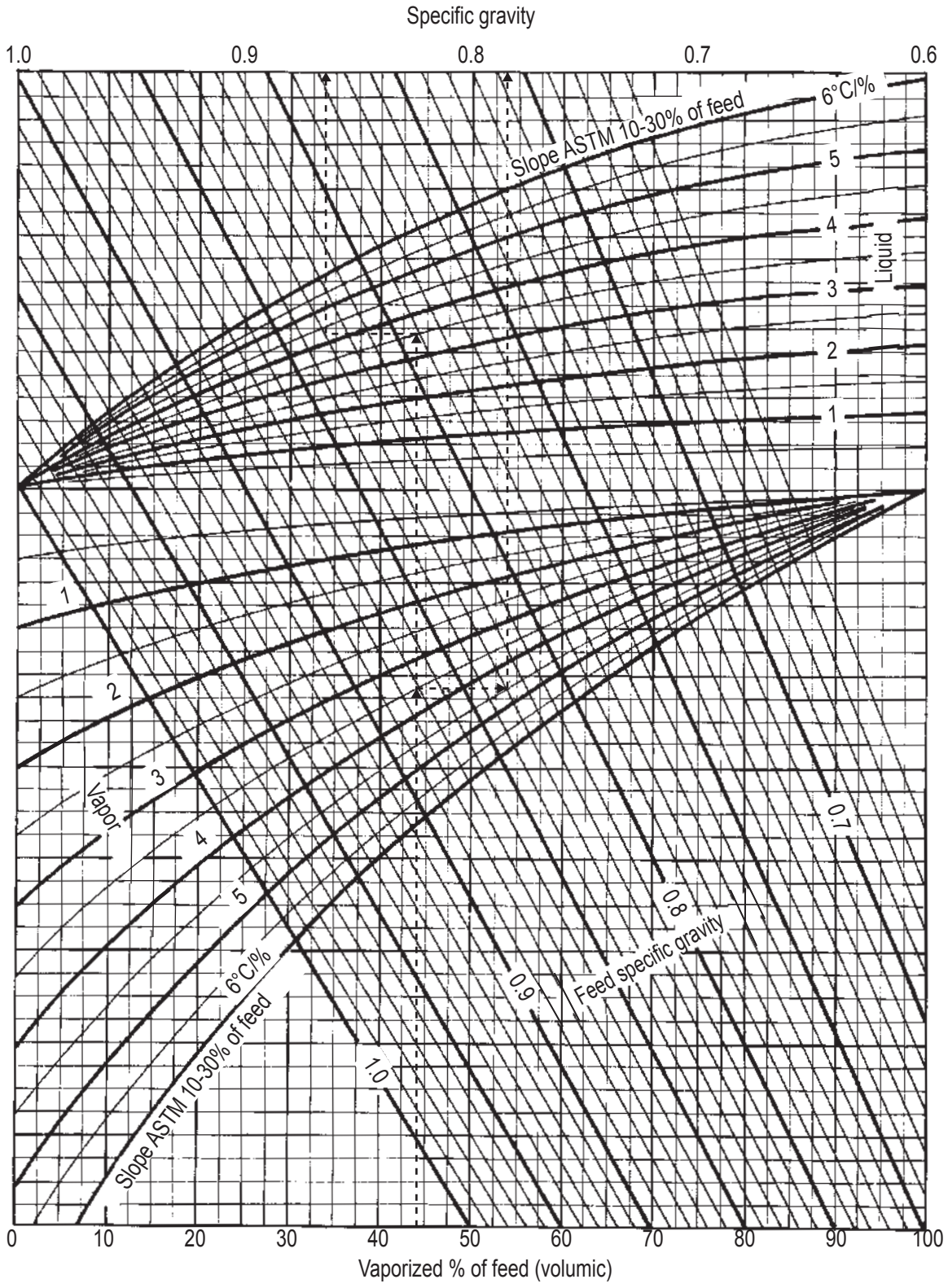




D TH 1636 D

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SPECIFIC GRAVITY OF EQUILIBRIUM LIQUID AND VAPOR PHASES



D TH 1637 D

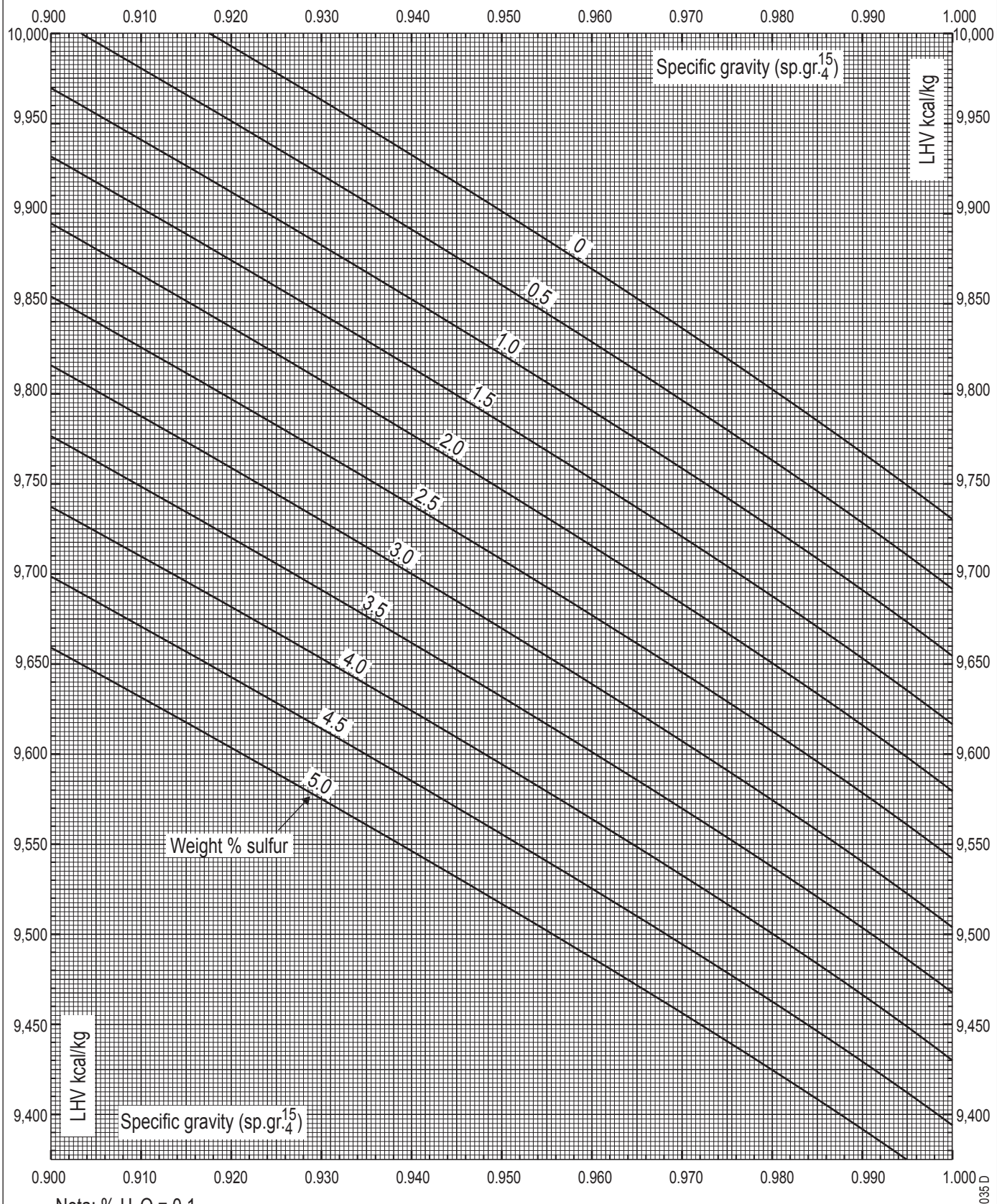
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REACTION HEAT AND CALORIC VALUE OF PURE HYDROCARBONS

Fuel	Chemical equation of combustion	C/H	Physical state of fuel	Physical state of produced water	Heat of reaction kcal/mol	Calorific value kcal/kg	Calorific value kcal/Nm^3	Stoichiometric Fuel / Air ratio Nm^3 air/kg	Stoichiometric Fuel / Air ratio Nm^3 air/Nm^3
Carbon	$C + O_2 \rightarrow CO_2$	–	S	–	94.05	7840	–	8.90	–
Carbon	$C + 1/2\ O_2 \rightarrow CO$	–	S	–	26.45	2205	–	4.45	–
Carbon monoxide	$CO + 1/2\ O_2 \rightarrow CO_2$	–	G	–	67.6	2413	–	1.91	2.39
Hydrogen	$H_2 + 1/2\ O_2 \rightarrow H_2O$	–	**G**	**V** L	**57.7** 68.5	**28640** 33975	**2575** 3060	**26.50**	**2.39**
Sulfur	$S + O_2 \rightarrow SO_2$	–	L	–	70.7	2205	–	3.33	–
Hydrogen sulfide	$H_2S + 3/2\ O_2 \rightarrow SO_2 + H_2O$	–	**G**	**V** L	**123.9** 134.7	**3635** 3955	**5530** 6015	**4.70**	**7.16**
Methane	$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$	3	**G**	**V** L	**191.8** 213.8	**11960** 13300	**8565** 9520	**13.32**	**9.54**
Ethane	$C_2H_6 + 7/2\ O_2 \rightarrow 2\ CO_2 + 3\ H_2O$	4	**G**	**V** L	**341.4** 373.6	**11355** 12125	**15240** 16680	**12.44**	**16.70**
Propane	$C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$	4.5	**G**	**V** L	**488.7** 531.7	**11085** 12060	**21815** 23735	**12.12**	**23.85**
n-Butane	$nC_4H_{10} + 13/2\ O_2 \rightarrow 4\ CO_2 + 5\ H_2O$	4.8	**G**	**V** L	**635.6** 689.3	**10935** 11860	**28375** 30770	**11.95**	**31.01**
n-Hexane	$nC_6H_{14} + 19/2\ O_2 \rightarrow 6\ CO_2 + 7\ H_2O$	5.14	**L**	**V** L	**921.6** 996.9	**10695** 11570	**–** –	**11.78**	**–**
Cyclohexane	$C_6H_{12} + 9\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$	6	**L**	**V** L	**874.1** 938.6	**10385** 11155	**–** –	**11.43**	**–**
Benzene	$C_6H_6 + 15/2\ O_2 \rightarrow 6\ CO_2 + 3\ H_2O$	12	**L**	**V** L	**749.6** 781.8	**9595** 10010	**–** –	**10.26**	**–**
n-Hexadecane	$nC_{16}H_{34} + 49/2\ O_2 \rightarrow 16\ CO_2 + 17\ H_2O$	5.65	**L**	**V** L	**2377.7** 2560.5	**10500** 11310	**–** –	**11.56**	**–**

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FUEL OIL LHV DETERMINATION

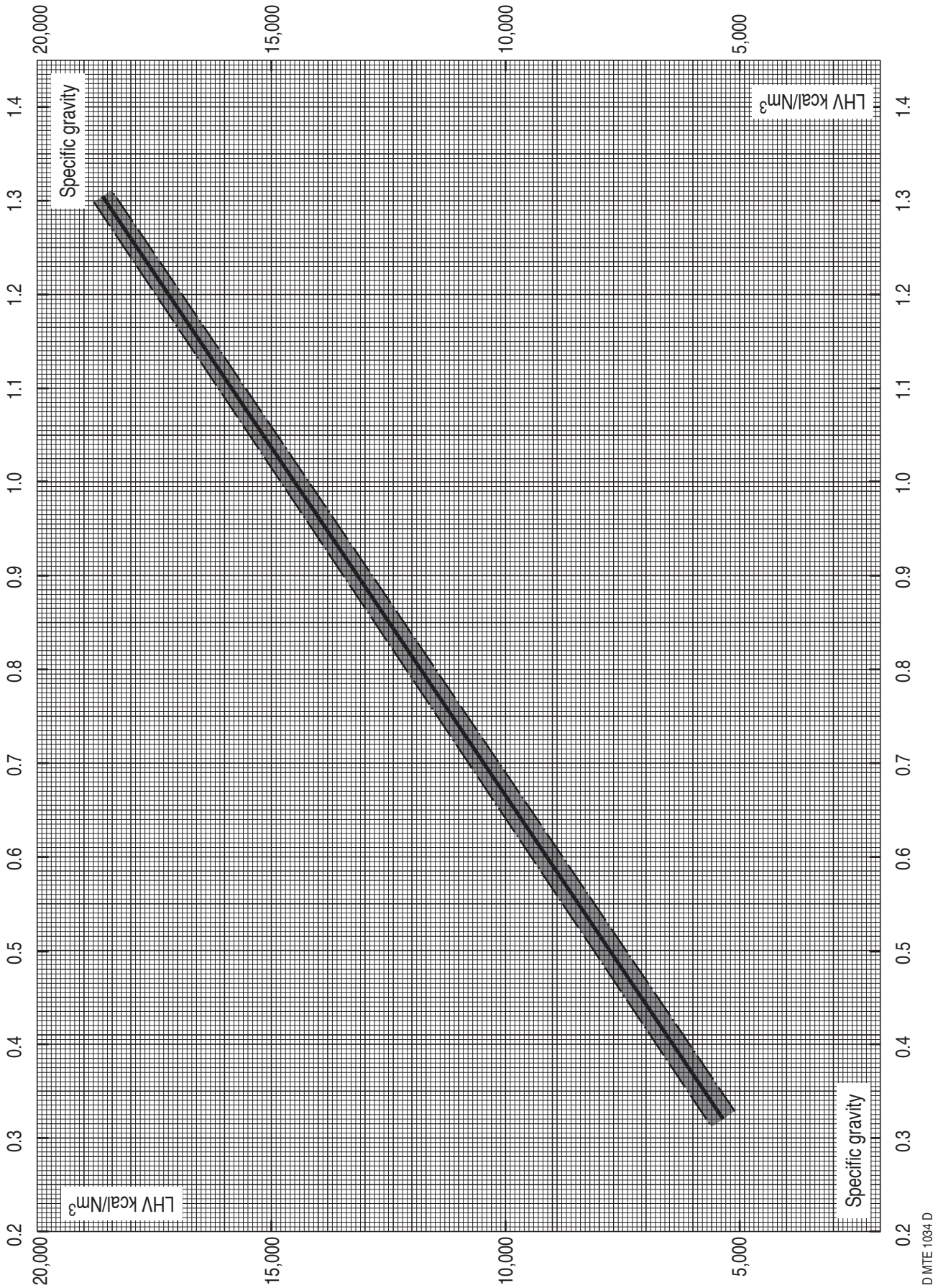


Nota: % H_2O = 0.1

D MTE1035 D

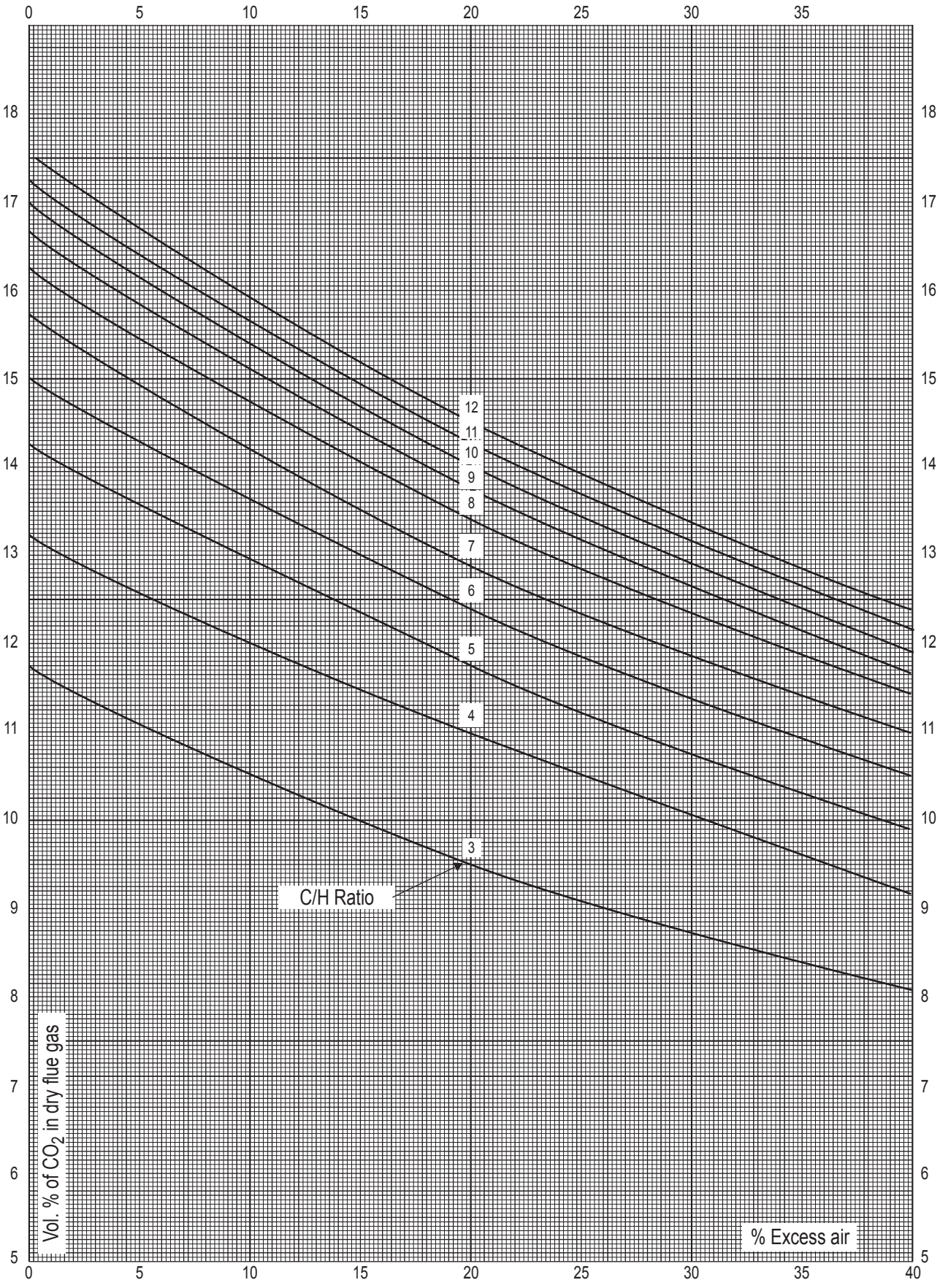
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FUEL-GAS LHV ESTIMATION



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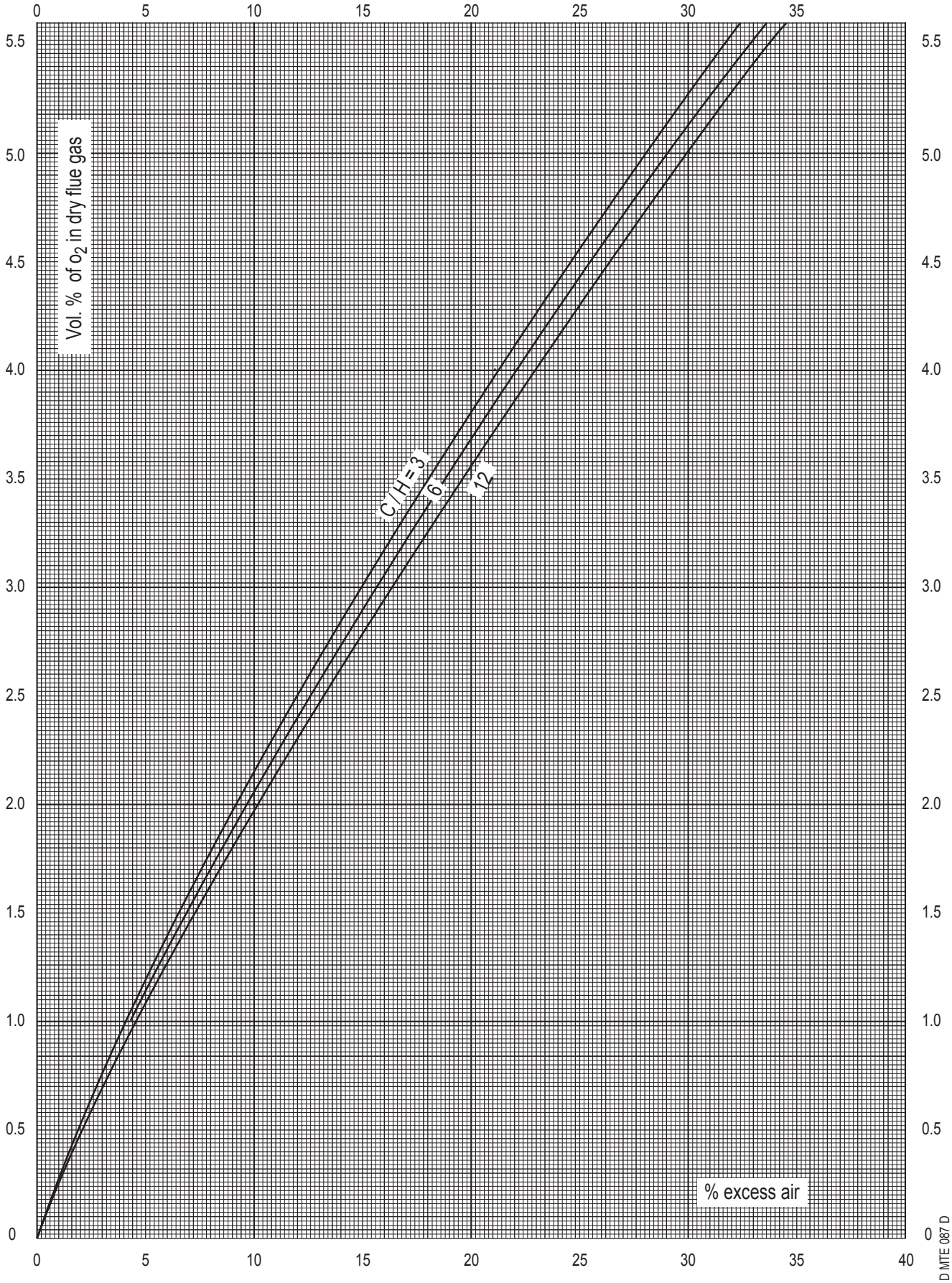
EXCESS AIR VERSUS CO_2 CONTENT IN DRY FLUE GAS



D MTE 094 D

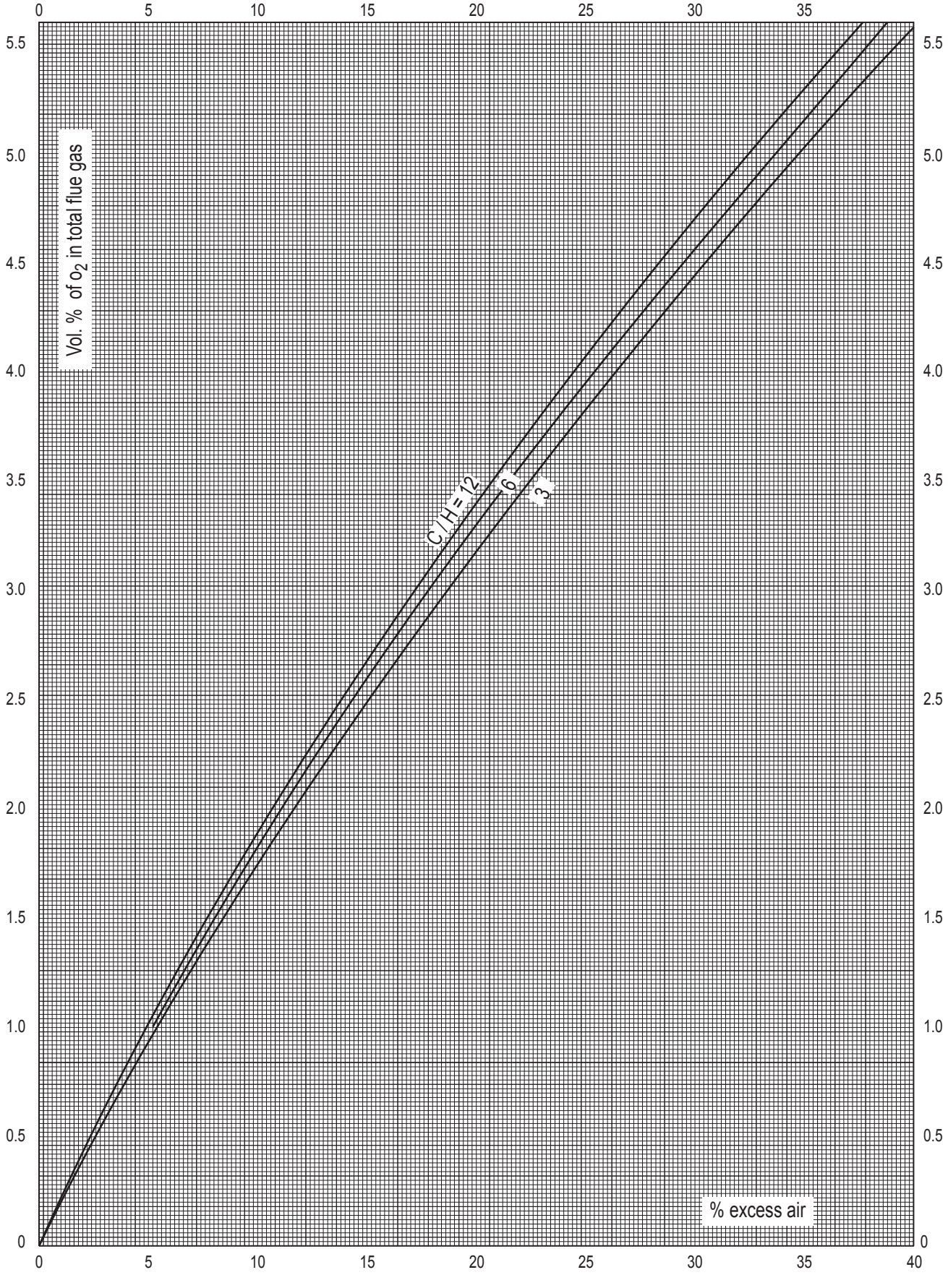
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EXCESS AIR VERSUS O2 CONTENT IN DRY FLUE GAS



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EXCESS AIR VERSUS O_2 CONTENT IN TOTAL FLUE GAS

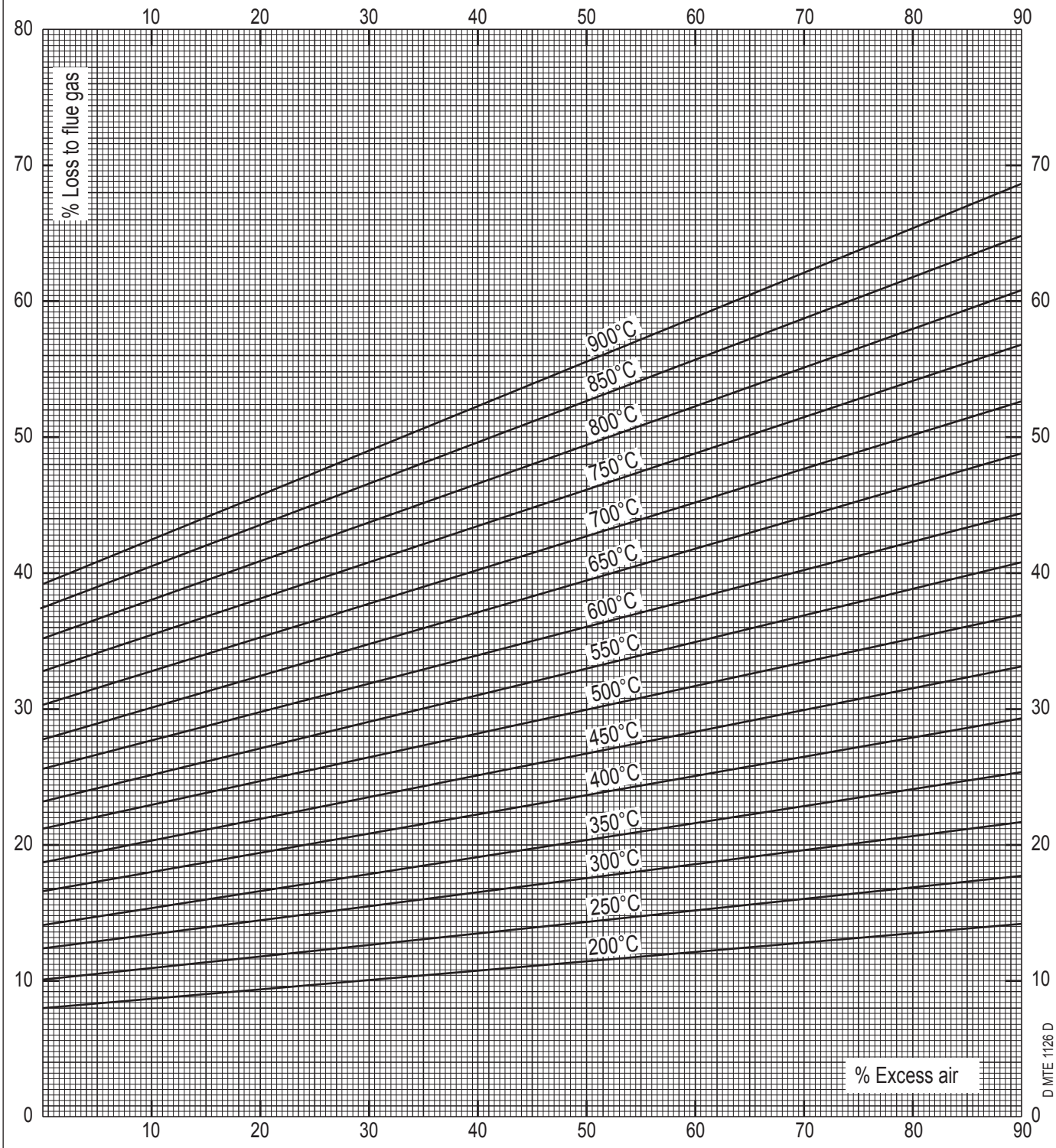


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FURNACE EFFICIENCY ESTIMATE
– Case of a fuel gas firing furnace –

Chart plotted for:

- Fuel gas { C/H = 3 d_{sh} = 0.540; LHV = 8530 kcal/kg; $t°_{gas}$ = 35°C
- Ambient air { 25°C; 64% humidity

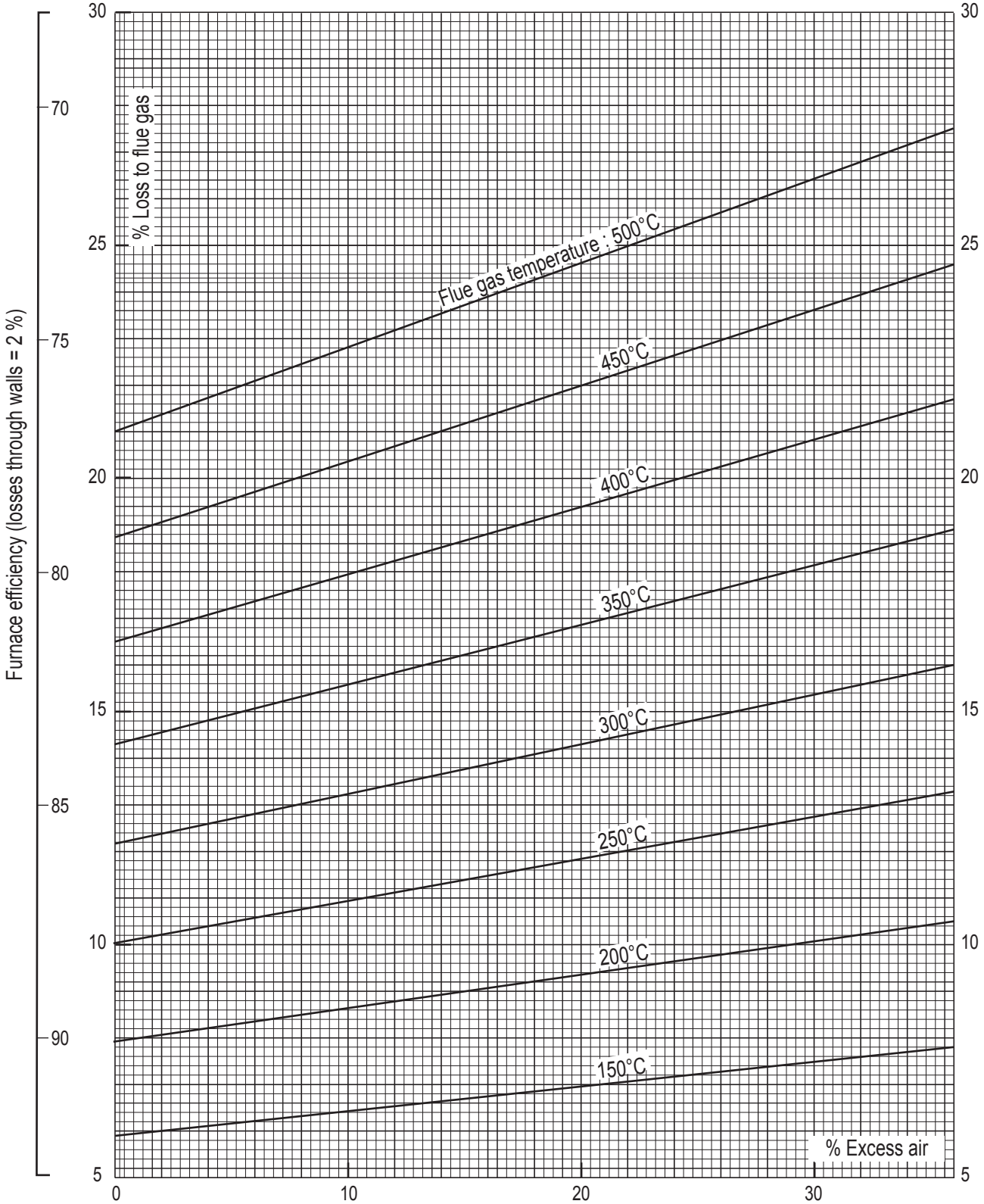


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FURNACE EFFICIENCY ESTIMATE
– Case of a fuel oil firing furnace –

Chart plotted for :

- Fuel oil { C/H = 7; Sp.gr.$^{15}_{4}$ = 0.970 ; %S = 3.5; t°$_{fuel}$ = 120°C
- Atomization steam { 200°C; 180 kg/ton of fuel
- Ambient air { 15°C; 72% humidity



D MTE 1127 D

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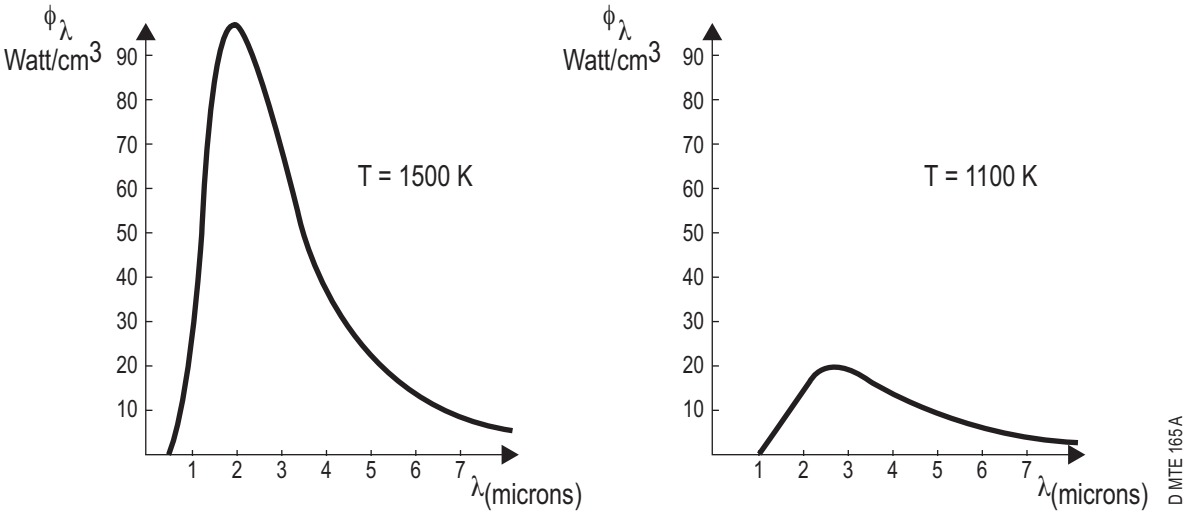
HEAT TRANSMISSION IN A FURNACE
– Black body properties –

MONOCHROMATIC RADIATION OF BLACK BODY

$$\Phi\lambda = a \cdot 10^{-8} \frac{\lambda^{-5}}{e^{1,44/\lambda T} - 1}$$

with	a = 3,17	with	a = 3,74
if	Φ en kcal/h.m^2.cm	if	Φ en W/m^2.cm
	λ in cm		λ in cm
	T in K		T in K

The following graphs show, for two different temperatures, the evolution of the thermal flux emited by black body versus the wavelenght carrying the energy.



OVERALL RADIATION OF BLACK BODY

Flux emited by black body $\Phi_{\text{black body}} = \sigma \cdot \left(\frac{T}{100}\right)^4$

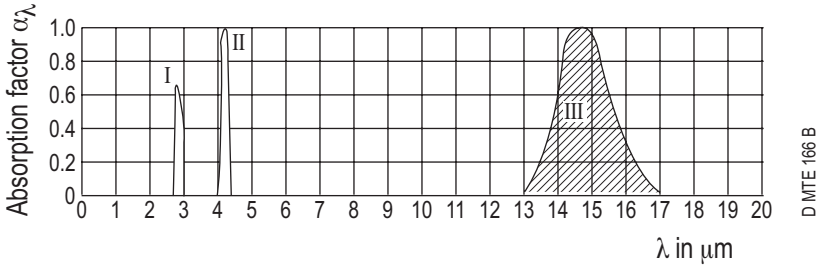
in W/m^2	σ = 5,67
in kcal/h.m^2	σ = 4,88

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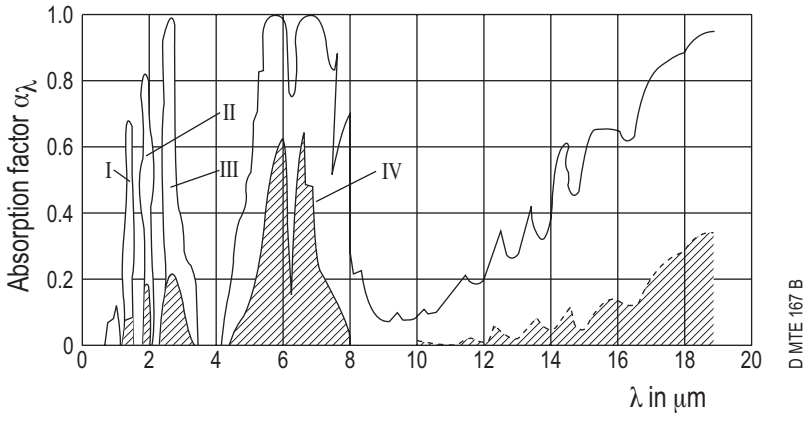
HEAT TRANSMISSION IN A FURNACE
– Emissivity coefficients for materials and gases –

MATERIALS	EMISSIVITY
Polished silver	0.04
Aluminium paints	0.3 à 0.5
Steels (new)	0.5 à 0.6
Steels (oxydized)	0.8 à 0.9
Refractory insulator	0.75 à 0.9
Smoke black	0.95 à 0.98
Black body	1 definition

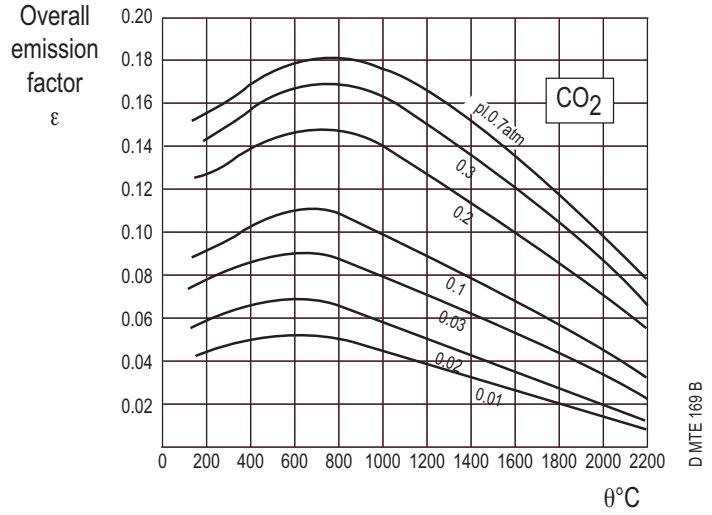
Absorption spectrum of CO_2 à 20°C (p.L = 6,3 . 10^{-2} atm.m)

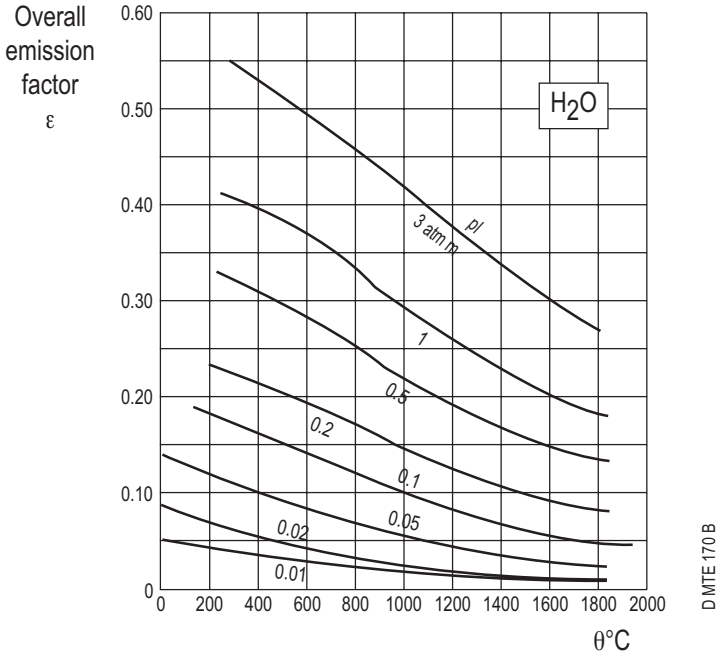


Absorption spectrum of steam at 127°C : p.L = 2,4 atm.m
p.L = 0,24 atm.m



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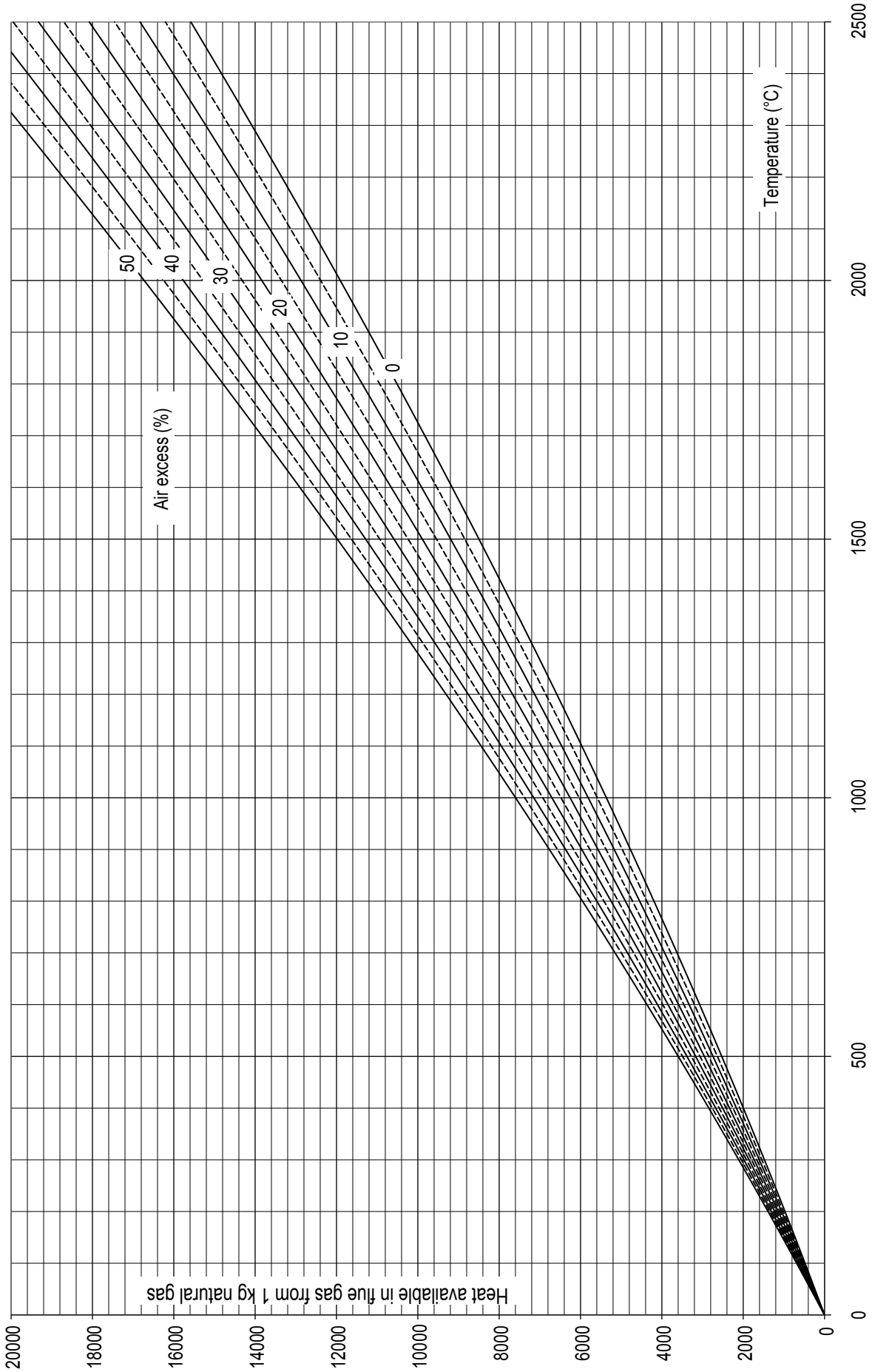
Thermal emission of steam for p.l = 0,2 atm.m

t°C	ε	$\varepsilon \sigma T^4$ kcal/h.m^2
400	0.22	2240
800	0.17	11180
1000	0.15	19540
1200	0.13	30360
1400	0.11	42740

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg NATURAL GAS

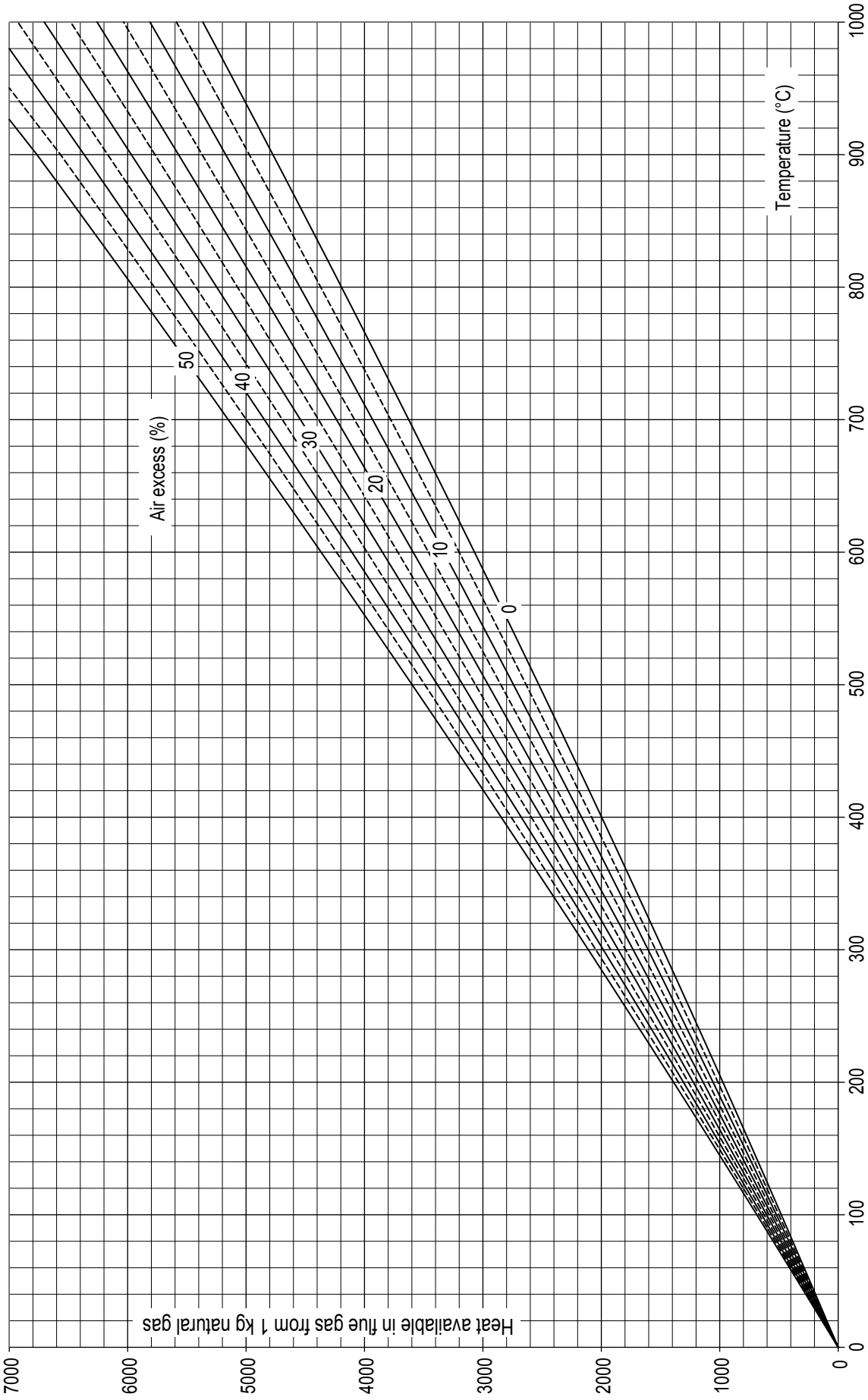
1/4



D MTE 1418 D

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg NATURAL GAS

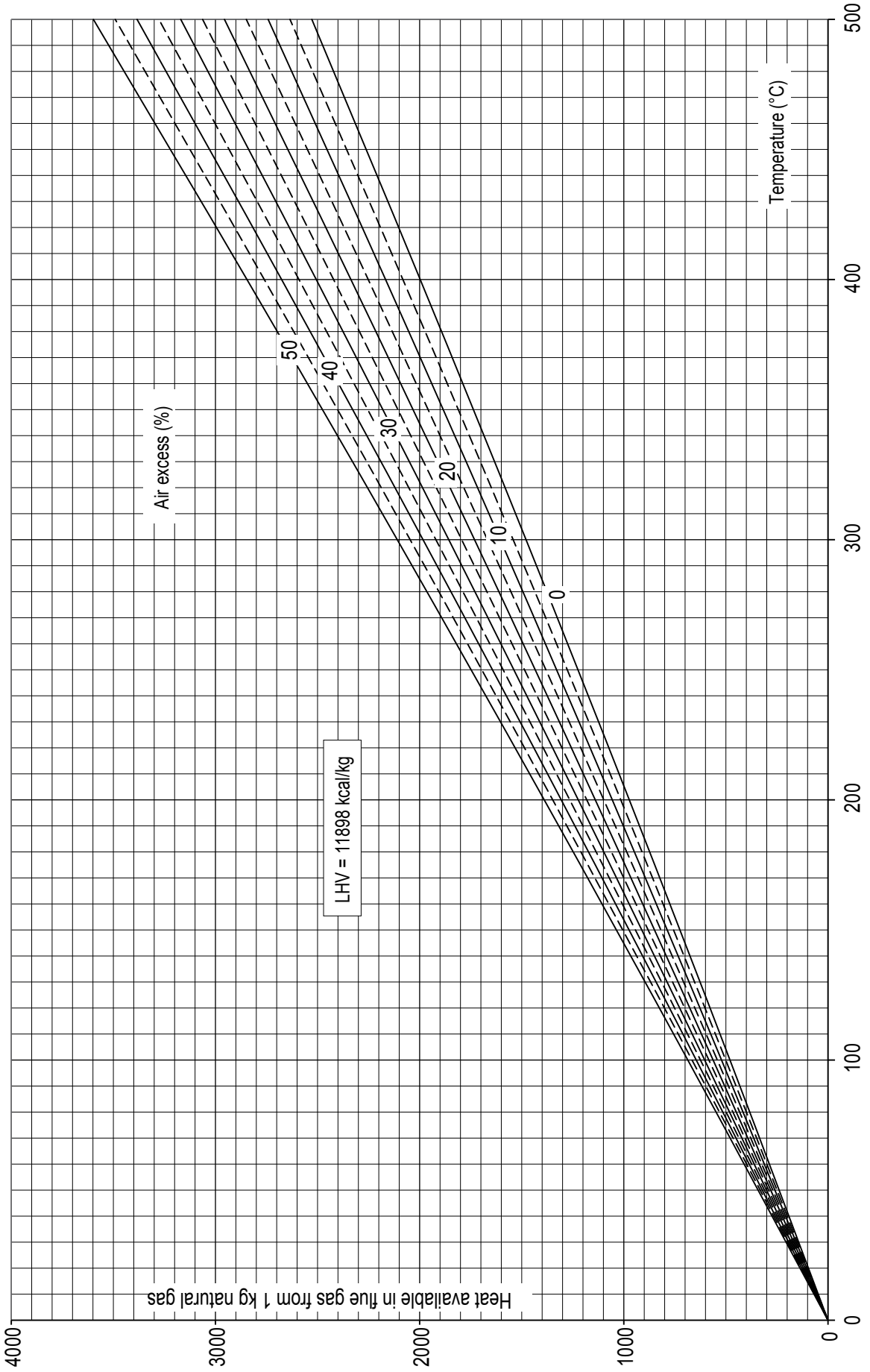


D MTE 1418 E

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg NATURAL GAS

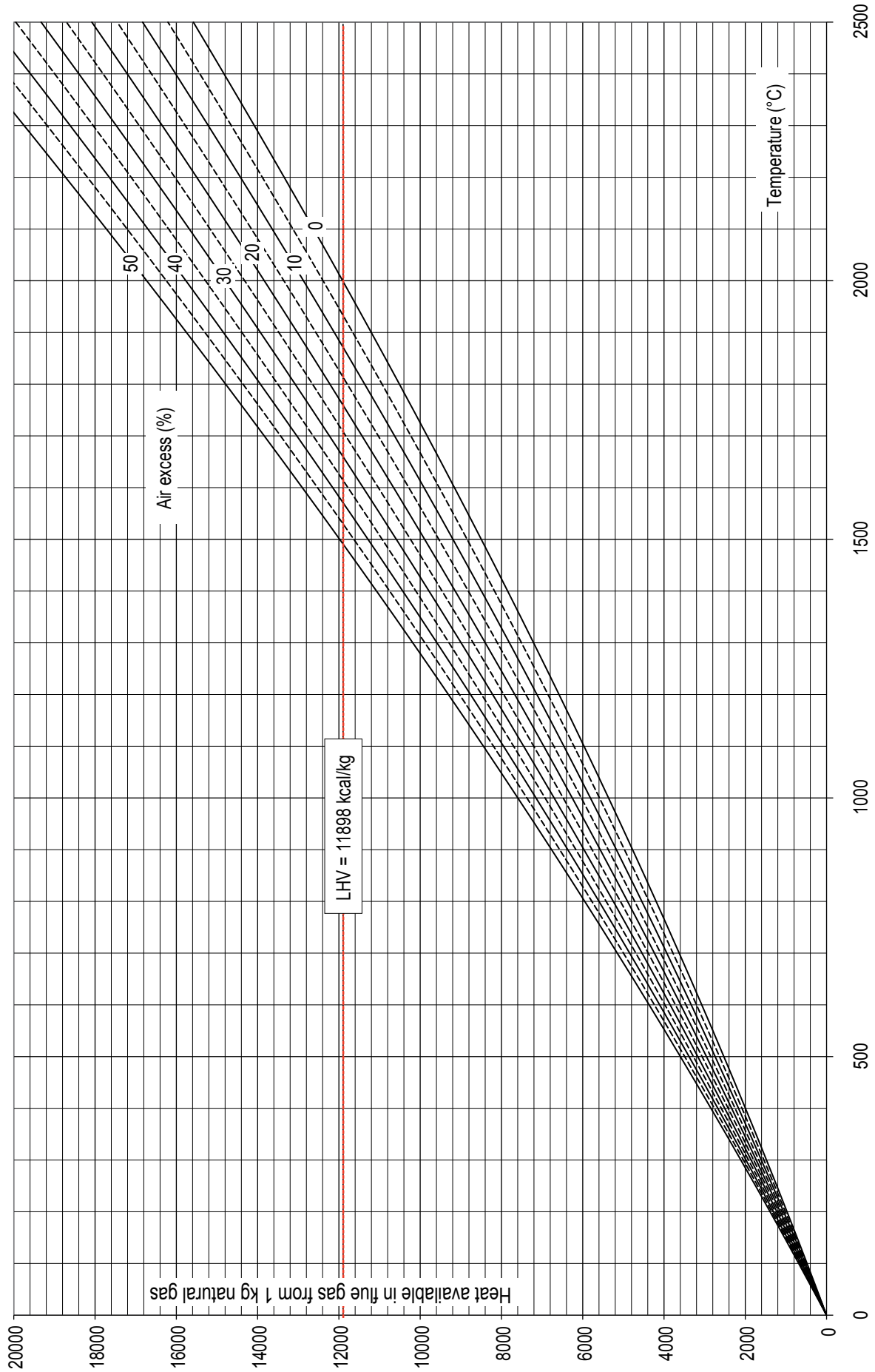
3/4



D MTE 1418 J

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg NATURAL GAS

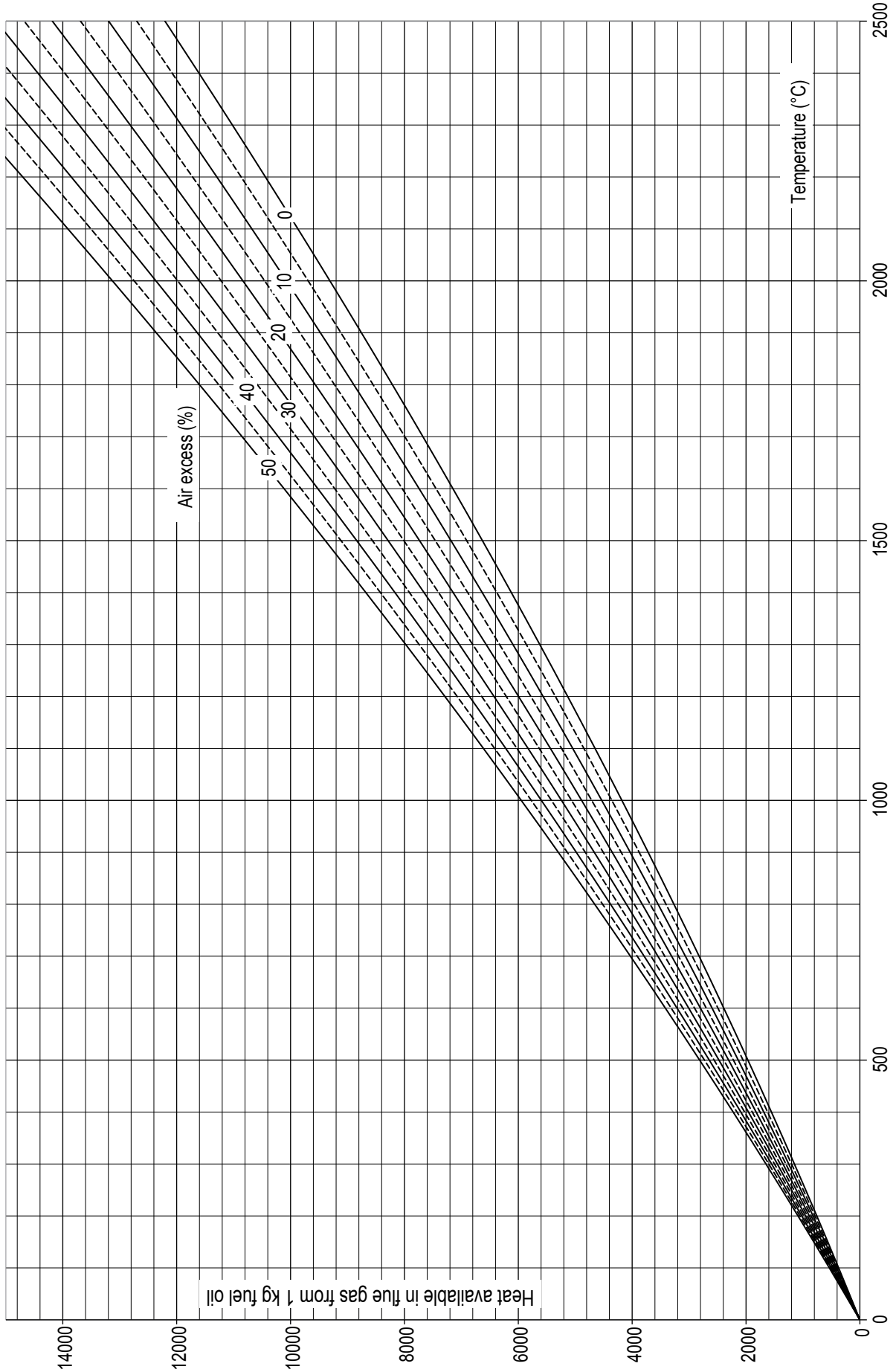


D MTE 1418 F

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg FUEL OIL

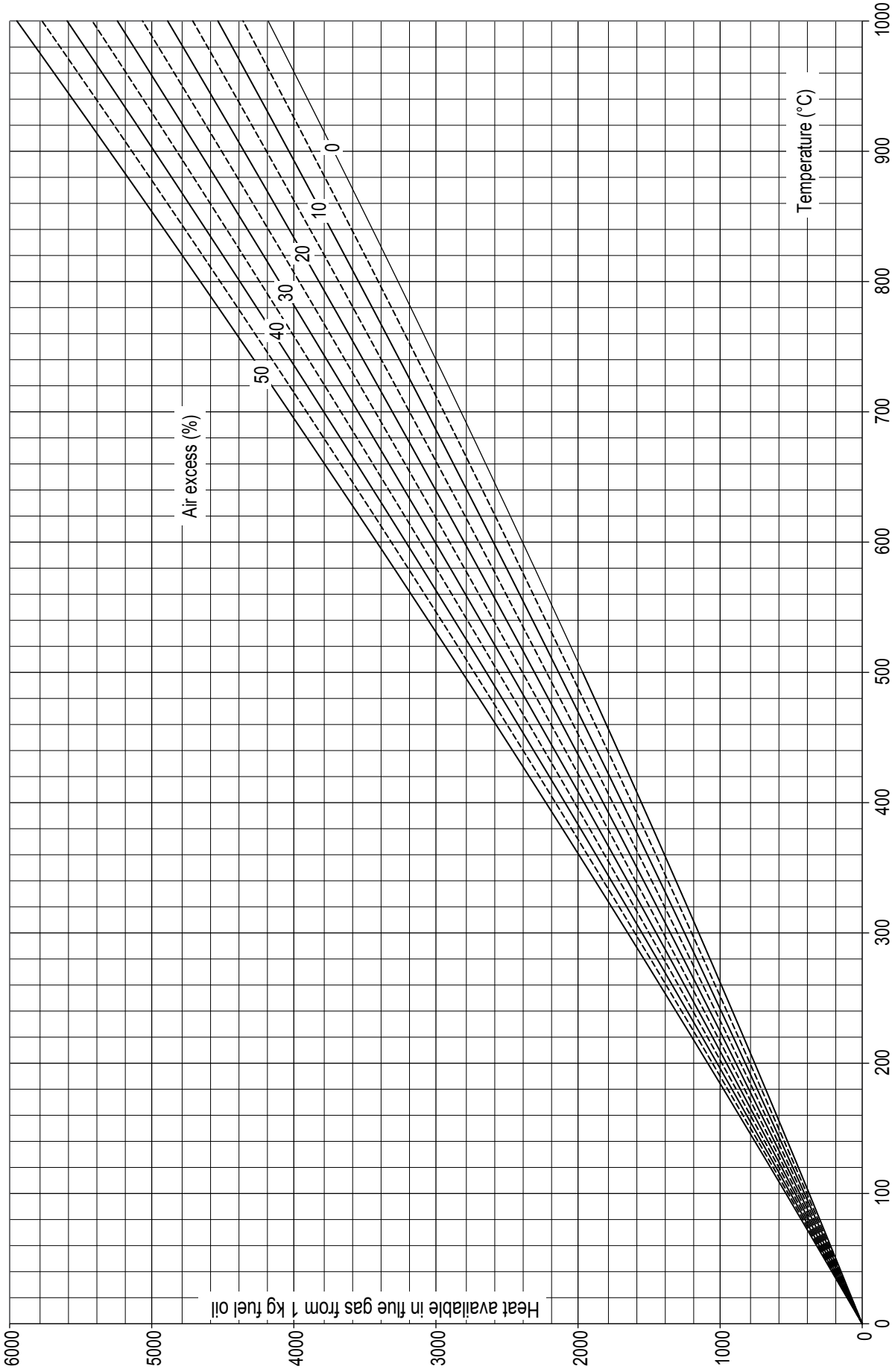
1/4



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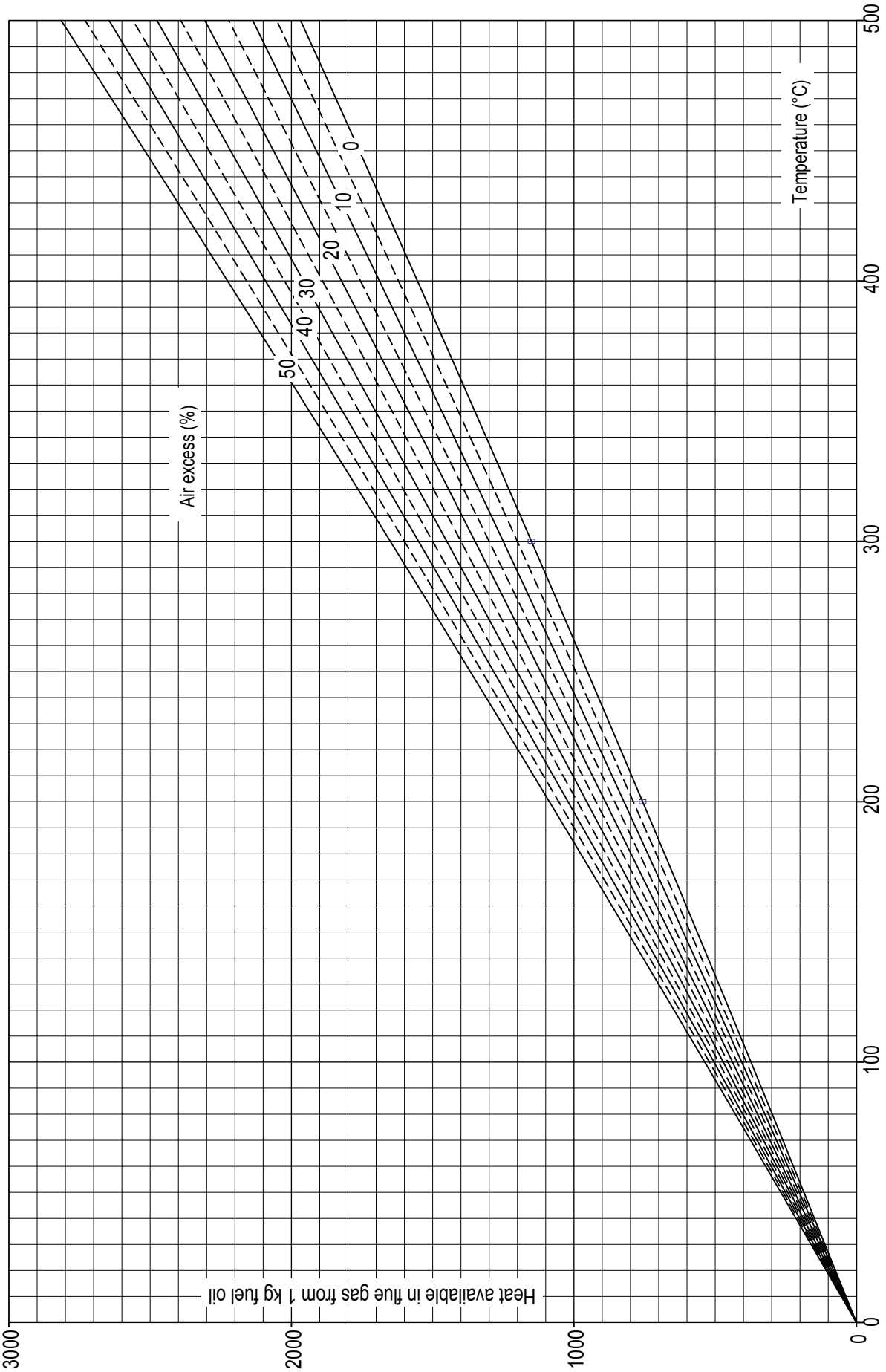
HEAT AVAILABLE IN FLUE GAS FROM 1 kg FUEL OIL



D MTE 1419 E

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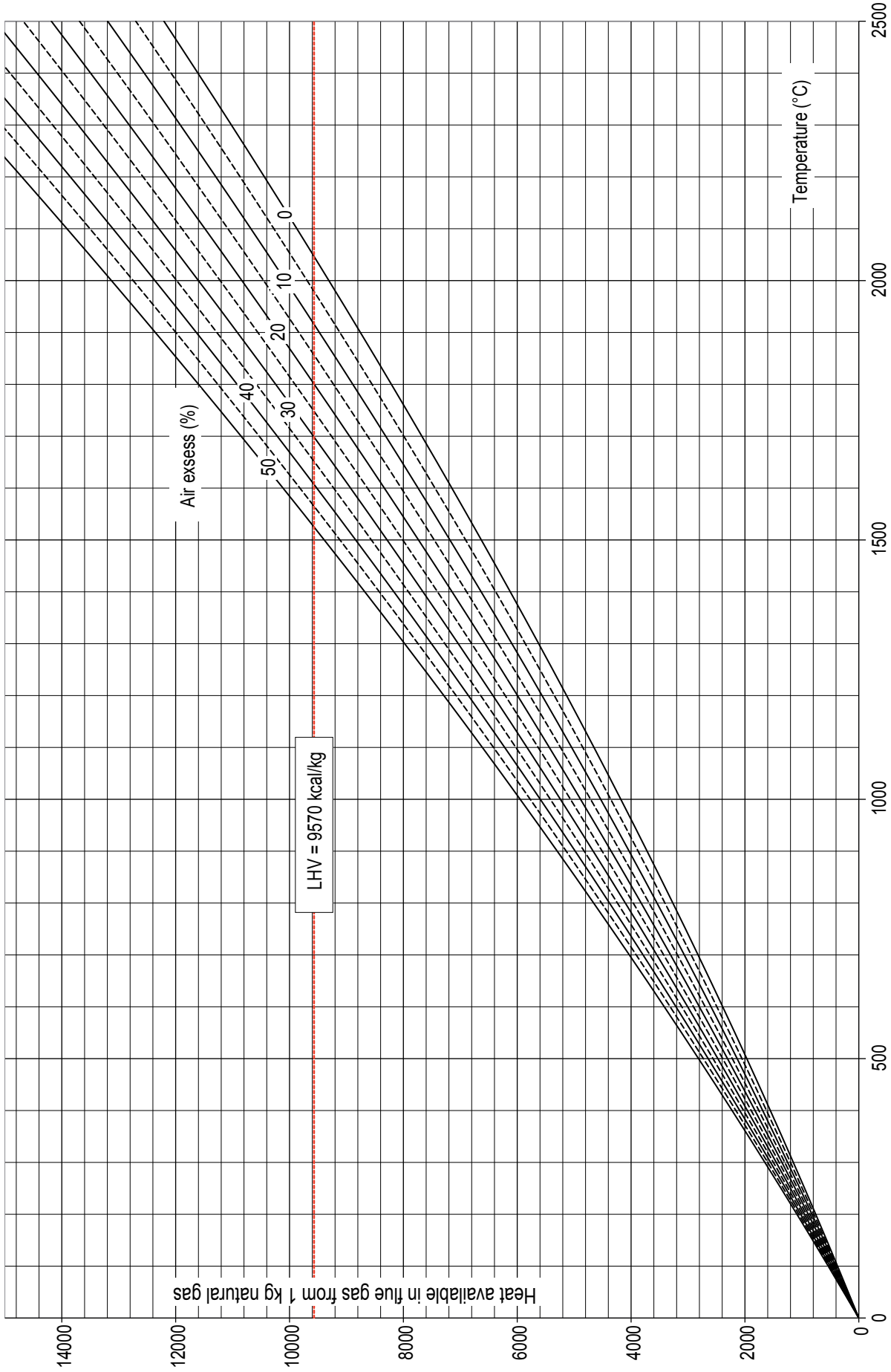
HEAT AVAILABLE IN FLUE GAS FROM 1 kg FUEL OIL



D MTE 1419 H

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HEAT AVAILABLE IN FLUE GAS FROM 1 kg FUEL OIL



D MTE 1419 F

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ISO 4200 - 1991 standard —

NOMINAL DIAMETER		OUTSIDE DIAMETER	THICKNESS SERIE 1	THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm		mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ANSI NPS	ISO DN					FREE	METAL	EXT.	INT.	METAL	WATER		
1/8 "	6	10,2	A, E	1,6	7	38,48	43,23	320,44	219,91	0,339	0,038	0,0413	0,081
1/8 "	*6*	*10,2*	*F*	*2,0*	*6,2*	*30,19*	*51,52*	*320,44*	*194,78*	*0,404*	*0,030*	*0,0459*	*0,090*
1/8 "	6	10,2	G	2,3	5,6	24,63	57,08	320,44	175,93	0,448	0,025	0,0483	0,095
1/4"	8	13,5	A, D	1,6	10,3	83,32	59,82	424,12	323,58	0,470	0,083	0,1078	0,160
1/4"	*8*	*13,5*	*E*	*2,0*	*9,5*	*70,88*	*72,26*	*424,12*	*298,45*	*0,567*	*0,071*	*0,1231*	*0,182*
1/4"	8	13,5	F	2,3	8,9	62,21	80,93	424,12	279,60	0,635	0,062	0,1322	0,196
1/4"	*8*	*13,5*	*G*	*2,6*	*8,3*	*54,11*	*89,03*	*424,12*	*260,75*	*0,699*	*0,054*	*0,1397*	*0,207*
3/8"	10	17,2	A, D	1,6	14	153,94	78,41	540,35	439,82	0,616	0,154	0,2410	0,280
3/8"	*10*	*17,2*	*E*	*2,0*	*13,2*	*136,85*	*95,50*	*540,35*	*414,69*	*0,750*	*0,137*	*0,2806*	*0,326*
3/8"	10	17,2	F	2,3	12,6	124,69	107,66	540,35	395,84	0,845	0,125	0,3059	0,356
3/8"	*10*	*17,2*	*G*	*2,6*	*12*	*113,10*	*119,25*	*540,35*	*376,99*	*0,936*	*0,113*	*0,3278*	*0,381*
1/2"	15	21,3	A	1,6	18,1	257,30	99,02	669,16	568,63	0,777	0,257	0,4835	0,454
1/2"	*15*	*21,3*	*D*	*1,8*	*17,7*	*246,06*	*110,27*	*669,16*	*556,06*	*0,866*	*0,246*	*0,5286*	*0,496*
1/2"	15	21,3	E	2,0	17,3	235,06	121,27	669,16	543,50	0,952	0,235	0,5707	0,536
1/2"	*15*	*21,3*	*F*	*3,2*	*14,9*	*174,37*	*181,96*	*669,16*	*468,10*	*1,428*	*0,174*	*0,7684*	*0,722*
1/2"	15	21,3	G	4,0	13,3	138,93	217,40	669,16	417,83	1,707	0,139	0,8568	0,805
3/4"	20	26,9	A	1,6	23,7	441,15	127,17	845,09	744,56	0,998	0,441	1,0216	0,760
3/4"	*20*	*26,9*	*D*	*1,8*	*23,3*	*426,38*	*141,94*	*845,09*	*731,99*	*1,114*	*0,426*	*1,1235*	*0,835*
3/4"	20	26,9	E	2,0	22,9	411,87	156,45	845,09	719,42	1,228	0,412	1,2203	0,907
3/4"	*20*	*26,9*	*F*	*3,2*	*20,5*	*330,06*	*238,26*	*845,09*	*644,03*	*1,870*	*0,330*	*1,7033*	*1,266*
3/4"	20	26,9	G	4,0	18,9	280,55	287,77	845,09	593,76	2,259	0,281	1,9439	1,445
1"	25	33,7	A	1,6	30,5	730,62	161,35	1058,72	958,19	1,267	0,731	2,0834	1,236
1"	*25*	*33,7*	*B, D*	*2,0*	*29,7*	*692,79*	*199,18*	*1058,72*	*933,05*	*1,564*	*0,693*	*2,5118*	*1,491*
1"	25	33,7	E	2,3	29,1	665,08	226,89	1058,72	914,20	1,781	0,665	2,8113	1,668
1"	*25*	*33,7*	*F*	*3,2*	*27,3*	*585,35*	*306,62*	*1058,72*	*857,65*	*2,407*	*0,585*	*3,6047*	*2,139*
1"	25	33,7	G	4,5	24,7	479,16	412,81	1058,72	775,97	3,241	0,479	4,5042	2,673
1 "1/4	32	42,4	A	1,6	39,2	1206,87	205,08	1332,04	1231,50	1,610	1,207	4,2739	2,016
1 "1/4	*32*	*42,4*	*B*	*2,0*	*38,4*	*1158,12*	*253,84*	*1332,04*	*1206,37*	*1,993*	*1,158*	*5,1915*	*2,449*
1 "1/4	32	42,4	D	2,3	37,8	1122,21	289,75	1332,04	1187,52	2,275	1,122	5,8432	2,756
1 "1/4	*32*	*42,4*	*E*	*2,6*	*37,2*	*1086,87*	*325,09*	*1332,04*	*1168,67*	*2,552*	*1,087*	*6,4645*	*3,049*
1 "1/4	32	42,4	F	3,6	35,2	973,14	438,82	1332,04	1105,84	3,445	0,973	8,3288	3,929
1 "1/4	*32*	*42,4*	*G*	*5,0*	*32,4*	*824,48*	*587,48*	*1332,04*	*1017,88*	*4,612*	*0,824*	*10,4553*	*4,932*
1"1/2	40	48,3	A	1,6	45,1	1597,51	234,74	1517,39	1416,86	1,843	1,598	6,4068	2,653
1"1/2	*40*	*48,3*	*B*	*2,0*	*44,3*	*1541,34*	*290,91*	*1517,39*	*1391,73*	*2,284*	*1,541*	*7,8098*	*3,234*
1"1/2	40	48,3	D	2,3	43,7	1499,87	332,38	1517,39	1372,88	2,609	1,500	8,8134	3,649
1"1/2	*40*	*48,3*	*E*	*2,6*	*43,1*	*1458,96*	*373,28*	*1517,39*	*1354,03*	*2,930*	*1,459*	*9,7765*	*4,048*
1"1/2	40 →	48,3	F	3,6	41,1	1326,70	505,55	1517,39	1291,19	3,969	1,327	12,7085	5,262
1"1/2	*40*	*48,3*	*G*	*5,0*	*38,3*	*1152,09*	*680,15*	*1517,39*	*1203,23*	*5,339*	*1,152*	*16,1527*	*6,689*
2"	50	60,3	A	1,6	57,1	2560,72	295,06	1894,38	1793,85	2,316	2,561	12,7179	4,218
2"	*50*	*60,3*	*B*	*2,0*	*56,3*	*2489,47*	*366,31*	*1894,38*	*1768,72*	*2,876*	*2,489*	*15,5814*	*5,168*
2"	50	60,3	C, D	2,3	55,7	2436,69	419,09	1894,38	1749,87	3,290	2,437	17,6504	5,854
2"	*50*	*60,3*	*E*	*2,9*	*54,5*	*2332,83*	*522,95*	*1894,38*	*1712,17*	*4,105*	*2,333*	*21,5924*	*7,162*
2"	50	60,3	F	4,0	52,3	2148,29	707,49	1894,38	1643,05	5,554	2,148	28,1729	9,344
2"	*50*	*60,3*	*G*	*5,6*	*49,1*	*1893,45*	*962,33*	*1894,38*	*1542,52*	*7,554*	*1,893*	*36,3696*	*12,063*
2"1/2	65	76,1	A	1,6	72,9	4173,93	374,48	2390,75	2290,22	2,940	4,174	25,9926	6,831
2"1/2	*65*	*76,1*	*B*	*2,3*	*71,5*	*4015,15*	*533,25*	*2390,75*	*2246,24*	*4,186*	*4,015*	*36,3395*	*9,550*
2"1/2	65	76,1	C, D	2,6	70,9	3948,05	600,36	2390,75	2227,39	4,713	3,948	40,5918	10,668
2"1/2	*65*	*76,1*	*E*	*2,9*	*70,3*	*3881,51*	*666,90*	*2390,75*	*2208,54*	*5,235*	*3,882*	*44,7376*	*11,758*
2"1/2	65	76,1	F	5,0	66,1	3431,57	1116,84	2390,75	2076,59	8,767	3,432	70,9220	18,639
2"1/2	*65*	*76,1*	*G*	*7,1*	*61,9*	*3009,34*	*1539,07*	*2390,75*	*1944,65*	*12,082*	*3,009*	*92,5635*	*24,327*
3"	80	88,9	A	2,0	84,9	5661,16	546,01	2792,88	2667,21	4,286	5,661	51,5679	11,601
3"	*80*	*88,9*	*B*	*2,3*	*84,3*	*5581,42*	*625,74*	*2792,88*	*2648,36*	*4,912*	*5,581*	*58,7013*	*13,206*
3"	80	88,9	C, D	2,9	83,1	5423,65	783,51	2792,88	2610,66	6,151	5,424	72,5182	16,315
3"	*80*	*88,9*	*E*	*3,2*	*82,5*	*5345,62*	*861,55*	*2792,88*	*2591,81*	*6,763*	*5,346*	*79,2059*	*17,819*
3"	80	88,9	F	5,6	77,7	4741,68	1465,49	2792,88	2441,02	11,504	4,742	127,6854	28,726
3"	*80*	*88,9*	*G*	*8,0*	*72,9*	*4173,93*	*2033,24*	*2792,88*	*2290,22*	*15,961*	*4,174*	*167,9661*	*37,788*
4"	100	114,3	A	2,0	110,3	9555,22	705,60	3590,84	3465,18	5,539	9,555	111,2671	19,469
4"	*100*	*114,3*	*B*	*2,6*	*109,1*	*9348,45*	*912,38*	*3590,84*	*3427,48*	*7,162*	*9,348*	*142,3731*	*24,912*
4"	100	114,3	C	2,9	108,5	9245,90	1014,92	3590,84	3408,63	7,967	9,246	157,5461	27,567
4"	*100*	*114,3*	*D*	*3,2*	*107,9*	*9143,93*	*1116,90*	*3590,84*	*3389,78*	*8,768*	*9,144*	*172,4695*	*30,178*
4"	100	114,3	E	3,6	107,1	9008,84	1251,99	3590,84	3364,65	9,828	9,009	191,9837	33,593
4"	*100*	*114,3*	*F*	*6,3*	*101,7*	*8123,29*	*2137,54*	*3590,84*	*3195,00*	*16,780*	*8,123*	*312,7138*	*54,718*
4"	100	114,3	G	8,8	96,7	7344,17	2916,65	3590,84	3037,92	22,896	7,344	408,6120	71,498

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ISO 4200 - 1991 standard —

NOMINAL DIAMETER		OUTSIDE DIAMETER	THICKNESS SERIE 1	THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm		mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ANSI NPS	ISO DN					FREE	METAL	EXT.	INT.	METAL	WATER		
5"	125	139,7	A	2,0	135,7	14462,71	865	4388,80	4263,14	6,792	14,463	205,11	29,364
5"	125	139,7	B	2,6	134,5	14208,05	1120	4388,80	4225,44	8,791	14,208	263,21	37,682
5"	125	139,7	C	3,2	133,3	13956	1372	4388,80	4187,74	10,772	13,956	319,78	45,780
5"	125	139,7	D	3,6	132,5	13789	1539	4388,80	4162,61	12,083	13,789	356,65	51,059
5"	125	139,7	E	4,0	131,7	13623	1705	4388,80	4137,46	13,386	13,623	392,86	56,243
5"	125	139,7	F	6,3	127,1	12688	2640	4388,80	3992,96	20,726	12,688	588,62	84,269
5"	125	139,7	G	10,0	119,7	11253	4075	4388,80	3760,49	31,986	11,253	861,89	123,392
6"	150	168,3	A	2,0	164,3	21201	1045	5287,30	5161,64	8,202	21,201	361,27	42,931
6"	150	168,3	B	2,6	163,1	20893	1353	5287,30	5123,94	10,625	20,893	464,63	55,215
6"	150	168,3	C	3,2	161,9	20587	1660	5287,30	5086,24	13,029	20,587	565,74	67,230
6"	150	168,3	D	4,0	160,3	20182	2065	5287,30	5035,97	16,208	20,182	697,09	82,839
6"	150	168,3	E	4,5	159,3	19931	2316	5287,30	5004,56	18,178	19,931	777,22	92,361
6"	150	168,3	F	7,1	154,1	18651	3596	5287,30	4841,19	28,226	18,651	1170,19	139,060
6"	150	168,3	G	11,0	146,3	16810	5436	5287,30	4596,15	42,672	16,810	1689,50	200,772
8"	200	219,1	A	2,0	215,1	36339	1364	6883,23	6757,57	10,708	36,339	803,72	73,366
8"	200	219,1	B	2,6	213,9	35934	1768	6883,23	6719,87	13,882	35,934	1036,26	94,593
8"	200	219,1	C	3,6	211,9	35266	2437	6883,23	6657,03	19,132	35,266	1415,22	129,185
8"	200	219,1	D	4,5	210,1	34669	3034	6883,23	6600,49	23,816	34,669	1747,24	159,492
8"	200	219,1	E	6,3	206,5	33491	4212	6883,23	6487,39	33,062	33,491	2386,14	217,813
8"	200	219,1	F	8,0	203,1	32397	5306	6883,23	6380,57	41,648	32,397	2959,63	270,163
8"	200	219,1	G	12,5	194,1	29590	8113	6883,23	6097,83	63,688	29,590	4344,58	396,584
10"	250	273,0	A	2,0	269,0	56832	1703	8576,55	8450,88	13,367	56,832	1563,22	114,522
10"	250	273,0	B	3,6	265,8	55488	3047	8576,55	8350,35	23,918	55,488	2764,60	202,535
10"	250	273,0	C	4,0	265,0	55155	3380	8576,55	8325,22	26,536	55,155	3058,25	224,047
10"	250	273,0	D	5,0	263,0	54325	4210	8576,55	8262,39	33,046	54,325	3780,81	276,983
10"	250	273,0	E	6,3	260,4	53256	5279	8576,55	8180,71	41,437	53,256	4695,82	344,016
10"	250	273,0	F	10,0	253,0	50273	8262	8576,55	7948,23	64,860	50,273	7154,09	524,109
10"	250	273,0	G	14,2	244,6	46990	11545	8576,55	7684,34	90,630	46,990	9694,97	710,254
12"	300	323,9	A	2,6	318,7	79773	2624	10176	10012	20,602	79,773	3386,83	209,128
12"	300	323,9	B	4,0	315,9	78377	4020	10176	9924	31,557	78,377	5143,17	317,577
12"	300	323,9	C	4,5	314,9	77882	4515	10176	9893	35,446	77,882	5759,22	355,617
12"	300	323,9	D	5,6	312,7	76797	5600	10176	9824	43,959	76,797	7094,02	438,037
12"	300	323,9	E	7,1	309,7	75331	7066	10176	9730	55,471	75,331	8869,35	547,660
12"	300	323,9	F	10,0	303,9	72536	9861	10176	9547	77,412	72,536	12158,34	750,747
12"	300	323,9	G	16,0	291,9	66920	15477	10176	9170	121,492	66,920	18389,93	1 135,531
14"	350	355,6	A	2,6	350,4	96431	2883	11172	11008	22,634	96,431	4491,39	252,609
14"	350	355,6	B	4,0	347,6	94896	4418	11172	10920	34,684	94,896	6828,46	384,053
14"	350	355,6	C	5,0	345,6	93807	5507	11172	10857	43,232	93,807	8463,58	476,017
14"	350	355,6	D	5,6	344,4	93157	6158	11172	10820	48,337	93,157	9431,12	530,434
14"	350	355,6	E	8,0	339,6	90579	8736	11172	10669	68,579	90,579	13201,37	742,485
14"	350	355,6	F	11,0	333,6	87406	11909	11172	10480	93,482	87,406	17694,60	995,197
14"	350	355,6	G	17,5	320,6	80727	18588	11172	10072	145,916	80,727	26631,49	1 497,834
16"	400	406,4	A	2,6	401,2	126419	3298	12767	12604	25,892	126,419	6722,80	330,846
16"	400	406,4	B	4,0	398,4	124660	5057	12767	12516	39,695	124,660	10236,15	503,748
16"	400	406,4	C	5,0	396,4	123412	6305	12767	12453	49,496	123,412	12700,75	625,037
16"	400	406,4	D	6,3	393,8	121798	7919	12767	12372	62,163	121,798	15849,43	779,992
16"	400	406,4	E	8,8	388,8	118725	10992	12767	12215	86,288	118,725	21731,73	1 069,475
16"	400	406,4	F	12,5	381,4	114249	15468	12767	11982	121,427	114,249	30030,67	1 477,887
16"	400	406,4	G	20,0	366,4	105439	24278	12767	11511	190,584	105,439	45432,14	2 235,834
18"	450	457,0	A	3,2	450,6	159468	4562	14357	14156	35,812	159,468	11744,24	513,971
18"	450	457,0	B	4,0	449,0	158337	5693	14357	14106	44,687	158,337	14603,21	639,090
18"	450	457,0	C	5,0	447,0	156930	7100	14357	14043	55,735	156,930	18134,20	793,619
18"	450	457,0	D	6,3	444,4	155109	8920	14357	13961	70,024	155,109	22654,16	991,429
18"	450	457,0	E	10,0	437,0	149987	14043	14357	13729	110,237	149,987	35091,32	1 535,725
18"	450	457,0	F	14,2	428,6	144276	19754	14357	13465	155,066	144,276	48463,80	2 120,954
18"	450	457,0	G	22,2	412,6	133705	30324	14357	12962	238,047	133,705	71847,58	3 144,314
20"	500	508,0	A	3,2	501,6	197608	5075	15959	15758	39,837	197,608	16165,36	636,431
20"	500	508,0	B	5,0	498,0	194782	7901	15959	15645	62,024	194,782	24990,60	983,882
20"	500	508,0	C	5,6	496,8	193844	8839	15959	15607	69,384	193,844	27890,14	1 098,037
20"	500	508,0	D	6,3	495,4	192753	9930	15959	15563	77,948	192,753	31246,49	1 230,177
20"	500	508,0	E	11,0	486,0	185508	17175	15959	15268	134,824	185,508	53055,99	2 088,819
20"	500	508,0	F	16,0	476,0	177952	24731	15959	14954	194,135	177,952	74909,04	2 949,175
20"	500	508,0	G	25,0	458,0	164748	37935	15959	14388	297,788	164,748	110918,31	4 366,863

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ISO 4200 - 1991 standard —

NOMINAL DIAMETER		OUTSIDE DIAMETER	THICKNESS SERIE 1	THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm		mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ANSI NPS	ISO DN					FREE	METAL	EXT.	INT.	METAL	WATER		
24"	600	610,0	A	3,2	603,6	286146	6100	19164	18963	47,9	286,1	28078	920,6
24"	*600*	*610,0*	*B*	*5,6*	*598,8*	*281613*	*10633*	*19164*	*18812*	*83,5*	*281,6*	*48558*	*1592,1*
24"	600	610,0	C, D	6,3	597,4	280298	11948	19164	18768	93,8	280,3	54439	1784,9
24"	*600*	*610,0*	*E*	*12,5*	*585,0*	*268783*	*23464*	*19164*	*18378*	*184,2*	*268,8*	*104755*	*3434,6*
24"	600	610,0	F	17,5	575,0	259672	32574	19164	18064	255,7	259,7	143068	4690,7
24"	*600*	*610,0*	*G*	*30,0*	*550,0*	*237583*	*54664*	*19164*	*17279*	*429,1*	*237,6*	*230476*	*7556,6*
28"	700	711,0	A	4,0	703,0	388151	8884	22337	22085	69,7	388,2	55513	1561,5
28"	*700*	*711,0*	*B*	*6,3*	*698,4*	*383068*	*13947*	*22337*	*21941*	*109,5*	*383,1*	*86586*	*2435,6*
28"	700	711,0	C, D	7,1	696,8	381335	15701	22337	21891	123,3	381,3	97251	2735,6
28"	*700*	*711,0*	*E*	*14,2*	*682,6*	*365951*	*31085*	*22337*	*21445*	*244,0*	*366,0*	*188735*	*5309,0*
28"	700	711,0	F	20,0	671,0	353618	43417	22337	21080	340,8	353,6	259351	7295,4
28"	*700*	*711,0*	*G*	*32,0*	*647,0*	*328775*	*68261*	*22337*	*20326*	*535,8*	*328,8*	*394260*	*11090,3*
32"	800	813,0	A	4,0	805,0	508958	10166	25541	25290	79,8	509,0	83172	2046,0
32"	*800*	*813,0*	*B*	*7,1*	*798,8*	*501148*	*17976*	*25541*	*25095*	*141,1*	*501,1*	*145947*	*3590,3*
32"	800	813,0	C, D	8,0	797,0	498892	20232	25541	25038	158,8	498,9	163901	4032,0
32"	*800*	*813,0*	*E*	*16,0*	*781,0*	*479062*	*40062*	*25541*	*24536*	*314,5*	*479,1*	*318222*	*7828,3*
32"	800	813,0	F	22,2	768,6	463971	55153	25541	24146	433,0	464,0	431474	10614,4
32"	*800*	*813,0*	*G*	*36,0*	*741,0*	*431247*	*87877*	*25541*	*23279*	*689,8*	*431,2*	*664594*	*16349,2*
36"	900	914,0	A	4,0	906,0	644683	11435	28714	28463	89,8	644,7	118373	2590,2
36"	*900*	*914,0*	*B*	*8,0*	*898,0*	*633348*	*22770*	*28714*	*28212*	*178,7*	*633,3*	*233651*	*5112,7*
36"	900	914,0	C	8,8	896,4	631093	25025	28714	28161	196,4	631,1	256341	5609,2
36"	*900*	*914,0*	*D*	*10,0*	*894,0*	*627718*	*28400*	*28714*	*28086*	*222,9*	*627,7*	*290147*	*6349,0*
36"	900	914,0	E	17,5	879,0	606831	49288	28714	27615	386,9	606,8	495352	10839,2
36"	*900*	*914,0*	*F*	*25,0*	*864,0*	*586297*	*69822*	*28714*	*27143*	*548,1*	*586,3*	*690317*	*15105,4*
36"	900	914,0	G	40,0	834,0	546288	109830	28714	26201	862,2	546,3	1050904	22995,7
40"	1000	1016,0	A	4,0	1008,0	798015	12717	31919	31667	99,8	798,0	162805	3204,8
40"	*1000*	*1016,0*	*B*	*8,8*	*998,4*	*782887*	*27845*	*31919*	*31366*	*218,6*	*782,9*	*353120*	*6951,2*
40"	1000	1016,0	C, D	10,0	996,0	779128	31604	31919	31290	248,1	779,1	399850	7871,1
40"	*1000*	*1016,0*	*E*	*20,0*	*976,0*	*748151*	*62581*	*31919*	*30662*	*491,3*	*748,2*	*776324*	*15282,0*
40"	1000	1016,0	F	28,0	960,0	723823	86909	31919	30159	682,2	723,8	1061298	20891,7
40"	*1000*	*1016,0*	*G*	*45,0*	*926,0*	*673460*	*137272*	*31919*	*29091*	*1077,6*	*673,5*	*1621294*	*31915,2*
42"	-	1067,0	B	8,8	1049,4	864912	29255	33521	32968	229,7	864,9	409521	7676,1
42"	-	*1067,0*	*C*	*10,0*	*1047,0*	*860961*	*33207*	*33521*	*32892*	*260,7*	*861,0*	*463792*	*8693,4*
42"	-	1067,0	D	11,0	1045,0	857674	36493	33521	32830	286,5	857,7	508735	9535,8
44"	-	1118,0	B	8,8	1100,4	951023	30665	35123	34570	240,7	951,0	471628	8437,0
44"	-	*1118,0*	*C*	*10,0*	*1098,0*	*946879*	*34809*	*35123*	*34495*	*273,2*	*946,9*	*534213*	*9556,6*
44"	-	1118,0	D	11,0	1096,0	943433	38255	35123	34432	300,3	943,4	586055	10484,0
48"	1200	1219,0	B	10,0	1199,0	1129089	37982	38296	37668	298,2	1129,1	694014	11386,6
48"	*1200*	*1219,0*	*C*	*11,0*	*1197,0*	*1125326*	*41745*	*38296*	*37605*	*327,7*	*1125,3*	*761534*	*12494,4*
48"	1200	1219,0	D	12,5	1194,0	1119692	47379	38296	37511	371,9	1119,7	862181	14145,7
56"	1400	1422,0	B	12,5	1397,0	1532790	55351	44673	43888	434,5	1532,8	1374673	19334,4
56"	*1400*	*1422,0*	*C, D*	*14,2*	*1393,6*	*1525338*	*62803*	*44673*	*43781*	*493,0*	*1525,3*	*1556020*	*21885,0*
64"	1600	1626,0	B	14,2	1597,6	2004592	71903	51082	50190	564,4	2004,6	2335153	28722,7
64"	*1600*	*1626,0*	*C, D*	*16,0*	*1594,0*	*1995568*	*80927*	*51082*	*50077*	*635,3*	*1995,6*	*2622409*	*32256,0*
72"	1800	1829,0	B	14,2	1800,6	2546387	80959	57460	56568	635,5	2546,4	3333198	36448,3
72"	*1800*	*1829,0*	*C*	*16,0*	*1797,0*	*2536215*	*91131*	*57460*	*56454*	*715,4*	*2536,2*	*3744614*	*40947,1*
72"	1800	1829,0	D	17,5	1794,0	2527754	99592	57460	56360	781,8	2527,8	4085578	44675,5
80"	2000	2032,0	B	16,0	2000,0	3141593	101335	63837	62832	795,5	3141,6	5148477	50674,0
80"	*2000*	*2032,0*	*C*	*17,5*	*1997,0*	*3132175*	*110753*	*63837*	*62738*	*869,4*	*3132,2*	*5618657*	*55301,7*
80"	2000	2032,0	D	20,0	1992,0	3116510	126418	63837	62581	992,4	3116,5	6397595	62968,5
88"	2200	2235,0	B	17,5	2200,0	3801327	121913	70215	69115	957,0	3801,3	7494038	67060,7
88"	*2200*	*2235,0*	*C*	*20,0*	*2195,0*	*3784068*	*139173*	*70215*	*68958*	*1092,5*	*3784,1*	*8535844*	*76383,4*
88"	2200	2235,0	D	22,2	2190,6	3768912	154328	70215	68820	1211,5	3768,9	9446764	84534,8
100"	-	2540,0	B	20,0	2500,0	4908739	158336	79796	78540	1242,9	4908,7	12569525	98972,6
100"	-	*2540,0*	*C*	*22,2*	*2495,6*	*4891475*	*175600*	*79796*	*78402*	*1378,5*	*4891,5*	*13915868*	*109573,8*
100"	-	2540,0	D	25,0	2490,0	4869547	197528	79796	78226	1550,6	4869,5	15619128	122985,3

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Anc. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
6	1/8"	10,3	...	...	...	10S	1,24	7,82	48	35	324	246	0,28	0,05	0,0369	0,072
6	*1/8"*	*10,3*	*5L*	*STD*	*40*	*40S*	*1,73*	*6,84*	*37*	*47*	*324*	*215*	*0,37*	*0,04*	*0,0445*	*0,086*
6	1/8"	10,3	5L	XS	80	80S	2,41	5,48	24	60	324	172	0,47	0,02	0,0508	0,099
8	1/4"	13,7	...	...	...	10S	1,65	10,40	85	62	430	327	0,49	0,08	0,1155	0,169
8	*1/4"*	*13,7*	*5L*	*STD*	*40*	*40S*	*2,24*	*9,22*	*67*	*81*	*430*	*290*	*0,63*	*0,07*	*0,1375*	*0,201*
8	1/4"	13,7	5L	XS	80	80S	3,02	7,66	46	101	430	241	0,80	0,05	0,1560	0,228
10	3/8"	17,1	...	...	...	10S	1,65	13,80	150	80	537	434	0,63	0,15	0,2417	0,283
10	*3/8"*	*17,1*	*5L*	*STD*	*40*	*40S*	*2,31*	*12,48*	*122*	*107*	*537*	*392*	*0,84*	*0,12*	*0,9006*	*0,362*
10	3/8"	17,1	5L	XS	80	80S	3,20	10,70	90	140	537	336	1,10	0,09	0,3554	0,416
15	1/2"	21,3	...	...	...	5S	1,65	18,00	254	102	669	565	0,80	0,25	0,4951	0,465
15	*1/2"*	*21,3*	...	...	...	*10S*	*2,11*	*17,08*	*229*	*127*	*669*	*537*	*1,00*	*0,23*	*0,6926*	*0,556*
15	1/2"	21,3	5L	STD	40	40S	2,77	15,76	195	161	669	495	1,27	0,20	0,7076	0,664
15	*1/2"*	*21,3*	*5L*	*XS*	*80*	*80S*	*3,73*	*13,84*	*150*	*206*	*669*	*436*	*1,62*	*0,15*	*0,8303*	*0,780*
15	1/2"	21,3	...	...	160	...	4,78	11,74	108	248	669	369	1,95	0,11	0,9171	0,861
15	*1/2"*	*21,3*	*5L*	*XXS*	...	...	*7,47*	*6,36*	*32*	*325*	*669*	*200*	*2,55*	*0,03*	*1,0024*	*0,941*
20	3/4"	26,7	...	...	...	5S	1,65	23,40	430	130	839	735	1,03	0,43	1,0229	0,766
20	*3/4"*	*26,7*	...	...	...	*10S*	*2,11*	*22,48*	*397*	*163*	*839*	*706*	*1,28*	*0,40*	*1,2411*	*0,930*
20	3/4"	26,7	5L	STD	40	40S	2,87	20,96	345	215	839	658	1,69	0,35	1,5473	1,159
20	*3/4"*	*26,7*	*5L*	*XS*	*80*	*80S*	*3,91*	*18,88*	*280*	*280*	*839*	*593*	*2,20*	*0,28*	*1,8710*	*1,401*
20	3/4"	26,7	...	...	160	...	5,56	15,58	191	369	839	489	2,90	0,19	2,2055	1,652
20	*3/4"*	*26,7*	*5L*	*XXS*	...	...	*7,82*	*11,06*	*96*	*464*	*839*	*347*	*3,64*	*0,10*	*2,4212*	*1,814*
25	1"	33,4	...	...	...	5S	1,65	30,10	712	165	1049	946	1,30	0,71	2,0794	1,245
25	*1"*	*33,4*	...	...	...	*10S*	*2,77*	*27,86*	*610*	*267*	*1049*	*875*	*2,09*	*0,61*	*3,1515*	*1,887*
25	1"	33,4	5L	STD	40	40S	3,38	26,64	557	319	1049	837	2,50	0,56	3,6365	2,178
25	*1"*	*33,4*	*5L*	*XS*	*80*	*80S*	*4,55*	*24,30*	*464*	*412*	*1049*	*763*	*3,24*	*0,46*	*4,3972*	*2,633*
25	1"	33,4	...	...	160	...	6,35	20,70	337	540	1049	650	4,24	0,34	5,2075	3,118
25	*1"*	*33,4*	*5L*	*XXS*	...	...	*9,09*	*15,22*	*182*	*694*	*1049*	*478*	*5,45*	*0,18*	*6,8454*	*3,500*
32	1"1/4	42,2	...	...	...	5S	1,65	38,90	1188	210	1328	1222	1,65	1,19	4,3275	2,051
32	*1"1/4*	*42,2*	...	...	...	*10S*	*2,77*	*36,66*	*1056*	*943*	*1326*	*1162*	*2,70*	*1,06*	*6,7013*	*3,176*
32	1"1/4	42,2	5L	STD	40	40S	3,56	35,08	967	432	1326	1102	3,39	0,97	8,1338	3,855
32	*1"1/4*	*42,2*	*5L*	*XS*	*80*	*80S*	*4,85*	*32,50*	*830*	*569*	*1326*	*1021*	*4,47*	*0,83*	*10,0910*	*4,782*
32	1"1/4	42,2	...	...	160	...	6,35	29,50	683	715	1326	927	5,61	0,68	11,8500	5,616
32	*1"1/4*	*42,2*	*5L*	*XXS*	...	...	*9,70*	*22,80*	*408*	*990*	*1326*	*716*	*7,77*	*0,41*	*14,2410*	*6,749*
40	1"1/2	48,3	...	...	...	5S	1,65	45,00	1590	242	1517	1414	1,91	1,59	6,5863	2,727
40	*1"1/2*	*48,3*	...	...	...	*10S*	*2,77*	*42,76*	*1436*	*396*	*1517*	*1343*	*3,11*	*1,44*	*10,3047*	*4,267*
40	*1"1/2*	*48,3*	*5L*	*STD*	*40*	*40S*	*3,68*	*40,94*	*1316*	*516*	*1517*	*1286*	*4,05*	*1,32*	*12,9253*	*5,352*
40	*1"1/2*	*48,3*	*5L*	*XS*	*80*	*80S*	*5,08*	*38,14*	*1142*	*690*	*1517*	*1198*	*5,41*	*1,14*	*16,3261*	*6,761*
40	1"1/2	48,3	...	...	160	...	7,14	34,02	909	923	1517	1069	7,25	0,91	20,1400	8,340
40	*1"1/2*	*48,3*	*5L*	*XXS*	...	...	*10,15*	*28,00*	*616*	*1216*	*1517*	*880*	*9,56*	*0,62*	*23,6980*	*9,813*
50	2"	60,3	...	...	...	5S	1,65	57,00	2552	304	1894	1791	2,40	2,55	13,0825	4,338
50	*2"*	*60,3*	*5L*	...	...	...	*2,11*	*56,08*	*2470*	*386*	*1894*	*1762*	*3,03*	*2,47*	*16,3478*	*5,422*
50	2"	60,3	5L	...	...	10S	2,77	54,76	2355	501	1894	1720	3,93	2,36	20,7601	6,886
50	2"	60,3	5L	...	...	...	3,18	53,94	2285	571	1894	1695	4,48	2,29	23,3451	7,743
50	*2"*	*60,3*	*5L*	...	...	...	*3,58*	*53,14*	*2218*	*638*	*1894*	*1669*	*5,01*	*2,22*	*25,7560*	*8,543*
50	2"	60,3	5L	STD	40	40S	3,91	52,48	2163	693	1894	1649	5,44	2,16	27,6647	9,176
50	*2"*	*60,3*	*5L*	...	...	...	*4,37*	*51,56*	*2086*	*768*	*1894*	*1620*	*6,03*	*2,09*	*30,2078*	*10,019*
50	2"	60,3	5L	...	...	...	4,78	50,74	2022	834	1894	1594	6,54	2,02	32,3626	10,734
50	*2"*	*60,3*	*5L*	*XS*	*80*	*80S*	*5,54*	*49,22*	*1903*	*953*	*1894*	*1546*	*7,48*	*1,90*	*36,0696*	*11,970*
50	2"	60,3	5L	...	...	...	6,35	47,60	1780	1076	1894	1495	8,45	1,78	39,0993	13,167
50	*2"*	*60,3*	*5L*	...	...	...	*7,14*	*46,02*	*1663*	*1192*	*1894*	*1446*	*9,36*	*1,66*	*42,8823*	*14,223*
50	2"	60,3	...	...	160	...	8,74	42,82	1440	1416	1894	1345	11,11	1,44	48,3964	16,052
50	*2"*	*60,3*	...	*XXS*	...	...	*11,07*	*38,16*	*1144*	*1712*	*1894*	*1199*	*13,44*	*1,14*	*54,4903*	*18,073*
65	2"1/2	73,0	5L	...	...	5S	2,11	68,78	3715	470	2293	2161	3,69	3,72	29,5449	8,094
65	*2"1/2*	*73,0*	*5L*	...	...	...	*2,77*	*67,46*	*3574*	*611*	*2293*	*2119*	*4,80*	*3,57*	*37,7383*	*10,339*
65	2"1/2	73,0	...	...	...	10S	3,05	66,90	3515	670	2293	2102	5,26	3,52	41,0722	11,253
65	*2"1/2*	*73,0*	*5L*	...	...	...	*3,18*	*66,64*	*3488*	*698*	*2293*	*2094*	*5,48*	*3,49*	*42,5919*	*11,669*
65	2"1/2	73,0	5L	...	...	...	3,58	65,84	3405	781	2293	2068	6,13	3,40	47,1574	12,920
65	*2"1/2*	*73,0*	*5L*	...	...	...	*3,96*	*65,08*	*3326*	*859*	*2293*	*2046*	*6,74*	*3,33*	*51,3433*	*14,067*
65	2"1/2	73,0	5L	...	...	...	4,37	64,26	3243	942	2293	2019	7,40	3,24	55,6981	15,260
65	*2"1/2*	*73,0*	*5L*	...	...	...	*4,78*	*63,44*	*3161*	*1024*	*2293*	*1993*	*8,04*	*3,16*	*59,8894*	*16,408*
65	2"1/2	73,0	5L	STD	40	40S	5,16	62,58	3086	1100	2293	1969	8,63	3,09	63,6315	17,433
65	*2"1/2*	*73,0*	*5L*	...	...	...	*5,49*	*62,02*	*3021*	*1164*	*2293*	*1948*	*9,14*	*3,02*	*66,7727*	*18,294*
65	2"1/2	73,0	5L	...	...	...	6,35	60,30	2856	1330	2293	1894	10,44	2,86	74,5004	20,411
65	*2"1/2*	*73,0*	*5L*	*XS*	*80*	*80S*	*7,01*	*58,98*	*2732*	*1453*	*2293*	*1863*	*11,41*	*2,73*	*79,9992*	*21,918*
65	2"1/2	73,0	...	...	160	...	9,53	53,94	2285	1900	2293	1695	14,92	2,29	97,8455	26,807
65	*2"1/2*	*73,0*	*5L*	*XXS*	...	...	*14,02*	*44,96*	*1588*	*2598*	*2293*	*1412*	*20,39*	*1,59*	*119,3421*	*32,696*
80	3"	88,9	5L	...	...	5S	2,11	84,68	5632	575	2793	2660	4,51	5,63	54,2011	12,194
80	*3"*	*88,9*	*5L*	...	...	...	*2,77*	*83,36*	*5458*	*750*	*2793*	*2619*	*5,88*	*5,46*	*69,5748*	*15,652*
80	3"	88,9	...	...	...	10S	3,05	82,80	5385	823	2793	2601	6,45	5,38	75,8802	17,071
80	*3"*	*88,9*	*5L*	...	...	...	*3,18*	*82,54*	*5351*	*856*	*2793*	*2593*	*6,72*	*5,35*	*78,7645*	*17,720*
80	3"	88,9	5L	...	...	...	3,58	81,74	5248	960	2793	2568	7,53	5,25	87,4701	19,678
80	*3"*	*88,9*	*5L*	...	...	...	*3,96*	*80,98*	*5150*	*1057*	*2793*	*2544*	*8,29*	*5,15*	*95,6069*	*21,486*
80	3"	88,9	5L	...	...	...	4,37	80,16	5047	1160	2793	2518	9,11	5,05	103,9281	23,381
80	*3"*	*88,9*	*5L*	...	...	...	*4,78*	*79,34*	*4944*	*1263*	*2793*	*2493*	*9,92*	*4,94*	*112,0948*	*25,218*
80	3"	88,9	5L	STD	40	40S	5,49	77,92	4769	1439	2793	2448	11,29	4,77	125,6504	28,268
80	*3"*	*88,9*	*5L*	...	...	...	*6,35*	*76,20*	*4560*	*1647*	*2793*	*2394*	*12,93*	*4,56*	*141,1066*	*31,745*
80	3"	88,9	5L	...	...	...	7,14	74,62	4373	1834	2793	2344	14,40	4,37	154,4117	34,738
80	*3"*	*88,9*	*5L*	*XS*	*80*	*80S*	*7,62*	*73,66*	*4261*	*1946*	*2793*	*2314*	*15,27*	*4,26*	*162,0938*	*36,467*
80	3"	88,9	...	...	160	...	11,13	66,64	3488	2719	2793	2094	21,35	3,49	209,7957	47,198
80	*3"*	*88,9*	*5L*	*XXS*	...	...	*15,24*	*58,42*	*2680*	*3627*	*2793*	*1836*	*27,68*	*2,68*	*249,4271*	*56,114*

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— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Anc. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
...	3"1/2	101,6	5L	...	...	5S	2,11	97,38	7448	659	3192	3059	5,18	7,45	81,6349	16,070
...	*3"1/2*	*101,6*	*5L*	...	...	...	*2,77*	*96,06*	*7247*	*860*	*3192*	*3018*	*6,75*	*7,25*	*105,0865*	*20,686*
...	3"1/2	101,6	...	...	...	10S	3,05	95,50	7163	944	3192	3000	7,40	7,16	114,7481	22,588
...	*3"1/2*	*101,6*	*5L*	...	...	...	*3,18*	*96,24*	*7124*	*983*	*3192*	*2992*	*7,72*	*7,12*	*119,1764*	*23,460*
...	3"1/2	101,6	5L	...	...	...	3,58	94,44	7006	1102	3192	2967	8,65	7,00	132,5783	26,098
...	*3"1/2*	*101,6*	*5L*	...	...	...	*3,96*	*93,68*	*6893*	*1215*	*3192*	*2943*	*9,53*	*6,89*	*144,9947*	*28,542*
...	3"1/2	101,6	5L	...	...	...	4,37	92,86	6772	1335	3192	2917	10,48	6,77	158,0588	31,114
...	*3"1/2*	*101,6*	*5L*	...	...	...	*4,78*	*92,04*	*6653*	*1454*	*3192*	*2892*	*11,41*	*6,65*	*170,7813*	*33,618*
...	3"1/2	101,6	5L	STD	40	40S	5,74	90,12	6379	1729	3192	2831	13,57	6,38	199,2684	39,226
...	*3"1/2*	*101,6*	*5L*	...	...	...	*6,35*	*88,90*	*6207*	*1900*	*3192*	*2793*	*14,92*	*6,21*	*216,4485*	*42,608*
...	3"1/2	101,6	5L	...	...	...	7,14	87,32	5988	2119	3192	2743	16,63	5,99	237,6710	46,786
...	*3"1/2*	*101,6*	*5L*	*XS*	*80*	*80S*	*8,08*	*85,44*	*5733*	*2374*	*3192*	*2684*	*18,63*	*5,73*	*261,4656*	*51,470*
100	4"	114,3	5L	...	...	5S	2,11	110,08	9517	744	3591	3458	5,84	9,52	117,0465	20,481
100	*4"*	*114,3*	*5L*	...	...	...	*2,77*	*108,76*	*9290*	*971*	*3591*	*3417*	*7,62*	*9,29*	*151,0020*	*26,422*
100	4"	114,3	5L	...	...	10S	3,05	108,20	9195	1066	3591	3399	8,36	9,19	165,0388	28,878
100	*4"*	*114,3*	*5L*	...	...	...	*3,18*	*107,94*	*9151*	*1110*	*3591*	*3391*	*8,71*	*9,15*	*171,4823*	*30,006*
100	4"	114,3	5L	...	...	...	3,58	107,14	9016	1245	3591	3365	9,77	9,02	191,0183	33,424
100	*4"*	*114,3*	*5L*	...	...	...	*3,96*	*106,38*	*8888*	*1373*	*3591*	*3342*	*10,78*	*8,89*	*209,1766*	*36,601*
100	4"	114,3	5L	...	...	...	4,37	105,56	8752	1509	3591	3316	11,85	8,75	228,3367	39,954
100	*4"*	*114,3*	*5L*	...	...	...	*4,78*	*104,74*	*8616*	*1645*	*3591*	*3291*	*12,91*	*8,62*	*247,0655*	*43,229*
100	4"	114,3	5L	...	...	...	5,16	103,98	8492	1769	3591	3267	13,89	8,49	264,0165	46,197
100	*4"*	*114,3*	*5L*	...	...	...	*5,56*	*103,18*	*8361*	*1899*	*3591*	*3241*	*14,91*	*8,36*	*281,4729*	*49,252*
100	4"	114,3	5L	STD	40	40S	6,02	102,26	8213	2048	3591	3213	16,07	8,21	301,0519	52,678
100	*4"*	*114,3*	*5L*	...	...	...	*6,35*	*101,60*	*8107*	*2154*	*3591*	*3192*	*16,90*	*8,11*	*314,7761*	*55,079*
100	4"	114,3	5L	...	...	...	7,14	100,02	7857	2404	3591	3142	18,87	7,86	346,5612	60,641
100	*4"*	*114,3*	*5L*	...	...	...	*7,92*	*98,46*	*7614*	*2647*	*3591*	*3093*	*20,78*	*7,61*	*376,5005*	*65,879*
100	4"	114,3	5L	XS	80	80S	8,56	97,18	7417	2844	3591	3053	22,32	7,42	400,0262	69,996
100	*4"*	*114,3*	*5L*	...	*120*	...	*11,13*	*92,04*	*6653*	*3607*	*3591*	*2892*	*28,32*	*6,65*	*485,5674*	*84,962*
100	4"	114,3	5L	...	160	...	13,49	87,32	5988	4272	3591	2743	33,54	5,99	552,4471	96,666
100	*4"*	*114,3*	*5L*	*XXS*	...	...	*17,12*	*80,06*	*5034*	*5227*	*3591*	*2515*	*41,03*	*5,03*	*636,1621*	*111,314*
125	5"	141,3	5L	...	...	...	2,11	137,08	14758	923	4439	4306	7,24	14,76	223,494	31,634
125	*5"*	*141,3*	...	...	...	*5S*	*2,77*	*135,76*	*14475*	*1206*	*4439*	*4255*	*9,47*	*14,48*	*289,298*	*40,948*
125	5"	141,3	5L	...	...	...	3,18	134,94	14301	1380	4439	4239	10,83	14,30	329,220	46,599
125	*5"*	*141,3*	...	...	...	*10S*	*3,40*	*134,50*	*14208*	*1473*	*4439*	*4226*	*11,57*	*14,21*	*350,345*	*49,589*
125	5"	141,3	5L	...	...	...	3,96	133,38	13972	1709	4439	4190	13,41	13,97	403,188	57,068
125	*5"*	*141,3*	*5L*	...	...	...	*4,78*	*131,74*	*13631*	*2050*	*4439*	*4139*	*16,09*	*13,63*	*478,199*	*67,686*
125	5"	141,3	5L	...	...	...	5,56	130,18	13310	2371	4439	4090	18,61	13,31	546,999	77,424
125	*5"*	*141,3*	*5L*	*STD*	*40*	*40S*	*6,55*	*128,20*	*12908*	*2773*	*4439*	*4028*	*21,77*	*12,91*	*630,830*	*89,269*
125	5"	141,3	5L	...	...	...	7,14	127,02	12672	3009	4439	3990	23,62	12,67	678,978	96,104
125	*5"*	*141,3*	*5L*	...	...	...	*7,92*	*125,46*	*12362*	*3319*	*4439*	*3941*	*26,06*	*12,36*	*740,604*	*104,827*
125	5"	141,3	5L	...	...	...	8,74	123,82	12041	3640	4439	3890	28,57	12,04	802,958	113,653
125	*5"*	*141,3*	*5L*	*XS*	*80*	*80S*	*9,53*	*122,24*	*11736*	*3945*	*4439*	*3840*	*30,97*	*11,74*	*860,732*	*121,830*
125	5"	141,3	5L	...	120	...	12,70	115,90	10550	5131	4439	3641	40,28	10,55	1071,030	151,597
125	*5"*	*141,3*	*5L*	...	*160*	...	*15,88*	*109,54*	*9424*	*6257*	*4439*	*3441*	*49,11*	*9,42*	*1250,023*	*176,932*
125	5"	141,3	5L	XXS	...	...	19,05	103,20	8365	7316	4439	3242	57,43	8,36	1399,978	198,157
150	6"	168,3	5L	...	...	...	2,11	164,08	21145	1102	5287	5155	8,65	21,14	360,388	45,204
150	*6"*	*168,3*	...	...	...	*5S*	*2,77*	*162,76*	*20806*	*1440*	*5287*	*5113*	*11,31*	*20,81*	*493,505*	*58,646*
150	6"	168,3	5L	...	...	...	3,18	161,94	20597	1650	5287	5087	11,57	20,60	562,402	66,833
150	*6"*	*168,3*	...	...	...	*10S*	*3,40*	*161,60*	*20485*	*1761*	*5287*	*5074*	*12,96*	*20,48*	*598,943*	*71,176*
150	6"	168,3	5L	...	...	...	3,58	161,14	20394	1853	5287	5062	14,54	20,39	628,818	74,702
150	*6"*	*168,3*	*5L*	...	...	...	*3,96*	*160,38*	*20202*	*2046*	*5287*	*5038*	*16,05*	*20,20*	*690,617*	*82,070*
150	6"	168,3	5L	...	...	...	4,37	159,56	19996	2251	5287	5013	17,67	20,00	756,528	89,902
150	*6"*	*168,3*	*5L*	...	...	...	*4,78*	*158,74*	*19791*	*2456*	*5287*	*4987*	*19,27*	*19,79*	*821,432*	*97,615*
150	6"	168,3	5L	...	...	...	5,16	157,98	19602	2645	5287	4963	20,76	19,60	880,694	104,658
150	*6"*	*168,3*	*5L*	...	...	...	*5,56*	*157,18*	*19404*	*2843*	*5287*	*4938*	*22,31*	*19,40*	*942,169*	*111,962*
150	6"	168,3	5L	...	...	...	6,35	155,60	19016	3231	5287	4888	25,36	19,02	1060,825	126,064
150	6"	168,3	5L	STD	40	40S	7,11	154,08	18646	3600	5287	4841	28,26	18,65	1171,62	139,230
150	*6"*	*168,3*	*5L*	...	...	...	*7,92*	*152,46*	*18256*	*3990*	*5287*	*4790*	*31,32*	*18,26*	*1286,16*	*152,841*
150	6"	168,3	5L	...	...	...	8,74	150,82	17865	4381	5287	4738	34,39	17,87	1398,44	166,184
150	*6"*	*168,3*	*5L*	...	...	...	*9,53*	*149,24*	*17493*	*4753*	*5287*	*4689*	*37,31*	*17,49*	*1503,21*	*178,636*
150	6"	168,3	5L	XS	80	80S	10,97	146,36	16824	5422	5287	4598	42,56	16,82	1685,81	200,333
150	*6"*	*168,3*	*5L*	...	...	...	*12,70*	*142,90*	*16098*	*6208*	*5287*	*4489*	*48,73*	*16,04*	*1891,37*	*224,761*
150	6"	168,3	5L	...	120	...	14,27	139,76	15341	6905	5287	4391	54,20	15,34	2065,43	245,447
150	*6"*	*168,3*	*5L*	...	...	...	*15,88*	*136,54*	*14642*	*7604*	*5287*	*4290*	*59,69*	*14,64*	*2232,16*	*265,259*
150	6"	168,3	5L	...	160	...	18,26	131,78	13639	8607	5287	4140	67,56	13,64	2457,92	292,087
150	*6"*	*168,3*	*5L*	...	...	...	*19,05*	*130,20*	*13314*	*8932*	*5287*	*4090*	*70,11*	*13,31*	*2527,65*	*300,374*
150	6"	168,3	...	XXS	...	...	21,95	124,40	12154	10092	5287	3908	79,22	12,15	2762,70	328,307
150	*6"*	*168,3*	*5L*	...	...	...	*22,23*	*123,84*	*12046*	*10201*	*5287*	*3891*	*80,07*	*12,06*	*2783,73*	*330,805*
200	8"	219,1	...	...	...	5S	2,77	213,56	35820	1883	6883	6709	14,79	35,82	1101,44	100,542
200	*8"*	*219,1*	*5L*	...	...	...	*3,18*	*212,74*	*35546*	*2157*	*6883*	*6683*	*16,93*	*35,55*	*1257,36*	*114,775*
200	8"	219,1	...	...	...	10S	3,76	211,58	35159	2544	6883	6647	19,96	35,16	1474,87	134,630
200	*8"*	*219,1*	*5L*	...	...	...	*3,96*	*211,18*	*35026*	*2676*	*6883*	*6634*	*21,01*	*36,03*	*1549,05*	*141,401*
200	8"	219,1	5L	...	...	...	4,78	209,54	34484	3218	6883	6583	25,26	34,48	1848,81	168,764
200	*8"*	*219,1*	*5L*	...	...	...	*5,16*	*208,78*	*34235*	*3468*	*6883*	*6559*	*27,22*	*34,23*	*1985,36*	*181,228*
200	8"	219,1	5L	...	...	...	5,56	207,98	33973	3730	6883	6534	29,28	33,97	2127,49	194,202
200	*8"*	*219,1*	*5L*	...	*20*	...	*6,35*	*206,40*	*33459*	*4244*	*6883*	*6484*	*33,31*	*33,46*	*2403,42*	*219,390*
200	8"	219,1	5L	...	30	...	7,04	205,02	33013	4690	6883	6441	36,81	33,01	2639,29	240,921
200	*8"*	*219,1*	*5L*	...	...	...	*7,92*	*203,26*	*32448*	*5254*	*6883*	*6386*	*41,24*	*32,45*	*2933,28*	*267,757*
200	8"	219,1	5L	STD	40	40S	8,18	202,74	32283	5420	6883	6369	42,55	32,28	3018,89	275,554
200	*8"*	*219,1*	*5L*	...	...	...	*8,74*	*201,62*	*31927*	*5776*	*6883*	*6334*	*45,34*	*31,93*	*3200,44*	*292,144*
200	8"	219,1	5L	...	...	...	9,53	200,04	31428	6274	6883	6284	49,25	31,43	3451,73	315,083
200	*8"*	*219,1*	...	...	*60*	...	*10,31*	*198,48*	*30940*	*6763*	*6883*	*6235*	*53,08*	*30,94*	*3694,07*	*337,204*
200	8"	219,1	5L	...	...	...	11,13	196,84	30431	7272	6883	6184	57,08	30,43	3942,75	359,904
200	*8"*	*219,1*	*5L*	*XS*	*80*	*80S*	*12,70*	*193,70*	*29468*	*8236*	*6883*	*6085*	*64,64*	*29,47*	*4401,84*	*401,811*

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

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NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Ano. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
200	8"	219,1	5L	...	...	...	14,27	190,56	28520	9163	6883	5987	72,08	28,52	4839,13	441,726
200	*8"*	*219,1*	...	...	*100*	...	*15,09*	*188,92*	*28031*	*9671*	*6883*	*5935*	*76,92*	*28,03*	*5059,10*	*461,807*
200	8"	219,1	5L	...	...	...	15,88	187,34	27565	10136	6883	5885	79,58	27,58	5265,67	480,664
200	*8"*	*219,1*	*5L*	...	*120*	...	*18,26*	*182,58*	*26182*	*11521*	*6883*	*5736*	*90,44*	*26,18*	*5857,15*	*534,656*
200	8"	219,1	5L	...	...	...	19,05	181,00	25730	11972	6883	5688	93,98	25,73	6043,53	551,669
200	*8"*	*219,1*	*5L*	...	*140*	...	*20,62*	*177,86*	*24846*	*12857*	*6883*	*5588*	*100,92*	*24,85*	*6399,72*	*584,183*
200	8"	219,1	5L	XXS	...	...	22,23	174,64	23954	13749	6883	5486	107,92	23,95	6745,91	615,783
200	*8"*	*219,1*	...	...	*160*	...	*23,01*	*173,08*	*23528*	*14175*	*6883*	*5437*	*111,27*	*23,53*	*6906,88*	*630,478*
200	8"	219,1	5L	...	...	...	25,40	168,30	22246	15457	6883	5287	121,33	22,25	7373,72	673,092
250	10"	273,0	...	...	...	5S	3,40	266,20	55655	2660	8577	8363	22,63	55,66	2616,76	191,71
250	*10"*	*273,0*	*5L*	...	...	...	*3,96*	*265,08*	*55188*	*3347*	*8577*	*8328*	*26,26*	*55,19*	*3029,00*	*221,90*
250	10"	273,0	...	...	...	10S	4,19	264,62	54997	3538	8577	8313	27,78	55,00	3196,80	234,20
250	*10"*	*273,0*	*5L*	...	...	...	*4,78*	*263,44*	*54507*	*4029*	*8577*	*8276*	*31,63*	*54,51*	*3623,26*	*265,44*
250	10"	273,0	5L	...	...	...	5,16	262,68	54193	4342	8577	8252	34,09	54,19	3894,91	285,34
250	*10"*	*273,0*	*5L*	...	...	...	*5,56*	*261,88*	*53863*	*4671*	*8577*	*8227*	*36,68*	*53,86*	*4178,32*	*306,10*
250	10"	273,0	5L	...	20	...	6,35	260,30	53216	5319	8577	8178	41,77	53,22	4730,47	346,55
250	*10"*	*273,0*	*5L*	...	...	...	*7,09*	*258,82*	*52612*	*5923*	*8577*	*8131*	*46,61*	*52,61*	*5238,64*	*383,78*
250	10"	273,0	5L	...	30	...	7,80	257,40	52036	6499	8577	8086	51,03	52,04	5718,08	418,91
250	*10"*	*273,0*	*5L*	...	...	...	*8,74*	*255,52*	*51279*	*7256*	*8577*	*8027*	*56,96*	*51,28*	*6340,75*	*464,52*
250	10"	273,0	5L	STD	40	40S	9,27	254,46	50854	7680	8577	7994	60,31	50,85	6685,82	489,80
250	*10"*	*273,0*	*5L*	...	...	...	*11,13*	*250,74*	*49378*	*9157*	*8577*	*7877*	*71,90*	*49,38*	*7863,14*	*576,06*
250	10"	273,0	5L	XS	60	80S*	12,70	247,60	48149	10386	8577	7779	81,55	48,15	8816,96	645,93
250	*10"*	*273,0*	*5L*	...	...	...	*14,27*	*244,46*	*46936*	*11599*	*8577*	*7680*	*91,08*	*46,94*	*9736,17*	*713,20*
250	10"	273,0	...	...	80	...	15,09	242,82	46308	12227	8577	7628	96,01	46,31	10200,88	747,32
250	*10"*	*273,0*	*5L*	...	...	...	*15,88*	*241,24*	*45708*	*12827*	*8577*	*7579*	*100,73*	*45,71*	*10640,73*	*779,64*
250	10"	273,0	5L	...	100	...	18,26	236,48	43922	14613	8577	7429	114,75	43,92	11914,56	872,86
250	*10"*	*273,0*	*5L*	...	...	...	*20,62*	*231,76*	*42186*	*16349*	*8577*	*7281*	*128,36*	*42,19*	*13103,97*	*960,00*
250	10"	273,0	...	...	120	...	21,44	230,12	41591	16944	8577	7229	133,06	41,59	13500,58	989,05
250	*10"*	*273,0*	*5L*	...	...	...	*22,23*	*228,54*	*41022*	*17513*	*8577*	*7180*	*137,52*	*41,02*	*13874,76*	*1016,47*
250	10"	273,0	5L	...	...	...	23,83	225,34	39881	18654	8577	7079	146,48	39,88	14608,17	1070,27
250	*10"*	*273,0*	*5L*	*XXS*	*140*	...	*25,40*	*222,20*	*38777*	*19758*	*8577*	*6981*	*155,15*	*38,78*	*15300,02*	*1120,88*
250	10"	273,0	...	...	160	...	28,58	215,84	36589	21946	8577	6781	172,33	36,59	16612,31	1217,02
250	*10"*	*273,0*	*5L*	...	...	...	*31,75*	*209,50*	*34471*	*24064*	*8577*	*6582*	*188,97*	*34,47*	*17809,97*	*1304,76*
300	12"	323,8	...	...	...	5S	3,96	315,88	78387	3979	10172	9924	31,25	78,37	5088,85	314,32
300	*12"*	*323,8*	*5L*	...	...	...	*4,37*	*315,06*	*77961*	*4385*	*10172*	*9898*	*34,43*	*77,96*	*5694,35*	*346,54*
300	12"	323,8	...	...	...	10S	4,57	314,66	77763	4583	10172	9885	36,00	77,76	5839,51	360,69
300	*12"*	*323,8*	*5L*	...	...	...	*4,78*	*314,24*	*77566*	*4791*	*10172*	*9872*	*37,62*	*77,56*	*6096,92*	*376,52*
300	12"	323,8	5L	...	...	...	5,16	313,48	77181	5165	10172	9848	40,56	77,18	6557,29	405,02
300	*12"*	*323,8*	*5L*	...	...	...	*5,56*	*312,68*	*76787*	*5559*	*10172*	*9823*	*43,65*	*76,79*	*7039,33*	*434,80*
300	12"	323,8	5L	...	20	...	6,35	311,10	76013	6333	10172	9773	49,73	76,01	7980,56	492,93
300	*12"*	*323,8*	*5L*	...	...	...	*7,14*	*309,52*	*75243*	*7103*	*10172*	*9724*	*55,77*	*75,24*	*8907,55*	*550,19*
300	12"	323,8	5L	...	...	...	7,92	307,96	74467	7860	10172	9675	61,71	74,49	9808,99	605,87
300	*12"*	*323,8*	*5L*	...	*30*	...	*8,38*	*307,04*	*74042*	*8304*	*10172*	*9646*	*65,20*	*74,04*	*10334,23*	*638,31*
300	12"	323,8	5L	...	...	...	8,74	306,32	73695	8651	10172	9623	67,93	73,70	10742	663,50
300	*12"*	*323,8*	*5L*	*STD*	...	*40S**	*9,53*	*304,74*	*72937*	*9409*	*10172*	*9574*	*73,88*	*72,94*	*11627*	*718,15*
300	12"	323,8	5L	...	40	...	10,31	303,18	72192	10154	10172	9525	79,73	72,19	12487	771,26
300	*12"*	*323,8*	*5L*	...	...	...	*11,13*	*301,54*	*71413*	*10933*	*10172*	*9473*	*85,84*	*71,41*	*13377*	*826,26*
300	12"	323,8	5L	XS	...	80S*	12,70	298,40	69934	12412	10172	9375	97,46	69,93	15041	929,05
300	*12"*	*323,8*	*5L*	...	*60*	...	*14,27*	*295,26*	*68470*	*13876*	*10172*	*9276*	*108,96*	*68,47*	*16654*	*1028,65*
300	12"	323,8	5L	...	...	...	15,88	292,04	66985	15362	10172	9175	120,62	66,98	18255	1127,54
300	*12"*	*323,8*	*5L*	...	*80*	...	*17,48*	*288,84*	*65525*	*16822*	*10172*	*9074*	*132,08*	*65,52*	*19794*	*1222,62*
300	12"	323,8	5L	...	...	...	19,05	285,70	64108	18238	10172	8976	143,21	64,11	21258	1312,90
300	*12"*	*323,8*	*5L*	...	...	...	*20,62*	*282,56*	*62706*	*19640*	*10172*	*8877*	*154,21*	*62,71*	*22670*	*1400,26*
300	12"	323,8	...	...	100	...	21,44	280,92	61981	20366	10172	8825	159,91	61,98	23390	1444,74
300	*12"*	*323,8*	*5L*	...	...	...	*22,23*	*279,34*	*61285*	*21061*	*10172*	*8776*	*165,37*	*61,29*	*24072*	*1486,86*
300	12"	323,8	5L	...	...	...	23,83	276,14	59889	22457	10172	8675	176,33	59,89	25418	1570,01
300	*12"*	*323,8*	*5L*	*XXS*	*120*	...	*25,40*	*273,00*	*58535*	*23811*	*10172*	*8577*	*186,97*	*58,53*	*26695*	*1648,84*
300	12"	323,8	5L	...	...	...	26,97	269,86	57196	25150	10172	8478	197,48	57,20	27928	1725,00
300	*12"*	*323,8*	*5L*	...	*140*	...	*28,58*	*266,64*	*55839*	*26507*	*10172*	*8377*	*208,14*	*55,84*	*29148*	*1800,38*
300	12"	323,8	5L	...	...	...	31,75	260,30	53216	29131	10172	8178	228,74	53,22	31425	1941,03
300	*12"*	*323,8*	*5L*	...	*160*	...	*33,32*	*257,16*	*51939*	*30407*	*10172*	*8079*	*238,76*	*51,94*	*32493*	*2006,98*
350	14"	355,6	...	...	...	5S	3,96	347,68	94940	4375	11172	10923	34,36	94,94	6762	380,34
350	*14"*	*355,6*	*5L*	...	...	*10S**	*4,78*	*346,04*	*94046*	*5268*	*11172*	*10871*	*41,36*	*94,05*	*8106*	*455,92*
350	14"	355,6	5L	...	...	...	5,16	345,28	93634	5681	11172	10847	44,59	93,63	8723	490,58
350	*14"*	*355,6*	*5L*	...	...	...	*5,33*	*344,94*	*93460*	*5865*	*11172*	*10837*	*46,04*	*93,45*	*8997*	*506,02*
350	14"	355,6	5L	...	...	...	5,56	344,48	93200	6114	11172	10822	47,99	93,20	9367	526,82
350	*14"*	*355,6*	*5L*	...	*10*	...	*6,35*	*342,90*	*92347*	*6967*	*11172*	*10773*	*54,69*	*92,35*	*10626*	*597,66*
350	14"	355,6	5L	...	...	...	7,14	341,32	91498	7816	11172	10723	61,35	91,50	11869	667,52
350	*14"*	*355,6*	*5L*	...	*20*	...	*7,92*	*339,76*	*90664*	*8651*	*11172*	*10674*	*67,90*	*90,66*	*13078*	*735,56*
350	14"	355,6	5L	...	...	...	8,74	338,12	89791	9524	11172	10622	74,78	89,79	14332	806,08
350	*14"*	*355,6*	*5L*	*STD*	*30*	...	*9,53*	*336,54*	*88964*	*10361*	*11172*	*10573*	*81,33*	*88,95*	*15523*	*873,06*
350	14"	355,6	5L	...	...	...	10,31	334,98	88131	11184	11172	10524	87,79	88,13	16682	938,27
350	*14"*	*355,6*	*5L*	...	*40*	...	*11,13*	*333,34*	*87270*	*12045*	*11172*	*10472*	*94,55*	*87,27*	*17884*	*1005,84*
350	14"	355,6	5L	...	...	...	11,91	331,78	86455	12860	11172	10423	100,94	86,46	19011	1069,21
350	*14"*	*355,6*	*5L*	*XS*	...	...	*12,70*	*330,20*	*85634*	*13681*	*11172*	*10374*	*107,99*	*85,63*	*20136*	*1132,48*
350	14"	355,6	5L	...	...	...	14,27	327,06	84013	15302	11172	10275	120,11	84,01	22324	1255,55
350	*14"*	*355,6*	...	...	*60*	...	*15,09*	*325,42*	*83172*	*16142*	*11172*	*10223*	*126,71*	*83,17*	*23442*	*1318,44*
350	14"	355,6	5L	...	...	...	15,88	323,84	82367	16948	11172	10174	133,03	82,37	24503	1378,13
350	*14"*	*355,6*	*5L*	...	...	...	*17,48*	*320,64*	*80747*	*18568*	*11172*	*10073*	*146,75*	*80,75*	*26606*	*1496,38*
350	14"	355,6	5L	...	80	...	19,05	317,50	79173	20142	11172	9975	158,10	79,17	28608	1609,02
350	*14"*	*355,6*	*5L*	...	...	...	*20,62*	*314,36*	*77615*	*21700*	*11172*	*9876*	*170,33*	*77,61*	*30653*	*1718,37*
350	14"	355,6	5L	...	...	...	22,23	311,14	76033	23282	11172	9775	182,75	76,03	32687	1827,15
350	*14"*	*355,6*	*5L*	...	*100*	...	*23,83*	*307,94*	*74477*	*24838*	*11172*	*9674*	*194,96*	*74,48*	*34650*	*1931,96*
350	14"	355,6	5L	...	...	...	25,40	304,80	72966	26349	11172	9575	206,83	72,97	36123	2031,88

D MEQ 1460 F

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Ano. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
350	14"	355,6	5L	...	...	...	26,97	301,66	71470	27844	11172	9477	218,57	71,47	37842	2128,36
350	14"	355,6	...	...	120	...	27,79	300,02	70695	28619	11172	9425	224,65	70,70	38719	2177,68
350	14"	355,6	5L	...	...	...	28,58	298,44	69959	29362	11172	9378	230,48	69,95	39550	2224,43
350	14"	355,6	...	...	140	...	31,75	292,10	67012	32303	11172	9177	253,56	67,01	42755	2404,68
350	14"	355,6	...	...	160	...	35,71	284,18	63427	35887	11172	8928	281,70	63,43	46476	2613,96
350	14"	355,6	...	...	...	...	50,80	254,00	50671	48644	11172	7980	381,83	50,67	58059	3265,40
350	14"	355,6	...	...	...	...	53,98	247,64	48166	51150	11172	7780	401,50	48,16	60030	3376,24
350	14"	355,6	...	...	...	...	55,88	243,84	46698	52617	11172	7660	413,01	46,70	61137	3438,52
350	14"	355,6	...	...	...	...	63,50	228,60	41043	58271	11172	7182	457,40	41,04	65085	3660,59
400	16"	406,4	...	...	...	5S	4,19	398,02	124423	5294	12767	12504	41,56	124,42	10707	526,93
400	16"	406,4	5L	...	...	10S*	4,78	396,84	123686	6031	12767	12467	47,34	123,69	12162	598,51
400	16"	406,4	5L	...	...	...	5,16	396,08	123213	6504	12767	12443	51,06	123,21	13092	644,27
400	16"	406,4	5L	...	...	...	5,56	395,28	122716	7002	12767	12418	54,96	122,72	14066	692,16
400	16"	406,4	5L	...	10	...	6,35	393,70	121736	7981	12767	12368	62,84	121,74	15969	785,89
400	16"	406,4	5L	...	...	...	7,14	392,12	120761	8956	12767	12319	70,30	120,76	17851	878,50
400	16"	406,4	5L	...	20	...	7,92	390,56	119802	9915	12767	12270	77,83	119,80	19687	968,84
400	16"	406,4	5L	...	...	...	8,74	388,92	118798	10919	12767	12218	86,71	118,80	21593	1062,66
400	16"	406,4	5L	STD	30	...	9,53	387,34	117835	11882	12767	12169	93,27	117,84	23407	1151,93
400	16"	406,4	5L	...	...	...	10,31	385,78	116888	12829	12767	12120	100,70	116,89	25176	1239,00
400	16"	406,4	5L	...	...	...	11,13	384,14	115896	13821	12767	12068	108,49	115,90	27013	1329,40
400	16"	406,4	5L	...	...	...	11,91	382,58	114957	14760	12767	12019	115,56	114,96	28739	1414,33
400	16"	406,4	5L	XS	40	...	12,70	381,00	114009	15708	12767	11969	123,30	114,01	30486	1499,30
400	16"	406,4	5L	...	...	...	14,27	377,86	112138	17579	12767	11871	137,99	112,14	33934	1665,04
400	16"	406,4	5L	...	...	...	15,88	374,64	110235	19482	12767	11770	152,93	110,23	37201	1830,77
400	16"	406,4	...	...	60	...	16,66	373,08	109319	20399	12767	11721	160,12	109,32	38802	1909,54
400	16"	406,4	5L	...	...	...	17,48	371,44	108360	21358	12767	11669	167,65	108,36	40463	1991,29
400	16"	406,4	5L	...	...	...	20,62	365,16	104726	24991	12767	11472	181,97	104,73	46624	2294,47
400	16"	406,4	...	...	80	...	21,44	363,52	103788	25929	12767	11420	196,10	103,79	48181	2371,12
400	16"	406,4	5L	...	...	...	22,23	361,94	102888	26830	12767	11371	203,53	102,89	49662	2443,98
400	16"	406,4	5L	...	...	...	23,83	358,74	101076	28641	12767	11270	210,60	101,08	52602	2588,66
400	16"	406,4	5L	...	...	...	25,40	355,60	99315	30402	12767	11172	224,82	99,31	55411	2726,91
400	16"	406,4	...	...	100	...	26,19	354,02	98434	31283	12767	11122	245,56	98,43	56797	2795,10
400	16"	406,4	5L	...	...	...	26,97	352,46	97568	32149	12767	11073	252,35	97,57	58147	2861,65
400	16"	406,4	5L	...	...	...	28,58	349,24	95794	33923	12767	10972	268,28	95,79	60877	2995,93
400	16"	406,4	5L	...	...	...	30,18	346,04	94046	35671	12767	10871	280,00	94,05	63517	3125,84
400	16"	406,4	...	...	120	...	30,96	344,48	93200	36517	12767	10822	286,64	93,20	64778	3187,86
400	16"	406,4	5L	...	...	...	31,75	342,90	92347	37370	12767	10773	293,33	92,35	66037	3249,86
400	16"	406,4	...	...	140	...	36,53	333,34	87270	42447	12767	10472	333,19	87,27	73295	3607,02
400	16"	406,4	...	...	160	...	40,49	325,42	83172	46546	12767	10223	366,35	83,17	78853	3880,54
450	18"	457,0	...	...	...	5S	4,19	448,62	158089	5980	14357	14094	46,81	158,07	15278	668,81
450	18"	457,0	5L	...	...	10S*	4,78	447,44	157239	6791	14357	14057	53,31	157,24	17361	759,80
450	18"	457,0	5L	...	...	...	5,56	445,88	156144	7885	14357	14008	61,90	156,14	20091	879,25
450	18"	457,0	5L	...	10	...	6,35	444,30	155040	8990	14357	13958	70,57	155,04	22826	998,97
450	18"	457,0	5L	...	...	...	7,14	442,72	153939	10091	14357	13908	79,21	153,94	25533	1117,41
450	18"	457,0	5L	...	20	...	7,92	441,16	152856	11174	14357	13859	87,71	152,86	28177	1233,12
450	18"	457,0	5L	...	...	...	8,74	439,52	151722	12308	14357	13808	96,81	151,72	30926	1353,44
450	18"	457,0	5L	STD	...	...	9,53	437,94	150633	13397	14357	13758	106,16	150,63	33546	1468,10
450	18"	457,0	5L	...	...	...	10,31	436,38	149561	14468	14357	13709	113,57	149,56	36105	1580,08
450	18"	457,0	5L	...	30	...	11,13	434,74	148439	15590	14357	13658	122,38	148,44	38766	1696,54
450	18"	457,0	5L	...	...	...	11,91	433,18	147376	16654	14357	13608	130,72	147,38	41269	1806,09
450	18"	457,0	5L	XS	...	...	12,70	431,60	146303	17727	14357	13559	139,15	146,30	43777	1916,85
450	18"	457,0	5L	...	40	...	14,27	428,46	144182	19848	14357	13460	155,80	144,18	48680	2130,42
450	18"	457,0	5L	...	...	...	15,88	426,24	142023	22007	14357	13359	172,74	142,02	53597	2346,62
450	18"	457,0	5L	...	...	...	17,48	422,04	139893	24136	14357	13259	189,46	139,89	58375	2554,89
450	18"	457,0	5L	...	60	...	19,05	418,90	137819	26210	14357	13160	206,74	137,82	62958	2755,26
450	18"	457,0	5L	...	...	...	20,62	415,76	135761	28269	14357	13061	221,89	135,76	67439	2951,36
450	18"	457,0	5L	...	...	...	22,23	412,54	133666	30363	14357	12960	238,34	133,67	71930	3147,94
450	18"	457,0	5L	...	80	...	23,83	409,34	131601	32429	14357	12860	254,55	131,60	76291	3338,76
450	18"	457,0	5L	...	...	...	25,40	406,20	129569	34440	14357	12761	270,34	129,59	80471	3521,71
450	18"	457,0	5L	...	...	...	26,97	403,06	127594	36436	14357	12663	286,00	127,59	84556	3700,46
450	18"	457,0	5L	...	...	...	28,58	399,84	125563	38466	14357	12561	301,94	125,56	88646	3879,48
450	18"	457,0	...	...	100	...	29,36	398,28	124585	39444	14357	12512	309,62	124,59	90593	3964,87
450	18"	457,0	5L	...	...	...	30,18	396,64	123561	40468	14357	12461	317,66	123,56	92615	4053,15
450	18"	457,0	5L	...	...	...	31,75	393,50	121613	42417	14357	12362	332,95	121,61	96416	4219,53
450	18"	457,0	...	...	120	...	34,93	387,14	117713	46316	14357	12162	363,56	117,71	103843	4544,54
450	18"	457,0	...	...	140	...	39,67	377,66	112019	52011	14357	11865	408,26	112,02	114253	5000,13
450	18"	457,0	...	...	160	...	45,24	366,52	105508	58522	14357	11515	459,37	105,51	125524	5493,38
500	20"	508,0	...	...	...	5S	4,78	498,44	195126	7557	15959	15659	59,25	195,13	23922	941,82
500	20"	508,0	5L	...	...	10S*	5,56	496,88	193907	8776	15959	15610	68,89	193,91	27697	1090,45
500	20"	508,0	5L	...	10	...	6,35	495,30	192676	10007	15959	15560	78,55	192,68	31485	1239,57
500	20"	508,0	5L	...	...	...	7,14	493,72	191448	11236	15959	15511	88,19	191,45	35237	1387,27
500	20"	508,0	5L	...	...	...	7,92	492,16	190240	12443	15959	15462	97,67	190,24	38906	1531,72
500	20"	508,0	5L	...	...	...	8,74	490,52	188975	13708	15959	15410	107,60	188,97	42725	1682,10
500	20"	508,0	5L	STD	20	...	9,53	488,94	187759	14924	15959	15361	117,15	187,76	46369	1825,55
500	20"	508,0	5L	...	...	...	10,31	487,38	186563	16120	15959	15311	126,53	186,56	49932	1965,84
500	20"	508,0	5L	...	...	...	11,13	485,74	185309	17374	15959	15260	136,37	185,31	53642	2111,87
500	20"	508,0	5L	...	...	...	11,91	484,18	184121	18562	15959	15211	146,70	184,12	57135	2249,42
500	20"	508,0	5L	XS	30	...	12,70	482,60	182921	19762	15959	15161	155,12	182,92	60639	2387,37
500	20"	508,0	5L	...	...	...	14,27	479,46	180649	22134	15959	15063	173,74	180,65	67602	2657,55
500	20"	508,0	5L	...	40	...	15,09	477,82	179318	23367	15959	15011	183,42	179,32	71033	2796,57
500	20"	508,0	5L	...	...	...	15,88	476,24	178132	24551	15959	14962	192,71	178,13	74400	2929,15
500	20"	508,0	5L	...	...	...	17,48	473,04	175746	26937	15959	14861	211,44	175,75	81119	3193,66
500	20"	508,0	5L	...	...	...	19,05	469,90	173421	29262	15959	14762	229,70	173,42	87680	3448,05
500	20"	508,0	5L	...	...	...	20,62	466,76	171111	31572	15959	14664	247,83	171,11	93914	3697,38
500	20"	508,0	5L	...	60	...	22,23	463,54	168758	33926	15959	14563	266,29	168,76	100277	3947,90
500	20"	508,0	5L	...	...	...	23,83	460,34	166436	36247	15959	14462	284,52	166,44	106470	4191,74

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm⁴	cm³
ISO DN	ANSI NPS		Spec. API	Anc. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
500	20"	508,0	5L	...	...	...	25,40	457,20	164173	38510	15959	14363	302,28	164,17	112423	4426,12
500	20"	508,0	...	...	80	...	26,19	455,62	163040	39643	15959	14314	311,17	163,04	115373	4542,24
500	20"	508,0	5L	...	...	...	26,97	454,06	161926	40757	15959	14265	319,92	161,93	118255	4655,72
500	20"	508,0	5L	...	...	...	28,58	450,84	159637	43046	15959	14184	337,89	159,64	124111	4886,27
500	20"	508,0	5L	...	...	...	30,18	447,64	157379	45304	15959	14063	355,61	157,38	129808	5110,55
500	20"	508,0	5L	...	...	...	31,75	444,50	155179	47504	15959	13964	372,88	155,18	135280	5326,00
500	20"	508,0	...	...	100	...	32,54	442,92	154078	48605	15959	13915	381,53	154,08	137990	5432,69
500	20"	508,0	5L	...	...	...	33,32	441,36	152995	49688	15959	13866	390,03	152,99	140638	5536,92
500	20"	508,0	5L	...	...	...	34,93	438,14	150770	51913	15959	13765	407,49	150,77	146015	5748,60
500	20"	508,0	...	...	120	...	36,10	431,80	146438	56245	15959	13585	441,49	146,44	156260	6151,96
500	20"	508,0	...	...	140	...	44,45	419,10	137951	64732	15959	13166	508,11	137,95	176467	6906,17
500	20"	508,0	...	...	160	...	50,01	407,98	130728	71955	15959	12817	564,81	130,73	190912	7516,21
...	22"	559,0	...	...	...	5S	4,78	549,44	237099	8323	17562	17261	65,24	237,10	31957	1143,36
...	22"	559,0	5L	...	...	10S[4]	5,56	547,88	235756	9667	17562	17212	75,88	235,76	37016	1324,37
...	22"	559,0	5L	...	10	...	6,35	546,30	234397	11025	17562	17163	86,54	234,40	42096	1506,12
...	22"	559,0	5L	...	...	...	7,14	544,72	233043	12379	17562	17113	97,17	233,04	47132	1686,30
...	22"	559,0	5L	...	...	...	7,92	543,16	231710	13712	17562	17064	107,63	231,71	52062	1862,67
...	22"	559,0	5L	...	...	...	8,74	541,52	230313	15109	17562	17012	118,60	230,31	57199	2046,46
...	22"	559,0	5L	STD	20	...	9,53	539,94	228971	16451	17562	16963	129,13	228,97	62103	2221,95
...	22"	559,0	5L	...	...	...	10,31	538,38	227660	17772	17562	16914	139,50	227,66	66904	2393,71
...	22"	559,0	5L	...	...	...	11,13	536,74	226265	19157	17562	16862	150,37	226,27	71906	2572,66
...	22"	559,0	5L	...	...	...	11,91	535,18	224952	20470	17562	16813	160,58	224,96	76622	2741,40
...	22"	559,0	5L	XS	30	...	12,70	533,60	223626	21798	17562	16764	171,09	223,63	81356	2910,78
...	22"	559,0	5L	...	...	...	14,27	530,46	221001	24421	17562	16665	191,69	221,00	90641	3242,98
...	22"	559,0	5L	...	...	...	15,88	527,24	218327	27095	17562	16564	212,89	218,33	99993	3577,58
...	22"	559,0	5L	...	...	...	17,48	524,04	215684	29738	17562	16463	233,43	215,68	109118	3904,05
...	22"	559,0	5L	...	...	...	19,05	520,90	213107	32315	17562	16365	253,65	213,11	117911	4218,65
...	22"	559,0	5L	...	...	...	20,62	517,76	210546	34876	17562	16266	273,76	210,55	126547	4527,52
...	22"	559,0	5L	...	60	...	22,23	514,54	207935	37487	17562	16165	294,25	207,94	135241	4838,68
...	22"	559,0	5L	...	...	...	23,83	511,34	205357	40066	17562	16064	314,49	205,36	143721	5142,07
...	22"	559,0	5L	...	...	...	25,40	508,20	202843	42579	17562	15966	334,23	202,84	151888	5434,26
...	22"	559,0	5L	...	...	...	26,97	505,06	200344	45078	17562	15867	353,84	200,34	159906	5721,13
...	22"	559,0	5L	...	80	...	28,58	501,84	197797	47625	17562	15766	373,83	197,80	167974	6009,78
...	22"	559,0	5L	...	...	...	30,18	498,64	195283	50139	17562	15665	393,57	195,28	176839	6291,19
...	22"	559,0	5L	...	...	...	31,75	495,50	192831	52591	17562	15567	412,81	192,83	183411	6562,11
...	22"	559,0	5L	...	...	...	33,32	492,36	190395	55027	17562	15468	431,94	190,39	190841	6827,93
...	22"	559,0	5L	...	100	...	34,93	489,14	187913	57509	17562	15367	451,42	187,91	198313	7095,28
...	22"	559,0	5L	...	...	...	36,53	486,94	185462	59960	17562	15266	470,66	185,46	206595	7366,80
...	22"	559,0	5L	...	...	...	38,10	482,80	183073	62349	17562	15168	489,41	183,07	212601	7606,47
...	22"	559,0	...	...	120	...	41,28	476,44	178282	67140	17562	14968	527,02	178,28	226379	8099,44
...	22"	559,0	...	...	140	...	47,63	463,74	168904	76518	17562	14569	600,63	168,90	252289	9026,43
...	22"	559,0	...	...	160	...	53,98	451,04	159779	85643	17562	14170	672,26	159,78	276164	9880,30
600	24"	610,0	...	...	...	5S	5,54	598,92	281726	10520	19164	18816	82,47	281,73	48052	1575,47
600	24"	610,0	5L	...	10	10S	6,35	597,30	280204	12042	19164	18766	94,53	280,20	54868	1798,61
600	24"	610,0	5L	...	...	...	7,14	595,72	278724	13523	19164	18715	106,15	278,72	61442	2014,51
600	24"	610,0	5L	...	...	...	7,92	594,16	277266	14981	19164	18666	117,59	277,27	67893	2225,99
600	24"	610,0	5L	...	...	...	8,74	592,52	275738	16509	19164	18615	129,59	275,74	74619	2446,53
600	24"	610,0	5L	STD	20	...	9,53	590,94	274269	17978	19164	18565	141,12	274,27	81047	2657,27
600	24"	610,0	5L	...	...	...	10,31	589,38	272823	19424	19164	18516	152,47	272,82	87343	2863,70
600	24"	610,0	5L	...	...	...	11,13	587,74	271307	20940	19164	18464	164,37	271,31	93908	3078,95
600	24"	610,0	5L	...	...	...	11,91	586,18	269868	22378	19164	18415	175,86	269,87	100102	3282,04
600	24"	610,0	5L	XS	...	...	12,70	584,60	268416	23831	19164	18366	187,06	268,42	106326	3486,08
600	24"	610,0	5L	...	30	...	14,27	581,46	265540	26707	19164	18267	209,64	265,54	118544	3886,70
600	24"	610,0	5L	...	...	...	15,88	578,24	262607	29640	19164	18166	232,56	262,61	130871	4290,85
600	24"	610,0	5L	...	40	...	17,48	575,04	259708	32538	19164	18065	255,41	259,71	142918	4685,85
600	24"	610,0	5L	...	...	...	19,05	571,90	256880	35367	19164	17967	277,61	256,88	154546	5067,09
600	24"	610,0	5L	...	...	...	20,62	568,76	254067	38180	19164	17868	299,69	254,07	165984	5442,10
600	24"	610,0	5L	...	...	...	22,23	565,54	251198	41049	19164	17767	322,21	251,20	177618	5820,26
600	24"	610,0	5L	...	...	...	23,83	562,34	248384	43863	19164	17666	344,46	248,36	188787	6189,73
600	24"	610,0	...	...	60	...	24,61	560,78	246987	45259	19164	17617	355,26	246,99	194211	6367,58
600	24"	610,0	5L	...	...	...	25,40	559,20	245598	46649	19164	17568	366,17	245,60	199659	6546,20
600	24"	610,0	5L	...	...	...	26,97	556,06	242847	49399	19164	17469	387,76	242,85	210350	6896,71
600	24"	610,0	5L	...	...	...	28,58	552,84	240043	52204	19164	17368	409,77	240,04	221126	7250,04
600	24"	610,0	5L	...	...	...	30,18	549,64	237272	54975	19164	17267	431,52	237,27	231651	7595,11
600	24"	610,0	...	...	80	...	30,96	548,08	235927	56320	19164	17218	442,08	235,93	236716	7761,16
600	24"	610,0	5L	...	...	...	31,75	546,50	234569	57678	19164	17169	452,74	234,57	241801	7927,90
600	24"	610,0	5L	...	...	...	33,32	543,36	231881	60366	19164	17070	473,84	231,88	251778	8255,00
600	24"	610,0	5L	...	...	...	34,93	540,14	229141	63106	19164	16969	495,35	229,14	261830	8584,60
600	24"	610,0	5L	...	...	...	36,53	536,94	226434	65813	19164	16868	516,80	226,43	271644	8906,37
600	24"	610,0	5L	...	...	...	38,10	533,80	223793	68453	19164	16770	537,33	223,79	281106	9216,56
600	24"	610,0	...	...	100	...	38,89	532,22	222470	69776	19164	16720	547,71	222,47	285803	9370,58
600	24"	610,0	5L	...	...	...	39,67	530,66	221168	71079	19164	16671	557,43	221,17	290400	9521,32
600	24"	610,0	...	...	120	...	46,02	517,96	210709	81538	19164	16272	640,03	210,71	326347	10699,90
600	24"	610,0	...	...	140	...	52,37	505,26	200502	91744	19164	15873	720,16	200,50	359745	11794,91
600	24"	610,0	...	...	160	...	59,54	490,92	189283	102964	19164	15423	808,22	189,28	394546	12935,93
...	26"	660,0	5L	...	...	...	6,35	647,30	329080	13040	20735	20336	102,36	329,08	69648	2110,55
...	26"	660,0	5L	...	...	...	7,14	645,72	327475	14644	20735	20286	114,95	327,48	78032	2364,58
...	26"	660,0	5L	...	10	...	7,92	644,16	325895	16226	20735	20237	127,36	325,89	86249	2613,59
...	26"	660,0	5L	...	...	...	8,74	642,52	324237	17882	20735	20185	140,37	324,24	94823	2873,42
...	26"	660,0	5L	STD	...	...	9,53	640,94	322645	19475	20735	20136	152,87	322,64	103021	3121,86
...	26"	660,0	5L	...	...	...	10,31	639,38	321076	21043	20735	20087	165,18	321,08	111057	3365,37
...	26"	660,0	5L	...	...	...	11,13	637,74	319431	22688	20735	20035	178,09	319,43	119442	3619,44
...	26"	660,0	5L	...	...	...	11,91	636,18	317870	24249	20735	19986	190,34	317,87	127357	3859,32
...	26"	660,0	5L	XS	20	...	12,70	634,60	316293	25826	20735	19937	202,72	316,29	135316	4100,47
...	26"	660,0	5L	...	...	...	14,27	631,46	313171	28948	20735	19838	227,29	313,17	150966	4574,41

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— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

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NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Ame. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
...	26"	660,0	5L	...	...	...	15,88	628,24	309985	32134	20735	19737	252,24	309,99	186753	5053,13
...	*26"*	*660,0*	*5L*	...	...	...	*17,48*	*625,04*	*306835*	*35284*	*20735*	*19636*	*276,96*	*306,84*	*182214*	*5521,65*
...	26"	660,0	5L	...	...	...	19,05	621,90	303760	38359	20735	19538	301,10	303,78	197158	5974,44
...	26"	660,0	5L	...	...	...	20,62	618,76	300701	41419	20735	19439	325,12	300,70	211874	6420,42
...	*26"*	*660,0*	*5L*	...	...	...	*22,23*	*615,54*	*297579*	*44540*	*20736*	*19338*	*349,62*	*297,58*	*226736*	*6870,76*
...	26"	660,0	5L	...	...	...	23,83	612,34	294493	47626	20735	19237	373,84	294,49	241275	7311,37
...	*26"*	*660,0*	*5L*	...	...	...	*25,40*	*609,20*	*291481*	*50639*	*20736*	*19139*	*397,49*	*291,48*	*256322*	*7737,04*
700	28"	711,0	5L	...	...	...	6,35	698,30	382978	14057	22337	21938	110,34	382,98	87255	2454,42
700	*28"*	*711,0*	*5L*	...	...	...	*7,14*	*696,72*	*381247*	*15788*	*22337*	*21888*	*123,93*	*381,25*	*97783*	*2760,57*
700	28"	711,0	5L	...	10	...	7,92	695,16	379542	17494	22337	21839	137,32	379,54	108107	3040,99
700	*28"*	*711,0*	*5L*	...	...	...	*8,74*	*693,52*	*377753*	*19282*	*22337*	*21788*	*151,36*	*377,75*	*118886*	*3344,20*
700	28"	711,0	5L	STD	...	...	9,53	691,94	376034	21002	22337	21738	164,85	376,03	129199	3634,30
700	*28"*	*711,0*	*5L*	...	...	...	*10,31*	*690,38*	*374340*	*22696*	*22337*	*21689*	*178,15*	*374,34*	*139313*	*3918,78*
700	28"	711,0	5L	...	...	...	11,13	688,74	372564	24472	22337	21637	192,09	372,56	149871	4215,77
700	*28"*	*711,0*	*5L*	...	...	...	*11,91*	*687,18*	*370878*	*26157*	*22337*	*21588*	*205,32*	*370,88*	*159844*	*4496,32*
700	28"	711,0	5L	XS	20	...	12,70	685,60	369174	27861	22337	21539	218,69	369,17	169877	4778,52
700	*28"*	*711,0*	*5L*	...	...	...	*14,27*	*682,46*	*365800*	*31236*	*22337*	*21440*	*245,18*	*365,80*	*189609*	*5333,59*
700	28"	711,0	5L	...	30	...	15,88	679,24	362357	34678	22337	21339	271,21	362,36	209564	5894,90
700	*28"*	*711,0*	*5L*	...	...	...	*17,48*	*676,04*	*358951*	*38085*	*22337*	*21238*	*298,95*	*358,95*	*229115*	*6444,87*
700	28"	711,0	5L	...	...	...	19,05	672,90	355624	41411	22337	21140	325,06	355,62	248032	6977,00
700	*28"*	*711,0*	*5L*	...	...	...	*20,62*	*669,76*	*352313*	*44723*	*22337*	*21041*	*351,05*	*352,31*	*266686*	*7501,72*
700	28"	711,0	5L	...	...	...	22,23	666,54	348933	48102	22337	20940	377,58	348,93	285545	8032,20
700	*28"*	*711,0*	*5L*	...	...	...	*23,83*	*663,34*	*345591*	*51444*	*22337*	*20839*	*403,81*	*345,59*	*304017*	*8551,83*
700	28"	711,0	5L	...	...	...	25,40	660,20	342327	54708	22337	20741	429,44	342,33	321886	9054,45
...	30"	762,0	5L	...	...	5S	6,35	749,30	440962	15075	23939	23540	118,33	440,96	107603	2824,24
...	*30"*	*762,0*	*5L*	...	...	...	*7,14*	*747,72*	*439104*	*16932*	*23939*	*23490*	*132,91*	*439,10*	*120614*	*3165,71*
...	30"	762,0	5L	...	10	10S	7,92	746,16	437274	18763	23939	23441	147,28	437,27	133378	3500,74
...	*30"*	*762,0*	*5L*	...	...	...	*8,74*	*744,52*	*435354*	*20683*	*23939*	*23390*	*162,36*	*435,35*	*146712*	*3850,70*
...	30"	762,0	5L	STD	...	...	9,53	742,94	433508	22528	23939	23340	176,84	433,51	159474	4185,67
...	*30"*	*762,0*	*5L*	...	...	...	*10,31*	*741,38*	*431690*	*24347*	*23939*	*23291*	*1911,11*	*431,69*	*171995*	*4514,31*
...	30"	762,0	5L	...	...	...	11,13	739,74	429782	26255	23939	23240	206,09	429,78	185074	4857,58
...	*30"*	*762,0*	*5L*	...	...	...	*11,91*	*738,18*	*427971*	*28066*	*23939*	*23191*	*220,30*	*427,97*	*197434*	*5181,99*
...	30"	762,0	5L	XS	20	...	12,70	736,60	426141	29896	23939	23141	234,67	426,14	209873	5508,46
...	*30"*	*762,0*	*5L*	...	...	...	*14,27*	*733,46*	*422616*	*33521*	*23939*	*23042*	*263,12*	*422,52*	*234356*	*6151,08*
...	30"	762,0	5L	...	30	...	15,88	730,24	418814	37223	23939	22941	292,18	418,81	259139	6801,55
...	*30"*	*762,0*	*5L*	...	...	...	*17,48*	*727,04*	*416161*	*40885*	*23939*	*22841*	*320,93*	*416,16*	*283446*	*7439,52*
...	30"	762,0	5L	...	...	...	19,05	723,90	411573	44464	23939	22742	349,02	411,57	306986	8057,38
...	*30"*	*762,0*	*5L*	...	...	...	*20,62*	*720,76*	*408010*	*48026*	*23939*	*22643*	*376,98*	*408,01*	*330223*	*8667,26*
...	30"	762,0	5L	...	...	...	22,23	717,54	404373	51664	23939	22542	405,54	404,37	353738	9284,46
...	*30"*	*762,0*	*5L*	...	...	...	*23,83*	*714,34*	*400774*	*55262*	*23939*	*22442*	*433,78*	*400,77*	*376795*	*9889,64*
...	30"	762,0	5L	...	...	...	25,40	711,20	397259	58778	23939	22343	461,38	397,26	399121	10476,82
...	*30"*	*762,0*	*5L*	...	...	...	*26,97*	*708,06*	*393759*	*62278*	*23939*	*22244*	*488,85*	*393,76*	*421153*	*11053,90*
...	30"	762,0	5L	...	...	...	28,58	704,84	390185	65851	23939	22143	516,90	390,19	443445	11638,97
...	*30"*	*762,0*	*5L*	...	...	...	*30,18*	*701,64*	*386650*	*69386*	*23939*	*22043*	*544,66*	*386,65*	*465297*	*12212,51*
...	30"	762,0	5L	...	...	...	31,75	698,50	383198	72839	23939	21944	571,75	383,20	486450	12767,73
800	32"	813,0	5L	...	...	...	6,35	800,30	503032	16092	25541	25142	126,31	503,03	130693	3220,00
800	*32"*	*813,0*	*5L*	...	...	...	*7,14*	*798,72*	*501048*	*18076*	*25541*	*25093*	*141,89*	*501,05*	*146748*	*3610,03*
800	32"	813,0	5L	...	10	...	7,92	797,16	499092	20032	25541	25044	157,24	499,09	162310	3992,66
800	*32"*	*813,0*	*5L*	...	...	...	*8,74*	*795,52*	*497041*	*22083*	*25541*	*24992*	*173,34*	*497,04*	*178571*	*4392,90*
800	32"	813,0	5L	STD	...	...	9,53	793,94	495068	24055	25541	24942	188,82	495,07	194144	4775,98
800	*32"*	*813,0*	*5L*	...	...	...	*10,31*	*792,38*	*493126*	*25999*	*25541*	*24893*	*204,06*	*493,12*	*209428*	*5151,97*
800	32"	813,0	5L	...	...	...	11,13	790,74	491066	28038	25541	24842	220,08	491,09	225398	5544,85
800	*32"*	*813,0*	*5L*	...	...	...	*11,91*	*789,18*	*489150*	*29974*	*25541*	*24793*	*235,28*	*489,15*	*240498*	*5916,31*
800	32"	813,0	5L	XS	20	...	12,70	787,60	487193	31931	25541	24743	250,64	487,19	255700	6290,29
800	*32"*	*813,0*	*5L*	...	...	...	*14,27*	*784,46*	*483316*	*35807*	*25541*	*24645*	*281,07*	*483,32*	*285642*	*7026,87*
800	32"	813,0	5L	...	30	...	15,88	781,24	479357	39767	25541	24543	312,15	479,36	315976	7773,08
800	*32"*	*813,0*	*5L*	...	*40*	...	*17,48*	*778,04*	*475438*	*43686*	*25541*	*24443*	*342,91*	*475,44*	*345752*	*8505,58*
800	32"	813,0	5L	...	...	...	19,05	774,90	471608	47516	25541	24344	372,98	471,61	374614	9215,60
800	*32"*	*813,0*	*5L*	...	...	...	*20,62*	*771,76*	*467794*	*51330*	*25541*	*24246*	*402,92*	*467,79*	*403128*	*9917,05*
800	32"	813,0	5L	...	...	...	22,23	768,54	463898	55225	25541	24144	433,49	463,90	432009	10627,53
800	*32"*	*813,0*	*5L*	...	...	...	*23,83*	*765,34*	*460043*	*59081*	*25541*	*24044*	*463,76*	*460,04*	*460353*	*11324,81*
800	32"	813,0	5L	...	...	...	25,40	762,20	456276	62848	25541	23945	493,32	456,28	487823	12000,56
800	*32"*	*813,0*	*5L*	...	...	...	*26,97*	*759,06*	*452524*	*66599*	*25541*	*23847*	*522,77*	*452,52*	*514955*	*12668,02*
800	32"	813,0	5L	...	...	...	28,58	755,84	448693	70430	25541	23745	552,85	448,69	542431	13343,93
800	*32"*	*813,0*	*5L*	...	...	...	*30,18*	*752,64*	*444902*	*74222*	*25541*	*23646*	*582,61*	*444,90*	*569390*	*14007,14*
800	32"	813,0	5L	...	...	...	31,75	749,50	441198	77926	25541	23546	611,68	441,20	595512	14649,74
...	34"	864,0	5L	...	...	...	6,35	851,30	569187	17109	27143	26744	134,30	569,19	157321	3641,70
...	*34"*	*864,0*	*5L*	...	...	...	*7,14*	*849,72*	*567076*	*19220*	*27143*	*26695*	*150,87*	*567,08*	*176408*	*4083,51*
...	34"	864,0	5L	...	10	...	7,92	848,16	564996	21300	27143	26646	167,20	565,00	195149	4517,33
...	*34"*	*864,0*	*5L*	...	...	...	*8,74*	*846,52*	*562813*	*23483*	*27143*	*26594*	*184,33*	*562,81*	*214739*	*4970,81*
...	34"	864,0	5L	STD	...	...	9,53	844,94	560714	25582	27143	26545	200,31	560,71	233506	5405,22
...	*34"*	*864,0*	*5L*	...	...	...	*10,31*	*843,38*	*558646*	*27651*	*27143*	*26496*	*217,05*	*558,65*	*251931*	*5831,75*
...	34"	864,0	5L	...	...	...	11,13	841,74	556475	29821	27143	26444	234,08	556,48	271192	6277,60
...	34"	864,0	5L	...	...	...	11,91	840,18	554414	31882	27143	26395	250,28	554,41	289410	6699,30
...	*34"*	*864,0*	*5L*	*XS*	*20*	...	*12,70*	*838,60*	*552331*	*33965*	*27143*	*26345*	*266,61*	*552,33*	*307757*	*7124,01*
...	34"	864,0	5L	...	...	...	14,27	835,46	548203	38094	27143	26247	299,02	548,20	343913	7960,96
...	*34"*	*864,0*	*5L*	...	*30*	...	*15,88*	*832,24*	*543985*	*42311*	*27143*	*26146*	*332,12*	*543,99*	*380570*	*8809,49*
...	34"	864,0	5L	...	40	...	17,48	829,04	539810	46487	27143	26045	364,90	539,81	416580	9643,05
...	*34"*	*864,0*	*5L*	...	...	...	*19,05*	*825,90*	*535729*	*50568*	*27143*	*25946*	*396,93*	*535,73*	*451511*	*10451,65*
...	34"	864,0	5L	...	...	...	20,62	822,76	531663	54634	27143	25846	428,85	531,66	486047	11251,08
...	*34"*	*864,0*	*5L*	...	...	...	*22,23*	*819,54*	*527609*	*58787*	*27143*	*25747*	*461,45*	*527,61*	*521054*	*12061,43*
...	34"	864,0	5L	...	...	...	23,83	816,34	523398	62899	27143	25646	493,72	523,40	555437	12857,33
...	*34"*	*864,0*	*5L*	...	...	...	*25,40*	*813,20*	*519379*	*66917*	*27143*	*25547*	*525,27*	*519,38*	*588785*	*13629,27*
...	34"	864,0	5L	...	...	...	26,97	810,06	515376	70921	27143	25449	556,69	515,38	621748	14392,32
...	*34"*	*864,0*	*5L*	...	...	...	*28,58*	*806,84*	*511287*	*75010*	*27143*	*25348*	*588,79*	*511,29*	*656156*	*15166,65*
...	34"	864,0	5L	...	...	...	30,18	803,64	507239	79057	27143	25247	620,56	507,24	687962	15925,05
...	*34"*	*864,0*	*5L*	...	...	...	*31,75*	*800,50*	*503283*	*83013*	*27143*	*25146*	*651,61*	*503,28*	*719775*	*16661,45*

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

10/12

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Anc. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
900	36"	914,0	5L	...	...	...	6,35	901,30	638012	18107	28714	28315	142,13	638,01	186470	4080,32
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*7,14*	*899,72*	*635777*	*20342*	*28714*	*28266*	*159,67*	*636,78*	*209125*	*4576,04*
900	36"	914,0	5L		10	...	7,92	898,16	633574	22545	28714	28217	176,96	633,57	231376	5062,93
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*8,74*	*896,52*	*631262*	*24856*	*28714*	*28165*	*196,11*	*631,26*	*254643*	*5572,06*
900	36"	914,0	5L	STD	...	...	9,53	894,94	629039	27079	28714	28115	212,56	629,04	276939	6059,93
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*10,31*	*893,38*	*626848*	*29270*	*28714*	*28066*	*229,76*	*626,85*	*298836*	*6539,09*
900	36"	914,0	5L	...	...	...	11,13	891,74	624549	31570	28714	28015	247,31	624,55	321734	7040,13
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*11,91*	*890,18*	*622366*	*33753*	*28714*	*27966*	*264,94*	*622,37*	*343397*	*7514,17*
900	36"	914,0	5L	XS	20	...	12,70	888,60	620158	35960	28714	27916	282,27	620,16	365223	7991,75
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*14,27*	*885,46*	*615783*	*40336*	*28714*	*27818*	*316,11*	*615,78*	*408253*	*8933,33*
900	36"	914,0	5L	...	30	...	15,88	882,24	611313	44806	28714	27716	351,70	611,31	451907	9888,56
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*17,48*	*879,04*	*606886*	*49232*	*28714*	*27616*	*386,45*	*606,89*	*494819*	*10827,55*
900	36"	914,0	5L	...	40	...	19,05	875,90	602558	53560	28714	27517	420,42	602,56	536473	11739,02
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*20,62*	*872,76*	*598246*	*57873*	*28714*	*27419*	*454,27*	*598,25*	*577652*	*12640,73*
900	36"	914,0	5L	...	...	...	22,23	869,54	593839	62279	28714	27317	488,86	593,84	619481	13555,37
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*23,83*	*866,34*	*589477*	*66642*	*28714*	*27217*	*523,11*	*589,48*	*660562*	*14454,32*
900	36"	914,0	5L	...	...	...	25,40	863,20	585211	70907	28714	27118	556,59	585,21	700434	15326,78
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*26,97*	*860,06*	*580962*	*75157*	*28714*	*27020*	*589,95*	*580,96*	*739873*	*16189,78*
900	36"	914,0	5L	...	...	...	28,58	856,84	576620	79499	28714	26918	624,03	576,62	779870	17064,99
900	*36"*	*914,0*	*5L*	*...*	*...*	*...*	*30,18*	*853,64*	*572321*	*83798*	*28714*	*26818*	*657,77*	*572,32*	*819176*	*17925,06*
900	36"	914,0	5L	...	...	...	31,75	850,50	568118	88001	28714	26719	690,76	568,12	857316	18759,64
...	38"	965,0	5L	...	...	...	7,92	949,16	707569	23814	30316	29819	186,92	707,57	272684	5651,48
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*8,74*	*947,52*	*706126*	*26257*	*30316*	*29767*	*206,10*	*706,13*	*300148*	*6220,69*
...	38"	965,0	5L	STD	...	...	9,53	945,94	702776	28606	30316	29718	224,54	702,78	328473	6788,28
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*10,31*	*944,38*	*700460*	*30922*	*30316*	*29669*	*242,72*	*700,46*	*352335*	*7302,29*
...	38"	965,0	5L	...	...	...	11,13	942,74	698029	33353	30316	29617	261,80	698,03	379386	7862,93
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*11,91*	*941,18*	*695721*	*35661*	*30316*	*29568*	*279,92*	*695,72*	*404987*	*8393,52*
...	38"	965,0	5L	XS	...	...	12,70	939,60	693387	37995	30316	29518	298,24	693,39	430787	8928,22
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*14,27*	*936,46*	*688761*	*42622*	*30316*	*29420*	*334,56*	*688,76*	*481674*	*9982,88*
...	38"	965,0	5L	...	...	...	15,88	933,24	684032	47350	30316	29319	371,68	684,03	533329	11053,45
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*17,48*	*930,04*	*679349*	*52033*	*30316*	*29218*	*408,43*	*679,35*	*584196*	*12106,45*
...	38"	965,0	5L	...	...	...	19,05	926,90	674770	56613	30316	29119	444,38	674,77	633484	13129,20
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*20,62*	*923,76*	*670206*	*61177*	*30316*	*29021*	*480,21*	*670,21*	*682332*	*14141,60*
...	38"	965,0	5L	...	...	...	22,23	920,54	665542	65841	30316	28920	516,82	665,54	731911	15169,13
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*23,83*	*917,34*	*660923*	*70460*	*30316*	*28819*	*553,08*	*660,92*	*780668*	*16179,65*
...	38"	965,0	5L	...	...	...	25,40	914,20	656406	74977	30316	28720	588,53	656,41	828018	17161,00
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*26,97*	*911,06*	*651904*	*79478*	*30316*	*28622*	*623,87*	*651,90*	*874883*	*18132,29*
...	38"	965,0	5L	...	...	...	28,58	907,84	647304	84078	30316	28521	659,97	647,30	922441	19117,94
...	*38"*	*965,0*	*5L*	*...*	*...*	*...*	*30,18*	*904,64*	*642749*	*88633*	*30316*	*28420*	*695,73*	*642,75*	*969205*	*20087,14*
...	38"	965,0	5L	...	...	...	31,75	901,50	638295	93088	30316	28321	730,69	638,29	1014612	21028,23
1000	40"	1016,0	5L	...	...	...	7,92	1000,16	786650	25082	31919	31421	196,89	785,65	318637	6272,39
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*8,74*	*998,52*	*783075*	*27657*	*31919*	*31368*	*217,09*	*783,08*	*350775*	*6905,02*
1000	40"	1016,0	5L	STD	...	...	9,53	996,94	780589	30133	31919	31320	236,53	780,60	381588	7511,57
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*10,31*	*995,38*	*778168*	*32574*	*31919*	*31271*	*255,69*	*778,16*	*411867*	*8107,61*
1000	40"	1016,0	5L	...	...	...	11,13	993,74	775596	35136	31919	31219	275,80	775,60	443545	8731,21
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*11,91*	*992,18*	*773163*	*37569*	*31919*	*31170*	*294,90*	*773,16*	*473533*	*9321,52*
1000	40"	1016,0	5L	XS	...	...	12,70	990,60	770702	40030	31919	31121	314,22	770,70	503782	9916,58
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*14,27*	*987,46*	*765824*	*44908*	*31919*	*31022*	*352,51*	*766,82*	*563409*	*11090,73*
1000	40"	1016,0	5L	...	...	...	15,88	984,24	760837	49894	31919	30921	391,65	760,84	623988	12283,23
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*17,48*	*981,04*	*755898*	*54834*	*31919*	*30820*	*430,42*	*755,90*	*683604*	*13456,77*
1000	40"	1016,0	5L	...	...	...	19,05	977,90	751067	59665	31919	30722	468,34	751,07	741538	14597,21
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*20,62*	*974,76*	*746252*	*64480*	*31919*	*30623*	*506,14*	*746,25*	*798917*	*15726,71*
1000	40"	1016,0	5L	...	...	...	22,23	971,54	741329	69403	31919	30522	544,78	741,33	857184	16873,71
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*23,83*	*968,34*	*736454*	*74278*	*31919*	*30421*	*583,05*	*736,45*	*914519*	*18002,34*
1000	40"	1016,0	5L	...	...	...	25,40	965,20	731686	79046	31919	30323	620,48	731,69	970228	19098,99
1000	*40"*	*1016,0*	*5L*	*...*	*...*	*...*	*26,97*	*962,06*	*726933*	*83799*	*31919*	*30224*	*657,78*	*726,93*	*1025397*	*20184,98*
1000	40"	1016,0	5L	...	...	...	28,58	958,84	722075	88657	31919	30123	695,92	722,07	1081413	21287,66
...	42"	1067,0	5L	...	...	...	8,74	1049,52	865110	29057	33521	32972	226,09	865,11	406797	7625,07
...	*42"*	*1067,0*	*5L*	*STD*	...	...	*9,53*	*1047,94*	*862507*	*31660*	*33521*	*32922*	*248,52*	*862,51*	*442580*	*8295,79*
...	42"	1067,0	5L	...	...	...	10,31	1046,38	859941	34226	33521	32873	268,66	859,94	477752	8955,06
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*11,13*	*1044,74*	*857248*	*36919*	*33521*	*32821*	*289,80*	*857,25*	*514558*	*9644,96*
...	42"	1067,0	5L	...	...	...	11,91	1043,18	854690	39478	33521	32772	309,86	854,69	549409	10298,19
...	*42"*	*1067,0*	*5L*	*XS*	...	...	*12,70*	*1041,60*	*852102*	*42065*	*33521*	*32723*	*330,19*	*852,10*	*584647*	*10956,83*
...	42"	1067,0	5L	...	...	...	14,27	1038,46	846973	47194	33521	32624	370,45	846,97	653905	12256,89
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*15,88*	*1035,24*	*841728*	*52439*	*33521*	*32523*	*411,52*	*841,73*	*724380*	*13577,88*
...	42"	1067,0	5L	...	...	...	17,48	1032,04	836533	57634	33521	32422	452,40	836,53	793769	14878,51
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*19,05*	*1028,90*	*831450*	*62717*	*33521*	*32324*	*492,30*	*831,45*	*861232*	*16143,05*
...	42"	1067,0	5L	...	...	...	20,62	1025,76	826383	67784	33521	32225	532,07	826,38	928080	17398,07
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*22,23*	*1022,54*	*821203*	*72964*	*33521*	*32124*	*572,73*	*821,20*	*995997*	*18669,11*
...	42"	1067,0	5L	...	...	...	23,83	1019,34	816071	78096	33521	32024	613,02	816,07	1062859	19922,38
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*25,40*	*1016,20*	*811051*	*83116*	*33521*	*31925*	*652,42*	*811,05*	*1127859*	*21140,74*
...	42"	1067,0	5L	...	...	...	26,97	1013,06	806047	88120	33521	31826	691,70	806,05	1192258	22347,86
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*28,58*	*1009,84*	*800931*	*93236*	*33521*	*31725*	*731,86*	*800,93*	*1257680*	*23574,13*
...	42"	1067,0	5L	...	...	...	30,18	1006,64	795863	98304	33521	31625	771,64	795,86	1322078	24781,22
...	*42"*	*1067,0*	*5L*	*...*	*...*	*...*	*31,75*	*1003,50*	*790906*	*103262*	*33521*	*31526*	*810,55*	*790,91*	*1384674*	*25954,53*
...	44"	1118,0	5L	...	...	...	8,74	1100,52	951230	30458	35123	34574	239,08	951,23	468488	8380,82
...	*44"*	*1118,0*	*5L*	*STD*	...	...	*9,53*	*1098,94*	*948501*	*33187*	*35123*	*34524*	*260,50*	*948,50*	*509749*	*9118,96*
...	44"	1118,0	5L	...	...	...	10,31	1097,38	945810	35878	35123	34475	281,62	945,81	550314	9844,62
...	*44"*	*1118,0*	*5L*	*...*	*...*	*...*	*11,13*	*1095,74*	*942985*	*38703*	*35123*	*34424*	*303,80*	*942,99*	*592774*	*10604,18*
...	44"	1118,0	5L	...	...	...	11,91	1094,18	940302	41386	35123	34375	324,80	940,30	632985	11323,52
...	*44"*	*1118,0*	*5L*	*XS*	...	...	*12,70*	*1092,60*	*937589*	*44100*	*35123*	*34326*	*346,16*	*937,59*	*673537*	*12048,96*
...	44"	1118,0	5L	...	...	...	14,27	1089,48	932207	49481	35123	34226	388,40	932,21	753607	13481,34
...	*44"*	*1118,0*	*5L*	*...*	*...*	*...*	*15,88*	*1086,24*	*926705*	*54983*	*35123*	*34125*	*431,59*	*926,70*	*835001*	*14937,41*
...	44"	1118,0	5L	...	...	...	17,48	1083,04	921253	60435	35123	34025	474,37	921,25	915176	16371,67

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MAIN CARACTERISTICS OF SEAMLESS AND WELDED PIPES
— ANSI/ASME B36 10M-1995 STANDARD for carbon steel and low-alloy steel pipes —
— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

11/12

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Anc. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
...	44"	1118,0	5L	...	...	...	19,05	1079,90	915919	65769	35123	33926	516,26	915,92	993160	17766,72
...	44"	1118,0	5L	...	...	...	20,62	1076,76	910600	71088	35123	33827	558,00	910,60	1070486	19149,66
...	44"	1118,0	5L	...	...	...	22,23	1073,54	905162	76526	35123	33726	600,69	905,16	1149043	20555,33
...	44"	1118,0	5L	...	...	...	23,83	1070,34	899774	81914	35123	33626	642,99	899,77	1226434	21939,79
...	44"	1118,0	5L	...	...	...	25,40	1067,20	894502	87186	35123	33527	684,37	894,50	1301703	23286,28
...	44"	1118,0	5L	...	...	...	26,97	1064,06	889246	92442	35123	33428	725,62	889,25	1376310	24620,93
...	44"	1118,0	5L	...	...	...	28,58	1060,84	883873	97815	35123	33327	767,80	883,87	1452135	25977,37
...	44"	1118,0	5L	...	...	...	30,18	1057,64	878548	103140	35123	33227	809,60	878,55	1526808	27313,20
...	44"	1118,0	5L	...	...	...	31,75	1054,50	873339	108349	35123	33126	850,48	873,34	1599425	28612,25
...	46"	1168,0	5L	...	...	...	8,74	1150,52	1039629	31830	36694	36145	249,85	1039,63	534735	9156,42
...	46"	1168,0	5L	STD	...	...	9,53	1148,94	1036775	34684	36694	36095	272,25	1036,78	581884	9963,77
...	46"	1168,0	5L	...	...	...	10,31	1147,38	1033962	37497	36694	36046	294,34	1033,96	628248	10757,64
...	46"	1168,0	5L	...	...	...	11,13	1145,74	1031008	40451	36694	35994	317,52	1031,01	676782	11588,74
...	46"	1168,0	5L	...	...	...	11,91	1144,18	1028202	43257	36694	35945	339,54	1028,20	722758	12375,99
...	46"	1168,0	5L	XS	...	...	12,70	1142,60	1025365	46094	36694	35896	361,82	1025,36	769131	13170,05
...	46"	1168,0	5L	...	...	...	14,27	1139,46	1019737	51722	36694	35797	406,00	1019,74	860722	14738,39
...	46"	1168,0	5L	...	...	...	15,88	1136,24	1013981	57478	36694	35696	451,17	1013,98	963863	16333,27
...	46"	1168,0	5L	...	...	...	17,48	1133,04	1008276	63181	36694	35596	495,94	1008,28	1045645	17904,87
...	46"	1168,0	5L	...	...	...	19,05	1129,90	1002697	68762	36694	35497	539,76	1002,70	1134952	19434,12
...	46"	1168,0	5L	...	...	...	20,62	1126,76	997132	74327	36694	35398	583,43	997,13	1223519	20950,66
...	46"	1168,0	5L	...	...	...	22,23	1123,54	991441	80018	36694	35297	628,10	991,44	1313576	22492,74
...	46"	1168,0	5L	...	...	...	23,83	1120,34	985802	85657	36694	35197	672,37	985,80	1402310	24012,15
...	46"	1168,0	5L	...	...	...	25,40	1117,20	980284	91175	36694	35098	715,68	980,28	1488644	25490,49
...	46"	1168,0	5L	...	...	...	26,97	1114,06	974781	96678	36694	34999	758,88	974,78	1574253	26956,39
...	46"	1168,0	5L	...	...	...	28,58	1110,84	969154	102305	36694	34898	803,04	969,15	1661295	28446,63
...	46"	1168,0	5L	...	...	...	30,18	1107,64	963579	107880	36694	34798	846,81	963,58	1747050	29915,23
...	46"	1168,0	5L	...	...	...	31,75	1104,50	958123	113336	36694	34699	889,63	958,12	1830477	31343,78
...	48"	1219,0	5L	...	...	...	8,74	1201,52	1133840	33231	38296	37747	260,85	1133,84	608457	9982,89
...	48"	1219,0	5L	STD	...	...	9,53	1199,94	1130860	36211	38296	37697	284,24	1130,86	662163	10864,04
...	48"	1219,0	5L	...	...	...	10,31	1198,38	1127922	39149	38296	37648	307,30	1127,92	714982	11730,63
...	48"	1219,0	5L	...	...	...	11,13	1196,74	1124837	42234	38296	37597	331,52	1124,84	770287	12638,01
...	48"	1219,0	5L	...	...	...	11,91	1195,18	1121906	45165	38296	37548	354,52	1121,91	822684	13497,68
...	48"	1219,0	5L	XS	...	...	12,70	1193,60	1118942	48129	38296	37498	377,79	1118,94	875543	14364,95
...	48"	1219,0	5L	...	...	...	14,27	1190,46	1113082	54009	38296	37399	423,94	1113,08	979973	16078,30
...	48"	1219,0	5L	...	...	...	15,88	1187,24	1107049	60022	38296	37298	471,14	1107,05	1086208	17821,29
...	48"	1219,0	5L	...	...	...	17,48	1184,04	1101090	65982	38296	37198	517,92	1101,09	1190930	19539,46
...	48"	1219,0	5L	...	...	...	19,05	1180,90	1095257	71814	38296	37099	563,70	1095,26	1292867	21211,92
...	48"	1219,0	5L	...	...	...	20,62	1177,76	1089440	77631	38296	37000	609,36	1089,44	1393994	22871,10
...	48"	1219,0	5L	...	...	...	22,23	1174,54	1083491	83580	38296	36899	656,06	1083,49	1496860	24558,82
...	48"	1219,0	5L	...	...	...	23,83	1171,34	1077596	89475	38296	36799	702,34	1077,60	1598253	26222,37
...	48"	1219,0	5L	...	...	...	25,40	1168,20	1071826	95245	38296	36700	747,63	1071,83	1696941	27841,52
...	48"	1219,0	5L	...	...	...	26,97	1165,06	1066072	100999	38296	36601	792,80	1066,07	1794835	29447,67
...	48"	1219,0	5L	...	...	...	28,58	1161,84	1060187	106884	38296	36500	838,99	1060,19	1894406	31061,31
...	48"	1219,0	5L	...	...	...	30,18	1158,64	1054355	112716	38296	36400	884,77	1054,36	1992541	32691,40
...	48"	1219,0	5L	...	...	...	31,75	1155,50	1048648	118423	38296	36301	929,56	1048,65	2088049	34258,39
...	52"	1321,0	5L	...	...	...	9,53	1301,94	1331287	39265	41500	40902	308,2	1331,29	844211	12781,39
...	52"	1321,0	5L	...	...	...	10,31	1300,38	1328099	42453	41500	40853	333,2	1328,10	911686	13802,97
...	52"	1321,0	5L	...	...	...	11,13	1298,74	1324751	45801	41500	40801	359,5	1324,75	982361	14872,99
...	52"	1321,0	5L	...	...	...	11,91	1297,18	1321571	48981	41500	40752	384,6	1321,57	1049340	15887,05
...	52"	1321,0	5L	...	...	...	12,70	1295,60	1318353	52199	41500	40702	409,6	1318,35	1116932	16910,39
...	52"	1321,0	5L	...	...	...	14,27	1292,46	1311971	58581	41500	40604	459,8	1311,97	1250527	18933,04
...	52"	1321,0	5L	...	...	...	15,88	1289,24	1305442	65110	41500	40503	511,1	1305,44	1386520	20991,97
...	52"	1321,0	5L	...	...	...	17,48	1286,04	1298969	71583	41500	40402	561,9	1298,97	1520661	23022,88
...	52"	1321,0	5L	...	...	...	19,05	1282,90	1292634	77918	41500	40303	611,6	1292,63	1651318	25001,03
...	52"	1321,0	5L	...	...	...	20,62	1279,76	1286314	84238	41500	40205	661,2	1286,31	1781019	26964,71
...	52"	1321,0	5L	...	...	...	22,23	1276,54	1279849	90703	41500	40104	712,0	1279,85	1913037	28963,46
...	52"	1321,0	5L	...	...	...	23,83	1273,34	1273440	97112	41500	40003	762,3	1273,44	2043248	30934,87
...	52"	1321,0	5L	...	...	...	25,40	1270,20	1267168	103384	41500	39905	811,5	1267,17	2170068	32854,93
...	52"	1321,0	5L	...	...	...	26,97	1267,06	1260910	109642	41500	39806	860,6	1260,91	2295950	34760,79
...	52"	1321,0	5L	...	...	...	28,58	1263,84	1254510	116042	41500	39705	910,9	1254,51	2424072	36700,55
...	52"	1321,0	5L	...	...	...	30,18	1260,64	1248165	122387	41500	39604	960,7	1248,17	2550431	38613,64
...	52"	1321,0	5L	...	...	...	31,75	1257,50	1241955	128597	41500	39506	1009,5	1241,96	2673489	40476,74
...	56"	1422,0	5L	...	...	...	9,53	1402,94	1545853	42288	44673	44075	331,9	1545,85	1054655	14833,40
...	56"	1422,0	5L	...	...	...	10,31	1401,38	1542417	45724	44673	44026	358,9	1542,42	1139094	16021,02
...	56"	1422,0	5L	...	...	...	11,13	1399,74	1538809	49332	44673	43974	387,2	1538,81	1227561	17265,27
...	56"	1422,0	5L	...	...	...	11,91	1398,18	1535381	52760	44673	43925	414,1	1535,38	1311424	18444,78
...	56"	1422,0	5L	...	...	...	12,70	1396,60	1531912	56229	44673	43875	441,4	1531,91	1396077	19635,39
...	56"	1422,0	5L	...	...	...	14,27	1393,46	1525032	63109	44673	43777	495,4	1526,03	1563459	21989,58
...	56"	1422,0	5L	...	...	...	15,88	1390,24	1517992	70149	44673	43676	550,6	1517,99	1733935	24387,28
...	56"	1422,0	5L	...	...	...	17,48	1387,04	1511012	77129	44673	43576	605,4	1511,01	1902183	26753,63
...	56"	1422,0	5L	...	...	...	19,05	1383,90	1504178	83963	44673	43477	659,1	1504,18	2066148	29059,75
...	56"	1422,0	5L	...	...	...	20,62	1380,76	1497360	90781	44673	43378	712,6	1497,36	2229001	31350,23
...	56"	1422,0	5L	...	...	...	22,23	1377,54	1490384	97757	44673	43277	767,4	1490,38	2394853	33682,89
...	56"	1422,0	5L	...	...	...	23,83	1374,34	1483468	104673	44673	43176	721,6	1483,47	2558527	35984,91
...	56"	1422,0	5L	...	...	...	25,40	1371,20	1476897	111444	44673	43078	875,8	1476,70	2718024	38228,18
...	56"	1422,0	5L	...	...	...	26,97	1368,06	1469942	118199	44673	42979	927,6	1469,94	2876430	40456,11
...	56"	1422,0	5L	...	...	...	28,58	1364,84	1463030	125111	44673	42878	982,1	1463,03	3037742	42724,92
...	56"	1422,0	5L	...	...	...	30,18	1361,64	1456178	131963	44673	42777	1036,9	1456,18	3196925	44963,79
...	56"	1422,0	5L	...	...	...	31,75	1358,50	1449470	138671	44673	42679	1088,5	1449,47	3352037	47145,38
...	60"	1524,0	5L	...	...	...	9,53	1504,94	1778805	45342	47878	47279	355,9	1778,80	1300026	17060,71
...	60"	1524,0	5L	...	...	...	10,31	1503,38	1775119	49028	47878	47230	384,9	1775,12	1404266	18428,69
...	60"	1524,0	5L	...	...	...	11,13	1501,74	1771248	52899	47878	47179	415,2	1771,25	1513503	19862,25
...	60"	1524,0	5L	...	...	...	11,91	1500,18	1767570	56577	47878	47130	444,1	1767,57	1617080	21221,52
...	60"	1524,0	5L	...	...	...	12,70	1498,60	1763849	60298	47878	47080	473,3	1763,85	1721656	22593,91

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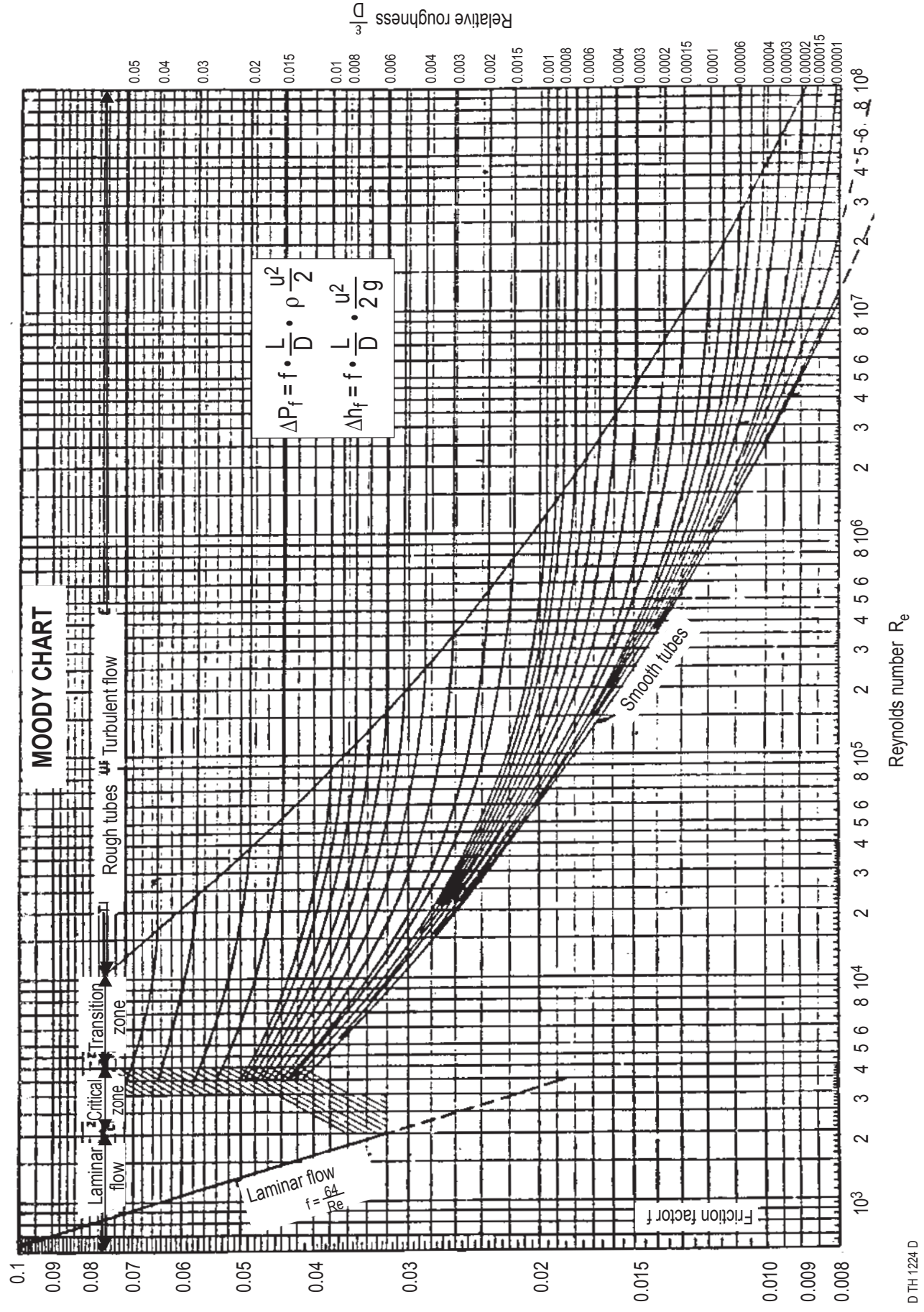
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— ANSI/ASME B36 10M-1995 STANDARD for hight-alloy steel pipes —

NOMINAL DIAMETER		OUTSIDE DIAMETER	STANDARD THICKNESS				THICKNESS	INSIDE DIAMETER	SECTION		LATERAL SURFACE		WEIGHT PER UNIT LENGTH		MOMENT OF INERTIA	MODULUS OF ELASTICITY
		mm					mm	mm	mm²		cm²/m		kg/m		cm4	cm3
ISO DN	ANSI NPS		Spec. API	Ano. API	SCH	SCH INOX			FREE	METAL	EXT.	INT.	METAL	WATER		
...	60*	1524,0	5L	...	...	...	14,27	1495,46	1756465	67682	47878	46981	531,3	1756,46	1926504	25308,45
...	60*	1524,0	5L	...	...	...	15,88	1492,24	1746909	75238	47878	46880	590,6	1748,91	2139273	28074,45
...	60*	1524,0	5L	...	...	...	17,48	1489,04	1741416	82731	47878	46780	649,4	1741,42	2347386	30805,59
...	60*	1524,0	5L	...	...	...	19,05	1485,90	1734080	90067	47878	46681	707,0	1734,08	2550298	33468,46
...	60*	1524,0	5L	...	...	...	20,62	1482,76	1726759	97388	47878	46582	764,5	1726,76	2751925	36114,50
...	60*	1524,0	5L	...	...	...	22,23	1479,54	1719267	104880	47878	46481	823,3	1719,27	2957364	38810,55
...	60*	1524,0	5L	...	...	...	23,83	1476,34	1711838	112309	47878	46381	881,6	1711,84	3160203	41472,48
...	60*	1524,0	5L	...	...	...	25,40	1473,20	1704564	119583	47878	46282	938,7	1704,56	3357960	44067,72
...	60*	1524,0	5L	...	...	...	26,97	1470,06	1697305	126841	47878	46183	995,6	1697,31	3554457	46646,42
...	60*	1524,0	5L	...	...	...	28,58	1466,84	1689878	134269	47878	46082	1054,0	1689,88	3754657	49273,71
...	60*	1524,0	5L	...	...	...	30,18	1463,64	1682513	141634	47878	45982	1111,8	1682,51	3952311	51867,60
...	60*	1524,0	5L	...	...	...	31,75	1460,50	1675302	148845	47878	45883	1168,4	1675,30	4145004	54396,36
...	64*	1626,0	5L	...	...	...	9,53	1606,94	2028099	48396	51082	50484	379,9	2028,10	1580778	19443,76
...	64*	1626,0	5L	...	...	...	10,31	1605,38	2024163	52332	51082	50435	410,8	2024,16	1707695	21004,86
...	64*	1626,0	5L	...	...	...	11,13	1603,74	2020030	56465	51082	50383	443,2	2020,03	1840723	22641,12
...	64*	1626,0	5L	...	...	...	11,91	1602,18	2016102	60393	51082	50334	474,1	2016,10	1966883	24192,90
...	64*	1626,0	5L	...	...	...	12,70	1600,60	2012128	64368	51082	50284	505,3	2012,13	2094286	25759,97
...	64*	1626,0	5L	...	...	...	14,27	1597,46	2004241	72256	51082	50186	567,2	2004,24	2346361	28860,52
...	64*	1626,0	5L	...	...	...	15,88	1594,24	1996169	80328	51082	50085	630,5	1996,17	2603319	32021,14
...	64*	1626,0	5L	...	...	...	17,48	1591,04	1988163	88332	51082	49984	693,4	1988,16	2857143	35143,21
...	64*	1626,0	5L	...	...	...	19,05	1587,90	1980324	96172	51082	49885	754,9	1980,32	3104724	38188,49
...	64*	1626,0	5L	...	...	...	20,62	1584,76	1972499	103996	51082	49787	816,3	1972,50	3350840	41215,75
...	64*	1626,0	5L	...	...	...	22,23	1581,54	1964492	112003	51082	49686	879,2	1964,49	3601712	44301,51
...	64*	1626,0	5L	...	...	...	23,83	1578,34	1956550	119946	51082	49586	941,5	1956,55	3849513	47349,48
...	64*	1626,0	5L	...	...	...	25,40	1575,20	1948773	127722	51082	49486	1002,6	1948,77	4091206	50322,34
...	64*	1626,0	5L	...	...	...	26,97	1572,06	1941012	135484	51082	49388	1063,5	1941,01	4331459	53277,48
...	64*	1626,0	5L	...	...	...	28,58	1568,84	1933068	143427	51082	49287	1125,8	1933,07	4576342	56289,57
...	64*	1626,0	5L	...	...	...	30,18	1565,64	1925190	151305	51082	49186	1187,7	1925,19	4818215	59264,63
...	64*	1626,0	5L	...	...	...	31,75	1562,50	1917476	159019	51082	49087	1248,2	1917,48	5054115	62166,23
...	68*	1727,0	5L	...	...	...	11,91	1703,18	2278300	64172	54255	53507	503,7	2278,30	2359681	27326,94
...	68*	1727,0	5L	...	...	...	12,70	1701,60	2274075	68398	54255	53457	536,9	2274,08	2512742	29099,50
...	68*	1727,0	5L	...	...	...	14,27	1698,46	2265690	76783	54255	53359	602,7	2265,69	2815663	32607,57
...	68*	1727,0	5L	...	...	...	15,88	1695,24	2257108	85365	54255	53258	670,1	2257,11	3124562	36184,86
...	68*	1727,0	5L	...	...	...	17,48	1692,04	2248594	93878	54255	53157	736,9	2248,59	3428804	39719,79
...	68*	1727,0	5L	...	...	...	19,05	1688,90	2240257	102216	54255	53058	802,4	2240,26	3727643	43169,00
...	68*	1727,0	5L	...	...	...	20,62	1685,76	2231934	110539	54255	52960	867,7	2231,93	4023828	46599,03
...	68*	1727,0	5L	...	...	...	22,23	1682,54	2223416	119057	54255	52859	934,5	2223,42	4325841	50096,60
...	68*	1727,0	5L	...	...	...	23,83	1679,34	2214966	127508	54255	52758	1000,9	2214,97	4624267	53552,60
...	68*	1727,0	5L	...	...	...	25,40	1676,20	2206691	135782	54255	52659	1066,8	2206,69	4915444	56924,66
...	68*	1727,0	5L	...	...	...	26,97	1673,06	2198431	144041	54255	52561	1130,7	2198,43	5204990	60277,82
...	68*	1727,0	5L	...	...	...	28,58	1669,84	2189977	152496	54255	52460	1197,0	2189,98	5500224	63696,86
...	68*	1727,0	5L	...	...	...	30,18	1666,64	2181592	160881	54255	52359	1262,8	2181,59	5791937	67075,12
...	68*	1727,0	5L	...	...	...	31,75	1663,50	2173379	169094	54255	52260	1327,9	2173,38	6076552	70371,18
...	72*	1829,0	5L	...	...	...	12,70	1803,60	2554879	72467	57460	56662	568,8	2554,88	2988461	32678,83
...	72*	1829,0	5L	...	...	...	14,27	1800,46	2545991	81355	57460	56563	638,6	2545,99	3349244	36623,77
...	72*	1829,0	5L	...	...	...	15,88	1797,24	2536892	90454	57460	56462	710,0	2536,89	3717263	40648,04
...	72*	1829,0	5L	...	...	...	17,48	1794,04	2527866	99480	57460	56361	780,9	2527,87	4081043	44626,95
...	72*	1829,0	5L	...	...	...	19,05	1790,90	2519025	108321	57460	56263	850,3	2519,03	4436114	48506,63
...	72*	1829,0	5L	...	...	...	20,62	1787,76	2510200	117146	57460	56164	919,5	2510,20	4789323	52370,94
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...	72*	1829,0	5L	...	...	...	26,97	1775,06	2474662	152684	57460	55765	1198,5	2474,66	6199036	67786,07
...	72*	1829,0	5L	...	...	...	28,58	1771,84	2465692	161654	57460	55664	1268,9	2465,69	6551685	71642,27
...	72*	1829,0	5L	...	...	...	30,18	1768,64	2456794	170552	57460	55563	1338,8	2456,79	6900244	75453,74
...	72*	1829,0	5L	...	...	...	31,75	1765,50	2448078	179268	57460	55465	1407,2	2448,08	7240434	79173,69
...	76*	1930,0	5L	...	...	...	12,70	1904,60	2849033	76497	60633	59835	600,5	2849,03	3515223	38427,18
...	76*	1930,0	5L	...	...	...	14,27	1901,46	2839646	85883	60633	59736	674,2	2839,65	3940133	40830,39
...	76*	1930,0	5L	...	...	...	15,88	1898,24	2830037	95493	60633	59635	749,6	2830,04	4373688	45323,19
...	76*	1930,0	5L	...	...	...	17,48	1895,04	2820504	105026	60633	59534	824,4	2820,50	4802369	49765,48
...	76*	1930,0	5L	...	...	...	19,05	1891,90	2811164	114365	60633	59436	897,7	2811,16	5220907	54102,87
...	76*	1930,0	5L	...	...	...	20,62	1888,76	2801841	123689	60633	59337	970,9	2801,84	5637367	58418,31
...	76*	1930,0	5L	...	...	...	22,23	1885,54	2792296	133234	60633	59236	1045,8	2792,30	6062285	62821,61
...	76*	1930,0	5L	...	...	...	23,83	1882,34	2782826	142704	60633	59136	1120,2	2782,83	6482413	67176,26
...	76*	1930,0	5L	...	...	...	25,40	1879,20	2773549	151980	60633	59037	1193,0	2773,55	6892586	71425,76
...	76*	1930,0	5L	...	...	...	26,97	1876,06	2764288	161241	60633	58938	1266,7	2764,29	7300708	75655,00
...	76*	1930,0	5L	...	...	...	28,58	1872,84	2754807	170722	60633	58837	1340,1	2754,81	7717105	79970,00
...	76*	1930,0	5L	...	...	...	30,18	1869,64	2745401	180128	60633	58736	1413,9	2745,40	8128793	84236,20
...	76*	1930,0	5L	...	...	...	31,75	1866,50	2736188	189342	60633	58636	1486,3	2736,19	8530713	88401,17
...	80*	2032,0	5L	...	...	...	14,27	2003,48	3152472	90456	63837	62941	710,0	3152,47	4803570	45310,73
...	80*	2032,0	5L	...	...	...	15,88	2000,24	3142347	100581	63837	62839	789,5	3142,35	5110771	50302,87
...	80*	2032,0	5L	...	...	...	17,48	1997,04	3132300	110627	63837	62739	868,4	3132,30	5812402	55240,16
...	80*	2032,0	5L	...	...	...	19,05	1993,90	3122458	120470	63837	62640	945,6	3122,46	6102288	60061,89
...	80*	2032,0	5L	...	...	...	20,62	1990,76	3112631	130296	63837	62542	1022,8	3112,63	6589866	64860,88
...	80*	2032,0	5L	...	...	...	22,23	1987,54	3102570	140358	63837	62440	1101,8	3102,57	7087475	69758,61
...	80*	2032,0	5L	...	...	...	23,83	1984,34	3092588	150340	63837	62340	1180,1	3092,59	7579604	74602,40
...	80*	2032,0	5L	...	...	...	25,40	1981,20	3082808	160120	63837	62241	1256,9	3082,81	8060196	79332,64
...	80*	2032,0	5L	...	...	...	26,97	1978,06	3073044	169884	63837	62143	1333,5	3073,04	8538509	84040,44
...	80*	2032,0	5L	...	...	...	28,58	1974,84	3063047	179881	63837	62041	1412,0	3063,05	9026648	88844,96
...	80*	2032,0	5L	...	...	...	30,18	1971,64	3053129	189799	63837	61941	1489,8	3053,13	9509395	93596,41
...	80*	2032,0	5L	...	...	...	31,75	1968,50	3043412	199516	63837	61842	1566,1	3043,41	9980812	98236,34

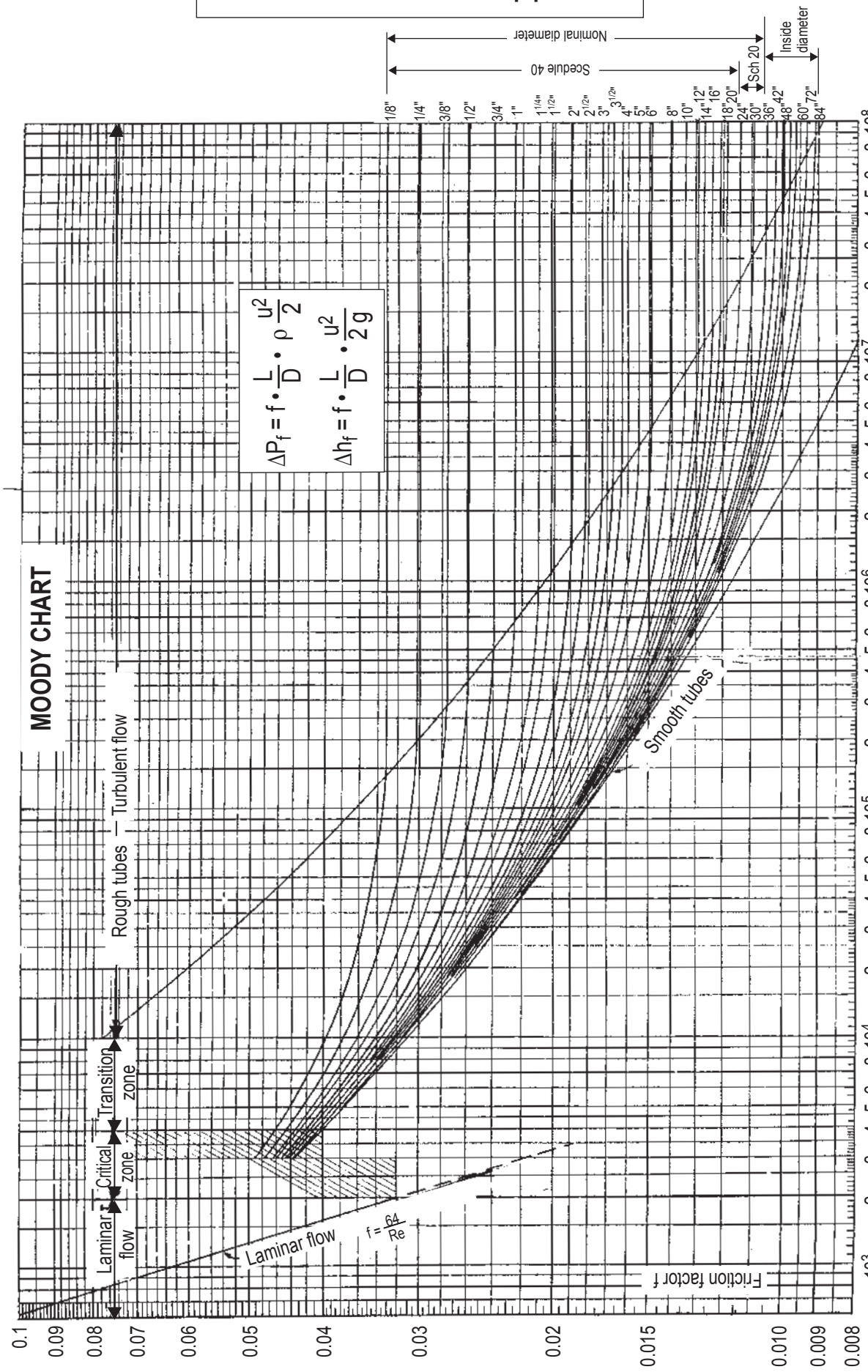
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HEAD LOSSES IN STRAIGHT TUBES
— Friction factor versus relative roughness —



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HEAD LOSSES IN STRAIGHT TUBES
— Friction factor for steel pipes —

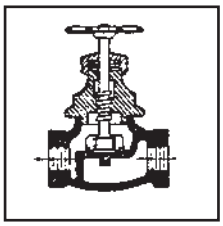


D TH 1223 D

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RESISTANCE OF VALVES AND FITTINGS TO FLUID FLOW
– K values –

$$h_f = K.\frac{u^2}{2g}\ (m)$$



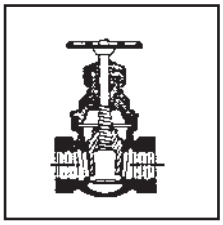
Globe valve

D	< 2"	2" to 4"	6" to 10"	> 12"
K	7.5	6.2	5	4.2



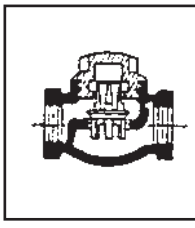
Swing check valve

D	< 2"	2" to 4"	6" to 10"	> 12"
K	2.2	1.8	1.5	1.3



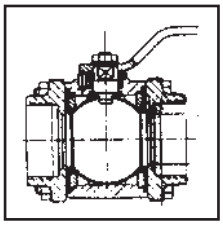
Gate valve (wedge ?)

D	< 2"	2" to 4"	6" to 10"	> 12"
K	0.18	0.15	0.12	0.1



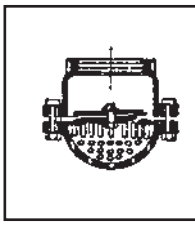
Lift check valve

D	< 2"	2" to 4"	6" to 10"	> 12"
K	13	10	8	7



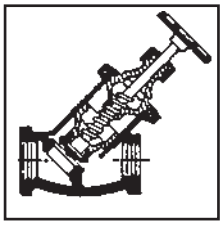
Ball valve

D	≤ 2"	> 2"
K	0.06	0.04



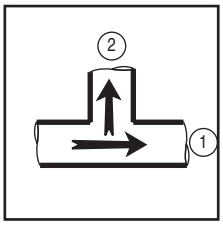
Foot valve

D	< 2"	2" to 4"	6" à 10"	> 12"
K	2	1.5	1.2	0.9



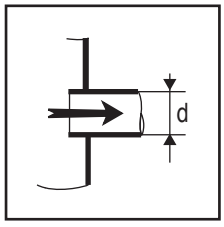
Y - blow off valve

D	< 2"	2" to 4"	6" to 10"	> 12"
K	1.2	1	0.8	0.7



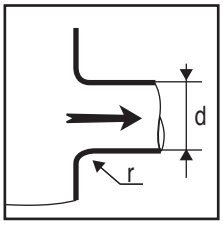
Standard tee

D	< 2"	2" à 4"	6" à 10"	> 12"
①K	0.5	0.4	0.3	0.3
②K	1.5	1.1	0.9	0.8



Borda entrance

K = 0.8



Ordinary entrance

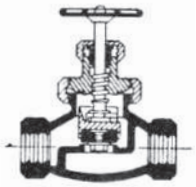
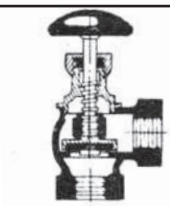
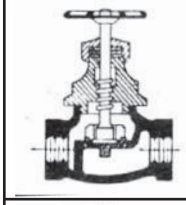
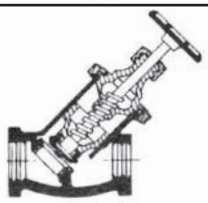
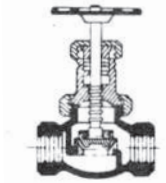
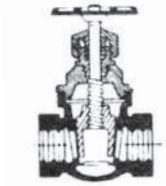
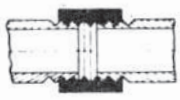
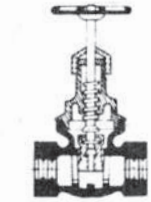
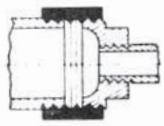
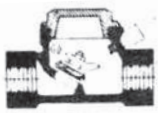
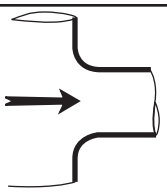
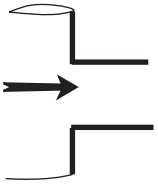
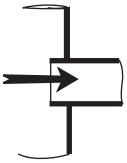
r/d	0	0.02	0.08	0.15	> 0.15
K	0.5	0.3	0.12	0.08	0.04

D MEQ 1341 D

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RESISTANCE OF VALVES AND FITTINGS TO FLUID FLOW
— K values —

$$h_f = K.\frac{u^2}{2g}\ (m)$$

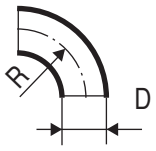
	GLOBE VALVE (removable disk) K = 5.2 - 7.3		ANGLE VALVE K = 2.1 - 3.1
	GLOBE VALVE (Inclined seat) K = 6.5 - 7.2		Y-BLOW OFF VALVE K = 2.9
	NEEDLE VALVE K = 7.2 - 10.3		FOOT VALVE K = 15
	GATE VALVE K = 0.05 - 0.19		COUPLING K = 0.02- 0.07
	DOUBLE GATE VALVE K = 0.08 - 0.13		COUPLING + REDUCTION K = 0.05 - 2.0
	SWING CHECK VALVE K = 0.6 - 2.3		SMOOTH ENTRANCE K = 0.04 - 0.05
	LIFT CHECK VALVE K = 8 - 12		STRAIGHT ENTRANCE K = 0.47 - 0.56
	BALL CHECK VALVE K = 65 - 70		BORDA ENTRANCE K = 0.62 - 1.0

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[illegible]

$$h_f = K.\frac{u^2}{2g}\ (m)$$

• [illegible]



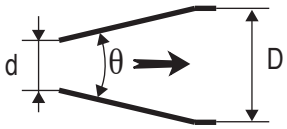
								Return bend
	R/D	0.75	1	1.5	2	4	8	r
K for	D ≤ 2	0.75	0.60	0.50	0.40	0.50	0.60	1.20
K for	2" < D ≤ 6"	0.50	0.40	0.35	0.30	0.30	0.40	0.90
K for	D > 6"	0.40	0.30	0.20	0.20	0.20	0.30	0.70

• [illegible]

$$K_d = \left(1 - \frac{d^2}{D^2}\right)^2 \qquad u \rightarrow \quad d \quad D \quad \rightarrow U \qquad K_D = \left(1 - \frac{d^2}{D^2}\right)^2 . \frac{D^4}{d^4}$$

$$h_f = K_d \frac{u^2}{2.g} \qquad h_f = K_D \frac{U^2}{2.g}$$

• [illegible]



$\theta < 45°$ $\quad A = 2.6 \sin \frac{\theta}{2}$

$45° < \theta < 180°$ $\quad A = 1$

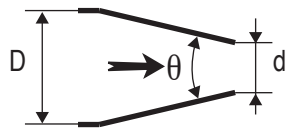
With $K_{gradual\ enlargement} = A\ .\ K_{sudden\ enlargement}$

• [illegible]

$$K_D = \frac{1}{2}\left(1 - \frac{d^2}{D^2}\right).\frac{D^4}{d^4} \qquad U \rightarrow \quad D \quad d \quad \rightarrow u \qquad K_d = \frac{1}{2}\left(1 - \frac{d^2}{D^2}\right)$$

$$h_f = K_D \frac{U^2}{2.g} \qquad h_f = K_d \frac{u^2}{2.g}$$

• [illegible]



$\theta < 45°$ $\quad A = 1.6 \sin \frac{\theta}{2}$

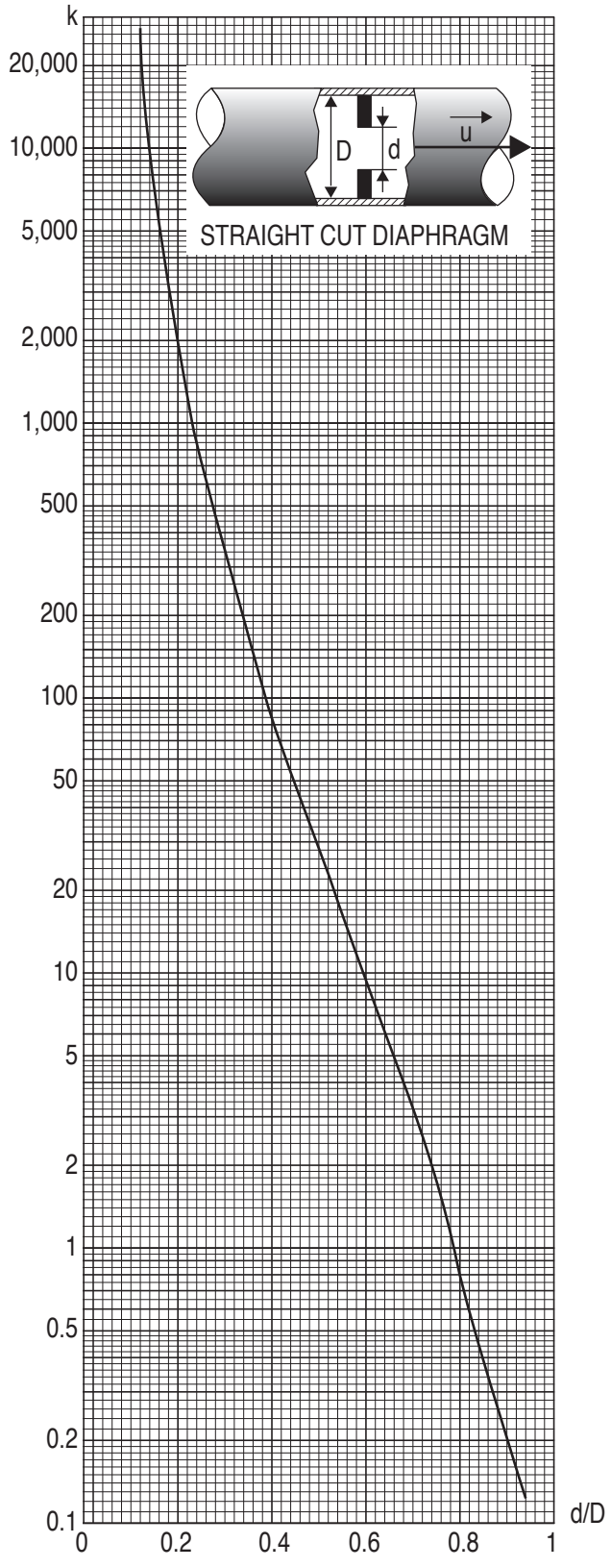
$45° < \theta < 180°$ $\quad A = \sqrt{\sin \frac{\theta}{2}}$

Avec $K_{gradual\ contraction} = A\ .\ K_{sudden\ contraction}$

D MEQ 1342 D

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[illegible]



$$h_f = K.\frac{u^2}{2g}\ (m)$$

With

h_f : losses in meters

K : factor read on the graph

u : liquid mean velocity (m/s)

[illegible]: standardized diaphragms generate lower pressure drops

D MEQ 1455 D

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RESISTANCE COEFFICIENT FOR COMPRESSOR'S ACCESSORY

Description	K
Clean filter (for a normal fouling max value can be 20)	4.0
Inter stage cooler	17.0
Scrubber/separator	7.0
Pulsation dampener vessel	
without any internal pipe	4.0
whith internal pipe	12.0
Suction compressor pipe	0.5
Lateral compressor suction pipe with diaphragm	1.0
Lateral compressor suction pipe between impellers	0.8
Discharge compressor pipe	0.5
Extraction compressor pipe	0.8

Source : Compressors : Selection & Sizing - Royce N. Brown - GPC

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RESISTANCE OF VALVES AND FITTINGS TO FLUID FLOW
— Equivalent straight pipe length —

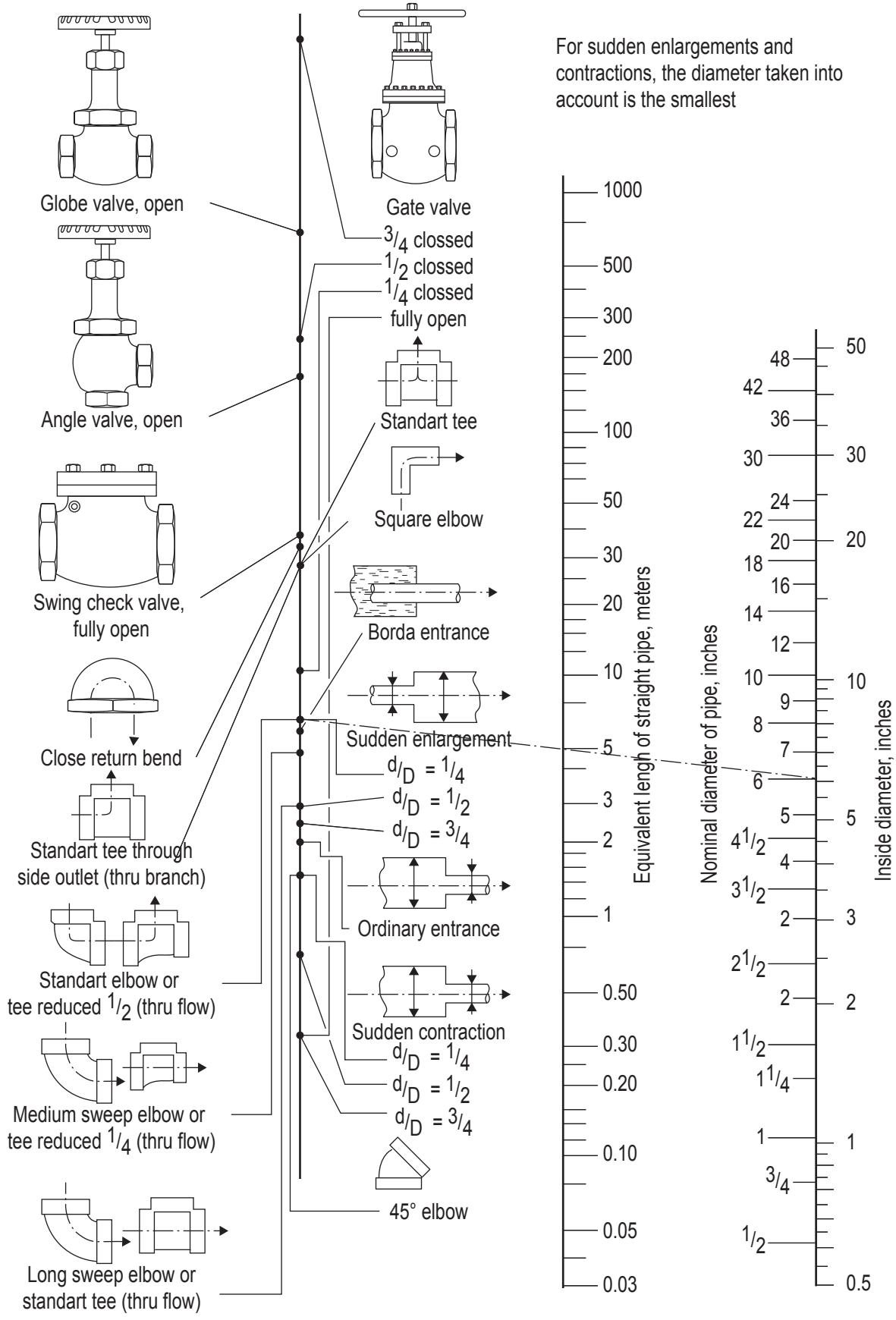
Diamètres en mm	50	100	150	200	250	300	350	400	450
90° elbow R/D = 2	1	1.7	2.5	3.2	4	5	6	7	8
90° elbow R/D = 1,5	1.5	2.5	4	5	6	7.5	9	11	12.5
Cast iron elbow	3.2	7.5	12.5	18	24	30	38	44	50
Plate elbow	7.5	17.5	29	42	56	70	87	102	115
Loop R/D = 6	4	9.5	14.5	20	27	33	41	48	54
Lyre à plis	5	12	18.5	26	34	42	52	61	69
Coude à plis R/D = 2	1.7	2.8	4.2	5.5	6.5	8.5	10	12	13.5
Coude à plis R/D = 1.5	2.4	4	6.5	8	9.5	12	14.5	17.5	20
Tee 	3.6	5.4	8	6.3	15.5	21	26	32	36
Tee 	4.5	7	9.5	14	19	25	31	38	43
Tee 	5	11.5	17.5	26	36	47	65	74	84
Tee 	4.5	9	14.5	20	26	34	41	47	54
Globe valve	13	31	50	73	100	130	160	200	250
Angle valve	10	20	32	45	61	77	95	115	130
Check valve	3.2	7.5	12.5	18	24	30	38	44	50
Parallel seats valve	0.6	1.5	2	3	4	5	6.5	7.5	8.5

The equivalent pipe lengths are given in meters

D CONS 1319 D

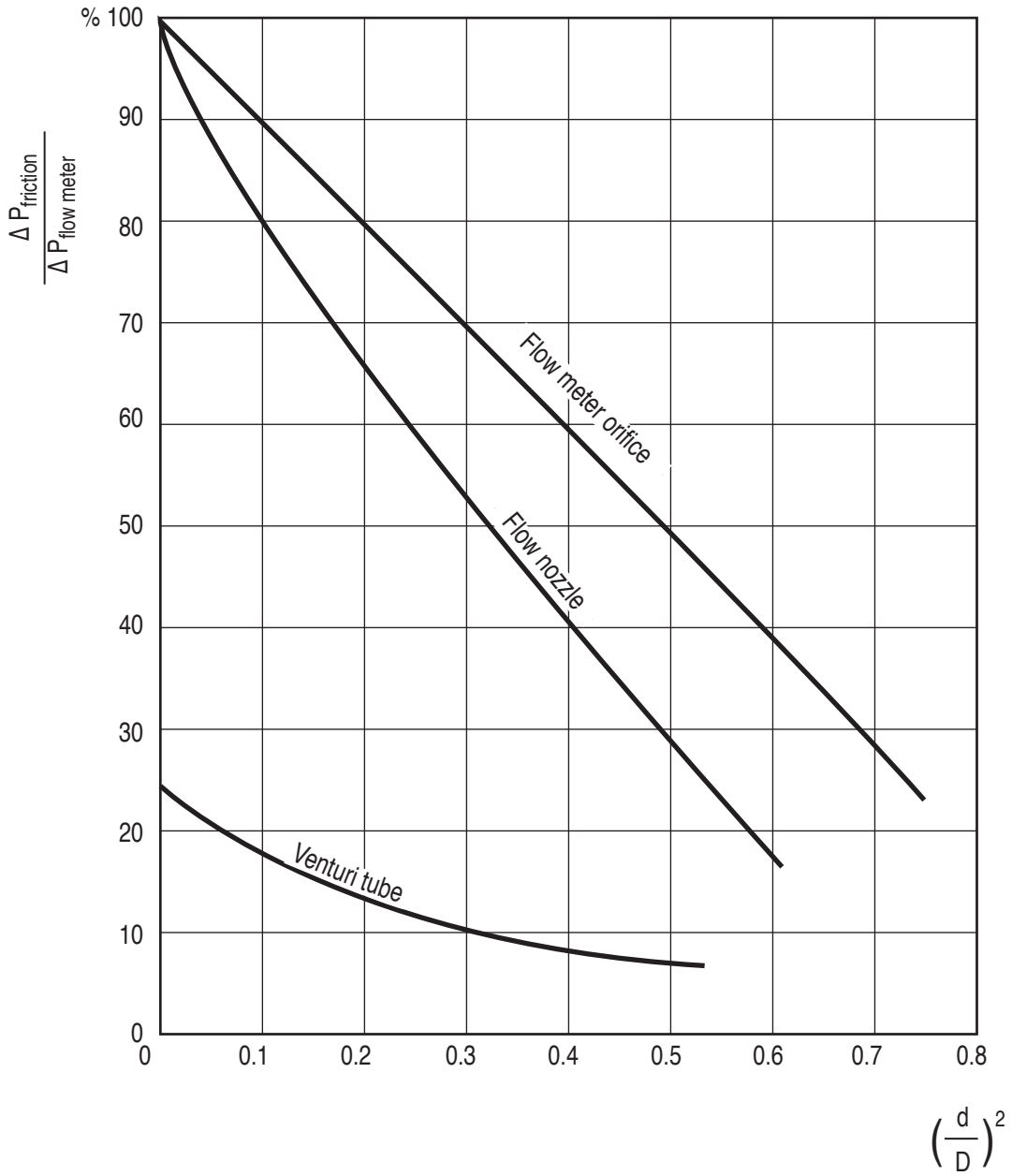
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RESISTANCE OF VALVES AND FITTINGS TO FLUID FLOW
— Equivalent straight pipe length —



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FRICTION LOSSES IN FLOW METERS



D MEQ 1456 D

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CORRECTIONS OF THE FLOWMETER INDICATIONS
— Mass flowrate indication —

The flowmeter has been calibrated under **reference conditions** (index "r"):

θ_r	reference operating temperature
$sp.gr_r$, den_r	density and specific gravity of reference fluid at θ_r

Under **usual operating conditions** (index "a"):

θ_a	actual operating temperature
$sp.gr_a$, den_a	density and specific gravity of fluid at θ_a

- ***Case of liquids***

$$\frac{Q_{M,a}}{Q_{M,r}} = \sqrt{\frac{den_a}{den_r}} = \sqrt{\frac{sp.gr_a}{sp.gr_r}}$$

- ***Case of gases and vapors***

$$\frac{Q_{M,a}}{Q_{M,r}} = \sqrt{\frac{den_a}{den_r}}$$

or

$$\frac{Q_{M,a}}{Q_{M,r}} = \sqrt{\frac{P_a}{P_r} \cdot \frac{T_r}{T_a} \cdot \frac{M_a}{M_r}}$$

where $\frac{M_a}{M_r} = \frac{sp.gr_a}{sp.gr_r}$

T_a and T_r	absolute temperatures
M_a and M_r	molecular weights

D TH 1232 B

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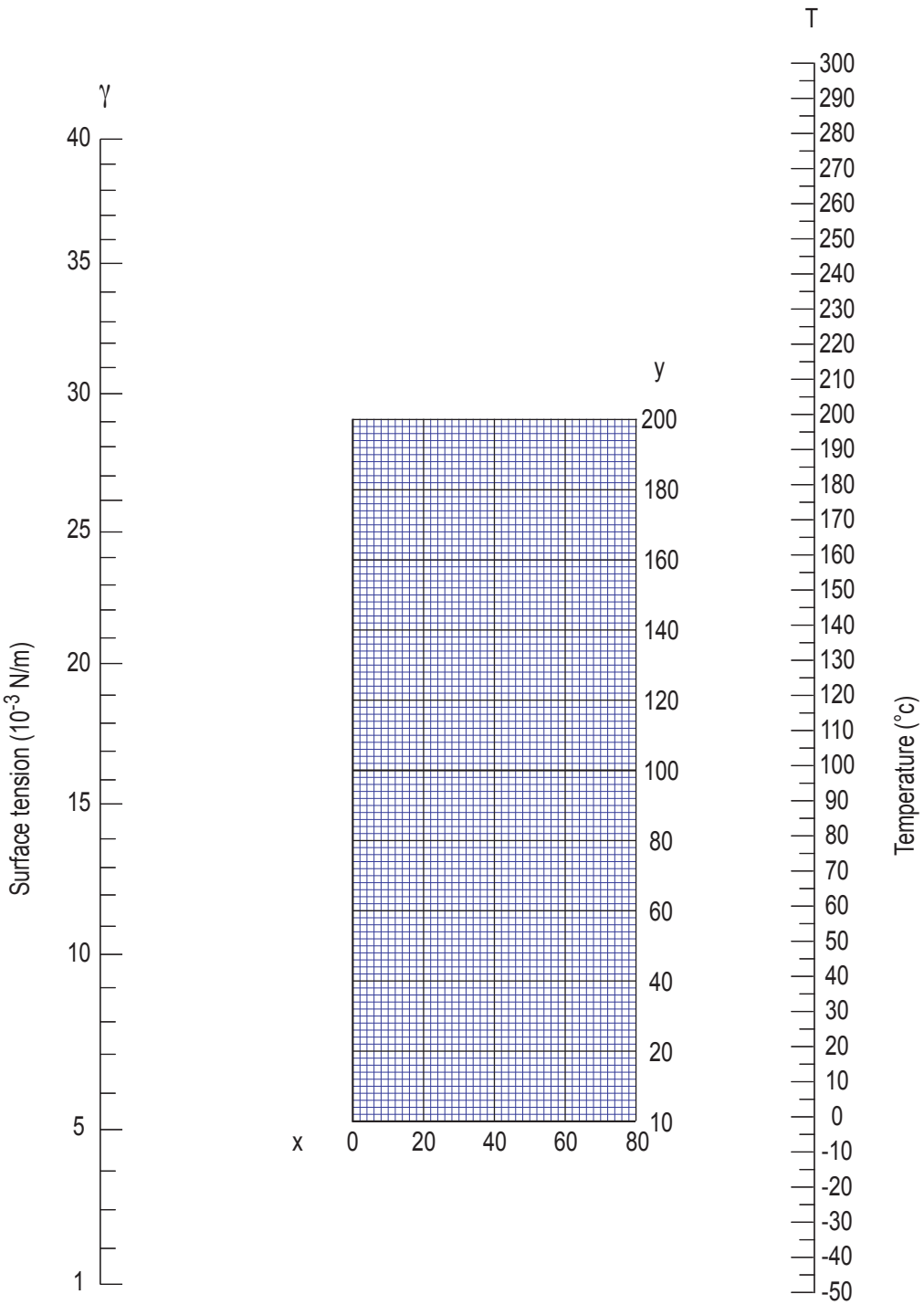
SURFACE TENSION OF SOME CHEMICAL COMPOUNDS

Compound	X	Y	Compound	X	Y
Acetal	19	88	Ethyl isobutyrate	20.9	93.7
Acetaldehyde	33	78	Ethyl ether	27.5	64
Acetaldoxime	23.5	127	Ethyl formate	30.5	88.8
Acetamide	17	192.5	Ethyl iodide	28	113.2
Acetic acid	17.1	116.5	Ethyl mercaptan	35	81
Acetic anhydride	25	129	Ethyl propionate	22.6	97
Acetone	28	91	Ethylamine	11.2	83
Acetonitrile	33.5	111	Ethylbenzene	22	118
Acetophenone	18	163	Ethylene chloride	32	120
Allyl alcohol	12	111.5	Ethylene oxide	42	83
Anethol	13	158.1	n-Hexane	22.7	72.2
Aniline	22.9	171.8	Hydrogen cyanide	30.6	66
Anisole	24.4	138.9	Mesitylene	17	119.8
Ammonia	56.2	63.5	Methyl acetate	34	90
Isoamyl-acétate	16.4	103.1	Methyl alcohol	17	93
Isoamyl-alcohol	6	106.8	Methyl n-Butyrate	25	88
Benzene	30	110	Methyl isobutyrate	24	93.8
Benzonitrile	19.5	159	Methyl chloride	45.8	53.2
Benzylamine	25	156	Methyl ether	44	37
Bromobenzene	23.5	145.5	Methyl ethyl ketone	23.6	97
Isobutyl-acetate	16	97.2	Methyl formate	38.5	88
n-butyl alcohol	9.6	107.5	Methyl propionate	29	95
Isobutyl alcohol	5	103	Methylamine	42	58
n-Butyric acid	14.5	115	Naphtalene	22.5	165
Isobutyric acid	14.8	107.4	Nitroethane	25.4	126.1
n-Butyronitrile	20.3	113	Nitromethane	30	139
Carbon disulfide	35.8	117.2	Nitrosyle chloride	38.5	93
Carbon tetrachloride	26	104.5	Nitrous oxide	62.5	0.5
Chloral	30	113	n-octane	17.7	90
Chlorine	45.5	59.2	Paraldehyde	22.3	103.8
Chlorobenzene	23.5	132.5	Phenetole	20	134.2
p-Chlorobromobenzene	14	162	Phenol	20	168
Chloroform	32	101.3	Phosphorus oxychlorine	26	125.2
p-Chlorotoluene	18.7	134	Piperidine	24.7	120
o-Cresol	20	161	Propionic acid	17	112
m-Cresol	13	161.2	Propionitrile	23	106.6
p-Cresol	11.5	160.5	n-propyl acetate	23	97
Cyclohexane	42	86.7	n-propyl alcohol	8.2	105.2
Diethyl ketone	20	101	n-propyl formate	24	97
Diethyl oxalate	20.5	130.8	n-propylamine	25.5	87.2
Diethyl sulfate	19.5	139.5	p-isopropyl toluene	12.8	121.2
Diethylaniline	17	142.6	Pyridine	34	138.2
Dimethyl sulfate	23.5	158	Quinoline	19.5	183
Dimethylamine	16	66	Thiophene	35	121
Dimethylaniline	20	149	Toluene	24	113
Ethyl acetate	27.5	92.4	Triethylamine	20.1	83.9
Ethyl acetoacetate	21	132	Trimethylamine	21	57.6
Ethyl alcohol	10	97	Tripalmitin	2	151
Ethyl benzoate	14.8	151	Triphenyl methane	12.5	182.7
Ethyl bromide	31.6	90.2	m-xylene	20.5	118
Ethyl n-butyrate	17.5	102	p-xylene	19	117

The surface tension is obtained through x and y values by the next chart.

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SURFACE TENSION OF SOME CHEMICAL COMPOUNDS



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BAKER FLOW PATTERN
— For horizontal pipes and cocurrent liquid - gas flow —

B_y = Baker paramètre (ordinate)

Q_m = Mass flow (kg/s) ; Q_{ml} : liquid alone ; Q_{mg} : gas alone ; Q_{mt} : total flow

G : Gas mass velocity (kg/s.m^2.s) : $G = \frac{Q_{mt}}{S}$

S = Pipe section (m^2)

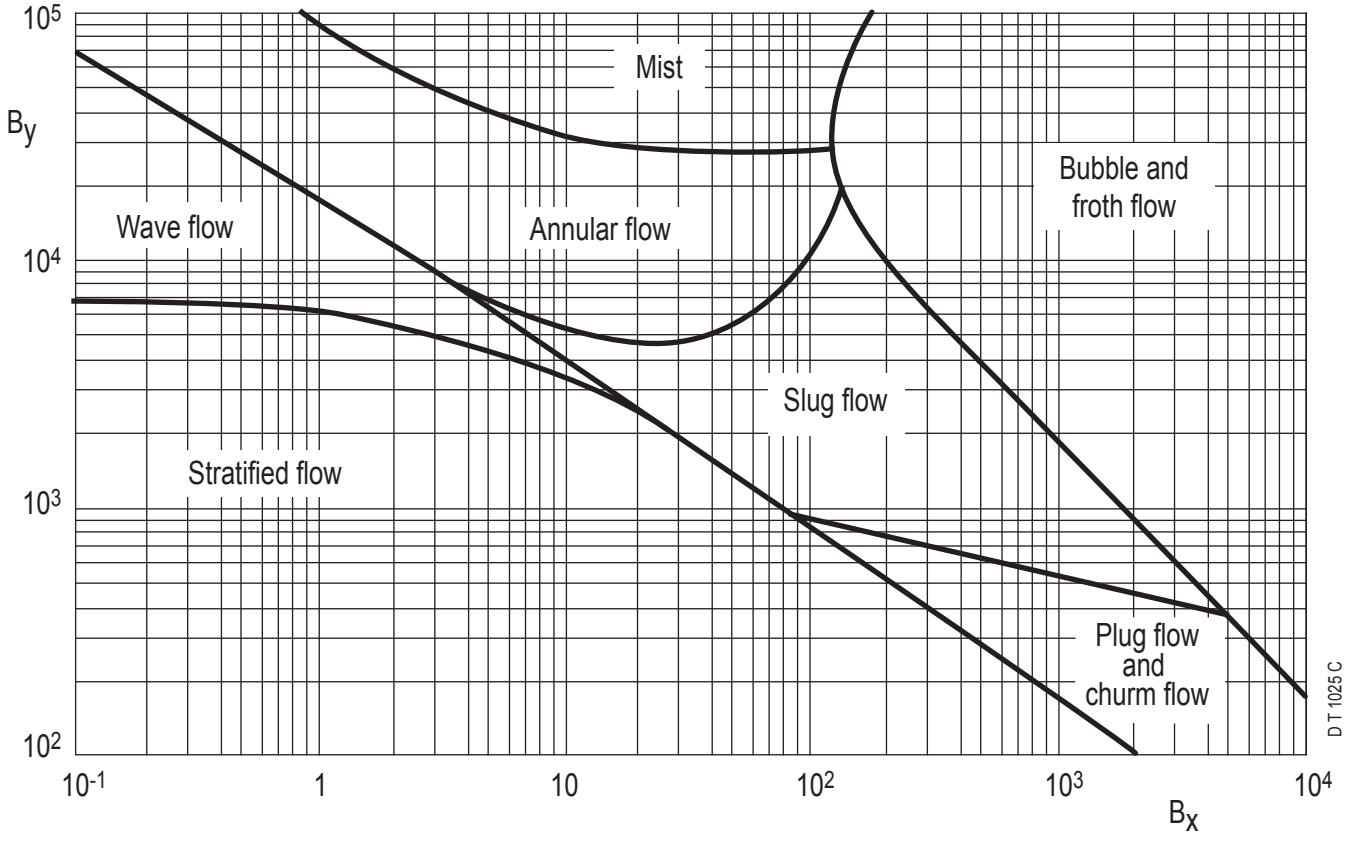
ρ_l ρ_g = Liquid and gas density (kg/m^3)

B_x = Baker paramètre (abscissae)

μ_l = Dynamic viscosity of liquid (Pa.s)

σ = Surface tension (N/m) (voir planche L9, 1 et 2)

$$B_x = 2.1 \frac{G_l}{G_g} \cdot \frac{1}{\sigma} \cdot \left(\frac{\mu_l}{\rho_l^2}\right)^{1/3} \cdot \left(\rho_g.\rho_l\right)^{1/2} \qquad B_y = 25.513.10^3.G_g\,(\rho_g.\rho_l)^{-1/2}$$



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TWO PHASE FLOW PATTERN
— In upflow through vertical pipes —

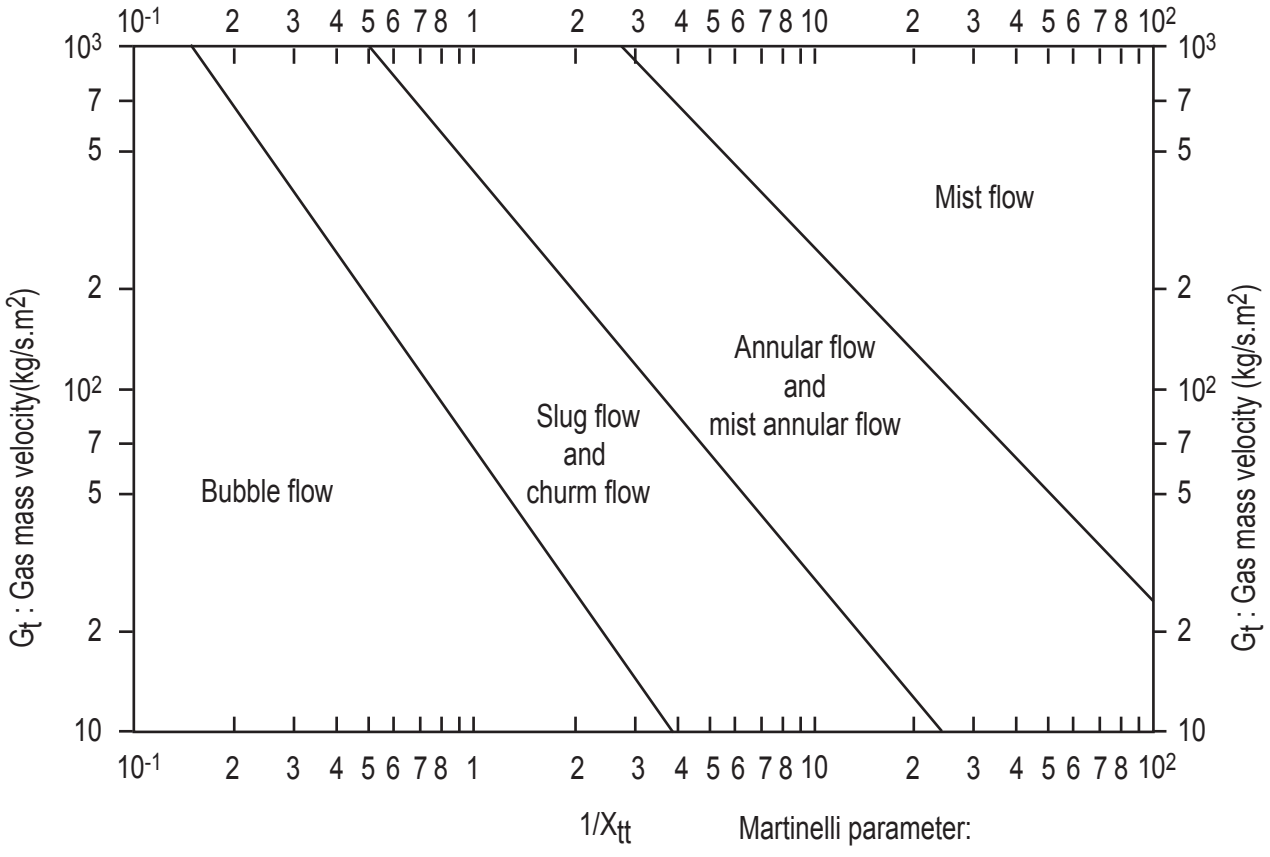
y = Gas mass fraction = $\frac{Q_{mg}}{Q_{mt}}$

X_{tt} = Martinelli paramèter : $X_{tt} = \left(\frac{1-y}{y}\right)^{0.9} \cdot \left(\frac{\rho_g}{\rho_l}\right)^{0.5} \cdot \left(\frac{\mu_l}{\mu_g}\right)^{0.1}$

μ_l μ_g = Dynamic viscosity of liquid and gas (Pa.s)

ρ_l ρ_g = Density of liquid and gas (kg/m^3)

G_t : Gas mass velocity : $G_t = \frac{Q_{mt}}{S}$ (kg/s.m^2)



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CORRECTION OF THE PRESSURE DROPS WITH DIAMETER

Table I - Laminar flow

D/Do	0	1	2	3	4	5	6	7	8	9
1.00	1.000	1.004	1.008	1.012	1.016	1.020	1.025	1.029	1.033	1.037
1.01	1.041	1.045	1.049	1.053	1.057	1.062	1.066	1.070	1.074	1.078
1.02	1.082	1.086	1.091	1.095	1.100	1.104	1.108	1.113	1.117	1.122
1.03	1.126	1.130	1.135	1.139	1.144	1.148	1.152	1.157	1.161	1.166
1.04	1.170	1.175	1.179	1.184	1.188	1.193	1.198	1.202	1.207	1.211
1.05	1.216	1.221	1.225	1.230	1.235	1.240	1.244	1.249	1.254	1.258
1.06	1.263	1.269	1.273	1.277	1.282	1.287	1.292	1.297	1.301	1.306
1.07	1.311	1.316	1.321	1.326	1.331	1.336	1.340	1.345	1.350	1.355
1.08	1.360	1.365	1.370	1.376	1.381	1.386	1.391	1.396	1.402	1.407
1.09	1.412	1.417	1.422	1.428	1.433	1.438	1.443	1.448	1.454	1.459
1.10	1.464	1.469	1.475	1.480	1.486	1.491	1.496	1.502	1.507	1.513
1.11	1.518	1.524	1.529	1.535	1.540	1.546	1.552	1.557	1.563	1.568
1.12	1.574	1.580	1.585	1.591	1.597	1.602	1.608	1.614	1.619	1.625
1.13	1.630	1.636	1.642	1.648	1.654	1.660	1.666	1.671	1.677	1.683
1.14	1.689	1.695	1.701	1.707	1.713	1.719	1.725	1.731	1.737	1.743
1.15	1.749	1.755	1.762	1.768	1.774	1.780	1.786	1.793	1.799	1.805
1.16	1.812	1.818	1.824	1.880	1.837	1.843	1.849	1.855	1.862	1.868
1.17	1.874	1.880	1.887	1.894	1.900	1.907	1.913	1.919	1.926	1.933
1.18	1.939	1.946	1.952	1.959	1.965	1.972	1.979	1.985	1.992	1.998
1.19	2.005	2.012	2.019	2.026	2.033	2.039	2.046	2.053	2.060	2.067
1.20	2.074	2.080	–	–	–	–	–	–	–	–

Table II - Turbulent flow

1.00	1.000	1.005	1.010	1.015	1.020	1.025	1.031	1.036	1.041	1.046
1.01	1.051	1.056	1.061	1.066	1.072	1.077	1.082	1.088	1.093	1.098
1.02	1.104	1.109	1.115	1.120	1.126	1.131	1.137	1.142	1.148	1.153
1.03	1.159	1.165	1.170	1.176	1.181	1.187	1.192	1.198	1.204	1.210
1.04	1.216	1.222	1.228	1.234	1.240	1.246	1.252	1.258	1.264	1.270
1.05	1.276	1.282	1.288	1.295	1.301	1.307	1.313	1.320	1.326	1.332
1.06	1.338	1.345	1.351	1.358	1.364	1.371	1.377	1.384	1.390	1.396
1.07	1.402	1.409	1.415	1.422	1.429	1.436	1.442	1.449	1.455	1.462
1.08	1.469	1.476	1.483	1.490	1.497	1.503	1.510	1.517	1.524	1.531
1.09	1.538	1.545	1.552	1.560	1.567	1.574	1.581	1.589	1.596	1.603
1.10	1.610	1.617	1.625	1.632	1.640	1.647	1.655	1.662	1.670	1.677
1.11	1.685	1.692	1.700	1.708	1.715	1.723	1.730	1.738	1.746	1.754
1.12	1.762	1.770	1.778	1.786	1.794	1.802	1.810	1.818	1.826	1.834
1.13	1.842	1.850	1.859	1.867	1.875	1.884	1.892	1.900	1.909	1.917
1.14	1.925	1.934	1.942	1.951	1.959	1.968	1.977	1.985	1.993	2.002
1.15	2.011	2.020	2.029	2.038	2.047	2.055	2.064	2.073	2.082	2.091
1.16	2.100	2.109	2.118	2.128	2.137	2.146	2.155	2.165	2.174	2.183
1.17	2.192	2.202	2.211	2.221	2.230	2.240	2.249	2.259	2.268	2.278
1.18	2.228	2.298	2.308	2.317	2.327	2.337	2.347	2.356	2.366	2.376
1.19	2.386	2.396	2.406	2.417	2.427	2.437	2.447	2.458	2.468	2.478
1.20	2.488	–	–	–	–	–	–	–	–	–

Example : 6 in schedule 40 : D_o = 154.05 mm
6 in schedule 80 : D = 146.33 mm

$\frac{D_o}{D} = 1.053$

— Laminar flow

$$\frac{\Delta P}{\Delta P_o} = \left(\frac{D_o}{D}\right)^4 = (1.053)^4 = 1.230$$

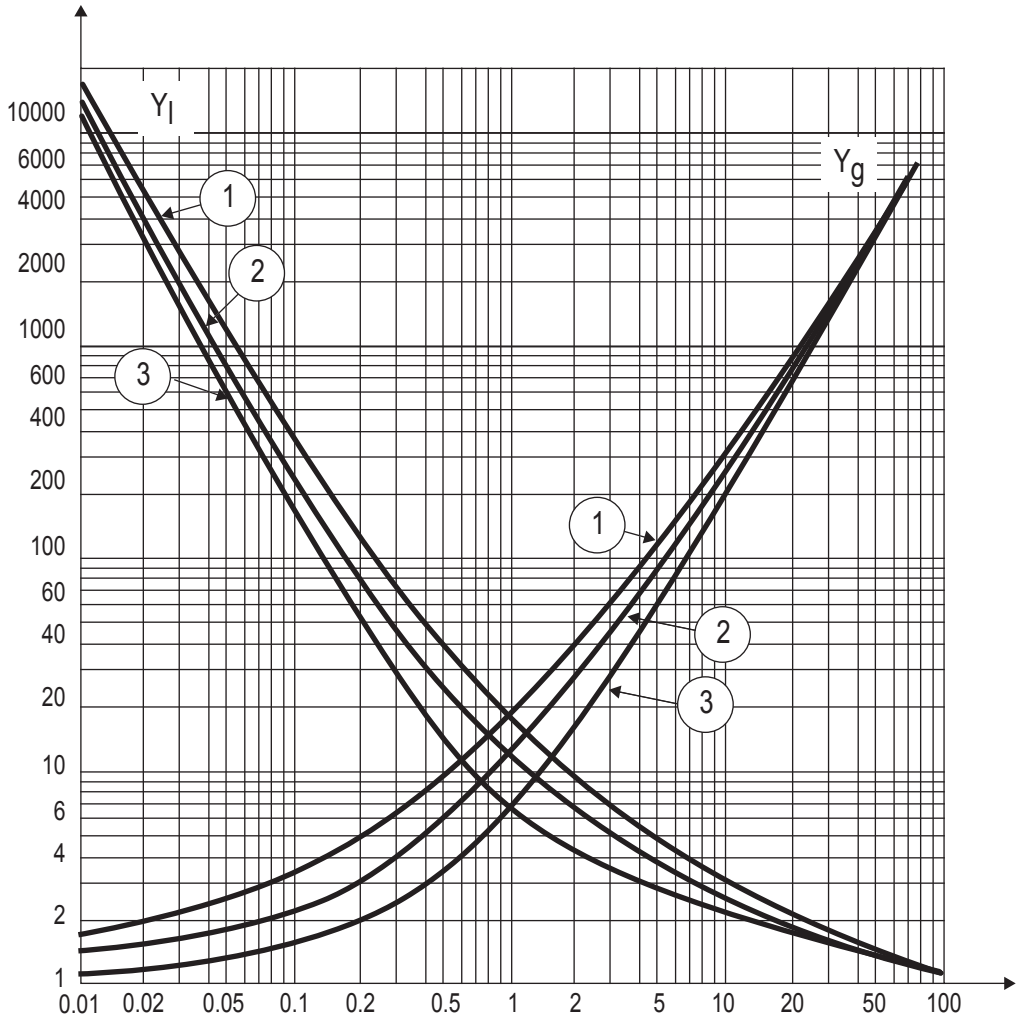
— Turbulent flow

$$\frac{\Delta P}{\Delta P_o} \approx \left(\frac{D_o}{D}\right)^5 = (1.053)^5 = 1.295$$

D TH 1225 B

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TWO PHASE FLOW PARAMETERS FOR PRESSURE DROP IN LIQUID - GAS FLOW - HORIZONTAL PIPES
— Based on Lockhart and Martinelli —



Flow conditions

1 - Both liquid and gas turbulent

2 - Combination of turbulent and viscous

3 - Both liquid and gas viscous

ΔP_{fl} et ΔP_{fg} Pressure drops of liquid and gas assuming that each phase is flowing alone in the pipe

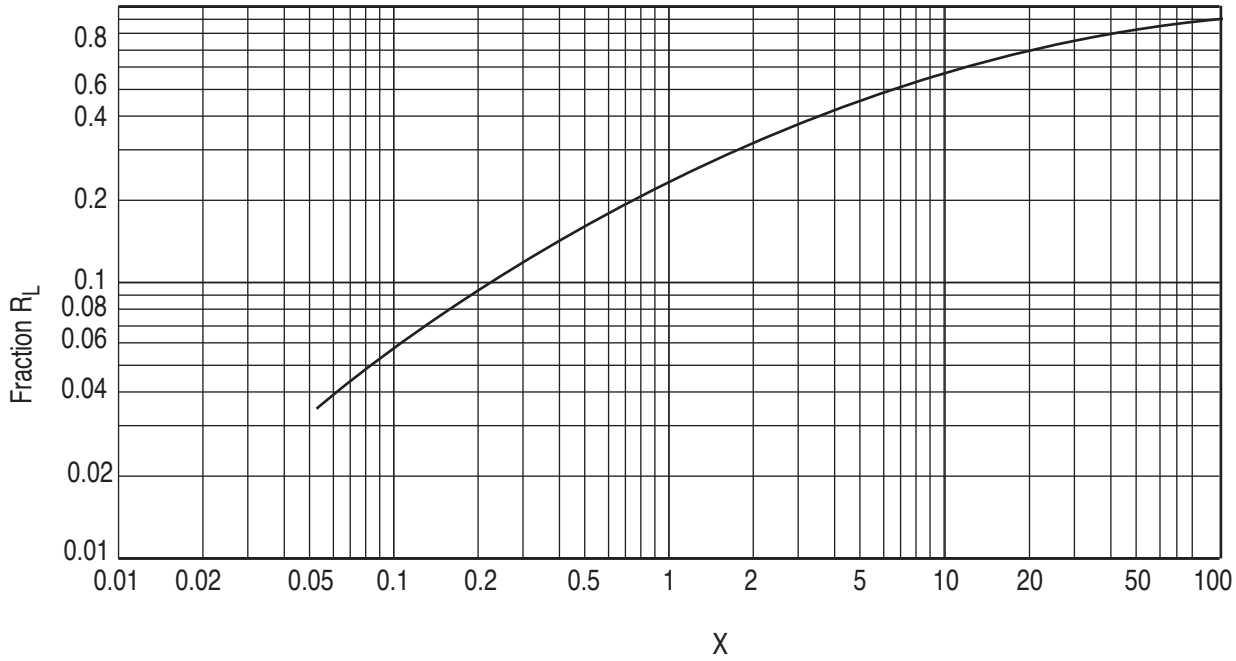
$$x = \sqrt{\frac{\Delta P_{fl}}{\Delta P_{fg}}}$$

$\Delta P_{\text{two phase pressure drop}} = Y_l \cdot \Delta P_{fl} = Y_g \cdot \Delta P_{fg}$

D T 1026 A

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LIQUID AND VAPOR HOLD-UP IN HORIZONTAL PIPE
— Lockhart and Martinelli correlation —



R_L volume fraction occupied by **liquid**

R_V volume fraction occupied by **vapor**

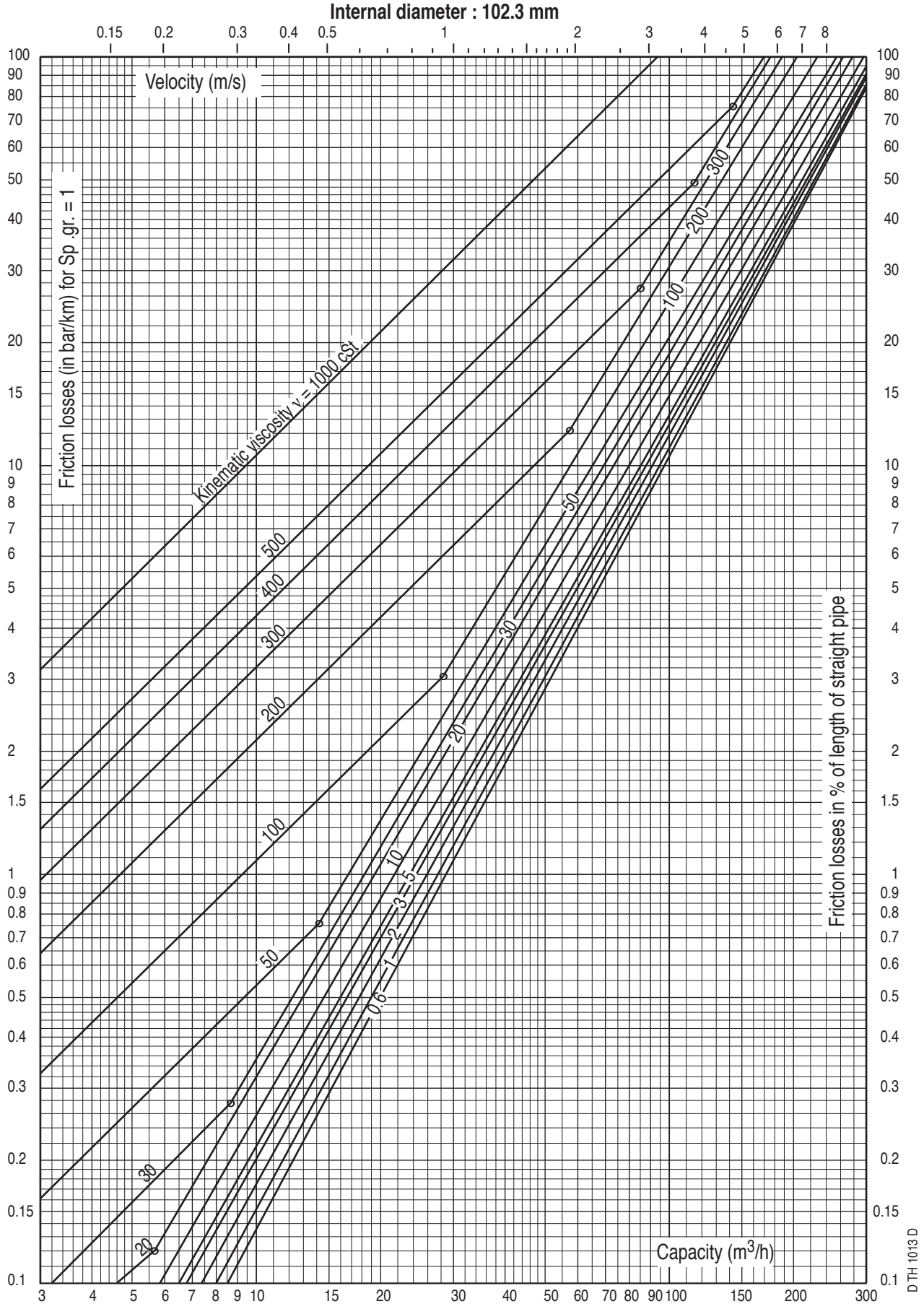
$$R_V = 1 - R_L$$

X Lockhart and Martinelli parameter

D T 1634 D

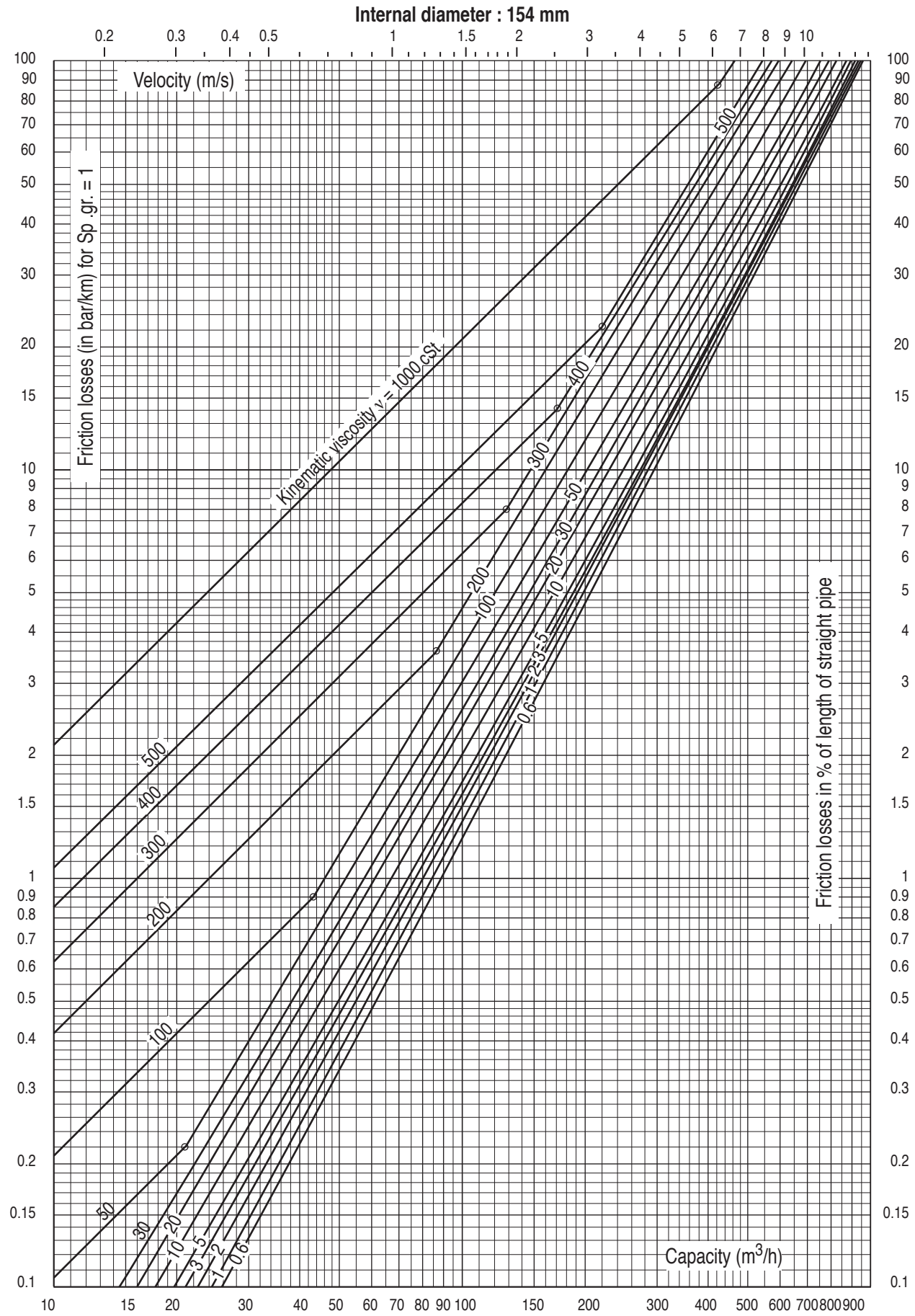
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FRICTION LOSSES FOR LIQUIDS IN 4" PIPES (AFTP)



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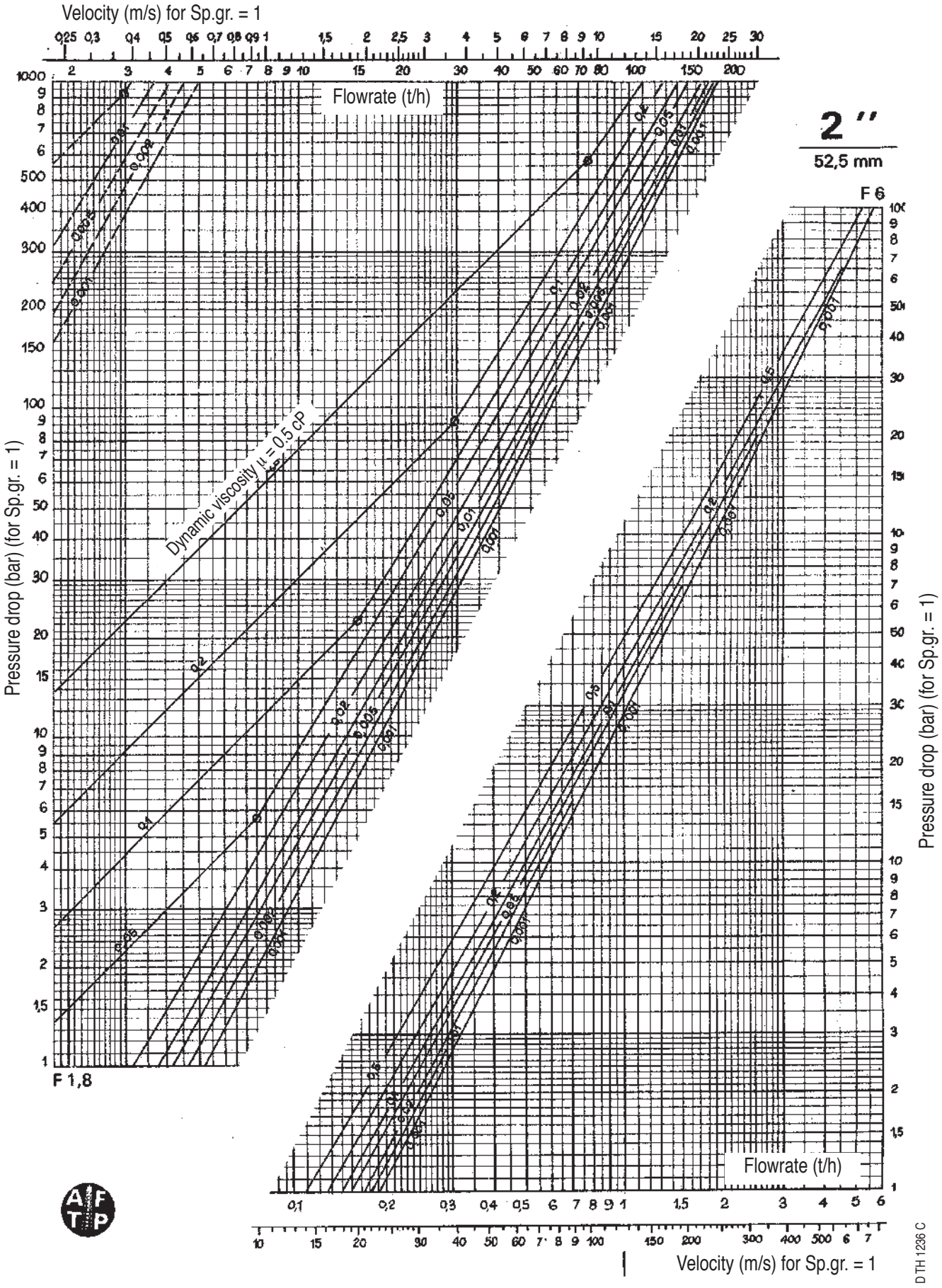
FRICTION LOSSES FOR LIQUIDS IN 6" PIPES (AFTP)



D TH 1014 D

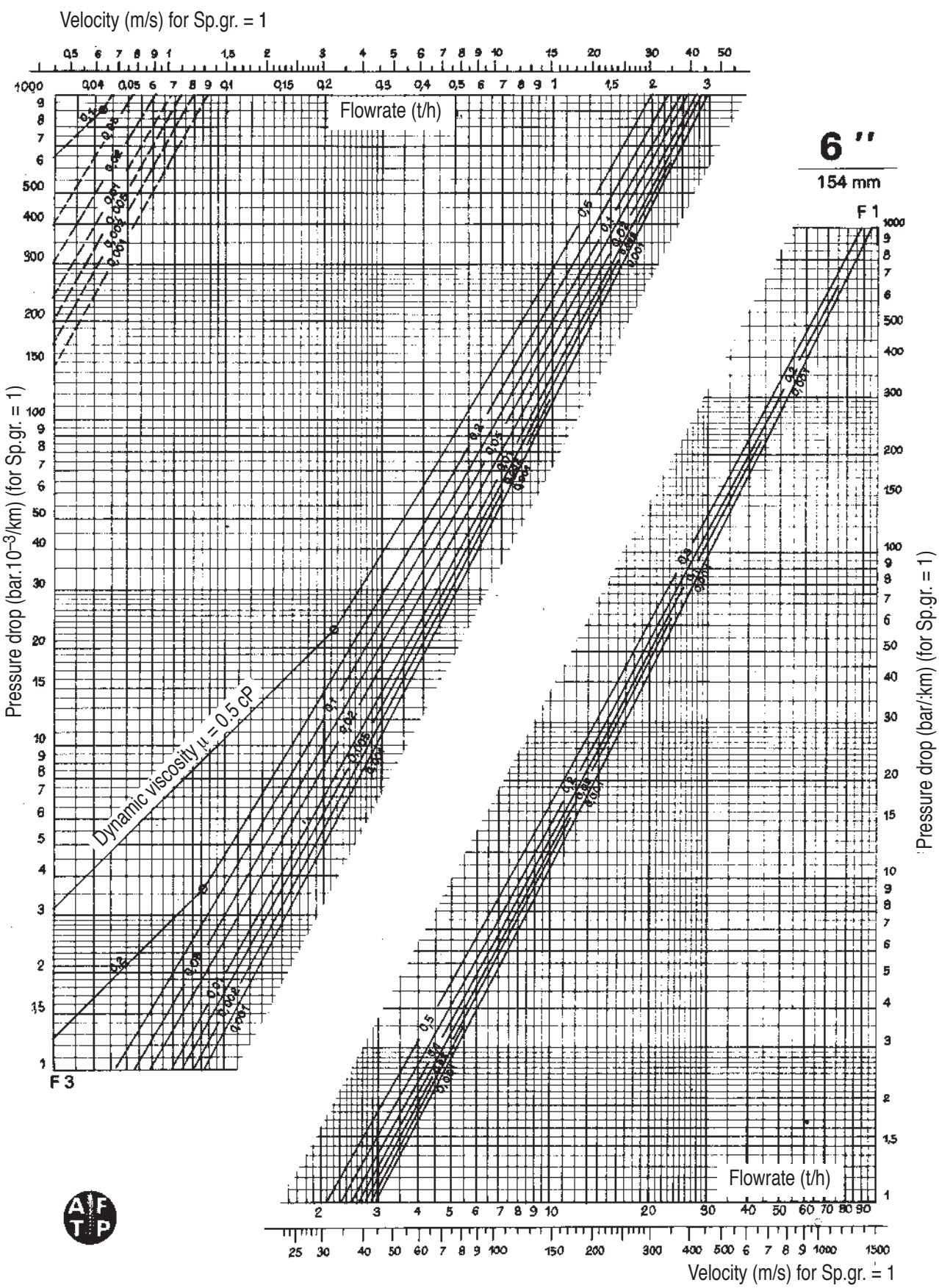
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PRESSURE DROPS FOR GASES IN 2" PIPES
— From AFTP —



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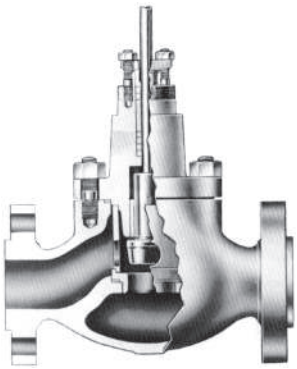
PRESSURE DROPS FOR GASES IN 6" PIPES
— From AFTP —



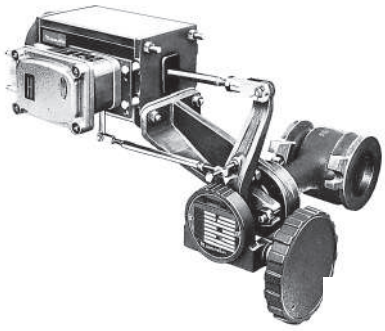
DTH 1222 D

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CONTROL VALVE
— Valve full open Cv —



Seat size (mm)		6.35							9.5	12.7	20.6	31.7	41.2	50.8	66.6	88.9	127.0
Valve size		Steam travel (mm)	Small capacity							Normal capacity	Large capacity						
mm	inches																
20	3/4	20.3	0.1	0.2	0.4	0.8	1.7	3.8	6.0	12	16						
25	1	20.3	0.1	0.2	0.4	0.8	1.7	3.8	6.0	12	16						
40	$1^{1/2}$	20.3					1.7	3.8	6.0	13	25	35					
50	2	20.3					1.7	3.8	6.0	15	26	46	55				
80	3	38.1									31	47	65	100			
100	4	38.1										49	66	113	195		
150	6	20.3											68	126	203	400	



Valve size		Seat size				C_V	
		Small size (0.4)		Normal size		Small size (0.4)	Normal size
mm	inches	mm	inches	mm	inches		
25	1	12.7	0.500	18.2	0.719	5.6	14
40	$1^{1/2}$	19.0	0.750	28.5	1.125	12.0	30
50	2	25.4	1.000	36.4	1.437	20.0	50
80	3	38.1	1.500	58.7	2.312	54.0	135
100	4	50.8	2.000	76.2	3.000	92.0	230
150	6	76.2	3.000	114.3	4.500	200.0	500
200	8			152.4	6.000		850
250	10			190.5	7.500		1300
300	12			234.9	9.250		1750

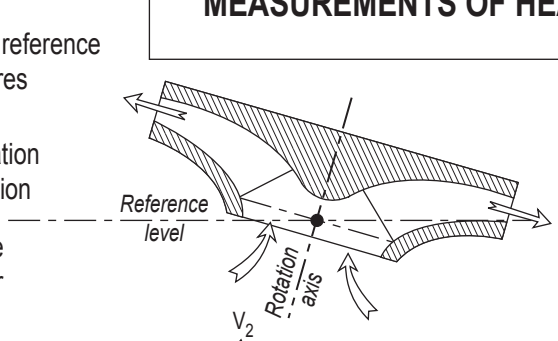
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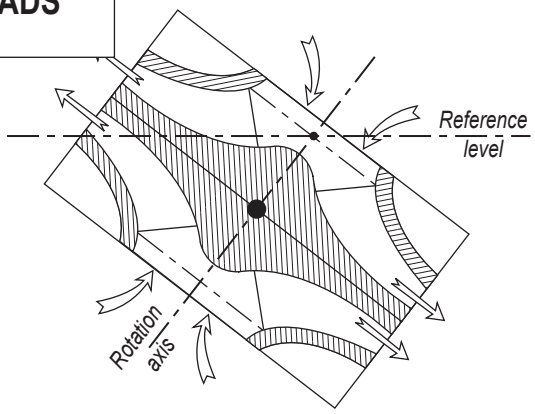
MEASUREMENTS OF HEADS

Horizontal plane is the reference level for pressures

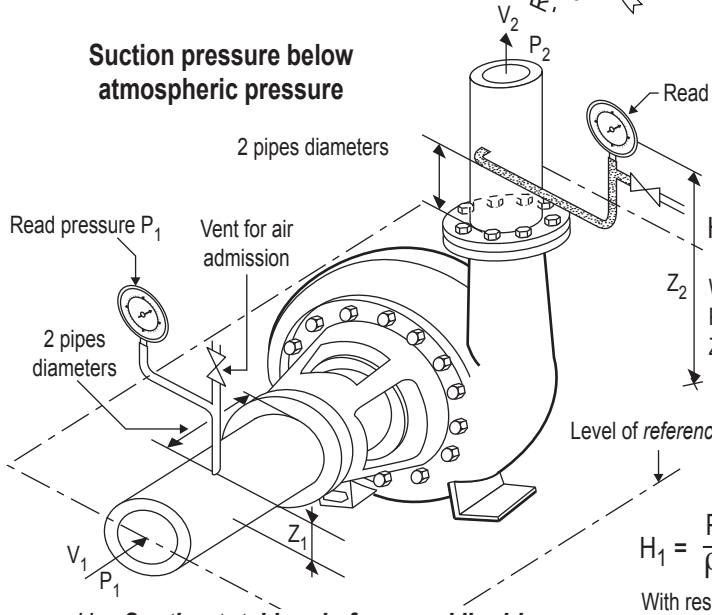
The axis of rotation is at any direction

Suction side first impeller





Suction pressure below atmospheric pressure



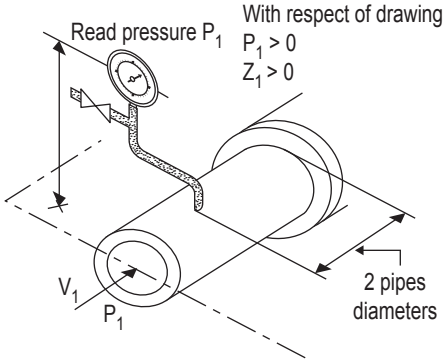
$$H_2 = \frac{P_2}{\rho_2 g} + Z_2 + \frac{V_2^2}{2g}$$

$$H_1 = \frac{P_1}{\rho g} + Z_1 + \frac{V_1^2}{2g}$$

With respect of drawing
$P_1 < 0$
$Z_1 > 0$

Suction pressure above atmospheric pressure

$$H_1 = \frac{P_1}{\rho_1 g} + Z_1 + \frac{V_1^2}{2g}$$



H_1 : ***Suction total head of pumped liquid***

H_2 : ***Discharge total head of pumped liquid***

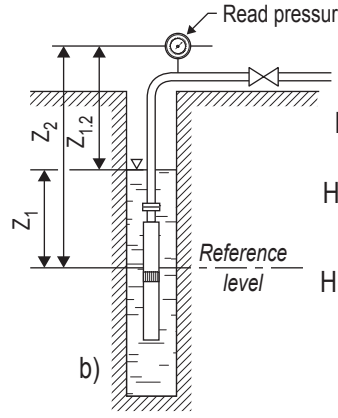
H : ***Total head of the pump***

$$H = Z_2 - Z_1 + \frac{P_2 - P_1}{\rho g} + \frac{V_2^2 - V_1^2}{2g}$$

Pratical formula with P in bar, v in m/s

$$H = Z_2 - Z_1 + \frac{10{,}2\,(P_2 - P_1)}{\text{sp.gr.}} + \frac{V_2^2 - V_1^2}{19.62}$$

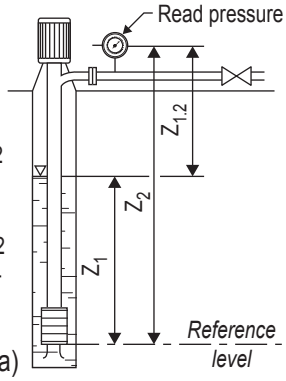
sp.gr : Specific gravity at pumping temperature



$$H_1 = Z_1$$

$$H_2 = \frac{P_2}{\rho g} + Z_2 + \frac{V_1^2}{2g}$$

$$H = \frac{P_2}{\rho g} + Z_{1.2} + \frac{V_2^2}{2g}$$



NPSHA : Net Positive Suction Head Available

$$NPSHA = H1 + \frac{P_b}{\rho g} - \frac{P_v}{\rho g}$$

with P_b Atmospheric pressure, P_v Vapor pressure (abs.), ρ density, g gravity

Pu = Hydraulic power $Pu = Q\rho gH = \frac{Q.H.sp\,gr}{367}$ (Pratical formula) with Q : m^3/h, H : m, Pu : kW

P = Shaft power $P = \frac{Q.H.sp\,gr}{367\,\eta}$

Correction law for a rotation speed variation of 20% around the specified rotation speed :

$$Q = Qsp\,\frac{N}{N_{sp}} \qquad H = Hsp\,(\frac{N}{N_{sp}})^2 \qquad P = Psp\,(\frac{N}{N_{sp}})^3$$

$\eta \cong \eta_{sp}$ $\qquad NPSHr = NPSHrsp\,(\frac{N}{N_{sp}})^2$

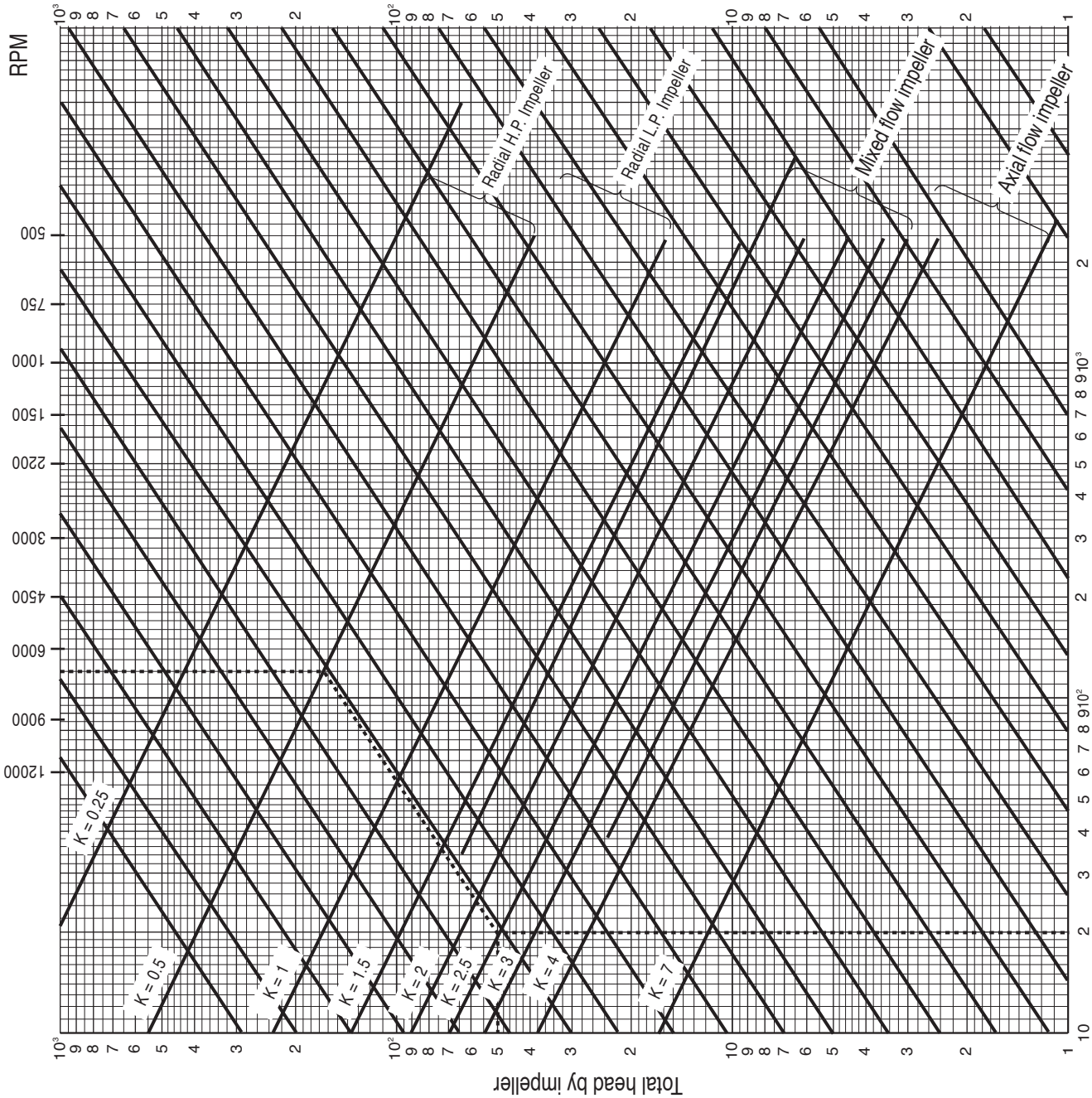
N : rotation speed (RPM)

η : efficency

D T 3237 D

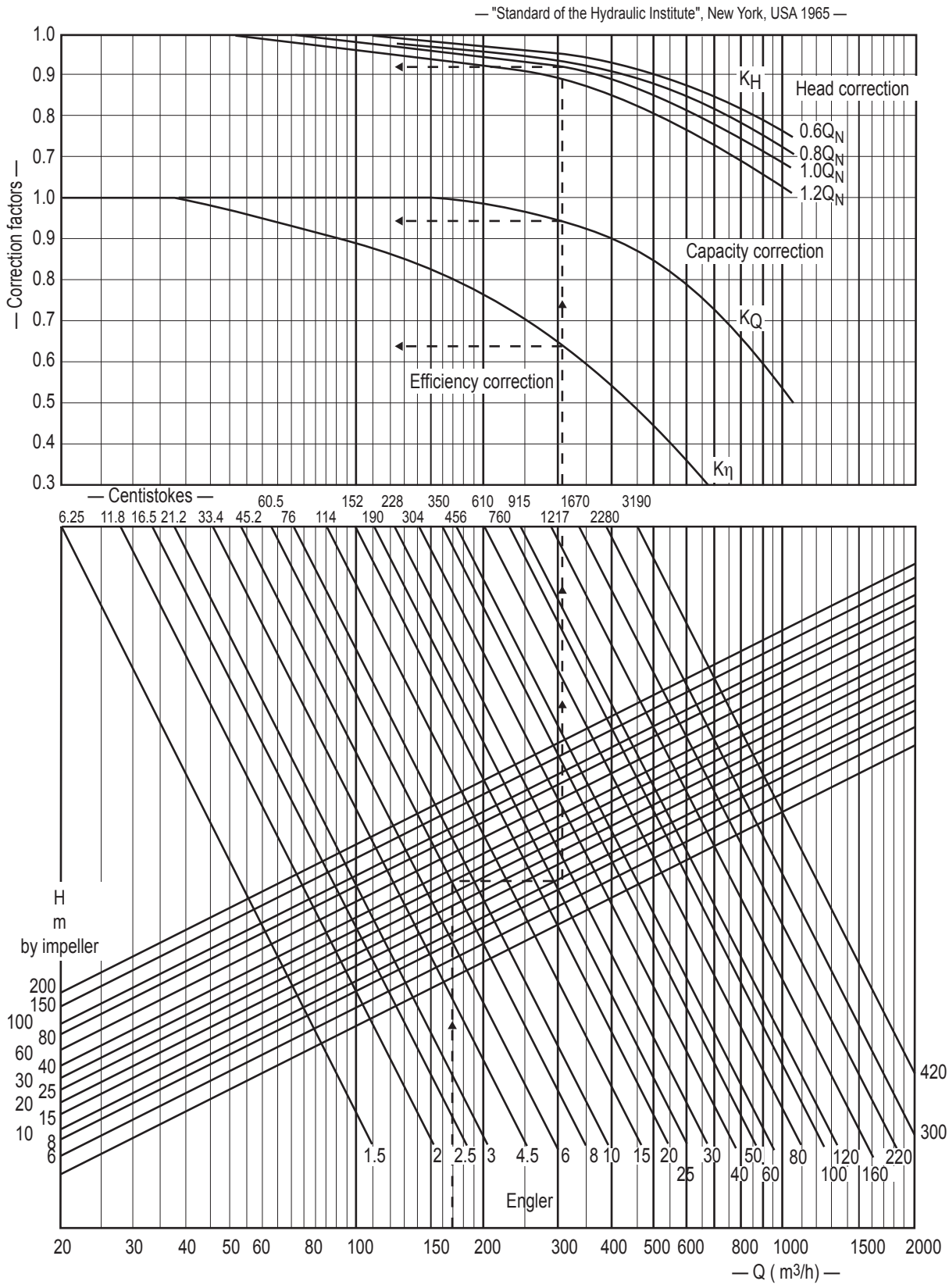
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TYPICAL IMPELLER SELECTION
— According to K number —



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PERFORMANCE CORRECTION CHART



D T 404 B

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CONVECTION HEAT TRANSFER COEFFICIENTS

$$\frac{kcal}{hr\ m^2\ °C}$$

Natural convection

Still air	10
40km/h wind	40
Furnace exhaust gases	50
Gaseous hydrocarbons (reheated in storage)	30* (μ = 1000 cP) 50* (μ = 100 cP)
(* multiply these values by 2 if a stirrer is used)	
Water	50 - 250 (depending on speed and temperature)

Forced convection

Laminar flow

Hydrocarbons	50 - 80 (μ > 10 cP) 200 (μ = 3 cP)
Water	150 - 400

Turbulent flow

Gaseous hydrocarbons (MW = 60)	1,700 (at 35 bar for ΔP = 0.5 bar) 500 (at 10 bar for ΔP = 0.2 bar) 200 (at 2 bar for ΔP = 0.06 bar)
Gaseous hydrocarbons (MW = 60) + 80%+ H	1,700 → 2,500 500 → 1,000 200 → 300
Gasoline	1,000 - 1,500
Water (40°C and 1m/s)	5,000

Condensation

Hydrocarbons	1,200 - 1,700
Water	5,000 - 10,000
Steam ($\Delta\theta$ = 10°C)	10,000
Steam + 1% air	5,000
+ 2% air	3,850

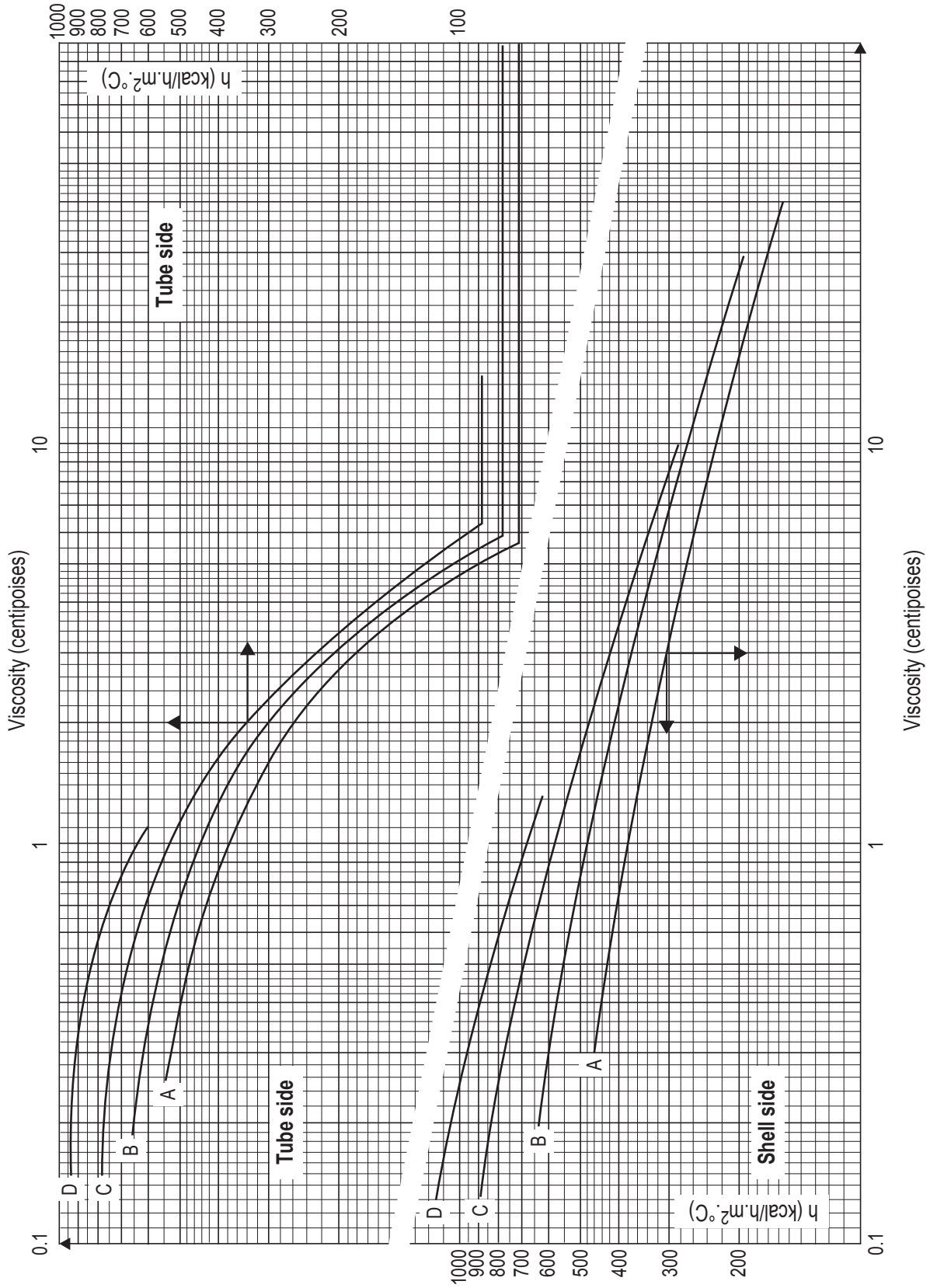
Vaporisation

Hydrocarbons	1,500 - 2,500
Water	5,000 - 10,000

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FILM COEFFICIENT

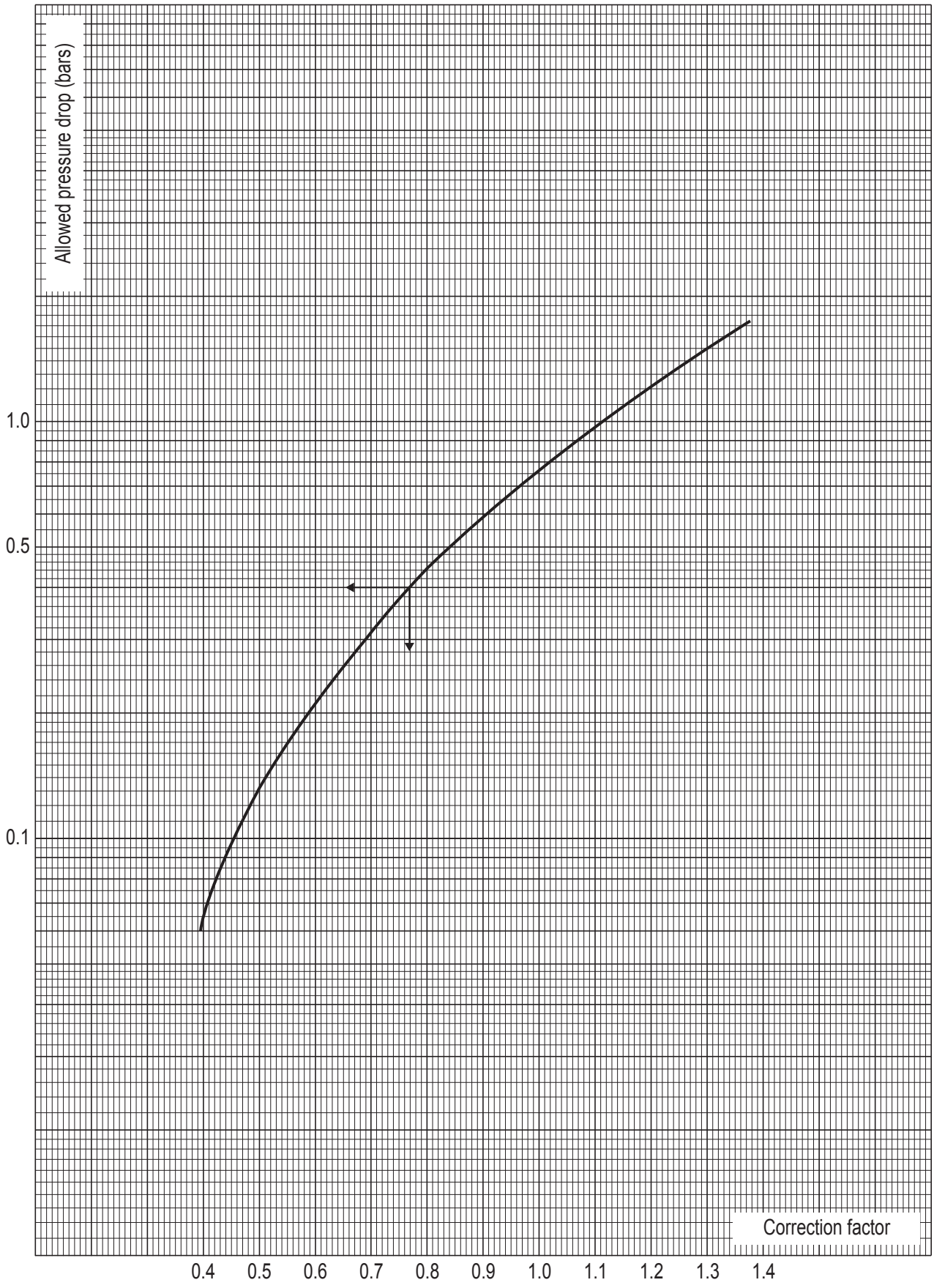
Group	Specific gravity	Products	Fouling resistance used for the estimation of coefficient
A	from 1.040 to 0.930	Asphalts - Residues	0.0010 (h.m^2.°C/kcal)
B	from 0.930 to 0.880	Heavy gas-oil	0.0006 (h.m^2.°C/kcal)
C	from 0.880 to 0.780	Ligh gas-oil - heavy naphtha	0.0004 (h.m^2.°C/kcal)
D	< 0.780	Gasoline and light products	0.0002 (h.m^2.°C/kcal)



D MTE 1321 D

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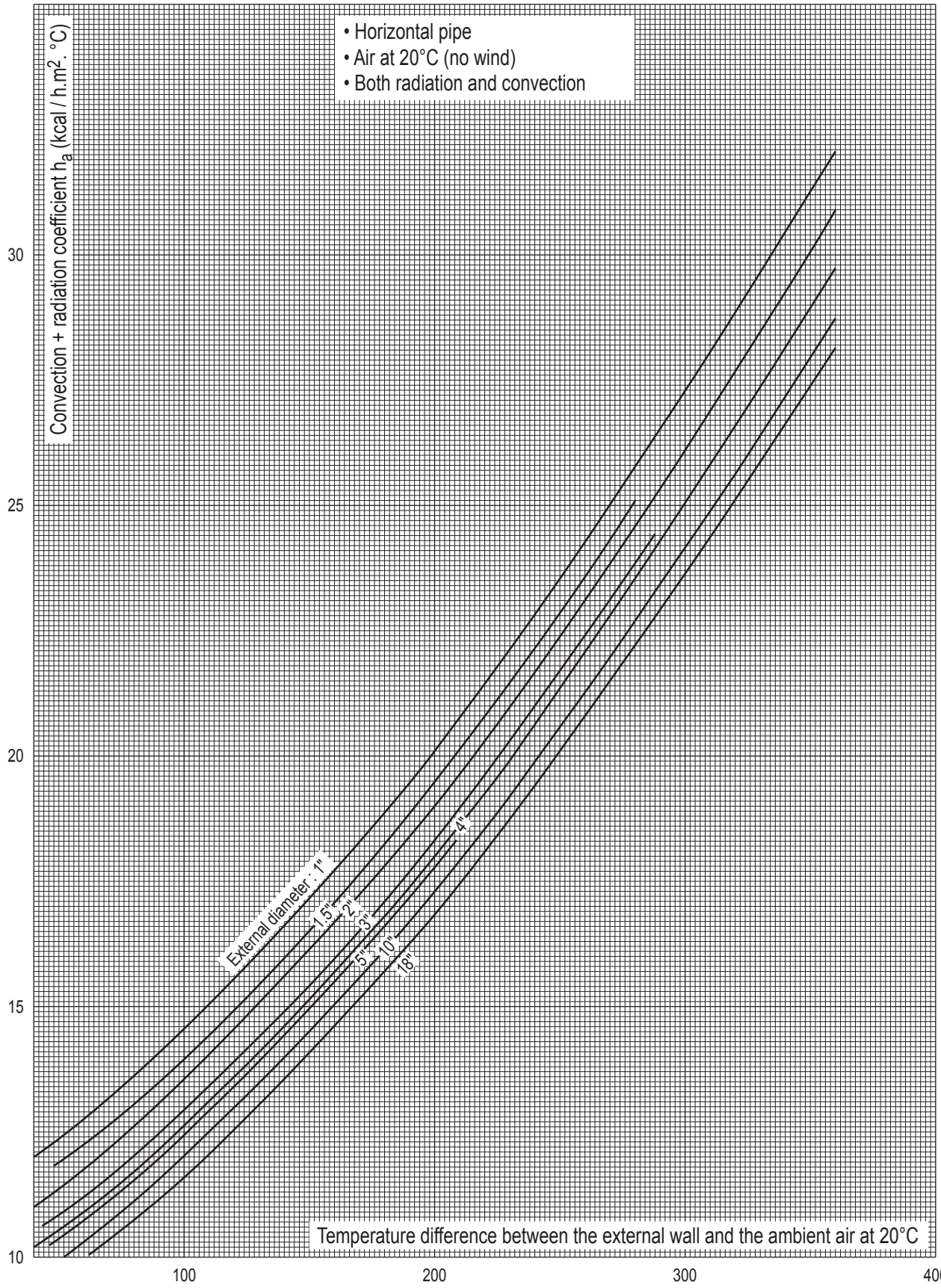
CORRECTION OF THE HEAT TRANSFER COEFFICIENT WITH THE ALLOWED PRESSURE DROP



D MTE 3077 D

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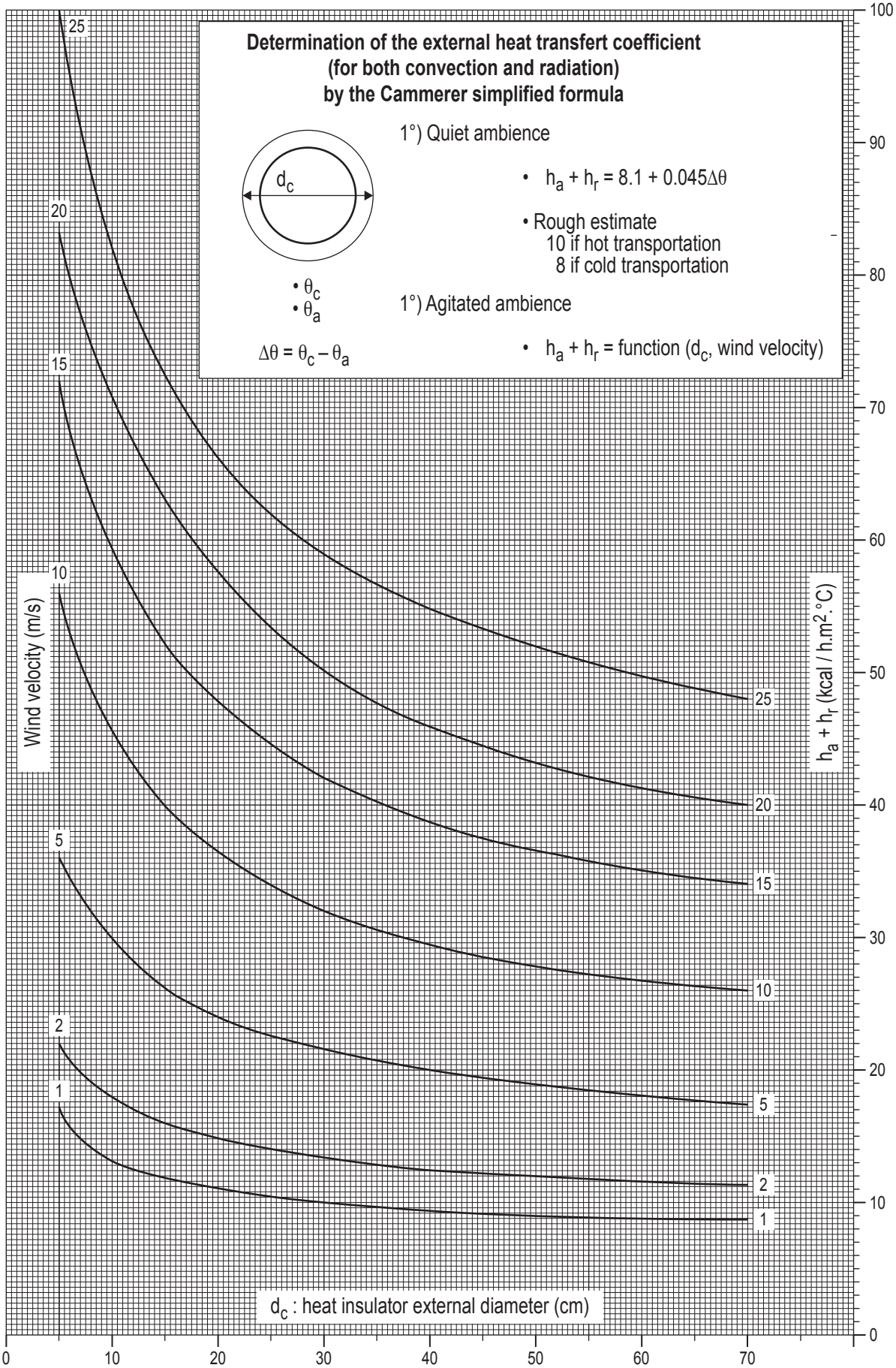
LOSSES DUE TO CONVECTION



D MTE 3078 D

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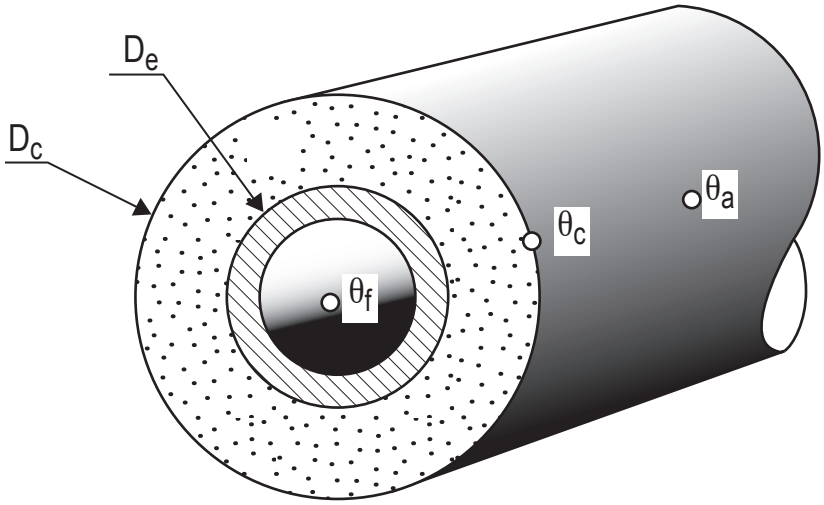
HEAT LOSSES FOR AN INSULATED PIPE



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HEAT INSULATOR : EXTERNAL SURFACE TEMPERATURE ESTIMATE

Insulation of a pipe for safety reasons



$$\theta_c - \theta_a = -\left(50 + \frac{10}{J.D_c}\right) + \sqrt{\left(50 + \frac{10}{J.D_c}\right)^2 + \frac{20\,(\theta_f - \theta_a)}{J.D_c}}$$

with $$J = \frac{1}{2\,\lambda_c} \cdot \ln\left(\frac{D_c}{D_e}\right)$$

θ_c : surface temperature of the heat insulator (°C)

θ_a : air ambiant temperature (°C)

θ_f :temperature of the fluid circulating in the pipe (°C)

λ_c: average thermal conductivity of the hest insulator (kcal/h.m.°C)

D_c : heat insulator diameter (m)

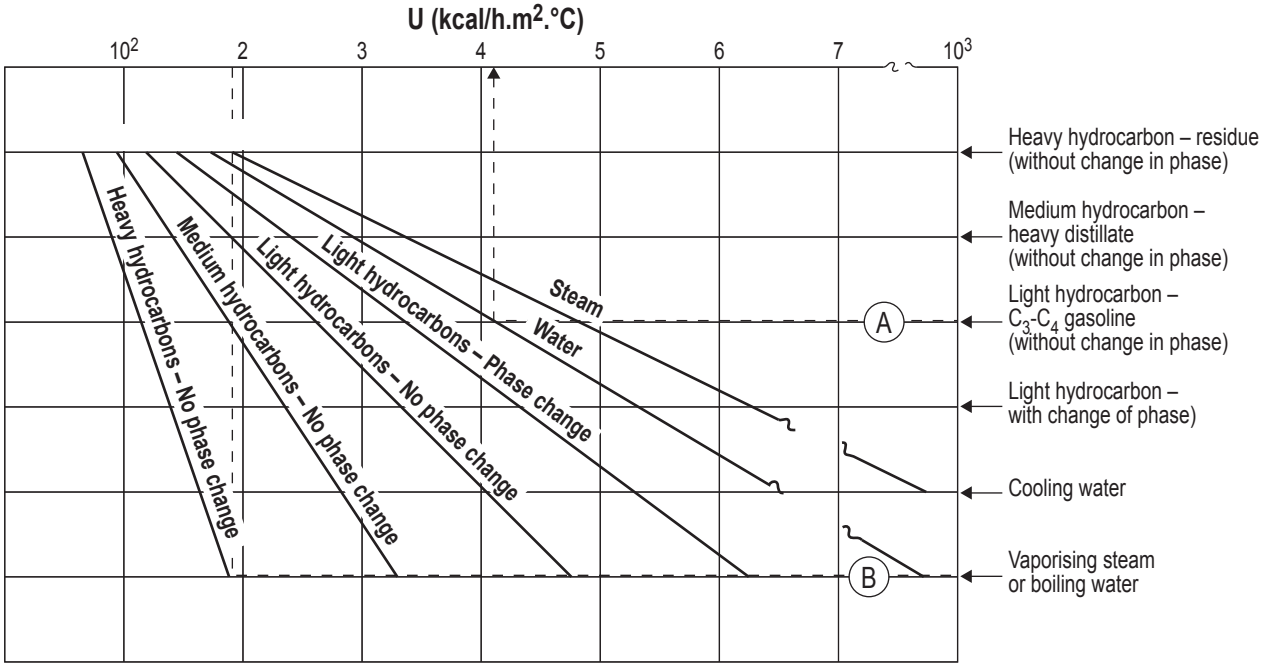
D_e : pipe external diameter (m)

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ESTIMATION OF THE OVERALL HEAT TRANSFER COEFFICIENT

FLUID		*U*	FLUID		*U*
Tube side	Shell side		Tube side	Shell side	
1. Cooler			**2. Condensers**		
Butadiene	Steam	60	Butadiene	Propylene (cooling)	330–390
Olefinic C_4	Propylene (vaporisation)	65–90	Olefinic C_4	Propylene	290–330
Gaseous ethylene	Condensate and steam	450-600	Ethylene	Propylene	300–450
Gaseous ethylene	Cold water	250–400	Light olefinics hydrocarbons	Propylene	250–300
Liquid ethylene	Gaseous ethylene	50–100	HCl	Propylene	300–550
Gazeous propane	Liquid propane	30–75	Light ends and chloroethanes	Propylene	75–125
Light olefinic hydrocarbons CO, CO_2, H_2	Steam	50–100	Olefinic, chlorinated hydrocarbons	Water	450–600
Light, chlorinated hydrocarbons	Steam	60–150	Chlorinated hydrocarbons	Water	100–150
Ethanol amine	Steam	75–125	Solvent & non–condens.	Water	125–75
Solvent	Propylene (vaporisation)	150–200	Water	Gaseous propylene	650–750
Solvent	Solvent	170–200	Water	Propylene	300–500
Solvent	Cold water	170–350	Water	Steam	600–1100
Oil	Oil	300–400	Feed water	Steam	100–150
Condensate	Propylene	300–600	Oil	Steam	350–550
25% calcium chloride	Chlorinated C_1	200–300	Cold water	Air–Cl_2 (partial condensation)	40–75
Steam	Air (mixture)	50–100	Water	Propylene (refrigeration andcondensation) (1)	
Steam	Styrene and tar	250–300	Water	Light hydrocarbons (refrigeration and condensation) (1)	200–450
Cold water	Freon 12	500–600	Water	Ammonia	700–800
Cold water	Transformation	200–350	Air–steam	Freon	50–250
Water	Feed water (35–45°C)	500–600	**3. Reboilers**		
Water	Feed water (100–35°C)	800–1100	Olefinic C_4	Steam	450–550
Water	Light, chlorinated C_2 hydrocarbons	30–50	Chlorinated hydrocarbons	Steam	170–120
Water	Heavy chlorinated C_2 hydrocarbons	220–150	Olefinic, chlorinated hydrocarbons	Steam	500–700
Water	Perchloroethylene	270–170	Dichloroethane	Steam	350–450
Water	Air–Cl_2	40–90	Heavy solvent	Steam	350–550
Water	HCl	35–75	Mono– and diethanol amine	Steam	750–1000
Water	Air and steam	100–170	Organic acids–water	Steam	300–500
Water	Absorbing oil	400–560	Amine–water	Steam	600–700
			Steam	Naphtha	75–100
			Propylene	Ethane–ethylene	600–700
			Propylene–butadiene	Butadiene–olefins	75–90

(1) Depending on the temperature range : 550–750 (–10 at – 5°C) ; 125–250 (0 at 7°C)



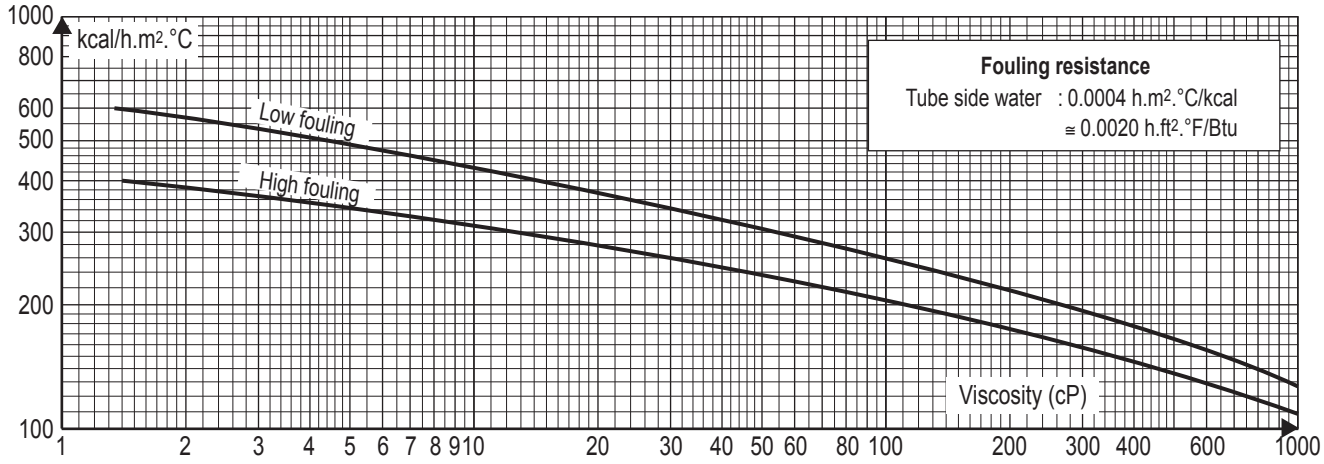
From "Manuel d'évaluation économique des procédés", A. CHAUVEL and B. LEFEBVRE

D MTE 3080 D

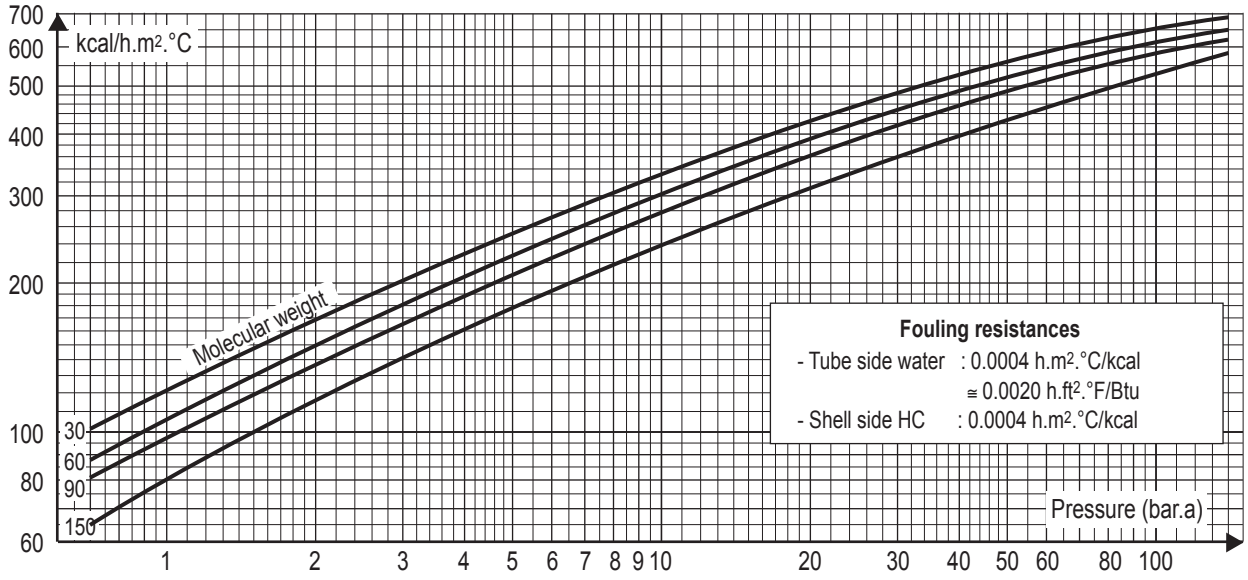
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ESTIMATION OF THE OVERALL HEAT TRANSFER COEFFICIENT
— Water - Hydrocarbons —

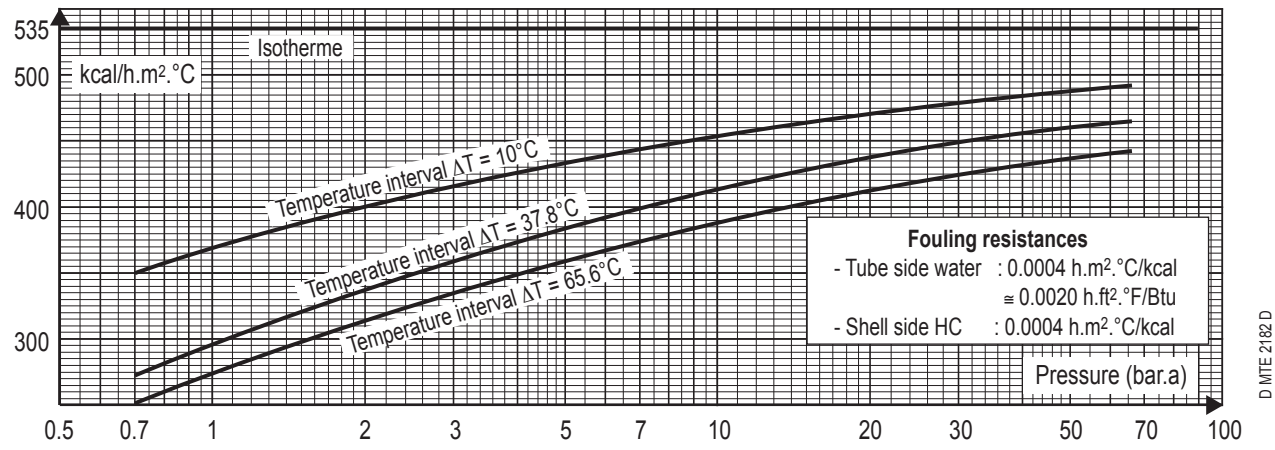
LIQUID HYDROCARBON COOLER



GASEOUS HYDROCARBON COOLER



TOTAL HYDROCARBON CONDENSER



D MTE 2182 D

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FOULING RESISTANCE

RGP-T-2 FOULING

RGP-T-2.1 TYPES OF FOULING

Currently five different types of fouling mechanisms are recognized. They are individually complex, often occurring simultaneously, and their effects may increase pressure drop, accelerate corrosion and decrease the overall heat transfer coefficient.

(1) Precipitation Fouling

Crystallization is one of the most common types of precipitation fouling. It occurs in many process streams, cooling water and chemical streams. Crystallization scale forms as the result of over-saturation of a relatively insoluble salt. The most common, calcium carbonate, forms on heat transfer surfaces as a result of the thermal decomposition of the bicarbonate ion and the subsequent reaction with calcium ions.

(2) Particulate Fouling

Sedimentation is the most common form of particulate fouling. Particles of clay, sand, silt, rust, etc. are initially suspended in the fluid and form deposits on the heat transfer surfaces. Sedimentation is frequently superimposed on crystallization and possibly acts as a catalyst for certain types of chemical reaction fouling.

(3) Chemical Reaction Fouling

Surface temperatures and the presence of oxidation promoters are known to significantly influence the rate of build up of this fouling type. Coking, the hard crust deposit of hydrocarbons formed on high temperature surfaces, is a common form of this type of fouling.

(4) Corrosion Fouling

Iron oxide, the most common form of corrosion product, is the result of an electro-chemical reaction and forms as a scale on iron-containing, exposed surfaces of the heat exchanger. This scale produces an added thermal resistance to the base metal of the heat transfer surface.

(5) Biological Fouling

Organic material growth develops on heat transfer surfaces in contact with untreated water such as sea, river, or lake water. In most cases, it will be combined or superimposed on other types of fouling such as crystallization and sedimentation. Biological growth such as algae, fungi, slime, and corrosive bacteria represent a potentially detrimental form of fouling. Often these micro-organisms provide a sticky holding medium for other types of fouling which would otherwise not adhere to clean surfaces.

RGP-T-2.2 EFFECT OF FOULING

There are different approaches to provide an allowance for anticipated fouling in the design of shell and tube heat exchangers. The net result is to provide added heat transfer surface area. This generally means that the exchanger is oversized for clean operation and barely adequate for conditions just before it should be cleaned. Although many heat exchangers operate for years without cleaning, it is more common that they must be cleaned periodically. Values of the fouling resistances to be specified are intended to reflect the values at the point in time just before the exchanger is to be cleaned. The major uncertainty is the assignment of realistic values of the fouling resistances. Further, these thermal resistances only address part of the impact of fouling as there is an increase in the hydraulic resistance as well; however, this is most often ignored. Fouling is complex, dynamic, and in time, degrades the performance of a heat exchanger.

The use of thermal resistance permits the assignment of the majority of the fouling to the side where fouling predominates. It also permits examination of the relative thermal resistance introduced by the different terms in the overall heat transfer coefficient equation. These can signal, to the designer, where there are potential design changes to reduce the effect of fouling. It also permits the determination of the amount of heat transfer surface area that has been assigned for fouling. Higher fouling resistances are sometimes inappropriately specified to provide safety factors to account for uncertainties in the heat transfer calculation, the actual operating conditions, and/or possible plant expansion. These uncertainties may well exist and should be reflected in the design, but they should not be masked in the fouling resistances. They should be clearly identified as appropriate factors in the design calculations.

From Doc TEMA

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FOULING RESISTANCE

Another inappropriate approach to heat exchanger design is to arbitrarily increase the heat transfer surface area to allow for fouling. This over-surfacing avoids the use of the appropriate fouling resistances. In effect, the fouling for the exchanger is combined and no longer can be identified as belonging to one side or the other.

In order to examine the effect of fouling on the pressure drop, it is necessary for the purchaser to supply the anticipated thicknesses of each of the fouling layers.

RGP-T-2.31 PHYSICAL CONSIDERATIONS

A) Properties Of Fluids And Usual Propensity For Fouling

The most important consideration is the fluid and the conditions when it produces fouling. At times, a process modification can result in conditions that are less likely to cause fouling.

B) Surface And Bulk Temperatures

For many kinds of fouling, as the temperatures increase, the amount of fouling increases. Lower temperatures produce slower fouling build-up and deposits that often are easier to remove.

C) Local Velocities

Normally, keeping the velocities high reduces the tendency to foul. Velocities on the tube side are limited by erosion, and on the shell side by flow-induced vibration. Stagnant and recirculation regions on the shell side lead to heavy fouling.

D) Tube Material, Configuration And Surface Finish

The selection of tube material is significant when it comes to corrosion. Some kinds of biological fouling can be lessened by copper-bearing tube materials. There can be differences between finned and plain tubing. Surface finish has been shown to influence the rate of fouling and the ease of cleaning.

E) Heat Exchanger Geometry And Orientation

The geometry of a particular heat exchanger can influence the uniformity of the flows on the tube side and the shell side. The ease of cleaning can be greatly influenced by the orientation of the heat exchanger.

F) Heat Transfer Process

The fouling resistances for the same fluid can be considerably different depending upon whether heat is being transferred through sensible heating or cooling, boiling, or condensing.

G) Fluid Purity And Freedom From Contamination

Most fluids are prone to have inherent impurities that can deposit out as a fouling layer, or act as catalysts to the fouling processes. It is often economically attractive to eliminate the fouling constituents by filters.

H) Fluid Treatment To Prevent Corrosion And Biological Growth

Fluid treatment is commonly carried out to prevent corrosion and/or biological growth. If these treatments are neglected, rapid fouling can occur.

I) Fluid Treatment To Reduce Fouling

There are additives that can disperse the fouling material so it does not deposit. Additives may also alter the structure of the fouling layers that deposit so that they are easily removed. The use of these treatments is a product quality and economic decision.

J) Cathodic Protection

One of the effective ways to reduce the possibility of corrosion and corrosion fouling is to provide cathodic protection in the design.

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K) Planned Cleaning Method And Desired Frequency

It is important that the cleaning method be planned at the design stage of the heat exchanger. Considerations in design involving cleaning are whether it will be done on-line, off-line, bundle removed or in place, whether it will involve corrosive fluids, etc.. Access, clearances, valving, and piping also must be considered to permit ease of cleaning. The cleaning method may require special safety requirements, which should be incorporated in the design.

L) Place The More Fouling Fluid On The Tube Side

There are two benefits from placing the more fouling fluid on the tube side. There is less danger of low velocity or stagnant flow regions on the tube side, and, it is generally easier to clean the tube side than the shell side. It is often possible to clean the tube side with the exchanger in place while it may be necessary to remove the bundle to clean the shell side.

RGP-T-2.32 ECONOMIC CONSIDERATIONS

Planned fouling prevention, maintenance and cleaning make possible lower allowances for fouling, but do involve a commitment to ongoing costs. The amount and frequency of cleaning varies considerably with user and operation.

The most significant parameters involved in deciding upon the amount of fouling allowance that should be provided are the operational and economic factors that change with time. New fluid treatments, changing first costs and operating costs, different cleaning procedures and the degree of payback for longer periods of being on stream should be some of the items evaluated in determining an appropriate fouling resistance. Failure to include the economic considerations may lead to unnecessary monetary penalties for fouling.

Companies concerned about fouling continually monitor the performance of their heat exchangers to establish fouling experience and develop their own guidelines for determining the appropriate fouling resistance to specify when purchasing new equipment.

Almost every source of cooling water needs to be treated before it is used for heat exchanger service. The treatment ranges from simple biocide addition to control biological fouling, to substantial treatment of brackish water to render it suitable for use. The amount of treatment may be uneconomical and substitute sources of cooling must be sought. With today's technology, the quality of water can be improved to the point that fouling should be under control as long as flow velocities are maintained and surface temperatures controlled.

RGP-T-2.4 DESIGN FOULING RESISTANCES (HR FT^2 ° F/Btu)

The purchaser should attempt to select an optimal fouling resistance that will result in a minimum sum of fixed, shutdown and cleaning costs. The following tabulated values of fouling resistances allow for oversizing the heat exchanger so that it will meet performance requirements with reasonable intervals between shutdowns and cleaning. These values do not recognize the time related behavior of fouling with regard to specific design and operational characteristics of particular heat exchangers.

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$$1 \ \frac{hr.ft^2.°F}{Btu} = 0.2049 \qquad \frac{h.m^2.°C}{kcal} = 0.1763 \ \frac{m^2.K}{W}$$

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FOULING RESISTANCE

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$1 \frac{hr.ft^2.°F}{Btu} = 0.2049 \quad \frac{h.m^2.°C}{kcal} = 0.1763 \frac{m^2.K}{W}$

Fouling Resistances For Industrial Fluids

Oils:	
Fuel Oil #2	0.002
Fuel Oil #6	0.005
Transformer Oil	0.001
Engine Lube Oil	0.001
Quench Oil	0.004
Gases And Vapors:	
Manufactured Gas	0.010
Engine Exhaust Gas	0.010
Steam (Non-Oil Bearing)	0.0005
Exhaust Steam (Oil Bearing)	0.0015-0.002
Refrigerant Vapors (Oil Bearing)	0.002
Compressed Air	0.001
Ammonia Vapor	0.001
CO_2 Vapor	0.001
Chlorine Vapor	0.002
Coal Flue Gas	0.010
Natural Gas Flue Gas	0.005
Liquids:	
Molten Heat Transfer Salts	0.0005
Refrigerant Liquids	0.001
Hydraulic Fluid	0.001
Industrial Organic Heat Transfer Media	0.002
Ammonia Liquid	0.001
Ammonia Liquid (Oil Bearing)	0.003
Calcium Chloride Solutions	0.003
Sodium Chloride Solutions	0.003
CO_2 Liquid	0.001
Chlorine Liquid	0.002
Methanol Solutions	0.002
Ethanol Solutions	0.002
Ethylene Glycol Solutions	0.002

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FOULING RESISTANCE

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$1\ \frac{hr.ft^2.°F}{Btu} = 0.2049 \qquad \frac{h.m^2.°C}{kcal} = 0.1763\ \frac{m^2.K}{W}$

Fouling Resistances For Chemical Processing Streams

Gases And Vapors:	
Acid Gases	0.002-0.003
Solvent Vapors	0.001
Stable Overhead Products	0.001
Liquids:	
MEA And DEA Solutions	0.002
DEG And TEG Solutions	0.002
Stable Side Draw And Bottom Product	0.001-0.002
Caustic Solutions	0.002
Vegetable Oils	0.003

Fouling Resistances For Natural Gas-Gasoline Processing Streams

Gases And Vapors:	
Natural Gas	0.001-0.002
Overhead Products	0.001-0.002
Liquids:	
Lean Oil	0.002
Rich Oil	0.001-0.002
Natural Gasoline And Liquified Petroleum Gases	0.001-0.002

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FOULING RESISTANCE

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$1\ \frac{hr.ft^2.°F}{Btu} = 0.2049 \qquad \frac{h.m^2.°C}{kcal} = 0.1763\ \frac{m^2.K}{W}$

Fouling Resistances For Oil Refinery Streams

Crude And Vacuum Unit Gases And Vapors:						
Atmospheric Tower Overhead Vapors						0.001
Light Naphthas						0.001
Vacuum Overhead Vapors						0.002
Crude And Vacuum Liquids:						
Crude Oil						
	0 to 250 ° F VELOCITY FT/SEC			250 to 350 ° F VELOCITY FT/SEC		
	<2	2-4	>4	<2	2-4	>4
DRY	0.003	0.002	0.002	0.003	0.002	0.002
SALT*	0.003	0.002	0.002	0.005	0.004	0.004
	350 to 450 ° F VELOCITY FT/SEC			450 ° F and over VELOCITY FT/SEC		
	<2	2-4	>4	<2	2-4	>4
DRY	0.004	0.003	0.003	0.005	0.004	0.004
SALT*	0.006	0.005	0.005	0.007	0.006	0.006
*Assumes desalting @ approx. 250 ° F						
Gasoline						0.002
Naphtha And Light Distillates						0.002-0.003
Kerosene						0.002-0.003
Light Gas Oil						0.002-0.003
Heavy Gas Oil						0.003-0.005
Heavy Fuel Oils						0.005-0.007
Asphalt And Residuum:						
Vacuum Tower Bottoms						0.010
Atmosphere Tower Bottoms						0.007
Cracking And Coking Unit Streams:						
Overhead Vapors						0.002
Light Cycle Oil						0.002-0.003
Heavy Cycle Oil						0.003-0.004
Light Coker Gas Oil						0.003-0.004
Heavy Coker Gas Oil						0.004-0.005
Bottoms Slurry Oil (4.5 Ft/Sec Minimum)						0.003
Light Liquid Products						0.002

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FOULING RESISTANCE

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$1 \frac{hr.ft^2.°F}{Btu} = 0.2049 \qquad \frac{h.m^2.°C}{kcal} = 0.1763 \frac{m^2.K}{W}$

Fouling Resistances For Oil Refinery Streams- continued

Catalytic Reforming, Hydrocracking And Hydrodesulfurization Streams:	
Reformer Charge	0.0015
Reformer Effluent	0.0015
Hydrocracker Charge And Effluent*	0.002
Recycle Gas	0.001
Hydrodesulfurization Charge And Effluent*	0.002
Overhead Vapors	0.001
Liquid Product Over 50 ° A.P.I.	0.001
Liquid Product 30 - 50 ° A.P.I.	0.002
*Depending on charge, characteristics and storage history, charge resistance may be many times this value.	
Light Ends Processing Streams:	
Overhead Vapors And Gases	0.001
Liquid Products	0.001
Absorption Oils	0.002-0.003
Alkylation Trace Acid Streams	0.002
Reboiler Streams	0.002-0.003
Lube Oil Processing Streams:	
Feed Stock	0.002
Solvent Feed Mix	0.002
Solvent	0.001
Extract*	0.003
Raffinate	0.001
Asphalt	0.005
Wax Slurries*	0.003
Refined Lube Oil	0.001
*Precautions must be taken to prevent wax deposition on cold tube walls.	
Visbreaker:	
Overhead Vapor	0.003
Visbreaker Bottoms	0.010
Naphtha Hydrotreater:	
Feed	0.003
Effluent	0.002
Naphthas	0.002
Overhead Vapors	0.0015

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FOULING RESISTANCE

Fouling resistance in $\frac{hr.ft^2.°F}{Btu}$

$$1\ \frac{hr.ft^2.°F}{Btu} = 0.2049 \qquad \frac{h.m^2.°C}{kcal} = 0.1763\ \frac{m^2.K}{W}$$

Fouling Resistances for Oil Refinery Streams - continued

Catalytic Hydro Desulfurizer:	
Charge	0.004-0.005
Effluent	0.002
H.T. Sep. Overhead	0.002
Stripper Charge	0.003
Liquid Products	0.002
HF Alky Unit:	
Alkylate, Deprop. Bottoms, Main Fract. Overhead Main Fract. Feed	0.003
All Other Process Streams	0.002

Fouling Resistances For Water

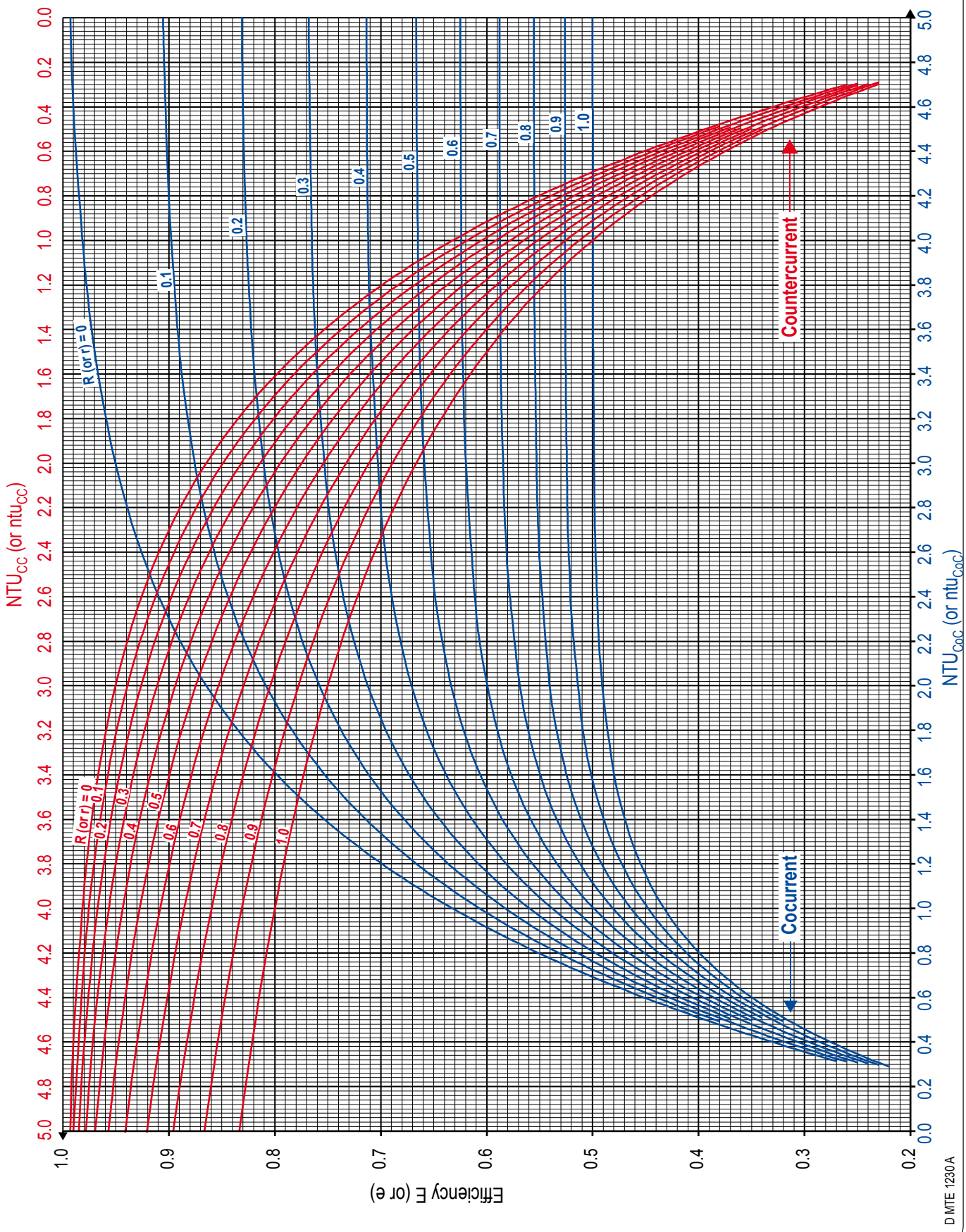
Temperature Of Heating Medium	Up To 240° F		240 to 400° F	
Temperature Of Water	125 ° F		Over 125° F	
	Water Velocity Ft/Sec		Water Velocity Ft/Sec	
	3 and Less	Over 3	3 and Less	Over 3
Sea Water	0.0005	0.0005	0.001	0.001
Brackish Water	0.002	0.001	0.003	0.002
Cooling Tower And Artificial Spray Pond:				
Treated Make Up	0.001	0.001	0.002	0.002
Untreated	0.003	0.003	0.005	0.004
City Or Well Water	0.001	0.001	0.002	0.002
River Water:				
Minimum	0.002	0.001	0.003	0.002
Average	0.003	0.002	0.004	0.003
Muddy Or Silty	0.003	0.002	0.004	0.003
Hard (Over 15 Grains/Gal.)	0.003	0.003	0.005	0.005
Engine Jacket	0.001	0.001	0.001	0.001
Distilled Or Closed Cycle				
Condensate	0.0005	0.0005	0.0005	0.0005
Treated Boiler Feedwater	0.001	0.0005	0.001	0.001
Boiler Blowdown	0.002	0.002	0.002	0.002

If the heating medium temperature is over 400 ° F and the cooling medium is known to scale, these ratings should be modified accordingly.

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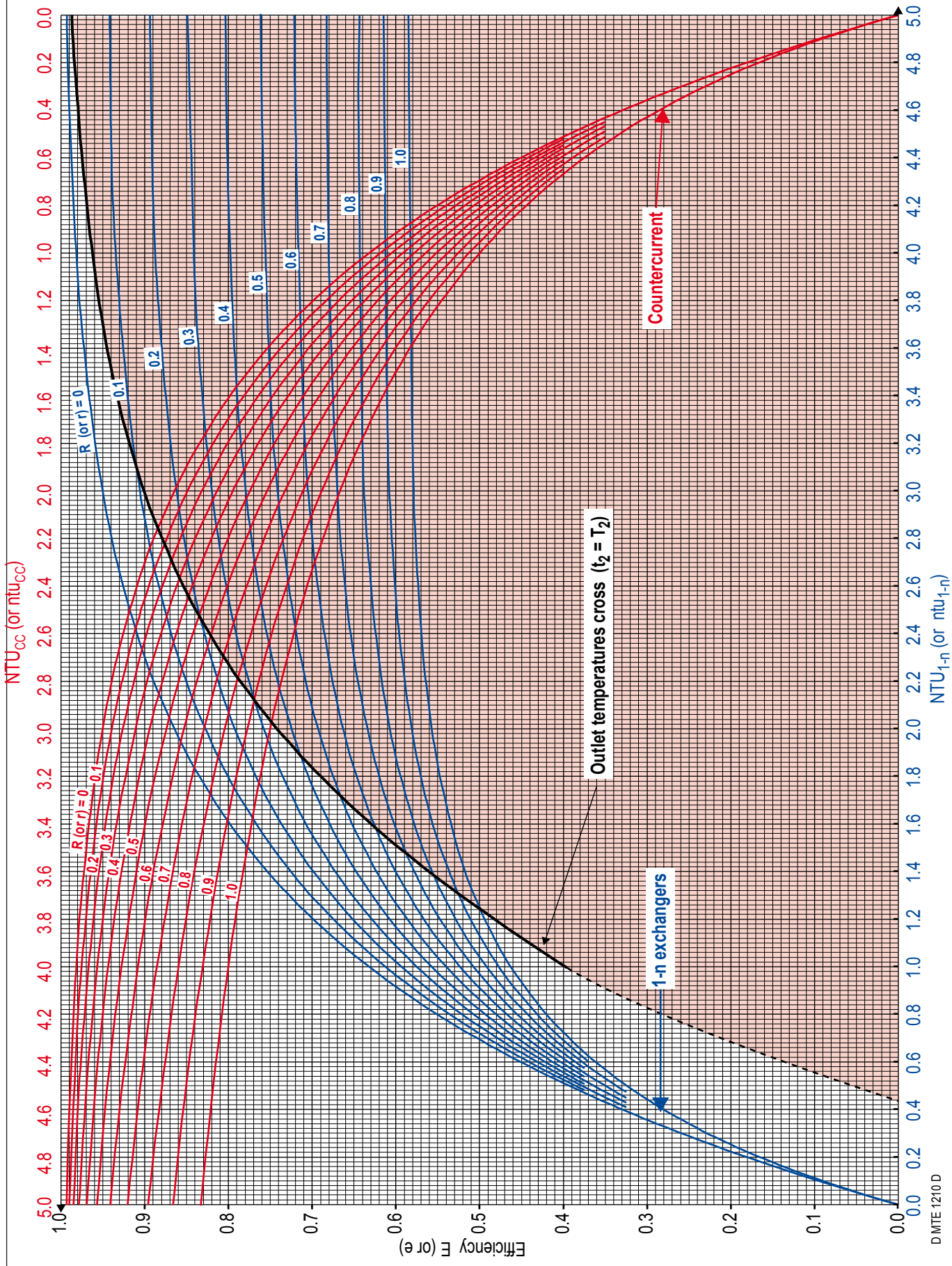
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HEAT EXCHANGER EFFICIENCY
— Cocurrent and Countercurrent —



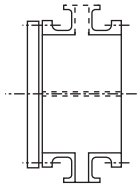
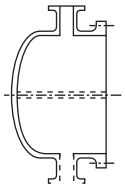
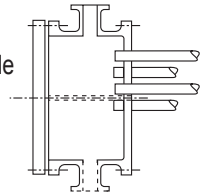
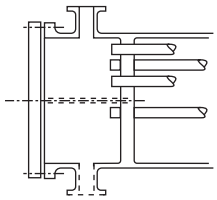
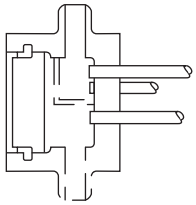
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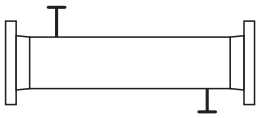
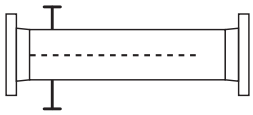
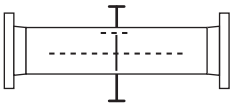
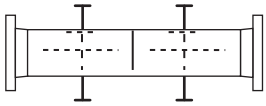
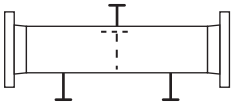
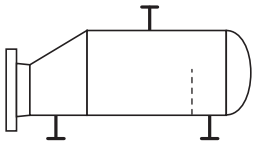
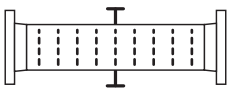
HEAT EXCHANGER EFFICIENCY
— Countercurrent and 1-n exchangers —

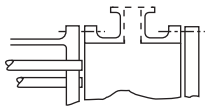
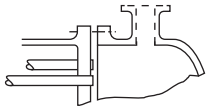
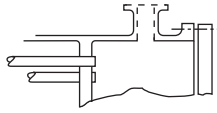
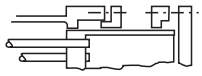
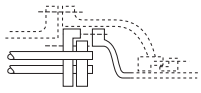
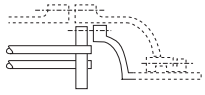
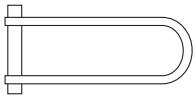
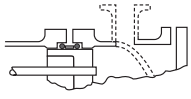


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HEAT EXHANGER NOMENCLATURE
— TEMA standards —

	Front end stationary head types
A	 Channel and removable cover
B	 Bonnet (integral cover)
C	Removable tube bundle only  Channel integral with tubesheet and removable cover
N	 Channel integral with tubesheet and removable cover
D	 Special high pressure closure

	Shell types
E	 One pass shell
F	 Two pass shell with longitudinal baffle
G	 Split flow
H	 Double split flow
J	 Divided flow
K	 Kettle type reboiler
X	 Cross flow

	Rear end head types
L	 Fixed tubesheet like "A" stationary head
M	 Fixed tubesheet like "B" stationary head
N	 Fixed tubesheet like "N" stationary head
P	 Outside packed floating head
S	 Floating head with backing device
T	 Pull through floating head
U	 U-tube bundle
W	 Externally sealed floating tubesheet

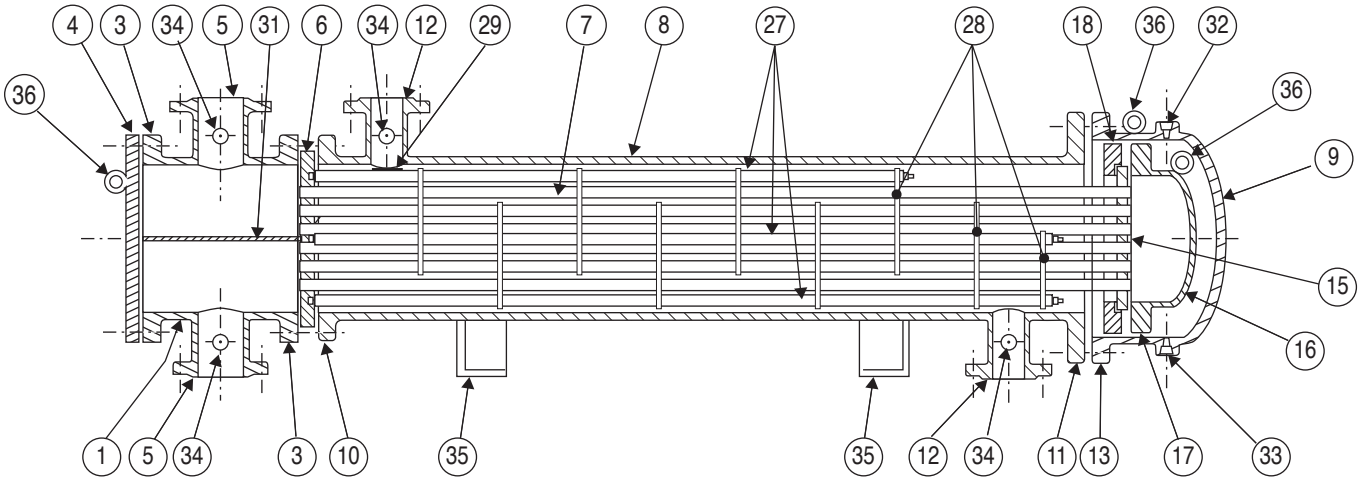
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DIFFERENT HEAT EXCHANGES
— From table TEMA —

For the purpose of establishing standard terminology, Figure illustrates various types of heat exchangers. Typical parts and connections, for illustrative purposes only, are numbered for identification in Table

1. Stationary Head-Channel
2. Stationary Head-Bonnet
3. Stationary Head Flange-Channel or Bonnet
4. Channel cover
5. Stationary Head Nozzle
6. Stationary Tubesheet
7. Tubes
8. Shell
9. Shell Cover
10. Shell Flange-Stationary Head End
11. Shell Flange-Rear Head End
12. Shell Nozzle
13. Shell Cover Flange
14. Expansion Joint
15. Floating Tubesheet
16. Floating Head Cover
17. Floating Head Cover Flange
18. Floating Head Backing Device
19. Split Shear Ring
20. Slip-on Backing Flange
21. Floating Head Cover-External
22. Floating Tubesheet Skirt
23. Packing Box
24. Packing
25. Packing Gland
26. Lantern Ring
27. Tierods and Spacers
28. Transverse Baffles or Support Plates
29. Impingement Plate
30. Longitudinal Baffle
31. Pass Partition
32. Vent Connection
33. Drain Connection
34. Instrument Connection
35. Support Saddle
36. Lifting Lug
37. Support Bracket
38. Weir
39. Liquid Level Connection

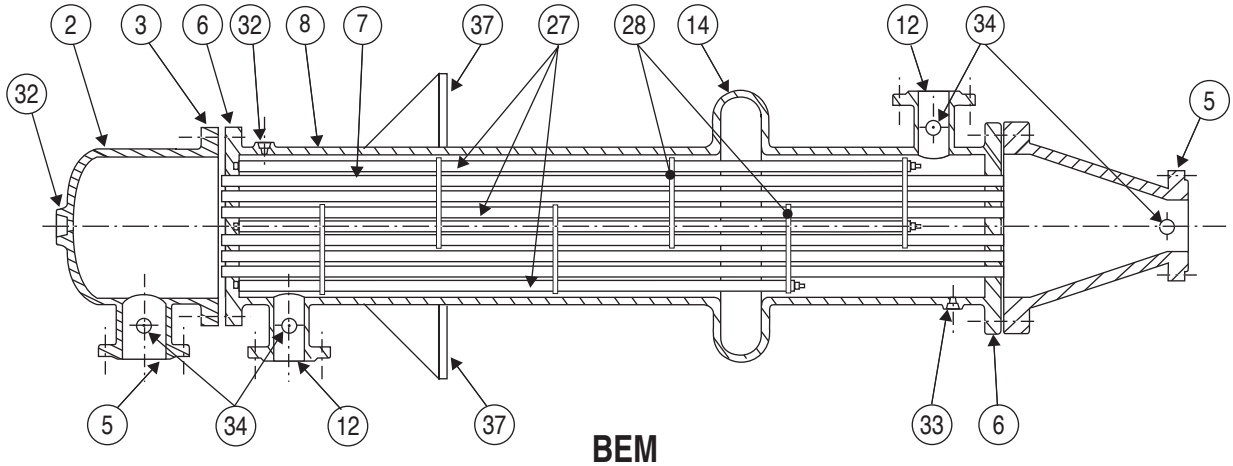


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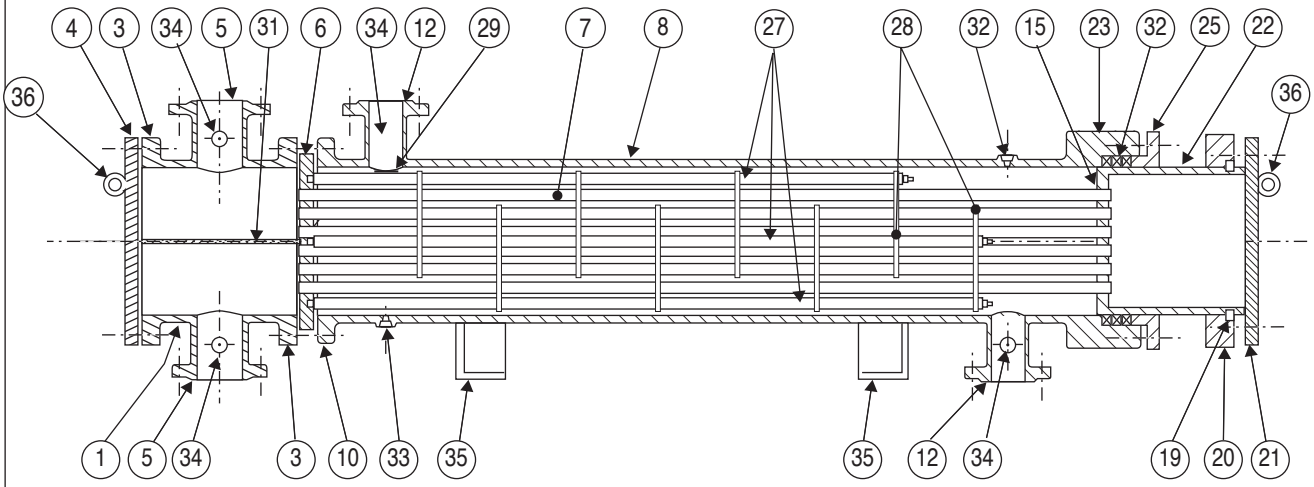
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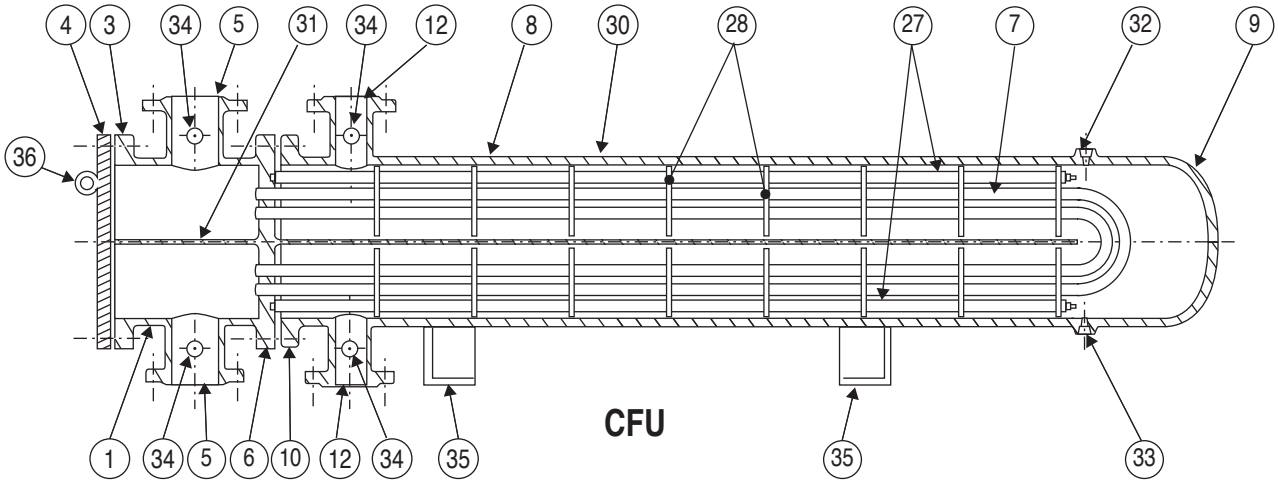
DIFFERENT HEAT EXCHANGERS



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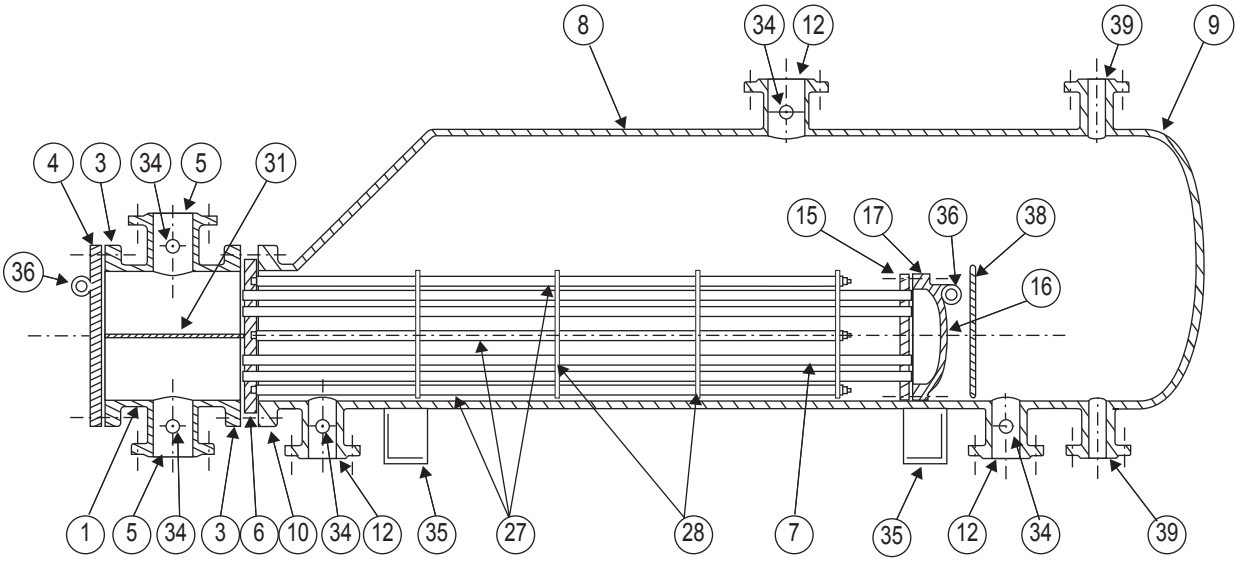


CFU

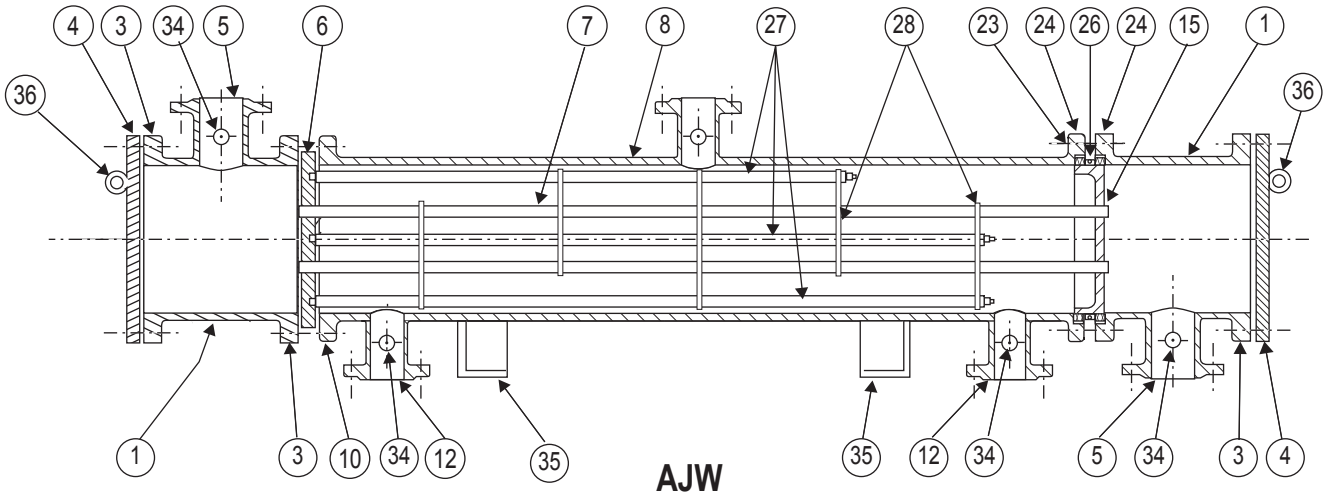
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DIFFERENT HEAT EXCHANGERS



AKT



AJW

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EXCHANGER TUBE STANDARDS

MECHANICAL STANDARDS TEMA CLASS R C B **SECTION 5**

* RCB-2 TUBES

RCB-2.1 Tube length

The following tube lengths for both straight and U-tube exchangers are commonly used: 96, 120, 144, 192 and 240 inches. Other lengths may be used.

RCB-2.2 Tube diameters and gages

RCB-2.21 Bare tubes

Table RCB-2.21 lists common tube diameters and gages for bare tubes of copper, steel and alloy. Other diameters and gages are acceptable.

Table RCB-2.21

BARE TUBE DIAMETERS AND GAGES			
O.D. Inches	Copper and Copper Alloys	Carbon Steel, Aluminium and Aluminium Alloys	Other Alloys
	B.W.G.	B.W.G.	B.W.G.
3/4	20 18 16	16 14 12	18 16 14
1	18 16 14	14 12 —	16 14 12
1-1/4	16 14	14 12	14 12
1-1/2	16 14	14 12	14 12
2	14 12	14 12	14 12

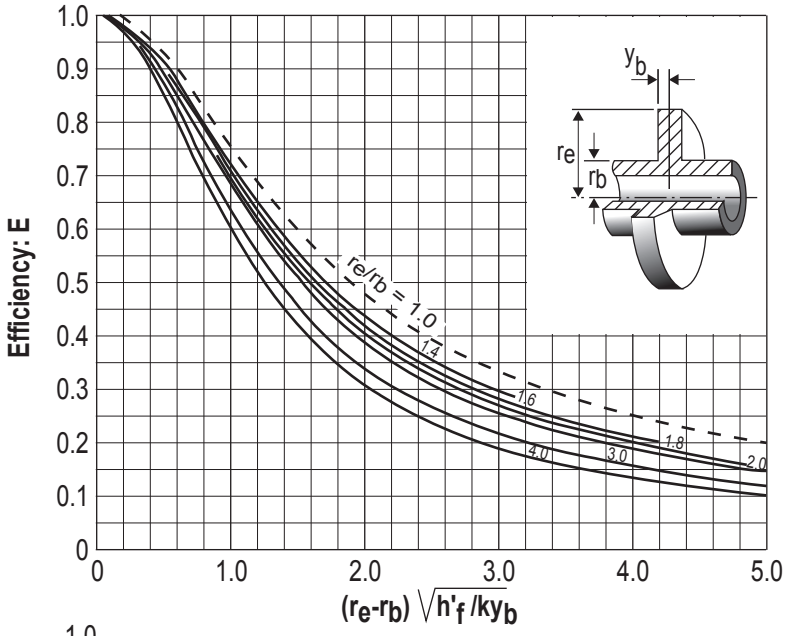
Wall thickness tolerances shall be specified as either minimum or average.

RCB-2.22 Circumferentially-finned tubes

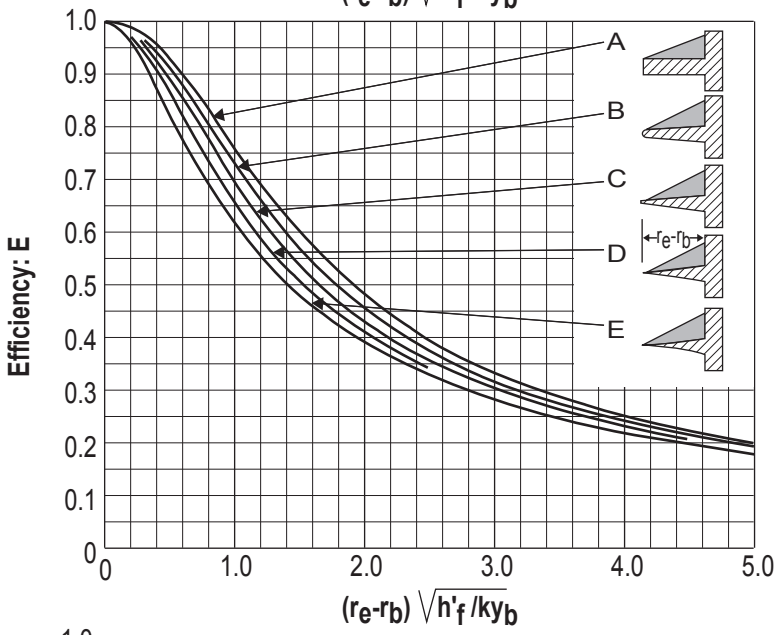
The nominal fin diameter shall not exceed the outside diameter of the unfinned section. Specified wall shall be based on the thickness at the root diameter.

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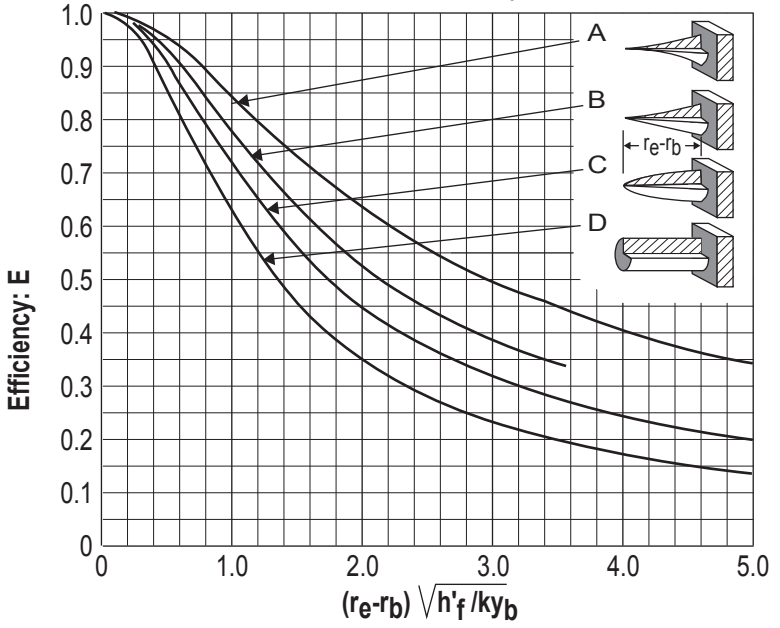
EXTENDED - SURFACE EFFICIENCY



ANNULAR FINS



STRAIGHT FINS



SPINES

D MTE 2192 D

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CONVECTION COEFFICIENT

$$\phi = \frac{\Delta\theta}{R} = h\Delta\theta$$

	UNIT SYSTEM		
	S.I.	METRIC	BRITISH
Units	$\frac{W}{m^2 . K}$	$\frac{kcal}{h . m^2 . °C}$	$\frac{BTU}{h . ft^2 . °F}$
$\frac{W}{m^2 . K}$	1	0.8601	0.1763
$\frac{kcal}{h . m^2 . °C}$	1.1626	1	0.2049
$\frac{BTU}{h . ft^2 . °F}$	5.6721	4.8804	1

The complexity of convection heat transfer and the diversity of cases lead to many equations for the calculation of the heat transfer coefficient. The equations are usually empirical, but purely theoretical equations which include multiple correction factors to enable experimental results to be taken into account do exist. However, the latter are often of doubtful accuracy. The aim here is to investigate the effect of the various parameters on the convection heat transfer coefficient, not to calculate it.

There are three types of parameters:

- the geometry of the system. For example, the internal diameter of a tube in a forced convection calculation, or the vertical height of a wall in a free convection calculation
- the characteristics of the fluid circulation
- the properties of the fluid

In all cases of convection heat transfer, the above parameters can be related by non-dimensional equations. For example:

$$Nu = a\, Re^n\, Pr^m$$

$$St = a\, Re^{n-1}\, Pr^{m-1}$$

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CONVECTION COEFFICIENT

Nu = Nusselt number = $\frac{hD}{\lambda}$

Re = Reynolds number = $\frac{uD\rho}{\mu} = \frac{DG}{\mu} = \frac{uD}{\nu}$

Pr = Prandtl number = $\frac{Cp\mu}{\lambda}$

St = Stanton number = $\frac{Nu}{Re\ Pr} = \frac{h}{Cp\ G}$

Pe = Peclet number = Re Pr = $\frac{D\ Cp\ G}{\lambda}$

Where

- D = characteristic geometrical dimension
- u = mean fluid velocity
- G = ρu = mass flow rate per unit area
- Cp = specific heat
- μ = dynamic viscosity (or = ν kinematic viscosity)
- λ = thermal conductivity

Examples of equations

- ***Inside a tube***

For example, if some minor corrections are neglected, the internal film coefficient for gases or organic liquids inside a tube, when convection is forced, can be estimated by the following expression:

$$Nu = 1.86 \left(\frac{D_i}{L} \cdot Re \cdot Pr\right)^{1/3}$$ in laminar flow (Re < 2100)

$$Nu = 0.023\ Re^{0.8}\ Pr^{1/3}$$ in turbulent flow (Re > 10,000)

D_i and L: inner diameter and tube length

- ***Outside a tube***

In shell and tube heat exchangers, the heat transfer coefficient outside the tubes can be estimated by the following equation:

$$N_u = 0.22 \cdot R_e^{0,6} \cdot P_r^{1/3}$$

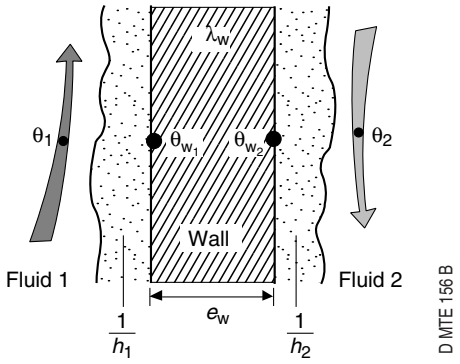
In the expression of Reynolds and Nusselt numbers, D is the outer diameter of the tubes.

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SIMULTANEOUS CONDUCTION AND CONVECTION

OVERALL HEAT TRANSFER COEFFICIENT FOR A PLANE WALL

Heat is transferred by convection followed by conduction and convection.



For a clean wall

$$\phi = \frac{\theta_1 - \theta_{w1}}{\frac{1}{h_1}} = \frac{\theta_{w1} - \theta_{w2}}{\frac{e_w}{\lambda_w}} = \frac{\theta_{w2} - \theta_2}{\frac{1}{h_2}}$$

$$= \frac{\theta_1 - \theta_2}{\frac{1}{h_1} + \frac{e_w}{\lambda_w} + \frac{1}{h_2}} = \frac{\theta}{R} = U_c\,\theta$$

The overall heat transfer coefficient for a clean wall is given by

$$\frac{1}{U_c} = \frac{1}{h_1} + \frac{e_w}{\lambda_w} + \frac{1}{h_2}$$

The overall fouling resistance is given by

$$R_F = R_{F_1} + R_{F_2}$$

The overall heat transfer coefficient for a fouled wall is given by

$$\frac{1}{U_F} = \frac{1}{U_c} + R_F$$

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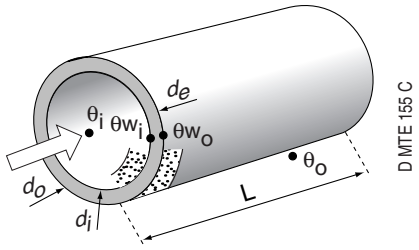
SIMULTANEOUS CONDUCTION AND CONVECTION

CYLINDRICAL WALL

$$Q = h_i \pi D_i \ell (\theta_i - \theta_{w_i})$$

$$= \frac{2 \pi \lambda_w \ell (\theta_{wi} - \theta_{w2})}{\ell n D_o / D_i}$$

$$= h_o \pi D_o \ell (\theta_{wo} - \theta_o)$$



If the reference heat transfer area is the external surface area of the cylinder, we obtain:

$$Q_o = \frac{Q}{\pi D_o \ell} = \frac{\theta_i - \theta_{w_i}}{\frac{1}{h_i \frac{D_i}{D_o}}} = \frac{\theta_{w_i} - \theta_{w_o}}{\frac{D_o}{2 \lambda_w} \ell n \frac{D_o}{D_i}} = \frac{\theta_{w_o} - \theta_o}{\frac{1}{h_o}} = \frac{\theta_i - \theta_o}{R} = U_{ce}(\theta_i - \theta_o)$$

$h_i \frac{D_i}{D_o} = h_{ie}$: internal film coefficient corrected to be related to external area.

For cylindrical wall, the clean global transfer coefficient is U_{ce} (related to external area):

$$\frac{1}{U_{ce}} = \frac{1}{h_{ie}} + \frac{D_o}{2 \lambda_w} \ell n \frac{D_o}{D_i} + \frac{1}{h_o}$$

Fouling resistance:

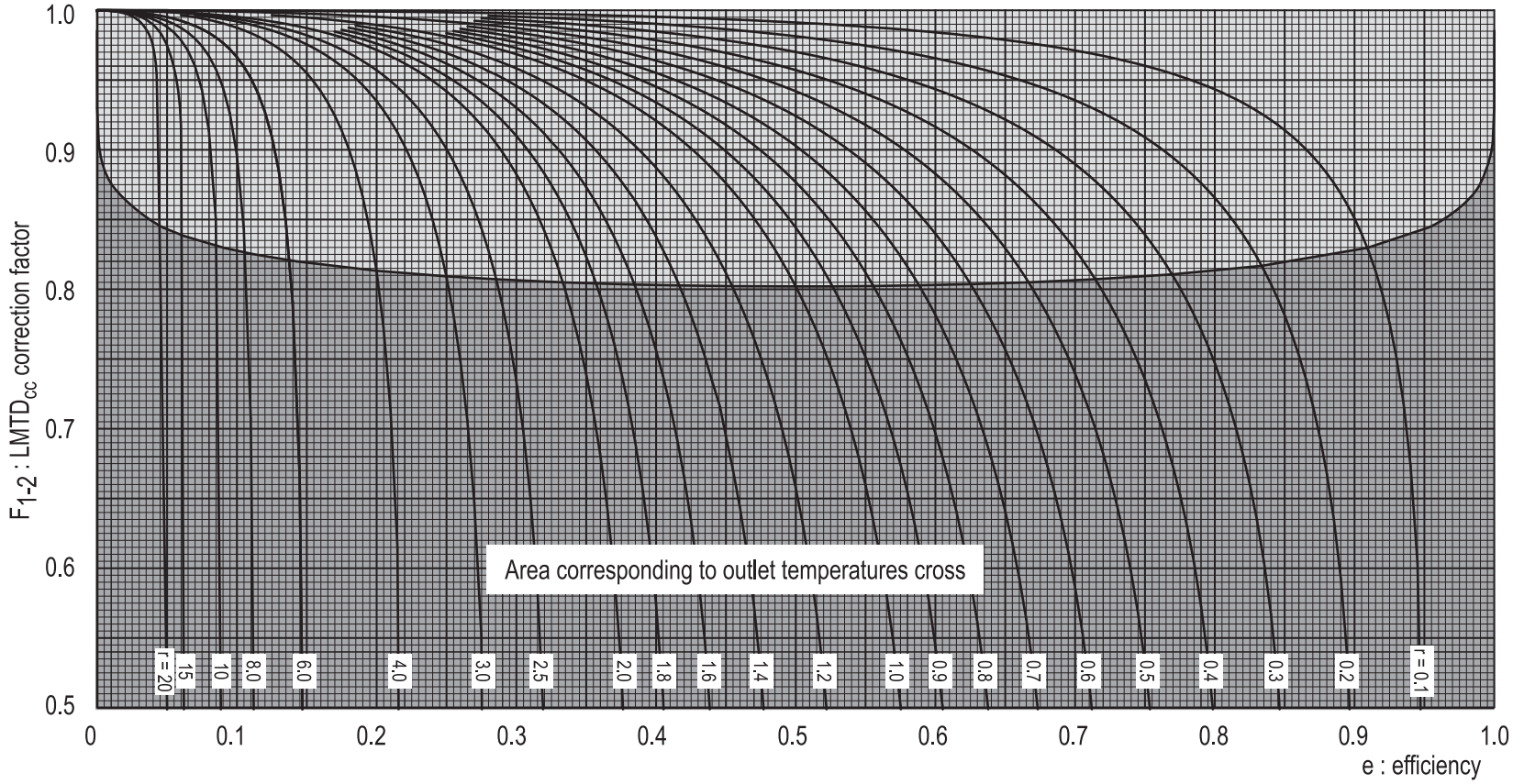
$$R_f = R_{fi} \frac{D_o}{D_i} + R_{fo}$$

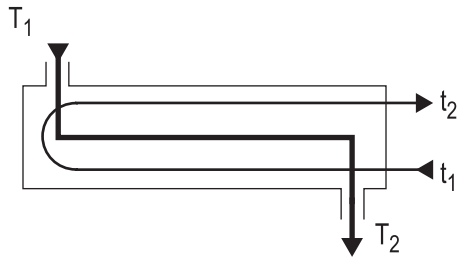
$R_{fi} \frac{D_o}{D_i} = R_{fie}$: internal fouling resistance corrected to be related to external area.

Overall transfer coefficient (including fouling) related to external area:

$$\frac{1}{U_{fe}} = \frac{1}{U_{ce}} + R_f$$

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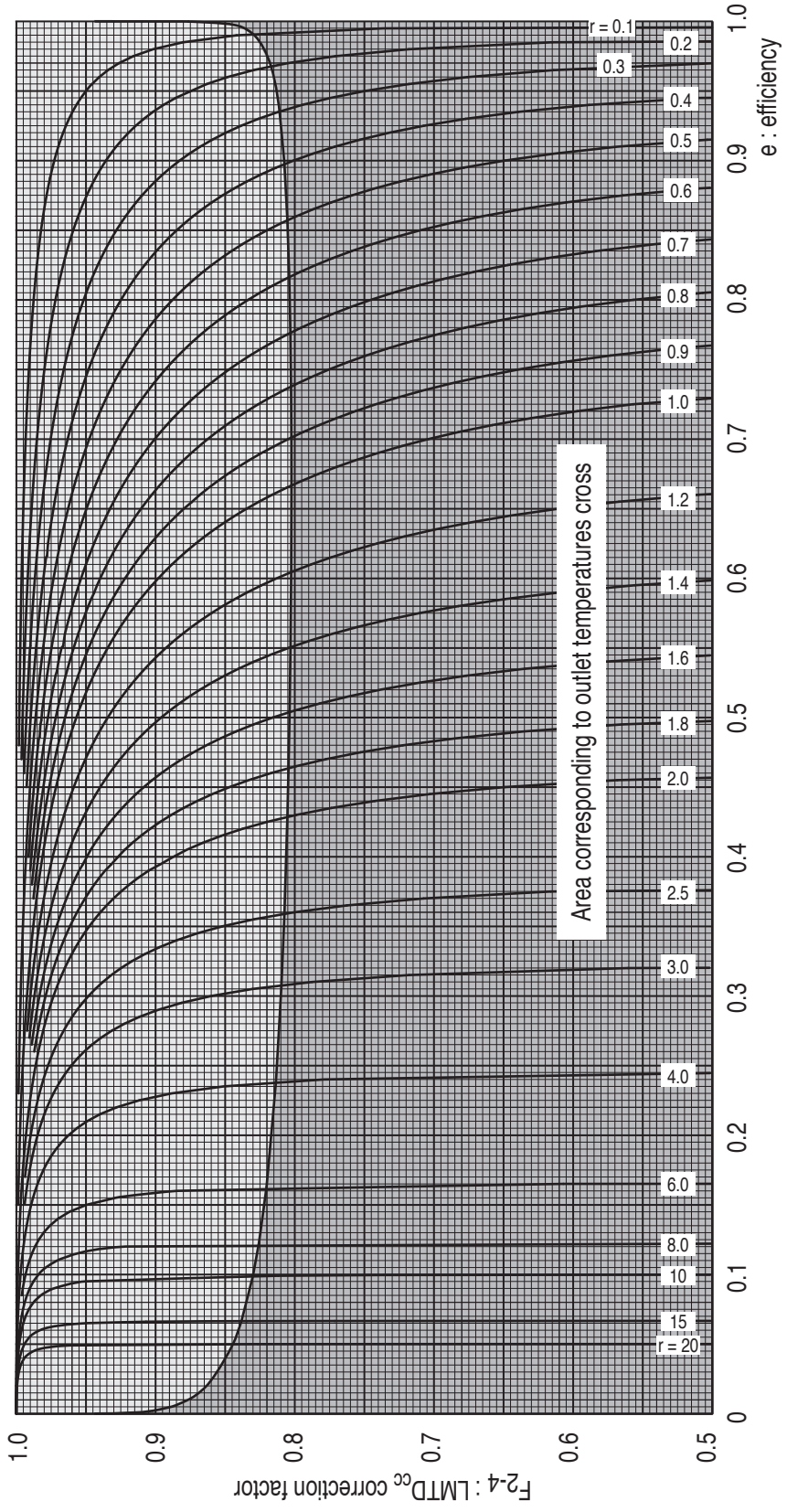




$LMTD_{cc}$ CORRECTION FACTOR
— 1 shell pass, 2 or 2 n tube passes —

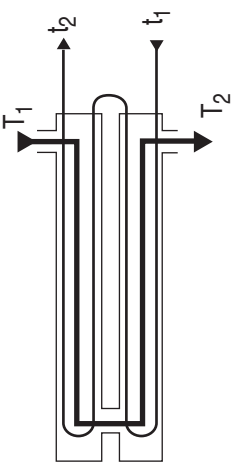
$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{1-2} = \frac{(\Delta t)_{1-2}}{LMTD_{cc}}$$

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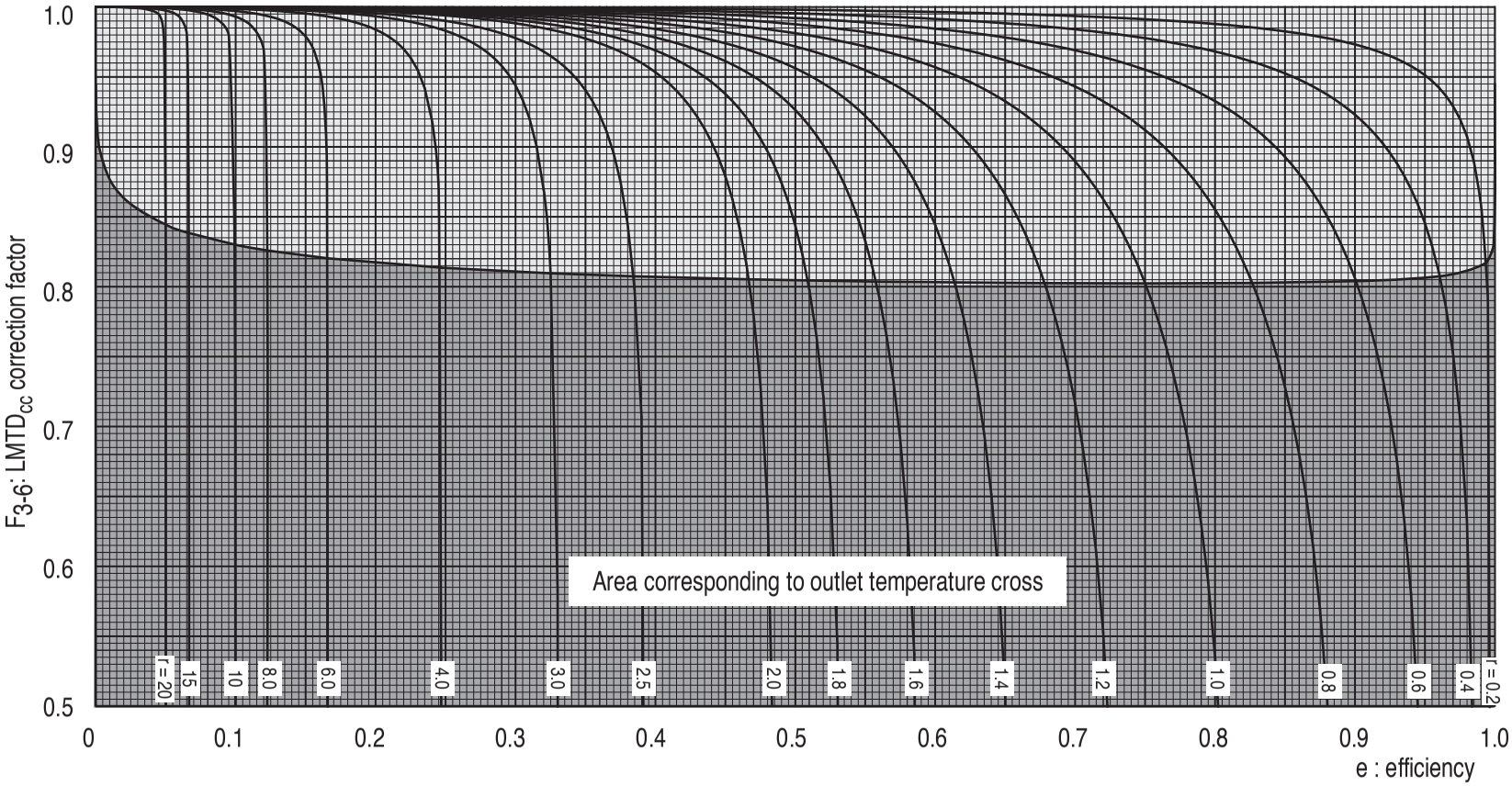
$LMTD_{cc}$ CORRECTION FACTOR
— 2 shell passes, 4 or 4 n tube passes —

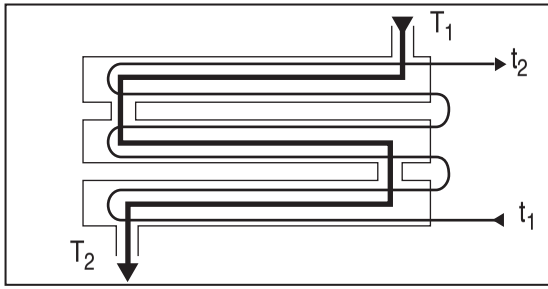
$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{2\text{-}4} = \frac{(\Delta t)_{2\text{-}4}}{LMTD_{cc}}$$



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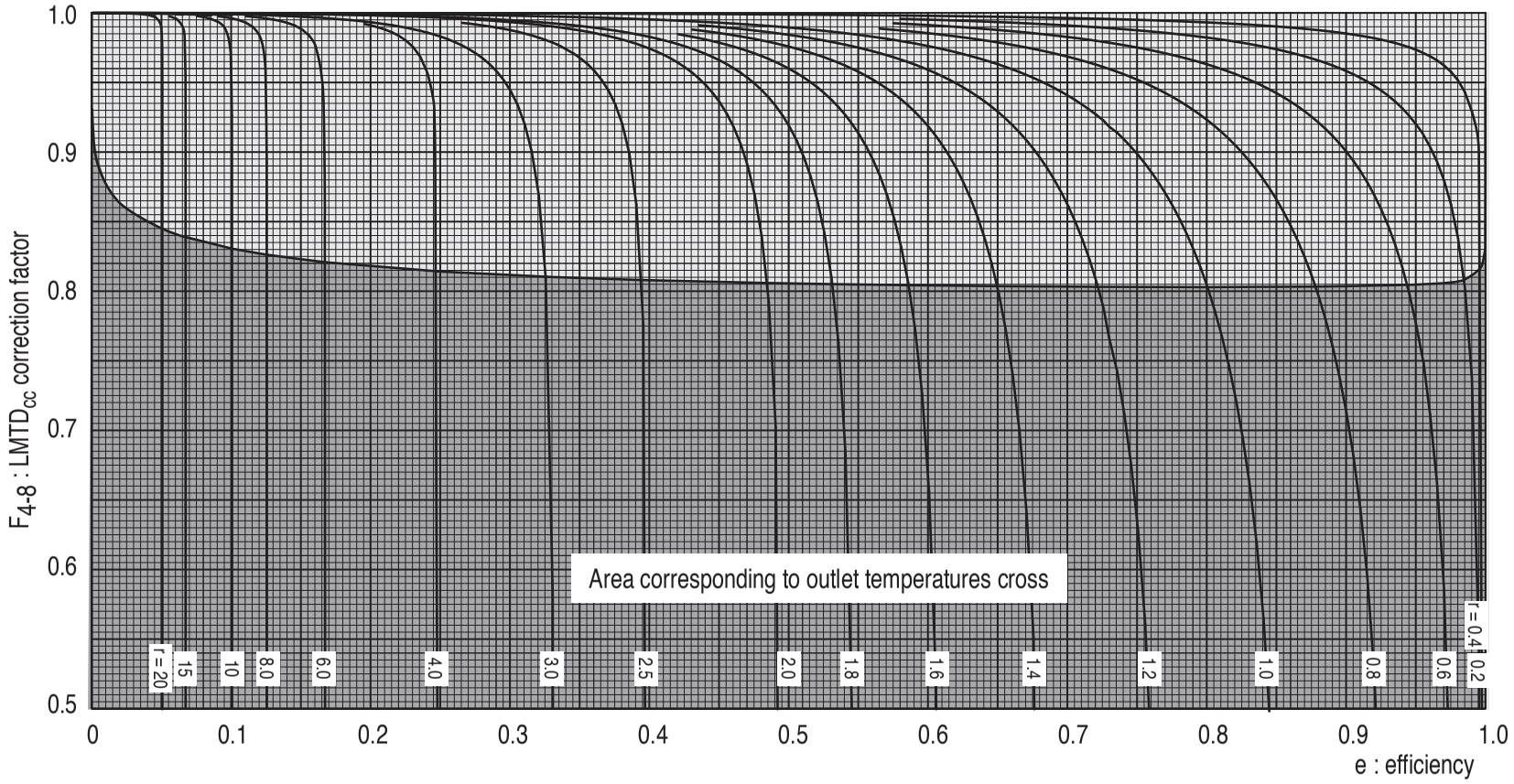


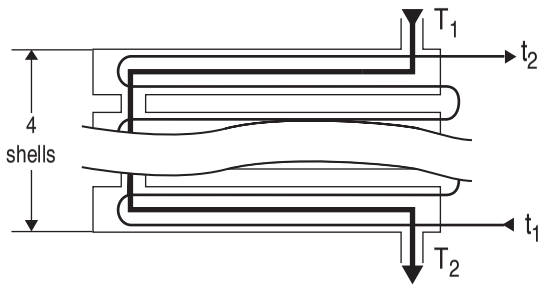


$LMTD_{cc}$ CORRECTION FACTOR
— 3 shell passes, 6 or 6 n tube passes —

$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{3-6} = \frac{(\Delta t)_{3-6}}{LMTD_{cc}}$$

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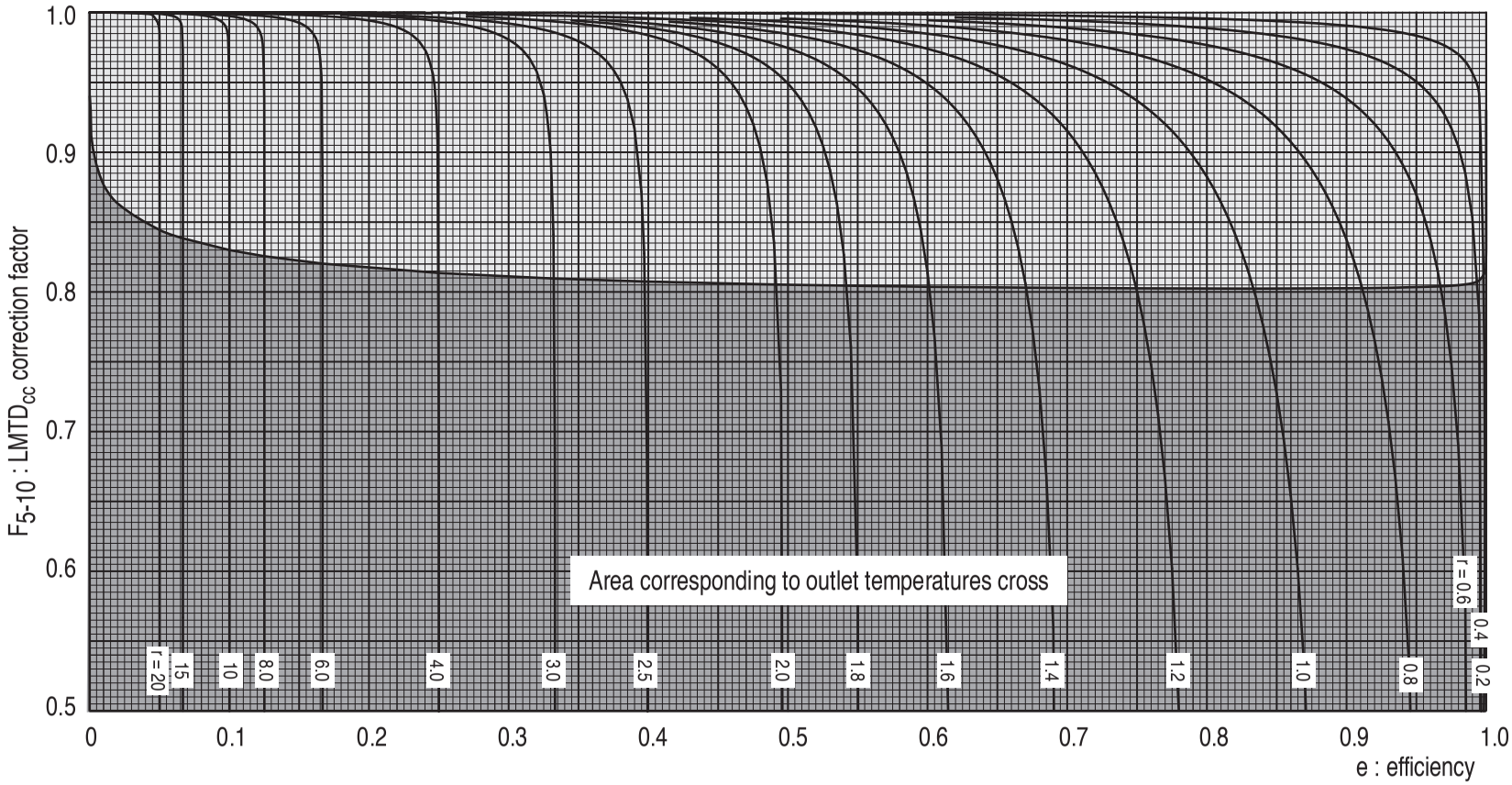


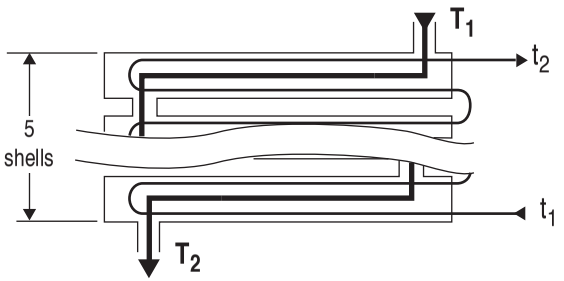


$LMTD_{cc}$ CORRECTION FACTOR
— 4 shell passes, 8 or 8 n tube passes —

$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{4-8} = \frac{(\Delta t)_{4-8}}{LMTD_{cc}}$$

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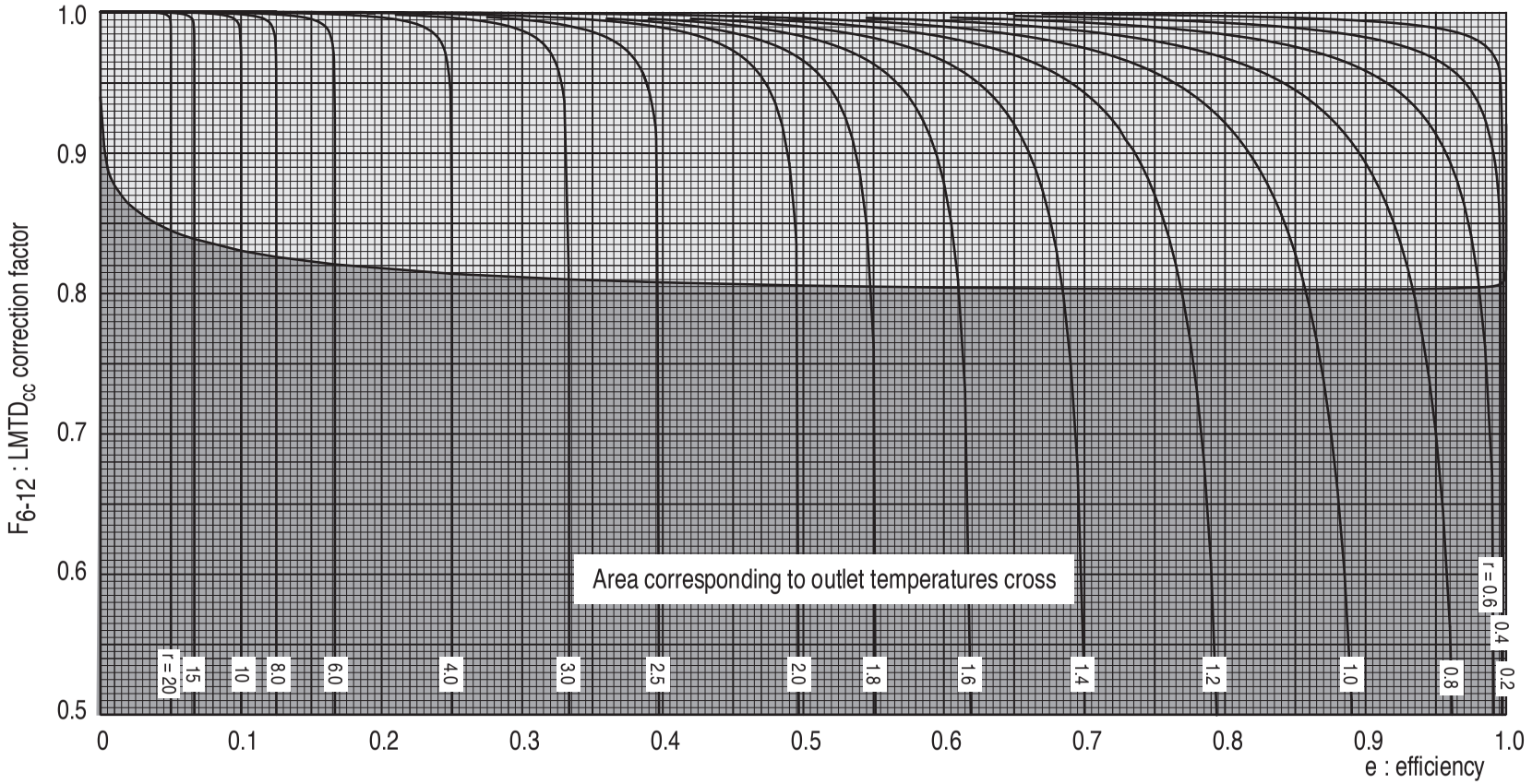


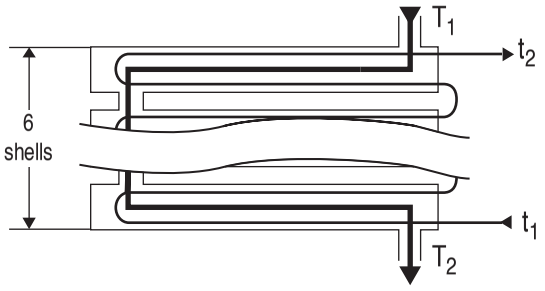


$LMTD_{cc}$ CORRECTION FACTOR
— 5 shell passes, 10 or 10 n tube passes —

$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{5-10} = \frac{(\Delta t)_{5-10}}{LMTD_{cc}}$$

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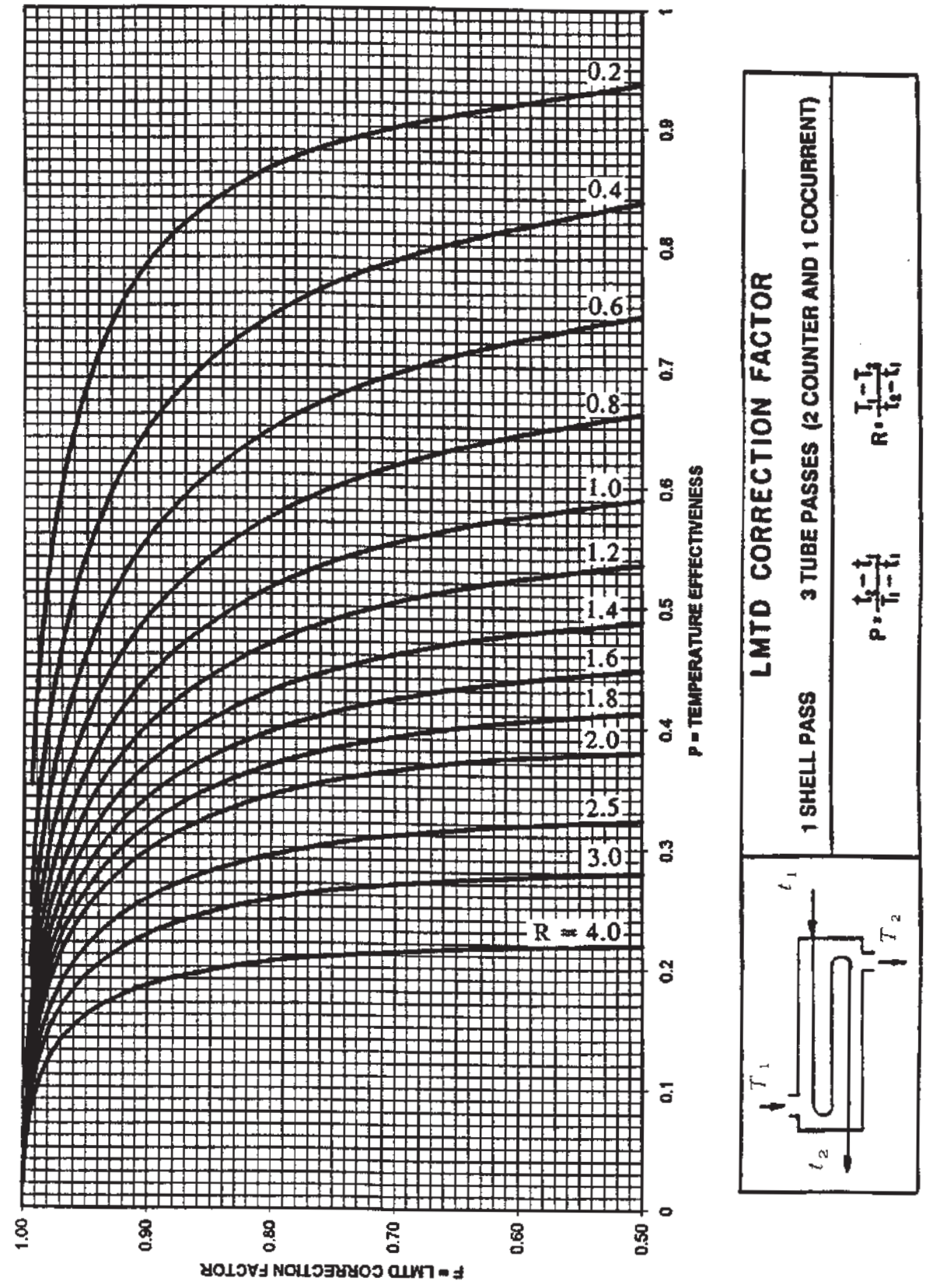




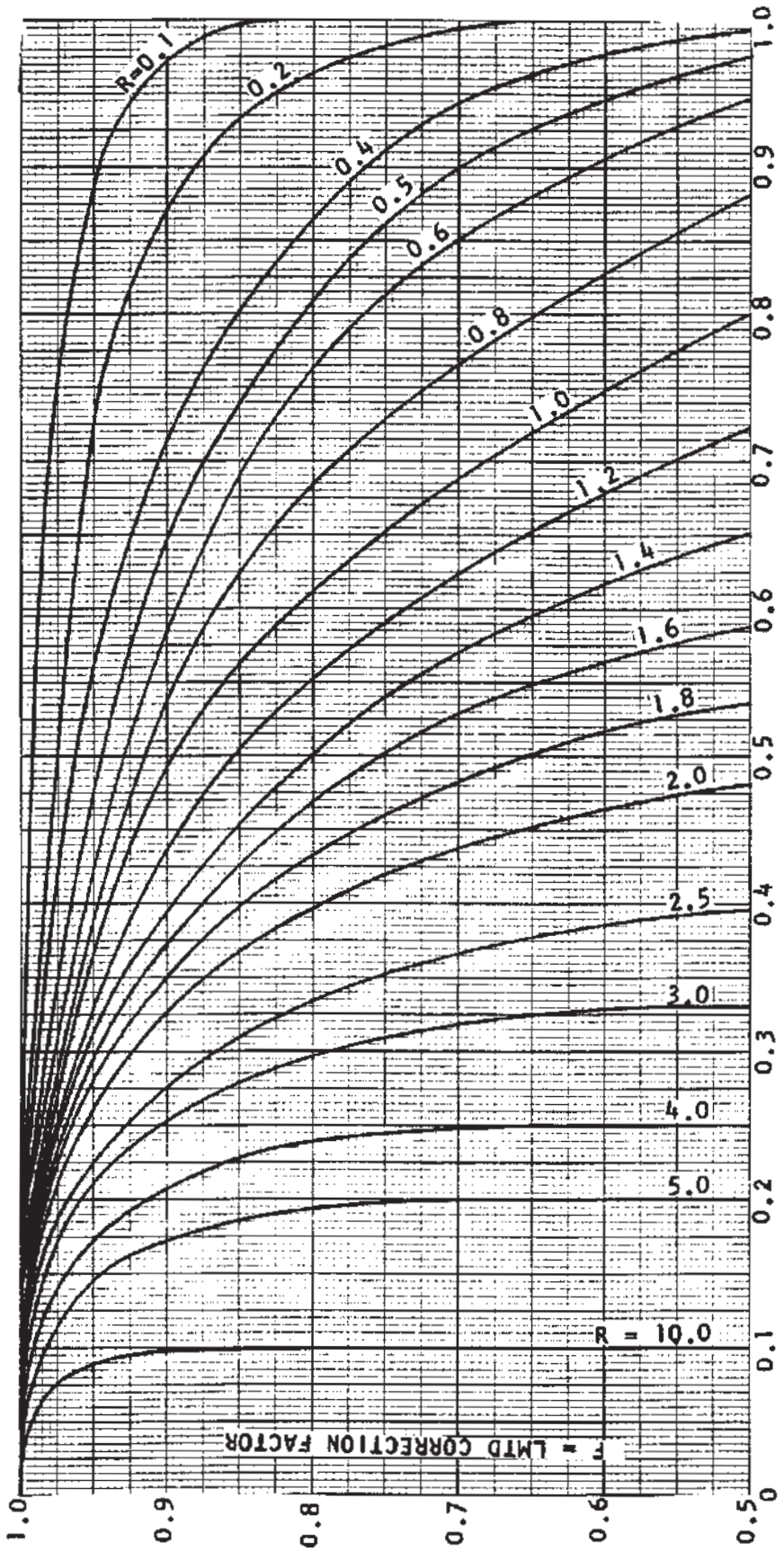
$LMTD_{cc}$ CORRECTION FACTOR
— 6 shell passes, 12 or 12 n tube passes —

$$e = \frac{t_2 - t_1}{T_1 - t_1} \qquad r = \frac{T_1 - T_2}{t_2 - t_1} \qquad F_{6-12} = \frac{(\Delta t)_{6-12}}{LMTD_{cc}}$$

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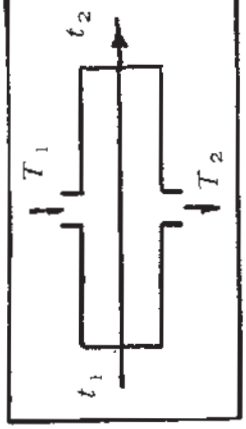


P = TEMPERATURE EFFECTIVENESS

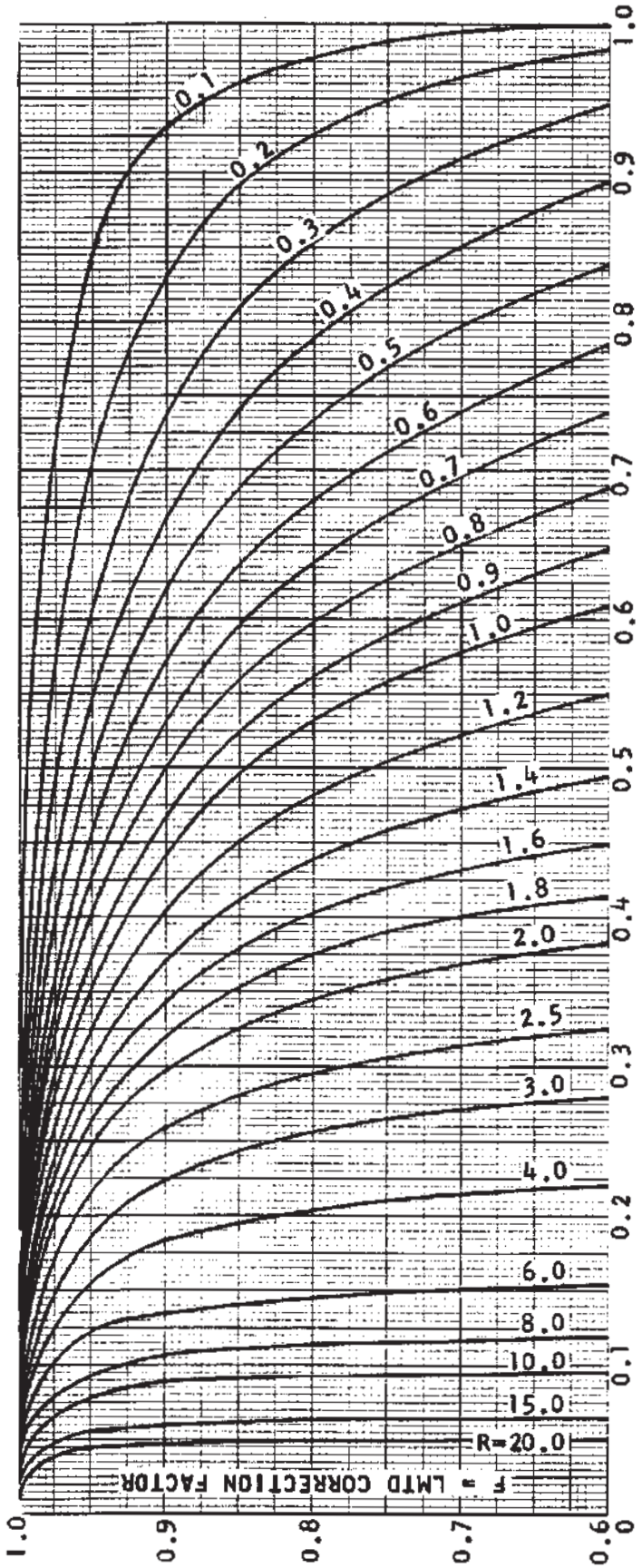
LMTD CORRECTION FACTOR

CROSS FLOW SHELL — **1 TUBE PASS**

$$P = \frac{t_2 - t_1}{T_1 - t_1} \qquad R = \frac{T_1 - T_2}{t_2 - t_1}$$



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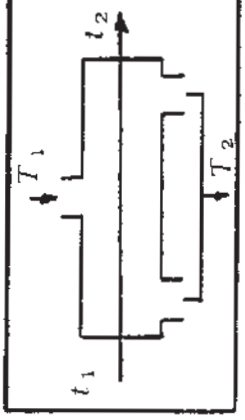


P = TEMPERATURE EFFECTIVENESS

LMTD CORRECTION FACTOR

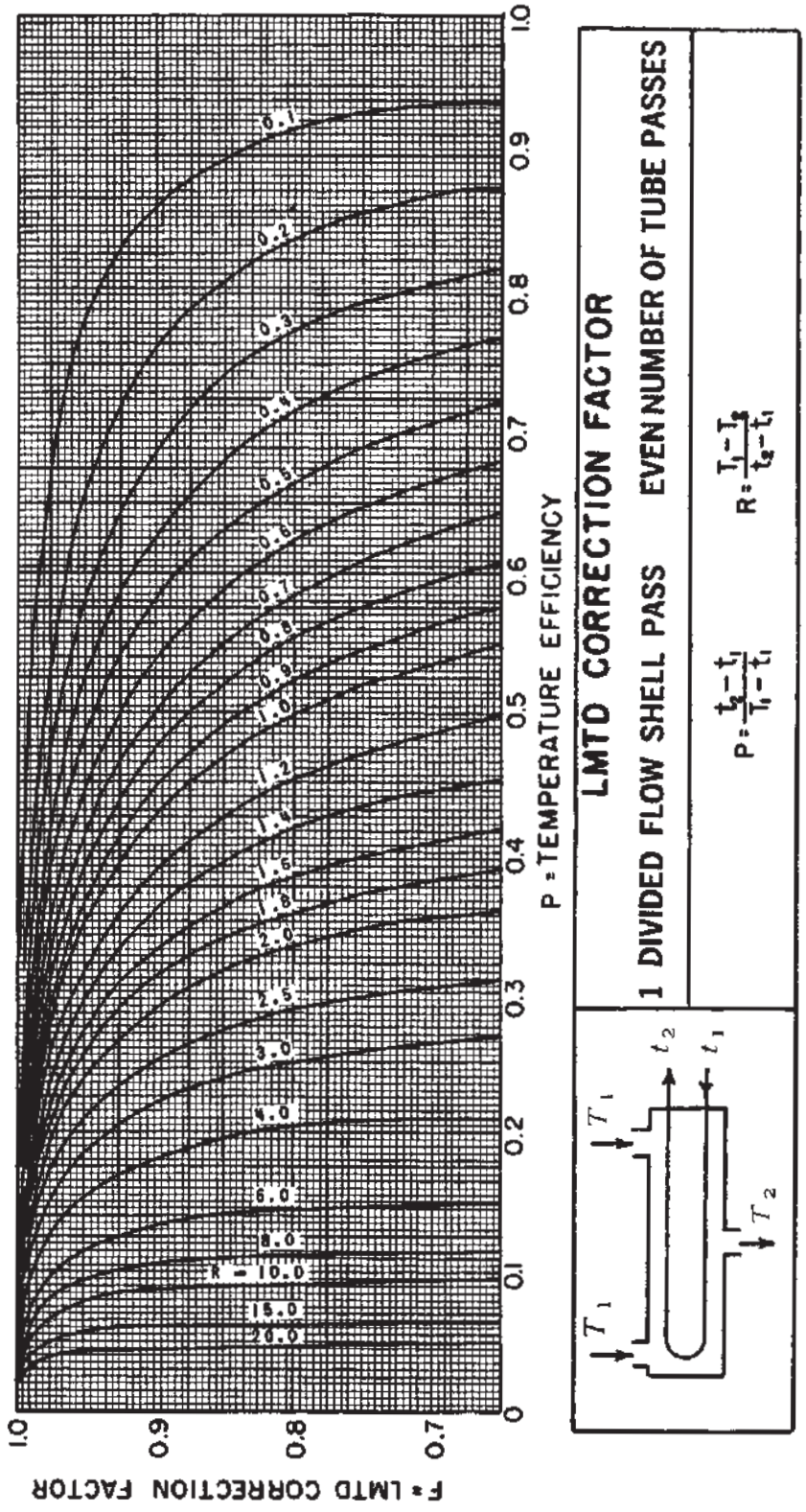
1 DIVIDED FLOW SHELL PASS

1 TUBE PASS

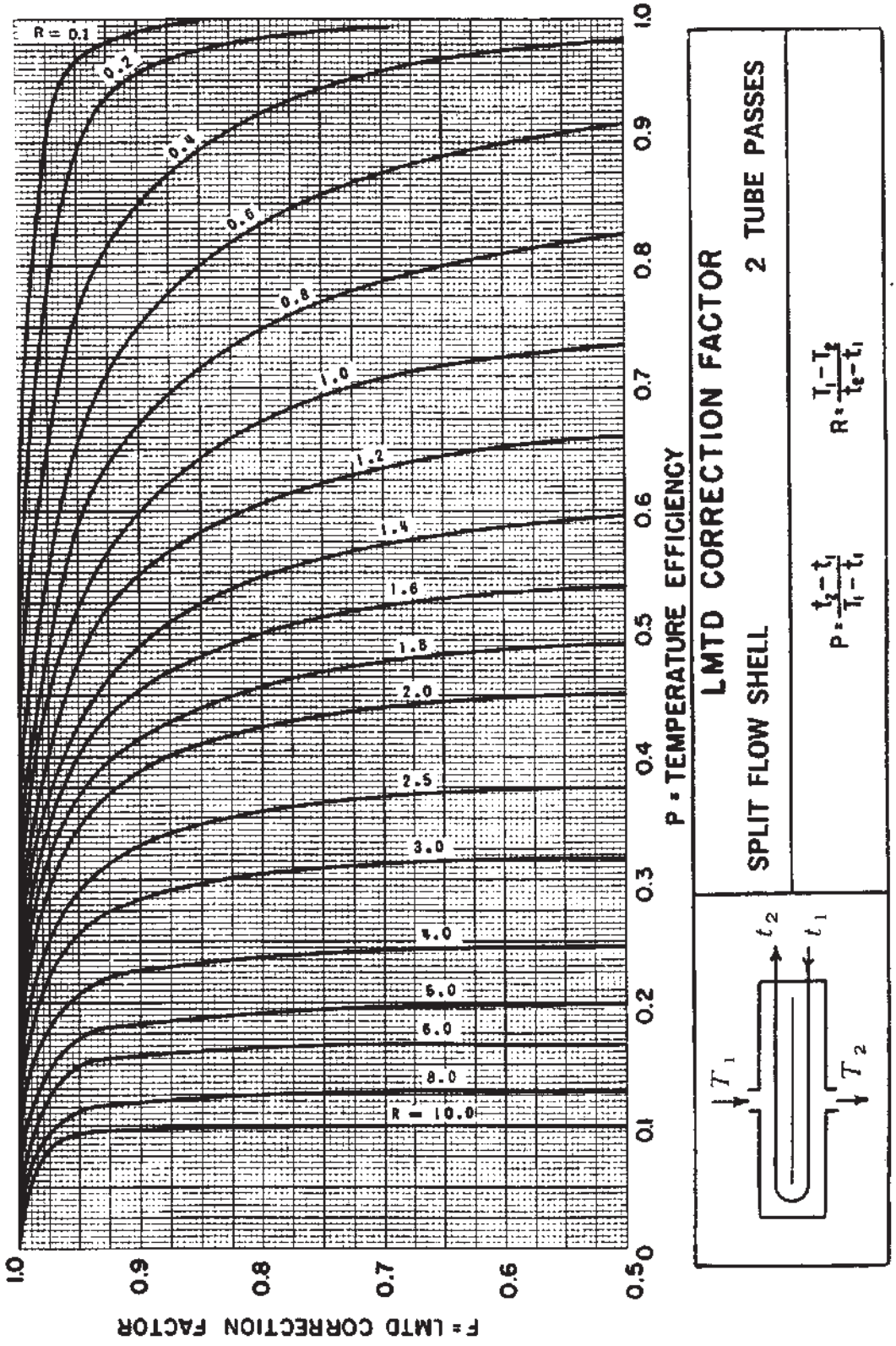


$$P = \frac{t_2 - t_1}{T_1 - t_1} \qquad R = \frac{T_1 - T_2}{t_2 - t_1}$$

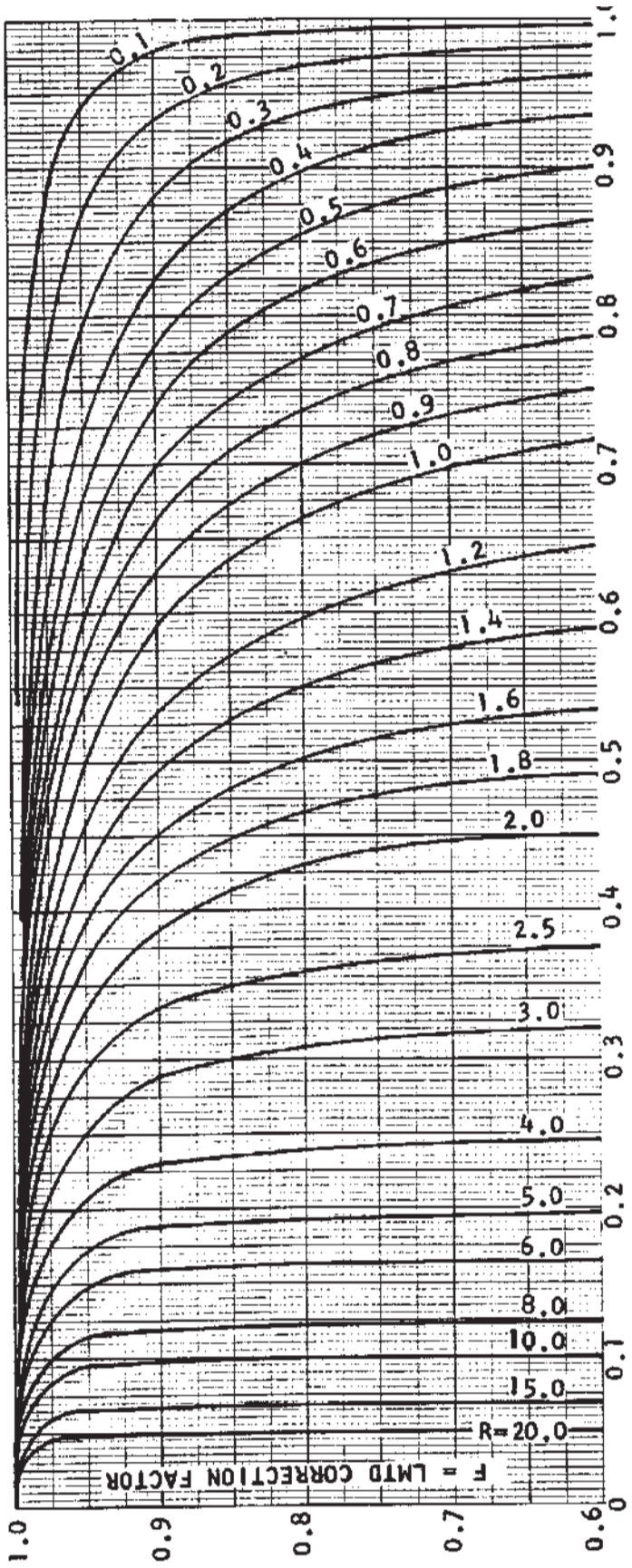
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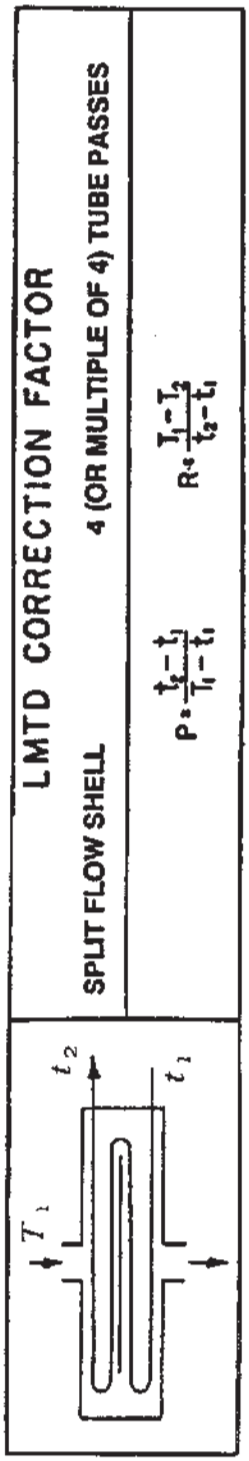


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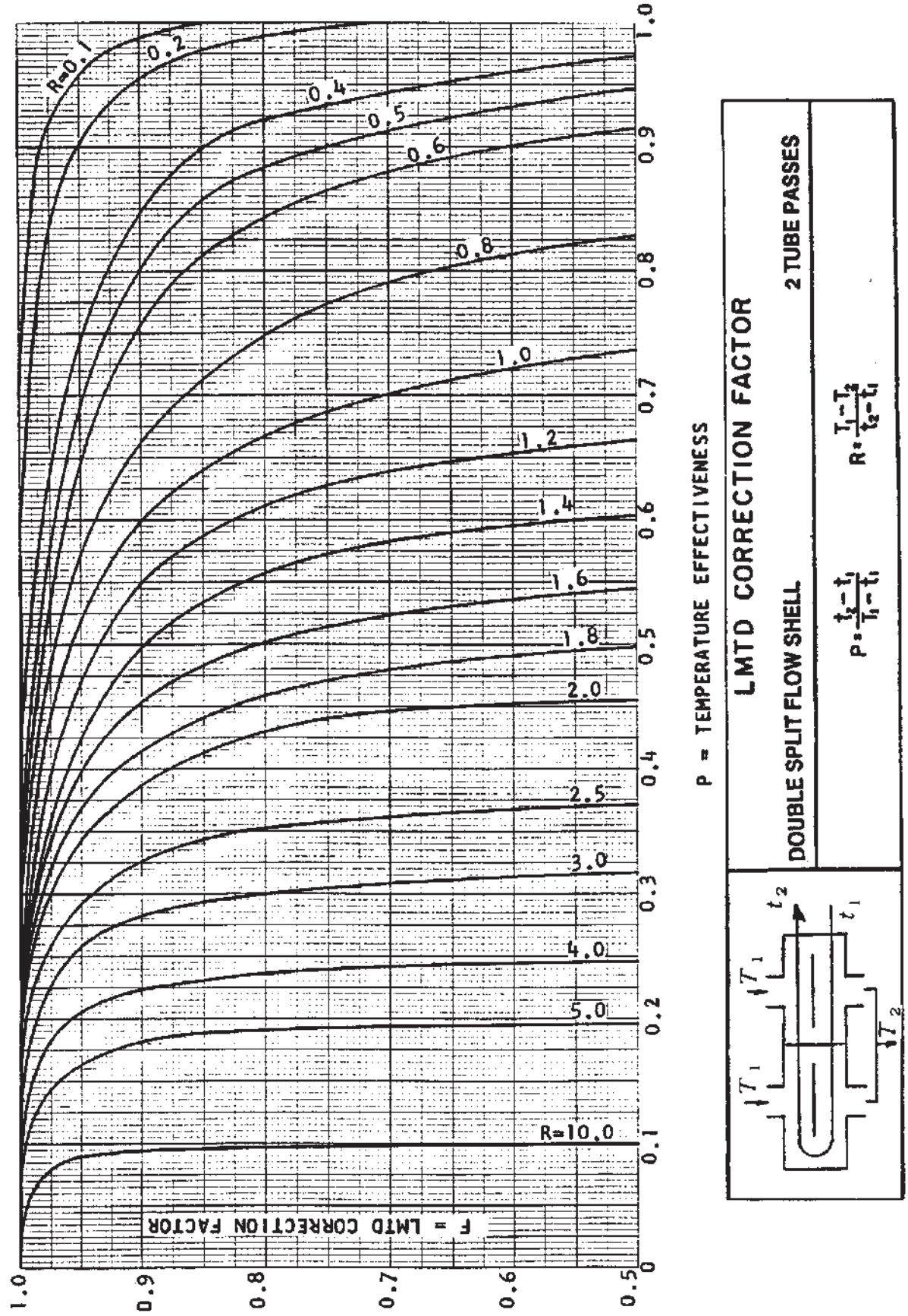


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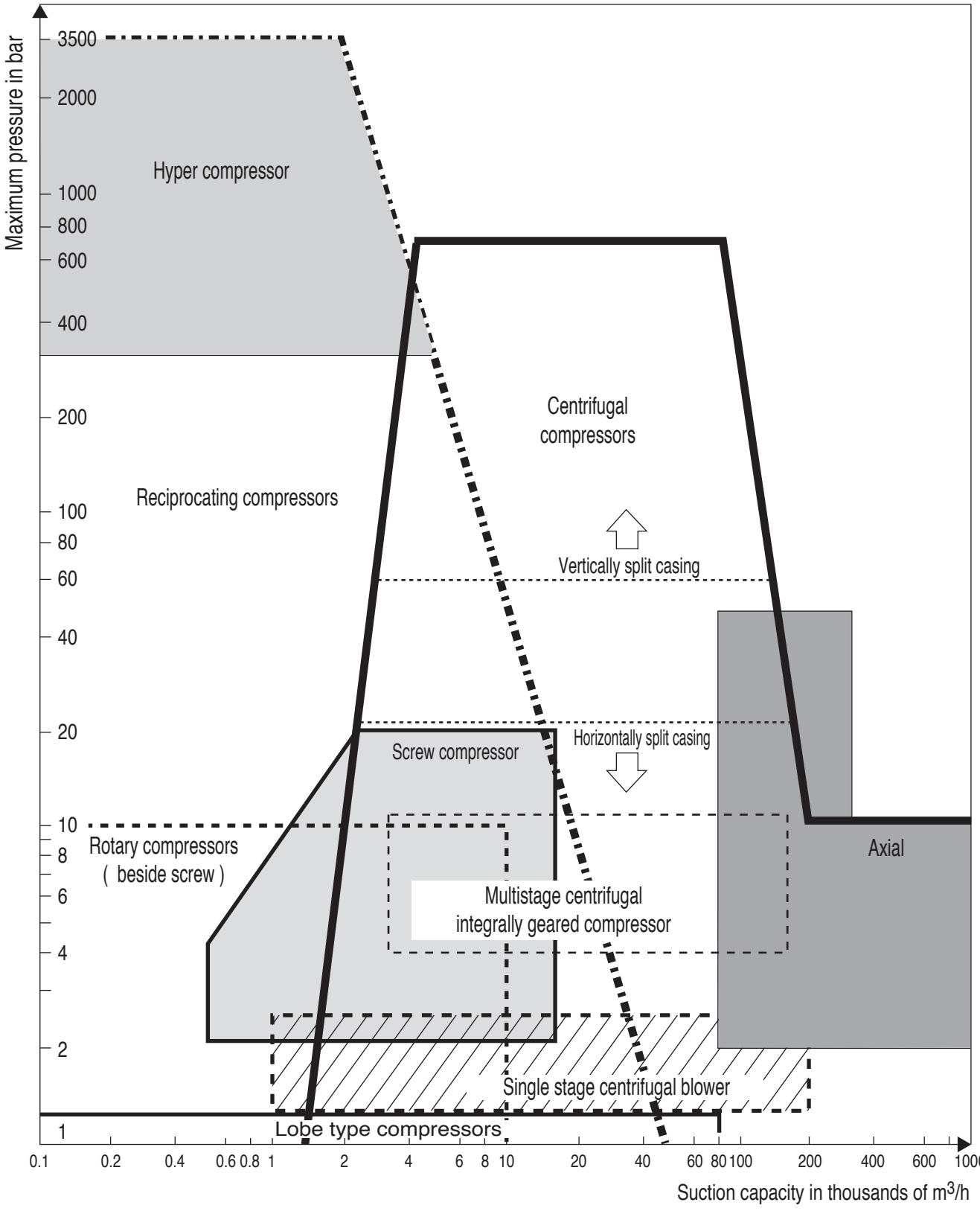
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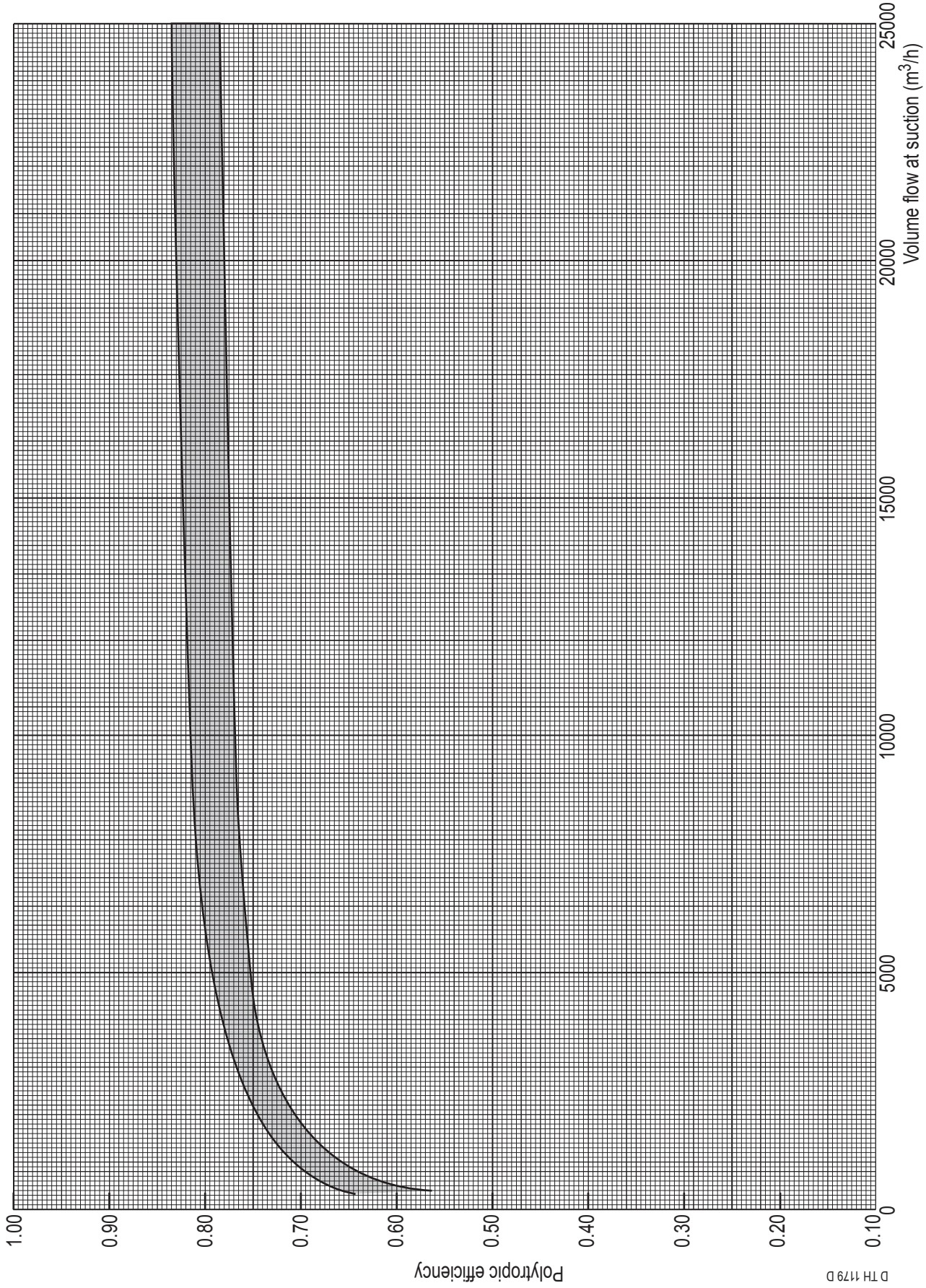
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DYNAMIC AND POSITIVE DISPLACEMENT COMPRESSORS
— General areas of application —



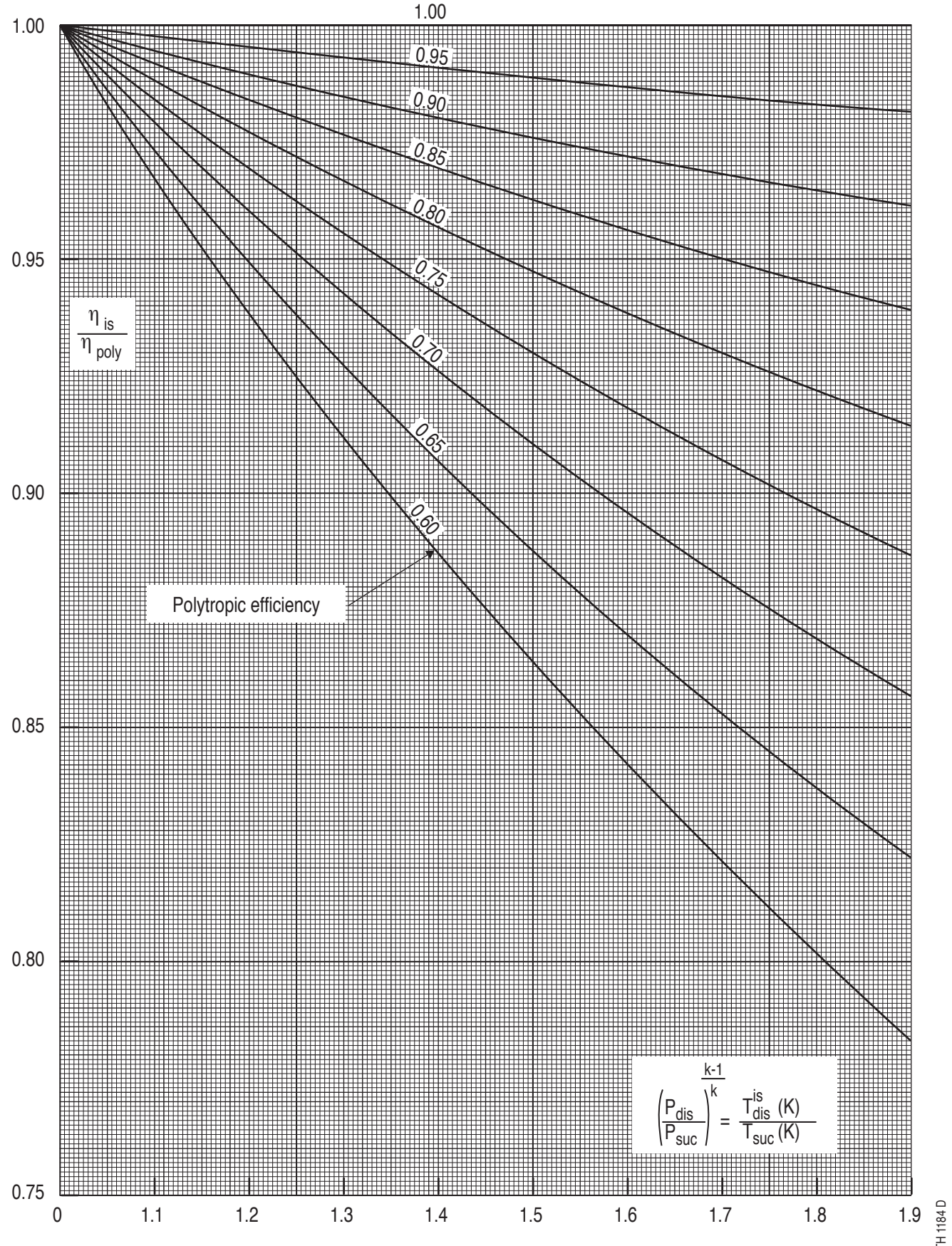
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CENTRIFUGAL COMPRESSOR
POLYTROPIC EFFICIENCY ESTIMATION



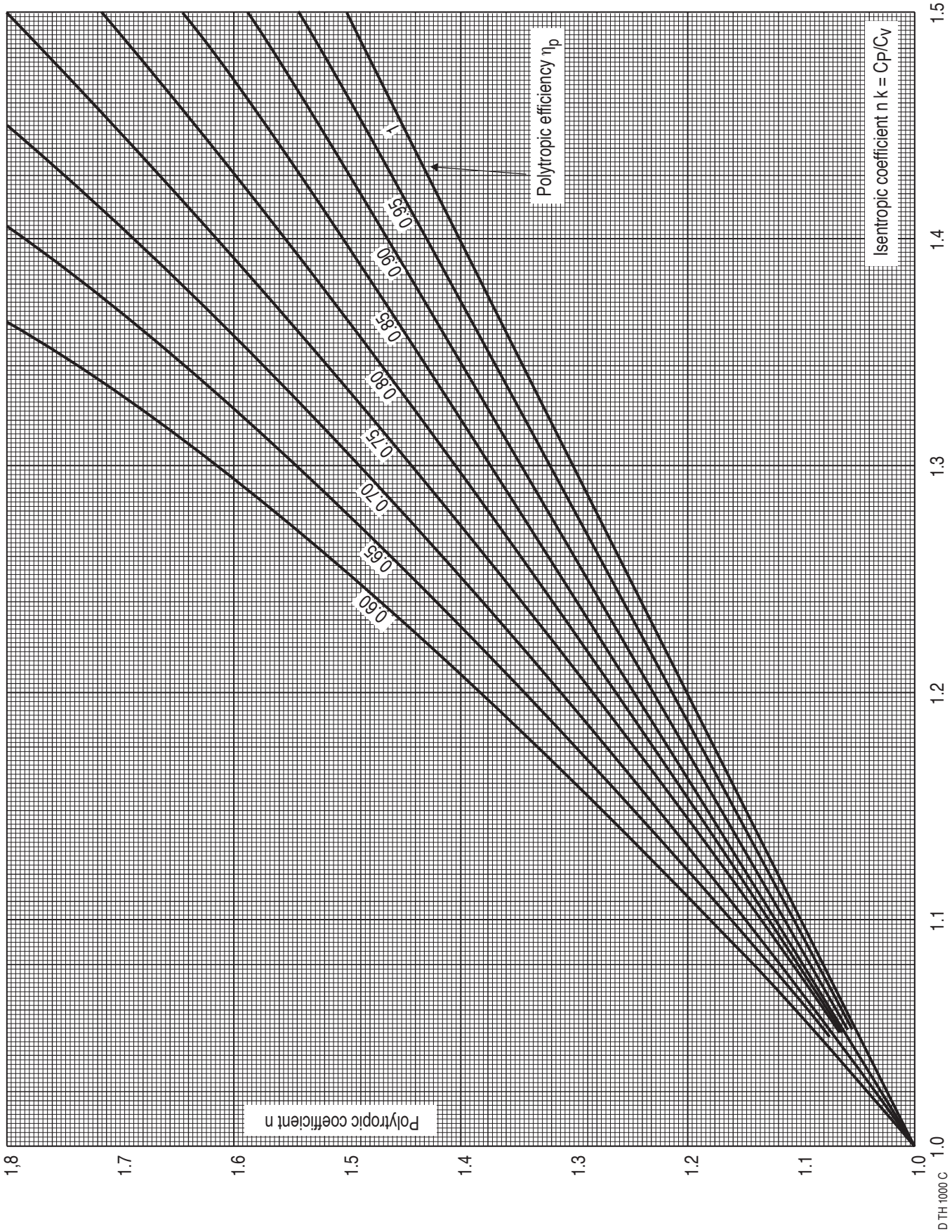
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DETERMINATION OF ISENTROPIC EFFICIENCY



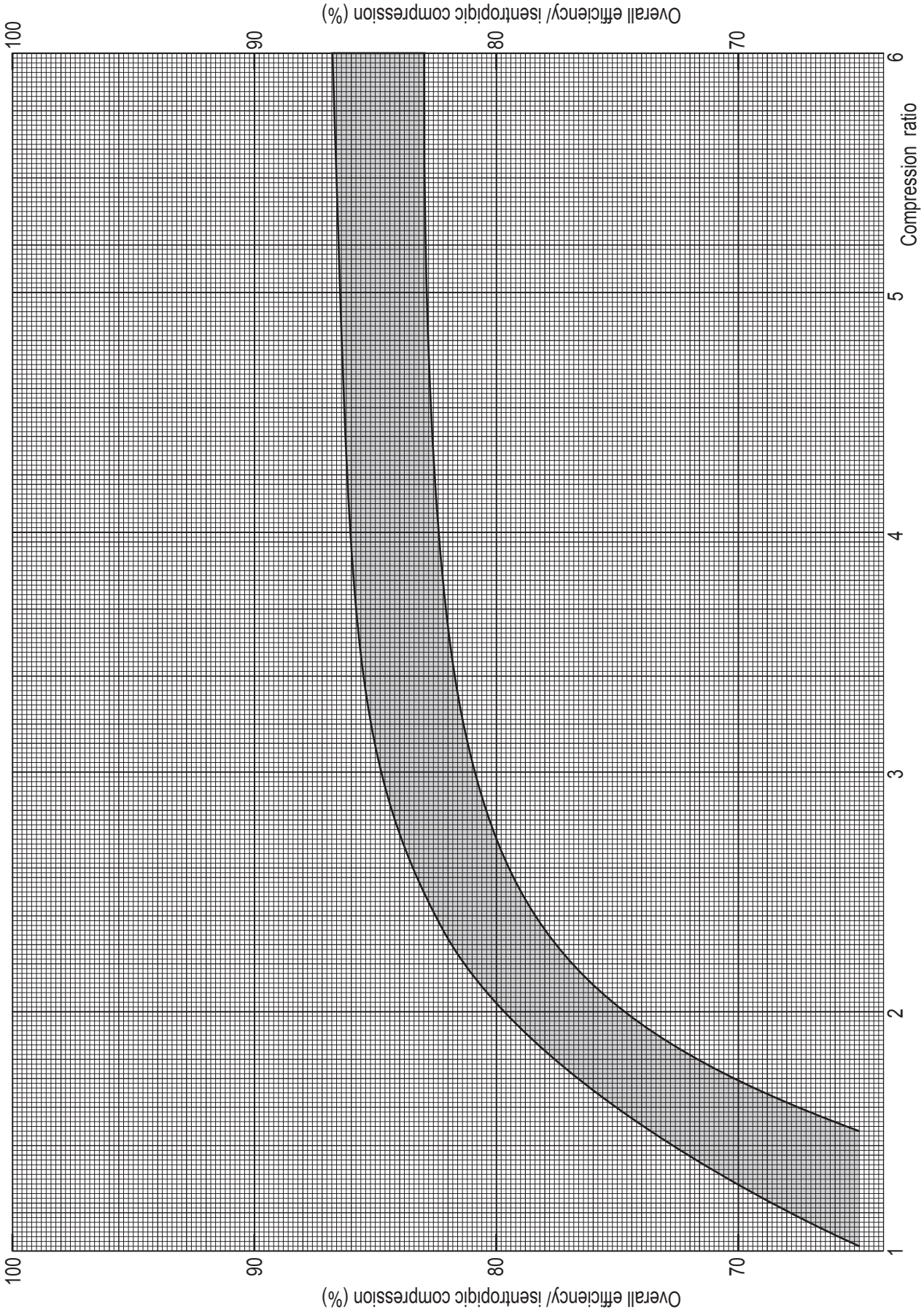
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POLYTROPIC COEFFICIENT CHART



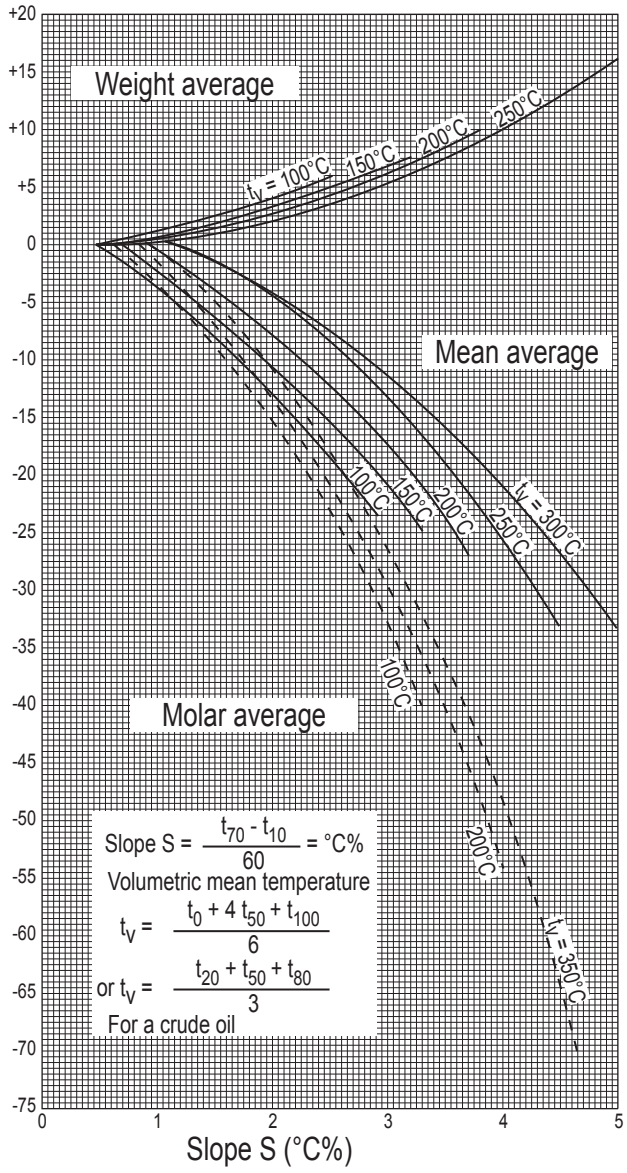
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RECIPROCATING COMPRESSOR ISENTROPIC AND OVERALL EFFICIENCY ESTIMATION

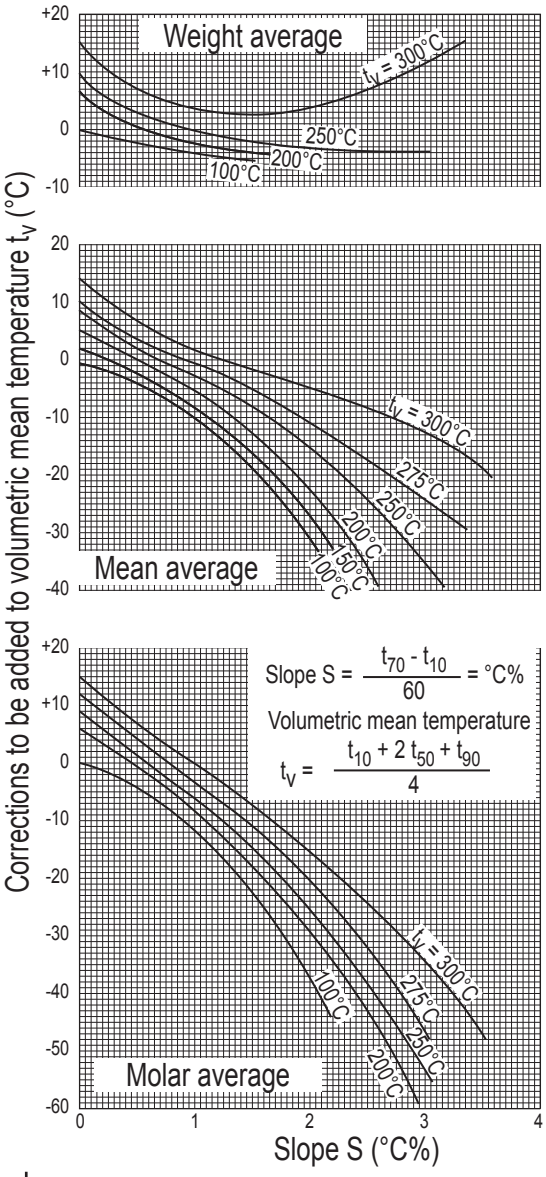


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MEAN BOILING POINT TEMPERATURES FOR PETROLEUM CUTS



From TBP distillation



From ASTM distillation

The "mean average" temperature ; t_{mav}
corresponds to the WEIGHTED MEAN

D TH 1405 D

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CHARACTERIZATION FACTOR OR HYDROCARBONS AND PETROLEUM CUTS (K_{UOP})

*Given the **extreme complexity** of the their composition, petroleum cuts cannot be characterized by a complete analysis of the molecules they contain. However, the characteristics of the hydrocarbons of a petroleum cut have a great influence on their **specific gravity** and **volatility**.*

On **Page 2**, some pure hydrocarbons can be represented on a diagram - **specific gravity** versus **normal boiling point**. We notice that **hydrocarbons of each type** (Paraffins, Naphthenes, Aromatics, …) are situated along a **characteristic curve**.

Nelson, Watson and Murphy from *Universal Oil Products (U.O.P.)* found a mathematical relation between specific gravity and boiling point, in order to obtain **parametric curves** best fitting those of the different hydrocarbon types.

$$K_{UOP} = \frac{\sqrt[3]{T_{bp}}}{spgr_{60}^{60}}$$

where T_{bp} denotes the normal boiling point (R)

The value of the KUOP parameter characterizes the chemical hydrocarbon type. Hence:

- **K_{UOP} = 13** : the parametric curve corresponds to **Paraffins**
- **K_{UOP} = 11** : the parametric curve corresponds to **Naphthenes**
- **K_{UOP} = 10** : the parametric curve corresponds to **Aromatics**

On the other hand, we notice that the different series of hydrocarbons with the same number of carbon atoms are situated along some curves. This remark allows to identify curves with **constant molecular weight**. See **Page 3**.

This diagram is not used only for pure hydrocarbons, but also for **petroleum cuts**. The specific gravity of a given cut can be **easily measured**. However, a hydrocarbon cut does not have a **single boiling point**. Starting from an idea on the volatility of a given petroleum cut (given by the **ASTM or TBP distillation**), J.B. MAXWELL found a statistic correlation that affects to the cut a **mean average temperature** (**tmav**) allowing to use the K_{UOP} diagram.

Hence, for a any petroleum cut, **two simple measurements** :

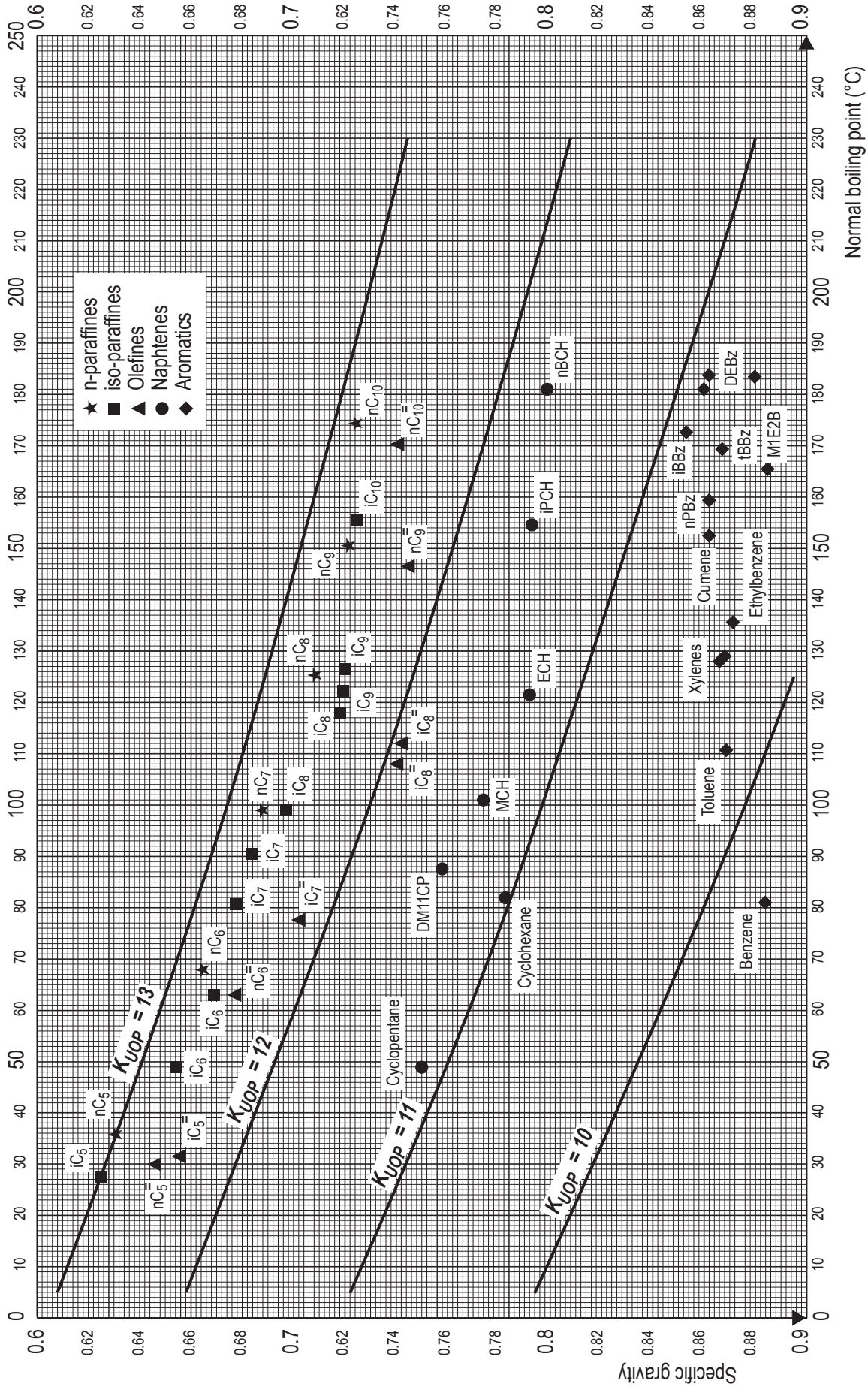
- **specific gravity** at liquid state
- **ASTM distillation** (or TBP distillation)

enable :

- to determine the **gross chemical type** of the cut by means of the K_{UOP} value
- to assess the **average molecular weight** of the hydrocarbons of the cut by using the constant molecular weight curves

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CHARACTERIZATION OF SOME HYDROCARBONS



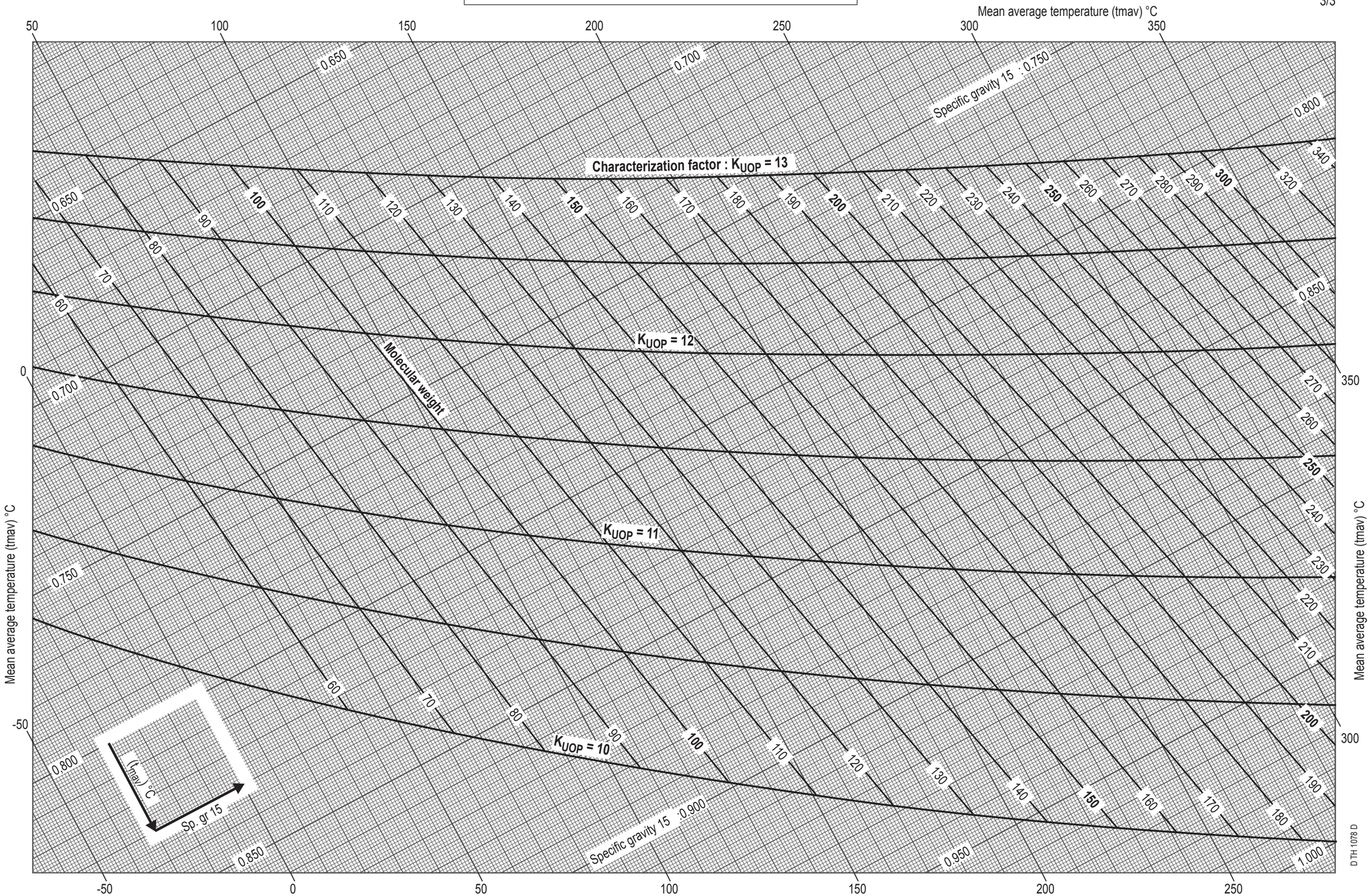
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K_{UOP} DIAGRAM - AVERAGE MOLECULAR WEIGHT

3/3



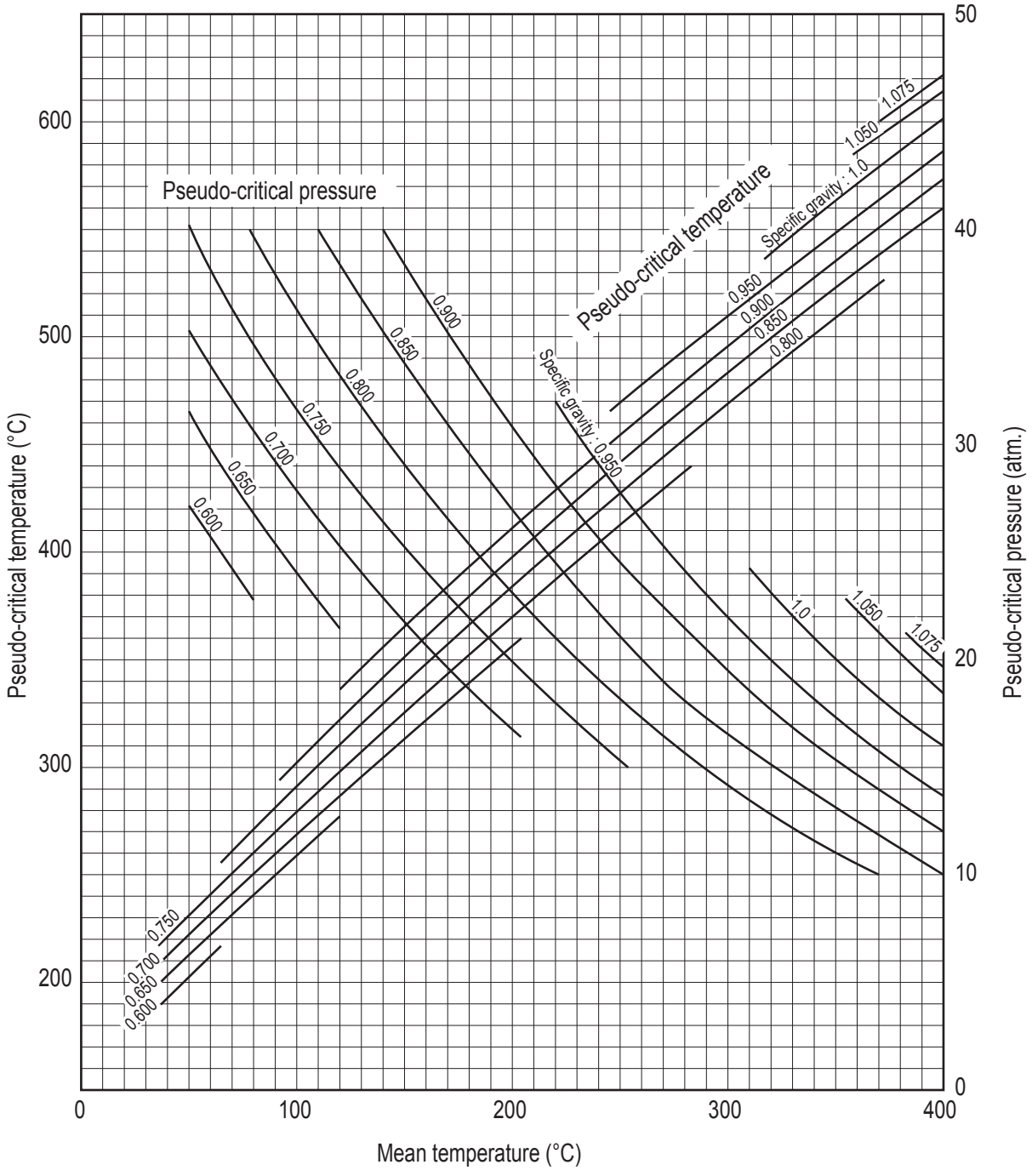
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PSEUDO-CRITICAL TEMPERATURE AND PRESSURE FOR PETROLEUM CUTS

Mean temperatures to be used

— for pseudo-critical pressure : t_{mav}

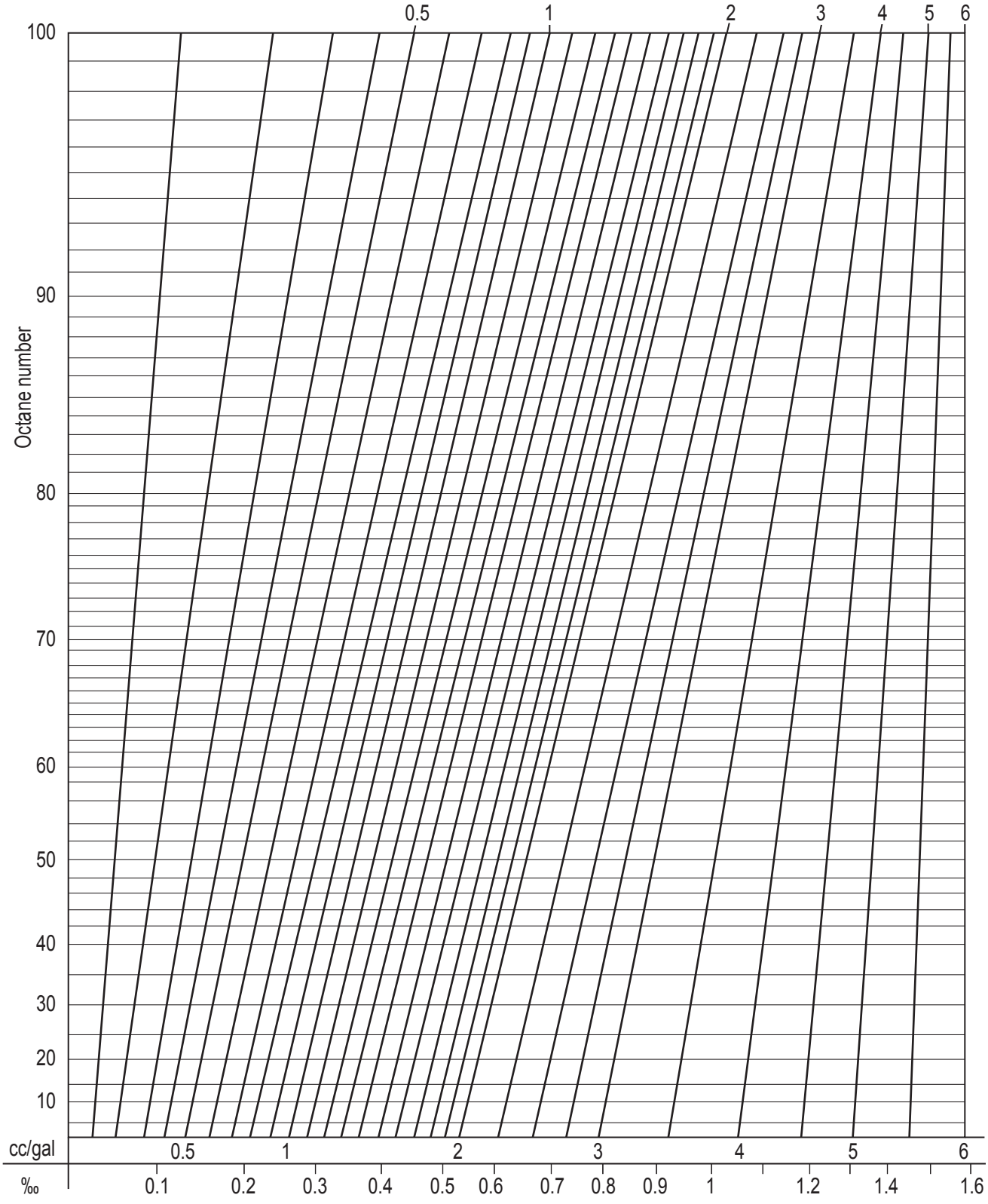
— for pseudo-critical température : $t_{molecular}$



D TH 1404 B

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AUTOMOTIVE FUEL SUSCEPTIBILITY TO TETRAETHYL LEAD
— ASTM-CFR Method - Hebl. Rendel & Garten —



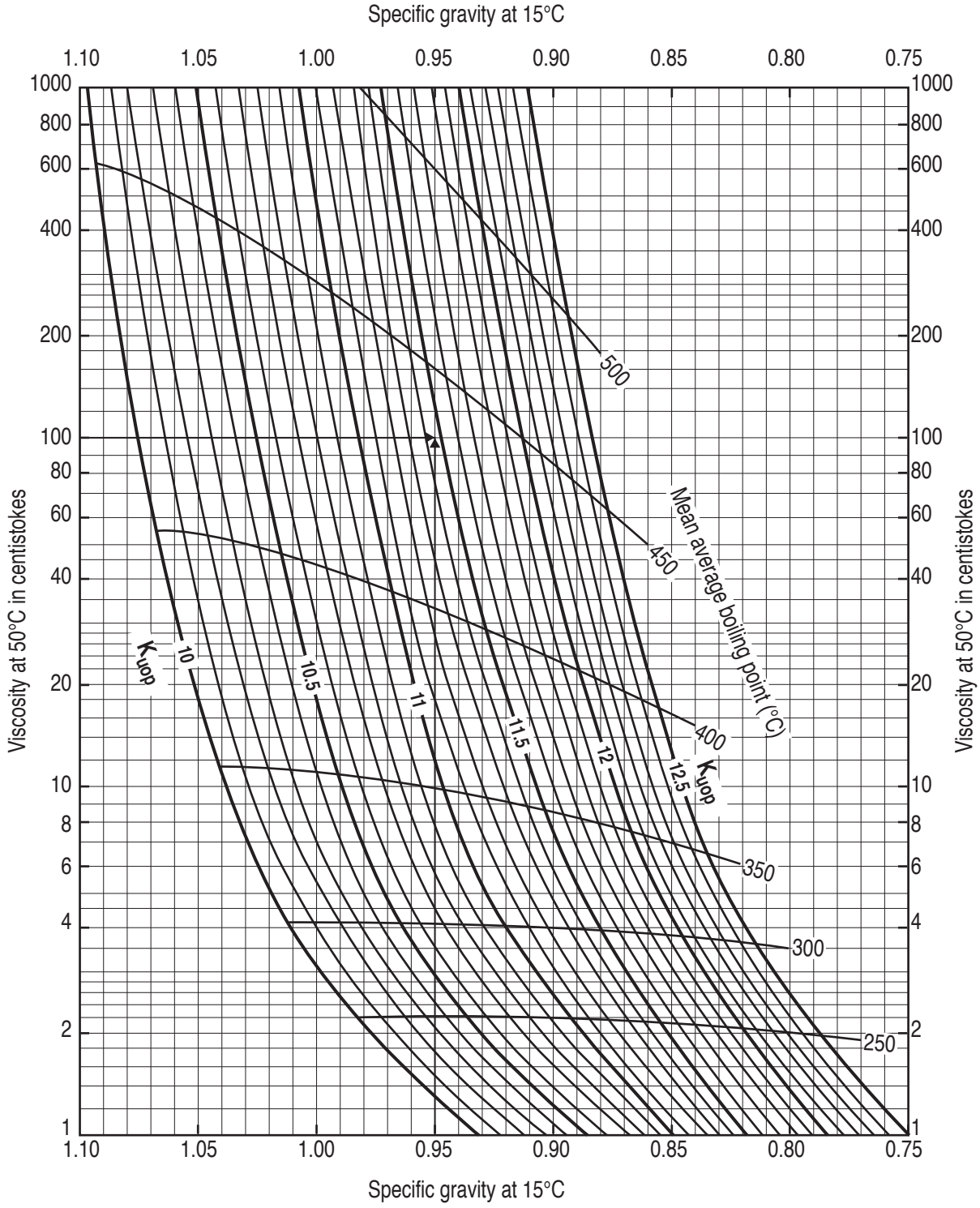
D TH 1622 D

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K_{UOP} DETERMINATION FOR HEAVY FEEDS

Example

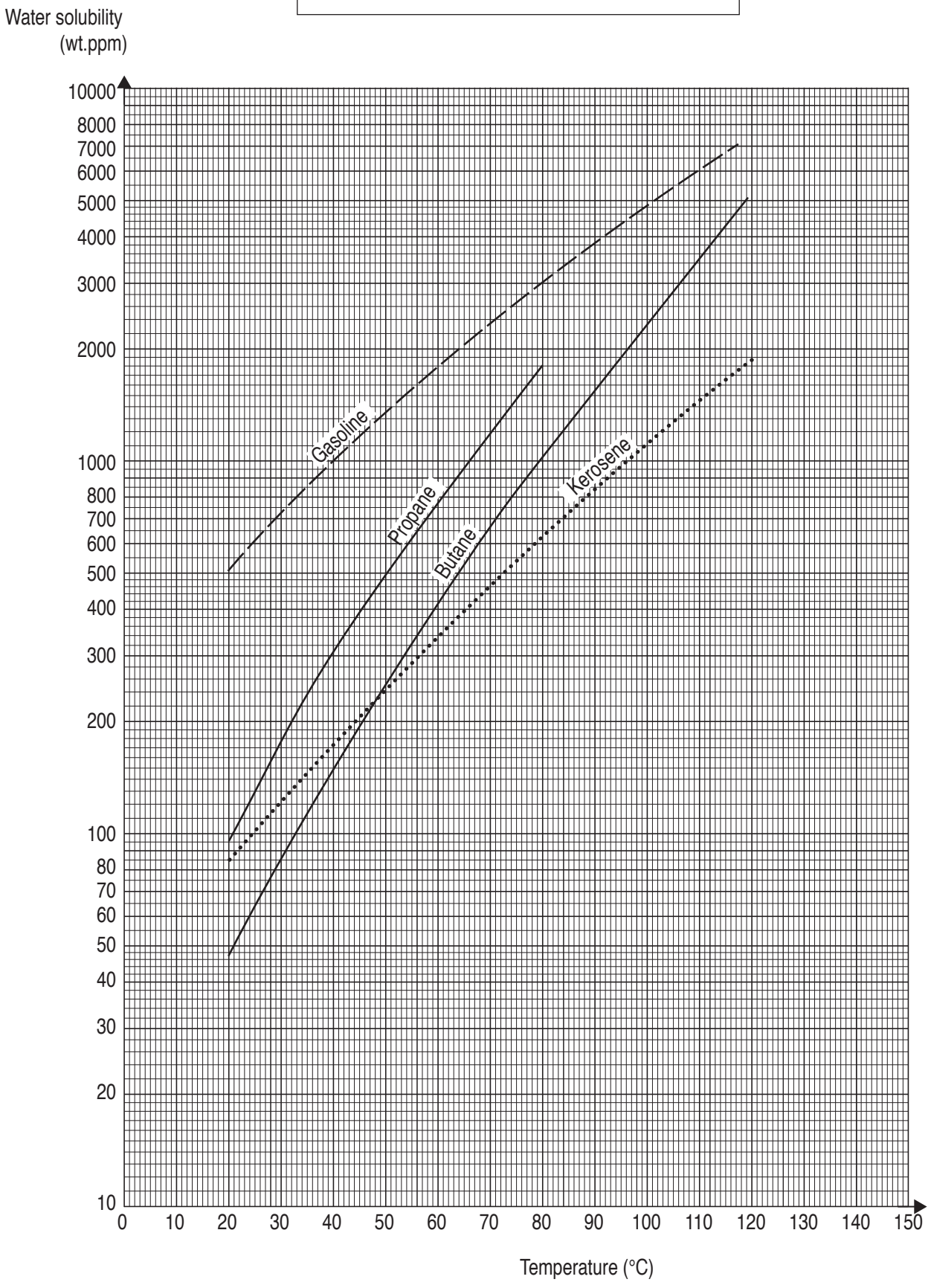
{ Specific gravity at 15°C = 0.950
Viscosity at 50°C = 100 cSt ⇒ **K_{uop} = 11.43**



D PCD 1352 D

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WATER SOLUBILITY
— In some liquid petroleum cuts —



From Handbook of natural gas engineering

D PPC 1415 D

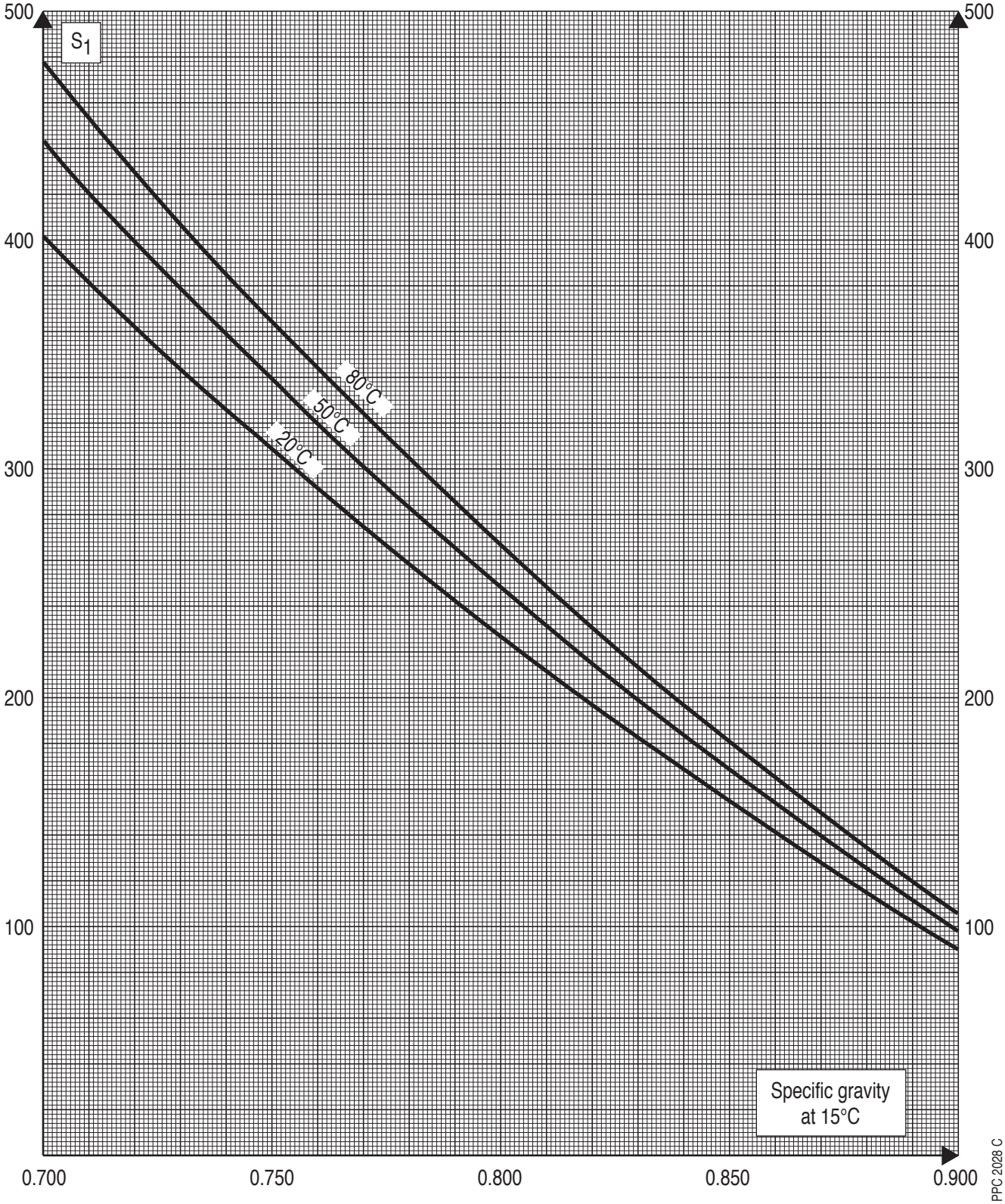
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AIR SOLUBILITY IN LIQUID PETROLEUM PRODUCTS — From NASA —

$$S_P = S_1 \times P$$

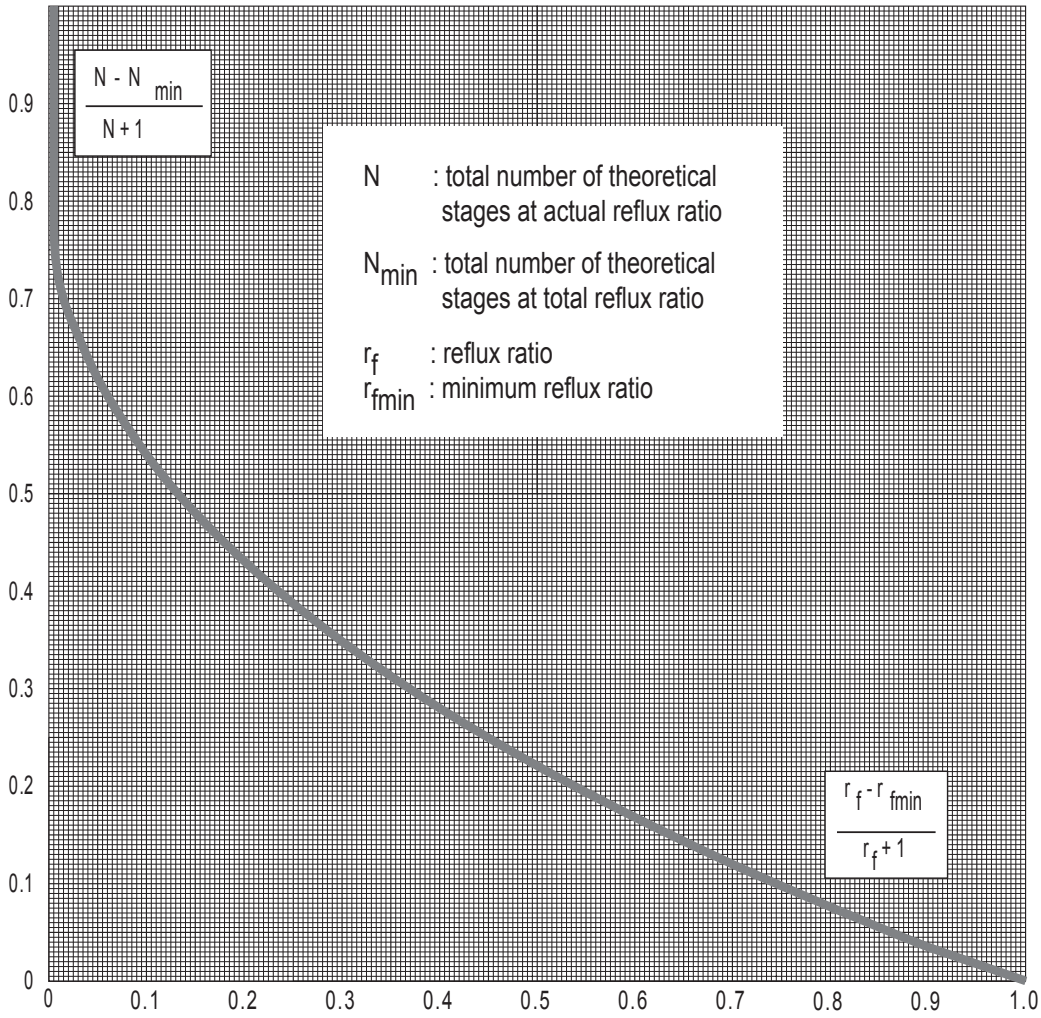
S_P : Air solubility in mass ppm at the pressure P (in abs.atm)

S_1 : Air solubility in mass ppm at atmospheric pressure



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GILLILAND'S CORRELATION



• Analytical forms

Eduljee (1975)

$$\frac{N-N_{min}}{N+1} = 0.75\left[1-\left(\frac{r_f-r_{fmin}}{r_f+1}\right)^{0.5668}\right]$$ Valid for > 0.01

Chang (1981)

$$\frac{N-N_{m+1}}{N+1} = 1-\exp\left[1.490 + 0.315\left(\frac{r_f-r_{fmin}}{r_f+1}\right) - \frac{1.805}{\left(\frac{r_f-r_{fmin}}{r_f+1}\right)^{0.1}}\right]$$

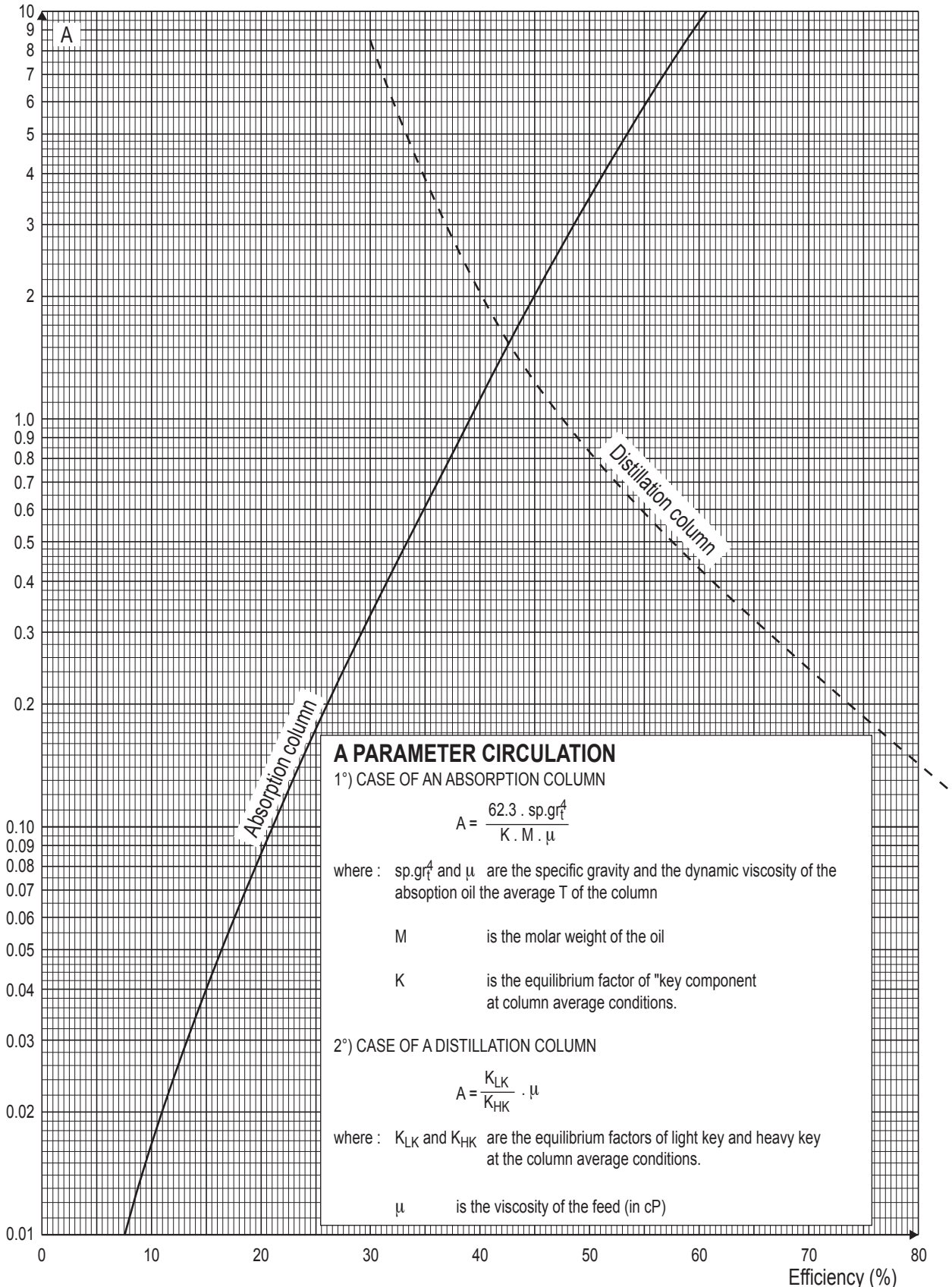
Molokanov (1972)

$$\frac{N-N_{min}}{N+1} = 1-\exp\left[\frac{\left(1+54.4\ \frac{r_f-r_{fmin}}{r_f+1}\right)}{\left(11+117.2\ \frac{r_f-r_{fmin}}{r_f+1}\right)} \cdot \frac{\left(\frac{r_f-r_{fmin}}{r_f+1}-1\right)}{\left(\frac{r_f-r_{fmin}}{r_f+1}\right)^{0.5}}\right]$$

D POD 2236 B

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TRAY EFFICIENCY
– For distillation columns and absorption columns –
(O'Connell)



A PARAMETER CIRCULATION

1°) CASE OF AN ABSORPTION COLUMN

$$A = \frac{62.3 \,.\, sp.gr_t^4}{K \,.\, M \,.\, \mu}$$

where : $sp.gr_t^4$ and μ are the specific gravity and the dynamic viscosity of the absoption oil the average T of the column

M is the molar weight of the oil

K is the equilibrium factor of "key component at column average conditions.

2°) CASE OF A DISTILLATION COLUMN

$$A = \frac{K_{LK}}{K_{HK}} \,.\, \mu$$

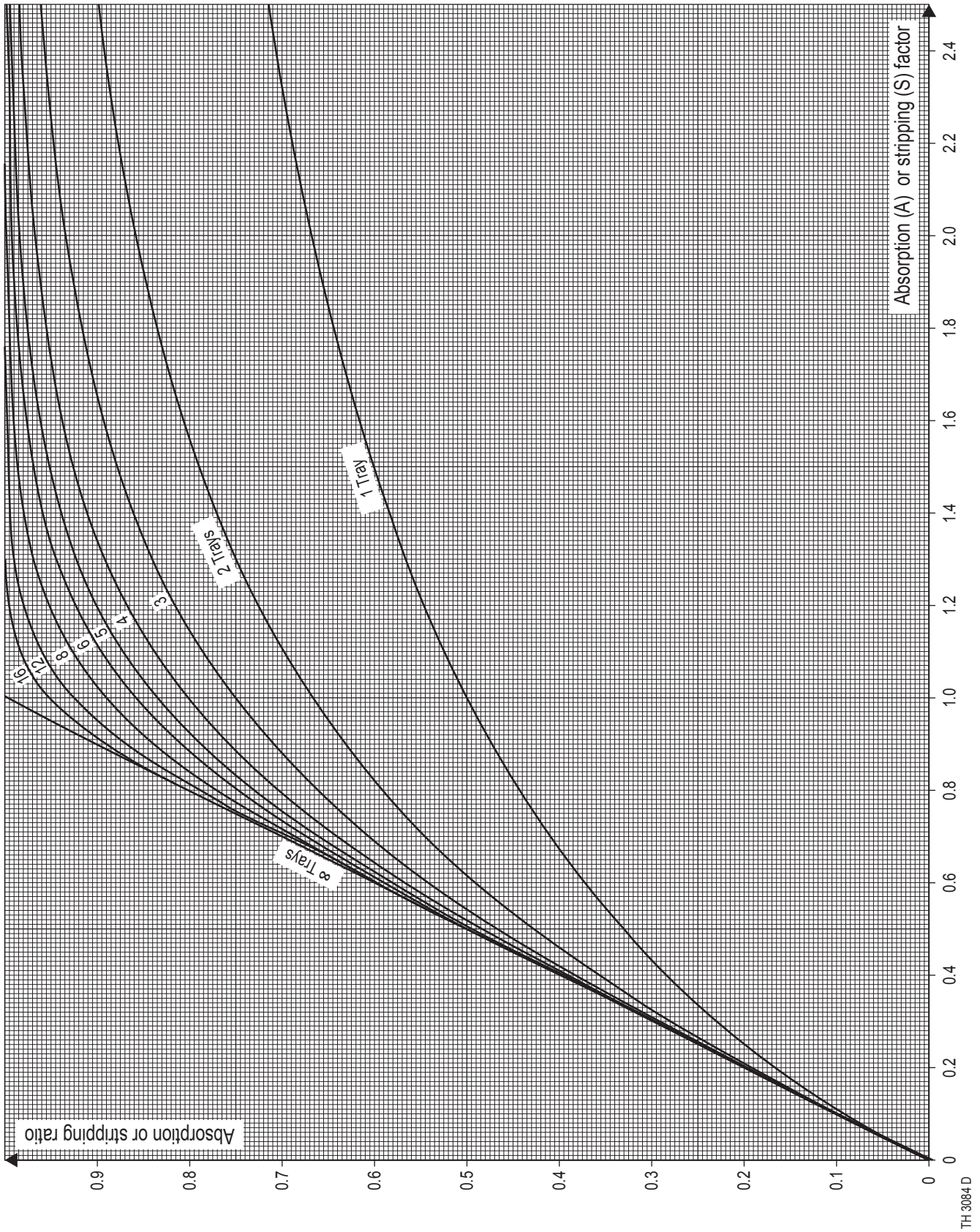
where : K_{LK} and K_{HK} are the equilibrium factors of light key and heavy key at the column average conditions.

μ is the viscosity of the feed (in cP)

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ABSORPTION (OR STRIPPING) RATIO VERSUS ABSORPTION (OR STRIPPING) FACTOR



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FLAMMABILITY

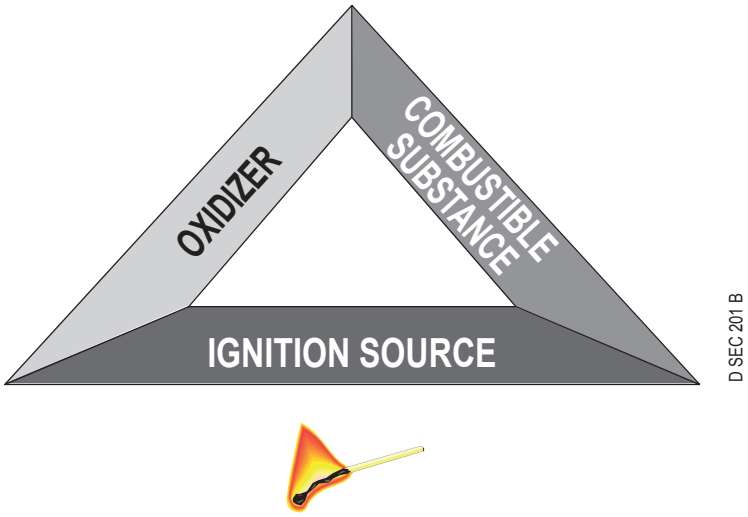
1 - THE FIRE TRIANGLE

Combustion is the high temperature oxidation of a **gaseous** substance. When the reaction is initiated it propagates with a heat release.

But combustion can only occur if three elements are put together:

- an oxidizer
- a combustible substance
- an ignition source

This is illustrated by the fire triangle.



2 - OXIDIZERS

Examples

Nitrous oxide	N_20	Fluorine	F_2	Sodium nitrate	$NaN0_2$
Nitrogen monoxide	NO	Chlorine	Cl_2	Hydrogen peroxide	H_20_2
Chlorine trifluoride	ClF3	Iodine	I_2		
Nitric acid	HNO_3	Bromine	Br_2		

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3 - COMBUSTIBLE SUBSTANCES

a - Necessary conditions

- combustible must be **gaseous; liquids and solids do not burn**!
- combustible and oxidant must be mixed in given proportions

The **flash point** of a combustible liquid and the **flammability limits** for gases characterize these conditions.

b - Flash point temperature

Products	Flash point
Crude oil	< 55°C
Gasoline	approx. – 40°C
Jet petroleum	> 38°C
Diesel oil	> 55°C and < 120°C
Lube oil	> 160°C
Benzene	– 1°C
Hexane	– 22°C
Steam cracking tars	80 to 120°C

The flash point (FP) determines storage conditions:

FP < 55°C: floating roof tank

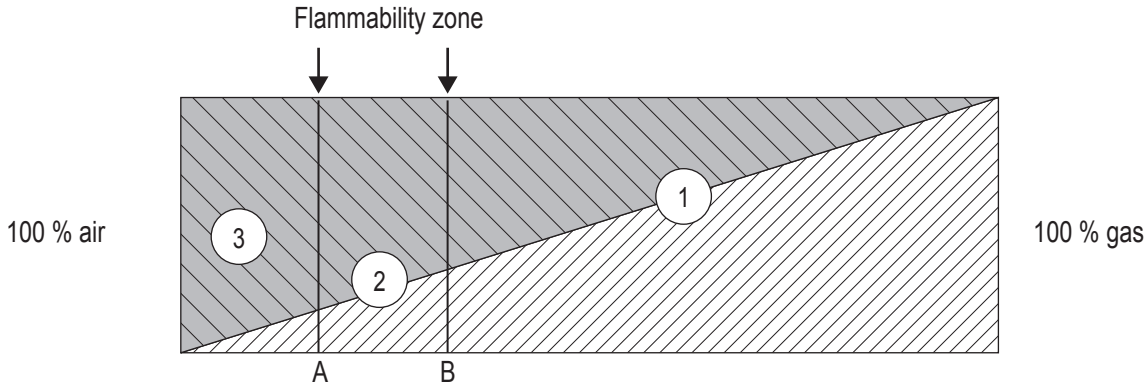
55°C < FP < 100°C fixed roof tanks with N_2 or steam blanketing

FP > 100°C :fixed roof tanks

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c - Flammability limits

The diagram below illustrates the flammability limits (% vol. in air).



Zone 3 : The air-gaseous fuel mixture is too lean, combustion is impossible
Zone 1 : The air-gaseous fuel mixture is too rich, combustion is impossible
Zone 2 : The air-gaseous fuel mixture is in the right proportions, combustion occurs
A - LFL Low Flammability Limit
B - UFL Upper Flammability Limit

Carbon monoxide	12,5	74
Hydrogen	4	75
Hydrogen sulfide	4,3	45
Methane	5,3	14
Ethylene	3,1	32
Propane	2,2	9,5
N-Butane	1,9	8,5
Butadiene	2	12
Methyl chloride	10,7	17
Hexane	1,1	
Gasoline vapors	1,5	8
Jet vapors	1	6
Diesel-oil vapors	6	14
Solvant	0,8	3,7
Tar vapors	6	14

D SEC 2037 C

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d - Solid fuels

Many solid organic and inorganic dusts can ignite or explode if they form a cloud.

As it is for gases, each product has a minimum concentration in the air below which explosion will not occur.

Knowing **maximum concentration** of dust in the atmosphere is rarely of any interest. The values ranges approximately from 1 to 3 kg/m^3.

Product	Minimum. concentration in air g/m^3	Minimum ignition energy
Sulfur dust	35	I5 mJ
PTFE dust	80	I5 mJ
Polyethylene dust	20	30 mJ
Urea-formaldehyde diameter 10 microns	85	80 mJ
Phenol-formaldehyde diameter 10 microns	25	15 mJ
PVC diameter 200 microns	?	200 to 300 mJ

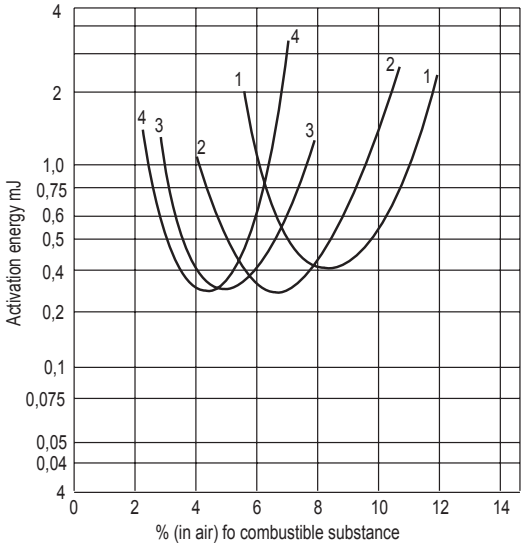
The risks of such dust clouds are present in operations such as grinding, pneumatic transportation, handling (emptying or filling silos, shoveling, shaking in sieves, etc.).

4 - IGNITION SOURCES

a - Minimum ignition energy

When flammable vapors and air are closely mixed within the explosive range, an energy input may trigger the combustion mechanism.

The graph below indicates the minimum ignition energy for the more common hydrocarbons versus their concentration in the air.



Ref	Fuel	% Stoechio-métrique
1	CH_4	9,5
2	C_2H_6	5,68
3	C_3H_8	4,03
4	C_4H_{10}	3,14

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Products (gas)	Minimum ignition energy (mJ)
Acetylene	0.017
Ethylène	0.07
Hydrogen	0.017
Carbon sulfide	0.009
Ethylene oxide	0.1

Minimum ignition energy for gaseous combustibles

Products (dust)	Minimum ignition energy (mJ)
Sulfur	15
Polyethylene	30
Phtalic anhydride	15
Flour	60
Wood	40

Minimum ignition energy for dust

b - Miscellaneous ignition sources

- **flames** produced by welding, matches, lighters, etc.
- **rise in temperature** that can bring the mixture to its **self-ignition temperature**. The product ignites spontaneously in the presence of air, without any flame or spark being present

The rise in temperature may be due to:

- hot equipment: furnaces, heat exchangers, piping, etc.
- hot spots created by maintenance works (mechanical sawing,)
- non safety electric equipment
- abnormal heating because of friction or seizing of machine parts: bearings, packing box, packing, etc.
- exothermic chemical reactions

Combustible	Self-ignition temperature (°C)
Carbon monoxide	610
Hydrogen	585
Hydrogen sulfide	260
Methane	535
Ethylene	450
Ethane	515
Propane	465
Butane	405

Sparks caused by:

- impact of metal on metal or metal on dry concrete (tools, cutting with a grinding wheel, hose nozzles, hob-nailed shoes);
- electric welding sets
- electric equipment other than the "Safety" kind used in explosive atmospheres
- static electricity: the main cause of accidents on tanks (storage tank, tank car, tank truck, etc.) during loading, unloading, sampling, …

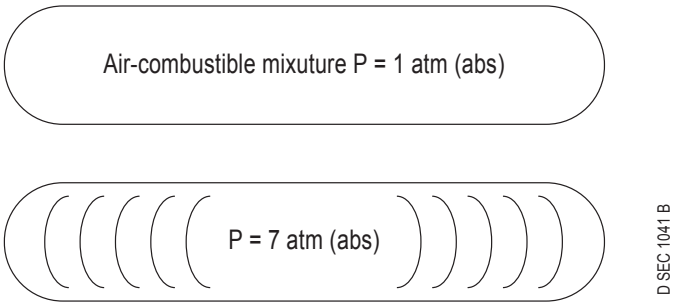
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5 - DIFFERENT TYPES OF COMBUSTION

Combustion reactions are generally triggered by a pinpoint ignition source for gaseous mixtures, at one point in the mixture. Combustion is then propagated little by little in the fresh gas mixture, while the reaction products raised to a high temperature and pressure, accumulate. These reactions give rise to an explosion.

• Deflagration

The flame velocity is subsonic (from a few meters per second up to 300 m/s).



Effect of a deflagration in a confined space

This simple and practical method represents reality with some accuracy, but the pressure wave propagation caused by the explosion of TNT as not exactly the same as that caused by a VCE.

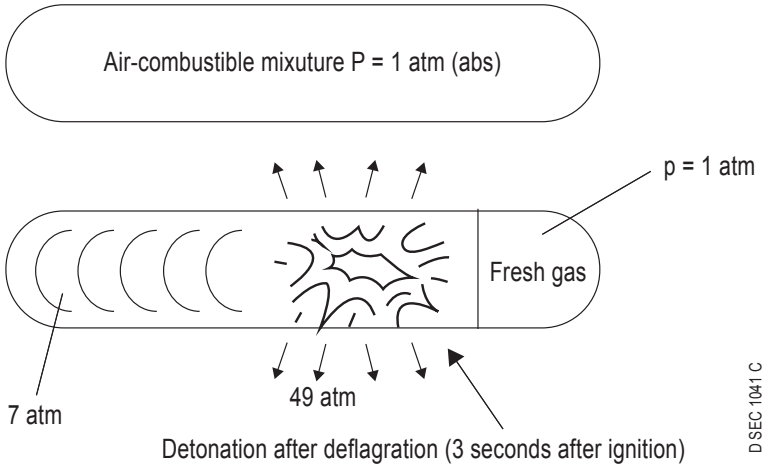
Other more complex approaches can be used to get a better grasp of the phenomenon, though they can not quantify them exactly.

• Detonation

The propagation velocity of the flame is supersonic (from 300 m/s up to 3200 m/s).

Detonation:

- is possible only in a confined or congested areas
- generates very high short-term pressures

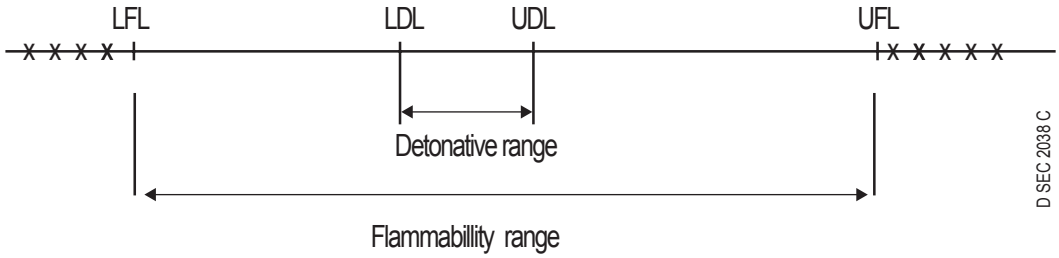


Initial conditions

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- requires the following conditions:

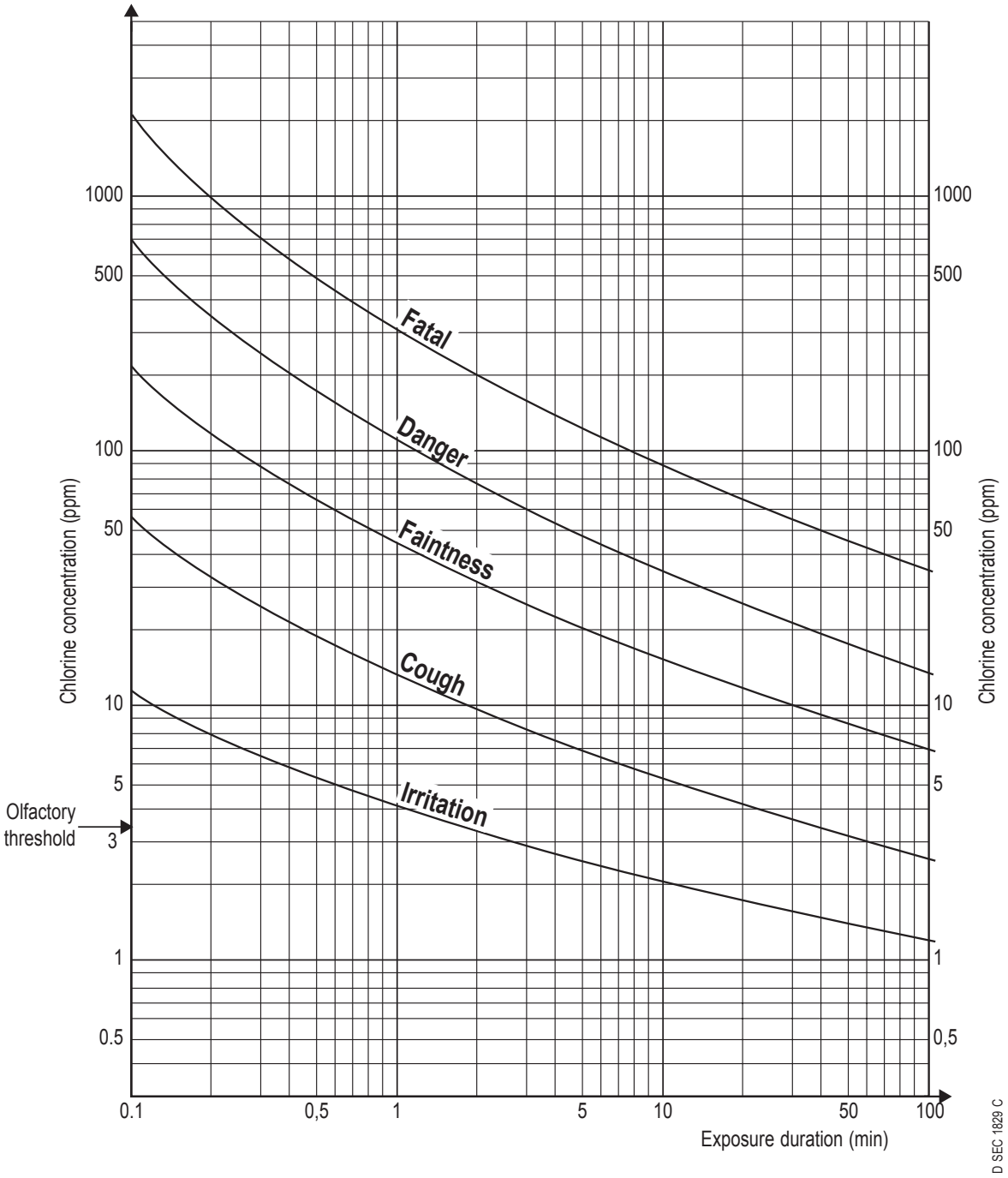
  - high minimum ignition energy much more than deflagration
  - composition of gaseous mixture within detonability limits



LDL : Lower detonability limit
UDL : Upper detonability limit

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PHYSIOLOGICAL EFFECTS OF CHLORINE



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TEMPERATURE CONVERSION TABLE

°C	T	°F	°C	T	°F	°C	T	°F	°C	T	°F
-273.0	**-459.4**		-33.9	**-29**	-20.2	1.1	**34**	93.2	36.1	**97**	206.6
-267.8	**-450**		-33.3	**-28**	-18.4	1.7	**35**	95	36.7	**98**	208.4
-262.2	**-440**		-32.8	**-27**	-16.6	2.2	**36**	96.8	37.2	**99**	210.2
-256.7	**-430**		-32.2	**-26**	-14.8	2.8	**37**	98.6	37.8	**100**	212
-251.1	**-420**		-31.7	**-25**	-13	3.3	**38**	100.4	38.3	**101**	213.8
-245.6	**-410**		-31.1	**-24**	-11.2	3.9	**39**	102.2	38.9	**102**	215.6
-240.0	**-400**		-30.6	**-23**	-9.4	4.4	**40**	104	39.4	**103**	217.4
-234.4	**-390**		-30.0	**-22**	-7.6	5.0	**41**	105.8	40.0	**104**	219.2
-228.9	**-380**		-29.4	**-21**	-5.8	5.6	**42**	107.6	40.6	**105**	221
-223.3	**-370**		-28.9	**-20**	-4	6.1	**43**	109.4	41.1	**106**	222.8
-217.8	**-360**		-28.3	**-19**	-2.2	6.7	**44**	111.2	41.7	**107**	224.6
-212.2	**-350**		-27.8	**-18**	-0.4	7.2	**45**	113	42.2	**108**	226.4
-206.7	**-340**		-27.2	**-17**	1.4	7.8	**46**	114.8	42.8	**109**	228.2
-201.1	**-330**		-26.7	**-16**	3.2	8.3	**47**	116.6	43.3	**110**	230
-195.6	**-320**		-26.1	**-15**	5	8.9	**48**	118.4	43.9	**111**	231.8
-190.0	**-310**		-25.6	**-14**	6.8	9.4	**49**	120.2	44.4	**112**	233.6
-184.4	**-300**		-25.0	**-13**	8.6	10.0	**50**	122	45.0	**113**	235.4
-178.9	**-290**		-24.4	**-12**	10.4	10.6	**51**	123.8	45.6	**114**	237.2
-173.3	**-280**		-23.9	**-11**	12.2	11.1	**52**	125.6	46.1	**115**	239
-169.4	**-273**	-459.4	-23.3	**-10**	14	11.7	**53**	127.4	46.7	**116**	240.8
-167.8	**-270**	-454	-22.8	**-9**	15.8	12.2	**54**	129.2	47.2	**117**	242.6
-162.2	**-260**	-436	-22.2	**-8**	17.6	12.8	**55**	131	47.8	**118**	244.4
-156.7	**-250**	-418	-21.7	**-7**	19.4	13.3	**56**	132.8	48.3	**119**	246.2
-151.1	**-240**	-400	-21.1	**-6**	21.2	13.9	**57**	134.6	48.9	**120**	248
-145.6	**-230**	-382	-20.6	**-5**	23	14.4	**58**	136.4	49.4	**121**	249.8
-140.0	**-220**	-364	-20.0	**-4**	24.8	15.0	**59**	138.2	50.0	**122**	251.6
-134.4	**-210**	-346	-19.4	**-3**	26.6	15.6	**60**	140	50.6	**123**	253.4
-128.9	**-200**	-328	-18.9	**-2**	28.4	16.1	**61**	141.8	51.1	**124**	255.2
-123.3	**-190**	-310	-18.3	**-1**	30.2	16.7	**62**	143.6	51.7	**125**	257
-117.8	**-180**	-292	-17.8	**0**	32	17.2	**63**	145.4	52.2	**126**	258.8
-112.2	**-170**	-274	-17.2	**1**	33.8	17.8	**64**	147.2	52.8	**127**	260.6
-106.7	**-160**	-256	-16.7	**2**	35.6	18.3	**65**	149	53.3	**128**	262.4
-101.1	**-150**	-238	-16.1	**3**	37.4	18.9	**66**	150.8	53.9	**129**	264.2
-95.6	**-140**	-220	-15.6	**4**	39.2	19.4	**67**	152.6	54.4	**130**	266
-90.0	**-130**	-202	-15.0	**5**	41	20.0	**68**	154.4	55.0	**131**	267.8
-84.4	**-120**	-184	-14.4	**6**	42.8	20.6	**69**	156.2	55.6	**132**	269.6
-78.9	**-110**	-166	-13.9	**7**	44.6	21.1	**70**	158	56.1	**133**	271.4
-73.3	**-100**	-148	-13.3	**8**	46.4	21.7	**71**	159.8	56.7	**134**	273.2
-67.8	**-90**	-130	-12.8	**9**	48.2	22.2	**72**	161.6	57.2	**135**	275
-62.2	**-80**	-112	-12.2	**10**	50	22.8	**73**	163.4	57.8	**136**	276.8
-56.7	**-70**	-94	-11.7	**11**	51.8	23.3	**74**	165.2	58.3	**137**	278.6
-51.1	**-60**	-76	-11.1	**12**	53.6	23.9	**75**	167	58.9	**138**	280.4
-45.6	**-50**	-58	-10.6	**13**	55.4	24.4	**76**	168.8	59.4	**139**	282.2
-45.0	**-49**	-56.2	-10.0	**14**	57.2	25.0	**77**	170.6	60.0	**140**	284
-44.4	**-48**	-54.4	-9.4	**15**	59	25.6	**78**	172.4	60.6	**141**	285.8
-43.9	**-47**	-52.6	-8.9	**16**	60.8	26.1	**79**	174.2	61.1	**142**	287.6
-43.3	**-46**	-50.8	-8.3	**17**	62.6	26.7	**80**	176	61.7	**143**	289.4
-42.8	**-45**	-49	-7.8	**18**	64.4	27.2	**81**	177.8	62.2	**144**	291.2
-42.2	**-44**	-47.2	-7.2	**19**	66.2	27.8	**82**	179.6	62.8	**145**	293
-41.7	**-43**	-45.4	-6.7	**20**	68	28.3	**83**	181.4	63.3	**146**	294.8
-41.1	**-42**	-43.6	-6.1	**21**	69.8	28.9	**84**	183.2	63.9	**147**	296.6
-40.6	**-41**	-41.8	-5.6	**22**	71.6	29.4	**85**	185	64.4	**148**	298.4
-40.0	**-40**	-40	-5.0	**23**	73.4	30.0	**86**	186.8	65.0	**149**	300.2
-39.4	**-39**	-38.2	-4.4	**24**	75.2	30.6	**87**	188.6	65.6	**150**	302
-38.9	**-38**	-36.4	-3.9	**25**	77	31.1	**88**	190.4	66.1	**151**	303.8
-38.3	**-37**	-34.6	-3.3	**26**	78.8	31.7	**89**	192.2	66.7	**152**	305.6
-37.8	**-36**	-32.8	-2.8	**27**	80.6	32.2	**90**	194	67.2	**153**	307.4
-37.2	**-35**	-31	-2.2	**28**	82.4	32.8	**91**	195.8	67.8	**154**	309.2
-36.7	**-34**	-29.2	-1.7	**29**	84.2	33.3	**92**	197.6	68.3	**155**	311
-36.1	**-33**	-27.4	-1.1	**30**	86	33.9	**93**	199.4	68.9	**156**	312.8
-35.6	**-32**	-25.6	-0.6	**31**	87.8	34.4	**94**	201.2	69.4	**157**	314.6
-35.0	**-31**	-23.8	0.0	**32**	89.6	35.0	**95**	203	70.0	**158**	316.4
-34.4	**-30**	-22	0.6	**33**	91.4	35.6	**96**	204.8	70.6	**159**	318.2

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TEMPERATURE CONVERSION TABLE

°C	T	°F	°C	T	°F	°C	T	°F	°C	T	°F
71.1	160	320	101.1	214	417.2	221.1	430	806	521.1	970	1778
71.7	161	321.8	101.7	215	419	226.7	440	824	526.7	980	1796
72.2	162	323.6	102.2	216	420.8	232.2	450	842	532.2	990	1814
72.8	163	325.4	102.8	217	422.6	237.8	460	860	537.8	1000	1832
73.3	164	327.2	103.3	218	424.4	243.3	470	878	543.3	1010	1850
73.9	165	329	103.9	219	426.2	248.9	480	896	548.9	1020	1868
74.4	166	330.8	104.4	220	428	254.4	490	914	554.4	1030	1886
75.0	167	332.6	105.0	221	429.8	260.0	500	932	560.0	1040	1904
75.6	168	334.4	105.6	222	431.6	265.6	510	950	565.6	1050	1922
76.1	169	336.2	106.1	223	433.4	271.1	520	968	571.1	1060	1940
76.7	170	338	106.7	224	435.2	276.7	530	986	576.7	1070	1958
77.2	171	339.8	107.2	225	437	282.2	540	1004	582.2	1080	1976
77.8	172	341.6	107.8	226	438.8	287.8	550	1022	587.8	1090	1994
78.3	173	343.4	108.3	227	440.6	293.3	560	1040	593.3	1100	2012
78.9	174	345.2	108.9	228	442.4	298.9	570	1058	598.9	1110	2030
79.4	175	347	109.4	229	444.2	304.4	580	1076	604.4	1120	2048
80.0	176	348.8	110.0	230	446	310.0	590	1094	610.0	1130	2066
80.6	177	350.6	110.6	231	447.8	315.6	600	1112	615.6	1140	2084
81.1	178	352.4	111.1	232	449.6	321.1	610	1130	621.1	1150	2102
81.7	179	354.2	111.7	233	451.4	326.7	620	1148	626.7	1160	2120
82.2	180	356	112.2	234	453.2	332.2	630	1166	632.2	1170	2138
82.8	181	357.8	112.8	235	455	337.8	640	1184	637.8	1180	2156
83.3	182	359.6	113.3	236	456.8	343.3	650	1202	643.3	1190	2174
83.9	183	361.4	113.9	237	458.6	348.9	660	1220	648.9	1200	2192
84.4	184	363.2	114.4	238	460.4	354.4	670	1238	654.4	1210	2210
85.0	185	365	115.0	239	462.2	360.0	680	1256	660.0	1220	2228
85.6	186	366.8	115.6	240	464	365.6	690	1274	665.6	1230	2246
86.1	187	368.6	116.1	241	465.8	371.1	700	1292	671.1	1240	2264
86.7	188	370.4	116.7	242	467.6	376.7	710	1310	676.7	1250	2282
87.2	189	372.2	117.2	243	469.4	382.2	720	1328	682.2	1260	2300
87.8	190	374	117.8	244	471.2	387.8	730	1346	687.8	1270	2318
88.3	191	375.8	118.3	245	473	393.3	740	1364	693.3	1280	2336
88.9	192	377.6	118.9	246	474.8	398.9	750	1382	698.9	1290	2354
89.4	193	379.4	119.4	247	476.6	404.4	760	1400	704.4	1300	2372
90.0	194	381.2	120.0	248	478.4	410.0	770	1418	710.0	1310	2390
90.6	195	383	120.6	249	480.2	415.6	780	1436	715.6	1320	2408
91.1	196	384.8	121.1	250	482	421.1	790	1454	721.1	1330	2426
91.7	197	386.6	126.7	260	500	426.7	800	1472	726.7	1340	2444
92.2	198	388.4	132.2	270	518	432.2	810	1490	732.2	1350	2462
92.8	199	390.2	137.8	280	536	437.8	820	1508	737.8	1360	2480
93.3	200	392	143.3	290	554	443.3	830	1526	743.3	1370	2498
93.9	201	393.8	148.9	300	572	448.9	840	1544	748.9	1380	2516
94.4	202	395.6	154.4	310	590	454.4	850	1562	754.4	1390	2534
95.0	203	397.4	160.0	320	608	460.0	860	1580	760.0	1400	2552
95.6	204	399.2	165.6	330	626	465.6	870	1598	765.6	1410	2570
96.1	205	401	171.1	340	644	471.1	880	1616	771.1	1420	2588
96.7	206	402.8	176.7	350	662	476.7	890	1634	776.7	1430	2606
97.2	207	404.6	182.2	360	680	482.2	900	1652	782.2	1440	2624
97.8	208	406.4	187.8	370	698	487.8	910	1670	787.8	1450	2642
98.3	209	408.2	193.3	380	716	493.3	920	1688	793.3	1460	2660
98.9	210	410	198.9	390	734	498.9	930	1706	798.9	1470	2678
99.4	211	411.8	204.4	400	752	504.4	940	1724	804.4	1480	2696
100.0	212	413.6	210.0	410	770	510.0	950	1742	810.0	1490	2714
100.6	213	415.4	215.6	420	788	515.6	960	1760	815.6	1500	2732

Interpolation table

°C	0.56	1.11	1.67	2.22	2.78	3.33	3.89	4.44	5.00	5.56	6.11	6.67	7.22	7.78	8.33	8.89	9.44	10.00	10.56	11.1
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
°F	1.8	3.6	5.4	7.2	9	10.8	12.6	14.4	16.2	18	19.8	21.6	23.4	25.2	27	28.8	30.6	32.4	34.2	36

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RELATION SPECIFIC GRAVITY - API DEGREES

Tenth of API degrees

°API	0	1	2	3	4	5	6	7	8	9
0	1.0760	1.0752	1.0744	1.0736	1.0728	1.0720	1.0712	1.0703	1.0695	1.0687
1	1.0679	1.0671	1.0663	1.0655	1.0647	1.0639	1.0631	1.0623	1.0615	1.0607
2	1.0599	1.0591	1.0583	1.0575	1.0568	1.0560	1.0552	1.0544	1.0536	1.0528
3	1.0520	1.0513	1.0505	1.0497	1.0489	1.0481	1.0474	1.0466	1.0458	1.0451
4	1.0443	1.0435	1.0427	1.0420	1.0412	1.0404	1.0397	1.0389	1.0382	1.0374
5	1.0366	1.0359	1.0351	1.0344	1.0336	1.0328	1.0321	1.0313	1.0306	1.0298
6	1.0291	1.0283	1.0276	1.0269	1.0261	1.0254	1.0246	1.0239	1.0231	1.0224
7	1.0217	1.0209	1.0202	1.0195	1.0187	1.0180	1.0173	1.0165	1.0158	1.0151
8	1.0143	1.0136	1.0129	1.0122	1.0114	1.0107	1.0100	1.0093	1.0086	1.0078
9	1.0071	1.0064	1.0057	1.0050	1.0043	1.0035	1.0028	1.0021	1.0014	1.0007
10	1.0000	0.9993	0.9986	0.9979	0.9972	0.9965	0.9958	0.9951	0.9944	0.9937
11	0.9930	0.9923	0.9916	0.9909	0.9902	0.9895	0.9888	0.9881	0.9874	0.9868
12	0.9861	0.9854	0.9847	0.9840	0.9833	0.9826	0.9820	0.9813	0.9806	0.9799
13	0.9792	0.9786	0.9779	0.9772	0.9765	0.9759	0.9752	0.9745	0.9738	0.9732
14	0.9725	0.9718	0.9712	0.9705	0.9698	0.9692	0.9685	0.9679	0.9672	0.9665
15	0.9659	0.9652	0.9646	0.9639	0.9632	0.9626	0.9619	0.9613	0.9606	0.9600
16	0.9593	0.9587	0.9580	0.9574	0.9567	0.9561	0.9554	0.9548	0.9541	0.9535
17	0.9529	0.9522	0.9516	0.9509	0.9503	0.9497	0.9490	0.9484	0.9478	0.9471
18	0.9465	0.9459	0.9452	0.9446	0.9440	0.9433	0.9427	0.9421	0.9415	0.9408
19	0.9402	0.9396	0.9390	0.9383	0.9377	0.9371	0.9365	0.9358	0.9352	0.9346
20	0.9340	0.9334	0.9328	0.9321	0.9315	0.9309	0.9303	0.9297	0.9291	0.9285
21	0.9279	0.9273	0.9267	0.9260	0.9254	0.9248	0.9242	0.9236	0.9230	0.9224
22	0.9218	0.9212	0.9206	0.9200	0.9194	0.9188	0.9182	0.9176	0.9170	0.9165
23	0.9159	0.9153	0.9147	0.9141	0.9135	0.9129	0.9123	0.9117	0.9111	0.9106
24	0.9100	0.9094	0.9088	0.9082	0.9076	0.9071	0.9065	0.9059	0.9053	0.9047
25	0.9042	0.9036	0.9030	0.9024	0.9018	0.9013	0.9007	0.9001	0.8996	0.8990
26	0.8984	0.8978	0.8973	0.8967	0.8961	0.8956	0.8950	0.8944	0.8939	0.8933
27	0.8927	0.8922	0.8916	0.8911	0.8905	0.8899	0.8894	0.8888	0.8883	0.8877
28	0.8871	0.8866	0.8860	0.8855	0.8849	0.8844	0.8838	0.8833	0.8827	0.8822
29	0.8816	0.8811	0.8805	0.8800	0.8794	0.8789	0.8783	0.8778	0.8772	0.8767
30	0.8762	0.8756	0.8751	0.8745	0.8740	0.8735	0.8729	0.8724	0.8718	0.8713
31	0.8708	0.8702	0.8697	0.8692	0.8686	0.8681	0.8676	0.8670	0.8665	0.8660
32	0.8654	0.8649	0.8644	0.8639	0.8633	0.8628	0.8623	0.8618	0.8612	0.8607
33	0.8602	0.8597	0.8591	0.8586	0.8581	0.8576	0.8571	0.8565	0.8560	0.8555
34	0.8550	0.8545	0.8540	0.8534	0.8529	0.8524	0.8519	0.8514	0.8509	0.8504
35	0.8498	0.8493	0.8488	0.8483	0.8478	0.8473	0.8468	0.8463	0.8458	0.8453
36	0.8448	0.8443	0.8438	0.8433	0.8428	0.8423	0.8418	0.8413	0.8408	0.8403
37	0.8398	0.8393	0.8388	0.8383	0.8378	0.8373	0.8368	0.8363	0.8358	0.8353
38	0.8348	0.8343	0.8338	0.8333	0.8328	0.8324	0.8319	0.8314	0.8309	0.8304
39	0.8299	0.8294	0.8289	0.8285	0.8280	0.8275	0.8270	0.8265	0.8260	0.8256
40	0.8251	0.8246	0.8241	0.8236	0.8232	0.8227	0.8222	0.8217	0.8212	0.8208
41	0.8203	0.8198	0.8193	0.8189	0.8184	0.8179	0.8174	0.8170	0.8165	0.8160
42	0.8156	0.8151	0.8146	0.8142	0.8137	0.8132	0.8128	0.8123	0.8118	0.8114
43	0.8109	0.8104	0.8100	0.8095	0.8090	0.8086	0.8081	0.8076	0.8072	0.8067
44	0.8063	0.8058	0.8054	0.8049	0.8044	0.8040	0.8035	0.8031	0.8026	0.8022
45	0.8017	0.8012	0.8008	0.8003	0.7999	0.7994	0.7990	0.7985	0.7981	0.7976
46	0.7972	0.7967	0.7963	0.7958	0.7954	0.7949	0.7945	0.7941	0.7936	0.7932
47	0.7927	0.7923	0.7918	0.7914	0.7909	0.7905	0.7901	0.7896	0.7892	0.7887
48	0.7883	0.7879	0.7874	0.7870	0.7865	0.7861	0.7857	0.7852	0.7848	0.7844
49	0.7839	0.7835	0.7831	0.7826	0.7822	0.7818	0.7813	0.7809	0.7805	0.7800
50	0.7796	0.7792	0.7788	0.7783	0.7779	0.7775	0.7770	0.7766	0.7762	0.7758
51	0.7753	0.7749	0.7745	0.7741	0.7736	0.7732	0.7728	0.7724	0.7720	0.7715
52	0.7711	0.7707	0.7703	0.7699	0.7694	0.7690	0.7686	0.7682	0.7678	0.7674
53	0.7669	0.7665	0.7661	0.7657	0.7653	0.7649	0.7645	0.7640	0.7636	0.7632
54	0.7628	0.7624	0.7620	0.7616	0.7612	0.7608	0.7603	0.7599	0.7595	0.7591
55	0.7587	0.7583	0.7579	0.7575	0.7571	0.7567	0.7563	0.7559	0.7555	0.7551
56	0.7547	0.7543	0.7539	0.7535	0.7531	0.7527	0.7523	0.7519	0.7515	0.7511
57	0.7507	0.7503	0.7499	0.7495	0.7491	0.7487	0.7483	0.7479	0.7475	0.7471
58	0.7467	0.7463	0.7459	0.7455	0.7451	0.7447	0.7443	0.7440	0.7436	0.7432
59	0.7428	0.7424	0.7420	0.7416	0.7412	0.7408	0.7405	0.7401	0.7397	0.7393
60	0.7389	0.7385	0.7381	0.7377	0.7374	0.7370	0.7366	0.7362	0.7358	0.7354
61	0.7351	0.7347	0.7343	0.7339	0.7335	0.7332	0.7328	0.7324	0.7320	0.7316
62	0.7313	0.7309	0.7305	0.7301	0.7298	0.7294	0.7290	0.7286	0.7283	0.7279
63	0.7275	0.7271	0.7268	0.7264	0.7260	0.7256	0.7253	0.7249	0.7245	0.7242
64	0.7238	0.7234	0.7230	0.7227	0.7223	0.7219	0.7216	0.7212	0.7208	0.7205

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RELATION SPECIFIC GRAVITY - API DEGREES

Tenth of API degrees

°API	0	1	2	3	4	5	6	7	8	9
65	0.7201	0.7197	0.7194	0.7190	0.7186	0.7183	0.7179	0.7175	0.7172	0.7168
66	0.7165	0.7161	0.7157	0.7154	0.7150	0.7146	0.7143	0.7139	0.7136	0.7132
67	0.7128	0.7125	0.7121	0.7118	0.7114	0.7111	0.7107	0.7103	0.7100	0.7096
68	0.7093	0.7089	0.7086	0.7082	0.7079	0.7075	0.7071	0.7068	0.7064	0.7061
69	0.7057	0.7054	0.7050	0.7047	0.7043	0.7040	0.7036	0.7033	0.7029	0.7026
70	0.7022	0.7019	0.7015	0.7012	0.7008	0.7005	0.7001	0.6998	0.6995	0.6991
71	0.6988	0.6984	0.6981	0.6977	0.6974	0.6970	0.6967	0.6964	0.6960	0.6957
72	0.6953	0.6950	0.6946	0.6943	0.6940	0.6936	0.6933	0.6929	0.6926	0.6923
73	0.6919	0.6916	0.6913	0.6909	0.6906	0.6902	0.6899	0.6896	0.6892	0.6889
74	0.6886	0.6882	0.6879	0.6876	0.6872	0.6869	0.6866	0.6862	0.6859	0.6856
75	0.6852	0.6849	0.6846	0.6842	0.6839	0.6836	0.6832	0.6829	0.6826	0.6823
76	0.6819	0.6816	0.6813	0.6809	0.6806	0.6803	0.6800	0.6796	0.6793	0.6790
77	0.6787	0.6783	0.6780	0.6777	0.6774	0.6770	0.6767	0.6764	0.6761	0.6757
78	0.6754	0.6751	0.6748	0.6745	0.6741	0.6738	0.6735	0.6732	0.6728	0.6725
79	0.6722	0.6719	0.6716	0.6713	0.6709	0.6706	0.6703	0.6700	0.6697	0.6693
80	0.6690	0.6687	0.6684	0.6681	0.6678	0.6675	0.6671	0.6668	0.6665	0.6662
81	0.6659	0.6656	0.6653	0.6649	0.6646	0.6643	0.6640	0.6637	0.6634	0.6631
82	0.6628	0.6625	0.6621	0.6618	0.6615	0.6612	0.6609	0.6606	0.6603	0.6600
83	0.6597	0.6594	0.6591	0.6588	0.6584	0.6581	0.6578	0.6575	0.6572	0.6569
84	0.6566	0.6563	0.6560	0.6557	0.6554	0.6551	0.6548	0.6545	0.6542	0.6539
85	0.6536	0.6533	0.6530	0.6527	0.6524	0.6521	0.6518	0.6515	0.6512	0.6509
86	0.6506	0.6503	0.6500	0.6497	0.6494	0.6491	0.6488	0.6485	0.6482	0.6479
87	0.6476	0.6473	0.6470	0.6467	0.6464	0.6461	0.6458	0.6455	0.6452	0.6449
88	0.6446	0.6444	0.6441	0.6438	0.6435	0.6432	0.6429	0.6426	0.6423	0.6420
89	0.6417	0.6414	0.6411	0.6409	0.6406	0.6403	0.6400	0.6397	0.6394	0.6391
90	0.6388	0.6385	0.6382	0.6380	0.6377	0.6374	0.6371	0.6368	0.6365	0.6362
91	0.6360	0.6357	0.6354	0.6351	0.6348	0.6345	0.6342	0.6340	0.6337	0.6334
92	0.6331	0.6328	0.6325	0.6323	0.6320	0.6317	0.6314	0.6311	0.6309	0.6306
93	0.6303	0.6300	0.6297	0.6294	0.6292	0.6289	0.6286	0.6283	0.6281	0.6278
94	0.6275	0.6272	0.6269	0.6267	0.6264	0.6261	0.6258	0.6256	0.6253	0.6250
95	0.6247	0.6244	0.6242	0.6239	0.6236	0.6233	0.6231	0.6228	0.6225	0.6223
96	0.6220	0.6217	0.6214	0.6212	0.6209	0.6206	0.6203	0.6201	0.6198	0.6195
97	0.6193	0.6190	0.6187	0.6184	0.6182	0.6179	0.6176	0.6174	0.6171	0.6168
98	0.6166	0.6163	0.6160	0.6158	0.6155	0.6152	0.6150	0.6147	0.6144	0.6141
99	0.6139	0.6136	0.6134	0.6131	0.6128	0.6126	0.6123	0.6120	0.6118	0.6115
100	0.6112	0.6110	0.6107	0.6104	0.6102	0.6099	0.6097	0.6094	0.6091	0.6089
101	0.6086	0.6083	0.6081	0.6078	0.6076	0.6073	0.6070	0.6068	0.6065	0.6063
102	0.6060	0.6057	0.6055	0.6052	0.6050	0.6047	0.6044	0.6042	0.6039	0.6037
103	0.6034	0.6032	0.6029	0.6026	0.6024	0.6021	0.6019	0.6016	0.6014	0.6011
104	0.6008	0.6006	0.6003	0.6001	0.5998	0.5996	0.5993	0.5991	0.5988	0.5986
105	0.5983	0.5981	0.5978	0.5976	0.5973	0.5970	0.5968	0.5965	0.5963	0.5960
106	0.5958	0.5955	0.5953	0.5950	0.5948	0.5945	0.5943	0.5940	0.5938	0.5935
107	0.5933	0.5930	0.5928	0.5925	0.5923	0.5921	0.5918	0.5916	0.5913	0.5911
108	0.5908	0.5906	0.5903	0.5901	0.5898	0.5896	0.5893	0.5891	0.5888	0.5886
109	0.5884	0.5881	0.5879	0.5876	0.5874	0.5871	0.5869	0.5867	0.5864	0.5862
110	0.5859	0.5857	0.5854	0.5852	0.5850	0.5847	0.5845	0.5842	0.5840	0.5837
111	0.5835	0.5833	0.5830	0.5828	0.5825	0.5823	0.5821	0.5818	0.5816	0.5813
112	0.5811	0.5809	0.5806	0.5804	0.5802	0.5799	0.5797	0.5794	0.5792	0.5790
113	0.5787	0.5785	0.5783	0.5780	0.5778	0.5776	0.5773	0.5771	0.5768	0.5766
114	0.5764	0.5761	0.5759	0.5757	0.5754	0.5752	0.5750	0.5747	0.5745	0.5743
115	0.5740	0.5738	0.5736	0.5733	0.5731	0.5729	0.5726	0.5724	0.5722	0.5719
116	0.5717	0.5715	0.5713	0.5710	0.5708	0.5706	0.5703	0.5701	0.5699	0.5696
117	0.5694	0.5692	0.5690	0.5687	0.5685	0.5683	0.5680	0.5678	0.5676	0.5674
118	0.5671	0.5669	0.5667	0.5665	0.5662	0.5660	0.5658	0.5655	0.5653	0.5651
119	0.5649	0.5646	0.5644	0.5642	0.5640	0.5637	0.5635	0.5633	0.5631	0.5628
120	0.5626	0.5624	0.5622	0.5620	0.5617	0.5615	0.5613	0.5611	0.5608	0.5606
121	0.5604	0.5602	0.5600	0.5597	0.5595	0.5593	0.5591	0.5588	0.5586	0.5584
122	0.5582	0.5580	0.5577	0.5575	0.5573	0.5571	0.5569	0.5566	0.5564	0.5562
123	0.5560	0.5558	0.5556	0.5553	0.5551	0.5549	0.5547	0.5545	0.5542	0.5540
124	0.5538	0.5536	0.5534	0.5532	0.5530	0.5527	0.5525	0.5523	0.5521	0.5519
125	0.5517	0.5514	0.5512	0.5510	0.5508	0.5506	0.5504	0.5502	0.5499	0.5497
126	0.5495	0.5493	0.5491	0.5489	0.5487	0.5484	0.5482	0.5480	0.5478	0.5476
127	0.5474	0.5472	0.5470	0.5468	0.5465	0.5463	0.5461	0.5459	0.5457	0.5455
128	0.5453	0.5451	0.5449	0.5446	0.5444	0.5442	0.5440	0.5438	0.5436	0.5434
129	0.5432	0.5430	0.5428	0.5426	0.5424	0.5421	0.5419	0.5417	0.5415	0.5413

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CONVERSION FACTORS
— Lenght, area and volume —

Length

Unit	Symbol	To convert in following unit, multiply by:				
		m	cm	in	ft	yd
meter	m	1	100	39.3701	3.2808	1.0936
centimeter	cm	0.01	1	0.3937	0.03281	0.01094
inch	in or "	0.0254	2.54	1	0.08333 (1/12)	0.02778 (1/36)
foot	ft or '	0.3048	30.48	12	1	0.33333 (1/3)
yard	yd	0.9144	91.44	36	3	1

Area

Unit	Symbol	To convert in following unit, multiply by:			
		m^2	cm^2	sq.in	sq.ft
square meter	m^2	1	10 000	1550.003	10.76391
square centimeter	cm^2	0.0001	1	0.15500	$1.07639.10^{-3}$
square inch	sq. in	$6.451.10^{-4}$	6.4516	1	$6.944.10^{-3}$ (1/144)
square foot	sq. ft	0.092903	929.0304	144	1

Volume

Unit	Symbol	To convert in following unit, multiply by:					
		m^3	l	cu.in	cu.ft	US gal	bbl
cubic meter	m^3	1	1000	61023.74	35.31467	264.1721	6.289811
liter	l	0.001	1	61.02374	0.035315	0.264172	$6.2898.10^{-3}$
cubic inch	cu.in	$1.6387.10^{-5}$	$1.6387.10^{-2}$	1	$5.7870.10^{-4}$ (1/1728)	$4.3290.10^{-3}$ (1/231)	$0.1032.10^{-3}$
cubic foot	cu.ft	0.028317	28.31685	1728	1	7.480519	0.1783
gallon fluid U.S.	US gal	$3.7854.10^{-3}$	3.78541	231	0.13368	1	0.02381 (1/42)
barrel petroleum	bbl	0.15899	158.99	9702.2	5.61458	42	1

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CONVERSION FACTORS
— Mass and density —

Mass

Unit	Symbol	To convert in following unit, multiply by:			
		kg	t	g	lb
kilogram	kg	1	0.001	1 000	2.2046
metric ton	t	1000	1	1 000 000	$2.2046.10^{3}$
gram	g	0.001	0.000 001	1	$2.2046.10^{-3}$
pound	lb	0.45359	$4.45359.10^{-3}$	453.59	1

Density

Unit	Symbol	To convert in following unit, multiply by:				
		kg/m^3	kg/l	g/cm^3	lb/cu.ft	lb/US gal
kilogram per cubic meter	kg/m^3	1	0.001	0.001	0.062428	$8.345.10^{-3}$
kilogram per liter	kg/l	1000	1	1	62.428	8.345
gram per cubic centimeter	g/cm_3	1000	1	1	62.428	8.345
pound per cubic foot	lb/cu.ft	16.0185	$16.0185.10^{-3}$	$16.0185.10^{-3}$	1	0.1337
pound per US gallon	lb/US gal	119.83	0.11983	0.11983	7.4805	1

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CONVERSION FACTORS
— Linear velocity, standard acceleration of gravity, volume flow rate, mass velocity (mass flux), force —

Linear velocity

Unit	Symbol	To convert in following unit, multiply by:				
		m/s	km/h	in/s	ft/s	mile/h
mètre par seconde	m/s	1	3.6	39.370	3.281	2.237
kilomètre par heure	km/h	0.2778	1	10.937	0.9113	0.6214
inch per second	in/s	0.0254	0.09144	1	0.08333	0.05683
foot per second	ft/s	0.3048	1.0973	12	1	0.682
mile per hour	mile/h	0.4469	1.609	17.596	1.4664	1

Standard acceleration of gravity

$g = 9.81\ m/s^2 = 127.14.106\ m/h^2 = 92.18\ ft/s^2 = 417.106\ ft/hr^2$

Volume flow rate

Unit	Symbol	To convert in following unit, multiply by:				
		m^3/h	GPM	cu.ft/s	cu.ft/hr	BPD
mètre cube par heure	m^3/h	1	4.4031	$9.81.10^{-3}$	35.3	151
gallon par minute	GPM	0.22711	1	$2.228.10^{-3}$	8.019	34.30
cubic foot per second	cu.ft/s	101.95	448.9	1	3600	15386
cubic foot per hour	cu.ft/hr	$28.32.10^{-3}$	0.1246	$277.8.10^{-6}$	1	4.272
barrel per day	BPD	$6.624.10^{-3}$	0.02917	$64.98.10^{-6}$	0.234	1

Mass velocity (mass flux)

Unit	Symbol	To convert in following unit, multiply by:		
		$kg/s.cm^2$	$kg/h.m^2$	lb/hr.sq.ft
kilogramme par seconde par centimètre carré	$kg/s.cm^2$	1	36.10^6	$7376.26.10^3$
kilogramme par heure par mètre carré	$kg/h.m^2$	$27.78.10^{-9}$	1	0.2048
pound per hour per square foot	lb/hr.sq.ft	$135.57.10^{-9}$	4.882	1

Force

Unit	Symbol	To convert in following unit, multiply by:		
		N	kgf	lbf
Newton	N	1	0.10197	0.22481
kilogram-force	kgf	9.80665	1	2.20462
pound-force	lbf	4.44822	0.45359	1

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CONVERSION FACTORS
— Pressure and energy —

Pressure

Unit	Symbol	To convert in following unit. multiply by :						
		Pa	bar	kgf/cm^2	atm	psi	mm Hg	mmH_2O
Pascal	Pa	1	10^{-5}	$1.0197.10^{-5}$	$9.869.10^{-6}$	$1.450.10^{-4}$	$7.500.10^{-4}$	0.10197
bar	bar	100 000	1	1.0197	0.9869	14.50	750.06	10197.2
kilogram-force per square centimeter	kgf/cm^2	98 066.5	0.98066	1	0.96784	14.6959	735.56	10 000
atmosphere	atm	101 325	1.01325	1.0332	1	14.6959	760	10332.3
pound per square inch	psi	6894.76	0.06895	0.0703	0.06805	1	51.715	703.07
millimeter mercury	mm Hg	133.32	1333.10^{-3}	$1.359.10^{-3}$	$1.316.10^{-3}$	0.01934	1	13.595
millimeter water	mmH_2O	9.80665	$9.8067.10^{-5}$	10^{-4}	$9.678.10^{-3}$	$1.422.10^{-3}$	0.0736	1

Energy

Unit	Symbol	To convert in following unit. multiply by :					
		J	kWh	kgf.m	kcal	th	BTU
Joule	J	1	$2.778.10^{-7}$	0.10197	$2.388.10^{-4}$	$2.388.10^{-7}$	$9.478.10^{-4}$
kilowatt hour	kWh	$3.6.10^{6}$	1	$3.671.10^{5}$	859.845	0.859845	3412.14
kilogram force meter	kgf.m	9.80665	$2.724.10^{-6}$	1	$2.342.10^{-3}$	$2.342.10^{-6}$	$9.295.10^{-3}$
kilocalorie	kcal	4185.8	$1.163.10^{-3}$	426.935	1	0.001	3.96832
British thermal unit	BTU	1055.056	$2.931.10^{-4}$	107.586	0.251996	$0.252.10^{-3}$	1

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CONVERSION FACTORS
— Power, temperature, specific heat capacity, heat flux and heat transfer rate —

Power

Unit	Symbol	To convert in following unit multiply by:			
		W	kgm/s	ch	hp
Watt	W	1	0.101972	$1.3596.10^{-3}$	$1.34102.10^{-3}$
kilogram meter per second	kgm/s	9.80665	1	$13.333.10^{-3}$	$13.151.10^{-3}$
cheval vapeur	ch	735.5	75	1	0.98632
horse power	hp	745.7	76.04	1.0139	1

Temperature

Temperature t expressed on scale:	Equivalence on scale:			
	Kelvin	Celsius	Fahrenheit	Rankine
Kelvin (K)	t	t - 273.15	9/5 (t - 273.115)+32	1,8.t
Celsius (°C)	t + 273.15	t	1,8.t + 32	1,8.t + 491,7
Fahrenheit (°F)	(t + 459.7)/1.8	(t - 32)/1.8	t	t + 459.7
Rankine (°R)	t/1.8	(t - 491.7)/1.8	t - 459.7	t

Specific heat capacity 1 cal/g.°C = 1 kcal/kg.°C = 1 th/t.°C = 1 BTU/lb.°F

Heat flux

Unit	Symbol	To convert in following unit multiply by:		
		kcal/h.m^2	W/m^2	BTU/h.sq.ft
kilocalorie par heure par mètre carré	kcal/h.m^2	1	1.1626	0.36877
watt par mètre carré	W/m^2	0.8601	1	0.31766
BTU per hour per square foot	BTU/h.sq.ft	2.7117	3.1527	1

Heat transfer rate

Unit	Symbol	To convert in following unit multiply by:		
		kcal/h.m^2.°C	W/m^2.K	BTU/h.sq.ft.°F
kilocalorie par heure par mètre carré par degré Celsius	kcal/h.m^2.°C	1	1.1626	0.36877
watt par mètre carré par Kelvin	W/m^2.K	0.8601	1	0.31766
BTU per hour per square foot per degree Fahrenheit	BTU/h.sq.ft.°F	2.7117	3.1527	1

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CONVERSION FACTORS
— Thermal conductivity, dynamic viscosity, kinematic viscosity, gas constant and molar volume under 1 atm —

Thermal conductivity

Unit	Symbol	To convert in following unit, multiply by:		
		kcal/h.m.°C	W/m.K	BTU/h.ft.°F
kilocalorie par heure par mètre par degré Celsius	kcal/h.m.°C	1	1.1626	0.6725
watt par mètre par Kelvin	W/m.K	0.8601	1	0.5784
BTU per hour per foot per degree Fahrenheit	BTU/h.ft.°F	1.4870	1.7288	1

Dynamic viscosity

Unit	Symbol	To convert in following unit, multiply by:			
		Pl	cP	kg/m.h	lb/ft.h
poiseuille (Pl)	Pl	1	1000	3600	2420
centipoise	cP	0.001	1	3.6	2.42
kilogramme par mètre par heure	kg/m.h	$0.2778.10^{-3}$	0.2778	1	0.67209
pound per foot per hour	lb/ft.h	$0.413.10^{-3}$	0.413	1.488	1

Kinematic viscosity

Unit	Symbol	To convert in following unit, multiply by:		
		cSt	m^2/h	sq.ft/h
centistokes	cSt	1	0.0036	0.0387
mètre carré par heure	m^2/h	277.8	1	10.76
square foot per hour	sq.ft/h	25.81	0.0929	1

Gas constant

n	V	T	P	R
lb-mole	cu.ft	R K	psi	10.73 19.314
g-mole	cm^3 cm^3 l l	K K K K	atm bar atm mm Hg	82.05 83.12 0.08205 62.36
kg-mole	l m^3	K K	atm bar	82.05 0.08314

Molar volume under 1 atm

Mole	Temperature	Molar volume
lb-mole	0°C	359 cu.ft
lb-mole	60°F	379 cu.ft
g-mole	0°C	22.4 litres
kg-mole	0°C	22.4 m^3

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Y1 - Specifications of Jet A-1 (AFQRJOS Issue 27) 5 pages

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RAFFINERIE / LABORATOIRE - REFINERY / LABORATORY

CERTIFICATE OF ANALYSIS
CERTIFICAT D'ANALYSE

"Aviation Fuel Quality Requirements for Jointly Operated Systems"

AFQRJOS ISSUE 27 - Check List Jet A-1 (February 2013)

Jet A-1

Embodying the most stringent requirements in the following specifications:
Incorporant les exigences les plus strictes des spécifications suivantes :
a) British MoD DEF STAN 91-91/Issue 7, Amendment 2 dated 1 December 2012, Jet A-1
b) ASTM D 1655-12, Jet A-1
The Aviation Fuel Quality Requirements for Jointly Operated System for Jet A-1 are defined in the following table, which should be read in conjunction with the notes that follow the table.
Les Exigences de Qualité du Carburant Aviation pour les Installations Communautaires (AFQRJOS) portant sur le Jet A-1 sont définies dans le tableau ci-après qui doit être lu en association avec les notes qui suivent qui le complète.
It should be specially noted that DEF STAN 91-91/7 Amendment2 requires traceability of product to point of manufacture
Il doit être particulièrement noté que la DEF STAN 91-91/7 Amendement2 exige la traçabilité du produit jusqu'à la fabrication

TEST CERTIFICATE n°
CERTIFICAT D'ANALYSE n°
BATCH n° TANK n°
LOT n° BAC n°
Volume du lot
Sample n° Sampling date
Echantillon n° Date d'échantillonnage

( ) See remarks on the reverse side - ( ) Voir les commentaires au verso

Methods Higlight method used	Méthodes Identifier la méthode	Properties	*Caractéristiques*		Results *Résultats*	Limits *Limites*
Visual	*Visuelle*	APPEARANCE Visual	ASPECT Visuelle		----------	Clear and Bright - *Clair et Limpide* (sans eau non dissoute ni sédiments à t° ambiante)
ASTM D156 or D 6045	IP	Colour (1)	Couleur (1)		----------	Report
D 5452	423	Particulate contamination (2)	Contamination particules (2)	mg/l	----------	1.0 max.
	564 or 565 or 577	Particulate, at point of manufacture, cumulative channel particle counts & Individual channel count / and ISO code (3)	Particules, à la raffinerie, Nbre cumulé par canal de comptage et comptage individuel par canal/ et code ISO (3)			
		≥ 4 µm(c)			--------/----	Report
		≥ 6 µm(c)			--------/----	Report
		≥ 14 µm(c)			--------/----	Report
		≥ 21 µm(c)			--------/----	Report
		≥ 25 µm(c)			--------/----	Report
		≥ 30 µm(c)			--------/----	Report
		COMPOSITION (4 & 5)	COMPOSITION (4 & 5)			
D 3242	354	Total Acidity	Acidité Totale	mg KOH/g	----------	0.015 max.
D 1319	156	Aromatics	Aromatiques	% vol.	----------	25.0 max.
D 6379	436	OR Total Aromatics (6)	OU Aromatiques totaux (6)	% vol.	----------	26.5 max
D 1266 or D2622 or D 4294 or D 5453	336	Sulphur , Total	Soufre Total	% mass.	----------	0.30 max.
D 3227	342	Sulphur , Mercaptans	Soufre Mercaptan	% mass.	----------	0.0030 max.
D 4952	30	OR Doctor Test (7)	OU Doctor Test (7)		----------	Negative
		Refinery Components at point of manufacture: (8) Composition du lot lors de la fabrication en raffinerie (8)				
		Non Hydroprocessed components in batch	% non Hydrotraité	% vol.	----------	Report (incl. "nil" or "100%")
		Mildly Hydroprocessed components in batch	% Moyennement Hydrotraité	% vol.	----------	Report (incl. "nil" or "100%")
		Severely hydroprocessed components	% Sévèrement hydrotraité	% vol.	----------	Report (incl. "nil" or "100%")
		Synthetic components (4)	% Synthétique	% vol.	----------	Report (incl. "nil" or "50%")
		VOLATILITY	VOLATILITE			
D 86 See note (9) or D 2887 See note (10)	123 See note (9) or 406 See note (10)	Distillation Initial Boiling Point	Point Initial	°C	----------	Report
		10% vol. rec. at	10% vol. réc. à	°C	----------	205.0 max.
		50% vol. rec. at	50% vol. réc. à	°C	----------	Report
		90% vol. rec. at	90% vol. réc. à	°C	----------	Report
		End Point	Point Final	°C	----------	300.0 max.
		Residue	Résidu	% vol.	----------	1.5 max.
		Loss	Perte	% vol.	----------	1.5 max.
D 3828 or D56	170 or 523	Flash Point (11)	Point d'Eclair (10)	°C	----------	38.0 min.
D 1298 or D 4052	160 or 365	Density at 15°C	Masse Volumique à 15°C	kg/m³	----------	775.0 min. to 840.0 max.
		FLUIDITY	FLUIDITE			
D 2386 or D 5972 or D 7153 or D 7154	16 ou 435 ou 529 ou 528	Freezing Point (12)	Point de disparition des cristaux (12)	°C	----------	minus 47.0 max.
D445	71	Viscosity at -20°C	Viscosité à -20°C	mm²/s (cSt)	----------	8.000 max.
		COMBUSTION	COMBUSTION			
D 3338 or D 4809		Specific Energy, net (13)	Pouvoir Calorifique inf. (13)	MJ / kg	----------	42.80 min.
D 1322	598	Smoke Point (14)	Point de fumée	mm	----------	25.0 min.
D 1322	598	OR Smoke Point (14)	OU Point de fumée	mm	----------	19.0 min.
D 1840		AND Naphtalenes	ET Naphtalènes	% vol.	----------	3.00 max.
		CORROSION	CORROSION			
D 130	154	Corrosion, Copper strip, classification (2 hours +/- 5 min.at 100°C +/- 1°C)	Corrosion Cuivre (2 heures +/- 5 min.at 100°C +/- 1°C)	Classification	----------	1 max.
		THERMAL STABILITY (JFTOT)	STABILITE THERMIQUE (JFTOT)			
		Control temperature	Température de contrôle	°C	----------	260°C min.
D 3241	323	JFTOT Filter Pressure Differential	Delta P du Filtre	mm Hg	----------	25.0 max.
D 3241	323	Tube Deposing Rating (visual) (15)	Cotation Tube (visuelle) (15)		----------	< 3 max.
		no 'Peacock' or 'Abnormal' colour deposits	sans irisation ou dépôts anormalement colorés			
		CONTAMINANTS	CONTAMINANTS			
D 381	540	Existent Gum	Gommes Actuelles	mg/100 cm³	----------	7 max.
D 3948		Microseparometer (MSEP) ratings: (16)	Cotation (MSEP) : (16)			
		Fuel with Static Dissipator Additive	Carburant additivé		----------	70 min.
		OR	OU			
		Fuel without Static Dissipator Additive	Carburant non additivé		----------	85 min.
		CONDUCTIVITY	CONDUCTIVITE			
D 2624	274	Electrical Conductivity (17)	Conductivité électrique (17)	pS/m	----------	50 min. to 600 max.
		LUBRICITY	LUBRIFIANCE			
D 5001		BOCLE wear scar diameter (18)	diamètre d'usure BOCLE (18)	mm	----------	0,85 max.
CODE ADDITIF DEF STAN 91-91 RDE/A/xxx		ADDITIVES (24) Names and approval codes from DEF STAN 91-91/7 should be quoted on quality certs)	ADDITIFS (24) Les Noms et codes d'approbation selon DEF STAN 91-91/7 doivent être précisés sur le certificat de qualité)			
RDE/A/...		Antioxidant in Hydroprocessed & synthetic Fuels (Mandatory) (19)	Antioxydant dans produit hydrotraité et de synthèse (obligatoire) (19)	mg/l	----------	17.0 min. to 24.0 max.
RDE/A/...		Antioxidant in non Hydroprocessed Fuels (Optional)	Antioxydant (prod. non hydrotraité) (optionnel)	mg/l	----------	24.0 max.
RDE/A/650		Metal Deactivator (Optional) (20)	Désactivant des métaux (optionnel) (20)			
		First doping	Première additivation	mg/l	----------	2,0 max.
		Cumulative after field redoping	Cumul après ré-additivation	mg/l	----------	5.7 max.
RDE/A/621		Static Dissipator Additive	Additif Anti-statique			
		First doping	Première additivation	mg/l	----------	3.0 max.
		Cumulative after field redoping	Cumul après ré-additivation	mg/l	----------	5.0 max.
RDE/A/...		LIA (18)	LIA (18)	mg/l	----------	

Fuel System Icing Inhibitor is not permitted unless agreed by all the participants in a joint system (see also Note 21).
L'additif anti-glace n'est pas autorisé sans l'accord de tous les participants (voir aussi note 21)

See Note (22) about - requirements for management of change in refineries
Voir Note (22) relative aux - exigences de management du changement dans les raffineries

The chief chemist certifies that the product complies with the specifications detailled above.
Le chef de laboratoire certifie que le produit analysé est conforme aux spécifications détaillées ci-dessus
Except for electrical conductivity / Excepté pour la conductivité électrique : Delete if NA / Rayer le cas échéant - Note 16

DATE
SIGNATURE

IMPORTANT NOTICE

Conformance to AFQRJOS requires conformance to the detail of both specifications listed above, not just to the table. See Note 23 for guidance on statements declaring conformance to these specifications. If necessary, airports operated to JIG Standards can supply jet fuel to either of the parent specifications listed above provided the participants agree.
Also, it should be specifically noted that DEF STAN 91-91/7 Amendment 2 requires traceability of product to point of manufacture and requirements applicable to fuels containing synthetic or renewable components. See Annexes J and D of DEFSTAN 91-91/7 Amendment 2 for more information. Although they have not been changed significantly, the requirements in both of these Annexes have been clarified in Amendment 2.

Main Table Notes (Issue 27 – February 2013)

(1) The requirement to report Saybolt Colour shall apply at point of manufacture, thus enabling a colour change in distribution to be quantified. Where the colour of the fuel precludes the use of the Saybolt Colour test method, then the visual colour shall be reported. Unusual or atypical colours should also be noted and investigated. For further information on the significance of colour see Annex E in DEF STAN 91-91/7 Amendment 2.

(2) This limit shall apply at point of manufacture only. For more information on particulate contamination refer to Annex F of DEF STAN 91-91 Issue 7 Amendment 2 . For guidance on contamination limits for into-plane fuelling refer to 5th Edition IATA Guidance Material (Part III).

(3) This requirement shall apply at point of manufacture only. Both the number of particles and the number of particles as a scale number as defined by Table 1 of ISO 4406:1999 shall be reported. It is the Specification Authority's intention to replace the gravimetric Millipore test with Particle Counting at the earliest opportunity.

(4) Attention is drawn to DEF STAN 91-91 Issue 7 Amendment 2, which approves both Semi-Synthetic and Fully Synthetic Jet Fuel produced by SASOL . It also approves two generic components a) up to 50% of Synthetic Paraffinic Kerosine certified as meeting the requirements of ASTM D7566 Annex A1 and b) up to 50% Hydrogenated Esters and Fatty Acids as meeting the requirements of ASTM D7566 Annex A2. For these fuels, additional testing requirements apply and reference should be made to DEF STAN 91-91/7 Amendment 2 Annex D. These particular semi- and fully synthetic fuels may be certified against this Issue of Check List. The volume percentage of each synthetic blending component shall be recorded along with its corresponding release Specification and Annex number, product originator and originator's Certificate of Quality number. From the point of manufacture to the point of blending to meet this specification, the synthetic component shall be handled, transported and documented in the same manner as finished jet fuel in order to maintain product integrity. Special care shall be taken to ensure homogeneity when blending semi synthetic jet fuel, particularly where the component densities are significantly different. DEF STAN 91-91/7 Amendment 2 also states that blending of synthetic fuels shall take place upstream of airports.See also note 10 in DEF STAN 91-91/7 Amendment 2 on minimum aromatic content.

(5) Concentrations of FAME (Fatty Acid Ethyl Ester) greater than or equal to 5.0 mg/kg are not approved. See section 5.6 and Annex G of DEFSTAN 91-91/7 Amendment 2 for details

(6) Round robin testing has demonstrated the correlation between total aromatics content measured by IP 156/ASTM D 1319 and IP 436/ASTM D6379. Bias between the two methods necessitates different equivalence limits as shown. Testing laboratories are encouraged to measure and report total aromatics content by the two methods to assist verification of the correlation. In cases of dispute IP 156 / ASTM D 1319 shall be the referee method. It is the intention of the DEF STAN /91.Technical Authority to change the referee method to IP 436 at a later date.

(7) The Doctor Test is an alternative requirement to the Sulphur Mercaptan Content. In the event of conflict between the Sulphur Mercaptan and Doctor Test results, the Sulphur Mercaptan result shall prevail.

(8) The need to report the %v/v of non hydroprocessed, mildly hydroprocessed, severely hydroprocessed and synthetic components (including "nil" or "100%" as appropriate) on refinery Certificates of Quality for Jet A-1 to Check List derives from DEF STAN 91-91/7 Amendment 2. Each refinery component used in the make up of the batch shall be reported on the certificate of quality as a percentage by volume of the total fuel in the batch. Mildly hydroprocessed components are defined as those petroleum derived hydrocarbons that have been subjected to a hydrogen partial pressure less than 7000 kPa (70 bar or 1015 psi) during manufacture. Severely hydroprocessed components are defined as those petroleum derived hydrocarbons that have been subjected to a hydrogen partial pressure of greater than 7000 kPa (70 bar or 1015 psi) during manufacture.The total of non-hydroprocessed plus mildly hydroprocessed plus severely hydroprocessed plus synthetic components shall equal 100%

(9) In methods IP 123 and ASTM D 86 all fuels certified to this specification shall be classed as group 4, with a condenser temperature of zero to 4°C

(10) If IP 406 or ASTM D 2887 are used to produce IP123 equivalent or ASTM D 86 correlated data, there is no requirement to report residue or loss..

(11) Subject to a minimum of 40°C, results obtained by method ASTM D 56 (Tag) may be accepted. The referee test method is IP 170

(12) These automatic methods are permitted; IP16/ASTM D2386 remains the referee method

(13) ASTM D 4529/IP 381 may be used where local regulations permit

(14) DEF STAN 91-91/7 Amendment 2 now references IP 598 instead of IP 57.The new IP 598 test for smoke point includes both the standard manual method and an automatic method. Note 12 in DEF STAN 91-91/7 Amd 2 states its intention to make the automatic method the referee method in January 2014.

(15) Examination of the heater tube to determine the Visual Tube Rating using the Visual Tuberator shall be carried out within 120 minutes of completion of the test. It is the Visual Method that shall be reported. Attention is drawn to Note **15** in DEF STAN 91-91/7 Amendment 2 , which stresses that only approved heater tubes shall be used and lists JFTOT tubes from PAC-Alcor and Falex as being technically suitable.

(16) Attention is drawn to Note 17 of DEF STAN 91-91/7 Amendment 2 that states "Where SDA is added at point of manufacture the MSEP limit of 70 shall apply. No precision data are available for fuels containing SDA; if MSEP testing is carried out during downstream distribution, no specification limits apply and the results are not to be used as the sole reason for rejection of a fuel. A protocol giving guidelines on possible actions to be taken following failed MSEP testing can be found in the Joint Inspection Group's Bulletin Number 14, MSEP Protocol at www.jigonline.com under 'Fuel Quality'". Where SDA is added downstream of point of manufacture, it is acknowledged that MSEP results may be less than 70

.(17) Due to the requirements of DEF STAN 91-91/7 Amendment 2, conductivity limits are mandatory for product to meet this specification. However it is acknowledged that in some manufacturing and distribution systems it is more practical to inject SDA further downstream. In such cases, the Certificate of Quality for the batch should be annotated thus: "Product meets the requirements of AFQRJOS Check List 27 except for electrical conductivity". In some situations, the conductivity can decrease rapidly and the fuel can fail to respond to additional dosing with Stadis 450. In such cases, fuel may be released with conductivity down to a minimum of 25pS/m provided that the fuel is fully tested against the specification and the Tank Release Note is annotated with the explanation "Product released below 50pS/m due to conductivity loss as per Annex H of DEFSTAN 91-91/7 Amendment 2"

(18) This requirement comes from DEF STAN 91-91/7 Amendment 2. The requirement to determine lubricity applies only to fuels whose composition is made up of a) at least 20% of severely hydroprocessed components and less than 5% non-hydrotreated components or b) includes synthetic fuel components. The limit applies only at the point of manufacture. For important advisory information on the lubricity of aviation turbine fuels see Annex B of DEF STAN 91-91/7 Amendment 2. CI/LI additive (also known as LIA) may be used to improve lubricity; only those additives listed in Table 2 of ASTM D1655-12/ Annex A of DEFSTAN 91-91/7 Amendment 2 are permitted. Note that two additionnal additives have been added to the list in DEFSTAN 91-91/7 Amendment 2. Refer also to Appendix A.5 of DEF STAN 91-91/7 Amendment 2 for advice on point of addition. When injecting CI/LI downstream of point of manufacture, care shall be taken to ensure that maximum dose rates are not exceeded.
Corrosion Inhibitor/Lubricity Improver (CI/LI) additive may be added to the fuel without prior consent of the joint system participants

(19) Antioxidants are mandatory in hydroprocessed fuels and synthetic fuels and shall be added immediately afterhydroprocessing or synthesising and prior to the product or component being passed into storage in order to prevent peroxidation and gum formation after manufacture .
Approved antioxidant additives are listed in Annex A.2.4 of DEF STAN 91-91/7 Amendment 2, together with the appropriate RDE/A/XXX-Qualification Reference for quoting on refinery Certificates of Quality. Refer also to Annex A 2.6 for requirements for reporting additions on the CoQ

(20) The approved Metal Deactivator Additive (MDA) RDE/A/650 appears in Annex A.3 of DEF STAN 91-91/7 Amendment 2. See also Annex A.3.1 about the need to report thermal stability before and after using when contamination of Jet A-1 by any of the trace metals listed in this Annex is unproven. Note also in A 3.3 that maximum doping at point of manufacture or on initial doping is limited to 2 mg/l.

(21) Concentrations of Fuel System Icing Inhibitor (FSII) less than 0.02% by volume can be considered negligible and do not require agreement/notification. The assent to allow these small quantities of FSII without agreement/notification is to facilitate the changeover from fuels containing FSII to those not containing FSII where the additive may remain in the fuel system for a limited time. This does not allow the continuous addition of FSII at these low concentrations. Attention is drawn to the new note in Annex A.6 in DEF STAN 91-91/7 Amendment 2 highlighting that filter monitors cannot be used with fuel containing FSII.

(22) Attention is drawn to the guidance in DEF STAN 91-91/7 Amendment 2and ASTM D 1655-12 concerning the need for appropriate management of change measures in refineries manufacturing jet fuel. The implications of any changes to feedstock, processing conditions or process additives on finished product quality and performance shall be considered (for example, experience has shown that some process additives might be carried over in trace quantities into aviation fuels).

(23) It is normal to certify conformance to specifications with statements like "It is certified that the samples have been tested using the Test Methods stated and that the Batch represented by the samples conforms to AFQRJOS Checklist Issue **27**". Where applicable, Batch Certificates may also confirm, specifically, compliance with DEF STAN 91-91 (latest issue) and/or ASTM D 1655 (latest issue). It is normal to certify conformance to a primary specification.
The following statements should be used.
"It is certified that the samples have been tested using the Test Methods stated and that the Batch represented by the samples conforms with DEF STAN 91-91 Issue 7 Amendment 2 and AFQRJOS Checklist Issue 27".
Or
"It is certified that the samples have been tested using the Test Methods stated and that the Batch represented by the samples conforms with ASTM D 1655 -12 and AFQRJOS Checklist Issue 27".
The minimum requirements of information to be included on the fuel's refinery batch certificate of quality shall be :
- Specification name, issue and any amendment number;
- Batch number or unique identifier;
- Quantity of fuel in the batch;
- Properties tested and including specification limit, test method and result of test;
- Additives, including qualification reference and quantity added;
- Name and position of authorised test certificate signatory or an electronic signature;
- Date of certification.

(24) The types and concentrations of all additives used shall be shown on the refinery Certificates of Quality and other quality documents. When additives are diluted (with hydrocarbon solvent only) to improve handling properties prior to addition, it is the concentration of the additive itself (prior to dilution) that shall be reported. See Annex **A** of DEF STAN 91-91/7 Amendment 2 for detailed advice.

REMARQUE IMPORTANTE

La conformité à l'AFQRJOS requiert la conformité à l'intégralité de chacune des deux spécifications citées plus haut dans le tableau, et pas seulement au tableau. Voir la note 23 relative à la déclaration de conformité aux spécifications. Si nécessaire, les aéroports opérés selon les standards JIG peuvent fournir du carburant à jet selon l'une ou l'autre des spécifications mentionnées ci-dessus sous réserve d'un accord des participants.
De plus, il est expressément rappelé que la DEF STAN 91-91/7 Amendement 2 exige la traçabilité des produits depuis le lieu defabrication et que les exigences sont applicables aux carburants contenant des composants synthétiques ou renouvelables. Voir les annexes J et D de DEFSTAN 91-91/7 Amendement 2 pour plus d'informations. Bien qu'elles n'aient pas été changées de manière significative, les exigences dans ces deux annexes ont été précisées dans l'amendement 2.

Tableau des Principales Notes (Edition 27 – Février 2013)

(1) L'exigence de reporter la Couleur Saybolt s'applique au point de fabrication, permettant ainsi d'évaluer un changement de couleur au cours de la distribution. Dans le cas ou la couleur du Jet A-1 empêche l'utilisation de la méthode Couleur Saybolt, la couleur visuelle sera reportée. Une couleur atypique ou inhabituelle doit également être notée et les raisons de la coloration recherchées. Pour plus d'information sur la signification de la couleur se reporter à l'annexe E de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2.

(2) Cette limite s'applique au point de fabrication uniquement. Pour des informations complémentaires sur la contamination par des particules se reporter à l'annexe F de la DEF STAN 91-91 Issue 7 Amendement 2. Pour des indications relatives aux limites de contamination à la mise bord avion se reporter à la 5ème édition du IATA Guidance material (Part 3).

(3) Cette exigence s'applique au point de fabrication uniquement. Aussi bien le nombre global de particules que le nombre de particules par numéro d'échelle tel que défini par le tableau 1 de la norme ISO 4406:1999 doivent être déclarés. L'Autorité en charge des Spécifications a l'intention de remplacer le test Millipore gravimétrique par un comptage de particules dans les meilleurs délais.

(4) L'attention est portée à la DEF STAN 91-91 Issue 7 Amendement 2, qui approuve les carburéacteurs semi-synthétiques et totalement synthétiques produits par SASOL. Elle approuve également 2 composants génériques a) jusqu'à 50% de kérosène synthétique paraffinique certifié conforme aux exigences de la norme ASTM D7566 annexe A1 et b) jusqu'à 50% d'esters et d'acides gras hydrogénés répondant aux exigences de l'ASTM D7566 Annexe 2. Pour ces carburants, les exigences des tests supplémentaires s'appliquent en se référant à la DEF STAN 91-91 ISSUE 7 Amendement 2 annexe D. Ces carburants particuliers semi- et entièrement synthétiques peuvent être certifiés selon cette version de la « check list ». Le pourcentage en volume de chaque composant synthétique du mélange doit être enregistré en même temps que sa référence numérotée à la spécification et à l'annexe, la référence numérotée du producteur et la référence numérotée du certificat de qualité du producteur. Du point de fabrication jusqu'au mélange assurant la conformité à la spécification, le composant de synthèse doit être manipulé, transporté et documenté de la même manière que le carburéacteur fini afin de maintenir l'intégrité du produit. Un soin particulier doit être pris pour assurer l'homogénéité du mélange de carburant semi synthétique, en particulier lorsque les densités des composants sont sensiblement différentes. La DEF STAN 91-91 ISSUE 7 Amendement 2 stipule que le mélange des carburants synthétiques doit s'effectuer en amont des aéroports. Voir également la note 10 dans la DEF STAN 91-91 ISSUE 7 amendement 2 sur le contenu minimum en aromatiques.

(5) Des concentrations d'EMAG (Ester Méthylique d'Acides Gras) supérieures ou égales à 5.0 mg/kg ne sont pas accordées. Voir Section 5.6 et ANNEX G de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 pour plus de détails.

(6) Des essais circulaires ont établi la corrélation entre les teneurs en aromatiques totaux mesurées selon l'IP 156/ASTM D 1319 et selon l'IP 436/ASTM D 6379. Un biais entre les deux méthodes entraîne différentes limites comme indiqué. Les laboratoires de mesures sont encouragés à déterminer et à préciser les teneurs en aromatiques totaux par les deux méthodes pour aider à la vérification de la corrélation. En cas de litige l'IP 156/ASTM D1319 devra être la méthode de référence. L'Autorité Technique a l'intention de changer dans le futur la méthode de référence par l'IP 436.

(7) Le "Doctor Test" est une exigence alternative à la teneur en soufre mercaptan. En cas de contradiction entre le résultat sur le soufre mercaptan et le "Doctor Test", le résultat soufre mercaptan prévaudra.

(8) La nécessité d'indiquer le % en volume de bases non hydrotraitées, moyennement et sévèrement hydrotraitées (y compris "nul" ou "100%" le cas échéant) sur le Certificat de Qualité de la Raffinerie pour le Jet A-1 découle de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. Chaque composant utilisé dans la fabrication d'un lot sera mentionné sur le certificat de qualité comme un pourcentage en volume du total de la fabrication.
Les composants moyennement hydrotraités sont définis comme étant des hydrocarbures pétroliers ayant été soumis à une pression partielle d'hydrogène inférieure à 7 000 kPa (70 bar ou 1015 psi) lors de la fabrication.
Les composants fortement hydrotraités sont définis comme étant des hydrocarbures pétroliers ayant été soumis à une pression partielle d'hydrogène supérieure à 7 000 kPa (70 bar ou 1015 psi) lors de la fabrication. Le total des pourcentages des composants non hydrotraités, moyennement hydrotraités , fortement hydrotraités, et synthétiques doit être égal à 100%

(9) Pour les méthodes IP123 et ASTM D86 tous les carburéacteurs certifiés par cette spécification doivent être classés dans le groupe 4, avec une température de condenseur de 0 à 4°C.

(10) Si l'IP406 ou ASTM D 2887 sont les méthodes utilisées pour générer les données équivalentes à celles de l'IP 123 ou corrélées à celles de l'ASTM 86, il n'y a pas besoin de faire mention du résidu ou de la perte.

(11) Soumis à une exigence minimum de 40 °C, les résultats obtenus par la méhode ASTM D56 (Tag) sont acceptables. La méthode de référence est l'IP 170

(12) Ces méthodes automatiques sont permises ; l'IP16/ASTM D2386 reste la méthode d'arbitrage.

(13) L'ASTM D 4529/IP 381 peut être utilisée si la règlementation locale l'autorise.

(14) La DEF STAN 91-91/7 Amendement 2 référence désormais l'IP 598 au lieu de l'IP 57. La nouvelle méthode de mesure du point de fumée IP 598 inclue à la fois la méthode classique manuelle et une méthode automatique. La note 12 de la DEF STAN 91-91/7 Amendement 2 indique l'intention de rendre la méthode automatique comme référence en Janvier 2014

(15) L'examen du tube pour déterminer la cotation visuelle avec le "Visual Tuberator" doit être effectué dans les 120 minutes suivant la fin du test. C'est la cotation visuelle du tube qui doit être reportée. L'attention est attirée sur la note **15** de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 qui stipule que seul des tubes approuvés peuvent être utilisés et liste les tubes JFTOT de PAC-Aclor et Falex comme techniquement appropriés

(16) L'attention est attirée sur la note 17 de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 qui stipule que "la limite de 70 concernant le test MSEP s'appliquera à l'endroit où l'Additif Dissipateur de Charges Electrostatiques (SDA) est ajouté à la fabrication. Aucune valeur de précision n'est connue pour les carburants contenant du SDA : si les tests du MSEP sont effectués en un point de la distribution "aval", aucune limite de spécification ne s'applique et les résultats ne peuvent pas constituer, à eux seuls, une justification du refus du produit". Un protocole donnant des recommandations sur la conduite à tenir dans le cas d'un test MSEP non satisfaisant est disponible dans le Bulletin n°14 du Joint Inspection Group, MSEP Protocol à www.jigonline.com rubrique 'fuel quality'".Si le SDA est ajouté en aval du point de fabrication, il est admis que les résultats des tests MSEP sont inférieurs à 70.

(17) Suite aux exigences de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2, les limites de conductivité sont impératives pour que le produit réponde à cette spécification. Toutefois, il est admis que dans certaines configurations de production et de distribution, il est plus pratique d'injecter l'Additif Dissipateur de Charges Electrostatiques (SDA) plus en aval. Dans de tels cas, il convient d'annoter le Certificat de Qualité couvrant le lot de la façon suivante : "Le produit répond aux exigences de l'AFQRJOS "Check List **27**" excepté pour la conductivité électrique". Dans certains cas, la conductivité peut chuter rapidement et le carburant ne plus réagir à un ajout complémentaire de Stadis 450. Dans de tels cas, le carburant peut être mis en livraison avec une conductivité réduite jusqu'à un minimum de 25pS/m sous réserve que le carburant ait passé tous les tests de l a spécification et que l'Autorisation de Mise en Livraison du réservoir soit annotée avec l'explication : « Produit mis en livraison au dessous de 50pS/m du fait de pertes de conductivité conformément à l'Annexe H de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 ».

(18) L'origine de cette exigence vient de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. L'exigence de détermination de la lubrifiance s'applique seulement aux carburants contenant a) au moins 20 % de produits fortement hydrotraités et moins de 5% de composants non hydrotraités ou lorsqu'ils contiennent des composants synthétiques. La limite s'applique seulement au point de production. Voir une remarque importante sur la lubrifiance des carburéacteurs dans l'annexe B de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. Les additifs CI/LI (aussi dénommés LIA) peuvent être utilisés pour améliorer la lubrifiance; seuls ceux listés dans la Table 2 de l'ASTM D1655-12/ Annexe A de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 sont permis. Noter que deux nouveaux additifs ont été ajoutés à la liste de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. Voir aussi les recommandations sur le point d'additivation de l'annexe A.5 de DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. S i le CI/LI est ajouté en aval du point de fabrication, un soin doit être pris pour s'assurer de ne pas dépasser la la dose maximale.
Les inhibiteurs de corrosion et améliorants de lubrifiance peuvent être ajoutés sans le consentement des participants

(19) Les additifs antioxydants sont obligatoires pour les carburéacteurs hydrotraités et synthétiques et doivent être ajoutés imédiatement après l'hydrotraitement ou la synthèse et avant que le produit ne soit stocké afin d'empêcher la peroxydation et la formation de gommes après la fabrication.
La liste des additifs antioxydants approuvés figure dans l'annexe A.2.4 de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 tout comme les références appropriées de qualification RDE / A / XXX- Celles ci doivent être mentionnées sur le Certificat de Qualité. Se reporter aussi à l'Annexe A2.6 fixant les exigences d'indication sur les CQ des concentrations en additif antioxydant.

(20) L'additif de désactivant de métal (MDA), RDE/A/650 approuvé est cité dans l'annexe A.3 de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2. Voir aussi l'annexe A.3.1 au sujet de la nécessité de noter la stabilité thermique avant et après utilisation du MDA au cas où la contamination du Jet A-1 par des traces d'un des métaux listés dans cette annexe ne serait pas prouvée. Noter que dans A.3.3 le maximum d'additivation au point de fabrication ou en cas de première additivation est de 2 mg/litre.

(21) Les concentrations en additif anti glace (FSII) inférieures à 0,02 % en volume peuvent être considérées comme négligeables et ne nécessitent pas d'agrément ou de notification. L'assentiment sur la présence de petites quantités d'anti glace sans agrément ou notification vise à faciliter le remplacement de Jets contenant de l'anti glace par d'autres qui en sont exempts lorsqu'un produit additivé a pu séjourner dans les installations un temps limité. Cet accord n'autorise pas l'additivation d'anti glace en continu à faible concentration. L'attention est portée à la nouvelle note de l'annexe A.6 de la DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 stipulant que les filtres monitors ne doivent pas être utilisés avec des carburants contenant de l'additif anti glace.

(22) La DEF STAN 91-91 ISSUE 7 AMENDEMENT 2 et l'ASTM D 1655-12 attirent l'attention sur la nécessité de mettre en place un Management du Changement dans les raffineries produisant du Jet A-1. Les conséquences de tout changement de matière première, de mode opératoire ou d'additif de procédés, sur la qualité et sur les performances du produit fini, doivent obligatoirement être étudiées (par exemple, l'expérience a montré que des traces de certains additifs de procédés pouvaient se retrouver dans les carburants aviation).

(23) Il est normal de certifier la conformité à une spécification primaire. Les expressions suivantes devraient être utilisées:
"Il est certifié que les échantillons ont été testés suivant les Méthodes en vigueur et que le Lot représenté par les échantillons est conforme à la DEF STAN 91-91 ISSUE 7 AMENDEMENT2 et à la Checklist de l'AFQRJOS issue **27**.
Ou
« "Il est certifié que les échantillons ont été testés suivant les Méthodes en vigueur et que le Lot représenté par les échantillons est conforme à l'ASTM D1655-12 et à la Checklist de l'AFQRJOS issue **27**.

Les exigences minimales d'informations à inclure sur le certificat de qualité associé au lot de carburéacteur fabriqué en raffinerie doivent être :
Intitulé de la spécification, source et numéro de version
Nom et adresse du laboratoire
Numéro du lot et identifiant unique
Quantité de carburéacteur du lot
Propriétés testées, incluant les limites de spécification, la méthode d'essai et le résultat de l'essai
Additifs, incluant la référence de l'agrément et la quantité ajoutée
Nom et position du signataire valideur autorisé du certificat ou signature électronique
Date de certification

(24) Les types et les concentrations de tous les additifs utilisés doivent être notés sur les Certificats de Qualité et autres documents relatifs à la qualité fournis par la raffinerie
Lorsque les additifs sont dilués (avec des solvants hydrocarbures uniquement) pour faciliter leur mise en œuvre avant additivation, c'est la concentration de l'additif lui-même (avant dilution) qui doit être mentionnée.
Voir l'Annexe A de la DEF STAN 91-91/7 amendement 2 pour plus de précisions.